



Л. И. ВОЛКОВЫСКИЙ, Г. Л. ЛУНЦ,
И. Г. АРАМАНОВИЧ

СБОРНИК ЗАДАЧ
ПО ТЕОРИИ ФУНКЦИЙ
КОМПЛЕКСНОГО
ПЕРЕМЕННОГО

ИЗДАТЕЛЬСТВО «НАУКА»

PROBLEMS IN THE THEORY OF FUNCTIONS OF A COMPLEX VARIABLE

L. VOLKOVYISKY, G. LUNTS,
I. ARAMANOVICH

Translated from the Russian
by
VICTOR SHIFFER

MIR PUBLISHERS • MOSCOW

First published 1972

Second edition 1977

Revised from the 1975 Russian edition

На английском языке

© Издательство << Наука >>, 1975

© English translation, Mir Publishers, 1977

CONTENTS

Introduction	9
------------------------	---

CHAPTER I

COMPLEX NUMBERS AND FUNCTIONS OF A COMPLEX VARIABLE

§ 1.	Complex Numbers	11
	The Geometric Interpretation of Complex Numbers	11
	Stereographic Projection	14
§ 2.	Elementary Transcendental Functions	16
§ 3.	Sequences and Number Series	19
§ 4.	Functions of a Complex Variable	21
	Complex Functions of a Real Variable	21
	Functions of a Complex Variable	21
	Continuity	22
§ 5.	Analytic and Harmonic Functions	23
	Cauchy-Riemann Conditions	23
	Formal Cauchy Derivatives	24
	Harmonic Functions	26
	Geometric Meaning of the Modulus and Argument of a Derivative	29

CHAPTER II

CONFORMAL MAPPINGS CONNECTED WITH ELEMENTARY FUNCTIONS

§ 1.	Linear Functions	30
	Entire Linear Functions	30
	Linear-Fractional Functions	31
§ 2.	Supplementary Problems in the Theory of Linear Transformations	35
	Canonical Forms of Linear Transformations	35
	Some Approximate Formulas for Linear Transformations	37
	Mapping Elementary Doubly Connected Domains	38

Group Properties of Linear-Fractional Transformations	39
Linear Transformations and Lobachevskian Geometry	41
§ 3. Rational and Algebraic Functions	42
Riemann-Schwarz Symmetry Principle	42
Principle of Corresponding Boundaries	42
Mapping Circular Lunes and Domains with Cuts	44
Zhukovsky's Function	45
Applying the Symmetry Principle	47
Simple Multivalent Mappings	49
§ 4. Elementary Transcendental Functions	50
Basic Transcendental Functions	50
Mappings That Reduce to Mappings of Strips and Semistrips	53
Applying the Symmetry Principle	53
Elementary Multivalent Mappings	56
§ 5. Boundaries of Univalence, Convexity and Starlikeness	57

CHAPTER III

INTEGRALS AND POWER SERIES

§ 1. Integration of Functions of a Complex Variable	59
§ 2. Cauchy's Integral Theorem	63
§ 3. Cauchy's Integral Formula	65
§ 4. Power Series	67
Finding the Radius of Convergence	67
Behaviour on the Boundary of the Circle of Convergence	67
Abel's Second Theorem	68
§ 5. Taylor's Series	69
Expanding Functions into Taylor's Series	69
Generating Functions of Systems of Polynomials	71
Solving Differential Equations	73
§ 6. Some Applications of Cauchy's Integral Formula and of Power Series	73
Zeros of Analytic Functions	73
Uniqueness Theorem	74
Expressing an Analytic Function in Terms of Its Real and Imaginary Part	75
Cauchy Inequalities	75
Area Theorems for Univalent Functions	76
Maximum-Modulus Principle	77

CHAPTER IV

THE LAURENT SERIES.
SINGULAR POINTS OF SINGLE-VALUED
ANALYTIC FUNCTIONS.
RESIDUES AND THEIR APPLICATIONS

§ 1. The Laurent Series	78
§ 2. Singular Points of Single-Valued Analytic Functions	80
§ 3. Evaluating Residues	83

§ 4. Evaluating Integrals	85
Direct Application of the Theorem of Residues	85
Definite Integrals	87
Integrals Associated with the Inverse Formula of the Laplace Transformation	99
Asymptotic Behaviour of Integrals	102
§ 5. Distribution of Zeros. Inversion of Series	106
Rouché's Theorem	106
Argument Principle	107
Series Inversion	110

CHAPTER V

VARIOUS SERIES OF FUNCTIONS.
PARAMETRIC INTEGRALS

§ 1. Series of Functions	112
§ 2. Dirichlet Series	115
§ 3. Parametric Integrals	117
Convergence of Integrals	117
Laplace's Integral	119

CHAPTER VI

INFINITE PRODUCTS.
ENTIRE AND MEROMORPHIC FUNCTIONS

§ 1. Infinite Products	121
§ 2. Partial Fraction and Infinite Product Expansions. -Summation of Series	124
§ 3. Characteristics of the Growth of Entire Functions	128

CHAPTER VII

CAUCHY-TYPE INTEGRALS.
INTEGRAL FORMULAS OF POISSON AND SCHWARZ

§ 1. Cauchy-Type Integrals	132
§ 2. Dirichlet's Integral, Harmonic Functions, Logarithmic Potential and Green's Function	138
§ 3. Poisson's Integral, Schwarz' Formula, Harmonic Measure	141

CHAPTER VIII

ANALYTIC CONTINUATION.
SINGULARITIES OF MANY-VALUED CHARACTER.
RIEMANN SURFACES

§ 1. Analytic Continuation	147
§ 2. Singularities of Many-Valued Character. Riemann Surfaces	153

CHAPTER IX
CONFORMAL MAPPINGS (CONTINUED)

§ 1. Schwarz-Christoffel Formula	161
§ 2. Conformal Mappings Involving the Use of the Elliptic Functions	177

CHAPTER X
APPLICATIONS TO MECHANICS
AND PHYSICS

§ 1. Applications to Hydrodynamics	186
§ 2. Applications to Electrostatics	197
§ 3. Applications to the Plane Problem of Heat Conduction	209

CHAPTER XI
GENERALIZATION OF ANALYTIC FUNCTIONS

§ 1. Quasi-Conformal Mappings	211
§ 2. Generalized Analytic Functions	217
§ 3. Certain Integral Relations and Double Integrals	219
ANSWERS AND SOLUTIONS	222

INTRODUCTION

The present book is meant primarily for the students of mechano-mathematical and physico-mathematical departments of universities, teacher colleges, and technical colleges with advanced programs in mathematics. It also includes a number of problems outside the standard curriculum.

We believe that the text will also be used to advantage by students majoring in fluid dynamics, the theory of elasticity, and electrical engineering.

To acquire the necessary theoretical background one should make a thorough study of a relevant text-book such as *The Theory of Functions of a Complex Variable* by A. Sveshnikov and A. Tikhonov, Mir Publishers, Moscow, 1971.

All hints to solutions are given in the text. The most difficult problems, marked with asterisks, are solved in the last part of the book.

CHAPTER I

COMPLEX NUMBERS AND FUNCTIONS OF A COMPLEX VARIABLE

In this chapter and throughout the book the following notation is accepted unless otherwise stated: $z = x + iy = re^{i\varphi}$, $w = u + iv = \rho e^{i\theta}$ ($x, y, u, v, \rho, \varphi$, and θ real numbers, $r \geq 0, \rho \geq 0$), $\operatorname{Re} z = x$, $\operatorname{Im} z = y$, $\operatorname{Arg} z = \varphi$, $|z| = r$, $\bar{z} = x - iy$. If no additional instructions are given, the principal value of the argument $\arg z$ is specified by the inequalities $-\pi < \arg z \leq \pi$; the complex plane whose points are represented by the complex numbers z is called here the z -plane; ordinarily the terms "the complex number z " and "the point z " are synonymous.

§ 1. COMPLEX NUMBERS

THE GEOMETRIC INTERPRETATION OF COMPLEX NUMBERS

1. Perform the following operations:

1) $\frac{1}{i}$; 2) $\frac{1-i}{1+i}$; 3) $\frac{2}{1-3i}$; 4) $(1+i\sqrt{3})^3$.

2. Find the moduli and arguments of the following complex numbers (a and b real numbers):

1) $3i$; 2) -2 ; 3) $1+i$; 4) $-1-i$; 5) $2+5i$; 6) $2-5i$;
7) $-2+5i$; 8) $-2-5i$; 9) bi ($b \neq 0$); 10) $a+bi$ ($a \neq 0$).

3. Solve the equation $\bar{z} = z^{n-1}$ ($n \neq 2$ a natural number).

4. Find all the values of the following roots and plot them:

1) $\sqrt[3]{1}$; 2) $\sqrt[3]{i}$; 3) $\sqrt[4]{-1}$; 4) $\sqrt[6]{-8}$; 5) $\sqrt[8]{1}$; 6) $\sqrt{1-i}$;
7) $\sqrt{3+4i}$; 8) $\sqrt[3]{-2+2i}$; 9) $\sqrt[5]{-4+3i}$.

5. Prove that both values of $\sqrt{z^2-1}$ lie on a straight line passing through the origin of coordinates and parallel to the

bisector of the interior angle of a triangle with vertices at the points -1 , 1 and z which passes through the vertex z .

6. Let m and n be integers. Show that $(\sqrt[n]{z})^m$ has $n/(n, m)$ distinct values, where (n, m) is the greatest common divisor of the numbers m and n . Verify that the sets of values $(\sqrt[n]{z})^m$ and $\sqrt[n]{z^m}$ coincide if and only if $(n, m) = 1$, i. e. if n and m are relatively prime.

7. Arguing geometrically, prove the following inequalities:

$$1) |z_1 + z_2| \leq |z_1| + |z_2|; \quad 2) |z_1 - z_2| \geq ||z_1| - |z_2||.$$

Prove the same inequalities algebraically. In each case determine when the equality sign holds.

8. Arguing geometrically, prove the following inequalities:

$$1) \left| \frac{z}{|z|} - 1 \right| \leq |\arg z|; \quad 2) |z - 1| \leq |z - 1| + |z| |\arg z|.$$

9. Prove the identity:

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$$

and ascertain its geometric meaning.

10. Prove the identity:

$$|1 - \bar{z}_1 z_2|^2 - |z_1 - z_2|^2 = (1 - |z_1|^2)(1 - |z_2|^2).$$

11. Prove the inequality:

$$|z_1 + z_2| \geq \frac{1}{2}(|z_1| + |z_2|) \left| \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right|.$$

12. Let z_1 and z_2 be arbitrary complex numbers, and a_1 and a_2 real numbers ($a_1^2 + a_2^2 \neq 0$). Prove the inequalities:

$$|z_1|^2 + |z_2|^2 - |z_1^2 + z_2^2| \leq 2 \frac{|a_1 z_1 + a_2 z_2|^2}{a_1^2 + a_2^2} \leq |z_1|^2 + |z_2|^2 + |z_1^2 + z_2^2|.$$

Hint: Introduce an auxiliary angle α such that $\tan \alpha = a_1/a_2$; represent the expression to be estimated as: $A + B \sin 2\alpha + C \cos 2\alpha$, and find its maximum and minimum values.

13. Prove the following identities:

$$1) (n-2) \sum_{k=1}^n |a_k|^2 + \left| \sum_{k=1}^n a_k \right|^2 = \sum_{1 \leq k < s \leq n} |a_k + a_s|^2;$$

$$2) n \sum_{k=1}^n |a_k|^2 - \left| \sum_{k=1}^n a_k \right|^2 = \sum_{1 \leq k < s \leq n} |a_k - a_s|^2.$$

14. Prove the following:

1) If $z_1 + z_2 + z_3 = 0$ and $|z_1| = |z_2| = |z_3| = 1$, then the points z_1, z_2, z_3 are the vertices of an equilateral triangle inscribed in the unit circle.

2) If $z_1 + z_2 + z_3 + z_4 = 0$ and $|z_1| = |z_2| = |z_3| = |z_4|$, then the points z_1, z_2, z_3, z_4 are either the vertices of a rectangle or coincide in pairs.

15. Find the vertices of a regular n -gon if its centre is located at the point $z=0$ and one of its vertices z_1 is known.

16. Points z_1 and z_2 are adjacent vertices of a regular n -gon. Find the vertex z_3 adjacent to z_2 ($z_3 \neq z_1$).

17. Given: three vertices of a parallelogram z_1, z_2, z_3 . Required: to find the fourth vertex z_4 , which is opposite to z_2 .

18. Under what condition are the three pairwise noncoincident points z_1, z_2, z_3 collinear?

19. Under what condition do the four pairwise noncoincident points z_1, z_2, z_3, z_4 lie on one circle or on one straight line?

20*. Points $z_1, z_2, \dots, z_n$ lie on one side of a straight line passing through the origin of coordinates. Prove that the points $1/z_1, 1/z_2, \dots, 1/z_n$ possess a similar property (indicate the line relative to which this is so) and that

$$z_1 + z_2 + \dots + z_n \neq 0, \quad \frac{1}{z_1} + \frac{1}{z_2} + \dots + \frac{1}{z_n} \neq 0.$$

21. Prove that if $z_1 + z_2 + \dots + z_n = 0$, then any straight line passing through the origin of coordinates separates the points $z_1, z_2, \dots, z_n$, provided only that they do not lie on this line.

22. Prove that any straight line passing through the centre of gravity of a system of material points $z_1, z_2, \dots, z_n$ having masses $m_1, m_2, \dots, m_n$ separates these points, provided only that they do not lie on this line.

In Problems 23 to 34 it is required to determine the geometric meaning of the indicated relations.

23. $|z - z_0| < R$; $|z - z_0| > R$; $|z - z_0| = R$.

24. $|z - 2| + |z + 2| = 5$. 25. $|z - 2| - |z + 2| > 3$.

26. $|z - z_1| = |z - z_2|$. 27. 1) $\operatorname{Re} z \geq C$; 2) $\operatorname{Im} z < C$.

28. $0 < \operatorname{Re}(iz) < 1$.

29. $\alpha < \arg z < \beta$; $\alpha < \arg(z - z_0) < \beta$ ($-\pi < \alpha < \beta \leq \pi$).

30. $|z| = \operatorname{Re} z + 1$. 31. $\operatorname{Re} z + \operatorname{Im} z < 1$.

32. $\operatorname{Im} \frac{z - z_1}{z - z_2} = 0$; $\operatorname{Re} \frac{z - z_1}{z - z_2} = 0$. 33. $|2z| > |1 + z^2|$.

34. 1) $|z| < \arg z$ if $0 \leq \arg z < 2\pi$;

2) $|z| < \arg z$ if $0 < \arg z \leq 2\pi$.

In Problems 35 to 38 it is required to determine families of lines in the z -plane specified by the appropriate equations.

35. 1) $\operatorname{Re} \frac{1}{z} = C$; 2) $\operatorname{Im} \frac{1}{z} = C$ ($-\infty < C < \infty$).

36. 1) $\operatorname{Re} z^2 = C$; 2) $\operatorname{Im} z^2 = C$ ($-\infty < C < \infty$).

37. $\left| \frac{z-z_1}{z-z_2} \right| = \lambda$ ($\lambda > 0$). 38. $\arg \frac{z-z_1}{z-z_2} = \alpha$ ($-\pi < \alpha \leq \pi$).

39. 1) A family of lines in the z -plane is given by the following equation:

$$|z^2 - 1| = \lambda \quad (\lambda > 0).$$

For what values of λ will the lines of the family consist of one simple curve and for what values will they decompose?

2) Answer the same questions concerning the following family:

$$|z^2 + az + b| = \lambda \quad (\lambda > 0).$$

40. Determine the maximum and the minimum distances from the origin of coordinates to the points of the given line ($a > 0$):

1) $\left| z + \frac{1}{z} \right| = a$; 2) $\left| z + \frac{b}{z} \right| = a$.

41. The function $\arg z$ is defined uniquely at any point $z \neq 0$ if we put $|z| - 2\pi < \arg z \leq |z|$. What is the locus of points at which continuity of the function $\arg z$ thus defined breaks down?

42. What is the locus of points at which continuity of a function $\arg z$, uniquely defined at any $z \neq 0$ by the inequalities $\ln |z| - 2\pi < \arg z \leq \ln |z|$, breaks down?

43. The initial value of $\operatorname{Arg} f(z)$ at $z=2$ is assumed equal to zero. The point z performs one complete revolution (counterclockwise) about a circle with centre at the origin and returns to the point $z=2$. Assuming that $\operatorname{Arg} f(z)$ varies continuously as z moves, indicate the value of $\operatorname{Arg} f(2)$ after the indicated circuit if:

1) $f(z) = \sqrt{z-1}$; 2) $f(z) = \sqrt[3]{z-1}$; 3) $f(z) = \sqrt{z^2-1}$;

4) $f(z) = \sqrt{z^2+2z-3}$; 5) $f(z) = \sqrt{\frac{z-1}{z+1}}$.

STEREOGRAPHIC PROJECTION

44. Deduce the formulas of a stereographic projection expressing the coordinates (ξ, η, ζ) of the point P of a sphere of diameter 1 tangent to the z -plane at the origin in terms of the coordinates (x, y) of the corresponding point z . Likewise express x and y in terms of ξ, η, ζ (the axes ξ and η are assumed to be coincident with the respective axes x and y).

Note. In Problem 44 a correspondence is set up between the points of the complex plane and those of a sphere of radius $1/2$ tangent to this plane. There is another method of correspondence

in which the sphere has a radius equal to 1 and the z -plane passes through its centre.

45. On a sphere, what are the images of the points 1 , -1 , i , $(1-i)/\sqrt{2}$?

46. In a plane, what is the image of a parallel of latitude β ($-\frac{\pi}{2} < \beta < \frac{\pi}{2}$)? What do the "north" and "south" poles correspond to?

47. On a sphere find the images of

1) the rays $\arg z = \alpha$; 2) the circles $|z| = r$.

48. On a sphere, what is the location of images of pairs of symmetrical points:

1) relative to the point $z=0$,

2) relative to the real axis,

3) relative to the unit circle?

49. Under what condition are the points z_1 and z_2 stereographic projections of two diametrically opposite points of a sphere?

50. Under what transformation of a sphere does the image of a point z go into that of the point $1/z$?

51. On a sphere, find the images of domains defined by the following inequalities:

1) $\operatorname{Im} z > 0$; 2) $\operatorname{Im} z < 0$; 3) $\operatorname{Re} z > 0$;

4) $\operatorname{Re} z < 0$; 5) $|z| < 1$; 6) $|z| > 1$.

52. What does a family of parallel straight lines in a plane correspond to on a sphere?

53. Prove that, in stereographic projection, circles situated on a sphere are projected into circles or straight lines in a plane. What circles on a sphere correspond to straight lines?

54. Let K be a circle in a plane corresponding to the circle K' on a sphere, let N be the north pole of the sphere and S the vertex of a cone in contact with the sphere along K' (it is assumed that K' is not a great circle). Prove that the centre of the circle K lies on the ray NS . Consider the case when K' is a great circle.

55. Prove that, in stereographic projection, angles between curves on a sphere are equal to the angles between their images in a plane.

56. Find the length $k(z, a)$ of the chord joining the points on a sphere corresponding to the points z and a . Also consider the case when a is the point at infinity.

57. Given: the points z_1 and z_2 (one of the two may be at infinity). Find the locus of points in the z -plane to which there corresponds on a sphere a circle equidistant from the images of the given points.

§ 2. ELEMENTARY TRANSCENDENTAL FUNCTIONS

By definition:

$$\exp z = e^z = e^x (\cos y + i \sin y);$$

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}; \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i}; \quad \tan z = \frac{\sin z}{\cos z}; \quad \cot z = \frac{\cos z}{\sin z};$$

$$\cosh z = \frac{e^z + e^{-z}}{2}; \quad \sinh z = \frac{e^z - e^{-z}}{2}; \quad \tanh z = \frac{\sinh z}{\cosh z}; \quad \coth z = \frac{\cosh z}{\sinh z}.$$

58. Using the definition of e^z , prove that:

1) $e^{z_1} \times e^{z_2} = e^{z_1 + z_2}$; 2) $e^{z + 2\pi i} = e^z$.

3) If $e^{z+\omega} = e^z$ for any z , then

$$\omega = 2\pi ki \quad (k = 0, \pm 1, \pm 2, \dots).$$

The relation $\exp i\varphi = e^{i\varphi} = \cos \varphi + i \sin \varphi$ (Euler's formula) enables us to use the exponential form of a complex number, $z = re^{i\varphi}$, instead of its trigonometrical form $z = r(\cos \varphi + i \sin \varphi)$. From now on, φ is usually understood as the principal value of the argument, i.e. $-\pi < \varphi \leq \pi$.

59. Give the following numbers in exponential notation: 1, -1, i , $-i$, $1+i$, $1-i$, $-1+i$, $-1-i$.

60. Find $e^{\pm \pi i/2}$; $e^{k\pi i}$ ($k = 0, \pm 1, \pm 2, \dots$).

61. Find the moduli and the principal values of the arguments of the following complex numbers:

$$e^{a+i}; e^{a-3i}; e^{3+4i}; e^{-3-4i}; -ae^{i\varphi} \quad (a > 0, |\varphi| \leq \pi);$$

$$e^{-i\varphi} \quad (|\varphi| \leq \pi); e^{i\alpha} - e^{i\beta} \quad (0 \leq \beta < \alpha \leq 2\pi).$$

62. Find the following sums:

1) $1 + \cos x + \cos 2x + \dots + \cos nx$;

2) $\sin x + \sin 2x + \dots + \sin nx$;

3) $\cos x + \cos 3x + \dots + \cos (2n-1)x$;

4) $\sin x + \sin 3x + \dots + \sin (2n-1)x$;

5) $\sin x - \sin 2x + \dots + (-1)^{n-1} \sin nx$.

63. Find the following sums:

1) $\cos \alpha + \cos (\alpha + \beta) + \dots + \cos (\alpha + n\beta)$;

2) $\sin \alpha + \sin (\alpha + \beta) + \dots + \sin (\alpha + n\beta)$.

64. Proceeding from the definition of the appropriate functions, prove the following:

1) $\sin^2 z + \cos^2 z = 1$; 2) $\sin z = \cos \left(\frac{\pi}{2} - z \right)$;

3) $\sin (z_1 + z_2) = \sin z_1 \cos z_2 + \cos z_1 \sin z_2$;

4) $\cos (z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2$;

5) $\tan 2z = \frac{2 \tan z}{1 - \tan^2 z}$; 6) $\cosh (z_1 + z_2) = \cosh z_1 \cosh z_2 + \sinh z_1 \sinh z_2$.

65. Prove that if $\cos(z + \omega) = \cos z$ for any z , then $\omega = 2\pi k$ ($k = 0, \pm 1, \pm 2, \dots$).

66. Prove that:

- 1) $\sin iz = i \sinh z$; 2) $\cos iz = \cosh z$;
3) $\tan iz = i \tanh z$; 4) $\cot iz = -i \coth z$.

67. Express the real and imaginary parts and also the moduli of the following functions in terms of trigonometric and hyperbolic functions of a real argument:

- 1) $\sin z$; 2) $\cos z$; 3) $\tan z$; 4) $\sinh z$; 5) $\cosh z$; 6) $\tanh z$.

68. Find the real and imaginary parts of the following values of the functions:

- 1) $\cos(2+i)$; 2) $\sin 2i$; 3) $\tan(2-i)$;
4) $\cot\left(\frac{\pi}{4} - i \ln 2\right)$; 5) $\coth(2+i)$; 6) $\tanh\left(\ln 3 + \frac{\pi i}{4}\right)$.

69. For each of the functions e^z , $\cos z$, $\sin z$, $\tan z$, $\cosh z$, $\coth z$ find a set of points z where the function has

- 1) real values,
2) purely imaginary values.

70. Find all values of z for which

- 1) $|\tan z| = 1$; 2) $|\tanh z| = 1$.

By definition $\operatorname{Ln} z = \ln r + i\varphi + 2\pi ik$ ($k = 0, \pm 1, \pm 2, \dots$), $\ln z = \ln r + i\varphi$ ($-\pi < \varphi \leq \pi$) ($\ln z$ is called the *principal value* of the quantity $\operatorname{Ln} z$).

71. Evaluate:

- 1) $\operatorname{Ln} 4$, $\operatorname{Ln}(-1)$, $\ln(-1)$; 2) $\operatorname{Ln} i$, $\ln i$;
3) $\operatorname{Ln} \frac{1+i}{\sqrt{2}}$; 4) $\operatorname{Ln}(2-3i)$, $\operatorname{Ln}(-2+3i)$.

72. Find the error in the reasoning leading to Bernoulli's paradox: $(-z)^2 = z^2$, and so $2 \operatorname{Ln}(-z) = 2 \operatorname{Ln} z$; hence $\operatorname{Ln}(-z) = \operatorname{Ln} z$ (!).

73. The initial value of $\operatorname{Im} f(z)$ at $z=2$ is taken to be zero. The point z performs one complete revolution anticlockwise around a circle centred at $z=0$ and returns to the point $z=2$. Assuming that $f(z)$ varies continuously as z moves, give the value of $\operatorname{Im} f(z)$ after the indicated circuit if:

- 1) $f(z) = 2 \operatorname{Ln} z$; 2) $f(z) = \operatorname{Ln} \frac{1}{z}$;
3) $f(z) = \operatorname{Ln} z - \operatorname{Ln}(z+1)$; 4) $f(z) = \operatorname{Ln} z + \operatorname{Ln}(z+1)$.

By definition, for any complex numbers $a \neq 0$ and α ,

$$a^\alpha = \exp \{\alpha \operatorname{Ln} a\} \quad (1)$$

or, if by e^z we assume, as usual¹⁾, $\exp z$, then $a^\alpha = e^{\alpha \operatorname{Ln} a}$.

¹⁾ According to (1), $e^z = \exp \{z \operatorname{Ln} e\} = \exp \{z(1+2\pi ik)\}$. But, unless otherwise stated, we shall assume $k=0$, i.e., as usual $e^z = \exp z$.

74. Find all the values of the following powers:

- 1) $1^{\sqrt{2}}$; 2) $(-2)^{\sqrt{2}}$; 3) 2^i ; 4) 1^{-i} ; 5) i^i ;
 6) $\left(\frac{1-i}{\sqrt{2}}\right)^{1+i}$; 7) $(3-4i)^{1+i}$; 8) $(-3+4i)^{1+i}$.

75. Show that in the case of a rational exponent ($\alpha = m/n$) the general definition of the power z^α coincides with the usual definition:

$$z^{\frac{m}{n}} = (\sqrt[n]{z})^m$$

(also see Problem 6).

76. Do the following sets of values coincide:

$$a^{2\alpha}, (a^\alpha)^2, (a^2)^\alpha?$$

By definition, the equality $w = \operatorname{Arc} \cos z$ is equivalent to the equality $z = \cos w$. The functions $\operatorname{Arc} \sin z$, $\operatorname{Arc} \tan z$, $\operatorname{Arc} \cot z$ and the inverse hyperbolic functions $\operatorname{Arc} \cosh z$, $\operatorname{Arc} \sinh z$, $\operatorname{Arc} \tanh z$, $\operatorname{Arccoth} z$ are defined similarly.

77. Prove the following equalities (all the values of the roots are taken):

- 1) $\operatorname{Arc} \cos z = -i \operatorname{Ln} (z + \sqrt{z^2 - 1})$;
 2) $\operatorname{Arc} \sin z = -i \operatorname{Ln} i (z + \sqrt{z^2 - 1})$;
 3) $\operatorname{Arc} \tan z = \frac{i}{2} \operatorname{Ln} \frac{1+z}{1-z} = \frac{1}{2i} \operatorname{Ln} \frac{1+iz}{1-iz}$;
 4) $\operatorname{Arc} \cot z = \frac{i}{2} \operatorname{Ln} \frac{z-i}{z+i}$; 5) $\operatorname{Arc} \cosh z = \operatorname{Ln} (z + \sqrt{z^2 - 1})$;
 6) $\operatorname{Arc} \sinh z = \operatorname{Ln} (z + \sqrt{z^2 + 1})$; 7) $\operatorname{Arc} \tanh z = \frac{1}{2} \operatorname{Ln} \frac{1+z}{1-z}$;
 8) $\operatorname{Arccoth} z = \frac{1}{2} \operatorname{Ln} \frac{z+1}{z-1}$.

78. Prove that for any value of $\operatorname{Arc} \cos z$ a value of $\operatorname{Arc} \sin z$ may be chosen such that the sum of these values is $\pi/2$. Prove a similar assertion for $\operatorname{Arc} \tan z$ and $\operatorname{Arc} \cot z$.

Note. The equalities $\operatorname{Arc} \sin z + \operatorname{Arc} \cos z = \pi/2$ and $\operatorname{Arc} \tan z + \operatorname{Arc} \cot z = \pi/2$ are always understood in the sense indicated in this problem.

79. Show that all the values of $\operatorname{Arc} \cos z$ are contained in the formula

$$\operatorname{Arc} \cos z = \pm i \operatorname{Ln} (z + \sqrt{z^2 - 1}),$$

where $\sqrt{z^2 - 1}$ is one of its values.

80. 1) For what values of z are all values of the functions $\operatorname{Arc} \cos z$, $\operatorname{Arc} \sin z$ and $\operatorname{Arc} \tan z$ real?

2) For what values of z has the function $\operatorname{Arc} \sinh z$ purely imaginary values?

81. Find all values of the following functions:

- 1) $\operatorname{Arc} \sin 1/2$; 2) $\operatorname{Arc} \cos 1/2$; 3) $\operatorname{Arc} \cos 2$; 4) $\operatorname{Arc} \sin i$;
 5) $\operatorname{Arc} \tan (1+2i)$; 6) $\operatorname{Arc} \cosh 2i$; 7) $\operatorname{Arc} \tanh (1-i)$.

82. Find all roots of the following equations:

- 1) $\sin z + \cos z = 2$; 2) $\sin z - \cos z = 3$;
 3) $\sin z - \cos z = i$; 4) $\cosh z - \sinh z = 1$;
 5) $\sinh z - \cosh z = 2i$; 6) $2 \cosh z + \sinh z = i$.

83. Find all roots of the following equations:

- 1) $\cos z = \cosh z$; 2) $\sin z = i \sinh z$; 3) $\cos z = i \sinh 2z$.

§ 3. SEQUENCES AND NUMBER SERIES

84. Prove that if $\sum_{n=1}^{\infty} c_n$ is a convergent series and $|\arg c_n| \leq \alpha < \frac{\pi}{2}$, then it converges absolutely.

85. Let $\sum_{n=1}^{\infty} c_n$ and $\sum_{n=1}^{\infty} c_n^2$ be convergent series. Prove that if $\operatorname{Re} c_n \geq 0$, then the series $\sum_{n=1}^{\infty} |c_n|^2$ will also converge.

86. The series $\sum_{n=1}^{\infty} c_n$ possesses the following property: its four parts, each consisting of terms contained in one and the same closed quadrant of the plane, converge. Prove that the given series is an absolutely convergent one.

87. Prove the following formula (Abel transformation):

$$\sum_{k=m}^n a_k b_k = \sum_{k=m}^{n-1} S_k (b_k - b_{k+1}) - S_{m-1} b_m + S_n b_n,$$

where $1 \leq m \leq n$, $S_k = a_1 + a_2 + \dots + a_k$ ($k \geq 1$), $S_0 = 0$.

88. Prove that for the series $\sum_{n=1}^{\infty} a_n b_n$, where $b_n > 0$, to be convergent, it is sufficient that the partial sums of the series $\sum_{n=1}^{\infty} a_n$ are bounded and the sequence of numbers $\{b_n\}$ tend monotonically to zero (Dirichlet's test).

Hint. Use the Abel transformation.

89. Prove that for the series $\sum_{n=1}^{\infty} a_n b_n$, where b_n are real numbers, to be convergent, it is sufficient that the series $\sum_{n=1}^{\infty} a_n$ converges and the sequence $\{b_n\}$ is monotonic and bounded (Abel's test).

90. Prove that for the series $\sum_{n=1}^{\infty} a_n b_n$ to be convergent it is sufficient to satisfy the following conditions:

- 1) $\lim_{n \rightarrow \infty} \sqrt[n]{n} b_n = 0$; 2) the series $\sum_{n=1}^{\infty} \sqrt[n]{n} |b_n - b_{n+1}|$ converges;
3) the sequence $\frac{S_n}{\sqrt[n]{n}}$, where $S_n = \sum_{k=1}^n a_k$, is bounded.

91. Let $\overline{\lim}_{n \rightarrow \infty} \sqrt[n]{|c_n|} = q$. Prove that $\sum_{n=1}^{\infty} c_n$ is an absolutely convergent series if $q < 1$, and a divergent one if $q > 1$.

92. By the examples of the series

$$1 + \frac{1}{2^\alpha} + \frac{1}{3^\alpha} + \frac{1}{4^\alpha} + \dots \quad (1 < \alpha < \beta),$$

$$\alpha + \beta^2 + \alpha^3 + \beta^4 + \dots \quad (0 < \alpha < \beta < 1).$$

show that the series $\sum_{n=1}^{\infty} c_n$ may converge even when $\overline{\lim}_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| > 1$.

93. Prove that if $\lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| = 1$, then for $\sum_{n=1}^{\infty} c_n$ to be an absolutely convergent series it is sufficient that $\overline{\lim}_{n \rightarrow \infty} n \left(\left| \frac{c_{n+1}}{c_n} \right| - 1 \right) < -1$ (Raabe's test).

94. Prove the Gauss test: if $\left| \frac{c_{n+1}}{c_n} \right| = 1 + \frac{a}{n} + o\left(\frac{1}{n}\right)$, where a is independent of n and $a < -1$, then the series converges absolutely.

In each of the following problems (95 to 104), investigate the series $\sum_{n=1}^{\infty} c_n$ for convergence.

95. $c_n = \frac{n}{(2i)^n}$. 96. $c_n = \frac{n!}{(in)^n}$. 97. $c_n = e^{in}$.

98. $c_n = \frac{e^{in}}{n}$. 99. $c_n = \frac{e^{in^2}}{n}$. 100. $c_n = \frac{e^{in}}{n^2}$. 101. $c_n = \frac{i}{n} e^{ni/n}$.

102. $c_n = \frac{\alpha(\alpha+1) \dots (\alpha+n-1) \beta(\beta+1) \dots (\beta+n-1)}{n! \gamma(\gamma+1) \dots (\gamma+n-1)}$ (hypergeometric series), $\operatorname{Re}(\alpha + \beta - \gamma) < 0$.

103. $c_n = \frac{\cos in}{2^n}$. 104. $c_n = \frac{n \sin in}{3^n}$.

105. Find the limit points of the following sets:

1) $z = 1 + (-1)^n \frac{n}{n+1}$ ($n = 1, 2, \dots$);

2) $z = \frac{1}{m} + \frac{i}{n}$ (m, n arbitrary integers);

3) $z = \frac{p}{m} + i \frac{q}{n}$ (m, n, p, q arbitrary integers);

4) $|z| < 1$.

106. Prove that a convergent subsequence may be chosen from an infinite bounded sequence of points $\{z_n\}$.

107. Prove the following:

1) Convergence of the sequence $\{z_n = x_n + iy_n\}$ is equivalent to a simultaneous convergence of the sequences $\{x_n\}$ and $\{y_n\}$.

2) For the limit $\lim_{n \rightarrow \infty} z_n \neq 0$ to exist it is necessary and sufficient that there exist the limits $\lim_{n \rightarrow \infty} |z_n| \neq 0$ and (for an appropriate definition of $\arg z_n$) $\lim_{n \rightarrow \infty} \arg z_n$. If $\lim_{n \rightarrow \infty} z_n$ is not a negative number, then we may, for instance, assume that $-\pi < \arg z_n \leq \pi$.

In what cases is convergence of the sequence $\{z_n\}$ equivalent to convergence of the sequence $\{|z_n|\}$ alone?

108. On the basis of the assertions of Problem 107, prove that:

1) $\lim_{n \rightarrow \infty} \left(1 + \frac{z}{n}\right)^n = e^x (\cos y + i \sin y)$;

2) $\lim_{n \rightarrow \infty} [n (\sqrt[n]{z} - 1)] = \ln r + i\varphi + 2\pi ik$ ($k = 0, 1, 2, \dots$).

§ 4. FUNCTIONS OF A COMPLEX VARIABLE

COMPLEX FUNCTIONS OF A REAL VARIABLE

In each of the following Problems (109 to 115) determine the lines specified by the indicated equations.

109. $z = 1 - it$, $0 \leq t \leq 2$. 110. $z = t + it^2$, $-\infty < t < \infty$.

111. $z = t^2 + it^4$, $-\infty < t < \infty$.

112. $z = a(\cos t + i \sin t)$, $\frac{\pi}{2} \leq t \leq \frac{3\pi}{2}$, $a > 0$.

113. $z = t + \frac{i}{t}$, $-\infty < t < 0$.

114. 1) $z = t + i\sqrt{1-t^2}$, $-1 \leq t \leq 1$; 2) $z = -t + i\sqrt{1-t^2}$, $-1 \leq t < 0$ (the positive square root is taken).

115. 1) $z = a(t + i - ie^{-it})$, $-\infty < t < \infty$, $a > 0$;

2) $z = ia + at - ibe^{-it}$, $0 \leq t \leq 2\pi$, $a > 0$, $b > 0$.

FUNCTIONS OF A COMPLEX VARIABLE

116. To map $w = z^2$ it is required:

1) to find the images of the lines $x = C$, $y = C$, $x = y$, $|z| = R$, $\arg z = \alpha$ and to determine which of them are transformed one-to-one;

2) to find the preimages (in the z -plane) of the lines $u=C$, $v=C$ ($w=u+iv$).

117. To map $w=1/z$ find:

- 1) the images of the lines $x=C$, $y=C$, $|z|=R$, $\arg z=\alpha$, $|z-1|=1$;
- 2) the preimages of the lines $u=C$, $v=C$.

118. To map $w=z+\frac{1}{z}$ and $w=z-\frac{1}{z}$ find the images of the circles $|z|=R$.

119. To transform $w=z+\frac{1}{z}$ find, in the z -plane, the preimage of the rectangular grid ($u=C$, $v=C$) of the w -plane.

120. Into what does the mapping $w=z/(1-z)^2$ carry the circle $|z|=1$?

121. To map $w=e^z$ find:

- 1) the images of the lines $x=C$, $y=C$, $x=y$;
- 2) the preimages of the line $\rho=\theta$ ($0\leq\theta<\infty$).

122. Find the transformation of the rectangular grid ($x=C$, $y=C$) of the z -plane with the aid of the indicated function:

- 1) $w=z^2+z$; 2) $w=\coth z$; 3) $w=e^{2z}$.

123. Into what are the segments of the straight lines $x=C$ and the straight lines $y=C$, lying in the strip $0\leq y\leq\pi$, transformed with the aid of the function $w=e^z+iz$?

124. What corresponds to the polar grid $|w|=R$, $\arg w=\alpha$ in the z -plane in the following transformations:

- 1) $w=e^{1/z}$; 2) $w=e^{z^2}$?

CONTINUITY

125. The function $f(z)$, defined in the neighbourhood of the point z_0 , is called Heine continuous at this point if for any sequence $\{z_n\}$ converging to z_0 the condition $\lim_{n\rightarrow\infty} f(z_n)=f(z_0)$ is satisfied.

The same function is called Cauchy continuous if for any $\varepsilon>0$ there exists a $\delta(\varepsilon)>0$ such that $|f(z)-f(z_0)|<\varepsilon$ follows from the inequality $|z-z_0|<\delta$. Prove that these definitions are equivalent.

126. The functions $\frac{\operatorname{Re} z}{z}$, $\frac{z}{|z|}$, $\frac{\operatorname{Re}(z^2)}{|z|^2}$, $\frac{z \operatorname{Re} z}{|z|}$ are defined for $z\neq 0$. Which of them may be redefined at $z=0$ so as to become continuous at this point?

127. Will the functions 1) $1/(1-z)$, 2) $1/(1+z^2)$ be continuous inside the unit circle ($|z|<1$)? Will they be uniformly continuous?

128. 1) Prove that the function $e^{-1/|z|}$ is uniformly continuous in the circle $|z|\leq R$ with point $z=0$ deleted.

2) Will the function e^{-1/z^2} be uniformly continuous in the same domain?

3) Will the function e^{-1/z^2} be uniformly continuous in the sector $0 < |z| \leq R$, $|\arg z| \leq \pi/6$?

129. The function $w = e^{-1/z}$ is defined everywhere, except at the point $z = 0$. Prove that:

1) in the semicircle $0 < |z| \leq 1$, $|\arg z| \leq \pi/2$ this function is bounded but not continuous;

2) inside the same semicircle the function is continuous but not uniformly continuous;

3) in the sector $0 < |z| \leq 1$, $|\arg z| \leq \alpha < \pi/2$ the function is uniformly continuous.

130. The function $f(z)$ is uniformly continuous in the circle $|z| < 1$. Prove that for any point ξ on the circle $|z| = 1$ and any sequence $z_n \rightarrow \xi$, $|z_n| < 1$ there exists the limit $\lim_{n \rightarrow \infty} f(z_n)$. Prove also that this limit does not depend on the choice of the sequence $\{z_n\}$ and that the function redefined on the boundary of the circle by a limit process will be continuous throughout the circle $|z| \leq 1$.

§ 5. ANALYTIC AND HARMONIC FUNCTIONS

CAUCHY-RIEMANN CONDITIONS

131. Check whether the Cauchy-Riemann conditions are satisfied for the functions z^n , e^z , $\cos z$, $\operatorname{Ln} z$ and prove that

$$(z^n)' = nz^{n-1}, (e^z)' = e^z, (\cos z)' = -\sin z, (\operatorname{Ln} z)' = \frac{1}{z}.$$

132. Find the constants a , b , c for which the function $f(z)$ is analytic:

1) $f(z) = x + ay + i(bx + cy)$;

2) $f(z) = \cos x (\cosh y + a \sinh y) + i \sin x (\cosh y + b \sinh y)$.

133. Find the domains in which the function

$$f(z) = |x^2 - y^2| + 2i|xy|$$

is analytic.

134. $f(z) = u + iv = \rho e^{i\theta}$ is an analytic function. Prove that if one of the functions u , v , ρ , θ is identically equal to a constant, then the function $f(z)$ is also constant.

135. Let $z = re^{i\varphi}$ and $f(z) = u(r, \varphi) + iv(r, \varphi)$. Write the Cauchy-Riemann equations in polar coordinates.

136. Prove that if $f(z) = u + iv$ is an analytic function and s and n are perpendicular vectors, the vector s rotated counterclockwise through a right angle to the vector n , then

$$\frac{\partial u}{\partial s} = \frac{\partial v}{\partial n} \quad \text{and} \quad \frac{\partial u}{\partial n} = -\frac{\partial v}{\partial s}$$

($\frac{\partial}{\partial s}$ and $\frac{\partial}{\partial n}$ are derivatives of functions of two real variables with respect to the corresponding direction).

137. Prove that the function $f(z) = \bar{z}$ is nowhere differentiable.

138. Prove that the function $w = z \operatorname{Re} z$ is differentiable only at the point $z = 0$; find $w'(0)$.

139. Prove that for the function $f(z) = \sqrt{|xy|}$ the Cauchy-Riemann conditions are satisfied at the point $z = 0$, but a derivative does not exist.

140. Prove the following assertions:

1) If at the point z the function $w = f(z)$ has the limit

$$\lim_{\Delta z \rightarrow 0} \left[\operatorname{Re} \frac{\Delta w}{\Delta z} \right],$$

then the partial derivatives u_x and v_y exist and are equal.

2) If the limit $\lim_{\Delta z \rightarrow 0} \left[\operatorname{Im} \frac{\Delta w}{\Delta z} \right]$ exists then the partial derivatives u_y and v_x exist and $u_y = -v_x$.

3) Assuming beforehand that the functions u and v are differentiable, the existence of either of the above limits (see (1) and (2)) ensures the existence of the other limit and, consequently, the differentiability of the function $f(z)$.

141. The function $w = f(z)$ possesses the following properties at the point z : 1) u and v are differentiable; 2) there exists a limit, $\lim_{\Delta z \rightarrow 0} \left| \frac{\Delta w}{\Delta z} \right|$. Prove that either $f(z)$ or $\overline{f(z)}$ is differentiable at the point z .

142. The function $w = f(z)$ possesses the following properties at the point z : 1) the functions u and v are differentiable; 2) there exists a limit $\lim_{\Delta z \rightarrow 0} \arg \frac{\Delta w}{\Delta z}$. Prove that $f(z)$ is differentiable at the point z .

143. Let $w = f(z) = u + iv$ and $u(x, y)$ and $v(x, y)$ be differentiable at the point z . Prove that the set of all possible limiting values of the ratio $\frac{\Delta w}{\Delta z}$ as $\Delta z \rightarrow 0$ is either a point or a circle.

FORMAL CAUCHY DERIVATIVES

If in the relation

$$w = w(z) = f(x, y) = f\left(\frac{z+\bar{z}}{2}, \frac{z-\bar{z}}{2i}\right) = \varphi(z, \bar{z})$$

z and $\bar{z}$ are considered to be independent variables, the derivatives

with respect to these variables are

$$\frac{\partial}{\partial z} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right), \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right).$$

We introduce the following notation:

$$\frac{\partial w}{\partial z} = w_z, \quad \frac{\partial w}{\partial \bar{z}} = w_{\bar{z}}, \quad \text{and so on.}$$

144. Prove the following relations:

$$1) dw = w_z dz + w_{\bar{z}} d\bar{z}; \quad 2) w_z = \frac{1}{2} [(u_x + v_y) + i(-u_y + v_x)];$$

$$3) w_{\bar{z}} = \frac{1}{2} [(u_x - v_y) + i(u_y + v_x)].$$

145. Prove that the Cauchy-Riemann equations are equivalent to the equation $w_{\bar{z}} = 0$.

146. Prove that Laplace's equation $\Delta u = 0$ may be written in the form $\frac{\partial^2 u}{\partial z \partial \bar{z}} = 0$.

147. Prove that $\overline{dw} = d\bar{w}$, $\overline{w_z} = \overline{w_{\bar{z}}}$, $\overline{w_{\bar{z}}} = \overline{w_z}$ (the long dash means that conjugates are taken after differentiation).

148. Prove that for the function $z(w)$, which is inverse to $w(z)$,

$$dz = \frac{\overline{w_{\bar{z}}}}{|w_z|^2 - |\overline{w_{\bar{z}}}|^2} dw + \frac{-\overline{w_z}}{|w_z|^2 - |\overline{w_{\bar{z}}}|^2} d\bar{w}.$$

149. Prove that the Jacobian of the transformation $w(z)$ is equal to

$$J_{w/z} = \frac{\partial(u, v)}{\partial(x, y)} = |w_z|^2 - |\overline{w_{\bar{z}}}|^2.$$

150. Prove the following equalities:

$$1) \frac{dw}{dz} = w_z + w_{\bar{z}} e^{-2i\alpha}, \quad \text{where } \alpha = \arg dz;$$

$$2) \max_{\alpha} \left| \frac{dw}{dz} \right| = |w_z| + |\overline{w_{\bar{z}}}|;$$

$$3) \min_{\alpha} \left| \frac{dw}{dz} \right| = ||w_z| - |\overline{w_{\bar{z}}}||.$$

151. Prove that if $\alpha = \arg dz$, $\alpha' = \arg dw$, then:

$$1) \frac{d\alpha'}{d\alpha} = \frac{J_{w/z}}{(u_x \cos \alpha + u_y \sin \alpha)^2 + (v_x \cos \alpha + v_y \sin \alpha)^2} = \left| \frac{dz}{dw} \right|^2 \frac{1}{J_{z/w}};$$

$$2) \max_{\alpha} \frac{d\alpha'}{d\alpha} = p, \quad \min_{\alpha} \frac{d\alpha'}{d\alpha} = \frac{1}{p}, \quad \text{where } p = \frac{\max \left| \frac{dw}{dz} \right|}{\min \left| \frac{dw}{dz} \right|} = \frac{|w_z| + |\overline{w_{\bar{z}}}|}{||w_z| - |\overline{w_{\bar{z}}}||}.$$

HARMONIC FUNCTIONS

A function $u(x, y)$, which has continuous partial derivatives of first and second orders in some region and which satisfies Laplace's equation

$$\Delta u \equiv \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0,$$

is called a *harmonic function*. Two harmonic functions $u(x, y)$ and $v(x, y)$ are said to be *conjugate harmonic* if they satisfy the Cauchy-Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

152. Prove the following propositions:

1) A linear combination of harmonic functions $\sum_{i=1}^n c_i u_i(x, y)$ is a harmonic function.

2) If the arguments of a harmonic function $u(x, y)$ are transformed thus $x = \xi/(\xi^2 + \eta^2)$, $y = \eta/(\xi^2 + \eta^2)$ (inversion transformation), the transformed function is harmonic.

3) If the arguments of a harmonic function $u(x, y)$ are transformed thus $x = \varphi(\xi, \eta)$, $y = \Psi(\xi, \eta)$, where φ and Ψ are conjugate harmonic functions, then the transformed function is harmonic. (Whence, in particular, follows the preceding assertion).

4) Let $u(x, y)$ and $v(x, y)$ be conjugate harmonic functions and let the Jacobian $\frac{\partial(u, v)}{\partial(x, y)}$ be nonzero in some region. Then the inverse functions $x(u, v)$ and $y(u, v)$ are also harmonic and conjugate.

153. 1) Prove that any function $u(x, y)$ harmonic in a singly connected domain G has a family of conjugate harmonic functions which differ from one another by a constant term:

$$v(x, y) = \int_{(x_0, y_0)}^{(x, y)} -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy + C.$$

2) Prove that if a domain G is multiply connected and bounded by an exterior contour Γ_0 and interior contours $\Gamma_1, \Gamma_2, \dots, \Gamma_n$ (Fig. 1) (each of which may degenerate into a point), then the function $v(x, y)$ may prove to be multiple-valued and the general formula for its values will have the form

$$v(x, y) = \int_{(x_0, y_0)}^{(x, y)} -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy + \sum_{k=1}^n m_k \pi_k + C.$$

The integral is taken along the path lying in the domain G , m_k are integers, and

$$\pi_k = \int_{\gamma_k} -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy,$$

where γ_k are simple closed contours, each containing one connected part of the boundary (Γ_k) (the numbers π_k are called the *periods of the integral* or *cyclic constants*).

For the function $v(x, y)$ to be single-valued, it is necessary and sufficient that all the numbers π_k equal zero.

Note. The contour Γ_0 may not exist if the function $u(x, y)$ is harmonic at the point at infinity. By definition this means that the function $U(\xi, \eta)$ obtained from the function $u(x, y)$ by inversion [see Problem 152 (2)] will be harmonic at the origin of coordinates. It can be proved that

in this case $\sum_{k=1}^n \pi_k = 0$.

154. Since an analytic function is infinitely differentiable, prove the following theorems:

1) The real part and imaginary part of the analytic function $f(z) = u + iv$ are conjugate harmonic functions.

2) Derivatives (of any order) of a harmonic function are also harmonic functions.

155. 1) Will u^2 be a harmonic function if u is harmonic?

2) Let u be a harmonic function. For what functions f will the function $f(u)$ also be harmonic?

156. Will the functions $|f(z)|$, $\arg f(z)$, $\ln |f(z)|$ be harmonic if $f(z)$ is an analytic function?

157. Transform the Laplace operator $\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ to the polar system of coordinates (r, φ) and find a solution to Laplace's equation $\Delta u = 0$ that depends only on r .

158. For $n = 1, 2, 3, 4$ write out the harmonic polynomials $p_n(x, y)$ and $q_n(x, y)$ defined by the equality $z^n = p_n + iq_n$. Give the general form of p_n and q_n in the polar system of coordinates.

Using the formulas of Problem 153, in each of the following problems (159 to 163) find the functions which are conjugate to the given harmonic functions in the indicated domains.

159. $u(x, y) = x^2 - y^2 + x$, $0 \leq |z| < \infty$.

160. $u(x, y) = \frac{x}{x^2 + y^2}$, $0 < |z| \leq \infty$.

161. $u(x, y) = \frac{1}{2} \ln(x^2 + y^2)$ a) in the domain obtained from a plane by removing the semiaxis: $y = 0$, $-\infty < x \leq 0$; b) in a plane with the origin ($0 < |z| < \infty$) deleted.

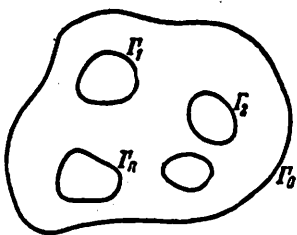


FIG. 1

(162). $u(x, y) = \frac{1}{2} \{ \ln(x^2 + y^2) - \ln[(x-1)^2 + y^2] \}$; a) in a plane with points $z=0$ and $z=1$ deleted; b) in the plane with the segment of the real axis $y=0$, $0 \leq x \leq 1$ deleted; c) in the plane with the ray $y=0$, $1 \leq x < \infty$ deleted.

163. $u(x, y) = \frac{1}{2} \sum_{k=1}^n \alpha_k \ln[(x-x_k)^2 + (y-y_k)^2]$; a) in the plane with the points $z_1, z_2, \dots, z_n$ ($z_k = x_k + iy_k$; $z_i \neq z_j$) deleted; b) in the plane with the deletion of a simple polygonal line (one without self-intersections) joining the given points.

164. Is there an analytic function $f(z) = u + iv$ for which
1) $u = \frac{x^2 - y^2}{(x^2 + y^2)^2}$; 2) $v = \ln(x^2 + y^2) - x^2 + y^2$; 3) $u = e^{y/x}$?

In each of the following problems (165 to 168) find analytic functions $f(z) = u + iv$ from the given real or imaginary part.

165. $u = x^2 - y^2 + 5x + y - \frac{y}{x^2 + y^2}$.

166. $u = e^x (x \cos y - y \sin y) + 2 \sin x \sinh y + x^3 - 3xy^2 + y$.

167. $v = 3 + x^2 - y^2 - \frac{y}{2(x^2 + y^2)}$.

168. $v = \ln(x^2 + y^2) + x - 2y$.

In Problems 169 to 176, determine whether there exist harmonic functions of the indicated kind (other than constant) and find them if they do.

169. $u = \varphi(x)$. 170. $u = \varphi(ax + by)$ (a and b real numbers).

171. $u = \varphi\left(\frac{y}{x}\right)$. 172. $u = \varphi(xy)$. 173. $u = \varphi(x^2 + y^2)$.

174. $u = \varphi\left(\frac{x^2 + y^2}{x}\right)$. 175. $u = \varphi(x + \sqrt{x^2 + y^2})$.

176. $u = \varphi(x^2 + y)$.

In Problems 177 to 180 prove the existence of analytic functions $f(z)$ and find them, given the modulus or the argument.

177. $\rho = (x^2 + y^2)e^x$.

178. $\rho = e^{r^2 \cos 2\varphi}$.

179. $\theta = xy$. 180. $\theta = \varphi + r \sin \varphi$.

181. Prove: for the family of lines $\varphi(x, y) = C$, where φ is a twice continuously differentiable function, to be the family of level lines of some harmonic function, it is necessary and sufficient that the ratio $\Delta\varphi/(\text{grad } \varphi)^2$ be dependent on φ alone.

Hint. First establish that the desired harmonic function has the form $u = f[\varphi(x, y)]$.

In Problems 182 to 186, find analytic functions such that the real part or the imaginary part or the modulus or the argument preserves a constant value along any line of the appropriate family.

182. $x = C$. 183. $y = C$. 184. $y = Cx$.

185. $x^2 + y^2 = C$. 186. $x^2 + y^2 = Cx$.

GEOMETRIC MEANING OF THE MODULUS AND ARGUMENT OF A DERIVATIVE

187. Mapping is accomplished with the aid of the functions $w = z^2$ and $w = z^3$. Find the angle of rotation (θ) of the direction issuing from the point z_0 , and find the stretching factor (k) at the following points:

1) $z_0 = 1$; 2) $z_0 = -\frac{1}{4}$; 3) $z_0 = 1 + i$; 4) $z_0 = -3 + 4i$.

188. What part of the plane is contracted and what part is stretched if the mapping is effected by the following functions:

1) $w = z^2$? 2) $w = z^2 + 2z$? 3) $w = \frac{1}{z}$? 4) $w = e^z$? 5) $w = \ln(z - 1)$?

189. A domain G is mapped by the function $f(z)$ conformally and one-to-one onto a domain G' . Give the formulas for calculating the area S of G' and the arc length L onto which is mapped a certain arc l belonging to G .

190. Find the length L of a helix onto which a segment $y = x$, $0 \leq x \leq 2\pi$, is mapped by the function e^z .

191. Find the area of the domain onto which the function e^z maps the rectangle $1 \leq x \leq 2$, $0 \leq y \leq 4$.

192. Find a domain D onto which the function e^z maps the rectangle $1 \leq x \leq 2$, $0 \leq y \leq 8$. Calculate the area of the domain D , using the formula obtained in solving Problem 189, and explain why this formula leads to an erroneous result.

CHAPTER II

CONFORMAL MAPPINGS CONNECTED WITH ELEMENTARY FUNCTIONS

§ 1. LINEAR FUNCTIONS

ENTIRE LINEAR FUNCTIONS

193. Find the entire linear function which maps a triangle with vertices at the points 0, 1 and i onto a similar triangle with vertices 0, 2 and $1+i$.

194. Find the entire linear transformation with a fixed point $1+2i$ which carries the point i into the point $-i$.

195. For the indicated transformations find the finite fixed point z_0 (if it exists), the angle of rotation about it, θ , and the stretching factor k . Reduce these transformations to canonical form $w - z_0 = \lambda(z - z_0)$.

1) $w = 2z + 1 - 3i$; 2) $w = iz + 4$; 3) $w = z + 1 - 2i$;

4) $w - w_1 = a(z - z_1)$ ($a \neq 0$); 5) $w = az + b$ ($a \neq 0$).

196. Find the general form of an entire linear transformation which maps:

1) the upper half-plane onto itself;

2) the upper half-plane onto the lower half-plane;

3) the upper half-plane onto the right half-plane;

4) the right half-plane onto itself.

Show that in all these cases the transformation is defined uniquely by giving one pair of corresponding interior points or two pairs of boundary points.

197. Find the general form of an entire linear transformation which maps:

1) the strip $0 < x < 1$ onto itself;

2) the strip $-2 < y < 1$ onto itself;

3) the strip bounded by the straight lines $y = x$ and $y = x - 1$, onto itself.

Find out which pairs of points can correspond to each other in these mappings and in what case this correspondence uniquely defines the mapping.

198. Find an entire linear function $w(z)$ which maps the strip contained between the given straight lines onto the strip $0 < u < 1$ with the given normalization:

- 1) $x=a$, $x=a+h$; $w(a)=0$;
- 2) $x=a$, $x=a+h$; $w\left(a+\frac{h}{2}\right)=\frac{1}{2}+i$, $\operatorname{Im} w\left(a+\frac{h}{2}+i\right)<1$.
- 3) $y=kx$, $y=kx+b$; $w(0)=0$;
- 4) $y=kx+b_1$, $y=kx+b_2$; $w(ib_1)=0$.

199. Find an entire linear function which maps the circle $|z|<1$ onto the circle $|w-w_0|<R$ so that the centres of the circles correspond to one another and the horizontal diameter is transformed into the diameter making an angle α with the direction of the real axis.

LINEAR-FRACTIONAL FUNCTIONS

200. For the function $w=\frac{1}{z}$ find the images of the following lines:

- 1) the family of circles $x^2+y^2=ax$;
- 2) the family of circles $x^2+y^2=by$;
- 3) the pencil of parallel lines $y=x+h$;
- 4) the pencil of straight lines $y=kx$;
- 5) the pencil of straight lines passing through a given point $z_0 \neq 0$;
- 6) the parabola $y=x^2$.

201. Find out into what the function $w=\frac{1}{z-z_0}+h$ transforms:

- 1) the rectangular grid $x=C$; $y=C$;
- 2) the polar grid $|z-z_0|=R$, $\arg(z-z_0)=\alpha$.

202. Given the function $w=\frac{z-z_1}{z-z_2}$.

1) Prove that the preimage of the family $|w|=\lambda$ ($0<\lambda<\infty$) is a family of circles (circles of Apollonius). For a given λ find the radius and location of the centre of the corresponding circle in the z -plane.

2) Find the preimages of the rays $\arg w=\theta$.

3) Construct a grid in the z -plane corresponding to a polar grid in the w -plane.

4) Find a domain in the z -plane corresponding to the semicircle $|w|<1$, $\operatorname{Im} w>0$.

In each of the following problems (203 to 207) determine how the indicated domains are transformed by the given mapping functions.

203. Quadrant $x > 0, y > 0$; $w = \frac{z-i}{z+i}$.

204. Semicircle $|z| < 1, \operatorname{Im} z > 0$; $w = \frac{2z-i}{2+iz}$.

205. Angle $0 < \varphi < \frac{\pi}{4}$; $w = \frac{z}{z-1}$.

206. Strip $0 < x < 1$; 1) $w = \frac{z-1}{z}$; 2) $w = \frac{z-1}{z-2}$.

207. Annulus $1 < |z| < 2$; $w = \frac{z}{z-1}$.

208. Map onto the vertical strip $0 < \operatorname{Re} w < 1$:

1) the half-plane $\operatorname{Re} z > 0$ with the circle $\left| z - \frac{d}{2} \right| \leq \frac{d}{2}$ deleted;

2) the lune formed by the intersection of the circles:

$$\left| z - \frac{d_1}{2} \right| = \frac{d_1}{2}, \quad \left| z - \frac{d_2}{2} \right| = \frac{d_2}{2}, \quad (d_1 < d_2);$$

3) the exterior of the circles $\left| z + \frac{d_1}{2} \right| = \frac{d_1}{2}, \left| z - \frac{d_2}{2} \right| = \frac{d_2}{2}$ so that $w(d_2) = 0$.

209. Find the linear-fractional functions which carry the points $-1, i, 1+i$ into the points: 1) $0, 2i, 1-i$; 2) $i, \infty, 1$, respectively.

210. Find the linear-fractional functions that transform the points $-1, \infty, i$ into the respective points:

1) $i, 1, 1+i$; 2) $\infty, i, 1$; 3) $0, \infty, 1$.

211. Find the linear-fractional functions, given the following conditions:

1) the points 1 and i are fixed, and the point 0 goes into -1 ;

2) the points $\frac{1}{2}$ and 2 are fixed, and the point $\frac{5}{4} + \frac{3}{4}i$ goes into ∞ ;

3) the point i is a double fixed point, and the point 1 goes into ∞ .

212. Find the linear-fractional function that transforms the points $-1, 0, 1$ into the points $1, i, -1$, respectively, and determine how the upper half-plane is transformed under this mapping.

213. Find the general form of a linear-fractional transformation which maps:

1) the upper half-plane onto itself;

2) the upper half-plane onto the lower half-plane;

3) the upper half-plane onto the right half-plane.

214. Find the mapping of the upper half-plane onto itself for the indicated normalization:

1) $w(0)=1$, $w(1)=2$, $w(2)=\infty$; 2) $w(0)=1$, $w(i)=2i$.

Note. For mappings of the upper half-plane onto itself for other normalizations see Problem 226.

215. Find the function $w(z)$ which maps the circle $|z| < R$ onto the right half-plane $\operatorname{Re} w > 0$ so that $w(R)=0$, $w(-R)=\infty$, $w(0)=1$. What is the image of the upper semicircle under this mapping?

Two points P_1 and P_2 are called symmetric with respect to a circle K with centre O and radius R if they lie on a common ray issuing from O and the product of their distances from the centre is equal to the square of the radius of the circle:

$$OP_1 \cdot OP_2 = R^2.$$

216. Find points symmetric to the point $2+i$ with respect to the circles 1) $|z|=1$; 2) $|z-i|=3$.

217. Find the symmetric image of the following lines with respect to the unit circle:

1) $|z| = \frac{1}{2}$; 2) $|z-1|=1$; 3) $y=2$;

4) $|z-z_0|=|z_0|$ ($z_0=x_0+iy_0$);

5) $|z-z_0| = \sqrt{|z_0|^2-1}$ ($|z_0| > 1$); 6) hyperbola $x^2 - y^2 = 1$;

7) boundary of a rectilinear triangle with vertices z_1, z_2, z_3 , ($z_i \neq 0$).

218. Prove that for the points P_1 and P_2 to be symmetric with respect to K it is necessary and sufficient that one of the following two conditions be satisfied:

1) any circle K_1 passing through the points P_1 and P_2 is orthogonal to K ;

2) $\frac{MP_1}{MP_2} = \text{const}$ for all points M on the circle K (i.e. K is a circle of Apollonius with respect to the points P_1 and P_2).

219. The function $w = e^{i\alpha} \frac{z-\beta}{z-\bar{\beta}}$ ($\beta = a+ib$, $b > 0$) maps the upper half-plane onto the unit circle.

1) Find $\arg w(x) = \theta(x)$; 2) find $w'(\beta)$; 3) find out what part of the upper half-plane is contracted in this mapping and what part is stretched.

220. Map the upper half-plane $\operatorname{Im} z > 0$ onto the unit circle $|w| < 1$ so that:

1) $w(i) = 0$, $\arg w'(i) = -\frac{\pi}{2}$; 2) $w(2i) = 0$, $\arg w'(2i) = 0$;

3) $w(a+bi) = 0$, $\arg w'(a+bi) = \theta$ ($b > 0$).

221. Map the upper half-plane $\operatorname{Im} z > 0$ onto the circle $|w - w_0| < R$ so that the point i goes into the centre of the circle and the derivative at this point is positive.

222. Map the circle $|z| < 2$ onto the half-plane $\operatorname{Re} w > 0$ so that $w(0) = 1$, $\arg w'(0) = \pi/2$.

223. Map the circle $|z - 4i| < 2$ onto the half-plane $v > u$ so that the centre of the circle goes into the point -4 , and the point $2i$ on the circle goes into the origin.

224. Find the general form of the linear-fractional function $w(z)$ which maps the circle $|z| < 1$ onto the right half-plane $\operatorname{Re} w > 0$ so that $w(z_1) = 0$, $w(z_2) = \infty$, where z_1 and z_2 are given points on the circle $|z| = 1$ such that $\arg z_1 < \arg z_2$.

Construct a family of lines inside the circle $|z| < 1$ which lines correspond to a polar grid in the half-plane $\operatorname{Re} w > 0$.

Hint. Use the general form of a linear-fractional transformation for three pairs of corresponding points and find $\arg \frac{e^{i\tau} - z_1}{e^{i\tau} - z_2}$.

225. Find the centre w_0 and the radius R of a circle onto which the function $w = \frac{z - z_1}{z - z_2}$ maps the real axis ($\operatorname{Im} z_2 \neq 0$).

226. Find the function that maps the upper half-plane onto itself so that $w(a) = b$, $\arg w'(a) = \alpha$ ($\operatorname{Im} a > 0$, $\operatorname{Im} b > 0$).

Hint. First map both sheets of the half-plane onto the unit circle with the corresponding normalization.

227. Map the upper half-plane onto the lower one so that $w(a) = a$ and $\arg w'(a) = -\frac{\pi}{2}$ ($\operatorname{Im} a > 0$).

228. For the function $w = e^{ia} \frac{z-a}{1-az}$ ($|a| < 1$) which maps the unit circle on itself:

1) find $\arg w(e^{i\varphi}) = \theta(\varphi)$; 2) find $w'(0)$ and $w'(a)$; 3) find out what part of the unit circle contracts and what part stretches under this mapping; 4) find $\max \left| \frac{dw}{dz} \right|$ and $\min \left| \frac{dw}{dz} \right|$ for $|z| \leq 1$.

229. Map the circle $|z| < 1$ onto the circle $|w| < 1$ so that:

1) $w\left(\frac{1}{2}\right) = 0$, $\arg w'\left(\frac{1}{2}\right) = 0$; 2) $w\left(\frac{i}{2}\right) = 0$, $\arg w'\left(\frac{i}{2}\right) = \frac{\pi}{2}$;

3) $w(0) = 0$, $\arg w'(0) = -\frac{\pi}{2}$; 4) $w(a) = a$, $\arg w'(a) = \alpha$.

230. Map the circle $|z| < R_1$ onto the circle $|w| < R_2$ so that $w(a) = b$, $\arg w'(a) = \alpha$ ($|a| < R_1$, $|b| < R_2$).

231. Map the circle $|z| < 1$ onto the circle $|w-1| < 1$ so that $w(0) = 1/2$ and $w'(1) = 0$.

232. Map the circle $|z-2| < 1$ onto the circle $|w-2i| < 2$ so that $w(2) = i$ and $\arg w'(2) = 0$.

233. Find the general form of the linear-fractional function $w(z)$ which maps the circle $|z| < R$ onto itself under the following conditions:

1) $w(a) = 0$ ($|a| < R$); 2) $w(a) = b$ ($|a| < R$, $|b| < R$); 3) $w(\pm R) = \pm R$.

234. Map the circle $|z| < 1$ onto itself so that the given points z_1, z_2 located inside the circle go into the points $\pm a$ ($0 < a < 1$); find a .

Hint. Use the result of Problem 233(2) and the identity of Problem 10.

235. Map the circle $|z| < 1$ onto itself so that the segment of the real axis: $y=0$, $0 \leq x \leq a$ ($a < 1$) is transformed into a segment of the real axis symmetric with respect to the origin. Find the length of the transformed segment.

236. Prove that in mapping a circle onto a circle the linear transformation is defined uniquely by giving the images of one interior point and one boundary point.

237. The unit circle is mapped onto itself so that the point $z_0 \neq 0$ goes into the centre of the circle. Prove that in this transformation the unit semicircle is mapped onto a unit semicircle if and only if its end-points lie on the diameter passing through the point z_0 .

238. Construct a mapping of the unit circle onto itself under which the preimage of the centre is located on the real axis and the arc $0 \leq \varphi \leq \frac{\pi}{2}$ of the unit circle is mapped onto the following arcs:

1) $0 \leq \theta \leq \frac{\pi}{2}$; 2) $0 \leq \theta \leq \frac{\pi}{3}$; 3) $\frac{\pi}{2} \leq \theta \leq \frac{7\pi}{8}$.

§ 2. SUPPLEMENTARY PROBLEMS IN THE THEORY OF LINEAR TRANSFORMATIONS

CANONICAL FORMS OF LINEAR TRANSFORMATIONS

A linear-fractional transformation with one fixed point z_0 is called *parabolic*. A parabolic transformation can be written in canonical form:

$$\frac{1}{w-z_0} = \frac{1}{z-z_0} + h \text{ if } z_0 \neq \infty,$$

or

$$w = z + h \quad \text{if } z_0 = \infty.$$

A linear-fractional transformation with two different fixed points z_1 and z_2 looks like:

$$\frac{w - z_1}{w - z_2} = k \frac{z - z_1}{z - z_2},$$

in canonical form if $z_1 \neq \infty$, $z_2 \neq \infty$; it takes the form $w - z_1 = k(z - z_1)$ if $z_2 = \infty$; a transformation with two different fixed points is called *hyperbolic* if $k > 0$, *elliptic* if $k = e^{i\alpha}$ and $\alpha \neq 0$, and *loxodromic* if $k = ae^{i\alpha}$, where $a \neq 1$ and $\alpha \neq 0$ (α and a real numbers, $a > 0$).

239. Prove the following assertions:

1) The general linear-fractional transformation $w = \frac{az + b}{cz + d}$ can be reduced to the form $w = \frac{\alpha z + \beta}{\gamma z + \delta}$, where $\begin{vmatrix} \alpha & \beta \\ \gamma & \delta \end{vmatrix} = 1$.

2) If $\alpha + \delta$ is a real number, the transformation is elliptic when $|\alpha + \delta| < 2$, hyperbolic when $|\alpha + \delta| > 2$, and parabolic when $|\alpha + \delta| = 2$.

3) If $\text{Im}(\alpha + \delta) \neq 0$, then the transformation is loxodromic.

240. Prove that if a linear transformation possesses two fixed points, then the product of the derivatives at these points equals unity.

241. Find the circles which, under the parabolic transformation $\frac{1}{w - z_0} = \frac{1}{z - z_0} + h$, go into themselves.

242. Find the general form of a parabolic transformation of the circle $|z| < R$ onto itself if the point R is fixed.

243. Prove the following properties of a hyperbolic transformation:

1) Any circle passing through two fixed points goes into itself, the direction of traversal being preserved.

2) Any circle that is orthogonal to circles passing through fixed points goes into a circle possessing the same property. [This property follows immediately from Property (1).]

Hint. First consider the case when the fixed points are 0 and ∞ .

244. Prove that in an elliptic transformation:

1) Any circle orthogonal to circles passing through two fixed points goes into itself, the direction of traversal being preserved.

2) An arc of a circle joining the fixed points goes into an arc of a circle that joins fixed points and makes an angle α with the first arc ($\alpha = \arg k$).

245. 1) Prove that under a loxodromic transformation the properties of the hyperbolic [see Problem 243 (2)] and elliptic (see Problem 244) transformations are preserved.

2) Prove that under a loxodromic transformation there exist no fixed circles provided only that $\alpha \neq \pi$ ($\alpha = \arg k$). Prove that if $\alpha = \pi$, then circles passing through the fixed points are transformed into themselves with the direction of traversal changed.

246. Prove that under the loxodromic transformation $w = ae^{i\alpha}z$ the logarithmic spirals $r = Ae^{\frac{\ln a}{\alpha}\varphi}$ ($A > 0$) go into themselves.

247. Prove that the linear transformation $w = e^{i\alpha} \frac{z-a}{1-\bar{a}z}$ ($a = |a|e^{i\alpha}$, $|a| < 1$), which carries the unit circle onto itself, can be of three kinds only: elliptic, parabolic, or hyperbolic. Find out at what values of a each of these transformations takes place. Find the fixed points of the transformation and reduce the latter to canonical form.

SOME APPROXIMATE FORMULAS FOR LINEAR TRANSFORMATIONS

248. The upper half-plane is mapped onto the unit circle so that the point $z = hi$ ($h > 0$) goes into the centre of the circle. Find the length Γ of the image of the segment $[0, a]$ of the real axis ($a > 0$) and obtain approximate linear formulas for Γ for small a/h and small h/a .

249. The unit circle is mapped onto itself so that the preimage of the centre of the circle (the point x_0) lies on the real axis. Find the length Γ of the image of the arc $0 \leq \varphi \leq \gamma$ of the unit circle ($\gamma \leq \pi$). How does the quantity Γ/γ vary depending on the sign of x_0 ?

250. With the conditions of Problem 249, obtain the following formulas:

$$1) \quad \Gamma = \frac{1+x_0}{1-x_0} \gamma + O(\gamma^3) \text{ for small } \gamma;$$

$$2) \quad \Gamma = \pi - \varepsilon \cot \frac{\gamma}{2} - \frac{\varepsilon^2}{2} \cot^2 \frac{\gamma}{2} + O(\varepsilon^3) \text{ for small } \varepsilon, \text{ where } \varepsilon = 1 - x_0.$$

251. The unit circle is mapped onto itself so that the point $z_0 = r_0 e^{i\varphi_0}$ goes into the centre. The points $z_1 = e^{i\varphi_1}$ and $z_2 = e^{i\varphi_2}$ lie on one side of the diameter passing through z_0 ($\varphi_0 < \varphi_1 < \varphi_2 \leq \varphi_0 + \pi$). Assuming that the point z_0 is situated close to the unit circle, prove that for the length Γ of the image of the arc $\varphi_1 \leq \varphi \leq \varphi_2$ of the unit circle the following formula holds:

$$\Gamma = \varepsilon \left[\cot \frac{\varphi_1 - \varphi_0}{2} - \cot \frac{\varphi_2 - \varphi_0}{2} \right] + \frac{\varepsilon^3}{2} \left[\cot^2 \frac{\varphi_1 - \varphi_0}{2} - \cot^2 \frac{\varphi_2 - \varphi_0}{2} \right] + O(\varepsilon^5),$$

where $\varepsilon = 1 - r_0$.

MAPPING ELEMENTARY DOUBLY CONNECTED DOMAINS

252. Prove that if a linear mapping of the circle $|z| < 1$ onto itself does not result in rotation, then no concentric annulus centred at the origin goes into a concentric one.

Note. This proposition is a particular case of the following theorem:

For a conformal mapping of the annulus $r_1 < |z| < r_2$ onto the annulus $R_1 < |w| < R_2$ to exist, it is necessary and sufficient that the condition $R_2/R_1 = r_2/r_1$ be satisfied. The mapping function may be of two kinds only: either $w = az$ or $w = a/z$. The mapping is defined uniquely by specifying one pair of boundary points that correspond to each other.

253. 1) Map the annulus $2 < |z| < 5$ onto the annulus $4 < |w| < 10$ so that $w(5) = -4$.

2) Map the annulus $1 < |z - 2i| < 2$ onto the annulus $2 < |w - 3 + 2i| < 4$ so that $w(0) = -1 - 2i$.

The following theorem holds:

Each doubly connected domain whose boundaries do not degenerate into points can be conformally mapped onto a concentric annulus with a definite ratio (μ) of the radius of the outer circle to the radius of the inner circle (μ the *modulus* of a doubly connected domain).

254. Map the half-plane $\operatorname{Re} z > 0$ with the circle $|z - h| < R$ ($h > R$) deleted onto the annulus $\rho < |w| < 1$ so that the imaginary axis is transformed into the circle $|w| = 1$. Find ρ .

Hint. Construct a circle orthogonal to the circle $|z - h| = R$ and with centre at the origin; then find the linear transformation that carries the real axis and the constructed circle into two orthogonally intersecting straight lines and verify that here the domain under consideration is mapped onto a concentric annulus. Prove that the centre of the annulus coincides with the origin if the points of intersection of the constructed circle and the real axis go into 0 and ∞ .

255. Map the half-plane $\operatorname{Re} z > 0$ with the circle $|z - h| < 1$, $h > 1$ deleted onto the annulus $1 < |w| < 2$. Find h .

256. Map an eccentric annulus bounded by the circles $|z - 3| = 9$ and $|z - 8| = 16$ onto the annulus $\rho < |w| < 1$. Find ρ .

257. Map a doubly connected domain bounded by the circles $|z - z_1| = r_1$ and $|z - z_2| = r_2$ ($|z_2 - z_1| > r_1 + r_2$ or $|z_2 - z_1| < |r_2 - r_1|$) onto a concentric circular annulus with centre at the origin. Find the modulus (μ) of the domain.

Hint. Find a pair of points mutually symmetric with respect to both circles, and map one of them onto 0 and the other onto ∞ .

Note. It is easy to see that the methods recommended for solving Problems 254 and 257 are the same.

258. Using the solution of the preceding problem, find the moduli of the doubly connected domains bounded by the given circles:

$$1) |z-i|=2, |z+i|=5; \quad 2) |z-3i|=1, |z-4|=2.$$

GROUP PROPERTIES OF LINEAR-FRACTIONAL TRANSFORMATIONS

The transformation $T(z) = T_2[T_1(z)]$ is called the *product* of the transformations T_1 and T_2 and is written in the form $T = T_2T_1$ (here the order is of importance; since, generally speaking, $T_2T_1 \neq T_1T_2$). A set G of transformations T constitutes a *group* if it contains the product of any two transformations belonging to it, and, together with the transformation T , contains the inverse transformation T^{-1} . The group consisting of the powers T^n and T^{-n} of the single transformation T is called a *cyclic group*. If the group G is made up of the transformations $T_1, T_2, \dots, T_n$ by constructing all inverse transformations and all possible products of the given and inverse transformations, then these transformations are called *generating transformations* of the group G . The points obtained from the fixed point z with the aid of all transformations of the group G are called *equivalent* or *congruent* with respect to the group G .

The *fundamental domain* of the group G is a domain (connected or disconnected) that does not contain a single pair of points which are equivalent to each other with respect to this group, and in the neighbourhood of each boundary point of which there exist points equivalent to the points of the domain.

259. Let T_i be linear transformations:

$$T_i(z) = \frac{a_i z + b_i}{c_i z + d_i}, \quad \Delta_i = \begin{vmatrix} a_i & b_i \\ c_i & d_i \end{vmatrix} \neq 0 \quad (i = 1, 2, \dots).$$

Prove the following assertions:

1) $T = T_1T_2$ is a linear transformation with the determinant $\Delta = \Delta_1\Delta_2$.

2) The product of the transformations is associative, i.e.

$$(T_3T_2)T_1 = T_3(T_2T_1).$$

3) Each transformation T_i possesses an inverse, T_i^{-1} , i.e.

$$T_i T_i^{-1} = T_i^{-1} T_i = J,$$

where $J(z) \equiv z$ is the identity transformation.

4) The product of the transformations is, generally speaking, noncommutative (give examples).

260. Prove that the transformations

$$T_1 = z, \quad T_2 = \frac{1}{z}, \quad T_3 = 1 - z, \quad T_4 = \frac{1}{1 - z}, \quad T_5 = \frac{z - 1}{z}, \quad T_6 = \frac{z}{z - 1}$$

form a group (the group of *anharmonic ratios*).

261. Prove that a set of linear transformations resulting from rotation of the plane about the origin through angles that are multiples of α form a cyclic group. In what case will this group consist of a finite number of transformations?

262. 1) Prove that a set of transformations of the form $w = \frac{az+b}{cz+d}$, where a, b, c and d are real integers and $ad-bc=1$, form a group (it is called a *modular group*).

2) Prove that if a, b, c and d are complex integers (i. e. numbers of the form $m+ni$, where m and n are real integers) satisfying the condition $ad-bc=1$, then the set of transformations from Item 1 also form a group (the *Picard group*).

263. Find fundamental domains for the groups generated by the following transformations:

1) $T(z) = e^{2\pi i/n} z$ (n a natural number); 2) $T_1(z) = e^{2\pi i/n} z$, $T_2 = \frac{1}{z}$; 3) $T(z) = z + \omega$; 4) $T_1(z) = z + \omega$, $T_2(z) = -z$; 5) $T_1(z) = z + \omega_1$, $T_2(z) = z + \omega_2$ ($\text{Im} \frac{\omega_2}{\omega_1} \neq 0$) (a *doubly periodic group*); 6) $T_1(z) = z + \omega_1$, $T_2(z) = z + \omega_2$, $T_3(z) = -z$; 7) $T_1(z) = z + \omega$, $T_2(z) = iz$; 8) $T_1(z) = z + \omega$, $T_2(z) = e^{2\pi i/3} z$; 9) $T_1(z) = z + \omega$, $T_2(z) = e^{2\pi i/6} z$.

264. Find the groups of linear transformations corresponding, under a stereographic projection, to a rotation of a sphere about:

- 1) the vertical diameter;
- 2) the diameter parallel to the real axis;
- 3) the diameter parallel to the imaginary axis;
- 4) the diameter the stereographic projection of one end of which is the point a .

Hint. If z_1, z_2 are the images of diametrically opposite points on the sphere, then $z_1 \bar{z}_2 = -1$ (see Problem 49).

265. 1) Prove that the group of linear transformations which correspond to a rotation of a sphere and carry the points with stereographic projections a and b into one another is given by the following relation:

$$\frac{\omega-b}{1+b\bar{\omega}} = e^{i\alpha} \frac{z-a}{1+a\bar{z}}.$$

2) Prove that the differential $ds = \frac{|dz|}{1+|z|^2}$ is invariant with respect to the transformations of this group and represents the spherical length of the element of arc dz (i.e. the length of the image of this element on the sphere).

LINEAR TRANSFORMATIONS AND LOBACHEVSKIAN GEOMETRY

When interpreting Lobachevskian geometry in the unit circle $|z| < 1$, the role of straight lines in this circle is played by arcs of circles orthogonal to the unit circle; the role of motion is played by linear transformations of the unit circle into itself; the distance between the points z_1 and z_2 is represented by the quantity $\rho(z_1, z_2) = \frac{1}{2} \ln (\alpha, \beta, z_2, z_1)$, where α and β are the points of intersection of the "straight line" passing through the points z_1 and z_2 with the unit circle (the order of points is: α, z_1, z_2, β); $(\alpha, \beta, z_2, z_1)$ is the anharmonic ratio of the indicated points. Angles are measured in the same way as in Euclidean geometry.

266. Prove that $\rho(z_1, z_2) > 0$ if $z_1 \neq z_2$ and $\rho(z, z) = 0$.

267. Prove that $\rho(z_1, z_3) \leq \rho(z_1, z_2) + \rho(z_2, z_3)$, the equality sign being taken if and only if the point z_3 lies on the "segment" joining the points z_1 and z_2 .

268. Prove that if one of the points z_1 and z_2 tends to a point on the unit circle (or if both of them tend to different points on the unit circle), then the non-Euclidean length $\rho(z_1, z_2)$ tends to infinity (i.e. points on the unit circle correspond to points at infinity in the non-Euclidean plane).

269. Prove that the differential $ds = \frac{|dz|}{1-|z|^2}$ ($|z| < 1$) is invariant with respect to the group of linear transformations which map the circle $|z| < 1$ onto itself and represents the non-Euclidean length of an element dz of arc.

Hint. Write the general form of the transformation of the circle $|z| < 1$ onto itself that maps the point a into the point b ($|a| < 1$, $|b| < 1$).

270. Indicate ways of constructing the following lines:

- 1) a pencil of "straight lines" passing through the point z_0 ;
- 2) a "straight line" passing through the points z_1 and z_2 ;
- 3) the equidistance of a "straight line" (i.e. the locus of points "equidistant" from the given "straight line");
- 4) the limiting lines (i.e. lines orthogonal to a pencil of "parallel straight lines").

271. 1) Prove that for a "rectilinear" triangle with the angles $\varphi_1, \varphi_2, \varphi_3$ the following inequality holds:

$$\varphi_1 + \varphi_2 + \varphi_3 < \pi.$$

2) Prove that a "rectilinear" triangle is determined to within "motion" by its angles $\varphi_1, \varphi_2, \varphi_3$. Construct a "rectilinear" triangle from its angles.

§ 3. RATIONAL AND ALGEBRAIC FUNCTIONS

The general mapping of a circle or half-plane onto a singly connected domain of the w -plane has the form $w = \varphi[l(z)]$, where $\varphi(z)$ is a particular mapping and l is an arbitrary linear-fractional mapping of the circle or half-plane onto itself (the inverse mapping is of the form $z = l[\psi(w)]$). This should be taken into consideration when finding a normed mapping, i.e. a mapping that satisfies certain supplementary conditions. If the normalization conditions are not given, then ordinarily one of the mapping functions is indicated in the answer.

In the actual construction of conformal mappings, an important role is played by certain general principles.

Riemann-Schwarz Symmetry Principle

Let the boundary of a domain D_1 contain an arc of a circle C (a straight-line segment in a particular case), and let the function $w = f_1(z)$ effect a conformal mapping of this domain onto a domain D_1^* such that the arc C again passes into a circular arc or a straight-line segment C^* . Then the function $f_2(z)$, which, at points symmetric about C , assumes values symmetric to those of $f_1(z)$ with respect to C^* ¹⁾ will be analytic in the domain D_2 which is symmetric to the domain D_1 with respect to C and will map it onto the domain D_2^* which is symmetric to D_1^* with respect to C^* .

The function

$$w = \begin{cases} f_1(z) & \text{in } D_1 \\ f_1(z) = f_2(z) & \text{on } C \\ f_2(z) & \text{in } D_2 \end{cases}$$

effects a conformal mapping of the domain $D_1 + C + D_2$ onto the domain $D_1^* + C^* + D_2^*$ ²⁾.

Principle of Corresponding Boundaries

Let D and D^* be singly connected domains with the boundaries C and C^* , and let the domain D^* lie wholly in a finite part of the plane. If the function $w = f(z)$ is analytic in D , continuous in $\overline{D}$ and effects a one-to-one mapping of C onto C^* preserving the direc-

¹⁾ If C and C^* are segments of real axes (this can always be attained with the aid of additional linear-fractional transformations), then $f_2(z) = \overline{f_1(\overline{z})}$.

²⁾ The mapping will be one-to-one provided the domains D_1 and D_2 , and also D_1^* and D_2^* do not intersect each other.

tion of traversal, then it effects a one-to-one conformal mapping of the domain D onto D^* .

In solving the problems of this and the next section in cases when a mapping is effected by a branch of a multivalued function, it is advisable to check the correspondence of the points on the boundaries of the domain being mapped and its image (this especially refers to problems involving the mapping of domains with cuts).

272. Using the function $w = z^2$ and its inverse, find the conformal mapping of the following domains:

1) the interior of the right-hand branch of the equilateral hyperbola $x^2 - y^2 = a^2$ onto the upper half-plane;

2) the exterior of the parabola $y^2 = 2px$, $p > 0$ (i.e. the domain bounded by this parabola and not containing its focus) onto the upper half-plane.

Note. For mapping domains bounded by quadric curves see also Problems 302, 303, 330 to 332, 367.

273. Using the functions indicated in the preceding problem, map:

1) the interior of the circle $r = a \cos \varphi$ ($a > 0$) onto the interior of the cardioid $\rho = \frac{1}{2}(1 + \cos \theta)$;

2) the interior of the same circle onto the interior of the right-hand branch of the lemniscate $\rho = \sqrt{\cos 2\theta}$;

3) the circle $|z| < 1$ onto the interior of the cardioid $\rho = A(1 + \cos \theta)$, $A > 0$, so that $w(0) = A/8$, $w'(0) > 0$.

274. Find the domain onto which the circle $|z| < 1$ is mapped by the function $w = R(z + mz^2)$, $R > 0$, $0 \leq m \leq 1/2$. Find the images of the polar grid of the z -plane.

275. Find the domain onto which the semicircle $|z| < 1$, $\operatorname{Re} z > 0$ is mapped by the function $w = z + z^2$.

276. 1) Find the domain onto which the circle $|z| < 1$ is mapped by the function $w = R\left(z + \frac{z^n}{n}\right)$, $R > 0$ (n an integer, $n > 1$).

2) Find the domain onto which the exterior of the unit circle $|z| > 1$ is mapped by the function $w = R\left(z + \frac{1}{nz^n}\right)$, $R > 0$, n an integer, $n > 1$.

Note. For the mappings effected by the function $w = R\left(z + \frac{1}{z}\right)$ (Zhukovsky's function) see Problem 298 and subsequent problems.

277. 1) Determine for which values of m the function $w = R(z + mz^n)$ (n a natural number) effects a conformal mapping of the circle $|z| < 1$ onto a certain domain; find this domain.

2) Determine the same in the mapping of the exterior of the circle $|z| < 1$ by the function $w = R \left(z + \frac{m}{z^n} \right)$, and of the interior of the same circle by the function $w = R \left(\frac{1}{z} + mz^n \right)$.

MAPPING CIRCULAR LUNES AND DOMAINS WITH CUTS

278. 1) Map the angle $0 < \arg z < \pi\alpha$ ($0 < \alpha \leq 2$) onto the upper half-plane.

2) Map the angle $-\frac{\pi}{4} < \arg z < \frac{\pi}{2}$ onto the upper half-plane so that $w(1-i) = 2$, $w(i) = -1$, $w(0) = 0$.

279. Find the function $w(z)$ that maps the semicircle $|z| < 1$, $\operatorname{Im} z > 0$ onto the upper half-plane under the following conditions:

1) $w(-1) = 0$, $w(0) = 1$, $w(1) = \infty$;

2) $w(\pm 1) = \mp 1$, $w(0) = \infty$;

3) $w\left(\frac{i}{2}\right) = i$, $\arg w'\left(\frac{i}{2}\right) = -\frac{\pi}{2}$.

280. Find the function $w(z)$ that maps the semicircle $|z| < 1$, $\operatorname{Im} z > 0$ onto the circle $|w| < 1$ under the following conditions:

1) $w(\pm 1) = \pm 1$, $w(0) = -i$; 2) $w\left(\frac{i}{2}\right) = 0$, $\arg w'\left(\frac{i}{2}\right) = \frac{\pi}{2}$.

281. Find the function $w(z)$ that maps the domain $|z| > 1$, $\operatorname{Im} z > 0$ onto the upper half-plane.

282. Map onto the upper half-plane:

1) the sector $|z| < R$, $0 < \arg z < \pi\alpha$ ($0 < \alpha \leq 2$);

2) the domain $|z| > R$, $0 < \arg z < \pi\alpha$ ($0 < \alpha \leq 2$).

283. Map onto the upper half-plane the following circular lunes:

1) $|z| < 1$, $|z-i| < 1$; 2) $|z| < 1$, $|z-i| > 1$;

3) $|z| > 1$, $|z-i| < 1$; 4) $|z| > 1$, $|z-i| > 1$;

5) $|z| > 2$, $|z - \sqrt{2}| < \sqrt{2}$.

284. Map onto the upper half-plane the exterior of the unit upper semicircle.

In Problems 285 to 297 map the given domains onto the upper half-plane.

285. A plane with a cut along the segment $[-1, 1]$.

286. A plane with a cut along the segment $[-i, i]$.

287. A plane with a cut along the segment $[z_1, z_2]$.

288. A plane with cuts along the rays $(-\infty, -R]$, $[R, \infty)$ ($R > 0$).

289. A plane with a cut along the ray situated in the first quadrant and emanating from the point i parallel to the straight line $u = x$.

290. A plane with a cut along the circular arc that joins the points -1 and 1 and passes through the point ih , where $0 < h < 1$.

291. A half-plane $\operatorname{Im} z > 0$ with a cut along the line segment $[0, ih]$, $h > 0$.

292. A half-plane $\operatorname{Im} z > 0$ with a cut from ih to ∞ along the positive imaginary semi-axis ($h > 0$).

293. A half-plane $\operatorname{Im} z > 0$ with a cut along the circular arc $|z| = 1$ from the point $z = 1$ to the point $z = e^{i\alpha}$, where $0 < \alpha < \pi$.

294. An angle $0 < \arg z < \pi\beta$, where $0 < \beta < 2$ with a cut along the circular arc $|z| = 1$ from the point $z = 1$ to the point $z = e^{i\alpha}$, where $0 < \alpha < \beta$.

295. The exterior of the unit upper semicircle with a cut along the segment $[0, -i]$ (the exterior of a vane).

Hint. This problem is reduced to the preceding one by means of a linear transformation.

296. 1) A circle $|z| < 1$ with a cut along the radius $[0, 1]$. 2) The exterior of the unit circle with a cut along the ray $[1, \infty)$.

297. Find the mapping of the circle $|z| < 1$ onto the w -plane with a cut along the ray $(-\infty, -\frac{1}{4}]$, provided $w(0) = 0, w'(0) > 0$.

ZHUKOVSKY'S FUNCTION

298. Find the transformation of the polar grid $|z| = R, \arg z = \alpha$ with the aid of Zhukovsky's function $w = \frac{1}{2} \left(z + \frac{1}{z} \right)$.

299. Find the domains onto which Zhukovsky's function maps:

- 1) circle $|z| < R < 1$; 2) domain $|z| > R > 1$; 3) circle $|z| < 1$;
- 4) domain $|z| > 1$; 5) half-plane $\operatorname{Im} z > 0$; 6) half-plane $\operatorname{Im} z < 0$;
- 7) semicircle $|z| < 1, \operatorname{Im} z > 0$; 8) semicircle $|z| < 1, \operatorname{Im} z < 0$;
- 9) domain $|z| > 1, \operatorname{Im} z > 0$; 10) domain $1 < |z| < R, \operatorname{Im} z > 0$;
- 11) domain $R < |z| < 1, \operatorname{Im} z > 0$; 12) domain $\frac{1}{R} < |z| < R, \operatorname{Im} z > 0, \operatorname{Re} z > 0$;
- 13) angle $\frac{\pi}{2} - \alpha < \arg z < \frac{\pi}{2} + \alpha$ ($0 < \alpha < \frac{\pi}{2}$).

300. Find the transformation of the polar grid with the aid of the following functions:

1) $w = \frac{1}{2} \left(z - \frac{1}{z} \right)$; 2) $w = \frac{1}{2} \left(z + \frac{a^2}{z} \right)$ ($a > 0$);

3) $w = \frac{1}{2} \left(z + \frac{c^2}{z} \right)$, $c = |c| e^{i\gamma}$ ($0 \leq \gamma < \pi$).

301. Using Zhukovsky's function, map:

- 1) the exterior of the segment $[-c, c]$, ($c > 0$) onto the exterior of the unit circle if $w(\infty) = \infty, \arg w'(\infty) = \alpha$;

2) the exterior of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ onto the exterior of the unit circle so that $w(\infty) = \infty$, $\arg w'(\infty) = 0$.

302. Map the upper half-plane with semi-ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$, $y > 0$ removed onto the upper half-plane.

303. Map the doubly connected domain bounded by the confocal ellipses $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $\frac{x^2}{a^2+k^2} + \frac{y^2}{b^2+k^2} = 1$ ($a > b$) onto a concentric annulus centred at the origin of coordinates and find the modulus (see page 38) of the given doubly connected domain.

304. Find the domain onto which Zhukovsky's function maps the circle $|z| < 1$ with a cut along the segment $[a, 1]$ ($-1 < a < 1$). Consider the cases $a > 0$ and $a < 0$.

In Problems 305 to 309 map the indicated domains onto the upper half-plane.

305. A circle $|z| < 1$ with a cut along the segment $[1/2, 1]$.

306. A circle $|z| < 1$ with cuts along the radius $[-1, 0]$ and the segment $[a, 1]$ ($0 < a < 1$).

307. The exterior of the unit circle with cuts along the segment $[-a, -1]$ and along the ray $[1, \infty)$, where $a > 1$.

308. The upper half of the circle $|z| < 1$ with a cut along the segment $[0, \alpha i]$ ($0 < \alpha < 1$).

309. The upper half of the circle $|z| < 1$ with a cut along the segment $[\alpha i, i]$ ($0 < \alpha < 1$).

310. Map the circle $|z| < 1$ with the segment $[(1-h)e^{i\alpha}, e^{i\alpha}]$ removed onto the unit circle of the w -plane.

311. Map the circle $|z| < 1$ with a cut along the segment $[a, 1]$, $0 < a < 1$ onto the circle $|w| < 1$ so that $w(0) = 0$, $w'(0) > 0$. Find $w'(0)$ and the length of the arc corresponding to the cut. At what value of a will the cut turn into a semicircle?

Hint. It is expedient first to map both the given domain and the circle $|w| < 1$ onto the exterior of the segment.

312. Map the circle $|z| < 1$ with cuts along the segments $[a, 1]$, $[-1, -b]$ ($0 < a < 1$, $0 < b < 1$) onto the circle $|w| < 1$ so that $w(0) = 0$, $w'(0) > 0$. Define $w'(0)$ and the length of the arcs corresponding to the cuts.

313. Representing Zhukovsky's function as $\frac{w-1}{w+1} = \left(\frac{z-1}{z+1}\right)^2$, find:

1) the image of the circle C passing through the points $z = \pm 1$ at an angle α ($-\pi < \alpha < \pi$) to the real axis at the point 1, and the domain onto which the exterior of such a circle is mapped;

2) the image of the circle C' passing through the point $z = 1$ at

an angle α to the real axis and containing within it the point -1 , and also the domain onto which the exterior of such a circle is mapped.

314. 1) Find the images of the circles and domains of the z -plane mentioned in Problem 313 if the mapping function $w(z)$ is given by the equation

$$\frac{w-1}{w+1} = \left(\frac{z-1}{z+1} \right)^{2-\delta} \quad (0 < \delta < 2, w > 0 \text{ for } z > 1).$$

2) What is the image of the interior of the circle C under this mapping?

315. Map the exterior of the unit circle $|z| > 1$ onto the w -plane with a cut along the arc $\arg \frac{w-1}{w+1} = \beta$ ($0 < |\beta| < \pi$) so that $w(\infty) = \infty$, $\arg w'(\infty) = \alpha$.

In Problems 316 to 319 find the domains obtained in mappings of the given domains by the indicated functions.

316. Circle $|z| < 1$; $w = \frac{z}{z^2+1}$.

317. Semicircle $|z| < 1$, $\operatorname{Im} z > 0$; $w = \frac{1}{z^2+1}$.

318. Angle $0 < \arg z < \frac{\pi}{n}$; $w = \frac{1}{2} \left(z^n + \frac{1}{z^n} \right)$.

319. Sector $-\frac{\pi}{n} < \arg z < \frac{\pi}{n}$, $|z| < 1$; $w = \frac{z}{(1+z^n)^{2/n}}$
 $(w(z) > 0 \text{ for } z > 0).$

Hint. Represent the mapping function in the form $w = F\{f[\varphi(z)]\}$, where $\varphi(t) = t^n$, $f(t) = \frac{t}{(1+t)^2}$, $F(t) = \frac{n}{\sqrt{t}}$.

APPLYING THE SYMMETRY PRINCIPLE

320. 1) Using the solution of Problem 319 and the symmetry principle, find the image of the unit circle under the mapping $w = \frac{z}{(1+z^n)^{2/n}}$.

2) Find the function that maps the interior (and the exterior) of the unit circle onto the exterior of a "star":

$$|w| \leq 1, \arg w = 2\pi k/n \quad (k=0, 1, 2, \dots, n-1).$$

321. Map onto the exterior of the unit circle:

1) the entire plane with cuts along the segments $[-1, 1]$ and $[-i, i]$ (the exterior of a cross);

2) the entire plane with cuts along the rays $(-\infty, -1]$, $[1, +\infty)$, $(-i\infty, -i]$ and $[i, +i\infty)$.

322. 1)* Applying the function of Problem 318, map the sector $|z| < 1$, $0 < \arg z < \frac{\pi}{n}$ (n an integer) onto itself so that the segments of the radii $|z| \leq \alpha$, $\arg z = 0$ and $|z| \leq \alpha$, $\arg z = \frac{\pi}{n}$ ($0 < \alpha < 1$) go into the corresponding radii.

2) Map the exterior of the unit circle with cuts along the segments $1 \leq |z| \leq \alpha$, $\arg z = 2k\frac{\pi}{n}$ ($k = 0, 1, 2, \dots, n-1$) onto the exterior of the unit circle.

323. Map onto the upper half-plane and the exterior of the unit circle the exterior of a cross consisting of a segment $[-a, b]$ of the real axis and a segment $[-ci, ci]$ of the imaginary axis ($a \geq 0$, $b \geq 0$, $c \geq 0$, $a^2 + b^2 + c^2 \neq 0$).

Hint. Find the function that maps the upper half-plane with a cut along the segment $[0, ci]$ onto the upper half-plane and apply the symmetry principle in virtue of which the exterior of the cross is mapped onto the entire plane with a cut along the segment of the real axis.

324. Map the plane with cuts along the ray $[-a, +\infty)$ ($a \geq 0$) and along the segment $[-ci, ci]$ ($c > 0$) onto the upper half-plane.

Hint. See the hint to Problem 323.

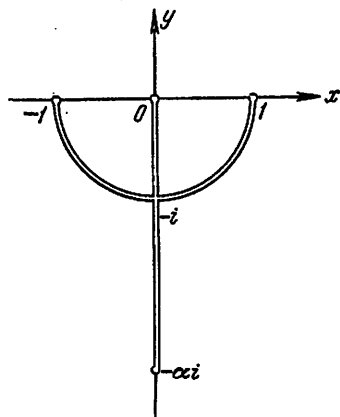


FIG. 2

325. Map the plane with cuts along the negative part of the imaginary axis and along the lower half of the unit circle onto the exterior of the unit circle.

Hint. This problem reduces to Problem 321 (1) via a linear transformation.

326. Map the plane with cuts along the segment $[-\alpha i, 0]$ ($\alpha > 1$) and along the lower half of the unit circle (Fig. 2) onto the upper half-plane.

Hint. This problem reduces to Problem 323 via a linear transformation.

327. Map onto the upper half-plane the plane with cuts along the segment $[-1, b]$ ($b > -1$) and along a circular arc with extremities at the points $e^{\pm i\alpha}$ passing through the point $z = -1$ (Fig. 3).

328. Map onto the upper half-plane the exterior of the unit circle with cuts along the segments: $[i, bi]$, $[-bi, -i]$, $[1, a]$, $[-a, -1]$ ($a > 1$, $b > 1$).

Hint. Zhukovsky's function maps the domain in question onto the domain of Problem 323.

329.* Map the exterior of the "star" shown in Fig. 4 onto the exterior of the unit circle.

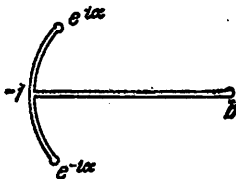


FIG. 3

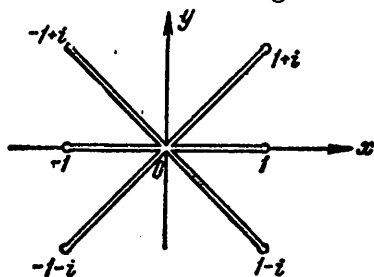


FIG. 4

330.* Map the interior of the right-hand branch of the hyperbola $\frac{x^2}{\cos^2 \alpha} - \frac{y^2}{\sin^2 \alpha} = 1$ onto the upper half-plane.

331. Map the exterior of the right-hand branch of the hyperbola $\frac{x^2}{\cos^2 \alpha} - \frac{y^2}{\sin^2 \alpha} = 1$ onto the upper half-plane.

332. Map the domain located between the branches of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ onto the upper half-plane.

SIMPLE MULTI-VALENT MAPPINGS

Problems 333 and 334 consider mappings leading to multivalent domains (Riemann surfaces).¹⁾

333. Find the domains obtained under a mapping by the function $w = z^2$ of:

- 1) a portion of the annulus $r_1 < |z| < r_2$, $0 < \arg z < \pi + \alpha$ ($0 < \alpha \leq \pi$);
- 2) a domain $|z^2 - 1| < a$ ($0 < a < \infty$).

334. Find the domains obtained when Zhukovsky's function $w = \frac{1}{2} \left(z + \frac{1}{z} \right)$ maps:

- 1) the circle $|z| < R$ ($R > 1$);

Hint. It is expedient first to consider the mapping of the circle $|z| < 1$ and the annulus $1 < |z| < R$ (see Problem 299).

- 2) the circle $\left| \frac{z-1}{z+1} \right| < R$ ($0 < R < \infty$).

In Problems 335 to 337 construct Riemann surfaces of the given functions.

¹⁾ We give here only a few of the most elementary problems of this kind. § 2 of Chapter 8 is devoted specially to Riemann surfaces.

335. 1) $w = \sqrt{\frac{z-1}{z+1}}$; 2) $w = \sqrt{z^2-1}$.

336. 1) $w = \sqrt{z(z^2+1)}$; 2) $w = \sqrt{\frac{z^2+1}{z}}$. 337. $w = \sqrt[3]{z^2-1}$.

§ 4. ELEMENTARY TRANSCENDENTAL FUNCTIONS

BASIC TRANSCENDENTAL FUNCTIONS

338. How are the following transformed under the mapping $w = e^z$:

- 1) orthogonal grid $x=C, y=C$;
- 2) straight lines $y=kx+b$;
- 3) strip $\alpha < y < \beta$ ($0 \leq \alpha < \beta \leq 2\pi$);
- 4) strip between straight lines $y=x, y=x+2\pi$;
- 5) semistrip $x < 0, 0 < y < \alpha \leq 2\pi$;
- 6) semistrip $x > 0, 0 < y < \alpha \leq 2\pi$;
- 7) rectangle $\alpha < x < \beta, \gamma < y < \delta$ ($\delta - \gamma \leq 2\pi$).

339. What is the preimage of the upper half-plane under the mapping $w = \left(1 + \frac{z}{n}\right)^n$? What is the limiting preimage of the upper half-plane as $n \rightarrow \infty$ (n natural number)?

340. How are the following transformed under mapping $w = \ln z$:

- 1) polar grid $|z|=R, \arg z=\vartheta$;
- 2) logarithmic spirals $r=Ae^{k\varphi}$ ($A > 0$);
- 3) angle $0 < \arg z < \alpha \leq 2\pi$;
- 4) sector $|z| < 1, 0 < \arg z < \alpha \leq 2\pi$;
- 5) annulus $r_1 < |z| < r_2$ with a cut along the segment $[r_1, r_2]$.

The real and imaginary parts of the function $w = \xi + i\eta = \ln \frac{a+z}{a-z}$ are called the *bipolar coordinates* of the point $z = x + iy$ with respect to the poles $\pm a$ ($a > 0$).

341. 1) Prove that the function w univalently maps the entire z -plane with cuts $(-\infty, -a]$ and $[a, \infty)$ onto a strip of the w -plane: $-\pi \leq \eta \leq \pi$, the straight line $\eta = \pi$ corresponding to the upper lips of the cut, and $\eta = -\pi$ to the lower ones (Fig. 5).

2) Ascertain the validity of the following:

$$x = \frac{a \sinh \xi}{\cosh \xi + \cos \eta}, \quad y = \frac{a \sin \eta}{\cosh \xi + \cos \eta},$$

$$\sqrt{x^2 + y^2} = r = a \sqrt{\frac{\cosh \xi - \cos \eta}{\cosh \xi + \cos \eta}}.$$

3) Prove that the preimages of the segments $\xi = \xi_0$, $-\pi \leq \eta \leq \pi$ are circles of Apollonius with respect to the points $\pm a$:

$$(x - a \coth \xi_0)^2 + y^2 = \left(\frac{a}{\sinh \xi_0} \right)^2$$

(the preimage of the segment $\xi = 0$, $-\pi \leq \eta \leq \pi$ is the axis of ordinates). (Fig. 6).

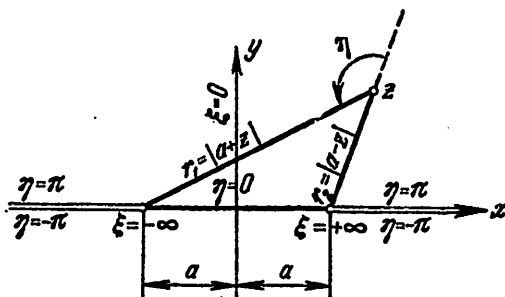


FIG. 5

4) Prove that the preimages of straight lines $\eta = \eta_0$ are arcs of circles passing through the points $\pm a$,

$$x^2 + (y + a \cot \eta_0)^2 = \left(\frac{a}{\sin \eta_0} \right)^2$$

lying in the upper half-plane for $\eta_0 > 0$ and in the lower one for $\eta_0 < 0$. The segment $[-a, a]$ corresponds to the line $\eta = 0$. The

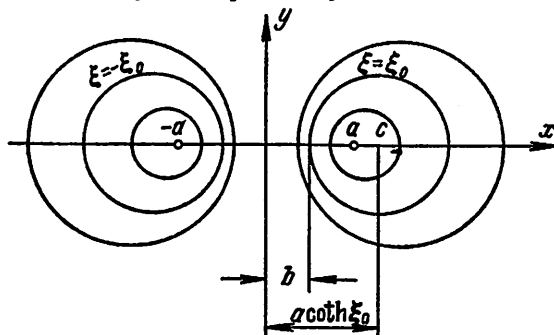


FIG. 6

arcs corresponding to the values $\eta = \eta_0$ and $\eta = \eta_0 - \pi$ ($\eta_0 > 0$) supplement each other to complete a circle (Fig. 7).

5) Find the magnitudes of the line segments b (see Fig. 6) and l (see Fig. 7).

Note. The coordinate grid thus constructed in the z -plane is called a *bipolar grid*.

- 1) orthogonal grid $x=C, y=C$;
- 2) semistrip $0 < x < \pi, y > 0$;
- 3) strip $0 < x < \pi$; 4) strip $0 < x < \frac{\pi}{4}$;
- 5) strip $-\frac{\pi}{4} < x < \frac{\pi}{4}$.

347. Into what are the following transformed under the mapping $w = \coth z$:

- 1) strip $0 < y < \pi$; 2) semistrip $0 < y < \pi, x > 0$.

MAPPINGS THAT REDUCE TO MAPPINGS OF STRIPS AND SEMISTRIPS

In Problems 348 to 355 map the indicated domains onto the upper half-plane.

348. The strip bounded by the straight lines $y=x, y=x+h$.

349. The semistrip $x < 1, 0 < y < h$.

350. The circular lune bounded by the circles $|z|=2, |z-1|=1$.

351. The domain bounded by the circles $|z|=2, |z-3|=1$ (plane with circles removed).

352. The domain defined by the inequalities:

$$|z-1| > 1, |z+1| > 1, \operatorname{Im} z > 0$$

(upper half-plane with semicircles removed).

353. The domain contained between the confocal parabolas $y^2 = 4(x+1), y^2 = 8(x+2)$.

Hint. See Problem 272 (2).

354. Find the function $w(z)$ that maps the domain bounded by the circle $|z|=1$ and the straight line $\operatorname{Im} z = 1$ (a half-plane $\operatorname{Im} z < 1$ with circle removed):

1) onto the circle $|w| < 1$ with normalization $w(-3i) = 0$, $\arg w'(-3i) = \frac{\pi}{3}$;

2) onto the circle $|w| < 1$ with normalization $w(-3i) = \frac{-1+i}{2}$, $\arg w'(-3i) = \frac{\pi}{2}$;

3) onto the upper half-plane with normalization $w(-3i) = 1+i$, $\arg w'(-3i) = \pi$.

APPLYING THE SYMMETRY PRINCIPLE

355. Map onto the upper half-plane:

- 1) the strip $0 < x < 1$ with a cut along the ray $x = \frac{1}{2}, h \leq y < \infty$;

2) the strip $0 < x < 1$ with cuts along the rays $x = \frac{1}{2}$, $h_1 \leq y < \infty$ and $x = \frac{1}{2}$, $-\infty < y \leq h_2$ ($h_2 < h_1$).

Hint. First map the strip $0 < x < \frac{1}{2}$ onto the upper half-plane. The mapping function will, in accordance with the symmetry principle, map the given domain onto the entire plane with a certain cut.

In Problems 356 to 366 map the indicated domains onto the upper half-plane.

356. The strip $0 < x < 1$ with a cut along the segment $0 \leq x \leq h$, $y = 0$ ($h < 1$).

357. The strip $0 < x < 1$ with cuts along the segments $0 \leq x \leq h_1$, $y = 0$ and $1 - h_2 \leq x \leq 1$, $y = 0$ ($h_1 + h_2 < 1$).

358. The semistrip $0 < x < \pi$, $y > 0$ with a cut along the segment $x = \frac{\pi}{2}$, $0 \leq y \leq h$.

359. The semistrip $0 < x < \pi$, $y > 0$ with a cut along the ray $x = \frac{\pi}{2}$, $h \leq y < \infty$ ($h > 0$).

360. The semistrip $0 < x < \pi$, $y > 0$ with cuts along the segment $x = \frac{\pi}{2}$, $0 \leq y \leq h_1$, and along the ray $x = \frac{\pi}{2}$, $h_2 \leq y < \infty$ ($h_2 > h_1$).

361. The domain bounded by the circles $|z - 1| = 1$, $|z + 1| = 1$ with a cut along the ray $2 \leq x < \infty$, $y = 0$.

362. The domain bounded by the circles $|z - 1| = 1$, $|z - 2| = 2$ with a cut along the segment $y = 0$, $2 \leq x \leq a$ ($a < 4$).

363. The domain bounded by the circles $|z - 1| = 1$, $|z - 2| = 2$ with cuts along the segments $y = 0$, $2 \leq x \leq a$ and $y = 0$, $b \leq x \leq 4$ ($a < b$).

364. The domain bounded by the imaginary axis and the circle $|z - 1| = 1$ with cuts along the segment $y = 0$, $2 \leq x \leq a$ and along the ray $y = 0$, $b \leq x < \infty$ ($a < b$).

365. The domain bounded by the circles $|z - 1| = 1$, $|z + 1| = 1$ with a cut along the segment $x = 0$, $-\alpha \leq y \leq \beta$ ($\alpha \geq 0$, $\beta \geq 0$).

366. The domain $|z - 1| > 1$, $|z + 1| > 1$, $\text{Im } z > 0$ (the upper half-plane with semicircles removed) with a cut along the segment $x = 0$, $0 \leq y \leq h$.

367. Map the interior of the parabola $y^2 = 4\alpha^2(x + \alpha^2)$ onto the upper half-plane and onto the unit circle.

Hint. Make a cut along the axis of symmetry of the parabola,

map the upper half of the parabola (by the function $\sqrt{z}$) onto the semistrip and then onto the half-plane; take advantage of the symmetry principle.

368.* Map the upper half-plane with cuts along the segments $0 \leq y \leq a$, $x = \frac{\pi}{2} + k\pi$ ($k=0, \pm 1, \pm 2, \pm \dots$) onto the upper half-plane (Fig. 8).

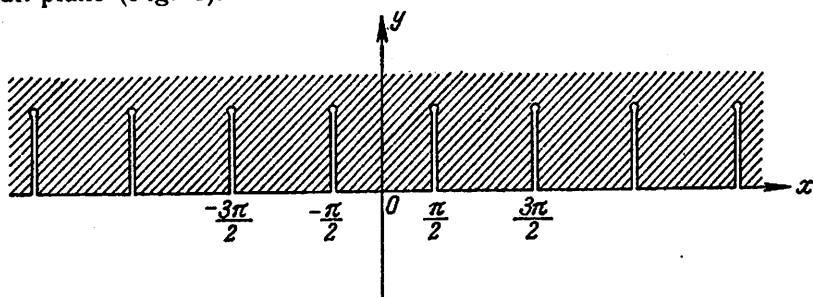


FIG. 8

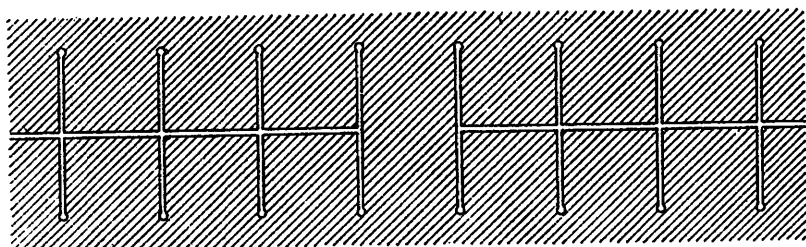


FIG. 9

369. Map the plane with parallel cuts $-a \leq x \leq a$, $y = \frac{\pi}{2} + k\pi$ ($k=0, \pm 1, \pm 2, \dots$) onto a plane with cuts along the real axis $[k\pi - b, k\pi + b]$ ($k=0, \pm 1, \pm 2, \dots$, $0 < b < \frac{\pi}{2}$).

Hint. Make an additional cut along the imaginary axis, map one of the domains thus obtained onto the upper half-plane and apply the symmetry principle.

370. Map a plane with cuts along the rays $(-\infty, -\frac{\pi}{2}]$, $[\frac{\pi}{2}, +\infty)$ and along the segments $-a \leq y \leq a$, $x = \frac{\pi}{2} + k\pi$ ($k=0, \pm 1, \pm 2, \dots$) onto the exterior of the unit circle (Fig. 9).

Hint. The function which yields a solution to Problem 368 maps the given domain onto a plane with cuts along the rays

$$\left(-\infty, -\frac{1}{\operatorname{arcsin} \frac{1}{\cosh a}}\right], \left[\frac{1}{\operatorname{arcsin} \frac{1}{\cosh a}}, +\infty\right).$$

371. Map a plane with cuts along the rays $(-\infty, p], [q, +\infty)$ $\left(-\frac{\pi}{2} \leq p < q \leq \frac{\pi}{2}\right)$ and along the segments $-a \leq y \leq a, x = \frac{\pi}{2} + k\pi$ ($k=0, \pm 1, \pm 2, \dots$) onto the upper half-plane (Fig. 10).

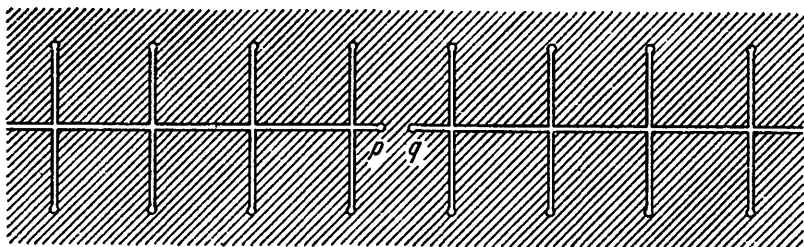


FIG. 10

372.* Map a plane with cuts along the rays $0 \leq y < \infty, x = \frac{k\pi}{2}$ ($k=0, \pm 1, \pm 2, \dots$) onto the upper half-plane.

ELEMENTARY MULTIVALENT MAPPINGS

In Problems 373 to 376 the mappings lead to multivalent domains (see footnote on page 49).

373. Find the domains onto which the following are mapped by the function $w = e^z$:

- 1) rectangle $0 < x < a, 0 < y < b$;
- 2) semistrip $0 < x < a, y > 0$; 3) strip $0 < x < a$.

374. Find the domains onto which the following are mapped by the function $w = \cos z$:

- 1) strip $-\pi/2 < x < \pi/2$;
- 2) strip $0 < x < 2\pi$.

375. Find the domain onto which the strip $0 < x < 2\pi$ is mapped with the aid of the function $w = \tan z$.

376. Construct the Riemann surface onto which the function $w = e^{1/z}$ maps the z -plane.

§ 5. BOUNDARIES OF UNIVALENCE, CONVEXITY AND STARLIKENESS

Let $w=f(z)$ be a function analytic at the origin of coordinates and $f(0)=0$.

In Problems 377 to 385, r_1 denotes the maximum radius of a circle, centred at the origin of coordinates, in which the function $w=f(z)$ is univalent; r_2 denotes the maximum radius of a circle centred at the origin of coordinates, which is mapped univalently onto a convex domain by the function $w=f(z)$, and r_3 denotes the maximum radius of a circle centred at the origin of coordinates, which is mapped univalently by the function $w=f(z)$ onto a domain that is starlike with respect to the point $w=0$. (A domain is called *starlike* with respect to a given point if any point of the domain can be joined to the given one by a linear segment wholly in the domain). It is obvious that $r_2 \leq r_3 \leq r_1$.

377. Find r_1, r_2, r_3 for the function $w = \frac{z}{1-z}$ and construct the images of the circles $|z| < r_1, |z| < r_2, |z| < r_3$.

378. Find r_1 for each of the following functions: 1) $w = z + z^2$, 2) $w = z + az^2$ (a a real number); 3) $w = z/(1-z)^2$.

379. Prove that under the mapping $w=f(z)$ the curvature of the image of the circle $|z|=r$ is expressed by the formula $k = \frac{1 + \operatorname{Re} \{z f''(z)/f'(z)\}}{|zf'(z)|}$.

380. Prove that the analytic function $f(z)$ maps a circle $|z|=r$ onto a convex curve if and only if $\frac{\partial}{\partial \varphi} \left[\frac{\pi}{2} + \varphi + \arg f'(z) \right] \equiv 1 + \operatorname{Re} \left[\frac{zf''(z)}{f'(z)} \right] \geq 0$ for all φ ($z = re^{i\varphi}$).

381. Prove that the circle $|z| < r$ is mapped by the analytic function $f(z)$ ($f(0)=0$) onto a domain which is starlike with respect to the point $w=0$ if and only if $\frac{\partial}{\partial \varphi} \arg f(z) = \operatorname{Re} \left[\frac{zf''(z)}{f'(z)} \right] \geq 0$ for all φ ($z = re^{i\varphi}$).

382. Prove that: 1) If the function $w=f(z)$ ($f(0)=0$) maps the circle $|z| < 1$ onto a domain which is starlike with respect to the point $w=0$, then the function $w_1 = \int_0^z \frac{f(t)}{t} dt$ will map the same circle onto a convex domain.

2) If $w=f(z)$ maps the circle $|z| < 1$ onto a convex domain, then $w_1 = zf'(z)$ maps this circle onto a domain that is starlike with respect to the point $w=0$.

383. Find r_z for each of the following functions: 1) $w = z + z^2$; 2) $w = z + az^2$ (a real); 3) $w = z/(1-z)^2$.

384. Find r_1 and r_2 for the function $w = e^z - 1$.

385. Find r_3 for each of the following functions: 1) $w = z + z^2$; 2) $w = z + az^2$ (a real); 3) $w = z/(1-z)^2$.

Hint. When solving Problem 385(3), it is more convenient to proceed from the inequality $\frac{\partial}{\partial \varphi} [\varphi + \arg f'(z)] \geq 0$.

CHAPTER III

INTEGRALS AND POWER SERIES

Here and henceforward, simple (i.e. without points of self-intersection) closed contours are traversed in the positive direction unless otherwise stated.

§ 1. INTEGRATION OF FUNCTIONS OF A COMPLEX VARIABLE

386. Prove the following equalities by direct summing.

$$1) \int_{z_0}^{z_1} dz = z_1 - z_0; \quad 2) \int_{z_0}^{z_1} z dz = \frac{1}{2} (z_1^2 - z_0^2).$$

387. Let C be a simple closed contour bounding an area S . Prove the following equalities:

$$1) \int_C x dz = iS; \quad 2) \int_C y dz = -S; \quad 3) \int_C \bar{z} dz = 2iS.$$

388. Evaluate the integrals $I_1 = \int x dz$, $I_2 = \int y dz$ along the following paths:

- 1) along the radius vector of the point $z = 2 + i$;
- 2) around the semicircle $|z| = 1$, $0 \leq \arg z \leq \pi$ (the starting point is $z = 1$);
- 3) around the circle $|z - a| = R$.

389. Evaluate the integral $\int |z| dz$ along the following paths:

- 1) along the radius vector of the point $z = 2 - i$;
- 2) around the semicircle $|z| = 1$, $0 \leq \arg z \leq \pi$ (the starting point is $z = 1$);
- 3) around the semicircle $|z| = 1$, $-\frac{\pi}{2} \leq \arg z \leq \frac{\pi}{2}$ (the starting point is $z = -i$);

4) around the circle $|z| = R$.

390. Evaluate the integral $\int_C |z| \bar{z} dz$, where C is a closed contour consisting of the upper semicircle $|z| = 1$ and the segment $-1 \leq x \leq 1$, $y = 0$.

391. Evaluate the integral $\int_C \frac{z}{z} dz$, where C is the boundary of the half-annulus shown in Fig. 11.

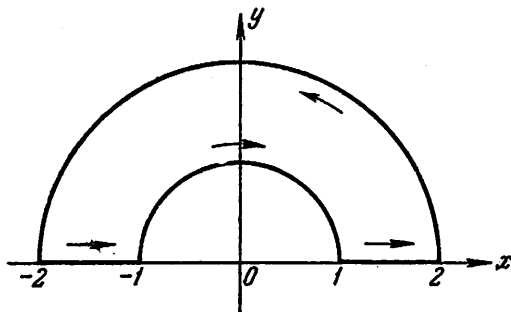


FIG. 11

392. Evaluate the integral $\int (z-a)^n dz$ (n an integer):

1) around the semicircle $|z-a| = R$, $0 \leq \arg(z-a) \leq \pi$ (the starting point is $z = a + R$);

2) around the circle $|z-a| = R$;

3) around the perimeter of a square centred at a point a and with sides parallel to the coordinate axes.

In Problems 393 to 396 the branch of the multi-valued function under the integration sign is chosen by specifying its value at some point of the contour of integration. If the contour is closed, then the starting point of the path of integration is always taken to be the point at which the value of the integrand is specified (note that the value of an integral may depend on the choice of the starting point).

393. Evaluate the integral $\int \frac{dz}{\sqrt{z}}$ along the following contours:

1) the semicircle $|z| = 1$, $y \geq 0$, $\sqrt{1} = 1$;

2) the semicircle $|z| = 1$, $y \geq 0$, $\sqrt{1} = -1$;

3) the semicircle $|z| = 1$, $y \leq 0$, $\sqrt{1} = 1$;

4) the circle $|z| = 1$, $\sqrt{1} = 1$;

5) the circle $|z| = 1$, $\sqrt{-1} = i$.

394. Evaluate the integral $\int_C \operatorname{Ln} z \, dz$, where:

- 1) C is the unit circle and $\operatorname{Ln} 1 = 0$;
- 2) C is the unit circle and $\operatorname{Ln} i = \frac{\pi i}{2}$;
- 3) C is the circle $|z| = R$ and $\operatorname{Ln} R = \ln R$;
- 4) C is the circle $|z| = R$ and $\operatorname{Ln} R = \ln R + 2\pi i$.

395. Evaluate the integral $\int_{|z|=1} z^n \operatorname{Ln} z \, dz$, where n is an integer and

- 1) $\operatorname{Ln} 1 = 0$, 2) $\operatorname{Ln}(-1) = \pi i$.

396. Evaluate the integral $\int_{|z|=1} z^\alpha \, dz$, where α is an arbitrary complex number and $1^\alpha = 1$.

397. Prove that $\int_{|z|=1} a^z \, dz = 0$ for any choice of the initial value of the function a^z .

398. For what values of α ($0 \leq \alpha < 2\pi$) do the following integrals exist: 1) $I_1 = \int e^{-\frac{1}{z}} \, dz$, 2) $I_p = \int e^{-\frac{1}{z^p}} \, dz$ (p a natural number) taken along the radius vector of the point $z = e^{i\alpha}$?

399. Prove that if $|a| \neq R$, then

$$\int_{|z|=R} \frac{|dz|}{|z-a||z+a|} < \frac{2\pi R}{|R^2 - |a|^2|}.$$

400. Prove the following assertions:

- 1) If $f(z)$ is continuous in the neighbourhood of the origin, then

$$\lim_{r \rightarrow 0} \int_0^{2\pi} f(re^{i\varphi}) \, d\varphi = 2\pi f(0).$$

- 2) If $f(z)$ is continuous in the neighbourhood of the point $z = a$, then

$$\lim_{r \rightarrow 0} \int_{|z-a|=r} \frac{f(z) \, dz}{z-a} = 2\pi i f(a).$$

401. Prove the following assertions:

- 1) If $f(z)$ is continuous in the semistrip $x \geq x_0$, $0 \leq y \leq h$ and there exists a limit $\lim_{x \rightarrow \infty} f(x + iy) = A$, independent of y and uni-

form in y , then $\lim_{x \rightarrow \alpha} \int_{\beta_x} f(z) dz = iAh$, where β_x is a segment of the vertical line $0 \leq y \leq h$ going upwards.

2) If $f(z)$ is continuous in the sector $0 < |z-a| \leq r_0$, $0 \leq \arg(z-a) \leq \alpha$ ($0 \leq \alpha \leq 2\pi$) and there exists a limit

$$\lim_{z \rightarrow a} [(z-a)f(z)] = A,$$

then

$$\lim_{r \rightarrow 0} \int_{\gamma_r} f(z) dz = iA\alpha,$$

where γ_r is, in the given sector, a positively traversed arc of the circle $|z-a| < r$.

3) If $f(z)$ is continuous in the domain $|z| \geq R_0$, $0 \leq \arg z \leq \alpha$ ($0 < \alpha \leq 2\pi$) and there exists a limit $\lim_{z \rightarrow \infty} zf(z) = A$, then

$$\lim_{R \rightarrow \infty} \int_{\Gamma_R} f(z) dz = iA\alpha,$$

where Γ_R is a positively traversed (relative to the origin) arc of the circle $|z| = R$ in the given domain.

402. Prove the following theorems:

1) If $f(z)$ is continuous in the domain $|z| \geq R_0$, $\operatorname{Im} z \geq a$ (a a fixed real number), and in this domain $f(z) \rightarrow 0$ as $z \rightarrow \infty$, then for any positive number m

$$\lim_{R \rightarrow \infty} \int_{\Gamma_R} e^{Im z} f(z) dz = 0,$$

where Γ_R is an arc of the circle $|z| = R$ lying in the indicated domain (Jordan's lemma).

Hint. When estimating the modulus of the integral around the semicircle $|z| = R$, $\operatorname{Im} z > 0$, use the inequality $\sin \theta \geq 2\theta/\pi$ for $0 \leq \theta \leq \pi/2$; when estimating it along arcs lying in the lower half-plane (in the case $a < 0$), make use of the fact that the length of each of them tends to $|a|$ as $R \rightarrow \infty$.

2) If $f(z)$ is continuous in the half-plane $\operatorname{Re} z \geq \sigma$ (σ a fixed real number) and in this half-plane $f(z) \rightarrow 0$ as $z \rightarrow \infty$, then for any negative number t

$$\lim_{R \rightarrow \infty} \int_{\Gamma_R} e^{zt} f(z) dz = 0,$$

where Γ_R is an arc of the circle $|z| = R$, $\operatorname{Re} z \geq \sigma$. If $f(z)$ is continuous in the half-plane $\operatorname{Re} z \leq \sigma$, then the assertion holds true if t is positive and Γ_R is an arc of the circle $|z| = R$, $\operatorname{Re} z \leq \sigma$.

§ 2. CAUCHY'S INTEGRAL THEOREM¹⁾

403. Show that if the path does not pass through the origin, then

$$\int_1^z \frac{d\zeta}{\zeta} = \ln r + i\varphi + 2\pi ik,$$

where k is an integer indicating how many times the path of integration winds round the origin ($z = re^{i\varphi}$).

404. Show that if the path does not pass through the points $\pm i$, then

$$\int_0^1 \frac{d\zeta}{1+\zeta^2} = \frac{\pi}{4} + k\pi,$$

where k is an integer.

405. Show that if C is an arbitrary simple closed contour which does not pass through the point a , and n is an integer, then

$$\int_C (z-a)^n dz = \begin{cases} 0 & \text{if } n \neq -1, \\ 2\pi i & \text{if } n = -1, a \text{ inside } C, \\ 0 & \text{if } n = -1, a \text{ outside } C. \end{cases}$$

406. Cauchy's integral theorem holds true in the following stronger form: if $f(z)$ is continuous in a closed domain G bounded by a simple rectifiable contour C and is analytic inside G , then $\int_C f(z) dz = 0$. Prove this for the case of a starlike contour²⁾.

Hint. Assuming C starlike with respect to the origin, consider the contours C_λ : $\zeta = \lambda z$ ($0 < \lambda < 1$, $z \in C$), and pass to the limit as $\lambda \rightarrow 1$.

407. Prove the following assertions:

1) If $f(z)$ is analytic in the strip $0 \leq y \leq h$, $\lim_{x \rightarrow \pm \infty} f(x+iy) = 0$

and the integral $\int_{-\infty}^{\infty} f(x) dx$ exists, then the integral $\int_{-\infty}^{\infty} f(x+ih) dx$ also exists and these integrals are equal.

2) If $f(z)$ is analytic in the angle $0 \leq \arg z \leq \alpha$ ($0 < \alpha \leq 2\pi$), $\lim_{z \rightarrow \infty} zf(z) = 0$ and the integral $\int_0^{\infty} f(x) dx$ exists, then the integral

¹⁾ Problems involving the calculation of certain integrals in this and the next section are mainly illustrative. Most problems of this kind are given in § 4 of Chapter IV, which is devoted to the application of the theory of residues.

²⁾ A contour is called starlike with respect to some point if each ray emanating from this point meets the contour at one point.

$\int f(z) dz$ along the ray $z = re^{i\alpha}$, $0 \leq r < \infty$, also exists and these integrals are equal.

Hint. Use the results of Problem 401.

408. Prove that
$$\int_0^{\infty} e^{-x^2} \cos 2bx \, dx = \frac{\sqrt{\pi}}{2} e^{-b^2}.$$

Hint. Integrate the function $f(z) = e^{-z^2}$ along the boundary of the rectangle $|x| \leq R$, $0 \leq y \leq b$ and make use of Poisson's integral

$$\int_{-\infty}^{\infty} e^{-t^2} dt = \sqrt{\pi}.$$

409. Prove the equalities
$$\int_0^{\infty} \cos x^2 \, dx = \int_0^{\infty} \sin x^2 \, dx = \frac{\sqrt{\pi}}{2\sqrt{2}}$$
 (Fresnel's integrals).

Hint. Integrate the function $f(z) = e^{iz^2}$ along the boundary of the sector $0 \leq |z| \leq R$, $0 \leq \arg z \leq \frac{\pi}{4}$ and use the result of Problem 402 (1) (putting $z^2 = \zeta$).

410. Prove that
$$\int_0^{\infty} \frac{\sin x}{x} \, dx = \frac{\pi}{2}$$
 (Dirichlet's integral).

Hint. Integrate the function $f(z) = \frac{e^{iz}}{z}$ along the boundary of the domain $r \leq |z| \leq R$, $0 \leq \arg z \leq \pi$ and use the results of Problems 401 and 402.

411. Prove that for $0 < s < 1$ the following equalities are valid:

$$1) \int_0^{\infty} x^{s-1} \cos x \, dx = \Gamma(s) \cos \frac{\pi s}{2}, \quad 2) \int_0^{\infty} x^{s-1} \sin x \, dx = \Gamma(s) \sin \frac{\pi s}{2}.$$

Hint. Integrate the function $f(z) = z^{s-1} e^{-iz}$ along the boundary of the domain $r \leq |z| \leq R$, $-\frac{\pi}{2} \leq \arg z \leq 0$; use the results of Problems 401 (2) and 402 (1) and the integral representation of the Γ -function:

$$\Gamma(t) = \int_0^{\infty} x^{t-1} e^{-x} \, dx.$$

§ 3. CAUCHY'S INTEGRAL FORMULA

Everywhere in the problems of this section, C means a simple closed rectifiable contour.

412. Evaluate the integral $\int_C \frac{dz}{z^2+9}$ if:

- 1) the point $3i$ lies inside the contour C , and the point $-3i$ outside it;
- 2) the point $-3i$ lies inside the contour C , and the point $3i$ outside it;
- 3) the points $\pm 3i$ lie inside the contour C .

413. Find all possible values of the integral $\int_C \frac{dz}{z(z^2-1)}$ for different positions of the contour C proceeding from the assumption that the contour C does not pass through any of the points $0, 1, -1$.

414. How many distinct values can the integral $\int_C \frac{dz}{\omega_n(z)}$ have if $\omega_n(z) = (z-z_1)(z-z_2)\dots(z-z_n)$ ($z_i \neq z_j$) and if the contour C does not pass through any of the points z_i ?

415. Evaluate the integral $\int \frac{z dz}{z^4-1}$, $a > 1$.

416. Evaluate the integral $\frac{1}{2\pi i} \int_C \frac{e^z dz}{z^2+a^2}$ if the contour C contains the circle $|z| \leq a$ within it.

417. Evaluate the integral $\frac{1}{2\pi i} \int_C \frac{ze^z dz}{(z-a)^3}$ if the point a lies inside the contour C .

Hint. Use the formulas for the derivatives of Cauchy's integral.

418. Evaluate the integral $\frac{1}{2\pi i} \int_C \frac{e^z dz}{z(1-z)^3}$ if:

- 1) the point 0 lies inside, and the point 1 outside the contour C ;
- 2) the point 1 lies inside, and the point 0 outside the contour C ;
- 3) the points 0 and 1 both lie inside the contour C .

419. The function $f(z)$ is analytic in the domain, bounded by a simple closed contour C containing the origin within it. Prove that for any choice of the branch of $\text{Ln } z$

$$\frac{1}{2\pi i} \int_C f'(z) \text{Ln } z dz = f(z_0) - f(0),$$

where z_0 is the starting point of integration.

Hint. Integrate by parts.

420. Evaluate the integral $\frac{1}{2\pi i} \int_C z^2 \operatorname{Ln} \frac{z+1}{z-1} dz$ if $\operatorname{Ln} a = \ln a$ for $a > 0$ and the contour C is:

- 1) the circle $|z|=2$;
- 2) the circle $|z-1|=1$, and $z=1+i$ is the starting point of integration.

421. By Liouville's theorem, a function $f(z)$, which is analytic and bounded throughout the whole plane, is constant. Prove this theorem after evaluating the integral $\int_{|z|=R} \frac{f(z) dz}{(z-a)(z-b)}$ ($|a| < R$, $|b| < R$) and taking its limit as $R \rightarrow \infty$.

422. Let $f(z)$ be analytic in a closed domain bounded by a contour C ; let $z_1, z_2, \dots, z_n$ be distinct arbitrary points inside C and $\omega_n(z) = (z-z_1)(z-z_2)\dots(z-z_n)$. Show that the integral

$$P(z) = \frac{1}{2\pi i} \int_C \frac{f(\xi) \omega_n(\xi) - \omega_n(z)}{\xi - z} d\xi$$

is a polynomial of degree $n-1$ that coincides with $f(z)$ at the points $z_1, z_2, \dots, z_n$ (the polynomial $P(z)$ is called Lagrange's interpolation polynomial).

423. Prove the following theorem (Cauchy's formula for an infinite domain):

Let C be a simple closed contour bounding a finite domain D . The function $f(z)$ is analytic in the exterior of the domain D , and $\lim_{z \rightarrow \infty} f(z) = A$. Then

$$\frac{1}{2\pi i} \int_C \frac{f(\xi)}{\xi - z} d\xi = \begin{cases} -f(z) + A & \text{if the point } z \text{ belongs to the exterior of } D, \\ A & \text{if the point } z \text{ belongs to } D. \end{cases}$$

The contour C is traversed in the positive direction with respect to the domain D .

Hint. First consider the case $A = 0$.

424. Let the function $f(z)$ and the contour C satisfy the conditions of the preceding problem.

Prove that if the origin of coordinates belongs to the domain D , then

$$\frac{1}{2\pi i} \int_C \frac{f(\xi)}{\xi z - \xi^2} d\xi = \begin{cases} 0 & \text{if } z \in D, \\ f(z)/z & \text{if } z \notin D. \end{cases}$$

§ 4. POWER SERIES

FINDING THE RADIUS OF CONVERGENCE

In Problems 425 to 436 determine the radii of convergence of the series.

$$425. \sum_{n=1}^{\infty} \frac{z^n}{n}. \quad 426. \sum_{n=0}^{\infty} \frac{z^n}{n!}. \quad 427. \sum_{n=1}^{\infty} n^n z^n.$$

$$428. \sum_{n=1}^{\infty} \frac{n}{2^n} z^n. \quad 429. \sum_{n=1}^{\infty} \frac{n!}{n^n} z^n. \quad 430. \sum_{n=0}^{\infty} z^{n!}.$$

$$431. \sum_{n=0}^{\infty} 2^n z^{n!}. \quad 432. \sum_{n=0}^{\infty} z^{2^n}. \quad 433. \sum_{n=0}^{\infty} [3 + (-1)^n]^n z^n.$$

$$434. \sum_{n=0}^{\infty} \cos in \cdot z^n. \quad 435. \sum_{n=0}^{\infty} (n + a^n) z^n.$$

$$436. 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha+1)\dots(\alpha+n-1)\beta(\beta+1)\dots(\beta+n-1)}{n! \gamma(\gamma+1)\dots(\gamma+n-1)} z^n.$$

437. The convergence radius of $\sum_{n=0}^{\infty} c_n z^n$ is R ($0 < R < \infty$). Determine the radii of convergence of the following series:

$$1) \sum_{n=0}^{\infty} n^k c_n z^n; \quad 2) \sum_{n=0}^{\infty} (2^n - 1) c_n z^n; \quad 3) \sum_{n=0}^{\infty} \frac{c_n}{n!} z^n; \quad 4) \sum_{n=1}^{\infty} n^n c_n z^n;$$

$$5) \sum_{n=0}^{\infty} c_n^k z^n; \quad 6) \sum_{n=0}^{\infty} (1 + z_0^n) c_n z^n.$$

438. The convergence radii of $\sum_{n=0}^{\infty} a_n z^n$ and $\sum_{n=0}^{\infty} b_n z^n$ are r_1 and r_2 , respectively. What can be said about the radii of convergence of the following series:

$$1) \sum_{n=0}^{\infty} (a_n \pm b_n) z^n; \quad 2) \sum_{n=0}^{\infty} a_n b_n z^n; \quad 3) \sum_{n=0}^{\infty} \frac{a_n}{b_n} z^n?$$

439. Sum the following series for $|z| < 1$:

$$1) \sum_{n=1}^{\infty} n z^n; \quad 2) \sum_{n=1}^{\infty} \frac{z^n}{n}; \quad 3) \sum_{n=0}^{\infty} \frac{z^{2n+1}}{2n+1}; \quad 4) \sum_{n=1}^{\infty} (-1)^{n+1} \frac{z^n}{n}.$$

BEHAVIOUR ON THE BOUNDARY OF THE CIRCLE OF CONVERGENCE

In Problems 440 to 446 investigate the behaviour of the power series on the boundary of the circle of convergence

$$440. \sum_{n=0}^{\infty} z^n. \quad 441. \sum_{n=1}^{\infty} \frac{z^n}{n}. \quad 442. \sum_{n=1}^{\infty} \frac{z^n}{n^2}.$$

$$443. \sum_{n=1}^{\infty} \frac{(-1)^n}{n} z^n. \quad 444. \sum_{n=1}^{\infty} \frac{z^{pn}}{n} \quad (p \text{ a natural number}).$$

$$445. \sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n} z^{3n-1}. \quad 446. \sum_{n=1}^{\infty} \frac{z^{n!}}{n^2}.$$

ABEL'S SECOND THEOREM

According to this theorem if $\sum_{n=0}^{\infty} c_n$ converges, then

$$\lim_{r \rightarrow 1} \sum_{n=0}^{\infty} c_n r^n = \sum_{n=0}^{\infty} c_n \quad (0 < r < 1).$$

447. Show that the converse Abel's second theorem is not true, i.e. give an example of a divergent series $\sum_{n=0}^{\infty} c_n$ for which there exists the limit $\lim_{r \rightarrow 1} \sum_{n=0}^{\infty} c_n r^n$.

448. Using Abel's second theorem and the solution of Problem 439, prove the following equalities:

$$1) \sum_{n=1}^{\infty} \frac{\cos n\varphi}{n} = -\ln \left| 2 \sin \frac{\varphi}{2} \right| \quad (0 < |\varphi| \leq \pi);$$

$$2) \sum_{n=1}^{\infty} \frac{\sin n\varphi}{n} = \frac{\pi - \varphi}{2} \quad (0 < \varphi < 2\pi);$$

$$3) \sum_{n=0}^{\infty} \frac{\cos (2n+1)\varphi}{2n+1} = \frac{1}{2} \ln \left| \cot \frac{\varphi}{2} \right| \quad (0 < |\varphi| < \pi);$$

$$4) \sum_{n=0}^{\infty} \frac{\sin (2n+1)\varphi}{2n+1} = \frac{\pi}{4} \quad (0 < \varphi < \pi);$$

$$5) \sum_{n=1}^{\infty} (-1)^{n+1} \frac{\cos n\varphi}{n} = \ln \left(2 \cos \frac{\varphi}{2} \right) \quad (-\pi < \varphi < \pi);$$

$$6) \sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sin n\varphi}{n} = \frac{\varphi}{2} \quad (-\pi < \varphi < \pi).$$

449.* Prove that the series $\sum_{n=1}^{\infty} \frac{(-1)^{[V\sqrt{n}]} z^n}{n}$ is nonabsolutely convergent at all points of the boundary of the circle of convergence.

Hint. If $z=1$, then split the series into groups of summands of the same sign and show that these groups satisfy the conditions of Leibniz' test for alternating series. If $|z|=1$ and $z \neq 1$, then use the theorem from Problem 90, putting $a_n = (-1)^{[Vn]} z^n$, $b_n = 1/n$.

450. Prove that if a sequence of real positive numbers $\{a_n\}$ tends monotonically to zero and the radius of convergence of the series

$\sum_{n=0}^{\infty} a_n z^n$ equals unity, then the series is everywhere convergent on the circle $|z|=1$ except perhaps at the point $z=1$.

Hint. Make use of Dirichlet's test for convergence (see Problem 88).

451. Prove that if the series $\sum_{n=0}^{\infty} c_n z^n$ converges at the point $\zeta = Re^{i\theta}$ on the circumference of the circle of convergence, then it converges uniformly in any closed domain G that belongs to the circle of convergence, lies inside an angle between any two chords of the circle $|z|=R$ emanating from the point ζ and does not contain any points of the circle $|z|=R$ except the point ζ .

Note. This assertion is a more general form of Abel's second theorem.

§ 5. TAYLOR'S SERIES

EXPANDING FUNCTIONS INTO TAYLOR'S SERIES

In Problems 452 to 466 expand the indicated functions into a power series $\sum_{n=0}^{\infty} c_n z^n$ and find the radius of convergence.

452. $\cosh z$. 453. $\sinh z$. 454. $\sin^2 z$. 455. $\cosh^2 z$.

456. $(a+z)^x$ ($a^x = e^{x \ln a}$). 457. $\sqrt{z+i}$ ($\sqrt{i} = \frac{1+i}{\sqrt{2}}$).

458. $\frac{1}{az+b}$ ($b \neq 0$). 459. $\frac{z}{z^2-4z+13}$. 460. $\frac{z^2}{(z+1)^3}$.

461. $\ln \frac{1+z}{1-z}$. 462. $\text{Arc tan } z$ ($\text{Arc tan } 0 = 0$).

463. $\text{Arc sinh } z$ ($\text{Arc sinh } 0 = 0$). 464. $\ln(z^2-3z+2)$.

465. $\int_0^z e^{z^2} dz$. 466. $\int_0^z \frac{\sin z}{z} dz$.

In Problems 467 to 472 expand the indicated functions into a series of powers of $(z-1)$ and find the radius of convergence.

467. $\frac{z}{z+2}$. 468. $\frac{z}{z^2-2z+5}$. 469. $\frac{z^2}{(z+1)^3}$.

470. $\sqrt[3]{z} \left(\sqrt[3]{1} = \frac{-1+i\sqrt{3}}{2} \right)$. 471. $\ln z$. 472. $\sin(2z - z^2)$.

In Problems 473 to 477 find the first five terms of the series expansion in powers of z of the indicated functions.

473. $e^{z \sin z}$. 474. $\sqrt{\cos z}$ ($\sqrt{\cos z} = 1$ for $z = 0$).

475. $(1+z)^z = e^{z \ln(1+z)}$. 476. e^{e^z} . 477. $e^z \ln(1+z)$.

478. 1) Give a series expansion of the function $\ln(1+e^z)$ in powers of z (find the recurrence relation between the coefficients of the series).

Hint. First find the expansion of the derivative of the given function.

2) Prove that the only term of expansion containing an odd power of z is $z/2$.

Hint. Take advantage of the identity $\ln(1+e^z) - \ln(1+e^{-z}) = z$.

In Problems 479 to 483 expand the indicated functions into series in powers of z by multiplication of series and substitution of one series into another.

479. $[\ln(1-z)]^2$. 480. $[\operatorname{Ln}(1-z)]^2$ ($\operatorname{Ln} 1 = 2\pi i$).

481. $(\operatorname{Arc} \tan z)^2$ ($\operatorname{Arc} \tan 0 = 0$).

482. $\operatorname{Arc} \tan z \ln(1+z^2)$ ($\operatorname{Arc} \tan 0 = 0$).

483. $e^{\frac{z}{1-z}}$.

484. Prove that if the expansion of the function $\frac{1}{\cos z}$ is written in the form $\frac{1}{\cos z} = \sum_{n=0}^{\infty} (-1)^n \frac{E_{2n}}{(2n)!} z^{2n}$, then the numbers E_{2n} (Euler numbers) satisfy the relations

$$E_0 = 1, \quad E_0 + \binom{2n}{2} E_2 + \dots + \binom{2n}{2n} E_{2n} = 0.$$

485. Prove that if the series expansion of the function $\frac{z}{e^z - 1}$ in powers of z is written as $\frac{z}{e^z - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} z^n$, then the numbers B_n (Bernoulli numbers) satisfy the relations

$$B_0 = 1, \quad \binom{n+1}{0} B_0 + \binom{n+1}{1} B_1 + \dots + \binom{n+1}{n} B_n = 0.$$

486. Prove that all Bernoulli numbers with odd indices, except B_1 , equal zero.

Hint. Take advantage of the identity $\frac{z}{e^z - 1} - \frac{(-z)}{e^{-z} - 1} = -z$.

487. Expand the function $z \cot z$ into a series of powers of z and find the radius of convergence of the series thus obtained.

Hint. Take advantage of the equality $z \cot z = iz + \frac{2iz}{e^{2iz} - 1}$ which follows from Euler's formulas.

488. Expand the given functions into series of powers of z and find the radii of convergence of the series thus obtained:

- 1) $\ln \frac{\sin z}{z}$; 2) $\tan z$; 3) $\ln \cos z$; 4) $\frac{z}{\sin z}$.

489. Prove that the coefficients c_n of the expansion

$$\frac{1}{1-z-z^2} = \sum_{n=0}^{\infty} c_n z^n$$

satisfy the relation $c_n = c_{n-1} + c_{n-2}$ ($n \geq 2$). Find c_n and the radius of convergence of the series.

Note. The numbers c_n are called *Fibonacci numbers*.

490. In the expansion $\frac{A+Bz+Cz^2}{\alpha+\beta z+\gamma z^2+\delta z^3} = \sum_{n=0}^{\infty} c_n z^n$ ($\alpha \neq 0$) find c_0 , c_1 , c_2 and also the recurrence relation between c_n , c_{n-1} , c_{n-2} and c_{n-3} ($n \geq 3$).

GENERATING FUNCTIONS OF SYSTEMS OF POLYNOMIALS

If in a circle $|t| < R$ we have the expansion

$$F(t, z) = \sum_{n=0}^{\infty} f_n(z) t^n,$$

then the function $F(t, z)$ is called the *generating function* of the sequence $\{f_n(z)\}$. Certain properties of the sequence of functions $\{f_n(z)\}$ can often be proved via the properties of its generating function.

491. The Bernoulli polynomials $\varphi_n(z)$ are defined by the expansion

$$t \frac{e^{tz} - 1}{e^t - 1} = \sum_{n=1}^{\infty} \frac{\varphi_n(z)}{n!} t^n.$$

Prove the following properties:

- 1) $\varphi_n(z+1) - \varphi_n(z) = nz^{n-1}$;
2) if m is a natural number, then

$$\frac{\varphi_{n+1}(m)}{n+1} = 1 + 2^n + 3^n + \dots + (m-1)^n;$$

3) $\varphi_n(z) = \sum_{k=0}^{n-1} \binom{n}{k} B_k z^{n-k}$, where B_k are Bernoulli numbers (see Problem 485).

492. The function $\frac{1}{\sqrt{1-2tz+t^2}}$ is the generating function of the Legendre polynomials $P_n(z)$:

$$\frac{1}{\sqrt{1-2tz+t^2}} = \sum_{n=0}^{\infty} P_n(z) t^n.$$

Prove the following relations:

$$1) (n+1) P_{n+1}(z) - (2n+1) z P_n(z) + n P_{n-1}(z) = 0,$$

$$2) P_n(z) = P'_{n+1}(z) - 2z P'_n(z) + P'_{n-1}(z);$$

$$3) (2n+1) P_n(z) = P'_{n+1}(z) - P'_{n-1}(z).$$

Hint. Differentiate the generating function with respect to t and to z .

493. Using the integral formula for the coefficients of Taylor's series, prove that if $-1 < s < 1$, then

$$P_n(s) = \frac{1}{2\pi i} \int_C \frac{\zeta^n d\zeta}{\sqrt{1-2s\zeta+\zeta^2}},$$

where C is a circle of radius $R > 1$ centred at the point $\zeta = 0$.

494. Prove that $\frac{4-t^2}{4-4tz+t^2}$ is the generating function of the Chebyshev polynomials:

$$T_n(z) = \frac{1}{2^{n-1}} \cos(n \arccos z).$$

Using the integral formulas for the coefficients of Taylor's series, establish that $4T_{n+1}(z) - 4zT_n(z) + T_{n-1}(z) = 0$ for $n \geq 2$.

495. The Hermite-Chebyshev polynomials $H_n(z)$ are defined by the expansion

$$e^{2tz-t^2} = \sum_{n=0}^{\infty} \frac{H_n(z)}{n!} t^n.$$

Prove the following relations:

$$1) H_{n+1}(z) - 2zH_n(z) + 2nH_{n-1}(z) = 0 \quad (n \geq 1);$$

$$2) H'_n(z) = 2nH_{n-1}(z) \quad (n \geq 1);$$

$$3) H''_n(z) - 2zH'_n(z) + 2nH_n(z) = 0 \quad (n \geq 0);$$

$$4) H_n(z) = (-1)^n e^{z^2} \frac{d^n(e^{-z^2})}{dz^n}.$$

496. The Chebyshev-Laguerre polynomials are given by

$$L_n(z) = e^z \frac{d^n(z^n e^{-z})}{dz^n}.$$

Find the generating function of the sequence $\{L_n(z)\}$ and use it to obtain a recurrence relation connecting

$$L_{n-1}(z), L_n(z), \text{ and } L_{n+1}(z).$$

Note. In Problems 492 to 496 we considered only certain particular properties of the indicated systems of polynomials.

SOLVING DIFFERENTIAL EQUATIONS

In Problems 497 to 499 find solutions to the given differential equations that satisfy the conditions $w(0)=0$, $w'(0)=1$.

497. $w'' - z^2 w = 3z^2 - z^4$.

498. $(1-z^2)w'' - 2zw' + n(n+1)w = 0$.

499. $(1-z^2)w'' - 4zw' - 2w = 0$.

500. Expand the function $\cos(m \arcsin z)$ ($\arcsin 0 = 0$) into a series of the form $\sum_{n=0}^{\infty} c_n z^n$ by setting up a differential equation for which this function is a solution.

501. The differential equation

$$z(1-z) \frac{d^2 w}{dz^2} + [c - (a+b+1)z] \frac{dw}{dz} - abw = 0$$

is called *hypergeometric*.

Find an analytic (at the point $z=0$) solution $w(z)$ to the hypergeometric equation that satisfies the condition $w(0)=1$, assuming that c is not equal to zero or to a negative integer.

502. Prove that the general solution of a hypergeometric equation has the form (c not equal to an integer)

$$w = C_1 F(a, b, c, z) + C_2 z^{1-c} F(a+1-c, b+1-c, 2-c, z),$$

where $F(a, b, c, z)$ is the function defined in the preceding problem (a hypergeometric series).

503*. Prove that if c is not equal to zero or to a negative integer, then

$$\frac{dF(a, b, c, z)}{dz} = \frac{ab}{c} F(a+1, b+1, c+1, z).$$

§ 6. SOME APPLICATIONS OF CAUCHY'S INTEGRAL FORMULA AND OF POWER SERIES

ZEROS OF ANALYTIC FUNCTIONS

504. Prove that the point z_0 is a zero of order k of an analytic function $f(z)$ if and only if in a certain neighbourhood of the point z_0 we have the equality $f(z) = (z-z_0)^k \varphi(z)$, where the function $\varphi(z)$ is analytic at the point z_0 , and $\varphi(z_0) \neq 0$.

505. Find the order of the zero $z=0$ for the following functions:

1) $z^2(e^{z^2}-1)$; 2) $6\sin z^3+z^3(z^6-6)$; 3) $e^{\sin z}-e^{\tan z}$.

506. The point z_0 is a zero of order k of the function $f(z)$ and a zero of order l of the function $\varphi(z)$. What is the point z_0 for the following functions:

1) $f(z)\varphi(z)$; 2) $f(z)+\varphi(z)$; 3) $\frac{f(z)}{\varphi(z)}$?

In Problems 507 to 521, find the orders of all zeros of the given functions.

507. z^3+9 . 508. $\frac{z^2+9}{z^4}$. 509. $z\sin z$.

510. $(1-e^z)(z^2-4)^3$. 511. $1-\cos z$.

512. $\frac{(z^2-\pi^2)^3\sin z}{z^7}$.

513. $\frac{1-\cot z}{z}$. 514. $e^{\tan z}$. 515. $\sin^3 z$.

516. $\frac{\sin^3 z}{z}$. 517. $\sin z^3$. 518. $\cos^3 z$.

519. $\cos z^3$. 520. $(\sqrt{z}-2)^3$. 521. $(1-\sqrt{2-2\cos z})^2$.

UNIQUENESS THEOREM

522. Can a sequence of zeros (or, generally, of A -points) of a function that is different from an identical constant and is analytic in the entire finite plane have a limit point?

523. Does a function exist that is analytic at the point $z=0$ and takes on the following values at the points $z=\frac{1}{n}$ ($n=1, 2, \dots$)

1) $0, 1, 0, 1, 0, 1, \dots, 0, 1, \dots$;

2) $0, \frac{1}{2}, 0, \frac{1}{4}, 0, \frac{1}{6}, \dots, 0, \frac{1}{2k}, \dots$;

3) $\frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{6}, \frac{1}{6}, \dots, \frac{1}{2k}, \frac{1}{2k}, \dots$;

4) $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \frac{6}{7}, \dots, \frac{n}{n+1}, \dots$?

524. Are there functions which are analytic at the point $z=0$ and which satisfy the following conditions (n natural):

1) $f\left(\frac{1}{n}\right)=f\left(-\frac{1}{n}\right)=\frac{1}{n^2}$;

2) $f\left(\frac{1}{n}\right)=f\left(-\frac{1}{n}\right)=\frac{1}{n^3}$?

525. The function $\sin \frac{1}{1-z}$ has an infinite sequence of zeros convergent at the point $z=1$, yet it is not a constant. Doesn't this contradict the uniqueness theorem?

EXPRESSING AN ANALYTIC FUNCTION IN TERMS OF
ITS REAL AND IMAGINARY PART

526.* The function $f(z) = u(x, y) + iv(x, y)$ is analytic at the point $z_0 = x_0 + iy_0$ and $f(z_0) = c_0$. Prove that

$$f(z) = 2u\left(\frac{z+\bar{z}_0}{2}, \frac{z-\bar{z}_0}{2i}\right) - \bar{c}_0.$$

527. Prove that under the conditions of the preceding problem

$$f(z) = 2iv\left(\frac{z+\bar{z}_0}{2}, \frac{z-\bar{z}_0}{2i}\right) + \bar{c}_0.$$

In Problems 528 to 531 find the analytic function $f(z) = u(x, y) + iv(x, y)$, given its real or imaginary part.

528. $u(x, y) = x^2 - y^2 + 2.$

529. $u(x, y) = e^x (x \cos y - y \sin y) - \frac{y}{x^2 + y^2}.$

530. $v(x, y) = x + y - 3.$ 531. $v(x, y) = \cos x \sinh y - \sinh x \sin y.$

CAUCHY INEQUALITIES

532. Let the expansion of the function $f(z)$ in the circle $|z| < R$ have the form

$$f(z) = \sum_{n=0}^{\infty} c_n z^n.$$

1) Prove that $\frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\varphi})|^2 d\varphi = \sum_{n=0}^{\infty} |c_n|^2 r^{2n} \quad (r < R).$

2) Prove that if $\max_{|z|=r} |f(z)| = M(r)$, then the coefficients c_n satisfy the inequalities (Cauchy inequalities)

$$|c_n| \leq \frac{M(r)}{r^n} \quad (r < R).$$

3) Prove that if at least one of the Cauchy inequalities becomes an equality, that is $|c_k| = M(r)/r^k$, then the function $f(z)$ has the form $f(z) = c_k z^k$.

Hint. Take advantage of the inequality following from item 1:

$$\sum_{n=0}^{\infty} |c_n|^2 r^{2n} \leq [M(r)]^2.$$

533. Using the Cauchy inequalities, prove Liouville's theorem: a function $f(z)$, analytic throughout the plane and bounded, is constant.

Note. An alternative proof of Liouville's theorem is given in Problem 421.

534. Prove that the distance between the point $z=0$ and the zero nearest to it of the function $f(z) = \sum_{n=0}^{\infty} c_n z^n$ is not less than $\frac{\rho |c_0|}{M + |c_0|}$, where ρ is any number not exceeding the radius of convergence of the series, and $M = M(\rho) = \max_{|z|=\rho} |f(z)|$.

Hint. Establish that the function $f(z)$ has no zeros in the domain where $|f(z) - c_0| < |c_0|$, and estimate $|f(z) - c_0|$ using the Cauchy inequalities.

535. The function $f(z) = \sum_{n=0}^{\infty} c_n z^n$ is analytic at $|z| \leq r$. Prove that the series $\varphi(z) = \sum_{n=0}^{\infty} \frac{c_n z^n}{n!}$ converges throughout the plane and for its sum the bounds $|\varphi(z)| < M e^{\frac{|z|}{r}}$, $|\varphi^{(k)}(z)| < \frac{M}{r^k} e^{\frac{|z|}{r}}$ (M constant) hold.

AREA THEOREMS FOR UNIVALENT FUNCTIONS

536. Let the function $f(z) = \sum_{n=0}^{\infty} c_n z^n$ be analytic in the circle $|z| \leq 1$ and let it map this circle univalently onto a domain G of area S . Prove that $S = \pi \sum_{n=1}^{\infty} n |c_n|^2$.

Hint. Write the formula for calculating the area S in polar coordinates.

Note. If we omit the condition of univalence, then the separate parts of the domain G are to be counted as many times as the appropriate values of the function $f(z)$ are assumed in the circle $|z| \leq 1$.

537. Prove that if in the preceding problem the function $f(z)$ is analytic only in the open circle $|z| < 1$, and if there exists a finite limit $\lim_{r \rightarrow 1} S_r = S$, where S_r is the area of the image of the

circle $|z| \leq r < 1$, then the series $\sum_{n=1}^{\infty} n |c_n|^2$ converges and its sum is $\frac{S}{\pi}$. Prove also that if $\lim_{r \rightarrow 1} S_r = \infty$ then $\sum_{n=1}^{\infty} n |c_n|^2$ is a divergent series.

538. 1) Using the solution of Problem 536, prove that if $f'(0) \neq 0$ and if the function $f(z)$ maps the circle $|z| \leq 1$ conformally and one-to-one onto a domain G , then the area of G is not less than that of the circle being mapped (the extremal property of mapping onto a circle).

2) Prove that of all functions $f(z)$ that are analytic in the circle $|z| \leq R$ and satisfy the condition $\int_0^{2\pi} |f(Re^{i\varphi})|^2 d\varphi = M$ a linear function maps the circle onto a domain of minimum area. Find this area if $f(0) = 0$.

MAXIMUM-MODULUS PRINCIPLE

In Problems 539 to 542 take advantage of the maximum-modulus principle.

539. Prove that if a function $f(z) \neq \text{const}$ is analytic in a domain G and does not vanish, then $|f(z)|$ cannot attain the minimum value inside the domain G .

540. 1) Prove that inside a domain that is bounded by a simple closed level line of the modulus of a function $f(z)$ (that is a line at all points of which $|f(z)| = \text{const}$) and is contained, together with its boundary, in the domain of analyticity of the function $f(z)$, at least one zero of this function [$f(z) \neq C$] will exist.

2) Prove that if $P(z)$ is a polynomial of degree n , then the level lines of its modulus $|P(z)| = C$ (lemniscates) can decompose into at most n connected components.

541. Prove the Schwarz lemma: if the function $f(z)$ is analytic in the circle $|z| < 1$, $f(0) = 0$ and $|f(z)| \leq 1$, then $|f(z)| \leq |z|$ throughout the circle.

Prove also that if $|f(z)| = |z|$ at least at one interior point of the circle, then $f(z) = e^{i\alpha} z$ (α real).

Hint. Consider the function $f(z)/z$ and apply the maximum-modulus principle.

542. Prove that if in the preceding problem the condition $f(0) = 0$ is replaced by $f(\alpha) = 0$ ($|\alpha| < 1$), then for $|z| \leq 1$ the inequality $|f(z)| \leq \left| \frac{z - \alpha}{1 - \bar{\alpha}z} \right|$ holds true.

Hint. Consider the function $\frac{1 - \bar{\alpha}z}{z - \alpha} f(z)$.

CHAPTER IV

THE LAURENT SERIES. SINGULAR POINTS OF SINGLE-VALUED ANALYTIC FUNCTIONS. RESIDUES AND THEIR APPLICATIONS

§ 1. THE LAURENT SERIES

In Problems 543 to 560 expand the given function in a Laurent series either within the indicated annulus or in the neighbourhood of the given point. In the latter case determine the domain where the expansion holds.

543. $\frac{1}{z-2}$ in the neighbourhood of the points $z=0$ and $z=\infty$.

544. $\frac{1}{(z-a)^k}$ ($a \neq 0$, k natural) in the neighbourhood of the points $z=0$ and $z=\infty$.

545. $\frac{1}{z(1-z)}$ in the neighbourhood of the points $z=0$, $z=1$, $z=\infty$.

546. $\frac{1}{(z-a)(z-b)}$ ($0 < |a| < |b|$) in the neighbourhood of the points $z=0$, $z=a$, $z=\infty$ and in the annulus $|a| < |z| < |b|$.

547. $\frac{z^2-2z+5}{(z-2)(z^2+1)}$ in the neighbourhood of the point $z=2$ and in the annulus $1 < |z| < 2$.

548. $\frac{1}{(z^2+1)^2}$ in the neighbourhood of the points $z=i$ and $z=\infty$.

549. $\sqrt{(z-a)(z-b)}$ ($|b| \geq |a|$) in the neighbourhood of the point $z=\infty$ (consider both branches of the function).

550. $f(z) = \sqrt{\frac{z}{(z-1)(z-2)}}$ ($\operatorname{Im} f\left(\frac{3}{2}\right) > 0$) in the annulus $1 < |z| < 2$.

551. $z^2 e^{1/z}$ in the neighbourhood of the points $z=0$ and $z=\infty$.

552. $e^{\frac{1}{1-z}}$ in the neighbourhood of the points $z=1$ and $z=\infty$.

553. $\cos \frac{z^2-4z}{(z-2)^2}$ in the neighbourhood of the point $z=2$.

554. $z^2 \sin \frac{1}{z-1}$ in the neighbourhood of the point $z=1$.

555. $e^{z+\frac{1}{z}}$ in the domain $0 < |z| < \infty$.

556. $\sin z \sin \frac{1}{z}$ in the domain $0 < |z| < \infty$.

557. $\sin \frac{z}{1-z}$ in the neighbourhood of the points $z=1$ and $z=\infty$ (in the latter case confine yourself to the first four terms of the series).

558. $\cot z$ in the neighbourhood of the point $z=0$ and in the annulus $\pi < |z| < 2\pi$.

559. $\ln \frac{z-a}{z-b}$ in the neighbourhood of the point $z=\infty$.

560. $\frac{1}{z-2} \ln \frac{z-i}{z+i}$ in the neighbourhood of the point $z=\infty$ and in the annulus $1 < |z| < 2$.

561. Find out whether the following functions can be expanded in a Laurent series in the neighbourhood of the given point:

1) $\cos \frac{1}{z}$, $z=0$; 2) $\cos \frac{1}{z}$, $z=\infty$; 3) $\sec \frac{1}{z-1}$, $z=1$;

4) $\cot z$, $z=\infty$; 5) $\tanh \frac{1}{z}$, $z=0$; 6) $\frac{z^2}{\sin \frac{1}{z}}$, $z=0$;

7) $\frac{z}{\sin z-3}$, $z=\infty$; 8) $\ln z$, $z=0$; 9) $\ln \frac{1}{z-1}$, $z=\infty$;

10) $\ln \frac{z-1}{z+i}$, $z=\infty$; 11) $z^z (=e^{z \ln z})$, $z=0$.

562. Find out whether the following multi-valued functions have single-valued branches admitting expansion in a Laurent series (a Taylor series, in particular) in the neighbourhood of the given point:

1) $\sqrt{z}$, $z=0$; 2) $\sqrt{z(z-1)}$, $z=\infty$;

3) $\sqrt{\frac{z}{(z-1)(z-2)}}$, $z=\infty$; 4) $\sqrt[3]{(z-1)(z-2)(z-3)}$, $z=\infty$;

5) $\sqrt[4]{z(z-1)^2}$, $z=\infty$; 6) $\sqrt{1+\sqrt{z}}$, $z=1$;

7) $\sqrt{1+\sqrt{z}}$, $z=0$; 8) $\sqrt{z+\sqrt{z^2-1}}$, $z=\infty$;

9) $\sqrt{z+\sqrt{z^2-1}}$, $z=1$; 10) $\sqrt{1+\sqrt[3]{\frac{z}{z+1}}}$, $z=\infty$;

11) $\text{Ln}[(z-1)(z-2)]$, $z=\infty$; 12) $\text{Ln} \frac{(z-\alpha)(z-\beta)}{(z-\gamma)(z-\delta)}$, $z=\infty$;

13) Arc sin z , $z=0$; 14) Arc tan $(1+z)$, $z=0$;15) Arc sinh $(i+z)$, $z=0$; 16) $\sqrt{\frac{\pi}{2} - \text{Arc sin } z}$, $z=1$;17) $\sqrt{\frac{\pi}{4} - \text{Arc sin } z}$, $z=\frac{\sqrt{2}}{2}$.

563. The function $f(z) = \sum_{n=-\infty}^{\infty} c_n z^n$ is analytic in the annulus $r \leq |z| \leq R$ and it maps this annulus univalently onto a certain domain D .

1) Prove that the area (S) of this domain is

$$S = \pi \sum_{n=-\infty}^{\infty} n |c_n|^2 (R^{2n} - r^{2n}).$$

2) Prove that the above formula also holds when the function $f(z)$ is analytic in the domain $r < |z| < R$ alone; then both parts of the equation may become infinite simultaneously. (see Problems 536 and 537.)

564. The function $f(z)$ is univalent in the domain $|z| > 1$ and, in this domain, expands in a Laurent series of the form $f(z) = z + \frac{c_{-1}}{z} + \frac{c_{-2}}{z^2} + \dots$

Prove that

$$\sum_{n=1}^{\infty} n |c_{-n}|^2 \leq 1,$$

and determine the geometric meaning of the inequality thus obtained.

Hint. Make use of the fact that for the area S_r bounded by the image of the circle $|z|=r > 1$ we have $(f(z)=u+iv)$

$$0 \leq S_r = \int_{|z|=r} u dv = \int_0^{2\pi} \frac{f+\bar{f}}{2} \frac{1}{2i} \left(\frac{\partial f}{\partial \varphi} - \frac{\partial \bar{f}}{\partial \varphi} \right) d\varphi$$

§ 2. SINGULAR POINTS OF SINGLE-VALUED ANALYTIC FUNCTIONS

In Problems 565 to 600 find the singular points of the given functions, determine their nature and investigate the behaviour of the functions at infinity.¹⁾

565. $\frac{1}{z-z^3}$. 566. $\frac{z^4}{1+z^4}$. 567. $\frac{z^5}{(1-z)^2}$. 568. $\frac{1}{z(z^2+4)^2}$.569. $\frac{e^z}{1+z^2}$. 570. $\frac{z^2+1}{e^z}$. 571. ze^{-z} . 572. $\frac{1}{e^z-1} - \frac{1}{z}$.

¹⁾ No distinction is made in the answers between removable singular points and regular points.

573. $\frac{z^2}{z(1-e^{-z})}$. 574. $\frac{1-e^z}{2+e^z}$. 575. $\frac{1}{z^3(2-\cos z)}$. 576. $\tanh z$.
 577. $e^{-\frac{1}{z^2}}$. 578. $ze^{\frac{1}{z}}$. 579. $e^{\frac{z}{1-z}}$. 580. $e^{z-\frac{1}{z}}$. 581. $\frac{1}{e^z-1}$.
 582. $\frac{1}{\sin z}$. 583. $\frac{\cos z}{z^2}$. 584. $\tan z$. 585. $\tan^2 z$. 586. $\frac{\cot z}{z^2}$.
 587. $\cot z - \frac{1}{z}$. 588. $\cot z - \frac{2}{z}$. 589. $\frac{1}{\sin z - \sin a}$. 590. $\frac{1}{\cos z + \cos a}$.
 591. $\sin \frac{1}{1-z}$. 592. $\frac{z^7}{(z^2-4)^2 \cos \frac{1}{z-2}}$. 593. $\cot \frac{1}{z}$.
 594. $\cot \frac{1}{z} - \frac{1}{z}$. 595. $\sin \frac{1}{z} + \frac{1}{z^2}$. 596. $e^{-z} \cos \frac{1}{z}$. 597. $e^{\cot \frac{1}{z}}$.
 598. $e^{\tan \frac{1}{z}}$. 599. $\sin\left(\frac{1}{\sin \frac{1}{z}}\right)$. 600. $\sin\left(\frac{1}{\cos \frac{1}{z}}\right)$.

In Problems 601 to 610 investigate the behaviour of each of the single-valued branches of the given multi-valued function at the indicated points (determine whether the point is regular for the appropriate branch or whether it is singular; in the latter case indicate the nature of singularity).

601. $\frac{z}{1+\sqrt{z-3}}$, $z=4$. 602. $\frac{1}{\sqrt{z}+\sqrt[3]{z}}$, $z=1$.
 603. $\frac{2z+3}{1+z-2\sqrt{z}}$, $z=1$. 604. $\cos \frac{1}{1+\sqrt{z}}$, $z=1$.
 605. $\frac{1}{(2+\sqrt{z})\sin(2-\sqrt{z})}$, $z=4$.
 606. $\cot \frac{1}{1+\sqrt{z}}$, $z=\left(1+\frac{1}{k\pi}\right)^2$, where $k=\pm 1, \pm 2, \dots$, and $z=1$.
 607. $\frac{1}{\sin\left(1+\sqrt{\frac{z}{z-2}}\right)}$, $z=\frac{2(1+k\pi)}{(1+k\pi)^2-1}$, where $k=\pm 1, \pm 2, \dots$, and $z=\infty$.
 608. $\sin \frac{1}{1+\sqrt{\frac{z}{z-2}}}$, $z=\infty$.
 609. 1) $\frac{1}{\sin \frac{\text{Ln } z}{2i}}$, $z=1$; 2) $\frac{1}{\sin \frac{\text{Ln } z}{4i}}$, $z=1$.
 610. $\sin\left(\cot \frac{\text{Ln } z}{4i}\right)$, $z=1$.

611. Let $P_n(z)$ and $Q_m(z)$ be polynomials of degree n and m , respectively. Characterize the behaviour of the following functions at infinity:

- 1) $P_n(z) + Q_m(z)$; 2) $\frac{P_n(z)}{Q_m(z)}$; 3) $P_n(z) Q_m(z)$.

612. Prove that the following two definitions are equivalent:

1) The point z_0 is called a pole of order n of the function $f(z)$ if in the Laurent expansion of $f(z)$ in the neighbourhood of z_0

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-z_0)^n, \quad c_{-n} \neq 0, \quad c_{-(n+1)} = c_{-(n+2)} = \dots = 0.$$

2) The point z_0 is called a pole of order n of the function $f(z)$ if in a certain neighbourhood of this point $f(z) = \frac{\varphi(z)}{(z-z_0)^n}$, where the function $\varphi(z)$ is analytic and $\varphi(z_0) \neq 0$.

613. Give examples of functions having only the following singularities in the extended plane:

1) a pole of second order at infinity; 2) a pole of second order at the point $z=0$, the principal part of the expansion being c_{-2}/z^2 , and a simple pole at infinity; 3) simple poles at the points $z_k = \omega^k$, where $\omega = e^{2\pi i/n}$ ($k=0, 1, 2, \dots, n-1$).

614. Find the general form of a function having in the extended plane only the following singularities:

- 1) one simple pole;
- 2) one pole of order n ;
- 3) a pole of order two at the point $z=0$, the principal part of the expansion being $1/z^2$;
- 4) a pole of order n at the point $z=0$ and a pole of order m at infinity;
- 5) n poles of order one.

615. Let $f(z)$ be a single-valued function with no singularities in the domain G other than poles. Prove that the function $\frac{f'(z)}{f(z)-A}$ (logarithmic derivative of the function $f(z)-A$) has simple poles at all poles of the function $f(z)$ and at all A -points of this function and has no other singular points.

616. What singularity does the function $F(z) = f[\varphi(z)]$ have at the point $z=z_0$ (the case $z_0 = \infty$ is admissible) if the function $\varphi(z)$ is analytic or has a pole at this point, and the point $\zeta_0 = \varphi(z_0)$ is one of the following singularities of the function $f(\zeta)$:

- 1) a removable singularity, 2) a pole of order n , 3) an essential singularity?

617. The point z_0 (the case $z_0 = \infty$ is admissible) is an isolated singularity of the function $f(z)$ which maps a circular arc (or a straight-line segment) γ onto a certain circular arc (or a straight-

line segment) γ' . What is the nature of the singularity of the function $f(z)$ at the point z_0^* , which is symmetric to z_0 with respect to γ (the function $f(z)$ is continued across γ by the symmetry principle) if for $f(z)$ the point z_0 is:

- 1) a pole of order n ; 2) an essential singularity?

618. Sokhotsky's theorem¹⁾ asserts that if a point z_0 is an essential singularity of a function $f(z)$, then whatever the complex number A (including $A = \infty$), there exists a sequence of points $\{z_n\}$ convergent at the point z_0 such that $\lim_{n \rightarrow \infty} f(z_n) = A$. Prove that Sokhotsky's theorem holds true for a nonisolated singularity which is a limiting point for the poles.²⁾ (Sometimes such a point is simply regarded as an essential singularity.)

619. Find the following limits:

$$1) \lim_{y \rightarrow \pm \infty} \cot^2 z; \quad 2) \lim_{y \rightarrow \pm \infty} \frac{1}{\sin z}; \quad 3) \lim_{x \rightarrow \pm \infty} \frac{1}{\cosh z}; \quad 4) \lim_{\substack{x=0 \\ y \rightarrow 0}} \frac{1}{\sin \frac{1}{z}}.$$

Does the existence of these limits contradict Sokhotsky's theorem?

The Picard theorem states that in the neighbourhood of an essential singularity an analytic function takes on any finite value infinitely many times with at most one exception, which is called the *Picard exceptional value*. If one considers meromorphic functions, the possible number of exceptional values (including ∞) does not exceed two.

620. Check the Picard theorem for the following functions:

$$1) e^z; \quad 2) e^{1/z}; \quad 3) \cos \frac{1}{z}; \quad 4) \tan z; \quad 5) \tan^2 z.$$

Find exceptional values for each of these functions and show that these values (if they exist) are asymptotic, i.e. that it is possible to indicate at least one line terminating at an essential singularity along which the function tends to the exceptional value.

§ 3. EVALUATING RESIDUES

In Problems 621 to 641 find the residues of the indicated functions with respect to all isolated singular points and with respect to the point at infinity (provided the latter is not a limiting point of the singularities).

$$621. \frac{1}{z^3 - z^5}. \quad 622. \frac{z^2}{(z^2 + 1)^2}. \quad 623. \frac{z^{2n}}{(1+z)^n} \quad (n \text{ a natural number}).$$

¹⁾ Also known as the Casorati-Weierstrass theorem.

²⁾ It is assumed that poles are the only singularities in the neighbourhood of the point in question.

624. $\frac{i}{z(1-z^2)}$. 625. $\frac{z^2+z-1}{z^2(z-1)}$. 626. $\frac{\sin 2z}{(z+1)^3}$. 627. $\frac{e^z}{z^2(z^2+9)}$.
 628. $\tan z$. 629. $\frac{1}{\sin z}$. 630. $\cot^2 z$.
 631. $\cot^3 z$. 632. 1) $\cos \frac{1}{z-2}$; 2) $z^3 \cos \frac{1}{z-2}$. 633. $e^{z+\frac{1}{z}}$.
 634. $\sin z \sin \frac{1}{z}$. 635. $\sin \frac{z}{z+1}$. 636. $\cos \frac{z^2+4z-1}{z+3}$.
 637. $\frac{i}{z(1-e^{-hz})}$ ($h \neq 0$). 638. $z^n \sin \frac{1}{z}$ (n an integer).
 639. $\frac{1}{\sin \frac{1}{z}}$. 640. $\frac{\sqrt{z}}{\sin \sqrt{z}}$. 641. $\frac{\tan z}{z^n}$ (n a natural number).

In Problems 642 to 649 find the residues of each of the single-valued branches of the respective multi-valued functions with respect to the indicated points.

642. $\frac{\sqrt{z}}{1-z}$ with respect to the point $z=1$.
 643. $\frac{1}{\sqrt{2-z}+1}$ with respect to the point $z=1$.
 644. $\frac{z^a}{1-\sqrt{z}}$ ($z^a = e^{a \operatorname{Ln} z}$) with respect to the point $z=1$.
 645. $\sqrt{(z-a)(z-b)}$ with respect to the point $z=\infty$.
 646. 1) $\operatorname{Ln} \frac{z-\alpha}{z-\beta}$ with respect to the point $z=\infty$;
 2) $e^z \operatorname{Ln} \frac{z-\alpha}{z-\beta}$ with respect to the point $z=\infty$.
 647. 1) $\operatorname{Ln} z \sin \frac{1}{z-1}$ with respect to the point $z=1$;
 2) $\operatorname{Ln} z \cos \frac{1}{z-1}$ with respect to the point $z=1$.
 648. $\frac{\operatorname{Arc} \tan z}{z}$ with respect to the points $z=0$ and $z=\infty$.
 649. $z^n \operatorname{Ln} \frac{z-\alpha}{z-\beta}$ (n an integer) with respect to the points $z=0$ and $z=\infty$ (when calculating the residue with respect to the point $z=0$ it is assumed that $\alpha \neq 0$, $\beta \neq 0$).
 650. The expansion of a function in the neighbourhood of the point at infinity has the form $f(z) = c_0 + \frac{c_1}{z} + \dots$. Find $\operatorname{res} \{[f(z)]^2\}_{z=\infty}$.
 651. Find $\operatorname{res} [\varphi(z)f(z)]_{z=a}$ if $\varphi(z)$ is analytic at the point a and $f(z)$ has, at this point,
 1) a simple pole with residue A ;

2) a pole of order k with the principal part $\frac{c_{-1}}{z-a} + \dots + \frac{c_{-k}}{(z-a)^k}$.

(652) Find $\operatorname{res} \left[\frac{f'(z)}{f(z)} \right]_{z=a}$ if:

- 1) a is a zero of order n of the function $f(z)$;
- 2) a is a pole of order n of the function $f(z)$.

653. Find $\operatorname{res} \left[\varphi(z) \frac{f'(z)}{f(z)} \right]_{z=a}$ if $\varphi(z)$ is analytic at the point a and:

- 1) a is a zero of order n of the function $f(z)$;
- 2) a is a pole of order n of the function $f(z)$.

654. Find $\operatorname{res} \{f[\varphi(z)]\}_{z=a}$ if the function $\varphi(z)$ is analytic at the point a and $\varphi'(a) \neq 0$, and $f(\zeta)$ has a simple pole at the point $\zeta = \varphi(a)$ with residue A .

655. The function $\varphi(z)$ has a simple pole with residue A at the point a , and the function $f(\zeta)$ has at infinity a pole of order one with the principal part $B\zeta$. Find $\operatorname{res} \{f[\varphi(z)]\}_{z=a}$.

656. The function $f(z)$, which takes on real values on the arc l of the circle $|z-a|=R$, is analytically continued across this arc by the symmetry principle. For $f(z)$ let the point $z=\beta$ ($\beta \neq a$) be a pole of order k with the principal part

$$\sum_{n=1}^k \frac{c_{-n}}{(z-\beta)^n}.$$

Find $\operatorname{res} [f(z)]_{z=\beta^*}$, where β^* is a point symmetric to the point $z=\beta$ with respect to l .

§ 4 EVALUATING INTEGRALS

DIRECT APPLICATION OF THE THEOREM OF RESIDUES

In Problems 657 to 666 evaluate the integrals assuming traversal of closed contours in the positive direction.

(657) $\int_C \frac{dz}{z^4+1}$, where C is the circle $x^2+y^2=2x$.

(658) $\int_C \frac{z dz}{(z-1)(z-2)^2}$, where C is the circle $|z-2|=\frac{1}{2}$.

(659) $\int_C \frac{dz}{(z-3)(z^5-1)}$, where C is the circle $|z|=2$.

Hint. Take advantage of the fact that the sum of the residues with respect to all singular points (including the point at infinity) is equal to zero.

660. $\int_C \frac{z^3 dz}{2z^4 + 1}$, where C is the circle $|z| = 1$.

661. $\int_C \frac{e^z}{z^2(z^2 - 9)} dz$, where C is the circle $|z| = 1$.

662. $\frac{1}{2\pi i} \int_C \sin \frac{1}{z} dz$, where C is the circle $|z| = r$.

663. $\frac{1}{2\pi i} \int_C \sin^3 \frac{1}{z} dz$, where C is the circle $|z| = r$.

664. $\frac{1}{2\pi i} \int_C z^n e^{z/2} dz$, where n is an integer and C is the circle $|z| = r$.

665. $\int_{|z|=3} (1+z+z^2) \left(e^{\frac{1}{z}} + e^{\frac{1}{z-1}} + e^{\frac{1}{z-2}} \right) dz$.

666. $\int_{|z|=5} \frac{z dz}{\sin z (1 - \cos z)}$.

667. Evaluate the integral $\frac{1}{2\pi i} \int_C \frac{f(z)}{zg(z)} dz$ if C is a simple closed contour bounding a domain G which contains the point $z=0$. The functions $f(z)$ and $g(z)$ are analytic in the closed domain $\bar{G}$, and the function $g(z)$ does not vanish on C and in G has only simple zeros $a_1, a_2, \dots, a_n$, none of which coincides with the origin of coordinates.

668. Let $f(z) = a_0 + a_1 z + \dots + a_n z^n$. Prove that

$$\frac{1}{2\pi i} \int_{|z|=R} z^{n-1} |f(z)|^2 dz = a_0 \bar{a}_n R^{2n}.$$

In Problems 669 to 672 evaluate the indicated integrals.

669. $\frac{1}{2\pi i} \int_C \frac{dz}{\sqrt{z^2 + z + 1}}$, where C is the circle $|z| = r \neq 1$.

670. $\frac{1}{2\pi i} \int_C \frac{dz}{(z^4 + 1)\sqrt{z^2 + 1}}$ ($\sqrt{1} = 1$), where C is the parabola $y^2 = x$ traversed in the direction of increasing y .

671. $\frac{1}{2\pi i} \int_C \frac{dz}{a^z \sin \pi z}$ ($a^z = e^{z \ln a}$), where $a > 0$ and C is the straight line $x = \alpha$, $0 < \alpha < 1$ traced upwards.

Hint. Consider $\frac{1}{2\pi i} \int_\gamma \frac{dz}{a^z \sin \pi z}$, where γ is the contour shown in Fig. 12; then take the limit as $\beta \rightarrow \infty$.

[672] $\frac{1}{2\pi i} \int_C \frac{e^z dz}{\cos z}$, where C is the contour of integration shown in Fig. 13.

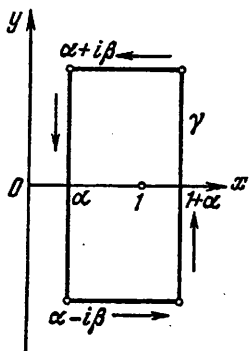


FIG. 12

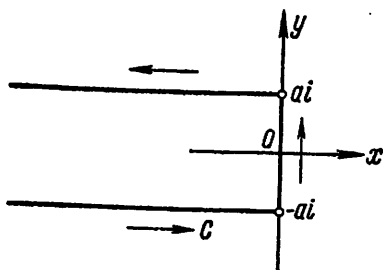


FIG. 13

DEFINITE INTEGRALS

If a function $f(x)$ goes to infinity for $x=c$ ($a < c < b$), then the Cauchy principal value of the integral $\int_a^b f(x) dx$ is

$$\lim_{\epsilon \rightarrow 0} \left[\int_a^{c-\epsilon} f(x) dx + \int_{c+\epsilon}^b f(x) dx \right].$$

This definition is naturally generalized to the case of a line integral.

If $f(x)$ is continuous over the entire number axis, then the principal value of the integral $\int_{-\infty}^{\infty} f(x) dx$ is $\lim_{N \rightarrow \infty} \int_{-N}^N f(x) dx$.

In Problems 673 to 680 find the definite integrals. If an integral is improper and divergent, find its principal value (if it exists).

673. $\int_0^{2\pi} \frac{d\varphi}{a + \cos \varphi} \quad (a > 1).$

Hint. Put $e^{i\varphi} = z$.

674. $\int_0^{2\pi} \frac{d\varphi}{(a + b \cos \varphi)^2} \quad (a > b > 0).$

675. $\int_0^{2\pi} \frac{d\varphi}{(a + b \cos^2 \varphi)^2} \quad (a > 0, b > 0).$

676. $\int_0^{2\pi} \frac{d\varphi}{1-2a \cos \varphi + a^2} \quad (a \text{ a complex number and } a \neq \pm 1).$

677. $\int_0^{2\pi} \frac{\cos^3 3\varphi d\varphi}{1-2a \cos \varphi + a^2} \quad (a \text{ a complex number and } a \neq \pm 1).$

678. $\int_0^{2\pi} e^{\cos \varphi} \cos(n\varphi - \sin \varphi) d\varphi \quad (n \text{ an integer}).$

679. $\int_0^{\pi} \tan(x+ia) dx \quad (a \text{ real}).$

680. $\int_0^{2\pi} \cot(x+a) dx \quad (a \text{ a complex number and } \operatorname{Im} a \neq 0).$

681. Prove that for $b > a > -1$

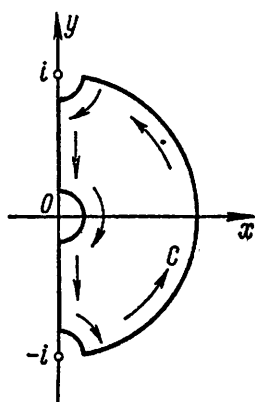


FIG. 14

$$\int_0^{\frac{\pi}{2}} \cos^a \varphi \cos b\varphi d\varphi = \frac{\pi \Gamma(a+1)}{2^{a+1} \Gamma\left(\frac{a+b}{2}+1\right) \Gamma\left(\frac{a-b}{2}+1\right)}.$$

Hint. Consider $\int_C \left(z + \frac{1}{z}\right)^a z^{b-1} dz$, where C

is the contour shown in Fig. 14; let the radii of the small circular arcs tend to zero. When calculating the integral along the vertical segment split it into two integrals and, by appropriate substitutions, reduce them

to Euler integrals of the first kind, also use the familiar ratio between the Euler integrals, $B(p, q) = \frac{\Gamma(p) \Gamma(q)}{\Gamma(p+q)}$, and the formula $\Gamma(p) \Gamma(1-p) = \frac{\pi}{\sin \pi p}$.

In Problems 682 to 688 evaluate the integrals with infinite limits.

682. $\int_{-\infty}^{\infty} \frac{x dx}{(x^2+4x+13)^2}$. 683. $\int_0^{\infty} \frac{x^2 dx}{(x^2+a^2)^2} \quad (a > 0).$

684. $\int_0^{\infty} \frac{dx}{(x^2+1)^n} \quad (n \text{ a natural number}).$

$$(685.) \int_{-\infty}^{\infty} \frac{dx}{(x^2+a^2)(x^2+b^2)} \quad (a > 0, b > 0). \quad 686. \int_0^{\infty} \frac{x^2+1}{x^4+1} dx.$$

$$687. \int_0^{\infty} \frac{dx}{1+x^n} \quad (n \geq 2, \text{ a natural number}).$$

Hint. Consider the integral $\int_C \frac{dz}{1+z^n}$, where C is a contour consisting of the rays $\arg z = 0$, $\arg z = 2\pi/n$ and a circular arc connecting them.

$$688. \int_0^{\infty} \frac{x^n}{1+x^{2n}} dx \quad (n \geq 2).$$

Note. The method of evaluating the integrals in Problems 687 and 688 can be extended to integrals of rational functions of the form $R(x^n)$.

689. Prove that

$$\frac{1}{2\pi i} \int_C \frac{d\tau}{\tau^n \tau^k} = \frac{i^{k-n-1} (n+k-2)!}{(2h)^{n+k-1} (k-1)! (n-1)!} \quad (n \text{ and } k \text{ natural numbers}),$$

where C is a straight line parallel to the real axis and intercepting on the imaginary axis a segment equal to h ($h > 0$).

690. Evaluate the integral

$$\frac{1}{2\pi i} \int_C \frac{d\tau}{\tau^k (\tau - z)} \quad (k \text{ a natural number}),$$

where C is the contour of the preceding problem.

In Problems 691 to 694 evaluate the indicated integrals using Jordan's lemma (see Problem 402).

$$(691.) 1) \int_{-\infty}^{\infty} \frac{x \cos x dx}{x^2 - 2x + 10}; \quad 2) \int_{-\infty}^{\infty} \frac{x \sin x dx}{x^2 - 2x + 10}.$$

$$692. \int_{-\infty}^{\infty} \frac{x \sin x dx}{x^2 + 4x + 20}.$$

$$693. \int_0^{\infty} \frac{\cos ax}{x^2 + b^2} dx \quad (a \text{ and } b \text{ positive}).$$

694. $\int_0^{\infty} \frac{x \sin ax}{x^2 + b^2} dx$ (a and b positive).

695. Let $f(z) = e^{imz}F(z)$, where $m > 0$ and the function $F(z)$ possesses the following properties:

1) it has a finite number of singular points $a_1, a_2, \dots, a_n$ in the upper half-plane;

2) it is analytic at all points of the real axis, except at the points $x_1, x_2, \dots, x_m$, which are simple poles;

3) $F(z) \rightarrow 0$ if $z \rightarrow \infty$ and $\text{Im } z \geq 0$.

Prove that

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \left\{ \sum_{k=1}^m \text{res } [f(z)]_{z=a_k} + \frac{1}{2} \sum_{k=1}^m \text{res } [f(z)]_{z=x_k} \right\},$$

where the integral is taken in the sense of the principal value (with respect to all points x_k and ∞).

In Problems 696 to 700 find the principal values of the indicated integrals (t a real number).

696. $\int_{-\infty}^{\infty} \frac{e^{itx}}{x} dx.$ 697. $\int_{-\infty}^{\infty} \frac{x \cos x dx}{x^2 - 5x + 6}.$

698. $\int_{-\infty}^{\infty} \frac{\sin x dx}{(x^2 + 4)(x - 1)}.$ 699. $\int_{-\infty}^{\infty} \frac{\cos tx}{1 + x^3} dx.$ 700. $\int_{-\infty}^{\infty} \frac{\cos tx}{1 - x^4} dx.$

In Problems 701 to 706 evaluate the indicated integrals (a and b positive numbers).

701. $\int_0^{\infty} \frac{x^2 - b^2 \sin ax}{x^2 + b^2} \frac{dx}{x}.$ 702. $\int_0^{\infty} \frac{\sin ax dx}{x(x^2 + b^2)}.$

703. $\int_0^{\infty} \frac{\sin ax dx}{x(x^2 + b^2)^2}.$ 704. $\int_0^{\infty} \frac{\cos 2ax - \cos 2bx}{x^2} dx.$ 705. $\int_0^{\infty} \frac{\sin^2 x}{x^2} dx.$

Hint. Use the integral $\int_C \frac{e^{2iz} - 1}{z^2} dz$, where C is the contour shown in Fig. 15.

706. $\int_0^{\infty} \frac{\sin^3 x}{x^3} dx.$

Hint. Use the integral $\int_C \frac{e^{3iz} - 3e^{iz} + 2}{z^3} dz$, where C is the contour shown in Fig. 15.

In Problems 707 to 710 evaluate the given integrals, assuming that $x^p > 0$ for $x > 0$ (this condition holds in all subsequent problems).

707. 1) $\int_0^{\infty} x^{p-1} \cos ax \, dx \quad (a > 0, 0 < p < 1);$

2) $\int_0^{\infty} x^{p-1} \sin ax \, dx \quad (a > 0, -1 < p < 1).$

Hint. Use the integral $\int_C z^{p-1} e^{-az} \, dz$, where C is the contour shown in Fig. 16.

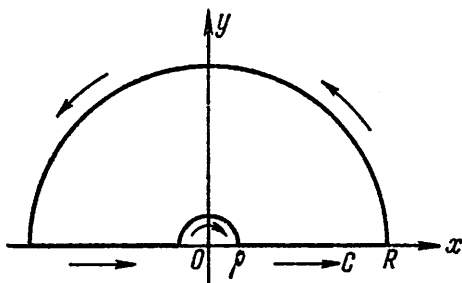


FIG. 15

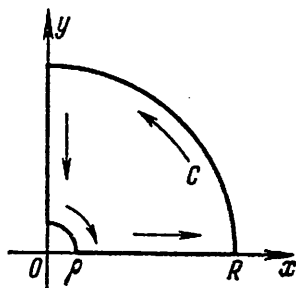


FIG. 16

708. $\int_0^{\infty} \cos x^p \, dx \quad (p > 1).$ 709. $\int_0^{\infty} \sin x^p \, dx \quad (|p| > 1).$

710. $\int_0^{\infty} \frac{\sin x^p}{x^p} \, dx \quad \left(p > \frac{1}{2}\right).$

711. Let the rational function $f(z)$ have poles $\alpha_1, \alpha_2, \dots, \alpha_n$, none of which lies on the positive real axis or is zero, and let p be a real number such that

$$\lim_{z \rightarrow 0} [z^{p+1} f(z)] = \lim_{z \rightarrow \infty} [z^{p+1} f(z)] = 0.$$

Prove that:

1) If p is not an integer, then

$$\int_0^{\infty} x^p f(x) \, dx = -\frac{\pi}{\sin \pi p} e^{-\pi p i} \sum_{k=1}^n \operatorname{res} [z^p f(z)]_{z=\alpha_k}.$$

2) If p is an integer, then

$$\int_0^{\infty} x^p f(x) \, dx = -\sum_{k=1}^n \operatorname{res} [z^p \operatorname{Ln} z \cdot f(z)]_{z=\alpha_k},$$

where

$$\operatorname{Ln} z = \ln |z| + i \arg z \quad \text{and} \quad 0 \leq \arg z < 2\pi.$$

Hint. Consider the integrals $\int_C z^p f(z) dz$ and $\int_C z^p \operatorname{Ln} z \cdot f(z) dz$, where C is the contour shown in Fig. 17.

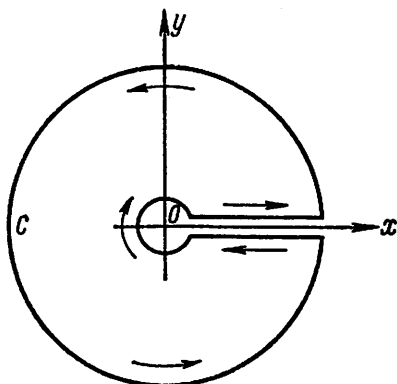


FIG. 17

712. Evaluate $\int_0^{\infty} \frac{dx}{x^p (x+1)}$ ($0 < p < 1$).

713. Prove that

$$\Gamma(a) \Gamma(1-a) = \frac{\pi}{\sin \pi a} \quad (0 < a < 1).$$

Hint. Take advantage of the familiar relation between Euler's integrals $\Gamma(a) \Gamma(b) = \Gamma(a+b) B(a, b)$, and, putting $x = y/(1+y)$, change the variable in the integral which defines the beta function

$$B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx.$$

Note. The relation, which is proved in the problem only for real numbers a in the interval $(0, 1)$, is valid for all complex numbers. For $z = -n$, where n is a natural number, both members of the equality become infinite.

In Problems 714 to 716 evaluate the indicated integrals.

714. $\int_0^{\infty} \frac{x^p dx}{1+x^2}$ ($-1 < p < 1$). 715. $\int_0^{\infty} \frac{x^p}{(1+x^2)^2}$ ($-1 < p < 3$).

716. $\int_0^{\infty} \frac{x^p dx}{x^2 + 2x \cos \lambda + 1}$ ($-1 < p < 1$, $-\pi < \lambda < \pi$).

717. On the positive real axis, let the rational function $f(z)$ have only simple poles $\beta_1, \beta_2, \dots, \beta_m$, none of the other poles $\alpha_1, \alpha_2, \dots, \alpha_n$ (if such exist) being equal to zero. Also let p be a real number such that

$$\lim_{z \rightarrow 0} [z^{p+1} f(z)] = \lim_{z \rightarrow \infty} [z^{p+1} f(z)] = 0.$$

Taking for the integral its principal value, prove that

1) If p is not an integer, then

$$\int_0^\infty x^p f(x) dx = -\frac{\pi}{\sin \pi p} e^{-\pi p i} \sum_{k=1}^n \operatorname{res} [z^p f(z)]_{z=\alpha_k} - \\ - \pi \cot \pi p \sum_{k=1}^m \beta_k^p \operatorname{res} [f(z)]_{z=\beta_k},$$

where $x^p > 0$ for $x > 0$.

2) If p is an integer, then

$$\int_0^\infty x^p f(x) dx = - \sum_{k=1}^n \operatorname{res} [z^p \operatorname{Ln} z \cdot f(z)]_{z=\alpha_k} - \\ - \sum_{k=1}^m \beta_k^p (\ln \beta_k + \pi i) \operatorname{res} [f(z)]_{z=\beta_k};$$

the branch of $\operatorname{Ln} z$ is chosen in the same manner as in Problem 711.

In Problems 718 to 721 find the principal values of the integrals.

$$718. \int_0^\infty \frac{x dx}{x^4 - 1}. \quad 719. \int_0^\infty \frac{x^2 dx}{x^4 - 1}.$$

$$720. \int_0^\infty \frac{x^p dx}{1-x} \quad (-1 < p < 0).$$

$$721. \int_{-\infty}^\infty \frac{e^{px} dx}{1-e^x} \quad (0 < p < 1).$$

In Problems 722 to 728 evaluate the indicated integrals.

$$722. \int_0^1 \frac{x^{1-p} (1-x)^p}{(1+x^2)} dx \quad (-1 < p < 2).$$

Hint. Consider $\int_C \frac{z^{1-p} (1-z)^p}{(1+z)^2} dz$, where C is the contour shown in

Fig. 18 that bounds a doubly connected domain; then take the limit as $R \rightarrow \infty$.

$$723. \int_0^1 \frac{x^{1-p} (1-x)^p}{1+x^2} dx \quad (-1 < p < 2).$$

Hint. Prove that

$\lim_{R \rightarrow \infty} \int_{C_R} \frac{z^{1-p}(1-z)^p}{1+z^2} dz = 2\pi i e^{-p\pi i}$, where C_R is the circle $|z|=R$ traversed in the positive direction.

724. $\int_0^1 \frac{x^{1-p}(1-x)^p dx}{(1+x)^2} \quad (-1 < p < 2).$

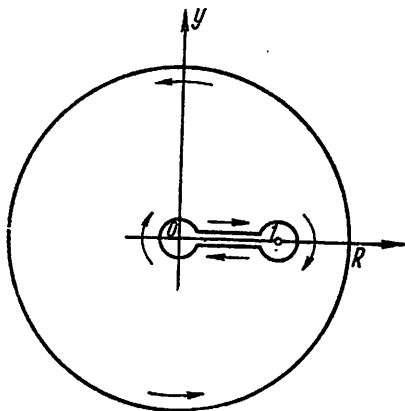


FIG. 18

725. $\int_0^1 \left(\frac{x}{1-x} \right)^p \frac{dx}{x+a} \quad (-1 < p < 1, a > 0).$

726. $\int_0^1 \left(\frac{x}{1-x} \right)^p \frac{dx}{(x+a)^2} \quad (-1 < p < 1, a > 0).$

727. $\int_{-1}^1 \frac{(1+x)^{1-p}(1-x)^p}{1+x^2} dx \quad (-1 < p < 2).$

728. $\int_0^1 \frac{dx}{(x+1) \sqrt[3]{x^2(1-x)}}.$

729. Evaluate the integral $\int_{-1}^1 \frac{dx}{(x-a) \sqrt{1-x^2}}$, where $\sqrt{1-x^2} > 0$

for $-1 < x < 1$, a is a complex number and $a \neq \pm 1$. Find, in particular, the values of the integral when $a = \pm e^{i\alpha}$ ($0 < \alpha < \pi$), $a = iy$, and $-1 < a < 1$ (principal value).

730. Evaluate the integral $\int_0^1 \frac{x^{p-1} (1-x)^{-p}}{b-x} dx$, where $0 < p < 1$, b is a complex number and $b \neq 0$, $b \neq 1$.

731. Evaluate the integral $\int_0^1 \frac{dx}{\sqrt[n]{1-x^n}}$ ($n = 2, 3, \dots$)

Hint. Consider the integral $\int_C \frac{dz}{\sqrt[n]{1-z^n}}$, where C is a contour consisting of cuts along radius vectors of the points $1, \omega, \omega^2, \dots, \omega^{n-1}$, where $\omega = e^{\frac{2\pi i}{n}}$, and of the circle $|z| = R > 1$. (This integral can also be evaluated with the aid of Euler's beta function.)

In Problems 732 to 737 evaluate the indicated integrals ($a > 0$).

732. $\int_0^\infty \frac{\ln x \, dx}{x^2 + a^2}$.

Hint. Use the integral $\int_C \frac{\ln z \, dz}{z^2 + a^2}$, where C is the contour shown in Fig. 19.

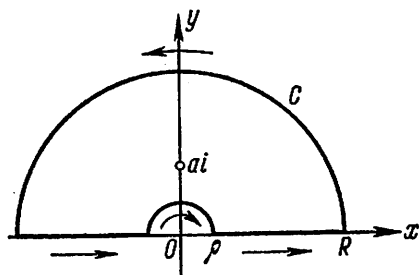


FIG. 19

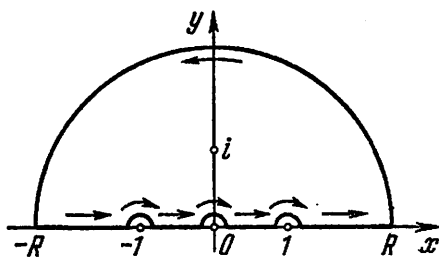


FIG. 20

733. $\int_0^\infty \frac{\ln^2 x \, dx}{x^2 + a^2}$. 734. $\int_0^\infty \frac{\ln x \, dx}{\sqrt{x} (x^2 + a^2)^2}$. 735. $\int_0^\infty \frac{\ln x \, dx}{\sqrt{x} (x+1)^2}$.

736. $\int_0^1 \ln \left(\frac{1}{x} - x \right) \frac{dx}{1+x^2}$.

Hint. Evaluate the real part of the integral

$$\int_C \ln \left(\frac{1}{z} - z \right) \frac{dz}{1+z^2},$$

where C is the contour shown in Fig. 20.

737. $\int_0^1 \ln \frac{1-x}{x} \frac{dx}{x+a}.$

738. Let $f(z)$ be a rational function that has no poles on the positive real axis and at the point $z=0$; $f(z)=O\left(\frac{1}{z}\right)$ as $z \rightarrow \infty$.

Prove that

$$\int_0^{\infty} \frac{f(x) dx}{\ln^2 x + \pi^2} = \sum_{k=1}^n \operatorname{res} \left[\frac{f(z)}{\operatorname{Ln} z - \pi i} \right]_{z=a_k},$$

where $a_1 = -1$, and $a_2, a_3, \dots, a_n$ are poles of the function $f(z)$ different from -1 , and $\operatorname{Ln} z = \ln |z| + i \arg z$, $0 \leq \arg z < 2\pi$.

Hint. Consider the integral $\frac{1}{2\pi i} \int_C \frac{f(z)}{\operatorname{Ln} z - \pi i} dz$, where C is the contour shown in Fig. 21.

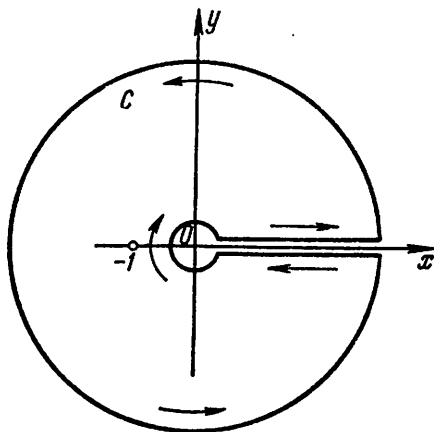


FIG. 21

In Problems 739 to 741 evaluate the indicated integrals, assuming that $a > 0$ and n is a natural number.

739. 1) $\int_0^{\infty} \frac{dx}{(x+a)(\ln^2 x + \pi^2)};$

2) $\int_0^{\infty} \frac{dx}{(x^2+a^2)(\ln^2 x + \pi^2)}.$

740. $\int_0^{\infty} \frac{dx}{(x^2+a^2)[\ln^2 x + (2n+1)^2 \pi^2]}.$

Hint. Use the integral

$$\int_C \frac{1}{z^2 + a^2} \left[\frac{1}{\operatorname{Ln} z - (2n+1)\pi i} + \frac{1}{\operatorname{Ln} z - (2n-1)\pi i} + \cdots + \frac{1}{\operatorname{Ln} z + (2n-1)\pi i} \right] dz,$$

where C is the contour shown in Fig. 21, and the branch of $\operatorname{Ln} z$ is chosen in the same manner as in Problem 738.

741.
$$\int_0^\infty \frac{dx}{(x^2 + a^2)(\ln^2 x + 4n^2\pi^2)}.$$

Hint. Use the integral

$$\int_C \frac{1}{z^2 + a^2} \left[\frac{1}{\operatorname{Ln} z - 2n\pi i} + \frac{1}{\operatorname{Ln} z - (2n-2)\pi i} + \cdots + \frac{1}{\operatorname{Ln} z + (2n-2)\pi i} \right] dz,$$

where C is the contour shown in Fig. 22, and the branch of $\operatorname{Ln} z$ is chosen in the same manner as in Problem 738.

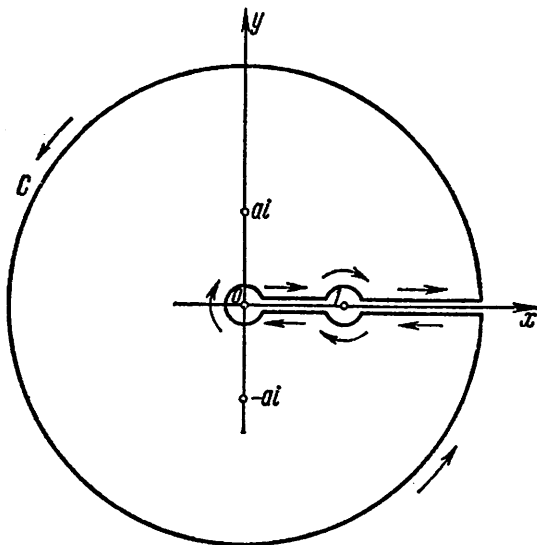


FIG. 22

742. Let $f(z)$ be a rational function that has no poles on the open contour C with starting point a and terminal point b .

Prove that

$$\int_C f(z) dz = \sum \operatorname{res} \left[f(z) \operatorname{Ln} \frac{z-b}{z-a} \right] + \operatorname{res} \left[f(z) \operatorname{Ln} \frac{z-b}{z-a} \right]_{z=\infty},$$

where the summation is over all poles of the function $f(z)$ different from ∞ (choice of a single-valued branch of the logarithm outside C is arbitrary).

Hint. Consider $\int_{\Gamma} f(z) \operatorname{Li} \frac{z-b}{z-a} dz$, where Γ (the contour bounding the doubly connected domain) is as shown in Fig. 23.

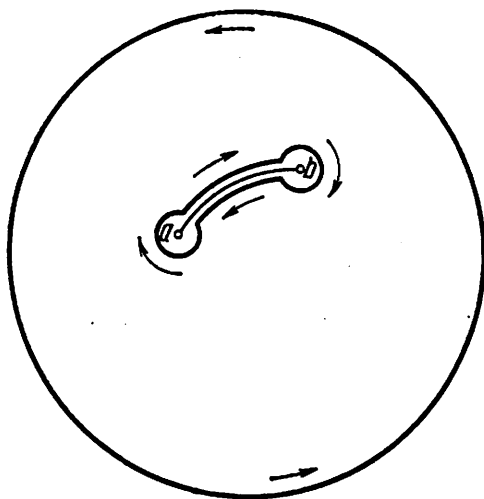


FIG. 23

In Problems 743 to 747 find the indicated integrals, assuming that a is a real number.

743. $\int_{-\infty}^{\infty} \frac{e^{ax} dx}{(e^x + 1)(e^x + 2)} \quad (0 < a < 2).$

Hint. Use the integral $\int_C \frac{e^{az}}{(e^z + 1)(e^z + 2)} dz$, where C is a rectangle with vertices $-R$, R , $R + 2\pi i$, $-R + 2\pi i$.

744. $\int_0^{\infty} \frac{\sin ax dx}{\sinh x}.$

Hint. Use the integral $\int_C \frac{e^{alz}}{\sinh z} dz$, where C is the contour shown in Fig. 24.

745. $\int_0^{\infty} \frac{\cos ax dx}{\cosh x}.$

746. $\int_0^{\infty} \frac{x \cos ax dx}{\sinh x}.$ 747. $\int_0^{\infty} \frac{\cosh ax}{\cosh \pi x} dx \quad (-\pi < a < \pi).$

Hint. Use the integral $\int_C \frac{e^{az} dz}{\cosh \pi z}$, where C is the boundary of the rectangle $-\alpha \leq \operatorname{Re} z \leq \alpha$, $0 \leq \operatorname{Im} z \leq 1$.

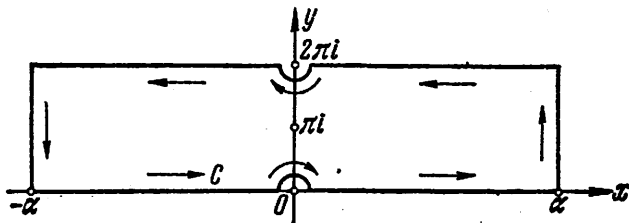


FIG. 24

$$748. \int_0^{\pi} \frac{x \sin x dx}{1 + a^2 - 2a \cos x} \quad (a > 0).$$

Hint. Use the integral $\int_C \frac{z dz}{a - e^{-iz}}$, where C is the boundary of the rectangle $-\pi \leq \operatorname{Re} z \leq \pi$, $0 \leq \operatorname{Im} z \leq h$, then take the limit as $h \rightarrow \infty$.

INTEGRALS ASSOCIATED WITH THE INVERSE FORMULA OF THE LAPLACE TRANSFORMATION

From now on we assume that $t > 0$, C_1 is the straight line $\operatorname{Re} z = \alpha > 0$ traced upwards, α being chosen so that all singular points of the integrand are located to the left of C_1 .

749. Prove that if $f(z) \rightarrow 0$ as $\operatorname{Im} z \rightarrow \pm \infty$, $\alpha_1 \leq \operatorname{Re} z \leq \alpha_2$, and the function $f(z)$ is analytic in the strip $\alpha_1 \leq \operatorname{Re} z \leq \alpha_2$, then the integral $\int_C f(z) dz$, where C is the straight line $\operatorname{Re} z = \alpha$, does not depend on the choice of α provided $\alpha_1 \leq \alpha \leq \alpha_2$.

In Problems 750 to 755 find the integrals (n a natural number).

$$750. 1) \frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} dz}{z^{n+1}}; 2) \frac{1}{2\pi i} \int_{C_1} \frac{t^z dz}{z^{n+1}} \quad (t^z = e^{z \ln t}).$$

$$751. \frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} dz}{(z-a)^{n+1}}. \quad 752. \frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} dz}{z^2 + 1}.$$

$$753. 1) \frac{1}{2\pi i} \int_{C_1} \frac{ze^{zt} dz}{z^2 + 1}; 2) \frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} dz}{z^2(z^2 + 1)}.$$

$$754. \frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} dz}{(z-a)(z-b)(z-c)}. \quad 755. \frac{1}{2\pi i} \int_{C_1} \frac{t^z dz}{z(z+1) \dots (z+n)}.$$

756. Using the identity $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}$ (see the note to Problem 713), prove that for $\operatorname{Re} v < 0$

$$\frac{1}{2\pi i} \int_{\gamma} \frac{e^z dz}{z^{v+1}} = \frac{1}{\Gamma(v+1)},$$

where γ is the contour shown in Fig. 25.

Note. Since the integral $\frac{1}{2\pi i} \int_{\gamma} \frac{e^z dz}{z^{v+1}}$ converges for $\operatorname{Re} v \geq 0$ as well, it continues the function $\frac{1}{\Gamma(v+1)}$ analytically over the entire plane.

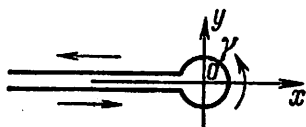


FIG. 25

757. Prove that for $\operatorname{Re} v > -1$

$$\frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} dz}{z^{v+1}} = \frac{t^v}{\Gamma(v+1)}.$$

In Problems 758 to 769 find the indicated integrals.

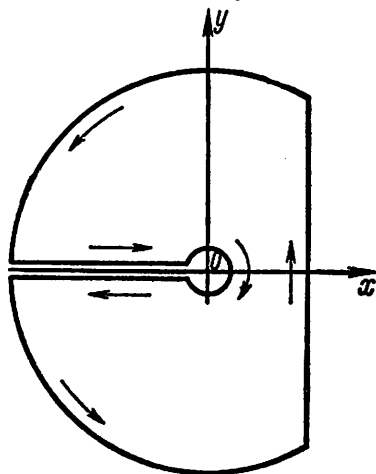


FIG. 26

758. 1) $\frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} dz}{\sqrt{1+z}}$; 2) $\frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} dz}{\sqrt{z+i}}$. 759. $\frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} dz}{z \sqrt{1+z}}$.

760. $\frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} dz}{(z+1) \sqrt{z+2}}$. 761. $\frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} (1+e^{-\pi z})}{z^2+1} dz$.

762. $\frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} (1-e^{-az})^2}{z} dz$ ($a > 0$). 763. $\frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} dz}{z(1-e^{-az})}$ ($a > 0$).

Hint. Use the expansion

$$\frac{1}{1-e^{-az}} = 1 + e^{-az} + e^{-2az} + \dots$$

764. $\frac{1}{2\pi i} \int_{C_1} \frac{e^{(z-x)\sqrt{z}}}{z} dz$ ($x > 0$).

Hint. Replace C_1 by the contour shown in Fig. 26.

765. $\frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} \sinh r \frac{1}{z}}{rz \sinh u \sqrt{z}} dz$ ($a > r > 0$).

$$766. \frac{1}{2\pi i} \int_{C_1} \frac{\ln(1+z) e^{zt}}{z} dz.$$

$$767. \frac{1}{2\pi i} \int_{C_1} e^{zt} \ln\left(\frac{z+1}{z-1}\right) dz.$$

$$768. \int_0^\infty dt \int_{C_1} \frac{e^{z-\frac{at}{z}}}{z^2} dz \quad (a > 0).$$

Hint. Change the order of integration.

$$769. \frac{1}{2\pi i} \int_{C_1} \frac{e^{bz}}{z} dz \int_0^\infty e^{-az \cosh x} dx \quad (a > 0, b \text{ a real number}).$$

Hint. Take advantage of the fact that $\int_{C_1} \frac{e^{-uz}}{z} dz = 0$ for $u > 0$.

770. Proceeding from the series expansion of Bessel's function

$$J_\nu(z) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+\nu+1)} \left(\frac{z}{2}\right)^{\nu+2k},$$

prove the following integral representations (γ is the contour in Problem 756):

$$1) \frac{1}{2\pi i} \int_{\gamma} \frac{e^{\zeta - \frac{z^2}{4\zeta}}}{\zeta^{\nu+1}} d\zeta = \left(\frac{z}{2}\right)^{\nu} J_{\nu}(z);$$

$$2) \frac{1}{2\pi i} \int_{C_1} \frac{e^{\zeta - \frac{z^2}{4\zeta}}}{\zeta^{\nu+1}} d\zeta = \left(\frac{z}{2}\right)^{\nu} J_{\nu}(z)$$

($\operatorname{Re} \nu > -1$).

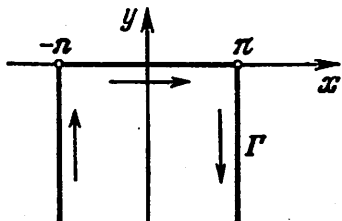


FIG. 27

Hint. Expand the function $e^{-\frac{z^2}{4\zeta}}$ into a series and make use of Problems 756 and 757.

771. Prove that for $\operatorname{Re} z > 0$

$$J_{\nu}(z) = \frac{1}{2\pi} \int_{\Gamma} e^{iz \sin \zeta - i\nu \zeta} d\zeta,$$

where Γ is the contour shown in Fig. 27, and ascertain therefrom that if n is an integer or zero, then

$$J_n(z) = \frac{1}{\pi} \int_0^\pi \cos(z \sin \zeta - n\zeta) d\zeta.$$

In Problems 772 to 774 find the integrals containing Bessel functions.

772. $\int_0^{\infty} e^{-zt} J_n(t) dt \quad (\operatorname{Re} z > 0, n \text{ an integer}).$

Hint. Use the integral representation of the preceding problem and change the order of integration.

773. 1) $\int_0^{\infty} J_0(at) \cos bt \, dt$; 2) $\int_0^{\infty} J_0(at) \sin bt \, dt$ (a and b positive numbers).

774. $\int_0^{\infty} \cos bx \frac{\sin t \sqrt{x^2 - a^2}}{\sqrt{x^2 - a^2}} dx \quad (t > |b|).$

Hint. Make use of the following relation:

$$\frac{\sin ut}{u} = \sqrt{\frac{\pi t}{2u}} J_{\frac{1}{2}}(ut) = \frac{\sqrt{\pi t}}{2} \frac{1}{2\pi i} \int_{C_1} \frac{e^{z - \frac{u^2 t^2}{4z}}}{z^{\frac{3}{2}}} dz$$

(see Problem 770) and change the order of integration.

ASYMPTOTIC BEHAVIOUR OF INTEGRALS

775. Let the analytic function $\varphi(z)$ have, on the left of C_1 , only a finite number of singular points, all of them being poles, and let $\varphi(z) \rightarrow 0$ as $z \rightarrow \infty$ and $\operatorname{Re} z \leq \alpha$. Put

$$f(t) = \frac{1}{2\pi i} \int_{C_1} e^{zt} \varphi(z) dz.$$

Find $\lim_{t \rightarrow \infty} f(t)$. Consider various arrangements of poles with respect to the imaginary axis.

Hint. Use Jordan's lemma (see Problem 402).

776. Let the analytic function $\varphi(z)$ have, on the left of C_1 , a finite number of singularities and let $\varphi(z) \rightarrow 0$ as $z \rightarrow \infty$ and $\operatorname{Re} z \leq \alpha$.

Prove that for large values of t we have the following asymptotic equality

$$\frac{1}{2\pi i} \int_{C_1} e^{zt} \varphi(z) dz \sim \sum \operatorname{res} [e^{zt} \varphi(z)],$$

where the sum is taken over all singularities of $\varphi(z)$ with a non-negative real part.

Note. The functions $f(t)$ and $F(t)$ are asymptotically equal as $t \rightarrow \infty$ ($f(t) \sim F(t)$) if $\lim_{t \rightarrow \infty} \frac{f(t)}{F(t)} = 1$.

777. Investigate the asymptotic behaviour (as $t \rightarrow \infty$) of the function

$$f(t) = \frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} dz}{z^2(z+a)^3} \quad (\operatorname{Re} a > 0).$$

778. Find an asymptotic expression (as $t \rightarrow \infty$) for the function

$$f(t) = \frac{1}{2\pi i} \int_{C_1} \frac{ze^{zt-1} \sqrt{z^2+2az}}{(z-\omega i) \sqrt{z^2+2az}} dz \quad (\omega > 0, a > 0),$$

where $\sqrt{z^2+2az} > 0$ for $z > 0$.

Hint. Replace the contour C_1 with the contour shown in Fig. 26 and prove that the integrals along circular arcs and along the negative real axis vanish as $t \rightarrow \infty$.

The series $\sum_{n=0}^{\infty} \frac{c_n}{z^n}$ is called the asymptotic expansion of the function $f(z)$ as $z \rightarrow \infty$ if

$$\lim_{z \rightarrow \infty} z^k \left[f(z) - \sum_{n=0}^k \frac{c_n}{z^n} \right] = 0 \quad (k=0, 1, 2, \dots).$$

(It does not follow herefrom that the series is convergent!).

Asymptotic expansions of a more general form are often considered as well. Let $\{q_n(z)\}$ be an arbitrary sequence of functions such that $\lim_{z \rightarrow \infty} \frac{q_{n+1}(z)}{q_n(z)} = 0$, and $\{\mu_n(z)\}$ a sequence satisfying the conditions:

$$\lim_{z \rightarrow \infty} \frac{\mu_{n+1}(z)}{q_n(z)} = 0, \quad \overline{\lim}_{z \rightarrow \infty} \left| \frac{\mu_n(z)}{q_n(z)} \right| > 0.$$

The series $\sum_{n=0}^{\infty} c_n \mu_n(z)$ is called an asymptotic expansion of the function $f(z)$:

$$f(z) \sim \sum_{n=0}^{\infty} c_n \mu_n(z),$$

$$\text{if } \lim_{z \rightarrow \infty} \frac{1}{q_k(z)} \left[f(z) - \sum_{n=0}^k c_n \mu_n(z) \right] = 0 \quad (k=0, 1, 2, \dots).$$

The sequence $\{1/z^{\alpha_n}\}$, where α_n are positive real numbers monotonically tending to infinity, is often chosen as the sequence $\{\mu_n(z)\}$.

779. Prove that for $x > 0$

$$\int_0^{\infty} \frac{e^{-xt}}{1+t^2} dt \sim \frac{1}{x} - \frac{2!}{x^3} + \frac{4!}{x^5} - \dots + (-1)^n \frac{(2n)!}{x^{2n+1}} + \dots$$

Hint. Use the equality

$$\frac{1}{1+t^2} = 1 - t^2 + t^4 - \dots + (-1)^n t^{2n} + \frac{(-1)^{n+1} t^{2n+2}}{1+t^2}$$

and estimate the residual term.

780. Prove that for $x > 0$

$$\int_x^\infty \frac{e^{x-t}}{t} dt \sim \frac{1}{x} - \frac{1}{x^2} + \frac{2}{x^3} - \dots + (-1)^n \frac{n!}{x^{n+1}} + \dots$$

Hint. Integrate by parts and estimate the residue.

781.* Prove that for $x > 0$

$$\int_{-x}^\infty \frac{e^{-x-t}}{t} dt \sim - \left(\frac{1}{x} + \frac{1}{x^2} + \frac{2}{x^3} + \dots + \frac{(n-1)!}{x^n} + \dots \right),$$

where the integral is taken to be its principal value.

782. Prove that at real values of x

$$\int_{-\infty}^\infty \frac{e^{-t^2}}{t-x} dt \sim -\sqrt{\pi} \sum_{n=0}^\infty \frac{(2n)!}{2^{2n} n!} \frac{1}{x^{2n+1}},$$

where the integral is taken to be its principal value.

783. Prove that for real values of x

$$\int_0^\infty \frac{t^\alpha e^{-t^\beta}}{t-x} dt \sim -\frac{1}{\beta} \sum_{n=1}^\infty \Gamma\left(\frac{\alpha+n}{\beta}\right) \frac{1}{x^n} \quad (\alpha > -1, \beta > 0),$$

where for $x > 0$ the integral is taken to be its principal value.

784.* Prove that

$$\int_0^z e^{z^2-t^2} dt \sim \pm \frac{\sqrt{\pi}}{2} e^{z^2} - \left(\frac{1}{2z} - \frac{1}{2^2 z^3} + \frac{1 \times 3}{2^3 z^5} - \frac{1 \times 3 \times 5}{2^4 z^7} + \dots \right),$$

the signs $+$ and $-$ depending on whether $\operatorname{Re} z > 0$ or $\operatorname{Re} z < 0$. If $\operatorname{Re} z = 0$, then the summand in front of the bracket may be omitted.

785. Find an asymptotic expansion of the function

$$f(t) = \frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} \sqrt{z} dz}{z^2 + \omega^2} \quad (\omega > 0)$$

Also find an expansion of $f(t)$ for small t .

Hint. Replace C_1 with the contour shown in Fig. 28. To obtain an asymptotic expansion of the integral $\int_0^\infty \frac{e^{-xt} \sqrt{x} dx}{x^2 + \omega^2}$, use the hint to Problem 779. For small t choose C_1 so that α is greater than ω , and expand $\frac{1}{z^2 + \omega^2}$ into a series.

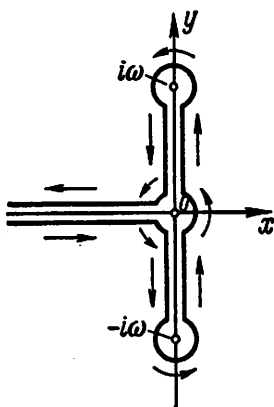


FIG. 28

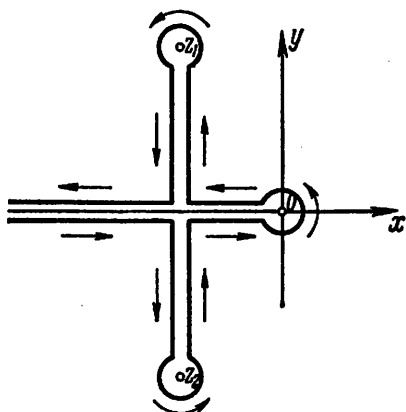


FIG. 29

786. Prove that

$$\frac{1}{2\pi i} \int_{C_1} \frac{e^{zt}}{\sqrt{z(z^2+1)}} dz - \sin\left(t - \frac{\pi}{4}\right) \sim \frac{2}{\sqrt{\pi}} \sum_{n=0}^{\infty} (-1)^n \frac{(4n)!}{(2n)!} \left(\frac{1}{2\sqrt{t}}\right)^{4n+1}.$$

787. Find an asymptotic expansion of the function

$$f(t) = \frac{1}{2\pi i} \int_{C_1} \frac{e^{zt} dz}{z \left(1 + z^{\frac{3}{2}}\right)} \quad \left(z^{\frac{3}{2}} > 0 \text{ for } z > 0\right).$$

Also obtain an approximate formula for $f(t)$ for small t .

Hint. To obtain an asymptotic expansion replace C_1 with the contour shown in Fig. 29, where $z_1 = -\frac{1}{2} + \frac{i\sqrt{3}}{2}$, $z_2 = -\frac{1}{2} - \frac{i\sqrt{3}}{2}$. For small t the abscissa of the straight line C_1 should be taken greater than unity.

§ 5. DISTRIBUTION OF ZEROS. INVERSION OF SERIES

ROUCHÉ'S THEOREM

In Problems 788 to 790 apply Rouché's theorem to determine the number of roots of the given equations that lie inside the circle $|z| < 1$.

788. $z^9 - 2z^6 + z^2 - 8z - 2 = 0$.

789. $2z^5 - z^3 + 3z^2 - z + 8 = 0$. 790. $z^7 - 5z^4 + z^2 - 2 = 0$.

791. Prove that if the inequality

$$|a_k z^k| > |a_0 + a_1 z + \dots + a_{k-1} z^{k-1} + a_{k+1} z^{k+1} + \dots + a_n z^n|$$

is valid at all points of a contour C , then the polynomial $a_0 + a_1 z + \dots + a_n z^n$ has k zeros inside the contour C if the point $z=0$ lies inside this contour, and has none if it lies outside C .

792. How many roots of the equation $z^4 - 5z + 1 = 0$ are found in the circle $|z| < 1$? in the annulus $1 < |z| < 2$?

793. How many roots of the equation $z^4 - 8z + 10 = 0$ are found in the circle $|z| < 1$? in the annulus $1 < |z| < 3$?

794. How many roots has the equation $z^n + \alpha_0 z^2 + \alpha_1 z + \alpha_2 = 0$ in the circle $|z| < 1$ if $|\alpha_0| > |\alpha_1| + |\alpha_2| + 1$ (n a natural number)?

795. How many roots has the equation $z = \varphi(z)$ in the circle $|z| < 1$ if at $|z| \leq 1$ the function $\varphi(z)$ is analytic and satisfies the inequality $|\varphi(z)| < 1$?

796. How many roots has the equation $e^z - 4z^n + 1 = 0$ in the circle $|z| < 1$ (n a natural number)?

797. How many roots has the equation $e^z = az^n$ (n a natural number) in the circle $|z| < R$ if $|a| > e^R/R^n$?

798. Prove that the equation $z = \lambda - e^{-z}$ ($\lambda > 1$) has a single (and real) root in the right half-plane.

799*. Prove that no matter how small $\rho > 0$, for a sufficiently large n all zeros of the function

$$f_n(z) = 1 + \frac{1}{z} + \frac{1}{2! z^2} + \dots + \frac{1}{n! z^n}$$

are found in the circle $|z| < \rho$.

800. Prove that if $\rho < 1$, then for n sufficiently large the polynomial

$$P_n(z) = 1 + 2z + 3z^2 + \dots + nz^{n-1}$$

has no roots in the circle $|z| < \rho$.

Hint. Apply the method of solving Problem 799.

ARGUMENT PRINCIPLE

801. The function $\varphi(z)$ is meromorphic in the domain G and analytic on its boundary C . Prove the following assertions:

1) If $|\varphi(z)| < 1$ on C , then the number of roots of the equation $\varphi(z) = 1$ in the domain G is equal to the number of poles of the function $\varphi(z)$ in that domain.

2) If $|\varphi(z)| > 1$ on C , then the number of roots of the equation $\varphi(z) = 1$ in the domain G is equal to the number of zeros of the function $\varphi(z)$ in that domain.

3) Assertions (1) and (2) hold true if the equation $\varphi(z) = 1$ is replaced by the equation $\varphi(z) = \alpha$, and $|\alpha| \geq 1$ in case (1) and $|\alpha| \leq 1$ in case (2).

802. Suppose none of the zeros of the polynomial

$$P_n(z) = z^n + a_1 z^{n-1} + \dots + a_n$$

lies on the imaginary axis.

Prove that when the point z traverses the imaginary axis downwards, then the increment of the argument of $P_n(z)$ equals $k\pi$, where k is an integer of the same parity as n , and $|k| \leq n$.

Prove that in this case the polynomial $P_n(z)$ has $(n+k)/2$ zeros in the right half-plane.

Hint. Represent $P_n(z)$ in the form

$$P_n(z) = z^n \left(1 + \frac{a_1}{z} + \dots + \frac{a_n}{z^n} \right)$$

and apply the argument principle to the semicircle $|z| < R$, $\operatorname{Re} z > 0$, for sufficiently large R .

803. How many roots has the polynomial

$$z^6 + z^5 + 6z^4 + 5z^3 + 8z^2 + 4z + 1$$

in the right half-plane?

804. How many roots has the equation

$$z^4 + 2z^3 + 3z^2 + z + 2 = 0$$

in the right half-plane and in the first quadrant?

805. How many roots has the equation

$$2z^4 - 3z^3 + 3z^2 - z + 1 = 0$$

in each quadrant?

806. In what quadrants are the roots of the equation $z^4 + z^3 + 4z^2 + 2z + 3 = 0$?

807. Prove that the number of roots of the equation

$$z^{2n} + \alpha z^{2n-1} + \beta = 0$$

(α and β real numbers, $\alpha \neq 0$, $\beta \neq 0$, n a natural number) that

have a positive real part is equal to n if n is even. If n is odd, their number is $n-1$ for $\alpha > 0$, and $n+1$ for $\alpha < 0$.

Hint. Consider the increment of $\arg(z^{2n} + \alpha z^{2n-1} + \beta^2)$ when the point z describes the boundary of a right semicircle of large radius.

If the coefficients of the polynomial

$$P_n(z) = z^n + a_1 z^{n-1} + \dots + a_{n-1} z + a_n$$

are continuously dependent on the real parameters α and β , then in order to find the dependence of the number of zeros of $P_n(z)$ located in the right half-plane upon the parameters, proceed as follows (bear in mind that each zero is continuously dependent on the coefficients of the polynomial):

In the plane α, β draw lines $P_n(i\tau) = 0$ (τ a real parameter), i.e. lines for whose points there are pure imaginary roots (or a zero) among the roots of the polynomial. These lines divide the plane α, β into domains each of which has a constant number of zeros of $P_n(z)$ with a positive real part. This number can be found by taking an arbitrary point of the appropriate domain and applying, for instance, the method in Problem 802 to it.

In Problems 808 to 810 determine the domains of the plane α, β in which the number of roots of the appropriate polynomial $P(z)$ with a positive real part is constant; find this number m for each domain.

808. $P(z) = z^3 + \alpha z^2 + \alpha z + \beta$. **809.** $P(z) = z^3 + \alpha z^2 + \beta z + 1$.

810. $P(z) = z^3 + (\alpha + \beta) z^2 + (\alpha - \beta) z + \alpha$.

811. Let $f(z) = P_n(z) + Q_m(z) e^{-\tau z}$, where $\tau > 0$; $P_n(z)$ and $Q_m(z)$ are relative prime polynomials, $n > m$ and $f(z)$ has no zeros on the imaginary axis. N is the number of zeros of the polynomial $P_n(z)$ in the right half-plane. Prove that for the function $f(z)$ to have no zeros in the right half-plane it is necessary and sufficient that the point $w = -\frac{Q_m(z)}{P_n(z)} e^{-\tau z}$ wind around the point $w = 1$ N times in the positive direction, while the point z traverses the entire imaginary axis upwards (if $P_n(z)$ has zeros on the imaginary axis, then, when moving along this axis, the point z should detour on the right round the zeros of $P_n(z)$ along semicircles of sufficiently small radii).

In Problems 812 to 814 apply the theorem of Problem 811 to find the domains in the space of coefficients a, b (that is, in the plane a, b) for which all zeros of the appropriate functions lie in

the left half-plane, assuming that $\tau > 0$ and a and b are real numbers.

$$812. z + a + be^{-z}. \quad 813. z^2 + az + be^{-z}.$$

$$814. z^2 + (az + b)e^{-z}.$$

815. Using the Rouché theorem, prove that if the function $w = f(z)$ has the expansion

$$f(z) = w_0 + c_k(z - z_0)^k + \dots \quad (c_k \neq 0, k \geq 1)$$

in the neighbourhood of the point z_0 , then for sufficiently small $r > 0$ there exists $\rho > 0$ such that any value $w \neq w_0$ of the small circle $|w - w_0| < \rho$ is taken on exactly k times in the small circle $|z - z_0| < r$ and at distinct points.

816. Using the result of the preceding problem, prove that an analytic function possesses the property of domain preservation.

817. Using the result of the preceding problem, prove the maximum-modulus principle for an analytic function. Prove that this principle is valid for arbitrary continuous domain-preserving mappings $w = f(z)$.

818. Prove that if, under the conditions of Problem 815, $k = 1$, i.e. $f'(z_0) \neq 0$, then the function $f(z)$ establishes a one-to-one conformal correspondence between a certain singly connected neighbourhood of the point z_0 and the small circle $|w - w_0| < \rho$.

Hint. Consider the function $z = f^{-1}(w)$ in the small circle $|w - w_0| < \rho$.

819. Prove that if, under the conditions of Problem 815, $k > 1$, then the function $w = f(z)$ maps one-to-one a certain singly connected neighbourhood of the point z_0 onto a k -valent circle centred at the point w_0 .

820. Extend the theorems proved in Problems 818 and 819 to the case when the point z_0 is a simple or multiple pole of the function $f(z)$.

821. Prove that if the expansion of the function $f(z)$ in the neighbourhood of the point at infinity has the form

$$f(z) = A_0 + \frac{A_1}{z} + \frac{A_2}{z^2} + \dots + \frac{A_k}{z^k} \dots,$$

then a certain neighbourhood of the point at infinity can be mapped one-to-one and conformally onto a univalent circle if $A_1 \neq 0$, and onto a k -valent circle if $A_1 = A_2 = \dots = A_{k-1} = 0$ and $A_k \neq 0$.

SERIES INVERSION

822. Let $F(z) = z - a - \omega f(z)$, the function $f(z)$ being analytic at the point $z = a$. Applying Rouché's theorem, prove that for sufficiently small $|\omega|$ there exists a circle K centred at the point $z = a$ in which the function $F(z)$ has only one zero (simple). Also show that if $f(a) \neq 0$, then for an appropriately chosen value of ω any point of a certain neighbourhood of the point $z = a$ can become a zero of the function $F(z)$.

823. Let $z = z(\omega)$ be a single-valued function defined by the equation $z - a - \omega f(z) = 0$ for $|\omega|$ sufficiently small, and let the function $f(z)$ be analytic at the point $z = a$, $f(a) \neq 0$. Prove that, for $|\omega|$ sufficiently small, for any function $\Phi(z)$ analytic at the point $z = a$, we have the following expansion:

$$\frac{\Phi(z)}{1 - \omega f'(z)} = \Phi(a) + \sum_{n=1}^{\infty} \frac{\omega^n}{n!} \frac{d^n}{da^n} \{\Phi(a) [f(a)]^n\}.$$

Hint. If we denote by C the circumference of the circle K in which the equation $z - a - \omega f(z) = 0$ has only one root (see Problem 822), then

$$\frac{\Phi(z)}{1 - \omega f'(z)} = \frac{1}{2\pi i} \int_C \frac{\Phi(\zeta)}{\zeta - a - \omega f(\zeta)} d\zeta.$$

Then expand the integrand into a series in powers of ω and estimate the residual term.

824. Using the notation of the preceding problem, prove the Lagrange formula

$$\Phi(z) = \Phi(a) + \sum_{n=1}^{\infty} \frac{\omega^n}{n!} \frac{d^{n-1}}{da^{n-1}} \{\Phi'(a) [f(a)]^n\}.$$

Obtain herefrom, in particular, an expansion of the function $z = z(\omega)$ itself into a Taylor series.

Hint. Apply the solution of the preceding problem to the function $\Phi(z) [1 - \omega f'(z)]$.

825. Expand into a power series of ω each branch of the function $z(\omega)$ defined by the equation $\omega = 2z + z^2$ (for one branch, $z(0) = 0$, for the other, $z(0) = -2$).

826. Expand into a power series of ω a branch of the function $z = z(\omega)$ defined by the equation $\omega = 2 \frac{z-a}{z^2-1}$ for which $z(0) = a$.

827. Proceeding from the definition of the Legendre polynomials $P_n(z)$ and using the generating function $\frac{1}{\sqrt{1-2zt+t^2}}$ (see Problem 492), prove that $P_n(z) = \frac{1}{2^n n!} \frac{d^n}{dz^n} [(z^2-1)^n]$.

Hint. Using the conditions of Problem 826, apply the Lagrange formula to the function $\Phi(z) = \frac{1}{\sqrt{1-2aw+w^2}} = \frac{1-z^2}{z^2-2az+1}$.

828. The function $z = z(w)$ is defined in the neighbourhood of the point $w=0$ by the equality $w = ze^{-az}$. Expand the following functions in a power series of w :

1) $z(w)$; 2) $e^{bz(w)}$.

829. Expand in powers of w the function $z = z(w)$ defined in the neighbourhood of the point $w=0$ by Kepler's equation:

$$z - a = w \sin z \quad (a \neq 0, \pm\pi, \pm 2\pi, \dots).$$

830.* Determine the radius of convergence of the expansion $z(w)$, obtained in the preceding problem, for the case $a = \frac{\pi}{2}$.

831. Prove the following generalization of Lagrange's theorem. Let $f(z)$ and $\varphi(z)$ be functions that are analytic in the neighbourhood of a point a , and let C be a circle of radius r centred at the point a such that at all its points

$$|\alpha f(z) + \beta \varphi(z)| < r.$$

If $\Phi(\zeta)$ is an analytic function of the single root of the equation $z - a - \alpha f(z) - \beta \varphi(z) = 0$, then

$$\Phi(\zeta) = \Phi(a) + \sum \frac{\alpha^m \beta^n}{m!n!} \frac{d^{m+n-1}}{da^{m+n-1}} \{ \Phi'(a) [f(a)]^m [\varphi(a)]^n \},$$

where the summation is extended over all m and n except $m=n=0$.

CHAPTER V

VARIOUS SERIES OF FUNCTIONS. PARAMETRIC INTEGRALS

§ 1. SERIES OF FUNCTIONS

In Problems 832 to 841 find the domains of convergence of the given series.

$$\begin{aligned} 832. \sum_{n=0}^{\infty} \left(z^n + \frac{1}{2^n z^n} \right). \quad 833. \sum_{n=0}^{\infty} \left(\frac{z^n}{n!} + \frac{n^2}{z^n} \right). \quad 834. \sum_{n=1}^{\infty} \left[\frac{z(z+n)}{n} \right]^n. \\ 835. \sum_{n=1}^{\infty} e^{z \ln n}. \quad 836. \sum_{n=1}^{\infty} \frac{\sin nz}{n}. \quad 837. \sum_{n=0}^{\infty} \frac{(-1)^n}{z+n}. \quad 838. \sum_{n=0}^{\infty} \frac{2^n}{z^{2n} + 1}. \\ 839. \sum_{n=1}^{\infty} \frac{z^n}{1-z^n}. \quad 840. \sum_{n=0}^{\infty} \frac{z^n}{1+z^{2n}}. \quad 841. \sum_{n=1}^{\infty} \frac{z^n}{(4+z)(4+z^2)\dots(4+z^n)}. \end{aligned}$$

842*. Prove that if the series $\sum_{n=1}^{\infty} a_n$ converges, then the series $\sum_{n=1}^{\infty} \frac{a_n z^n}{1-z^n}$ converges wherever $|z| \neq 1$; if the series $\sum_{n=1}^{\infty} a_n$ diverges, then the series $\sum_{n=1}^{\infty} \frac{a_n z^n}{1-z^n}$ converges inside the circle of convergence of the series $\sum_{n=1}^{\infty} a_n z^n$ and diverges outside this circle.

843. 1) Expand the series $\sum_{n=1}^{\infty} \frac{a_n z^n}{1-z^n}$ into a series of powers of z ; find the radius of convergence of the series obtained.

2) Prove that for $|z| < 1$, $\sum_{n=1}^{\infty} \varphi(n) \frac{z^n}{1-z^n} = \frac{z}{(1-z)^2}$, where $\varphi(n)$ is the number of those natural numbers less than n which are prime to n .

Hint. Make use of the familiar number-theory relation $\sum \varphi(n) = m$, where n runs through the values of all divisors of the number m , including 1 and m .

844. Expand the function $\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z} = \sum_{n=1}^{\infty} e^{-z \ln n}$ (Riemann zeta function) into a Taylor series in the neighbourhood of the point $z=2$ and find its radius of convergence.

In Problems 845 to 848 find the sums of the given series ($|z| \neq 1$).

$$845. \sum_{n=1}^{\infty} \left(\frac{1}{1+z^n} - \frac{1}{1+z^{n-1}} \right).$$

$$846. \sum_{n=1}^{\infty} \frac{z^n}{(1-z^n)(1-z^{n+1})}.$$

Hint. Multiply the numerator and denominator by $(1-z)$.

$$847. \sum_{n=0}^{\infty} \frac{z^{2^n}}{\prod_{k=0}^n (1+z^{2^k})}. \quad 848. \sum_{n=1}^{\infty} \frac{z^{2^n-1}}{z^{2^n}-1}.$$

849. Prove the following propositions:

1) For the series $\sum_{n=1}^{\infty} f_n(z)$ to converge uniformly on a set E , it is necessary and sufficient that for any $\varepsilon > 0$ there exist a number $N = N(\varepsilon)$ such that for all $n > N$, all $z \in E$, and any natural number p the inequality $\left| \sum_{k=n+1}^{n+p} f_k(z) \right| < \varepsilon$ is satisfied.

2) From the uniform convergence of the series $\sum_{n=1}^{\infty} |f_n(z)|$ on the set E follows the uniform convergence of the series $\sum_{n=1}^{\infty} f_n(z)$ on the same set.

850. Find sets on which the following sequences converge uniformly:

$$1) \left\{ \frac{1}{1+z^n} \right\}; \quad 2) \left\{ \frac{z^n}{1+z^{2n}} \right\}; \quad 3) \left\{ \frac{\sin nz}{n} \right\}.$$

851. Prove:

For a sequence of continuous functions $\{f_n(z)\}$ to converge uniformly on a closed bounded set E , it is necessary and sufficient that this sequence converge at all points of the set and that it converge continuously at all limit points of the set E , i.e., that for any sequence of points z_n belonging to the set E and converging to the point z_0

$$\lim_{n \rightarrow \infty} f_n(z_n) = f(z_0).$$

In Problems 852 to 856 find sets on which the given series converge uniformly.

$$852. \sum_{n=1}^{\infty} \frac{1}{n^2} \left(z^n + \frac{1}{z^n} \right). \quad 853. \sum_{n=0}^{\infty} e^{-nz}.$$

$$854. \sum_{n=1}^{\infty} e^{-z \ln n}. \quad 855. \sum_{n=1}^{\infty} \frac{\sin nz}{n^2}. \quad 856. \sum_{n=1}^{\infty} \frac{\sin nz}{n}.$$

857. Prove that the series $\sum_{n=1}^{\infty} \frac{z^n}{n^2}$ converges uniformly in the closed circle $|z| \leq 1$. Will a series obtained by termwise differentiation converge uniformly in the circle $|z| < 1$?

858. Prove that the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{z+n}$ converges uniformly in any finite part of the plane with deleted circles of arbitrarily small radius ρ and centred at the points $z = 0, -1, -2, \dots$

Also prove that this series does not converge absolutely at any point.

859. Prove that the series $\sum_{n=1}^{\infty} \frac{z^n}{n}$ converges uniformly in the interval $(-1, 0)$, and that in the same interval the series $\sum_{n=1}^{\infty} \left| \frac{z^n}{n} \right|$ converges but not uniformly. [Thus, in the interval $(-1, 0)$, the series $\sum_{n=1}^{\infty} \frac{z^n}{n}$ cannot be dominated by a convergent number series.]

Note. The present example shows that Weierstrass' sufficient test for uniform convergence is not a necessary one.

860. 1) The series $\sum_{n=0}^{\infty} \frac{z}{(1+z^2)^n}$ converges absolutely for $|z| \geq 0$, $|\arg z| \leq \frac{\pi}{4}$ (these values of z do not exhaust the domain of absolute convergence which, as can readily be seen, contains the point $z=0$ and the exterior of the lemniscate $|1+z^2|=1$). Prove that the series converges non-uniformly in the indicated domain.

Note. This shows that uniform convergence does not follow even from the absolute convergence of a series in a closed domain.

2) I prove that in the same domain the series $\sum_{n=0}^{\infty} \frac{(-1)^n z}{(1+z^2)^n}$ converges uniformly and absolutely, but not absolutely uniformly (i.e. a series of moduli does not converge uniformly).

861. Prove that if the series $\sum_{n=1}^{\infty} |f_n(z)|$ converges uniformly in any closed domain interior to G , then the series $\sum_{n=1}^{\infty} |f'_n(z)|$ possesses the same property.

§ 2. DIRICHLET SERIES

A series of the form $\sum_{n=1}^{\infty} a_n e^{-\lambda_n z}$, where a_n are complex coefficients and λ_n are positive numbers that satisfy the conditions

$$\lambda_1 < \lambda_2 < \dots \quad \text{and} \quad \lim_{n \rightarrow \infty} \lambda_n = \infty,$$

is called a *Dirichlet series* with positive exponents.

862. Prove that if the Dirichlet series converges at the point $z_0 = x_0 + iy_0$, then it converges at all points of the half-plane $\operatorname{Re} z > \operatorname{Re} z_0$, the convergence being uniform in each angle $|\arg(z - z_0)| \leq \theta < \frac{\pi}{2}$.

Hint. Apply Abel's transformation to the sum

$$\sum_{n=p}^q a_n e^{-\lambda_n z} = \sum_{n=p}^q a_n e^{-\lambda_n z_0} e^{-\lambda_n (z - z_0)}$$

and use the inequality ($a < b$, $z = x + iy$)

$$|e^{-az} - e^{-bz}| = \left| z \int_a^b e^{-zt} dt \right| \leq \frac{|z|}{x} (e^{-ax} - e^{-bx}).$$

863. Prove that if the Dirichlet series converges absolutely at the point $z = z_0$, then it converges absolutely and uniformly in the half-plane $\operatorname{Re} z \geq \operatorname{Re} z_0$.

From the theorems formulated in Problems 862 and 863 it follows that the domain of convergence of the Dirichlet series (if such exists) is the half-plane $\operatorname{Re} z > x_c$ ($x_c \geq -\infty$), and the domain of absolute convergence (if such exists), the half-plane $\operatorname{Re} z > x_a$ ($x_a \geq -\infty$), and the series either converges absolutely over the entire line $\operatorname{Re} z = x_a$ or does not converge absolutely at any point of the line. The numbers x_c and x_a are called the *abscissa of convergence* and the *abscissa of absolute convergence* of the Dirichlet series, respectively.

In Problems 864 to 870 find the abscissas of convergence (x_c) and the abscissas of absolute convergence (x_a) of the given series.

$$864. \sum_{n=0}^{\infty} e^{-n^3} e^{-zn^2}. \quad 865. \sum_{n=2}^{\infty} \frac{(-1)^n}{n} e^{-z \ln \ln n}.$$

$$866. \sum_{n=2}^{\infty} \frac{(-1)^n}{\sqrt{n}} e^{-z \ln \ln n}. \quad 867. \sum_{n=2}^{\infty} (-1)^n e^{-z \ln \ln n}.$$

$$868. \sum_{n=1}^{\infty} (-1)^n e^{-z \ln n}. \quad 869. \sum_{n=1}^{\infty} \frac{1}{n^2} e^{-z \ln n}.$$

$$870. \sum_{n=0}^{\infty} e^{e^n} e^{-zn^2}.$$

$$871. \text{ Prove that if } \lim_{n \rightarrow \infty} \frac{\ln n}{\lambda_n} = 0, \text{ then}$$

$$x_c = x_a = \overline{\lim}_{n \rightarrow \infty} \frac{\ln |a_n|}{\lambda_n}.$$

$$872. \text{ Prove that if } \overline{\lim}_{n \rightarrow \infty} \frac{\ln n}{\lambda_n} = l, \text{ then } x_a - x_c \leq l.$$

In problems 873 to 877 investigate the convergence of the Dirichlet series on the boundary of the half-plane of convergence.

$$873. \sum_{n=1}^{\infty} (-1)^n e^{-z \ln n}. \quad 874. \sum_{n=1}^{\infty} \frac{1}{n^3} e^{-z \ln n}.$$

$$875. \sum_{n=1}^{\infty} \frac{1}{n} e^{-zn}. \quad 876. \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} e^{-zn}. \quad 877. \sum_{n=1}^{\infty} \frac{(-1)^{[\sqrt{n}]}}{n} e^{-zn}.$$

Hint. See Problem 449.

In Problems 878 and 879 we consider the Dirichlet series with complex exponents.

878. Let the numbers λ_n satisfy the conditions

$$\lim_{n \rightarrow \infty} \frac{\ln n}{\lambda_n} = 0 \quad \text{and} \quad \overline{\lim}_{n \rightarrow \infty} \frac{\ln |a_n|}{|\lambda_n|} = k < \infty.$$

Prove that if $\alpha \leq \arg \lambda_n \leq \beta$, then the Dirichlet series converges absolutely at any point $z = x + iy$ for which the inequality $x \cos \varphi - y \sin \varphi - k > 0$ holds for all φ in $[\alpha, \beta]$, and diverges at the point for which

$$x \cos \varphi - y \sin \varphi - k < 0,$$

for all φ in $[\alpha, \beta]$.

879. Given an arbitrary Dirichlet series $\sum_{n=1}^{\infty} a_n e^{-\lambda_n z}$. Let

$k(\varphi, \alpha) = \overline{\lim}_{k \rightarrow \infty} \frac{\ln |a_{n_k}|}{|\lambda_{n_k}|}$ and $k(\varphi) = \lim_{\alpha \rightarrow 0} k(\varphi, \alpha)$, where $\{n_k\}$ is a sequence of all indices for which $\varphi - \alpha \leq \arg \lambda_{n_k} \leq \varphi + \alpha$ (if a subsequence $\{n_m\}$ such that $\lim_{m \rightarrow \infty} \arg \lambda_{n_m} = \varphi$ does not exist, then put $k(\varphi) = -\infty$).

Prove that if $\lim_{n \rightarrow \infty} \frac{\ln n}{\lambda_n} = 0$, then the series converges absolutely inside a domain G where points $z = x + iy$ satisfy, for arbitrary φ , the condition $x \cos \varphi - y \sin \varphi - k(\varphi) > 0$, and diverges at any point exterior to G .

§ 3. PARAMETRIC INTEGRALS

CONVERGENCE OF INTEGRALS

880. Prove the theorem:

Let C be a simple contour (closed or open) having a finite length and let $f(\tau, z)$ be a function analytic with respect to the variable z and continuous with respect to τ for all z in a certain domain D and for all points τ that belong to the contour C . Then the function given by the integral $F(z) = \int_C f(\tau, z) d\tau$ is an analytic function with respect to the variable z and

$$F'(z) = \int_C f'_z(\tau, z) d\tau.$$

If $\int_C f(\tau, z) d\tau$ is an improper integral, i.e. if the integrand has discontinuities at certain isolated values of $\tau \in C$ or the contour of integration contains a point at infinity, then the definition of convergence and uniform convergence of such an integral is completely analogous to the corresponding definitions given in courses of mathematical analysis.

881. Prove that for uniform convergence of the integral $\int_C f(\tau, z) d\tau$ on a set E with respect to some point $\tau_0 \neq \infty$ of a contour C , it is necessary and sufficient that for any $\varepsilon > 0$ there exist a number $\delta(\varepsilon)$ such that

$$\left| \int_{C_\delta} f(\tau, z) d\tau \right| < \varepsilon$$

for all points z in E and for any arc C_δ of the contour C that lies in the δ -neighbourhood of the point τ_0 and does not contain this point either inside it or at the extremities.

882. Formulate and prove a similar criterion for uniform convergence of the integral if $\tau_0 = \infty$. Consider the cases where the contour C is unbounded on one side, on both sides.

883. Prove that if $|f(\tau, z)| \leq |\psi(\tau)|$ for all points z in a set E and if $\int_C |\psi(\tau)| d\tau$ converges, then the integral $\int_C f(\tau, z) d\tau$ converges uniformly on the set E .

884. Let $f(\tau, z)$ be a function that is analytic with respect to z and continuous with respect to τ for points z belonging to a certain domain D and points τ belonging to a contour C with the exception of some isolated points where the conditions imposed on the function $f(\tau, z)$ are violated either for all points z or only for some of them.

Prove that if the improper integral

$$F(z) = \int_C f(\tau, z) d\tau$$

converges uniformly inside a domain D (i.e. in any closed subdomain of the domain D), then the function $F(z)$ is analytic and

$$F'(z) = \int_C \frac{\partial f}{\partial z} d\tau.$$

This integral converges uniformly inside D .

In Problems 885 to 892 find the sets on which the indicated integrals converge uniformly.

$$885. \Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt \quad (t^{z-1} = e^{(z-1) \ln t}). \quad 886. \int_0^\infty e^{-zt^2} dt.$$

$$887. \int_0^\infty \frac{\sin t}{t^z} dt. \quad 888. \int_0^\infty \frac{\cos t}{t^z} dt. \quad 889. \int_0^\infty \frac{\sin tz}{t} dt.$$

$$890. \int_{c-i\infty}^{c+i\infty} \frac{e^{zt}}{t} dt \quad (c \neq 0). \quad 891. \int_c^{c+i\infty} \frac{e^{zt}}{t} dt \quad (c \neq 0).$$

$$892. \int_{c-i\infty}^{c+i\infty} \frac{z^t}{t} dt \quad (c \neq 0, \quad z^t = e^{t \ln z}).$$

LAPLACE'S INTEGRAL

An integral of the form

$$\int_0^\infty e^{-tz} f(t) dt, \quad (1)$$

where the function $f(t)$ is integrable on the interval $[0, a]$ for any positive $a < \infty$, is called the *Laplace integral*.

893. Prove the following propositions:

1) If the integral (1) converges at the point $z = z_0$, then it converges in the half-plane $\operatorname{Re} z > \operatorname{Re} z_0$, the convergence being uniform in the angle $|\arg(z - z_0)| \leq \theta < \frac{\pi}{2}$.

2) If the integral (1) converges absolutely for $z = z_0$, then it converges absolutely and uniformly in the half-plane $\operatorname{Re} z \geq \operatorname{Re} z_0$.

3) If $\lim_{t \rightarrow \infty} \frac{\ln |f(t)|}{t} = \beta$, then the integral (1) converges absolutely in the half-plane $\operatorname{Re} z > \beta$ and uniformly in any half-plane $\operatorname{Re} z \geq \beta + \varepsilon$ ($\varepsilon > 0$) (construct an example of the Laplace integral converging absolutely throughout the plane and for which $\beta = \infty$).

4) If $\lim_{t \rightarrow \infty} \frac{\ln |f(t)|}{t} = \alpha$, then the integral (1) does not converge absolutely at any point of the half-plane $\operatorname{Re} z < \alpha$.

From the theorems formulated in Problem 893 it follows that the domains of convergence and absolute convergence of the Laplace integral (if such domains exist) are the half-planes $\operatorname{Re} z > x_c$ and $\operatorname{Re} z > x_a$; the number x_c is called the *abscissa of convergence*, and x_a the *abscissa of absolute convergence* of the Laplace integral.

In problems 894 to 900 find x_c and x_a for the integral $\int_0^\infty e^{-zt} f(t) dt$, where $f(t)$ is a given function.

$$894. f(t) = 1. \quad 895. f(t) = e^{-t^2}. \quad 896. f(t) = e^{t^2}.$$

$$897. f(t) = \begin{cases} e^{-t^2} & \text{for } 0 \leq t < \ln \ln 3 \text{ and } \ln \ln 2k \leq t < \ln \ln (2k+1) \\ & (k=2, 3, \dots) \\ -e^{-t^2} & \text{for } \ln \ln (2k+1) \leq t < \ln \ln (2k+2) \\ & (k=1, 2, \dots). \end{cases}$$

$$898. f(t) = \begin{cases} e^{\frac{1}{2}e^t} & \text{for } 0 \leq t < \ln \ln 3 \text{ and } \ln \ln 2k \leq t < \ln \ln (2k+1) \\ & (k=2, 3, \dots) \\ -e^{\frac{1}{2}e^t} & \text{for } \ln \ln (2k+1) \leq t < \ln \ln (2k+2) \\ & (k=1, 2, \dots). \end{cases}$$

$$899. f(t) = \begin{cases} e^{et} & \text{for } 0 \leq t < \ln \ln 3 \text{ and } \ln \ln 2k \leq t < \ln \ln (2k+1) \\ & (k=2, 3, \dots) \\ -e^{et} & \text{for } \ln \ln (2k+1) \leq t < \ln \ln (2k+2) \\ & (k=1, 2, \dots). \end{cases}$$

$$900. f(t) = \begin{cases} e^t & \text{for } \ln(2k-1) \leq t < \ln 2k \quad (k=1, 2, \dots) \\ -e^t & \text{for } \ln 2k \leq t < \ln(2k+1) \quad (k=1, 2, \dots). \end{cases}$$

In Problems 901 to 904 investigate the convergence of the Laplace integrals $\int_0^\infty e^{-zt} f(t) dt$ on the boundary of the half-plane of convergence.

901. $f(t) = 1$. 902. $f(t) = 0$ for $0 \leq t \leq 1$, $f(t) = 1/t^2$ for $t > 1$.

903. $f(t) = 0$ for $0 \leq t \leq 1$, $f(t) = 1/t$ for $t > 1$.

904. $f(t) = 0$ for $t = 0$, $f(t) = 1$ for $0 < t \leq 1$, and for $t > 1$ $f(t)$ is defined as follows:

$$f(t+1) = \begin{cases} f(t) + 1 & \text{if } (2k-1)^2 < t+1 \leq (2k)^2 \\ f(t) - 1 & \text{if } (2k)^2 < t+1 \leq (2k+1)^2 \quad (k=1, 2, \dots). \end{cases}$$

CHAPTER VI

INFINITE PRODUCTS. ENTIRE AND MEROMORPHIC FUNCTIONS

§ 1. INFINITE PRODUCTS

In Problems 905 to 911 prove the indicated equalities.

$$905. \prod_{n=2}^{\infty} \left(1 - \frac{1}{n^2}\right) = \frac{1}{2}. \quad 906. \prod_{n=1}^{\infty} \left(1 + \frac{1}{n(n+2)}\right) = 2.$$

$$907. \prod_{n=3}^{\infty} \frac{n^2-4}{n^2-1} = \frac{1}{4}. \quad 908. \prod_{n=2}^{\infty} \left(1 - \frac{2}{n(n+1)}\right) = \frac{1}{3}.$$

$$909. \prod_{n=2}^{\infty} \frac{n^3-1}{n^3+1} = \frac{2}{3}. \quad 910. \prod_{n=1}^{\infty} \left[1 + \frac{(-1)^{n+1}}{n}\right] = 1.$$

911. $\prod_{n=1}^{\infty} \frac{e^{1/n}}{1 + \frac{1}{n}} = e^C$, where $C = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{1}{k} - \ln n\right)$ is the Euler constant.

912. Prove that $\prod_{n=1}^{\infty} \cos \frac{\alpha}{2^n} = \frac{\sin \alpha}{\alpha}$.

Hint. First prove the identity $\sin \varphi = 2^k \sin \frac{\varphi}{2^k} \prod_{n=1}^k \cos \frac{\varphi}{2^n}$.

913. Using the solution of Problem 912, prove that

$$\frac{2}{\sqrt{2}} \frac{2}{\sqrt{2+\sqrt{2}}} \frac{2}{\sqrt{2+\sqrt{2+\sqrt{2}}}} \cdots = \frac{\pi}{2}.$$

914.* Prove Wallis' formula $\frac{\pi}{2} = \prod_{n=1}^{\infty} \left(\frac{2n}{2n-1} \frac{2n}{2n+1}\right)$.

915. Prove that if we assume, as usual, $-\pi < \arg p_n \leq \pi$, then the infinite product $\prod_{n=1}^{\infty} p_n$ converges and diverges simultaneously with the series $\sum_{n=1}^{\infty} \ln p_n$.

916. Find out whether the assertion of the preceding problem is valid if we agree that:

1) $0 \leq \arg p_n < 2\pi$; 2) $\alpha < \arg p_n \leq \alpha + 2\pi$ ($-2\pi < \alpha < 0$).

917. Prove that for absolute convergence of the infinite product $\prod_{n=1}^{\infty} (1 + a_n)$ [i.e. for absolute convergence of the series $\sum_{n=1}^{\infty} \ln(1 + a_n)$] it is necessary and sufficient that the series $\sum_{n=1}^{\infty} a_n$ converge absolutely.

918. The products $\prod_{n=1}^{\infty} p_n$ and $\prod_{n=1}^{\infty} q_n$ converge. Investigate the convergence of the following products:

1) $\prod_{n=1}^{\infty} (p_n + q_n)$; 2) $\prod_{n=1}^{\infty} (p_n - q_n)$; 3) $\prod_{n=1}^{\infty} p_n q_n$; 4) $\prod_{n=1}^{\infty} \frac{p_n}{q_n}$.

In Problems 919 to 923 investigate the given products for convergence and absolute convergence.

919. $\prod_{n=1}^{\infty} \left[1 + \frac{(-1)^{n+1}}{n} \right]$. 920. $\prod_{n=1}^{\infty} n^{(-1)^n}$.

921. $\prod_{n=1}^{\infty} \left[1 + \frac{(-1)^{n+1}}{n^p} \right]$ ($p > 0$). 922. $\prod_{n=1}^{\infty} \left(1 + \frac{1}{n^p} \right)$ ($p > 0$).

923. $\prod_{n=1}^{\infty} \cos z_n$ if it is known that the series $\sum_{n=1}^{\infty} |z_n|^2$ converges.

924. Prove that inside the unit circle

$$\prod_{n=0}^{\infty} (1 + z^{2^n}) = \frac{1}{1 - z},$$

where the product converges absolutely.

In Problems 925 to 934 find domains of convergence of the given products.

925. $\prod_{n=1}^{\infty} (1 - z^n)$. 926. $\prod_{n=1}^{\infty} \left(1 + \frac{z^n}{2^n} \right)$. 927. $\prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2} \right)$.

$$928. \prod_{n=2}^{\infty} \left[1 - \left(1 - \frac{1}{n} \right)^{-n} z^{-n} \right]. \quad 929. \prod_{n=1}^{\infty} \left[1 + \left(1 + \frac{1}{n} \right)^{n^2} z^n \right].$$

$$930. \prod_{n=1}^{\infty} \cos \frac{z}{n}. \quad 931. \prod_{n=1}^{\infty} \frac{\sin \frac{z}{n}}{\frac{z}{n}}. \quad 932. \prod_{n=1}^{\infty} \left(1 + \frac{z}{n} \right) e^{-\frac{z}{n}}.$$

933. $\prod_{n=1}^{\infty} (1 + c_n z)$ if it is known that $\sum_{n=1}^{\infty} |c_n|$ is a convergent series.

934. Prove that the product

$$\prod_{n=1}^{\infty} \left[1 + \frac{(-1)^{n+1}}{n^z} \right] \quad (n^z = e^{z \ln n})$$

converges in the half-plane $\operatorname{Re} z > 1/2$ and converges absolutely in the half-plane $\operatorname{Re} z > 1$.

935. $\{f_n(z)\}$ is a sequence of functions analytic in the domain G ; all these functions, except a finite number of them, do not vanish in the domain G . Prove that if $|f_n(z)| \leq \alpha_n$ for all $z \in G$, where α_n is independent of z and $\sum_{n=1}^{\infty} \alpha_n$ a convergent series, then the function $F(z) = \prod_{n=1}^{\infty} [(1 + f_n(z))]$ is analytic in G .

In Problems 936 to 939 we examine some properties of the Gamma function which follow from its definition as the limit of an infinite product.

936. Prove that the product

$$\Gamma(z+1) = \prod_{n=1}^{\infty} \frac{n}{z+n} \left(\frac{n+1}{n} \right)^z \quad \left(\left(\frac{n+1}{n} \right)^z = e^{z \ln \frac{n+1}{n}} \right)$$

converges absolutely in the entire plane, except for those values of z which are equal to negative integers, and represents a function analytic in the whole plane, except at the points $z = -1, -2, \dots$

937. Prove the Euler formula

$$\Gamma(z) = \lim_{n \rightarrow \infty} \frac{n! n^z}{z(z+1)(z+2)\dots(z+n)} \quad (n^z = e^{z \ln n})$$

and show that:

$$1) \Gamma(z+1) = z\Gamma(z);$$

$$2) \Gamma(m+1) = m! \text{ if } m \text{ is a natural number.}$$

938. Prove that

$$\prod_{n=1}^{\infty} \frac{n(\alpha + \beta + n)}{(\alpha + n)(\beta + n)} = \frac{\Gamma(\alpha + 1)\Gamma(\beta + 1)}{\Gamma(\alpha + \beta + 1)} \quad (\alpha, \beta \neq -1, -2, \dots).$$

939. Prove the Weierstrass formula

$$\frac{1}{\Gamma(z+1)} = e^{Cz} \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right) e^{-\frac{z}{n}},$$

where C is the Euler constant.

Hint. Use the solution of Problem 911.

940.* Let $p_1, p_2, \dots, p_n, \dots$ be a sequence of all the prime numbers ($p_1=2, p_2=3, p_3=5, \dots$), and let $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ ($n^{-s} = e^{-s \ln n}$) be the Riemann Zeta function analytic in the half-plane $\operatorname{Re} s > 1$ (see Problem 844). Prove that:

$$1) \quad \zeta(s) = \frac{1}{\prod_{n=1}^{\infty} (1 - p_n^{-s})};$$

2) the function $\zeta(s)$ has no zeros in the half-plane $\operatorname{Re} s > 1$.

941.* Prove that $\sum_{n=1}^{\infty} \frac{1}{p_n}$, where $\{p_n\}$ is a sequence of prime numbers, is a divergent series.

§ 2. PARTIAL FRACTION AND INFINITE PRODUCT EXPANSIONS. SUMMATION OF SERIES

942. Let $f(z)$ be a meromorphic function with simple poles at points $a_1, a_2, \dots, a_n, \dots$; $0 < |a_1| \leq |a_2| \leq \dots$, $\lim_{n \rightarrow \infty} a_n = \infty$.

Denote by A_n the residue of the function $f(z)$ with respect to the pole a_n ($n=1, 2, \dots$). Suppose there exists a sequence of closed contours C_m which satisfy the following conditions:

- 1) C_m do not pass through any of the points a_n ;
- 2) each contour C_m is enclosed by the contour C_{m+1} ;
- 3) the minimum distance from the contour C_m to the origin (we denote it by R_m) increases without bound as $m \rightarrow \infty$;
- 4) the ratio of the length L_m of the contour C_m to R_m remains bounded, i.e. $L_m = O(R_m)$;
- 5) $\max_{z \in C_m} |f(z)| = o(R_m)$ (this condition is apparently satisfied if the function $f(z)$ is bounded on all contours C_m).

Prove that under these conditions we have a decomposition of the function $f(z)$ into partial fractions

$$f(z) = f(0) + \sum_{n=1}^{\infty} A_n \left(\frac{1}{z-a_n} + \frac{i}{a_n} \right).$$

The series converges uniformly in any closed domain that does not contain the points a_n if under the summation sign we group summands relating to the poles lying between C_m and C_{m+1} ($m = 1, 2, \dots$).

Hint. Apply the theorem of residues to the integral

$$\frac{1}{2\pi i} \int_{C_m} \frac{f(\xi) d\xi}{\xi(\xi-z)}$$

and pass to the limit as $m \rightarrow \infty$.

In Problems 943 to 951 prove the validity of the given expansions.

$$943. \cot z = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2\pi^2}. \quad 944. \frac{1}{\sin z} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{(-1)^n 2z}{z^2 - n^2\pi^2}.$$

$$945. \tan z = \sum_{n=1}^{\infty} \frac{2z}{\left[\frac{(2n-1)\pi}{2} \right]^2 - z^2}.$$

$$946. \frac{1}{\cos z} = \pi \sum_{n=1}^{\infty} \frac{(-1)^n (2n-1)}{z^2 - \left[\frac{(2n-1)\pi}{2} \right]^2}.$$

$$947. \tanh z = \sum_{n=1}^{\infty} \frac{2z}{z^2 + \left[\frac{(2n-1)\pi}{2} \right]^2}.$$

$$948. \frac{1}{\sinh z} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{(-1)^n 2z}{z^2 + n^2\pi^2}.$$

$$949. \frac{z}{e^z - 1} = 1 - \frac{z}{2} + \sum_{n=1}^{\infty} \frac{2z^2}{z^2 + 4n^2\pi^2}.$$

$$950. \frac{1}{\sin^2 z} = \sum_{n=-\infty}^{\infty} \frac{1}{(z - n\pi)^2}.$$

951. Let $f(z)$ be an entire function with simple zeros at the points $a_1, a_2, \dots, a_n, \dots$; $0 < |a_1| \leq |a_2| \leq \dots$, $\lim_{n \rightarrow \infty} a_n = \infty$. Let us assume that there exists a system of contours $\{C_n\}$ satisfying conditions 1 to 4 of Problem 942 and on which

$$\max_{z \in C_n} \left| \frac{f'(z)}{f(z)} \right| = o(R_n).$$

Prove that the following expansion is valid throughout the plane:

$$f(z) = f(0) e^{\frac{f'(0)}{f(0)} z} \prod_{n=1}^{\infty} \left(1 - \frac{z}{a_n} \right) e^{\frac{z}{a_n}}.$$

In Problems 952 to 958 prove the validity of the expansions.

$$952. \sin z = z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2 \pi^2} \right).$$

$$953. \cos z = \prod_{n=0}^{\infty} \left\{ 1 - \left[\frac{2z}{(2n+1)\pi} \right]^2 \right\}.$$

$$954. \sinh z = z \prod_{n=1}^{\infty} \left(1 + \frac{z^2}{n^2 \pi^2} \right).$$

$$955. \cosh z = \prod_{n=0}^{\infty} \left\{ 1 + \left[\frac{2z}{(2n+1)\pi} \right]^2 \right\}.$$

$$956. e^z - 1 = z e^{\frac{z}{2}} \prod_{n=1}^{\infty} \left(1 + \frac{z^2}{4n^2 \pi^2} \right).$$

$$957. e^{az} - e^{bz} = (a-b) z e^{\frac{1}{2}(a+b)z} \prod_{n=1}^{\infty} \left[1 + \frac{(a-b)^2 z^2}{4n^2 \pi^2} \right].$$

$$958. \cosh z - \cos z = z^2 \prod_{n=1}^{\infty} \left(1 + \frac{z^4}{4n^4 \pi^4} \right).$$

959. Let $f(z)$ be a meromorphic function having a finite number of poles: $a_1, a_2, \dots, a_m$ which do not coincide with any of the points $z=0, \pm 1, \pm 2, \dots$. Prove that if there exists a sequence of contours $\{C_n\}$ which shrink to the point at infinity, and

$$\lim_{n \rightarrow \infty} \int_{C_n} f(z) \cot \pi z \, dz = 0, \quad (1)$$

then

$$\sum_{n=-\infty}^{\infty} f(n) = -\pi \sum_{k=1}^m \operatorname{res} [f(z) \cot \pi z]_{z=a_k}.$$

960. Prove that if, in the conditions of the preceding problem, requirement (1) is replaced by the condition $\lim_{n \rightarrow \infty} \int_{C_n} \frac{f(z) dz}{\sin \pi z} = 0$, then

$$\sum_{n=-\infty}^{\infty} (-1)^n f(n) = -\pi \sum_{k=1}^m \operatorname{res} \left[\frac{f(z)}{\sin \pi z} \right]_{z=a_k}$$

In Problems 961 to 968 find the sums of the series, assuming the number a to be such that none of the denominators vanishes.

$$961. \sum_{n=-\infty}^{\infty} \frac{1}{(a+n)^2}. \quad 962. \sum_{n=-\infty}^{\infty} \frac{(-1)^n}{(a+n)^2}. \quad 963. \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}.$$

$$964. \sum_{n=0}^{\infty} \frac{1}{n^2 + a^2}. \quad 965. \sum_{n=0}^{\infty} \frac{(-1)^n}{n^2 + a^2}. \quad 966. \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2}.$$

$$967. \sum_{n=1}^{\infty} \frac{1}{n^{2k}} \quad (k \text{ a natural number}).$$

Hint. First prove that if $z=0$ is a pole of the function $f(z)$, then the formula for the sum of the series of Problem 959 holds, provided the summation on the left is extended over all values of n from $-\infty$ to $+\infty$, except $n=0$.

In calculating $\operatorname{res} \left[\frac{\cot \pi z}{z^{2k}} \right]_{z=0}$ it is convenient to use the expansion in Problem 487.

$$968. \sum_{n=1}^{\infty} (-1)^n \frac{n \sin bn}{a^2 - n^2} \quad (-\pi < b < \pi).$$

Hint. Use the integral $\int_{C_n} \frac{ze^{ibz} dz}{(a^2 - z^2) \sin \pi z}$ (C_n the contour in Problem 960).

969. Prove that

$$\sum_{n=1}^{\infty} \frac{n}{(n^2-3)\sqrt{4n^2-3}} = \int_0^{\frac{\sqrt{3}}{2}} \frac{x \cot \pi x}{(3-x^2)\sqrt{3-4x^2}} dx + \frac{1}{6} \cot [\pi(2-\sqrt{3})].$$

Hint. Use the integral $\int_{C_n} \frac{z \cot \pi z dz}{(z^2-3)\sqrt{4z^2-3}}$, where C_n is the contour bounding the doubly connected domain in Fig. 30.

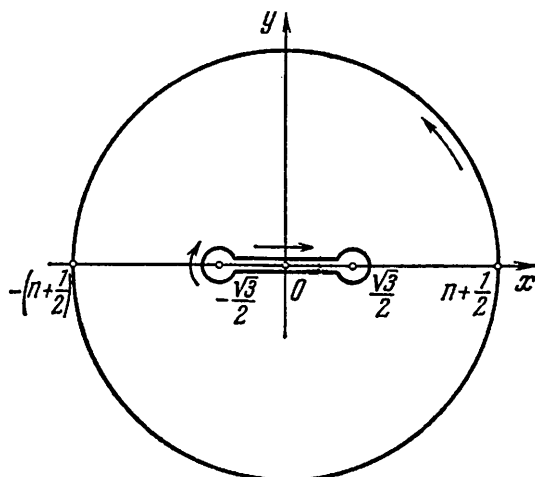


FIG. 30

§ 3. CHARACTERISTICS OF THE GROWTH OF ENTIRE FUNCTIONS

Let $f(z)$ be an entire function, and $M(r) = \max_{|z|=r} |f(z)|$. The number $\rho = \overline{\lim}_{r \rightarrow \infty} \frac{\ln \ln M(r)}{\ln r}$ is called the *order* of an entire function.

If $0 < \rho < \infty$, then the number $\sigma = \overline{\lim}_{r \rightarrow \infty} \frac{\ln M(r)}{r^\rho}$ is called the *type* of a function. If $\sigma = 0$, the function $f(z)$ is called a function of *zero type*; if $\sigma = \infty$, the function $f(z)$ is called a function of *maximum type*; if $0 < \sigma < \infty$, the function $f(z)$ is called a function of *finite type*.

970. Prove the following assertions:

1) If $\rho \neq \infty$ and $\sigma \neq \infty$ are respectively the order and the type of the function $f(z)$, then for arbitrary $\varepsilon > 0$ there is a number $R(\varepsilon)$ such that for $r > R$ the following inequalities hold:

$$M(r) < e^{r^{\rho+\varepsilon}}, \quad M(r) < e^{(\sigma+\varepsilon)r^2}.$$

There also exist sequences of numbers $\{r_n\}$ and $\{r'_n\}$, converging to infinity, for which

$$M(r_n) > e^{r_n^{\rho-\varepsilon}} \quad \text{and} \quad M(r'_n) > e^{(\sigma-\varepsilon)r_n^2}.$$

2) If for a natural number k

$$0 < \overline{\lim}_{r \rightarrow \infty} \frac{M(r)}{r^k} < \infty,$$

then $f(z)$ is a polynomial of degree k (and, hence, there exists $\lim_{r \rightarrow \infty} \frac{M(r)}{r^k}$).

Hint. Make use of Cauchy's inequalities for the coefficients of the power series $f(z) = \sum_{n=0}^{\infty} c_n z^n$.

3) If $f(z)$ is an entire transcendental function, then

$$\lim_{r \rightarrow \infty} \frac{\ln M(r)}{\ln r} = \infty.$$

In Problems 971 to 983 determine the orders and types of the indicated functions (n a natural number).

971. $c_0 z^n + c_1 z^{n-1} + \dots + c_n$. 972. e^{az^n} ($a > 0$). 973. $z^n e^{3z}$.

974. $z^2 e^{2z} - e^{3z}$. 975. $e^{5z} - 3e^{2z^3}$. 976. $e^{(2-i)z^2}$. 977. $\sin z$.

978. $\cosh z$. 979. $e^z \cos z$. 980. $\cos \sqrt{z}$.

981.* $\sum_{n=0}^{\infty} \frac{z^n}{(2^m \cdot n)!}$ (m a natural number).

982. e^{z^2} . 983.* $\int_0^1 e^{zt^2} dt$.

984. The entire function $f(z)$ is of order ρ and type σ ($0 \leq \sigma \leq \infty$). Prove that the function $P(z)f(z) + Q(z)$, where $P(z)$ and $Q(z)$ are any polynomials, is also of order ρ and type σ .

985. The entire functions $f_1(z)$ and $f_2(z)$ are respectively of orders ρ_1 and ρ_2 , ($\rho_1 \neq \rho_2$). What can be said about the order ρ^* of the functions $f_1(z)f_2(z)$ and $f_1(z) + f_2(z)$?

986. The entire functions $f_1(z)$ and $f_2(z)$ are of the same order ρ , and, respectively, of type σ_1 and σ_2 , ($\sigma_1 \neq \sigma_2$). What can be said about the order ρ^* and the type σ^* of the following functions:

1) $f_1(z)f_2(z)$, 2) $f_1(z) + f_2(z)$?

987. The entire functions $f_1(z)$ and $f_2(z)$ are of the same order ρ and the same type σ . What can be said about the order ρ^* and the type σ^* of the following functions:

1) $f_1(z)f_2(z)$, 2) $f_1(z) + f_2(z)$?

988.* $f(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{\lambda_n^2}\right)$, $\lambda_n > 0$ ($n = 1, 2, \dots$),

$$\lim_{n \rightarrow \infty} \frac{n}{\lambda_n} = \alpha > 0, \quad \overline{\lim}_{n \rightarrow \infty} \frac{n}{\lambda_n} = \beta < \infty.$$

Prove that $f(z)$ is an entire function of the first order and of type σ such that $\pi\alpha \leq \sigma \leq \pi\beta$.

989. $f(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^{2k}}{\lambda_n^2}\right)$, $\lambda_n > 0$ ($n = 1, 2, \dots$), $2k$ a natural number. Prove the following assertions:

1) if $\lim_{n \rightarrow \infty} \frac{n}{\lambda_n} = 0$, then $f(z)$ is an entire function increasing no faster than a function of order k and of zero type,

2) if $\lim_{n \rightarrow \infty} \frac{n}{\lambda_n} = \infty$ and $\sum_{n=1}^{\infty} \frac{1}{\lambda_n^2} < \infty$, then $f(z)$ is a function increasing no slower than a function of order k and of maximum type.

Hint. Take advantage of the method used in solving the preceding problem.

990*. Prove that the order and type of an entire function remain unchanged when the function is differentiated.

Solve Problems 991 to 998 proceeding from the following theorem.

If power series expansion of an entire function has the form $f(z) = \sum_{n=0}^{\infty} c_n z^n$, then the order ρ and the type σ of this function are defined by the equation

$$\rho = \overline{\lim}_{n \rightarrow \infty} \frac{n \ln n}{\ln |c_n|},$$

$$(\sigma \rho)^{\frac{1}{\rho}} = \overline{\lim}_{n \rightarrow \infty} \left(n^{\frac{1}{\rho}} \sqrt[\rho]{|c_n|} \right).$$

991. Prove that the entire function

$$f(z) = \sum_{n=0}^{\infty} \frac{(Az)^n}{\Gamma(\alpha n + 1)} \quad (A > 0, \alpha > 0)$$

is of order $\rho = 1/\alpha$ and of type $\sigma = A^{1/\alpha}$.

Hint. Use the Stirling formula

$$\Gamma(\alpha n + 1) = \left(\frac{\alpha n}{e}\right)^{\alpha n} \sqrt{2\pi\alpha n} \left[1 + O\left(\frac{1}{n}\right)\right].$$

In Problems 992 to 998 find the orders and types of the given functions

992. $f(z) = \sum_{n=1}^{\infty} \left(\frac{z}{n}\right)^n$. 993. $f(z) = \sum_{n=1}^{\infty} \left(\frac{\ln n}{n}\right)^{\frac{n}{a}} z^n \quad (a > 0)$.

994. $f(z) = \sum_{n=2}^{\infty} \left(\frac{1}{n \ln n}\right)^{\frac{n}{a}} z^n \quad (a > 0)$. 995. $f(z) = \sum_{n=0}^{\infty} e^{-n^2} z^n$.

$$996. f(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^{n^2+a}} \quad (a > 0). \quad 997. f(z) = \sum_{n=0}^{\infty} \frac{\cosh \sqrt{n}}{n!} z^n.$$

$$998. z^{-\nu} J_{\nu}(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{n! \Gamma(n+\nu+1)} \quad (\nu > -1; J_{\nu}(z) \text{ a Bessel function of the } \nu\text{th order}).$$

If ρ is the order of an entire function $f(z)$ ($0 < \rho < \infty$), then the function $h(\varphi) = \lim_{r \rightarrow \infty} \frac{\ln |f(re^{i\varphi})|}{r^{\rho}}$ is called the *indicator function* of the growth of the function $f(z)$.

In Problems 999 to 1005 find the order ρ and the indicator function $h(\varphi)$ of the given function.

$$999. e^z. \quad 1000. e^z + z^2. \quad 1001. \sin z. \quad 1002. \cos z. \quad 1003. \cosh z.$$

$$1004. e^{z^n}. \quad 1005. \frac{\sin \sqrt{z}}{\sqrt{z}}.$$

1006. The entire function $f(z)$ has an indicator function $h(\varphi)$, and $P(z)$ is a polynomial. What is the indicator function $h^*(\varphi)$ of the following functions:

$$1) f(z) + P(z), \quad 2) f(z)P(z)?$$

1007. $P(z)$ is an arbitrary polynomial. Give an example of an entire function $f(z)$ (of finite and nonzero order) whose indicator function $h(\varphi)$ in a certain interval is equal to zero, whereas the indicator function $f(z) + P(z)$ in the same interval is negative.

CHAPTER VII

CAUCHY-TYPE INTEGRALS. INTEGRAL FORMULAS OF POISSON AND SCHWARZ

§ 1. CAUCHY-TYPE INTEGRALS

An integral of the form

$$\frac{1}{2\pi i} \int_C \frac{\varphi(\xi) d\xi}{\xi - z},$$

where C is a smooth contour ¹⁾ (closed or open) and $\varphi(\xi)$ is a function continuous on C , except possibly at a finite number of points where it has an integrable discontinuity, is called a *Cauchy-type integral*. The function $\varphi(\xi)$ is called its *density*, and $\frac{1}{\xi - z}$, its *kernel*. A Cauchy-type integral represents a function $F(z)$ analytic in every domain not containing any points of the contour C . Here,

$$F^{(n)}(z) = \frac{n!}{2\pi i} \int_C \frac{\varphi(\xi) d\xi}{(\xi - z)^{n+1}}. \quad (1)$$

Let $\varphi(\xi)$ on the contour C satisfy a Lipschitz condition of order α ($0 < \alpha \leq 1$) (briefly $\varphi(\xi) \in \text{Lip } \alpha$), i.e.

$$|\varphi(\xi_1) - \varphi(\xi_2)| < k |\xi_1 - \xi_2|^\alpha,$$

where the points ξ_1 and ξ_2 belong to the contour C , and k is a constant. Then, if the point ξ_0 of the contour is not its end-point, there exists a *singular integral*

$$F(\xi_0) = \frac{1}{2\pi i} \int_C \frac{\varphi(\xi) d\xi}{\xi - \xi_0}$$

defined as the principal value of a Cauchy-type integral.

¹⁾ By a smooth contour is meant a simple (i.e. without points of self-intersection) curve with a continuously varying tangent and without cusps. Incidentally the conditions imposed on the contour C may be extended considerably. One may also consider compound contours consisting of a finite number of contours of the indicated type.

This principal value can be expressed in terms of an ordinary improper integral by the formula

$$F(\zeta_0) = \frac{1}{2\pi i} \int_C \frac{\varphi(\zeta) - \varphi(\zeta_0)}{\zeta - \zeta_0} d\zeta + \frac{1}{2} \varphi(\zeta_0) + \frac{\varphi(\zeta_0)}{2\pi i} \operatorname{Ln} \frac{b - \zeta_0}{a - \zeta_0}, \quad (2)$$

where the points a and b are the extremities of the contour C if it is open. The single-valued branch of Ln is chosen so that in the case of a closed contour ($a=b$) the term involving the logarithm vanishes and the formula takes the form

$$F(\zeta_0) = \frac{1}{2\pi i} \int_C \frac{\varphi(\zeta) - \varphi(\zeta_0)}{\zeta - \zeta_0} d\zeta + \frac{1}{2} \varphi(\zeta_0). \quad (2')$$

If we denote by $F^+(\zeta_0)$ and $F^-(\zeta_0)$ the limiting values of a Cauchy-type integral $F(z)$, as $z \rightarrow \zeta_0$, on the left of C and on the right of C , respectively, then, according to Sokhotsky's formulas,

$$\left. \begin{aligned} F^+(\zeta_0) &= F(\zeta_0) + \frac{1}{2} \varphi(\zeta_0) \\ F^-(\zeta_0) &= F(\zeta_0) - \frac{1}{2} \varphi(\zeta_0) \end{aligned} \right\} \quad (3)$$

or

$$F(\zeta_0) = \frac{1}{2} [F^+(\zeta_0) + F^-(\zeta_0)], \quad F^+(\zeta_0) - F^-(\zeta_0) = \varphi(\zeta_0). \quad (4)$$

If the contour C is closed and it is traversed in the usual direction, then $F^+(\zeta)$ are the limiting values of the function $F^+(z)$ defined **inside** the contour (domain D^+), and $F^-(\zeta)$, those of the function $F^-(z)$ defined **outside** the contour (domain D^-).¹⁾

1008. Prove that if C is a closed contour and the density of the Cauchy-type integral $\Phi(z) = \frac{1}{2\pi i} \int_C \frac{\varphi(\zeta)}{\zeta - z} d\zeta$ can be represented in the form $\varphi(\zeta) = \varphi^+(\zeta) + \varphi^-(\zeta)$, where $\varphi^+(\zeta)$ and $\varphi^-(\zeta)$ are the boundary values of functions analytic inside and outside the contour C , respectively, then

$$\Phi^+(z) = \varphi^+(z), \quad \Phi^-(z) = -\varphi^-(z) + \varphi^-(\infty).$$

Note. If it is given that one of the functions φ^- or φ^+ is identically zero, then the Cauchy-type integral turns into a Cauchy integral of the interior or exterior domain, respectively.

1009. Let C be a closed contour. Find $F^+(z)$ and $F^-(z)$ if the density of the Cauchy-type integral is one of the following functions (n a natural number):

$$1) \varphi(\zeta) = (\zeta - a)^n; \quad 2) \varphi(\zeta) = \frac{1}{(\zeta - a)^n} \quad (a \text{ inside } C);$$

¹⁾ The same is true for the case when the contour is an endless line dividing the plane into two domains

$$3) \varphi(\zeta) = \frac{1}{(\zeta - a)^n} \quad (a \text{ outside } C).$$

1010. Find $F^+(z)$ and $F^-(z)$ if:

1) the function $\varphi(\zeta)$ is the boundary value of a function analytic everywhere in D^+ except at a finite number of points a_k , where it has poles;

2) the function $\varphi(\zeta)$ is the boundary value of a function analytic everywhere in D^- except at a finite number of points a_k , where it has poles (a_k can also include the point $z = \infty$).

1011. Find $F^+(z)$ and $F^-(z)$ if

$$\varphi(\zeta) = \frac{\zeta^3 + i\zeta^2 - \zeta + 4i}{\zeta^4 - 3\zeta^2 - 4} + \frac{\ln \frac{\zeta - 2}{\zeta - 3}}{\zeta^2 - 4}$$

and C is the circle $|\zeta| = 3/2$.

1012. Find $F^+(z)$ and $F^-(z)$ if $\varphi(\zeta) = \cot \zeta$ and C is the circle $|\zeta| = 5$.

1013. Find $F^+(z)$ and $F^-(z)$ if $\varphi(\zeta) = \frac{\zeta^2}{\zeta^2 + 1}$ and C is the real axis traversed from left to right.

Note. By a Cauchy-type integral taken along the real axis,

$F(z) = \frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{\varphi(\tau) d\tau}{\tau - z}$, is meant its principal value if it diverges in the ordinary sense.

1014. Find $F^+(z)$ and $F^-(z)$ and also the limiting values $F_{\pm}(\zeta)$ on the contour of integration C if C is the circle $|\zeta| = R$ and

$\varphi(\zeta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\theta + b_n \sin n\theta)$ is a uniformly convergent

Fourier series of the real function $\psi(\theta) = \varphi(Re^{i\theta})$.

1015. 1) Let C be the circle $|\zeta| = \frac{\pi}{2}$ and $f(\zeta)$ a function analytic in the circle $|\zeta| \leq \frac{\pi}{2}$. Find the functions defined by the integrals

$$I_1(z) = \frac{1}{2\pi i} \int_C f(\zeta) \cot(\zeta - z) d\zeta, \quad I_2(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta) d\zeta}{\sin(\zeta - z)}$$

in domains whose points z possess the property that none of the points $z \pm k\pi$ (k an integer) lies on C .

2) Solve the problems formulated in item (1), taking C to be the circle $|\zeta| = \pi$.

1016. Let C be the interval $[-1, 1]$ traversed from left to right, and let $\varphi(\zeta) \equiv 1$. Find: $F(z)$ outside C , the limiting values $F_{\pm}(\zeta)$,

and the principal value $F(\zeta)$ on C . In particular, evaluate $F(\pm i)$, $F^\pm(0)$ and $F(0)$.

1017. Let C be the semicircle $|\zeta|=R$, $0 < \arg \zeta < \pi$ (commencing at the point R), and let $\varphi(\zeta) \equiv 1$. Find: $F(z)$ outside C , the limiting values $F^\pm(\zeta)$, and the principal value $F(\zeta)$ on C . In particular, evaluate $F(0)$, $F^\pm(iR)$ and $F(iR)$. Also find $F'(0)$.

1018. Let C be the semicircle $|\zeta|=R$, $-\pi < \arg \zeta < 0$ (commencing at the point R), and let $\varphi(\zeta) \equiv 1$. Find: $F(z)$ outside C , the limiting values $F^\pm(\zeta)$ on C , $F(0)$ and $F'(0)$.

1019. Let the density of the Cauchy-type integral be $\varphi(\zeta) = \frac{1}{\zeta^n}$. Find $F(z)$ outside C if the contour C is the indicated line:

- 1) boundary of the annulus $r < |z| < R$;
- 2) straight line $\operatorname{Im} \zeta = \pi$ traversed from left to right;
- 3) boundary of the strip $|\operatorname{Im} z| < \pi$;
- 4) semicircle $|\zeta|=R$, $0 < \arg \zeta < \pi$ (commencing at the point R);
- 5) semicircle $|\zeta|=R$, $-\pi < \arg \zeta < 0$ (commencing at the point R).

In items (4) and (5) find the limiting values $F^\pm(\zeta)$ on C and evaluate $F(0)$.

In Problems **1020** to **1025** find $F(z)$ outside C , assuming that the contour C is the arc joining the points a and b , and $\varphi(\zeta)$ is the indicated function.

1020. $\varphi(\zeta) \equiv 1$. **1021.** $\varphi(\zeta) = \zeta$. **1022.** 1) $\varphi(\zeta) = \zeta^n$;

2) $\varphi(\zeta) = \sum_{n=0}^{\infty} c_n \zeta^n$ —an entire function. **1023.** $\varphi(\zeta) = \frac{1}{\zeta - z_0}$ ($z_0 \in C$).

1024. $\varphi(\zeta) = \frac{1}{(\zeta - z_0)^n}$ ($z_0 \in C$). In particular, evaluate $F(z_0)$.

Hint. Use the equality

$$\frac{\varphi(\zeta)}{\zeta - z} = \frac{\varphi(\zeta) - \varphi(z)}{\zeta - z} + \frac{\varphi(z)}{\zeta - z}.$$

1025. Find $F^+(z)$, $F^-(z)$ and the limiting values $F^\pm(\zeta)$ if C is the circle $|\zeta|=R$ and $\varphi(\zeta)$ is a logarithmic function defined by the following conditions:

- 1) $\varphi(\zeta) = \ln \zeta = \ln R + i\varphi$, $-\pi < \varphi \leq \pi$;
- 2) $\varphi(\zeta) = \operatorname{Ln} \zeta = \ln R + i\varphi$, $0 \leq \varphi < 2\pi$.

Hint. Consider a contour consisting of the circle $|z|=R$ with a cut along the radius $[-R, 0]$ in the first case and $[0, R]$ in the second.

Note. If we do not fix the branch of the logarithm beforehand and require its continuous extension along the contour of integration, the integral will depend on the choice of the starting point of integration. See also the hint and examples on page 60.

1026. Find $F^+(z)$, $F^-(z)$ and the limiting values $F^\pm(\zeta)$ on C if $\varphi(\zeta) = \ln \frac{\zeta}{\zeta-1}$ and the contour C is:

- 1) the circle $|\zeta| = R$ ($R > 1$);
- 2) the straight line $\operatorname{Im} \zeta = 1$ traversed from left to right.

1027. Find $F(z)$ and the limiting values $F^\pm(\zeta)$ on C if $\varphi(\zeta) = \ln \frac{\zeta}{\zeta-1}$ and the contour C is the semicircle $|\zeta| = R$ ($R > 1$), $0 \leq \arg \zeta \leq \pi$ (commencing at the point R).

1028. Find $F^+(z)$ and $F^-(z)$ if $\varphi(\zeta) = \sqrt{\zeta}$ ($0 \leq \arg \sqrt{\zeta} < \pi$) and C is the circle $|\zeta| = 1$.

In Problems **1029** to **1031** find $F^\pm(z)$ if C is a closed contour, the points a and b lie within it, and $\varphi(\zeta)$ is a single-valued branch of a multi-valued function defined outside the cut connecting the points a and b and lying inside the contour C .

1029. $\varphi(\zeta) = \operatorname{Ln} \frac{\zeta-a}{\zeta-b}$ ($\operatorname{Ln} 1 = 0$).

1030. $\varphi(\zeta) = \sqrt{\frac{\zeta-a}{\zeta-b}}$ ($\varphi(\infty) = +1$).

Hint. To find the integral along the contour enclosing the cut, expand $\frac{\varphi(\zeta)}{\zeta-z}$ into a series of powers of $\frac{1}{\zeta}$.

1031. $\varphi(\zeta) = (\zeta-a)^\lambda (\zeta-b)^{1-\lambda}$ ($\left. \frac{\varphi(\zeta)}{\zeta-b} \right|_{\zeta=\infty} = 1$).

1032. Find $F^\pm(z)$ if C is a closed contour, the point a belongs to the domain D^+ , the point b to the domain D^- , and $\varphi(\zeta) = \operatorname{Ln} \frac{\zeta-a}{\zeta-b}$ ($\operatorname{Ln} 1 = 0$) is a single-valued branch defined outside the cut joining the points a and b and intersecting C at a single point ζ_0 .

Hint. Adjoin to C a cut along the arc $\zeta_0 a$ if z lies in the domain D^+ , and along $\zeta_0 b$ if z lies in the domain D^- .

1033. Prove the validity of the following equalities ($0 < \lambda < 1$):

- 1) $\int_0^1 \left(\frac{t}{1-t} \right)^\lambda \frac{dt}{t-z} = \frac{\pi}{\sin \lambda \pi} \left[1 - \left(\frac{z}{z-1} \right)^\lambda \right] \quad (z \notin [0, 1]);$
- 2) $\int_0^1 \left(\frac{t}{1-t} \right)^\lambda \frac{dt}{t-\tau} = \frac{\pi}{\sin \lambda \pi} \left[1 - \left(\frac{\tau}{1-\tau} \right)^\lambda \cos \lambda \pi \right] \quad (\tau \in (0, 1)).$

Hint. To obtain the first formula consider the Cauchy integral $\frac{1}{2\pi i} \int_C \left(\frac{\zeta}{\zeta-1} \right)^\lambda \frac{d\zeta}{\zeta-z}$, where $\left(\frac{\zeta}{\zeta-1} \right)^\lambda$ is a function single-valued in

a plane with a rectilinear cut $[0, 1]$ and equal to 1 at ∞ , and C is the boundary of a doubly connected domain: the circle $|\xi| < R$ ($R > 1$) with a cut along the segment $[0, 1]$. The second formula is obtained from the first by means of Sokhotsky's formulas.

1034. Evaluate the singular integral

$$\frac{1}{\pi} \int_{-1}^1 \sqrt{\frac{1-t}{1+t}} \frac{t^2+3}{t-x} dt \quad (-1 < x < 1).$$

1035. Find the following integrals:

$$1) \int_0^1 \ln \frac{1-t}{t} \frac{dt}{t-z} \quad (z \in [0, 1]); \quad 2) \int_0^1 \ln \frac{1-t}{t} \frac{dt}{t-\tau} \quad (\tau \in (0, 1)).$$

1036. Consider the singular integral

$$F(z, t_0) = \frac{1}{2\pi i} \int_C \frac{d\tau}{(\tau - t_0)^{\gamma} (\tau - z)} \quad (\gamma = \alpha + i\beta, \quad 0 \leq \alpha < 1),$$

where C is the arc joining a to b , t_0 is a point of this arc, and $(\tau - t_0)^{\gamma}$ is a single-valued branch in a plane with a cut connecting the points t_0 and ∞ . If the point t_0 coincides with one of the extremities of the arc C , then we shall consider that the cut goes along the whole contour C ; if t_0 is an interior point, then we make the cut along the arc (t_0, b) of C . Prove the following assertions:

1) In the neighbourhood of the point a

$$F(z, a) = \frac{e^{i\gamma\pi}}{2i \sin \gamma\pi} (z-a)^{-\gamma} + F_1(z) \quad (z \in C),$$

$$F(t, a) = \frac{\cot \gamma\pi}{2i} (t-a)^{-\gamma} + F_1(t) \quad (t \in C),$$

where $F_1(z)$ is a function analytic in the neighbourhood of the point a .

Hint. Consider the difference

$$F_1(z) = \frac{1}{2\pi i} \int_C \frac{d\tau}{(\tau-a)^{\gamma} (\tau-z)} - \frac{e^{i\gamma\pi}}{2i \sin \gamma\pi} (z-a)^{-\gamma}$$

and, using Sokhotsky's formulas, prove that $F_1(z)$ will be an analytic function in the neighbourhood of the point a .

2) In the neighbourhood of the point b

$$F(z, b) = -\frac{e^{-i\gamma\pi}}{2i \sin \gamma\pi} (z-b)^{-\gamma} + F_2(z) \quad (z \in C),$$

$$F(t, b) = -\frac{\cot \gamma\pi}{2i} (t-b)^{-\gamma} + F_2(t) \quad (t \in C),$$

where $F_2(z)$ is a function analytic in the neighbourhood of the point b .

3) In the neighbourhood of the interior point t_0 of the contour C

$$F(z, t_0) = (z - t_0)^{-1} + F_3(z) \text{ to the left of } C,$$

$$F(z, t_0) = F_4(z) \text{ to the right of } C,$$

$$F(t, t_0) = \frac{1}{2} (t - t_0)^{-1} + F_5(t) \text{ if } t \in C,$$

where $F_3(z)$, $F_4(z)$ and $F_5(z)$ are analytic in the neighbourhood of the point t_0 .

1037. Examine the behaviour of the Cauchy-type integral

$$\frac{1}{2\pi i} \int_C \ln \frac{\xi - z}{\xi - 1} \frac{d\xi}{\xi - z}$$

near the points $z = -R$ and $z = R$ if C is the semicircle $|\xi| = R$ ($R > 1$) lying in the upper half-plane (commencing at the point R).

Hint. See Problem 1027.

§ 2. DIRICHLET'S INTEGRAL, HARMONIC FUNCTIONS, LOGARITHMIC POTENTIAL AND GREEN'S FUNCTION

The integral

$$D(u) = \iint_G (u_x^2 + u_y^2) dx dy$$

is called *Dirichlet's integral*, and the integral

$$D(u, v) = \iint_G (u_x v_x + u_y v_y) dx dy,$$

the corresponding *bilinear form*.

1038. Prove the following properties of the Dirichlet integral and its corresponding bilinear form:

$$1) D(u) = \iint_G \left(u_r^2 + \frac{1}{r} u_\phi^2 \right) dx dy \quad (z = x + iy = re^{i\phi});$$

2) $D(u)$ and $D(u, v)$ are invariant under conformal mappings of the domain G ;

3) $D^2(u, v) \leq D(u) D(v)$ (the equality sign refers only to the case $u/v = \text{const}$);

4) if the function $f(z) = u + iv$ is analytic in the domain G , then

$$D(u) = D(v) = D(f) = \iint_G |f'(z)|^2 dx dy.$$

What is the geometric meaning of $D(f)$ in this case?

1039. Using Green's formula

$$\iint_G v \Delta u \, dx \, dy + D(u, v) = \int_C v \frac{\partial u}{\partial n} \, ds$$

(n the outer normal, Δ the Laplacian operator), prove the following properties of the harmonic functions u , u_1 , u_2 (v a function conjugate to u):

$$1) D(u) = \int_C u \frac{\partial u}{\partial n} \, ds = \int_C u \, dv; \quad 2) \int_C \frac{\partial u}{\partial n} \, ds = \int_C dv = 0;$$

$$3) \text{ if } u_1 = u_2 \text{ on } C, \text{ then } u_1 = u_2 \text{ in } G;$$

$$4) \text{ if } \frac{\partial u_1}{\partial n} = \frac{\partial u_2}{\partial n} \text{ on } C, \text{ then } u_1 - u_2 = \text{const in } G.$$

1040. Prove that if u is harmonic in the circle $|z| < R$ and continuous in the closed circle $|z| \leq R$, then

$$u(0) = \frac{1}{2\pi} \int_0^{2\pi} u(Re^{i\theta}) \, d\theta = \frac{1}{\pi R^2} \iint_{|z| < R} u(re^{i\varphi}) \, r \, dr \, d\varphi.$$

Hint. From the equality $\int_C \frac{\partial u}{\partial n} \, ds = 0$ it follows that the integral

$\int_{|z|=r} u \, d\varphi$ is independent of r ; find this value by passing to the limit as $r \rightarrow 0$ and then perform a limit operation as $r \rightarrow R$.

1041. Integrals of the form

$$\iint_C \rho(\zeta) \ln \frac{1}{|\zeta - z|} \, d\xi \, d\eta, \quad \int_C \rho(\zeta) \ln \frac{1}{|\zeta - z|} \, |d\zeta|,$$

$$\int_C v(\zeta) \frac{\partial}{\partial n} \ln \frac{1}{|\zeta - z|} \, |d\zeta|,$$

where $\zeta = \xi + i\eta$ and n is the normal to C , are called, respectively, the *logarithmic potential*, the *logarithmic potential of a simple layer*, and the *logarithmic potential of a double layer*.

Verify the following equalities:

$$1) \frac{1}{2\pi} \int_0^{2\pi} \ln \frac{1}{|\zeta - z|} \, d\theta = \begin{cases} \ln \frac{1}{|z|} & \text{if } |z| > R \\ \ln \frac{1}{R} & \text{if } |z| \leq R; \end{cases} \quad (\zeta = Re^{i\theta}).$$

$$2) \frac{1}{\pi R^2} \iint_{|z| < R} \ln \frac{1}{|\zeta - z|} \, d\xi \, d\eta = \begin{cases} \ln \frac{1}{|z|} & \text{if } |z| > R, \\ \ln \frac{1}{R} + \frac{1}{2} \left[1 - \left(\frac{|z|}{R} \right)^2 \right] & \text{if } |z| \leq R; \end{cases}$$

$$3) \int_C v(\zeta) \frac{\partial}{\partial n} \ln \frac{1}{|\zeta - z|} \, |d\zeta| = \int_C v(\zeta) \, d \arg(\zeta - z)$$

(in traversing C the normal n is taken on the left);

$$4) \int_{-a}^a \frac{\partial}{\partial \eta} \ln \frac{1}{|\xi - z|} d\xi = \begin{cases} \varphi & \text{if } \operatorname{Im} z > 0 \\ -\varphi & \text{if } \operatorname{Im} z < 0, \end{cases}$$

where φ ($0 < \varphi < \pi$) is the angle subtended by the segment $[-a, a]$ at the point z ($a > 0$).

Green's function of a domain G for Dirichlet's problem $g(x, y, \xi, \eta)$ (briefly, $g(z, \zeta)$, $z = x + iy$, $\zeta = \xi + i\eta$) is a harmonic function of both pairs of variables x, y and ξ, η , which is equal to zero on the boundary of G and has a singularity at $z = \zeta$, where

$$g(z, \zeta) = \ln \frac{1}{|z - \zeta|} + \text{harmonic function.}$$

Green's function is symmetric with respect to its arguments, i.e. $g(z, \zeta) \equiv g(\zeta, z)$.

1042. Formulate the Dirichlet problem for a harmonic function which is equivalent to finding Green's function $g(z, \zeta)$.

1043. Let the function $w = f(z, \zeta)$ map conformally a singly connected Jordan domain G onto the circle $|w| < 1$ so that $f(\zeta, \zeta) = 0$ ($\zeta \in G$). Prove the relations

$$g(z, \zeta) = -\ln |f(z, \zeta)|, \quad (1)$$

$$\bar{f}(z, \zeta) = e^{-(g + ih)}, \quad (2)$$

where $h(z, \zeta)$ is a harmonic function conjugate to $g(z, \zeta)$.

1044. Using relation (1) of Problem 1043, find Green's function $g(z, \zeta)$ for the following domains:

- 1) the circle $|z| < R$;
- 2) the semicircle $\operatorname{Im} z > 0$;
- 3) the strip $0 < \operatorname{Im} z < 1$.

1045. Prove the following assertions (n the inner normal, γ_r the circle $|z - \zeta| = r$):

- 1) If $u(z)$ is continuous near $z = \zeta$, then

$$\lim_{r \rightarrow 0} \int_{\gamma_r} u(r) \frac{\partial g(z, \zeta)}{\partial n} ds = 2\pi u(z).$$

- 2) If $u(z)$ is continuously differentiable near $z = \zeta$, then

$$\lim_{r \rightarrow 0} \int_{\gamma_r} g(z, \zeta) \frac{\partial u(z)}{\partial n} ds = 0.$$

- 3) If $u(z)$ is harmonic in G and continuously differentiable on C , then

$$u(\zeta) = \frac{1}{2\pi} \int_C u(z) \frac{\partial g(z, \zeta)}{\partial n} ds \quad (z \in G).$$

Hint. In the formula

$$\int_C \left(u \frac{\partial g}{\partial n} - g \frac{\partial u}{\partial n} \right) ds = \int_{\gamma_r} \left(u \frac{\partial g}{\partial n} - g \frac{\partial u}{\partial n} \right) ds$$

take the limit as $r \rightarrow 0$.

§ 3. POISSON'S INTEGRAL, SCHWARZ' FORMULA, HARMONIC MEASURE

If a real function $u(\zeta) = u(R, \theta)$ is defined and is piecewise continuous on the circle $\zeta = Re^{i\theta}$ ($0 \leq \theta < 2\pi$), then *Poisson's integral*

$$u(z) = u(r, \varphi) = \frac{1}{2\pi} \int_0^{2\pi} u(R, \theta) \frac{R^2 - r^2}{R^2 - 2Rr \cos(\theta - \varphi) + r^2} d\theta \quad (1)$$

defines, in the circle $|z| < R$ ($z = re^{i\varphi}$), a harmonic function with boundary values equal to $u(\zeta)$ at continuity points of $u(z)$:

$$\lim_{z \rightarrow \zeta} u(z) = u(\zeta)$$

($z \rightarrow \zeta$ along any nontangential paths). The corresponding function $f(z) = u + iv$, analytic in the circle $|z| < R$, is defined by the Schwarz formula:

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} u(\zeta) \frac{\zeta + z}{\zeta - z} d\theta + iv(0),$$

where $v(0)$ is an arbitrary real number.

1046. Prove the following assertions:

$$1) \frac{1}{2\pi} \int_0^{2\pi} \frac{R^2 - r^2}{R^2 - 2Rr \cos(\theta - \varphi) + r^2} dr = 1;$$

$$2) u(r, \varphi) - u(R, \theta_0) = \frac{1}{2\pi} \int_0^{2\pi} [u(R, \theta) - u(R, \theta_0)] \times \\ \times \frac{R^2 - r^2}{R^2 - 2Rr \cos(\theta - \varphi) + r^2} d\theta;$$

$$3) \text{ if } |u(R, \theta) - u(R, \theta_0)| < \varepsilon \text{ for } |\theta - \theta_0| < \alpha, \text{ then} \\ \frac{1}{2\pi} \int_{|\theta - \theta_0| < \alpha} \frac{|u(R, \theta) - u(R, \theta_0)|}{R^2 - 2Rr \cos(\theta - \varphi) + r^2} d\theta < \varepsilon;$$

$$4) \text{ if } |\theta - \theta_0| > \alpha \text{ and } |\varphi - \theta_0| < \frac{\alpha}{2}, \text{ then}$$

$$R^2 - 2Rr \cos(\theta - \varphi) + r^2 > 4Rr \sin^2 \frac{\alpha}{4};$$

5) if $|\varphi - \theta_0| < \frac{\alpha}{2}$ and the conditions of item (3) are satisfied, then

$$|u(r, \varphi) - u(R, \theta_0)| < \varepsilon + \frac{M(R^2 - r^2)}{2\pi A},$$

where $A = 4\pi Rr \sin^2 \frac{\alpha}{4}$, $M = \int_0^{2\pi} |u(R, \theta) - u(R, \theta_0)| d\theta$.

1047. Prove that if $\zeta = Re^{i\theta}$, $z = re^{i\varphi}$, then

$$\frac{\zeta + z}{\zeta - z} = \frac{(R^2 - r^2) + i2Rr \sin(\varphi - \theta)}{R^2 - 2Rr \cos(\theta - \varphi) + r^2}.$$

Using this equality, obtain the following expansions:

$$\begin{aligned} u(z) &= u(0) + \sum_{n=1}^{\infty} \left(\frac{r}{R}\right)^n (a_n \cos n\varphi + b_n \sin n\varphi), \\ v(z) &= v(0) + \sum_{n=1}^{\infty} \left(\frac{r}{R}\right)^n (-b_n \cos n\varphi + a_n \sin n\varphi), \\ f(z) &= f(0) + \sum_{n=1}^{\infty} c_n z^n, \quad f(0) = u(0) + i v(0), \end{aligned}$$

where

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_0^{2\pi} u(R, \theta) \cos n\theta d\theta, \quad b_n = \frac{1}{\pi} \int_0^{2\pi} u(R, \theta) \sin n\theta d\theta, \\ c_n &= \frac{a_n - ib_n}{R^n} = \frac{1}{\pi R^n} \int_0^{2\pi} u(R, \theta) e^{-in\theta} d\theta. \end{aligned}$$

1048. Prove that for a harmonic function $u(z)$ Dirichlet's integral

$$D(u) = \iint_{|z| < R} (u_x^2 + u_y^2) dx dy = \pi \sum_{n=1}^{\infty} n (a_n^2 + b_n^2)$$

(the coefficients a_n and b_n are defined in the preceding problem).

In this case both parts of the equality can go to infinity simultaneously.

Hint. Pass to polar coordinates (r, φ) and prove that if $r < R$,

$$\text{then } D_r(u) = \iint_{|z| < r} (u_x^2 + u_y^2) dx dy = \pi \sum_{n=1}^{\infty} n \left(\frac{r}{R}\right)^{2n} (a_n^2 + b_n^2).$$

1049. Prove that for the continuous function

$$u(R, \theta) = \sum_{n=1}^{\infty} \frac{\sin n! \theta}{n^2}$$

Dirichlet's integral defines a harmonic (in the circle $|z| < R$) function $u(z)$ with boundary values $u(R, 0)$ and the infinite Dirichlet integral $D(u) = \infty$.

1050. With the aid of Poisson's integral solve the exterior Dirichlet problem for the circle $|z| < R$: find the function $u(z)$, harmonic in the domain $|z| > R$, regular at infinity and having specified boundary values $u(\xi)$ on the circle $|\xi| = R$. Determine the value of $u(\infty)$.

Hint. Make the substitution $z_1 = R^2/\bar{z}$.

1051. Prove that for $|z| > R$ Schwarz' formula (see the introduction to § 3) defines the analytic function $f_1(z) = u_1(z) + iv_1(z)$ regular at infinity; the following expansions are valid:

$$f_1(z) = -\overline{f\left(\frac{R^2}{z}\right)} = -\overline{f(0)} - \sum_{n=1}^{\infty} \frac{c_{-n}}{z^n}, \quad c_{-n} = R^{2n} \bar{c}_n,$$

$$u_1(z) = -\left\{ u(0) + \sum_{n=1}^{\infty} \left(\frac{R}{r}\right)^n (a_n \cos n\varphi + b_n \sin n\varphi) \right\},$$

$$v_1(z) = -v(0) + \sum_{n=1}^{\infty} \left(\frac{R}{r}\right)^n (-b_n \cos n\varphi + a_n \sin n\varphi),$$

where a_n, b_n, c_n are defined in the same manner as in Problem 1047, and $\operatorname{Re} f_1(\xi) = -\operatorname{Re} f(\xi) = -u(\xi)$, $\operatorname{Im} f_1(\xi) = \operatorname{Im} f(\xi)$.

Hint. In the Schwarz formula for $|z| > R$ make the substitution $z = R^2/z_1$ and take advantage of the fact that $\xi = R^2/\bar{\xi}$ on the circle $|\xi| = R$.

In Problems 1052 to 1057 find the functions $f(z)$ ($|z| < R$) and $f_1(z)$ ($|z| > R$) defined by Schwarz' formula if $u(\xi)$ is the given function.

1052. 1) $u(\xi) = \operatorname{Re} [\varphi(\xi) + \psi(\xi)],$

2) $u(\xi) = \operatorname{Im} [\varphi(\xi) + \psi(\xi)],$

where $\varphi(z)$ is a function analytic for $|z| \leq R$, and $\psi(z)$ is a function analytic for $|z| \geq R$.

1053. $u(\xi) = \operatorname{Re} \xi^n.$ **1054.** $u(\xi) = \operatorname{Re} \frac{1}{\xi^n}.$

1055. $u(\xi) = \operatorname{Re} \ln \frac{\xi}{\xi-1} \quad (R > 1).$

1056. $u(\xi) = \operatorname{Re} \sqrt{\frac{\xi}{\xi-1}} \quad \left(\text{if } \xi > 1, \text{ then } \sqrt{\frac{\xi}{\xi-1}} > 0; R > 1 \right).$

1057. $u(\xi) = \operatorname{Re} \ln \xi.$

1058. Prove that Schwarz' formula can be written in the form

$$f(z) = \frac{1}{\pi i} \int_{|\xi|=R} \frac{u(\xi) d\xi}{\xi - z} - \overline{f(0)}$$

Hint. Use the equality $\frac{\xi+z}{\xi(\xi-z)} = \frac{2}{\xi-z} - \frac{1}{\xi}$.

1059. Obtain formulas similar to Poisson's integral and the Schwarz formula for the upper half-plane $\text{Im } z > 0$, i.e. express the harmonic function $u(z)$ and the analytic function $f(z) = u(z) + iv(z)$ in terms of $u(t)$ ($-\infty < t < \infty$).

Hint. Use a conformal mapping of the half-plane onto a circle.

1060. Derive the Schwarz formula for the strip $0 < \text{Im } z < 1$.

Hint. Use a conformal mapping of the strip onto a half-plane.

The *harmonic measure* $\omega(z, \alpha, G)$ of a boundary arc α at a point z with respect to a domain G is a bounded harmonic (in G) function equal to unity at interior points of the arc α and to zero at interior points of the rest of the boundary. The harmonic measure $\omega(z, \alpha, G)$ is invariant under conformal mappings.

In Problems 1061 to 1064 the domain G is the circle $|z| < 1$ and $\omega(z, \theta_1, \theta_2)$ is the harmonic measure of the arc $\alpha = (\theta_1, \theta_2)$: $\omega = e^{i\theta}$, $\theta_1 \leq \theta \leq \theta_2$.

1061. Using Poisson's integral prove that

$$\omega(z, \theta_1, \theta_2) = \frac{1}{2\pi} \int_{\theta_2}^{\theta_1} \frac{1-r^2}{1-2r \cos(\theta-\varphi) + r^2} d\theta,$$

in particular $\omega(0, \theta_1, \theta_2) = \frac{\theta_2 - \theta_1}{2\pi}$.

1062. Find the level lines of the function $\frac{d\omega(z, \theta_1, \theta)}{d\theta}$ for a fixed value of θ ($z = re^{i\varphi}$ a variable point).

Hint. Prove that

$$\frac{d\omega}{d\theta} = \frac{1}{2\pi} \frac{|w' - z|}{|w - z|},$$

where w' is the end of a chord emanating from w and passing through z .

1063. Let w' denote the end of a chord emanating from a point w and passing through a point z . Let α be the arc (θ_1, θ_2) , and let $\alpha'(z)$ be the arc described by the point w' when w traverses the arc α . Prove that the length of the arc $\alpha'(z)$ equals $2\pi\omega(z, \theta_1, \theta_2)$.

1064. Find the level lines of the harmonic measure $\omega(z, \theta_1, \theta_2)$ of the arc (θ_1, θ_2) , then prove that the integral defining the harmonic measure (see Problem 1061) actually has the limiting values 1 on (θ_1, θ_2) and 0 on the complement (interior points of the arcs are considered).

1065. Define the harmonic measure $\omega(z, a, b)$ of the segment $[a, b]$ of the ray $(-\infty; b)$ and of the ray (a, ∞) for the half-plane $\text{Im } z > 0$. What is the geometric meaning of these harmonic measures?

1066. Find the harmonic measure of the sides of the angle $0 < \arg z < \gamma$.

1067. Find the harmonic measures of the diameter Δ and the semicircular arc Γ for the semicircle $|z| < R, \text{Im } z > 0$. Also find the level lines of these harmonic measures.

1068. Find the harmonic measure of the boundary semicircle Γ of the domain $|z| > R, \text{Im } z > 0$.

1069. Find the harmonic measure of the boundary circle Γ of the domain $|z| > R, 0 < \arg z < 2\pi$.

1070. Find the harmonic measure of boundary circles of the annulus $r < |z| < R$.

In Problems 1071 to 1078 the domain G is bounded by a composite contour Γ consisting of n simple smooth contours Γ_v ($v=1, 2, \dots, n$). The contours are traversed in the positive direction with respect to G , the normal n is inner with respect to G . The integral

$$\int_{\Gamma_v} df(z)$$

is called the *period* of the analytic (in G) function along Γ_v .

1071. Prove that if the harmonic function $u(z)$ is single-valued in G , then the period of the analytic function $f(z) = u(z) + iv(z)$ along Γ_v equals $-i \int_{\Gamma_v} \frac{\partial u}{\partial n} ds$.

1072. Prove that for a complex Green's function $g + ih$ of the domain G [$g(z, \zeta)$ is Green's function of the domain G , $h(z, \zeta)$ is a harmonic function conjugate to g] the period along Γ_v is equal to $-2\pi i \omega_v(z)$, where $\omega_v(z)$ is the harmonic measure of Γ_v with respect to the domain G . Prove that $\sum_{v=1}^n \omega_v(\zeta) = 1$.

Hint. The function $u(\zeta)$ harmonic in G may be represented in the form $u(\zeta) = \frac{1}{2\pi} \int_{\Gamma} u(z) \frac{\partial g(z, \zeta)}{\partial n} ds$ (see Problem 1045).

1073. Express in terms of $\omega_v(z)$ a bounded function $u(z)$, harmonic in G , that has constant values c_v on Γ_v ($v=1, 2, \dots, n$).

1074. Prove that for a function $v(z)$, conjugate to the function $u(z)$ (single-valued and harmonic in $\bar{G}$), the periods p_v along Γ_v may be represented as

$$p_v = - \int_{\Gamma} u(z) \frac{\partial \omega_v(z)}{\partial n} ds.$$

1075. Let $\tilde{\omega}_v(z)$ be a harmonic function conjugate to $\omega_v(z)$, and let $p_{\mu v}$ be the period of the function $\tilde{\omega}_v$.

1) Prove that $p_{\mu v} = p_{v\mu}$ ($\mu, v=1, 2, \dots, n$).

Hint. Use the representation

$$p_{\mu v} = - \int_{\Gamma} \omega_v \frac{\partial \omega_\mu}{\partial n} ds.$$

2) Prove that $\sum_{v=1}^n p_{\mu v} = 0$ ($\mu=1, 2, \dots, n$).

1076. Let c_v ($v=1, 2, \dots, n$) be arbitrary real numbers.

1) Prove that if $p_{\mu v}$ are the numbers defined in problem 1075, then the quadratic form $\sum_{v, \mu=1}^n p_{\mu v} c_v c_\mu \geq 0$, the equality being valid only when all c_v are equal.

Hint. Apply the formula $D(\omega) = - \int_{\Gamma} \omega \frac{\partial \omega}{\partial n} ds$ to the harmonic function

$\omega(z) = \sum_{v=1}^n c_v \omega_v(z)$ (see Problem 1039; the sign is reversed, since now n is an inner normal).

2) Prove that the quadratic form $\sum_{v, \mu=1}^{n-1} p_{\mu v} c_v c_\mu$ is positive definite, i.e. it is positive for all sets of values $\{c_v\}$ except $c_1 = c_2 = \dots = c_{n-1} = 0$.

1077. Prove that the system of equations

$$\sum_{v=1}^{n-1} p_{\mu v} A_v = B_\mu \quad (\mu=1, 2, \dots, n-1)$$

(A_v unknown quantities) has a unique solution for any B_μ . Using this fact, prove that for any function $u(z)$ harmonic in $\bar{G}$ and, generally speaking, not single-valued, one can choose constants $A_1, A_2, \dots, A_{n-1}$ such that the harmonic function

$$u_1(z) = u(z) + \sum_{v=1}^{n-1} A_v \tilde{\omega}_v(z)$$

is single-valued in G .

CHAPTER VIII

ANALYTIC CONTINUATION. SINGULARITIES OF MANY-VALUED CHARACTER. RIEMANN SURFACES

§ 1. ANALYTIC CONTINUATION

1078. The function $f(z) = \sum_{n=0}^{\infty} z^n$ is expanded into a Taylor series in the neighbourhood of the point $z=a$ ($|a| < 1$). For what values of a does this expansion permit analytic continuations of the function $f(z)$?

1079. The sum of the power series $f(z) = \sum_{n=1}^{\infty} \frac{z^n}{n}$ is expanded into a Taylor series in the neighbourhood of the point $z = -1/2$. What is the domain into which the function $f(z)$ will be thus continued?

1080. Prove that the function $f(z) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{z^n}{n}$ can be continued into a larger domain by means of the series

$$\ln 2 - \frac{1-z}{2} - \frac{(1-z)^2}{2 \times 2^2} - \frac{(1-z)^3}{3 \times 2^3} - \dots$$

1081. The power series

$$f_1(z) = \sum_{n=1}^{\infty} \frac{z^n}{n} \quad \text{and} \quad f_2(z) = i\pi + \sum_{n=1}^{\infty} (-1)^n \frac{(z-2)^n}{n}$$

have no common domain of convergence. Prove, nevertheless, that the functions $f_1(z)$ and $f_2(z)$ are analytic continuations of one another.

1082. Prove that the functions defined by the series

$$1 + az + a^2 z^2 + \dots \quad \text{and} \quad \frac{1}{1-z} - \frac{(1-a)z}{(1-z)^2} + \frac{(1+a)^2 z^2}{(1-z)^3} - \dots$$

are analytic continuations of one another.

1083. Let the power series

$$f(z) = a_0 + a_1 z + \dots + a_n z^n + \dots$$

have radius of convergence $R=1$. Substituting the variable $z = \frac{Z}{1+Z}$, reduce it to the form

$$f(z) = f\left(\frac{Z}{1+Z}\right) = F(Z) = c_0 + c_1 Z + \dots + c_n Z^n + \dots$$

Denoting the radius of convergence of the series by ρ , prove the following assertions:

1) $\rho \geq 1/2$; if the point $z = -1$ is a singularity of the function $f(z)$, then $\rho = 1/2$.

2) If $\frac{1}{2} < \rho < 1$, then the equality $f(z) = F(Z) = F\left(\frac{z}{1-z}\right)$ permits analytic continuation of the function $f(z)$ into a domain exterior to the circle $|z| < 1$ and interior to the circle of Apollonius $\left|\frac{z}{1-z}\right| = \rho$.

3) If $\rho = 1$, then the equality indicated in item (2) analytically continues the function $f(z)$ into the half-plane $\operatorname{Re} z < 1/2$.

4) If $\rho > 1$, then the function $f(z)$ can be analytically continued into a domain exterior to the circle of Apollonius $\left|\frac{z}{z-1}\right| = \rho$.

1084. Prove that the power series $f(z) = \sum_{n=0}^{\infty} z^{2^n}$ represents a function analytic in the circle $|z| < 1$ and has the circumference $|z| = 1$ as its natural boundary [i.e. $f(z)$ is a function which cannot be continued beyond the unit circle].

Hint. Using the identity $f(z) = z^2 + z^4 + \dots + z^{2^k} + f(z^{2^k})$, prove that for any point of the form $\xi = \sqrt[k]{1}$ (k a natural number) $f(t\xi) \rightarrow \infty$ as $t \rightarrow 1$ ($0 < t < 1$).

In Problems 1085 and 1086 prove that the functions represented by the indicated power series cannot be continued beyond the unit circle.

1085. $f(z) = \sum_{n=0}^{\infty} z^{n!}$.

Hint. If p and q are relatively prime integers and $n \geq q$, then $\left(re^{\frac{2p\pi i}{q}}\right)^{n!} = r^{n!}$.

1086. $f(z) = \sum_{n=0}^{\infty} \frac{z^{n!}}{n^2}$.

Note. Problems 1084 to 1086 consider particular cases of Hadamard's general gap-theorem:

If the numbers of the non-zero coefficients of the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ form a sequence $n_1, n_2, \dots$, in which $n_{k+1} > (1 + \alpha)n_k$, where $\alpha > 0$, then the boundary of the circle of convergence of the series is the natural boundary of the function $f(z)$.

1087. Prove that the series $\sum_{n=1}^{\infty} \left(\frac{1}{1-z^{n+1}} - \frac{1}{1-z^n} \right)$ in the domains $|z| < 1$ and $|z| > 1$ represents two analytic functions which are not analytic continuations of each other (see also Problems 845 to 848).

1088. Let $f(z)$ and $\varphi(z)$ be arbitrary entire functions and let $S(z) = \sum_{n=1}^{\infty} \left(\frac{1-z^n}{1+z^n} - \frac{1-z^{n-1}}{1+z^{n-1}} \right)$. Prove that the expression

$$\Psi(z) = \frac{1}{2} [f(z) + \varphi(z)] + \frac{1}{2} S(z) [f(z) - \varphi(z)]$$

represents the function $f(z)$ in the domain $|z| < 1$, and the function $\varphi(z)$ in the domain $|z| > 1$.

1089. 1) Prove that if α is a real irrational number, then the series $\sum_{n=1}^{\infty} \frac{1}{2^n (z - e^{2i\alpha n})}$ represents, in the domains $|z| < 1$ and $|z| > 1$, analytic functions, for each of which the circle $|z| = 1$ is the natural boundary.

Hint. Prove that the sum of the series increases without bound along the radius vector as $z \rightarrow e^{2i\alpha n}$.

Note. This problem is a particular case of the following general theorem:

Let L be a curve, closed or open, that has a certain radius of curvature at each point. If $\sum_{n=1}^{\infty} c_n$ is an absolutely convergent series, and if all the points $a_1, a_2, \dots, a_n, \dots$ lie on L and are so distributed on it that any finite arc of L always contains infinitely many such points, then the series $F(z) = \sum_{n=1}^{\infty} \frac{c_n}{a_n - z}$ represents a function which is analytic in any domain containing no points of the curve L and for which this curve is a singular line.

2) Prove that if α is a real rational number, then the series in item (1) represents a rational function.

1090. Prove that the series

$$\sum_{n=1}^{\infty} \frac{1}{n^2 + n} \quad (n^2 = e^{z \ln n})$$

converges for $\operatorname{Re} z > 1$ and its sum has the straight line $\operatorname{Re} z = 1$ for its natural boundary.

1091. Prove that the function $f(z) = \sum_{m=0}^{\infty} e^{-m!z}$ is analytic for $\operatorname{Re} z > 0$ and has for its natural boundary the straight line $\operatorname{Re} z = 0$.

1092. Prove that the function $f(z)$ defined in the half-plane $\operatorname{Re} z > 0$ by the Dirichlet series $f(z) = \sum_{n=1}^{\infty} a_n e^{-\lambda_n z}$, where $a_n = (-1)^{n+1}$, $\lambda_{2k-1} = 2k$, $\lambda_{2k} = 2k + e^{-2k}$ ($k = 1, 2, \dots$), can be continued analytically into the half-plane $\operatorname{Re} z > -1$.

Hint. Write $f(z)$ in the form $f(z) = \sum_{k=1}^{\infty} (1 - e^{-ze^{-2k}}) e^{-2kz}$ and prove that in any finite domain $|1 - e^{-ze^{-2k}}| < M e^{-2k}$, where M is a constant for the domain under consideration.

Note. The present problem shows that on the straight line bounding the half-plane of convergence the sum of Dirichlet's series may have no singular points.

1093.* Continue analytically into the half-plane $\operatorname{Re} s > -1$ the function $f(s)$ defined in the half-plane $\operatorname{Re} s > 0$ by the Laplace integral $f(s) = \int_0^{\infty} e^{-st} e^t \sin e^t dt$.

1094. Euler's Gamma function is defined in the half-plane $\operatorname{Re} z > 0$ by means of the integral $\Gamma(z) = \int_0^{\infty} e^{-t} t^{z-1} dt$ ($t^{z-1} = e^{(z-1) \ln t}$). Integrating the right member of this equality by parts, show that the function $\Gamma(z)$ is continued analytically throughout the plane as a meromorphic function with simple poles $0, -1, -2, \dots, -n, \dots$, the residue with respect to the pole $-n$ being equal to $(-1)^n/n!$.

1095. Show that the Γ -function can be analytically continued by means of the formula $\Gamma(z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(z+n)} + \int_1^{\infty} e^{-t} t^{z-1} dt$.

Hint. In the integral $\int_0^1 e^{-t} t^{z-1} dt$ replace the function e^{-t} by its power-series expansion.

1096. In Problem 411 it was proved that for $0 < x < 1$

$$\int_0^x t^{x-1} \cos t dt = \Gamma(x) \cos \frac{\pi x}{2}, \quad \int_0^x t^{x-1} \sin t dt = \Gamma(x) \sin \frac{\pi x}{2}.$$

In what domains of the z -plane will these formulas be valid?

1097. Prove that $\Gamma(z)$ can be continued over the whole domain of its existence with the aid of the formula

$$\Gamma(z) = \frac{i}{2 \sin \pi z} \int_C e^{-\omega} (-\omega)^{z-1} d\omega \quad ((-\omega)^{z-1} = e^{(z-1) \ln(-\omega)}),$$

where the contour C consists of a cut along the positive real axis, the detour round the origin being counterclockwise.

1098. Let $\zeta(z)$ be the Riemann Zeta function (see Problem 844)

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z} \quad (\operatorname{Re} z > 1).$$

Prove that $\zeta(z) = \frac{1}{\Gamma(z)} \int_0^{\infty} \frac{\omega^{z-1}}{e^{\omega}-1} d\omega$ for $\operatorname{Re} z > 1$ and obtain the reform an analytic continuation of the function $\zeta(z)$ onto the whole plane, except for the point $z=1$; clarify the nature of the singularity of the function $\zeta(z)$ at $z=1$.

Hint. For analytic continuation consider the integral $\int_C \frac{(-\omega)^{z-1}}{e^{\omega}-1} d\omega$, where C is the contour in Problem 1097.

1099*. Let the function $f(z)$ be expanded into the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ with radius of convergence $R=1$. Let $\varphi(z)$ denote

the sum of the series $\varphi(z) = \sum_{n=0}^{\infty} \frac{a_n z^n}{n!}$ ($\varphi(z)$, conjugate according to Borel with the function $f(z)$, is an entire function, see Problem 535).

Prove that for $|z| < 1$ the following equality holds:

$$\int_0^{\infty} e^{-t} \varphi(zt) dt = f(z).$$

Also prove that the function $\int_0^{\infty} e^{-t} \varphi(zt) dt$ effects analytic continuation of the function $f(z)$ into a domain G defined in the following way: through each singular point of $f(z)$ draw a straight line perpendicular to the segment joining this point to the origin; G is a convex domain (containing the circle $|z| < 1$) whose boundary consists of points of the indicated straight lines; if the number of the straight lines is finite, then G is a polygon (Borel's method of continuation).

1100. Verify the Borel continuation method for the following series:

$$1) \sum_{n=0}^{\infty} z^n; \quad 2) \sum_{n=0}^{\infty} z^{2n}; \quad 3) \sum_{n=0}^{\infty} z^{4n}.$$

1101. In the Cauchy-type integral $F(z) = \frac{1}{2\pi i} \int_C \frac{\varphi(\xi) d\xi}{\xi - z}$ let C be a simple closed contour and $\varphi(\xi)$ a function continuous along C . Prove that for one of the functions $F^+(z)$, $F^-(z)$ (see page 133) to be an analytic continuation of the other across the arc $\gamma \in C$, it is necessary and sufficient that $f(z) \equiv 0$ on the arc γ .

1102. Prove that if the function $\varphi(\xi)$ is not analytic¹⁾ at any of the points of a simple open arc C , then all the points of the arc C are singular for the Cauchy-type integral $\frac{1}{2\pi i} \int_C \frac{\varphi(\xi) d\xi}{\xi - z}$.

Hint. Proceed from the Sokhotsky formulas for the limiting values of the Cauchy-type integral.

1103. Let γ be a simple closed contour traversed in the positive direction and consisting of arcs γ_1 and γ_2 with common end-points

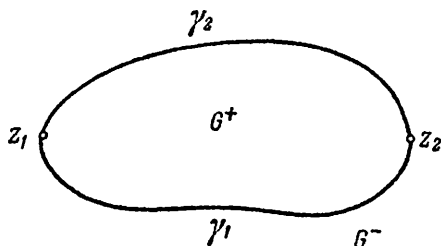


FIG. 31

z_1 and z_2 (Fig. 31), and let G^+ be a domain inside γ and G^- , a domain outside γ . Further, let

$$F(z) = \int_{\gamma} \frac{\varphi(\xi) d\xi}{\xi - z},$$

where $\varphi(\xi) = a$ if $\xi \in \gamma_1$, and $\varphi(\xi) = b$ if $\xi \in \gamma_2$ (a and b complex constants).

Find the functions $F^+(z)$ and $F^-(z)$ and continue the function $F^-(z)$ analytically into the domain G^+ :

a) across the arc γ_1 ; b) across the arc γ_2 .

¹⁾ That is, there is no analytic function that coincides with $\varphi(\xi)$ on any arc belonging to C .

1104. Let G be a doubly connected domain bounded by an inner contour γ and an outer contour Γ , and let $\varphi(z)$ be a function analytic in the closed domain $G + \gamma + \Gamma$.

Prove that the function

$$\varphi(z) - \frac{1}{2\pi i} \int_{\Gamma} \frac{\varphi(\zeta)}{\zeta - z} d\zeta$$

can be continued analytically into the whole exterior of the contour γ , and the function

$$\varphi(z) + \frac{1}{2\pi i} \int_{\gamma} \frac{\varphi(\zeta)}{\zeta - z} d\zeta,$$

into the whole interior of the contour Γ . The contours γ and Γ are traversed counterclockwise.

§ 2. SINGULARITIES OF MANY-VALUED CHARACTER. RIEMANN SURFACES

An isolated branch-point $z=a$ of order $k-1$ of the function $w(z)$ (k a natural number, $k \geq 2$) is characterized by the fact that there is a branch of $w(z)$ allowing for the following representation in the neighbourhood of the point $z=a$:

$$w = \sum_{n=-\infty}^{\infty} c_n (z-a)^{n/k} \quad (a \neq \infty)$$

or

$$w = \sum_{n=-\infty}^{\infty} c_n z^{-n/k} \quad (a = \infty).$$

If only a finite number of coefficients c_n with negative indices are different from zero, then the point $z=a$ (or $z=\infty$) is called an *algebraic branch-point*. Otherwise, that point is called a *transcendental branch-point* (an essential singularity of multi-valued character).

A function $w(z)$ can have, above one and the same point of the z -plane, at most a countable set of distinct algebraic and transcendental branch-points, regular points and singular points of single-valued character. On the Riemann surface of a function $w(z)$, such points have, above the z -plane, non-intersecting neighbourhoods.

In each such neighbourhood, w is a single-valued function of a local parameter t :

$$w = \sum_{n=-N}^{\infty} c_n t^n,$$

where

$$t = \begin{cases} (z-a)^{\frac{1}{k}} & (a \neq \infty), \\ z^{-\frac{1}{k}} & (a = \infty). \end{cases}$$

Logarithmic branch-points include the points $z=a$ and $z=\infty$ for which some branch $w(z)$ allows for unlimited analytic continuation in the domain $0 < |z-a| < r$ (or $R < |z| < \infty$) and is infinitely multi-valued there. Such a branch of $w(z)$ in the neighbourhood of a logarithmic branch-point becomes a single-valued analytic function when passing to the parameter $t = \text{Ln}(z-a)$, $\text{Re } t < \rho$ (or $t = \text{Ln } z$, $\text{Re } t > \rho$). It should be kept in mind that on a Riemann surface above one and the same point z there may be, besides distinct logarithmic branch-points, also other points of single- and multi-valued character.

1105. Find out for what values of z the values of $w(z)$ are the same on all sheets of its Riemann surface above the z -plane if:

1) $w = (z^2 - 9)\sqrt[3]{z}$; 2) $w = \sin z + (z^2 + 4)\text{Ln } z$; 3) $w = \sin z + (z^2 + 4)^2 \text{Ln } z$.

Are the values of $w'(z)$ identical at the same points?

1106. Verify that for each of the functions

1) $w = z\sqrt[3]{z}$, 2) $w = z^2 \text{Ln } z$, $w(0) = 0$

at the point $z=0$ there is a first derivative which is the same for all branches, and there is no finite second derivative.

In Problems 1107 to 1115 expand each of the indicated functions $w(z)$ into a series of powers of the local parameter t about all points of its Riemann surface situated above the given z -points; indicate the domains of convergence for the series obtained.

1107. $w = \frac{1}{1 + \sqrt[3]{2-z}}$, $z=1$, $z=2$.

1108. $w = \sqrt[3]{\sqrt[3]{z-1}-2}$, $z=1$, $z=5$, $z=\infty$.

1109. $w = \sqrt[3]{1 + \sqrt[3]{z-1}}$, $z=1$, $z=2$, $z=\infty$.

1110. $w = \sqrt[3]{1 + \sqrt[3]{z-1}}$, $z=1$, $z=2$, $z=\infty$.

1111. $w = \sqrt[3]{(\sqrt[3]{z-a})(\sqrt[3]{z-b})}$, $z=\infty$ ($a \neq b$).

1112. $w = e^{\frac{1}{\sqrt[3]{z}}}$, $z=0$. **1113.** $w = \frac{\sin \sqrt[3]{z}}{z^2}$, $z=0$.

1114. $w = \cot \sqrt[3]{z}$, $z=0$. **1115.** $w = \sqrt[3]{\sin z}$, $z=0$.

In Problems 1116 to 1122 find the points of the z -plane above which there is at least one singular point of the given multi-valued

function, and indicate the nature of all the points of the Riemann surface which lie above each of these points of the z -plane.

$$1116. \sin \frac{1}{\sqrt{z-1}}. \quad 1117. \sin \frac{1}{1+\sqrt{z}}. \quad 1118. \frac{1}{\sin \frac{1}{1+\sqrt{z}}}.$$

$$1119. \frac{1}{\sin \sqrt{z}}. \quad 1120. \sqrt[3]{\sqrt{z} + \sqrt{2-z}}.$$

$$1121. \tan(i \operatorname{Ln} z). \quad 1122. \tan\left(\frac{i}{4} \operatorname{Ln} z\right).$$

If a function $w=f(z)$ is single-valued and its inverse $z(w)$ multi-valued, then to determine the algebraic branch-points of $z(w)$ it is necessary to find the zeros of $f'(z)$, the multiple poles of $f(z)$ and to investigate the behaviour of $f(z)$ at infinity. In this case to the point $z_0 \neq \infty$ there corresponds an algebraic branch-point of order $k-1$ of the function $z(w)$ if in the neighbourhood of z_0 the Laurent expansion of $f(z)$ has the form

$$f(z) = w_0 + \sum_{n=k}^{\infty} c_n (z-z_0)^n \quad (c_k \neq 0)$$

or

$$f(z) = \sum_{n=-k}^{\infty} c_n (z-z_0)^n \quad (c_{-k} \neq 0).$$

If $z_0 = \infty$, then the above expansions must have the form

$$f(z) = \sum_{n=k}^{-\infty} c_n z^n \quad (c_k \neq 0)$$

or

$$f(z) = w_0 + \sum_{n=-k}^{-\infty} c_n z^n \quad (c_{-k} \neq 0).$$

In Problems 1123 to 1129 determine the singularities of $z(w)$ if w is the given function.

$$1123. w = z(1-z). \quad 1124. w = z^3 - 3z. \quad 1125. w = \frac{z}{(1+z)^2}.$$

$$1126. w = \left(\frac{z}{1-z}\right)^2. \quad 1127. w = \frac{z-a}{z^2-1} \quad (0 < a < 1).$$

$$1128. w = P_n(z) \text{ (polynomial of degree } n\text{)}.$$

$$1129. w = R(z) \text{ (rational function)}.$$

In Problems 1130 to 1138 investigate the mapping effected by the function $w(z)$, construct the Riemann surface R above the w -plane

and divide the z -plane into domains corresponding to the sheets or half-sheets of R .

1130. $w = \left(1 + \frac{z}{n}\right)^n$. Consider the limiting case $n \rightarrow \infty$.

1131. $w = \left(\frac{z-a}{z-b}\right)^n$. 1132. $w = \frac{1}{2} \left(z + \frac{1}{z}\right)$. 1133. $w = \frac{z}{(1-z)^2}$.

1134. $w = \frac{1}{2} \left(z^n + \frac{1}{z^n}\right)$. Also find the group of linear transformations with respect to which the function w is invariant and find out what transformations of the Riemann surface correspond to the transformations of the group.

1135. $w = \frac{z}{(1+z^n)^2}$. 1136. $w = z - \frac{z^n}{n}$. 1137. $w = \frac{1}{z} + \frac{z^n}{n}$.

1138. $w = T_n(z) = \frac{1}{2^{n-1}} \cos(n \arccos z)$, $n \geq 1$ ($T_n(z)$ Chebyshev's polynomials).

In Problems 1139 to 1142 find the singularities of the inverse of the given functions.

1139. $w = e^{\frac{1}{z}}$. 1140. $w = e^{t \left(z - \frac{1}{z}\right)}$ (t a complex number).

1141. $w = \cos z + \sin z$. 1142. $w = \frac{\sin z}{z}$.

In Problems 1143 to 1151 construct Riemann surfaces above the w -plane.

1143. $w = \cos z$. 1144. $w = \sin z$. 1145. $w = \tan z$.

1146. $w = \cot z$. 1147. $w = \cosh z$. 1148. $w = \sinh z$.

1149. $w = \tanh z$. 1150. $w = \coth z$. 1151. $w = z + e^z$.

1152. Assuming the Riemann surface above the w -plane of the rational function $w = R(z)$ to be known, construct Riemann surfaces of the functions $z(w)$ if

1) $w = R(e^z)$; 2) $w = R(\sin z)$.

In Problems 1153 to 1158 construct Riemann surfaces of the given functions (above the z -plane).

1153. 1) $w = \sqrt{(z-a)(z-b)}$; 2) $w = \sqrt{(z-a)(z-b)(z-c)}$;

3) $w = \sqrt[n]{\prod_{k=1}^n (z-a_k)}$ (consider the cases of even and odd n separately).

1154. 1) $w = \sqrt[3]{z-a}$; 2) $w = \sqrt[3]{(z-a)(z-b)}$;

3) $w = \sqrt[3]{(z-a)(z-b)(z-c)}$; 4) $w = \sqrt[n]{\prod_{k=1}^n (z-a_k)}$, $n > 3$.

$$1155. \omega = \sqrt[n]{(z-a)(z-b)(z-c)}, \quad n > 3.$$

$$1156. \omega = \sqrt[3]{\frac{z-a}{(z-b)^2}} + \sqrt{z-c}. \quad 1157. \omega = \sqrt[3]{\sqrt{z-1}}.$$

$$1158. \omega = \sqrt{\sin z}.$$

In Problems 1159 to 1164 investigate the mappings and construct Riemann surfaces for the algebraic functions $\omega(z)$ and $z(\omega)$.

$$1159. \omega^2 + z^2 = 1. \quad 1160. \omega^2 = \frac{z}{(1-z)^2}.$$

Hint. Use the solution of Problem 1133.

$$1161. \omega^3 + z^3 - 3\omega z = 0.$$

Hint. Use the parametric representation

$$z = \frac{3t}{1+t^3}, \quad \omega = \frac{3t^2}{1+t^3}.$$

$$1162. \omega^2 = \frac{z^3}{1-z}.$$

Hint. Use the parametric representation

$$z = \frac{t^2}{1+t^2}, \quad \omega = \frac{t^3}{1+t^2}.$$

$$1163. \omega^2 = z^2 \frac{1+z}{1-z}.$$

$$\text{Hint. Put } z = \frac{t^2-1}{t^2+1}, \quad \omega = t \frac{t^2-1}{t^2+1}.$$

$$1164. \omega^n = \frac{z^n}{(1+z^n)^2}.$$

Hint. Consider the cases of even and odd n separately.

If $z=a$ is a singular point for one of the branches $f(z)$ of the function $\omega(z)$, then the set of limiting values of $f(z)$ obtained from its values in the neighbourhood of $z=a$ above $|z-a| < r$ as $r \rightarrow 0$ is called the *domain of indeterminacy* of $f(z)$. For algebraic branch-points and poles the domain of indeterminacy consists of a single point. If a function is single-valued and the point a is an isolated essential singularity, then, according to Sokhotsky's theorem, the domain of indeterminacy covers the whole plane (see also Problem 618). For transcendental and logarithmic branch-points and also for non-isolated singular points the domain of indeterminacy may have a more complicated structure.

In Problems 1156 to 1176 determine the singularities of the given functions and find the domains of indeterminacy at transcendental and logarithmic branch-points.

$$1165. w = (\sqrt[n]{z})^4. \quad 1166. w = (\sqrt[n]{z})^m \quad (n, m \text{ natural numbers}).$$

$$1167. w = e^{\frac{1}{\sqrt[n]{z}-1}}. \quad 1168. w = \frac{\sin \sqrt[n]{z}}{\sqrt[n]{z}}. \quad 1169. w = \tan \frac{1-\sqrt[n]{z}}{1+\sqrt[n]{z}}.$$

$$1170. w = e^{(\operatorname{Ln} z)^n} \quad (n \text{ an integer}). \quad 1171. w = z^i = e^{i \operatorname{Ln} z}.$$

$$1172. w = \sin \operatorname{Ln} z. \quad 1173. w = \frac{1}{z} \operatorname{Ln} \frac{1}{1-z}. \quad 1174. w = z + \operatorname{Ln} z.$$

$$1175. w = \frac{1}{z} \operatorname{Arc} \sin z. \quad 1176. w = \frac{1}{z^2} + \operatorname{Arc} \tan z.$$

In Problems 1177 to 1180 construct Riemann surfaces of the given functions.

$$1177. w = z^a \quad (a \text{ complex}).$$

$$1178. w = \operatorname{Ln} [(z-a)(z-b)].$$

$$1179. w = \operatorname{Ln} [(z-a)(z-b)(z-c)].$$

$$1180. w = \operatorname{Ln} \sin z.$$

1181. Let $\xi = \varphi(z)$ be a single- or multi-valued analytic function, $R_{z, \varphi}$ its Riemann surface above the z -plane and $w = f(\xi)$ a single-valued analytic function with domain of definition G_ξ . In what domains on $R_{z, \varphi}$ does each of the expressions

$$1) w(z) = f[\varphi(z)], \quad 2) w(z) = f(z) + \varphi(z), \quad 3) w(z) = f(z) \varphi(z)$$

define a unique analytic function?

Consider, in particular, the case where φ is an algebraic function or the inverse of a meromorphic function, and f is a rational or transcendental meromorphic function.

In Problems 1182 to 1219 find out which of the indicated functions $w(z)$ decompose into distinct analytic functions and which do not; also determine their singularities and construct Riemann surfaces where required (n and m natural numbers).

$$1182. w = \sqrt[n]{z^2} \quad (\text{compare with } (\sqrt[n]{z})^2).$$

$$1183. w = \sqrt[n]{z^4} \quad (\text{compare with } (\sqrt[n]{z})^4).$$

$$1184. w = \sqrt[n]{z^m} \quad (\text{compare with } (\sqrt[n]{z})^m).$$

$$1185. w = \sqrt[n]{e^z}. \quad 1186. w = \sqrt[n]{\sin z}. \quad 1187. w = \operatorname{Ln} z^i.$$

$$1188. w = \operatorname{Ln} e^z. \quad 1189. w = \operatorname{Ln} \left(z - \frac{1}{z} \right).$$

$$1190. w = \operatorname{Ln}(e^z - 1). \quad \text{Construct the Riemann surface.}$$

$$1191. w = \operatorname{Ln} \sin z. \quad \text{Construct the Riemann surface.}$$

$$1192. w = \operatorname{Ln} \tan z. \quad \text{Construct the Riemann surface.}$$

1193. $w = \text{Arc cos}(\cos z)$ [compare with $\cos(\text{Arc cos } z)$].

1194. $w = \text{Arc tan}(\tan z)$ [compare with $\tan(\text{Arc tan } z)$].

1195. 1) $w = (z^{r_1})^{r_2}$ (r_1, r_2 rational numbers). Compare with $(z^{r_2})^{r_1}$. Consider, in particular, $r_1 = \frac{2}{3}, r_2 = \frac{3}{2}$; 2) $w = z^{r_1} z^{r_2}$; 3) $w = z^{r_1} + z^{r_2}$.

1196. $w = \sqrt[n]{z} \sqrt[m]{1-z}$. 1197. $w = \sqrt[n]{\sqrt[m]{z}-1}$.

1198. $w = \frac{\sqrt[n]{z}}{\sqrt[n]{z}+1}$.

1199. $w = \sqrt[n]{\text{Ln } z}$. Construct the Riemann surface.

1200. $w = \text{Ln Ln } z$. Construct the Riemann surface.

1201. $w = [\text{Ln}(z-1)]^i$.

1202. $w = \sqrt[n]{\text{Arc sin } z}$. Construct the Riemann surface.

1203. $w = \text{Arc sin Ln } z$. Construct the Riemann surface.

1204. $w = \text{Ln}(\sqrt[n]{z}-1)$. Construct the Riemann surface.

1205. $w = \text{Ln} \frac{1-\sqrt[n]{z}}{1+\sqrt[n]{z}}$. 1206. $w = \text{Arc sin} \frac{\sqrt[n]{z}}{1+z}$.

1207. $w = \text{Ln } z^\alpha$ (α real).

1208. $w = \sqrt[n]{z} + \text{Ln } z$. Construct the Riemann surface.

1209. $w = \text{Ln } z + \text{Ln}(-z)$. 1210. $w = \text{Arc sin } z + \text{Arc cos } z$.

1211. $w = \text{Arc tan } z + \text{Arc cot } z$. 1212. $w = \text{Artanh } z - \text{Arcoth } z$.

1213. Construct the Riemann surface of the function $w = (\text{Ln } z)^i$ and investigate the sets of limiting values of w for a single logarithmic branch-point situated above the point $z=0$ which are obtained for: 1) $r \rightarrow 0, \varphi = \text{const}$; 2) $r \rightarrow 0, \alpha < \varphi < \beta$; 3) $r = \text{const}, \varphi \rightarrow \pm \infty$; 4) $r \rightarrow 0, \varphi \rightarrow \pm \infty$.

1214. Let $\chi(z)$ be a single-valued analytic function in the circle $|z| < 1$ that cannot be continued beyond the circle $|z| = 1$.

Find out for which values of a the indicated functions decompose into distinct analytic functions, and for which they do not:

1) $w = \chi(z) + \sqrt[n]{z-a}$; 2) $w = \chi(z) \text{Ln}(z-a)$;

3) $w = \chi(a+z^n)$; 4) $w = \chi(a+e^z)$.

1215. Elucidate the decomposition of the following functions:

1) $w = \sqrt[n]{\zeta-a}$, 2) $w = \text{Ln}(\zeta-a)$, where $\zeta = \chi^{-1}(z)$, and χ is the function of Problem 1214.

1216. Investigate the behaviour of the separate analytic functions defined by the equalities

1) $w = \chi(z) (\text{Ln } z)^i$; 2) $w = \chi(z) [\text{Ln}(z-1)]^i$, where $\chi(z)$ is the function of Problem 1214.

In particular, find the domains of indeterminacy in the neighbourhood of the logarithmic branch-point.

Hint. Use the solution of Problem 1213.

1217. Let $f(z)$ be an entire function.

Construct Riemann surfaces of the following functions:

1) $w = \sqrt[n]{f(z)}$; 2) $w = \operatorname{Ln} f(z)$;

3) $w = [f(z)]^\alpha$ (α an irrational number).

1218. Construct the Riemann surface of the function

$$w = \sqrt{\sum_{n=1}^{\infty} z^{n!}}.$$

1219. Let $f(z) = z + \sum_{n=1}^{\infty} \frac{z^{2^n} + 2}{(2^n + 1)(2^n + 2)}$. Construct Riemann surfaces of the following functions:

1) $w = \sqrt[n]{f(z)}$; 2) $w = \operatorname{Ln} f(z)$; 3) $w = \operatorname{Ln} f\left(\frac{z}{2}\right) + \operatorname{Ln} f\left(\frac{1}{2z}\right)$.

Hint. First prove that the function $f(z)$ is univalent in the circle $|z| < 1$ and its natural boundary is the circle $|z| = 1$.

CHAPTER IX

CONFORMAL MAPPINGS (CONTINUED)

§ 1. SCHWARZ-CHRISTOFFEL FORMULA

Denote by P a bounded polygon in the w -plane, by A_k ($k = 1, 2, \dots, n$) its vertices enumerated in a positive sense with respect to the interior of P , and by $\alpha_k\pi$ its interior angles $\left(\sum_{k=1}^n \alpha_k = n - 2\right)$. The function $w = f(z)$ that maps the upper half-plane $\text{Im } z > 0$ onto the interior of P is defined by the Schwarz-Christoffel formula

$$w = f(z) = C \int_0^z \prod_{k=1}^n (z - a_k)^{\alpha_k - 1} dz + C_1, \quad (1)$$

where $-\infty < a_1 < a_2 < \dots < a_n < \infty$ are points on the x -axis corresponding to the vertices $A_1, A_2, \dots, A_n$ of the polygon P ; C and C_1 are complex constants.

The formula (1) includes the desired points a_k (the images of the given vertices A_k of the polygon) and the constants C and C_1 . Out of n points a_k three may be chosen arbitrarily, since they can be transformed into the three given points by a linear-fractional transformation of the upper half-plane onto itself. Determining the remaining $n - 3$ points and the complex constants C and C_1 (a total $n + 1$ real parameters) is the main difficulty when applying formula (1). In principle, the unknown parameters can be found by the following reasoning. The length l_i of the side $A_i A_{i+1}$ ($i = 1, 2, \dots, n - 1$) is equal to

$$l_i = \int_{a_i}^{a_{i+1}} |f'(x)| dx = |C| \int_{a_i}^{a_{i+1}} \prod_{k=1}^n |x - a_k|^{\alpha_k - 1} dx.$$

The length of the side $A_n A_1$ is

$$l_n = |C| \left[\int_{a_n}^{\infty} \prod_{k=1}^n |x - a_k|^{\alpha_k - 1} dx + \int_{-\infty}^{a_1} \prod_{k=1}^n |x - a_k|^{\alpha_k - 1} dx \right].$$

Forming the ratios of the length of $n-3$ sides to one of the three remaining sides, we obtain $n-3$ independent equations for determining $n-3$ points a_k . Then the function $t = \int_0^z \prod_{k=1}^n (z-a_k)^{\alpha_k-1} dz$

defines the mapping of the upper half-plane onto the polygon P' in the t -plane, which is similar to the given polygon. Now construct the linear transformation $w = Ct + C_1$ which carries P' into P .

The upper half-plane is mapped onto the exterior of the same polygon by the function

$$F(z) = C \int_0^z \prod_{k=1}^n (z-a'_k)^{\beta_k-1} \frac{dz}{(z-b)^2 (z-\bar{b})^2} + C_1, \quad (2)$$

where b is a point of the upper half-plane corresponding to the point at infinity of the w -plane; a'_k are points corresponding to the vertices A_k of the polygon (now $\infty > a'_1 > a'_2 > \dots > a'_n > -\infty$);

$\beta_k \pi$ are exterior angles of the polygon ($\beta_k = 2 - \alpha_k$, $\sum_{k=1}^n \beta_k = n + 2$).

1220. Prove that if one of the vertices of a polygon is an image of the point at infinity, say $a_n = \infty$, then the mapping function is of the form

$$f(z) = C \int_0^z \prod_{k=1}^{n-1} (z-a_k)^{\alpha_k-1} dz + C_1.$$

Hint. Carry out the transformation $\zeta = -\frac{1}{z}$ if all $a_k \neq 0$, and $\zeta = -\frac{1}{z-a}$, where $a \neq a_k$, if one of the points $a_k = 0$ ($k = 1, 2, \dots, n-1$).

1221. Prove that if one or several vertices A_k lie at infinity, then formula (1) holds true provided $\alpha_k \pi$ is understood as the angle (taken with the minus sign) between the corresponding rays at the finite point of their intersection.

Hint. Let $A_k = \infty$. If $\alpha_k < 1$, then consider the polygon P' cut from P by the segment $A'_k A''_k$, where A'_k and A''_k lie sufficiently far away on the sides $A_{k-1} A_k$ and $A_{k+1} A_k$ (Fig. 32, I), and in formula (1) for P' go to the limit as $A'_k \rightarrow \infty$, $A''_k \rightarrow \infty$. But if $\alpha_k \geq 1$, then connect $A'_k A''_k$ into P by means of a polygonal line (see Fig. 32, 2) and remove it to infinity by expanding it with similarity retained.

1222. Determine the quantities α_k in formula (1) for infinitely removed vertices of "polygons" formed by parallel rays (see Fig. 33).

Hint. Pass to the limit as suggested in the hint to Problem 1221.

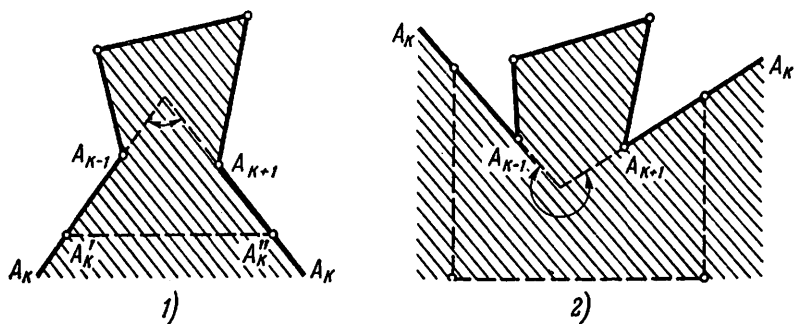


FIG. 32

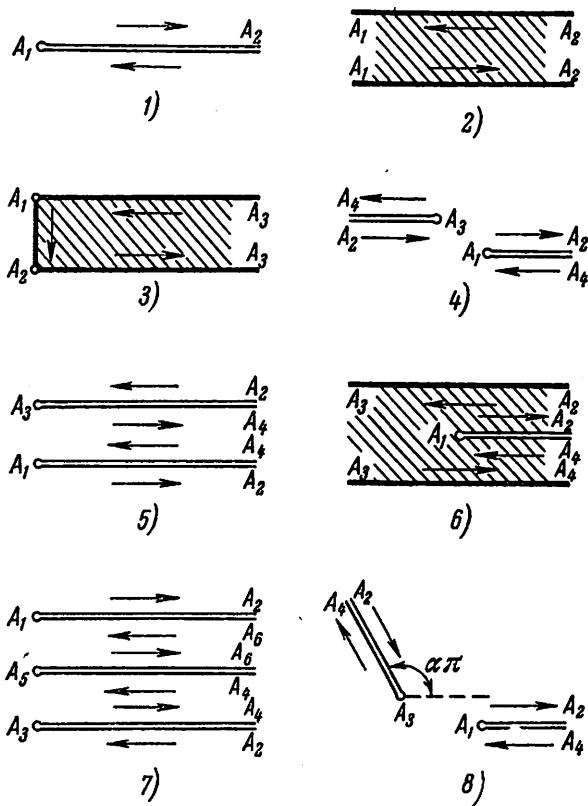


FIG. 33

1223. 1) Prove that when the unit circle $|z| < 1$ is mapped onto a polygon P located in a finite part of the plane, the mapping function has the form

$$f(z) = C \int_0^z \prod_{k=1}^n (z - a_k)^{\alpha_k - 1} dz + C_1,$$

where $a_k = e^{i\varphi_k}$ ($\varphi_1 < \varphi_2 < \dots < \varphi_n$) are points on the circle $|z| = 1$ corresponding to the vertices A_k traversed in the positive sense, and $\alpha_k \pi$ are interior angles of the polygon P .

(2) Prove that the function that maps the unit circle $|z| < 1$ onto the exterior of the same polygon P (provided the point $z = 0$ goes into the point $w = \infty$) has the form

$$f(z) = C \int_1^z \prod_{k=1}^n (z - a'_k)^{\beta_k - 1} \frac{dz}{z^2},$$

where $a'_k = e^{i\varphi'_k}$ ($\varphi'_1 > \varphi'_2 > \dots > \varphi'_n$) and $\beta_k \pi$ are exterior angles of P .

1224. Find all the cases of single-valued transformation of the Schwarz-Christoffel formula (1); that is, determine for which polygons P the inverse function $z = z(w)$ is defined and single-valued in the whole w -plane.

Hint. The polygons obtained from P by any even number of mirror reflections, with respect to the sides must cover the whole w -plane without gaps or overlappings.

In Problems 1225 to 1227 map the upper half-plane $\text{Im } z > 0$ onto the indicated domains of P located in the w -plane with the given correspondence between the vertices of P and points of the real axis. Also determine the period or periods of the inverse function $z(w)$, the group G of its invariant linear transformations of the w -plane and the fundamental domain B of this group (see page 39).

1225. 1) P a strip $0 < v < h$; $w(-\infty, \infty) \rightarrow z(0, \infty)$ ¹⁾.

2) P a strip $0 < v < h$; $w(-\infty, \infty) \rightarrow z(-1, 1)$.

1226. P a semistriple $0 < u < \pi$, $v < 0$; $w(0, \pi, -i\infty) \rightarrow z(1, -1, \infty)$.

1227. 1) P a right-angled triangle with acute angles

$$\frac{\pi}{3}, \frac{\pi}{6}; w\left(0, \omega, \omega + \frac{i\omega}{\sqrt{3}}\right) \rightarrow z(0, 1, \infty).$$

2) P a right isosceles triangle

$$w(0, \omega, \omega + i\omega) \rightarrow z(0, 1, \infty).$$

¹⁾ Here and elsewhere $w(A, B, \dots) \rightarrow z(a, b, \dots)$ denotes correspondence between the points of the w - and z -planes: $A \leftrightarrow a$, $B \leftrightarrow b$, ...

3) P an equilateral triangle

$$w(0, \omega, \omega \frac{1+i\sqrt{3}}{2}) \rightarrow z(0, 1, \infty).$$

1228. Find the domains of the w -plane onto which the function

$$w(z) = \int_0^z t^{\alpha-1} (1-t)^{\beta-1} dt$$

$$(\arg w'(x) = \theta, \text{ if } 0 < x < 1)$$

maps the upper half-plane $\text{Im } z > 0$, given:

1) $0 < \alpha \leq 2$, $0 < \beta \leq 2$; consider the cases: a) $\alpha + \beta < 1$, b) $\alpha + \beta = 1$, c) $\alpha + \beta > 1$, in particular $\alpha + \beta = 2$ and $\alpha = \beta = \frac{3}{2}$.

2) $1 \leq \alpha \leq 2$, $-1 \leq \beta \leq 0$, $\alpha + \beta \geq 1$; consider the cases:

a) $\alpha = 1$, $\beta = 0$, b) $\alpha + \beta = 1$, c) $\alpha = 2$, d) $\alpha = 2$, $\beta = -\frac{1}{2}$,

e) $\alpha = 2$, $\beta = -1$.

1229. 1) Find the domains of the w -plane onto which the function

$$w(z) = \int_0^z \frac{t-\lambda}{\sqrt{t(t-1)}} dt = \sqrt{z(z-1)} + (1-2\lambda) \left[\ln(\sqrt{z} + \sqrt{z-1}) - \frac{i\pi}{2} \right]$$

maps the upper half-plane. Consider the cases: $\lambda < 0$, $0 < \lambda < \frac{1}{2}$, $\lambda = \frac{1}{2}$; $\frac{1}{2} < \lambda < 1$ and $\lambda > 1$.

2) Find the domains of the w -plane onto which the function

$$w(z) = \frac{1}{2\pi i} \int_1^z \frac{t-\lambda}{\sqrt{t(t-1)}} \frac{dt}{t} = \frac{1}{\pi i} \left[\ln(\sqrt{z} + \sqrt{z-1}) - \lambda \sqrt{\frac{z-1}{z}} \right]$$

maps the upper half-plane. Consider the cases: $\lambda < 0$, $0 < \lambda < 1$ and $\lambda > 1$.

1230. Map the upper half-plane $\text{Im } z > 0$ onto the domains in the w -plane shown in Fig. 34 with the given correspondence of the points.

1) $w(A=0, B=1, C=\infty) \rightarrow z(0, 1, \infty)$;

2) $w(A=0, B=1, C=\infty) \rightarrow z(0, 1, \infty)$;

3) $w(A=0, B=\infty, C=\infty) \rightarrow z(0, 1, \infty)$;

4) $w(A=0, B=\infty, C=\infty) \rightarrow z(0, 1, \infty)$;

5) $w(A=0, B=ia, C=\infty) \rightarrow z(0, 1, \infty)$.

1231. Map the upper half-plane $\text{Im } z > 0$ on the domains in the w -plane shown in Fig. 35 ($0 < \theta < 1$).

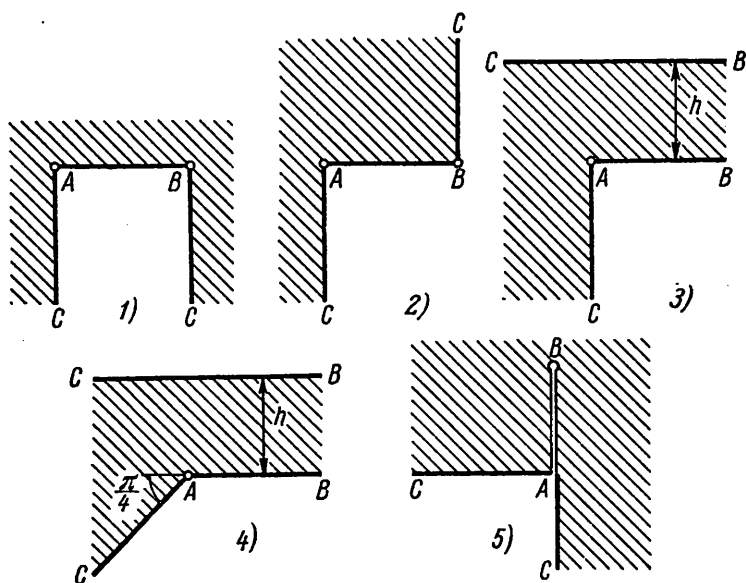


FIG. 34

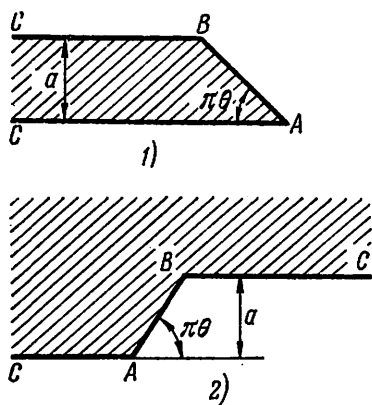


FIG. 35

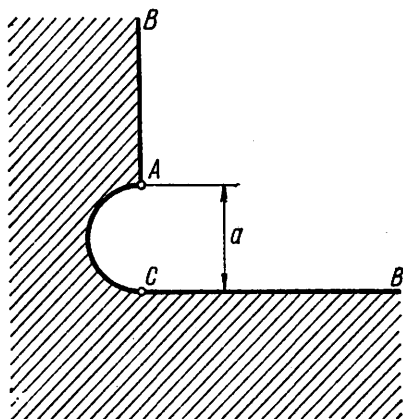


FIG. 36

1) $w(A, B=0, C=\infty) \rightarrow z(0, 1, \infty)$;

2) $w(A, B=0, C=\infty) \rightarrow z(0, 1, \infty)$.

If θ is a rational number ($\theta = p/q$), express the resulting integrals in terms of elementary functions.

1232. Map the upper half-plane onto the domain indicated in Fig. 36 (the arc AC is a semicircle)

$$w(A=ai, B=\infty, C=0) \rightarrow z(0, 1, \infty).$$

Hint. The mapping $\zeta = \frac{a}{w}$ reduces the problem to the particular case of Problem 1231(2).

1233. Map the upper half-plane $\text{Im } z > 0$ onto the domain of the w -plane shown in Fig. 37, given that

$$w(A=-h, B=\infty, C=h, D=\infty) \rightarrow z(-1, 0, 1, \infty).$$

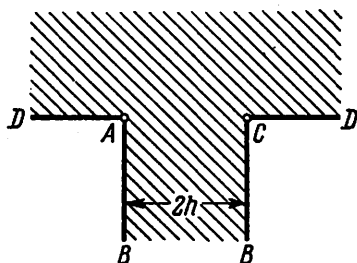


FIG. 37

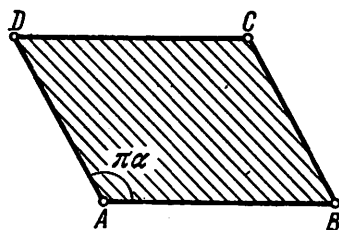


FIG. 38

(The possibility of this mapping follows from the symmetry principle.)

1234. Map the upper half-plane $\text{Im } z > 0$ onto a rhombus in the w -plane with angle $\pi \alpha$ at the vertex A and side d (Fig. 38).

The correspondence of the points is given by $w(A=0, B=d, C=d(1+e^{i\pi\alpha}), D=de^{i\pi\alpha}) \rightarrow z(0, 1, \infty, -1)$. Justify the possibility of such mapping.

1235. Map the upper half-plane $\text{Im } z > 0$ onto the domains of the w -plane shown in Fig. 39. The parameters a and b ($a > 0, b > 0$)—preimages of the corresponding vertices—cannot be specified arbitrarily and are to be found.

1) $w(A=0, B=\infty, C=H+ih, D=\infty) \rightarrow z(0, 1, a, \infty)$.

2) $w(B=\infty, C=H+ih, D=\infty, C'=-H+ih, B'=\infty) \rightarrow z(1, a, \infty, -a, -1)$.

Hint. Make an additional cut along the imaginary axis and use the symmetry principle.

3) $w(A=0, B=\infty, C=\infty, D=-h_3-i(h_2-h_1), E=\infty) \rightarrow z(0, 1, a, \infty, -b)$.

1236. 1) Find the function $w(z)$ that maps the circle $|z| < 1$ onto the interior of a regular n -gon with centre at the origin and one of the vertices at the point $w=1$, given that $w(0)=0$, $w'(0) > 0$.

2) Map the circle $|z| < 1$ onto the exterior of the same n -gon, given that $w(0)=\infty$, $w(x) > 0$ ($0 < x < 1$). Determine c_{-1} in the expansion $w(z) = \frac{c_{-1}}{z} + \dots$.

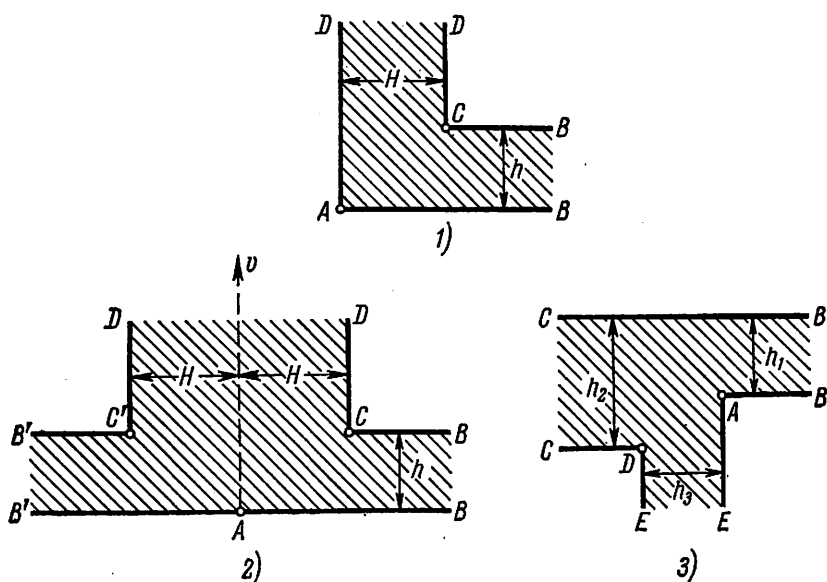


FIG. 39

1237. 1) Find the function $w(z)$ that maps the circle $|z| < 1$ onto the interior of a regular five-pointed star with centre at the origin and with one of the vertices at the point $w=1$; normalization conditions: $w(0)=0$, $w'(0) > 0$.

2) Map the circle $|z| < 1$ onto the exterior of the same star; $w(0)=\infty$, $w(x) > 0$ ($0 < x < 1$); determine c_{-1} in the expansion $w(z) = \frac{c_{-1}}{z} + \dots$.

1238. Find the domain onto which the function

$$w(z) = \int_0^z \frac{(1+t^n)^\lambda dt}{(1-t^n)^{\frac{2}{n}+\lambda}}, \quad w'(0) > 0, -1 < \lambda < 1 - \frac{2}{n}.$$

maps the unit circle $|z| < 1$.

1239. Map the upper half-plane $\text{Im } z > 0$ onto a regular n -gon in the w -plane with centre at the origin and with one of the vertices at the point $w = 1$; normalization conditions: $w(i) = 0$, $w(0) = 1$.

1240. Let the domain P in the w -plane be the exterior of a "star" consisting of n segments emanating from the point $w = 0$ (Fig. 40). Let A_k be the vertices of P at the origin and $\alpha_k \pi$ the corresponding angles $\left(\sum_{k=1}^n \alpha_k = 2\right)$, B_k the vertices of P at the ends

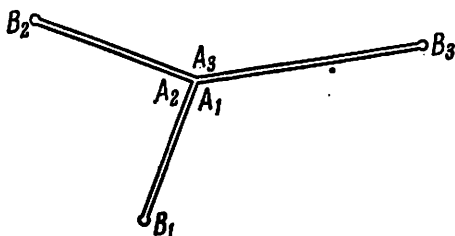


FIG. 40

of the segments of the star ($A_1, B_1, A_2, B_2, \dots$ are enumerated in a positive sense around P), and $l_k = \overline{A_k B_k}$ the lengths of the segments of the star. Prove that the function $w = f(z)$, $f(0) = \infty$ that maps the unit circle $|z| < 1$ onto P is of the form

$$f(z) = \frac{C}{z} \prod_{k=1}^n (z - a_k)^{\alpha_k},$$

where C is a complex constant and $a_k = e^{i\varphi_k}$ are points on the circle $|z| = 1$ corresponding to A_k . The points $b_k = e^{i\psi_k}$ corresponding to the vertices B_k are roots of the equation

$$\sum_{k=1}^n \frac{\alpha_k}{z - a_k} - \frac{1}{z} = 0. \quad (3)$$

How can the parameters C , a_k and b_k be determined?

Hint. When continued according to the symmetry principle the function $f(z)$ is multiplied by constant factors; therefore the function $\varphi(z) = \frac{f'(z)}{f(z)}$ is single-valued in the z -plane.

Note. The formula obtained at once provides solutions to a number of problems of Chapter II, for instance, 320(2); 321(1); 323; 329.

It is advisable to solve the indicated problems anew and determine the constants in the general formula.

1241. Map the exterior of the two-segment polygonal line in Fig. 41 onto the exterior of the circle $|z| > 1$, given that $\omega(\infty) = \infty$, $\omega'(\infty) > 0$.

1242. Prove that the function $\omega = f(z)$ that maps the upper half-plane $\text{Im } z > 0$ onto the exterior of the star in Problem 1240 has the form

$$f(z) = C \frac{\prod_{k=1}^n (z - a_k)^{\alpha_k}}{(z - z_0)(z - \bar{z}_0)},$$

where z_0 is a point of the upper half-plane that goes to infinity.

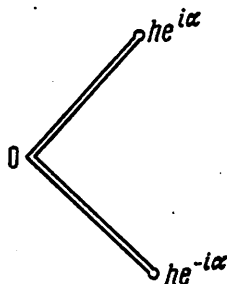


FIG. 41

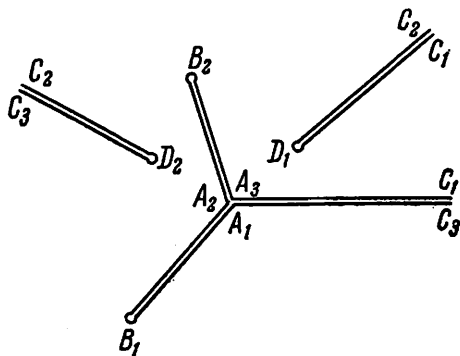


FIG. 42

1243. Let a domain P in the w -plane be bounded by the ray $[0, \infty)$, $n-1$ segments from $w=0$ to the points B_i ($i=1, 2, \dots, n-1$), and $m-1$ rays from the points D_s ($s=1, 2, \dots, m-1$) to ∞ such that their prolongations in the opposite direction meet the origin of coordinates (see Fig. 42). Let A_k ($k=1, 2, \dots, n$) be the vertices of P at the origin, $\alpha_k\pi$ the corresponding angles, C_j ($j=1, 2, \dots, m$) the vertices of P at ∞ , $\gamma_j\pi$ the corresponding angles, and let $A_1B_1A_2\dots A_nC_1D_1\dots C_m$ represent a traversal of the boundary of P in the positive sense. Prove that the function $w=f(z)$ that maps the upper half-plane $\text{Im } z > 0$ onto P is of the form

$$f(z) = C \frac{\prod_{k=1}^n (z - a_k)^{\alpha_k}}{\prod_{j=1}^m (z - c_j)^{\gamma_j}}, \quad (4)$$

where a_k and c_j are points on the x -axis corresponding to the vertices A_k and C_j . The points b_i and d_s on the x -axis corresponding

to the vertices B_i and D_s are roots of the equation

$$\sum_{k=1}^n \frac{\alpha_k}{z-a_k} - \sum_{j=1}^m \frac{\gamma_j}{z-c_j} = 0. \quad (5)$$

How can the parameters C , a_k , b_i , c_j and d_s be determined? What will change if one of the parameters a_k , b_i , c_j or d_s is equal to ∞ ?

Hint. Prove that

$$\frac{f'(z)}{f(z)} = \sum_{k=1}^n \frac{\alpha_k}{z-a_k} - \sum_{j=1}^m \frac{\gamma_j}{z-c_j} = \frac{\prod_{i=1}^{n-1} (z-b_i) \prod_{s=1}^{m-1} (z-d_s)}{\prod_{k=1}^n (z-a_k) \prod_{j=1}^m (z-c_j)}.$$

1244. Prove that formula (4) in Problem 1243 is also valid for the domains of P with a boundary containing two rays $[0, \infty)$ (Fig. 43). The vertex of the origin of coordinates is taken into account also when only two rays forming one straight line emanate from it.

In Problems 1245 to 1249 map onto the upper half-plane $\text{Im } z > 0$ the domains of the w -plane shown in the accompanying figures and determine the parameters a and b ($a > 0$, $b > 0$).

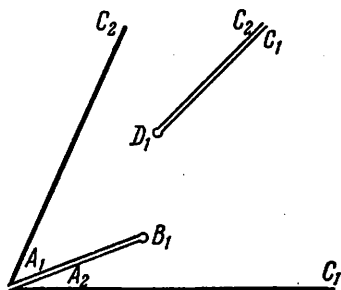


FIG. 43

1245. The domain shown in Fig. 44.

$$w(A_1=0, B=he^{i\pi\alpha}, A_2, C=\infty) \rightarrow z(-1, b, 1, \infty).$$

1246. The domain shown in Fig. 45,

$$w(A_1=0, B=ih, A_2, C=\infty, D=iH, C_2=\infty) \rightarrow z\left(-a, 0, a, \frac{1}{a}, \infty, -\frac{1}{a}\right).$$

1247. The domain shown in Fig. 46,

$$w(A_1=0, B_1=-he^{-i\pi\alpha}, A_2, B_2=he^{i\pi\alpha}, A_3, C=\infty) \rightarrow z(-1, -b, 0, b, 1, \infty).$$

1248. The domain shown in Fig. 47,

$$w(A_1=0, B=he^{i\pi\alpha}, A_2, C_1=\infty, D=He^{i\pi\alpha}, C_2=\infty) \rightarrow z\left(-a, 0, b, \frac{1}{b}, \infty, -\frac{1}{a}\right).$$

1249. 1) The domain shown in Fig. 48, 1,

$$w(\infty)=(\infty), w(\pm 1)=\pm 1.$$

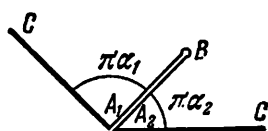


FIG. 44

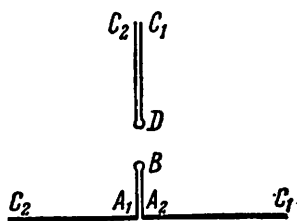


FIG. 45.



FIG. 46

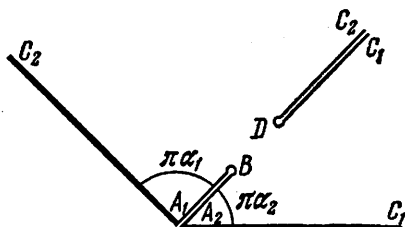


FIG. 47

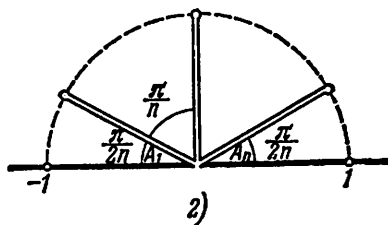
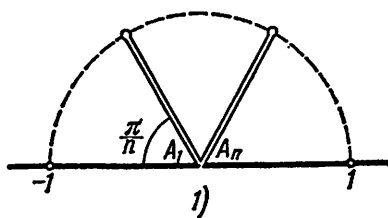


FIG. 48

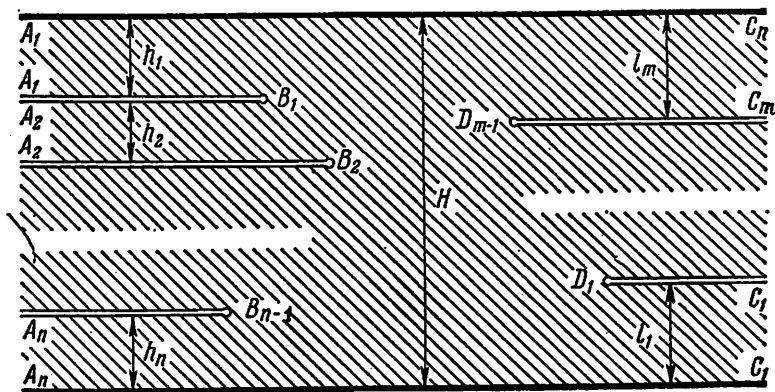


FIG. 49

2) The domain shown in Fig. 48,2 (the angles between the cuts $\frac{\pi}{n}$; the extreme segments form the angles $\frac{\pi}{2n}$ with the corresponding rays of the real axis),

$$\omega(\infty) = \infty, \quad \omega(A_1, A_n) \rightarrow z(-1, 1).$$

1250. Let the domain P in the w -plane be a horizontal strip of width H with cuts going leftwards to ∞ from the points B_i ($i=1, 2, \dots, n-1$) and rightwards from the points D_s ($s=1, 2, \dots, m-1$) (Fig. 49). Prove that the function $w=f(z)$ which maps the upper half-plane $\text{Im } z > 0$ onto the domain P has the form

$$\bar{f}(z) = \sum_{k=1}^n \frac{h_k}{\pi} \ln(z - a_k) - \sum_{j=1}^m \frac{l_j}{\pi} \ln(z - c_j) + C, \quad (6)$$

where a_k and c_j are points on the x -axis corresponding to the vertices A_k and C_j of the domain P , h_k are the distances between the cuts going leftwards, and l_j those going rightwards. The points b_i ($i=1, 2, \dots, n-1$) and d_s ($s=1, 2, \dots, m-1$) on the x -axis corresponding to the vertices B_i and D_s are roots of the equation

$$\sum_{k=1}^n \frac{h_k}{z - a_k} - \sum_{j=1}^m \frac{l_j}{z - c_j} = 0. \quad (7)$$

How can the parameters C , a_k , b_i , c_j and d_s be determined? What will change if one of the parameters a_k , b_i , c_j or d_s is equal to ∞ ?

Hint. The problem reduces to Problem 1244. It is also possible to find the function $f'(z)$, which is single-valued, or to proceed directly from the Schwarz-Christoffel formula.

1251. Let the domain P be the upper half-plane $\text{Im } w > 0$ with horizontal cuts going leftwards to ∞ from the points B_i ($i=1, 2, \dots, n$) and rightwards to ∞ from the points D_s ($s=1, 2, \dots, m$) (Fig. 50). Prove that the function $w=f(z)$ which maps the upper half-plane $\text{Im } z > 0$ onto the domain P has the form

$$f(z) = \sum_{k=1}^n \frac{h_k}{\pi} \ln(z - a_k) - \sum_{j=1}^m \frac{l_j}{\pi} \ln(z - c_j) + Az + B, \quad (8)$$

where a_k and c_j are points on the x -axis corresponding to the vertices A_k and C_j of the domain P , and h_k and l_j are the distances between cuts in one direction. The point A_0 goes to ∞ , $A > 0$ and

$\text{Im } B = \sum_{j=1}^m l_j$. The points b_i ($i=1, 2, \dots, n$) and d_s ($s=1, 2, \dots, m$) on the x -axis corresponding to the vertices B_i and D_s are roots of

the equation

$$\sum_{k=1}^n \frac{h_k}{z-a_k} - \sum_{j=1}^m \frac{l_j}{z-c_j} + A\pi = 0. \quad (9)$$

How can the parameters A , B , a_k , b_i , c_j and d_s be determined? Show that if A_0 goes into a point $a_0 \neq \infty$, then

$$f(z) = \sum_{k=1}^n \frac{h_k}{\pi} \ln(z-a_k) - \sum_{j=1}^m \frac{l_j}{\pi} \ln(z-c_j) + \frac{L-H}{\pi} \ln(z-a_0) + \frac{A}{z-a_0} + B, \quad (10)$$

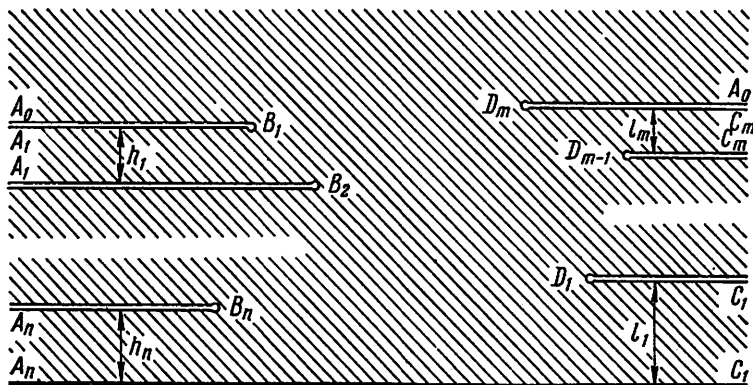


FIG. 50

where

$$H = \sum_{k=1}^n h_k, \quad L = \sum_{j=1}^m l_j, \\ A < 0 \text{ and } \operatorname{Im} B = L.$$

If the parameter a_k or c_j is equal to ∞ , then the corresponding summand in formula (10) drops out.

Hint. Use the Schwarz-Christoffel formula. To determine the coefficients of the logarithmic terms compare the increments of w in going round (along semicircles) the points a_0 , a_k and c_j calculated geometrically and by the formula for $f(z)$.

1252. Let the domain P be the w -plane with horizontal cuts leftwards to infinity from the points B_i ($i=1, 2, \dots, n$) (Fig. 51). Prove that the function $w=f(z)$ which maps the upper half-plane

$\text{Im } z > 0$ onto the domain P has the form

$$f(z) = -Az^2 + Bz + C + \sum_{k=1}^{n-1} \frac{h_k}{\pi} \ln(z - a_k),$$

where a_k are points on the x -axis corresponding to the vertices A_k of the domain P , and h_k are the distances between the cuts. Here the point A_0 goes to ∞ , $A > 0$, and B is a real number. The points b_i ($i = 1, 2, \dots, n$) corresponding to the vertices B_i are zeros of the derivative $f'(z)$. How can the parameters A , B , C , a_k and b_i be determined?

Hint. See the hint to Problem 1251.

1253. Let the domain P be the w -plane with horizontal cuts leftwards and rightwards to ∞ (Fig. 52). Prove that the function $\omega = f(z)$ which maps the upper half-plane $\text{Im } z > 0$ onto the domain P is of the form

$$f(z) = \sum_{k=0}^{n-1} \frac{h_k}{\pi} \ln(z - a_k) - \sum_{j=0}^{m-1} \frac{l_j}{\pi} \ln(z - c_j) + \frac{A}{z - a_0} + \frac{C}{z - c_0} + B, \quad (11)$$

where $A > 0$, $C > 0$, $h_0 = \text{Im}(D_{\infty} - B_1)$, $l_0 = \text{Re}(D_1 - B_n)$, the remaining parameters (including b_i and d_s for the vertices B_i and D_s) having the same values as in Problem 1251. How can the parameters A , B , C , a_k , b_i , c_j and d_s be determined? Show that if $a_0 = \infty$, then

$$f(z) = \sum_{k=1}^{n-1} \frac{h_k}{\pi} \ln(z - a_k) - \sum_{j=0}^{m-1} \frac{l_j}{\pi} \ln(z - c_j) - \frac{C}{z - c_0} + Az + B, \quad (12)$$

where $A > 0$ and $C > 0$. If the parameter a_k ($k \neq 0$) or c_j ($j \neq 0$) is equal to ∞ , then the corresponding summand in formula (12) drops out.

Hint. See the hint to Problem 1251.

1254. Map the upper half-plane $\text{Im } z > 0$ onto the domains in the w -plane shown in Fig. 53 (all dimensions are given in the appropriate figures) for the given conditions; find a and b ($a > 0, b > 0$).

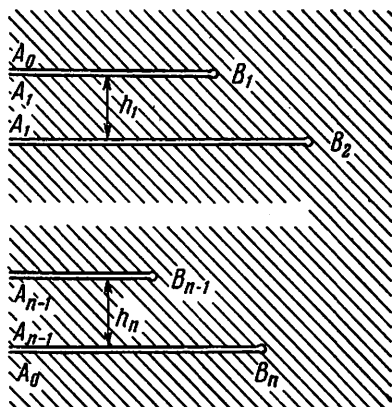


FIG. 51

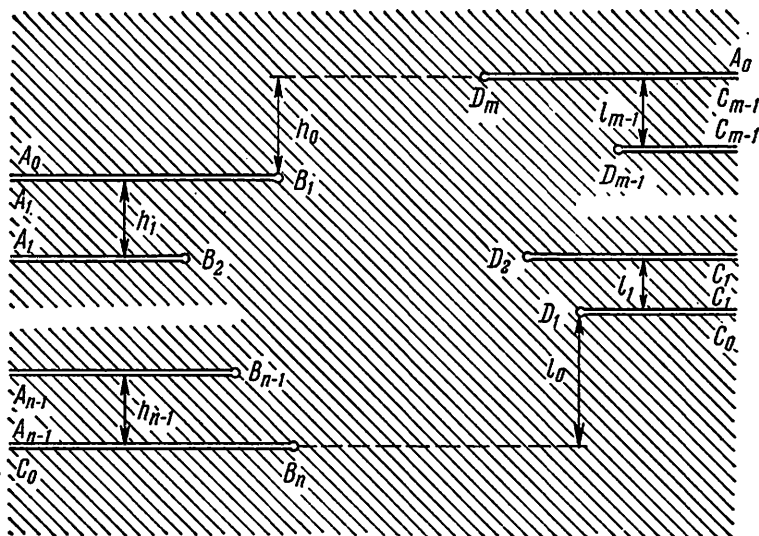


FIG. 52

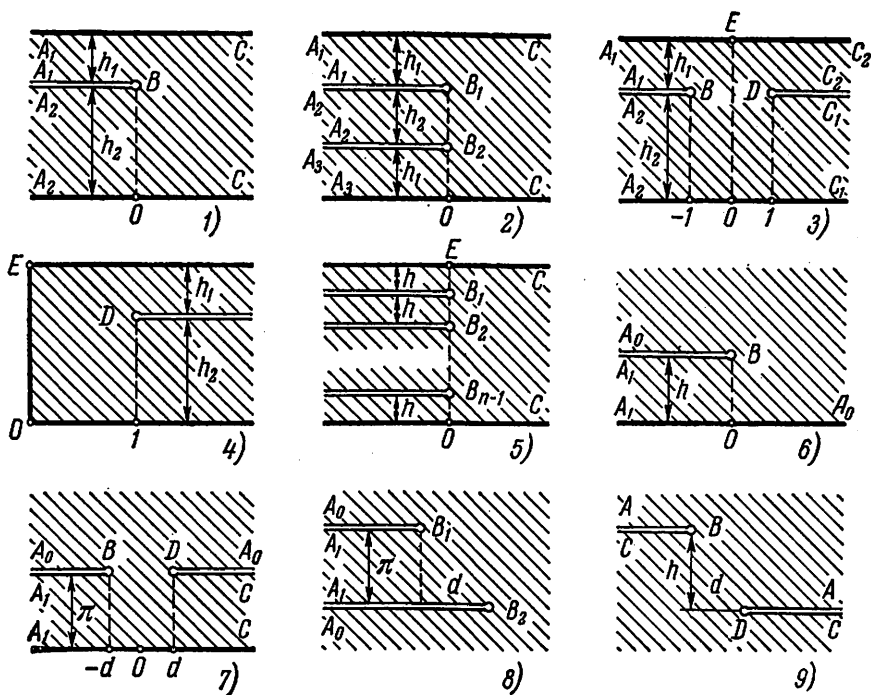


FIG. 53

- 1) $(A_1, B, A_2, C) \rightarrow (-1, b, 1, \infty)$;
- 2) $(A_1, B_1, A_2, B_2, A_3, C) \rightarrow (-1, -b, 0, b, 1, \infty)$;
- 3) $(E, O, D) \rightarrow (-1, 1, \infty)$; 4) $(E, O, D) \rightarrow (-1, 1, \infty)$;
- 5) $(E, O, C) \rightarrow (-1, 1, \infty)$; 6) $(A_0, B, A_1) \rightarrow (\infty, 0, 1)$;
- 7) $(A_1, O, C) \rightarrow (-1, 0, 1)$;
- 8) $(A_0, B_1, A_1, B_2) \rightarrow (\infty, -(1+a), -1, 0)$; $d = \operatorname{Re} (B_2 - B_1)$;
- 9) $(A, B, C, D) \rightarrow (\infty, -1, a, 1)$, $d = \operatorname{Re} (D - B)$; $h = \operatorname{Im} (B - D)$.

§ 2. CONFORMAL MAPPINGS INVOLVING THE USE OF THE ELLIPTIC FUNCTIONS

The integral

$$u(z, k) = \int_0^z \frac{dt}{V(1-t^2)(1-k^2t^2)}, \quad (1)$$

where the integrand is equal to unity for $t=0$, is called the *normal elliptic integral of the first kind in the Legendre form*. The parameter k is called the *modulus*; it is assumed henceforward that $0 < k < 1$.

Replacement of the independent variable $z = \sin \varphi$ and the substitution $t = \sin \psi$ bring this integral to the form

$$u(\sin \varphi, k) = F(\varphi, k) = \int_0^\varphi \frac{d\psi}{V1-k^2 \sin^2 \psi}. \quad (2)$$

The inverse function

$$z = \operatorname{sn}(u, k) \quad (3)$$

(or, in other notation, $\varphi = \operatorname{am} u$) is one of the fundamental *Jacobian elliptic functions* and is called the *Jacobian "sn"-function*. From the definition it follows that $\operatorname{sn}(0, k) = 0$. Related to the function $\operatorname{sn}(u, k)$ are two other functions

$$\operatorname{cn}(u, k) = V1 - \operatorname{sn}^2(u, k), \quad \operatorname{dn}(u, k) = V1 - k^2 \operatorname{sn}^2(u, k), \quad (4)$$

which are called the *Jacobian "cn"- and "dn"-functions*, respectively. The branches of the roots are defined by the conditions $\operatorname{cn}(0, k) = \operatorname{dn}(0, k) = 1$. If there is no need to indicate the modulus k , then we simply write $\operatorname{sn} u$, $\operatorname{cn} u$, $\operatorname{dn} u$.

It follows from (1) to (4) that

$$\left. \begin{aligned} \frac{d(\operatorname{sn} u)}{du} &= \operatorname{cn} u \operatorname{dn} u, & \frac{d(\operatorname{cn} u)}{du} &= -\operatorname{sn} u \operatorname{dn} u, \\ \frac{d(\operatorname{dn} u)}{du} &= -k^2 \operatorname{sn} u \operatorname{cn} u. \end{aligned} \right\} \quad (5)$$

The value of the function $u(z, k)$ for $z = i \left(\varphi = \frac{\pi}{2} \right)$, i.e. the integral

$$\int_0^1 \frac{dt}{V(1-t^2)(1-k^2t^2)} = K(k)$$

is called the *complete elliptic integral of the first kind*. The quantity $k' = \sqrt{1-k^2}$ is called the *complementary modulus*.

$K(k) = K$ and $K(k') = K'$ are called *bound elliptic integrals*.

In problems, use is often made of the following easily proved relations (transformations of complete elliptic integrals of the first kind):

$$\begin{aligned} K\left(\frac{1}{k}\right) &= k(K + iK'), & K\left(\frac{1}{k'}\right) &= k'(K' + iK), \\ K\left(\frac{ik}{k'}\right) &= k'K, & K\left(\frac{ik'}{k}\right) &= kK'. \end{aligned} \quad (6)$$

Also note that

$$\begin{aligned} \int_1^{1/k} \frac{dt}{V(t^2-1)(1-k^2t^2)} &= K' \quad (\text{substitution } k^2t^2 + k'^2\tau^2 = 1), \\ \int_0^\infty \frac{dt}{V(1+t^2)(1+k^2t^2)} &= K' \quad (\text{substitution } t = \tan \psi). \end{aligned}$$

1255. Prove that under a mapping by the normal elliptic integral of the first kind

$$u = \int_0^z \frac{dt}{V(1-t^2)(1-k^2t^2)},$$

the image of the upper half-plane $\text{Im } z > 0$ is a rectangle with vertices $\pm K$, $\pm K + iK'$ corresponding to points ± 1 , $\pm \frac{1}{k}$.

Continuing the mapping according to the symmetry principle, prove that the inverse function $z = \text{sn } u$ is a doubly periodic function with periods $4K$ and $2iK'$.

Consider the correspondence between the planes u and φ , where $u = F(\varphi, k)$.

Hint. Apply the principle of correspondence of boundaries. Special attention should be given to the variation of the argument $\frac{du}{dz}$ when z ranges over the real axis.

1256. Map the upper half-plane $\text{Im } z > 0$ onto a rectangle in the w -plane with vertices $\pm a$, $\pm a + ib$ ($a > 0$, $b > 0$).

1257. Find the image of the fourth quadrant of the z -plane under a mapping by the elliptic integral

$$u = \int_z^1 \frac{dt}{\sqrt{(1-t^2)(k'^2 + k^2 t^2)}}.$$

Continuing the mapping according to the symmetry principle, find the image of the entire z -plane with cuts along the rays $(-\infty, -1]$, $[1, \infty)$, $(-i\infty, -\frac{ik'}{k}]$, $[\frac{ik'}{k}, i\infty)$. Verify that the inverse function is $z = \operatorname{cn}(u, k)$ and prove that it is a doubly periodic function with periods $4K$, $2K + 2iK'$.

1258. Find the image of the fourth quadrant of the z -plane under a mapping by the elliptic integral

$$u = \int_z^1 \frac{dt}{\sqrt{(1-t^2)(t^2 - k'^2)}}.$$

Continuing the mapping according to the symmetry principle, find the image of the entire z -plane with cuts along the rays $(-\infty, -k']$, $[k', \infty)$. Verify that the inverse function is $z = \operatorname{dn}(u, k)$ and prove that it is a doubly periodic function with periods $2K$ and $4iK'$.

1259. Find the image of the first quadrant of the z -plane under a mapping by the elliptic integral of the first kind

$$u(z, k_1) = \int_0^z \frac{dt}{\sqrt{(1-t^2)(1-k_1^2 t^2)}},$$

where k_1 takes the values k , $\frac{1}{k}$, k' , $\frac{1}{k'}$, $\frac{ik}{k'}$, $\frac{ik'}{k}$.

Hint. Use formula (6).

The integral

$$v(z, k) = \int_0^z \sqrt{\frac{1-k^2 t^2}{1-t^2}} dt = \int_0^\varphi \sqrt{1-k^2 \sin^2 \psi} d\psi = E(\varphi, k) \quad (7)$$

($z = \sin \varphi$) is called the *normal elliptic integral of the second kind in the Legendre form*.

For $z = 1$ ($\varphi = \pi/2$) we obtain a *complete elliptic integral of the second kind*

$$E(k) = E = \int_0^1 \sqrt{\frac{1-k^2 t^2}{1-t^2}} dt. \quad (8)$$

Denoting $E(k') = E'$, it is easy to obtain the following relations

$$\begin{aligned} E\left(\frac{1}{k}\right) &= \frac{1}{k} [(E - k'^2 K) - i(E' - k'^2 K')], \\ E\left(\frac{1}{k'}\right) &= \frac{1}{k'} [(E' - k^2 K') - i(E - k'^2 K)], \\ E\left(\frac{ik}{k'}\right) &= \frac{1}{k'} E, \quad E\left(\frac{ik'}{k}\right) = \frac{1}{k} E'. \end{aligned} \quad (9)$$

Complete elliptic integrals of the first and second kinds corresponding to the complementary moduli k and k' are connected by the *Legendre relation*

$$EK' + E'K - KK' = \frac{\pi}{2}. \quad (10)$$

Substitution of the independent variable $z = \operatorname{sn} u$ in the integral defining $v(z, k)$ leads to *Jacobi's function*

$$E(u, k) \equiv E(u) = v(\operatorname{sn} u, k) = \int_0^u \operatorname{dn}^2 \tau \, d\tau, \quad (11)$$

which expresses the elliptic integral of the second kind as a function of the elliptic integral of the first kind. Related to the function $E(u)$ is the function $Z(u)$ defined by the formula

$$Z(u) = E(u) - \frac{E}{K} u. \quad (12)$$

1260. Find the image of the first quadrant of the z -plane under a mapping by the normal elliptic integral of the second kind

$$v(z, k_1) = \int_0^z \sqrt{\frac{1 - k_1^2 t^2}{1 - t^2}} \, dt,$$

where k_1 takes the values $k, \frac{1}{k}, k', \frac{1}{k'}, \frac{ik}{k'}, \frac{ik'}{k}$.

Hint. Use formula (9) and the fact that

$$\int_1^{1/k} \sqrt{\frac{1 - k^2 t^2}{1 - t^2}} \, dt = i(K' - E')$$

(this formula is obtained by substituting $k^2 t^2 + k'^2 \tau^2 = 1$).

1261.* Map the upper half-plane $\operatorname{Im} z > 0$ onto the upper half-plane $\operatorname{Im} w > 0$ with two vertical cuts along the line-segments $\operatorname{Re} w = \pm a$, $0 \leq \operatorname{Im} w \leq h$ with the normalization: $w(0) = 0$, $w(\infty) = 0$, $w'(\infty) > 0$.

Hint. Use the Schwarz-Christoffel formula in the form

$$w(z) = C \int_0^z \frac{b^2 - z^2}{\sqrt{(1 - z^2)(1 - k^2 z^2)}} \, dz = C_1 [(k^2 b^2 - 1) u(z, k) + v(z, k)],$$

where $C_1 = \frac{C}{k^2}$, $w(\pm 1) = \pm a$, $w(\pm b) = \pm a + ih$, and form equations to determine the parameters C_1 , k and b .

1262. Prove that the function

$$w = Z(u) + \frac{\operatorname{dn} u \operatorname{cn} u}{\operatorname{sn} u}$$

maps the rectangle $0 < \xi < K$, $0 < \eta < K'$ contained in the plane $u = \xi + i\eta$ onto the fourth quadrant of the w -plane with the cut $\left[-\frac{\pi i}{2K}, h - \frac{\pi i}{2K}\right]$, where $h - \frac{\pi i}{2K}$ is the image of the point $u = \xi + iK'$ for which $\frac{dw}{du} = 0$. Continuing the mapping according to the symmetry principle, show that the entire plane with the cut shown in Fig. 54 is the image of the rectangle $|\xi| < K$, $|\eta| < K'$.

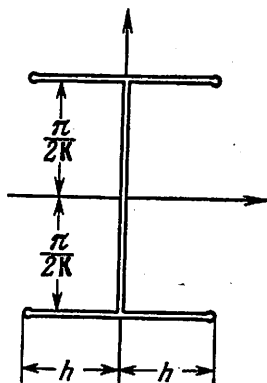


FIG. 54

The integral

$$\begin{aligned} w(z, v, k) &= \int_0^z \frac{dt}{(1+vt^2)\sqrt{(1-t^2)(1-k^2t^2)}} = \\ &= \int_0^\varphi \frac{d\varphi}{(1+v\sin^2\varphi)\sqrt{1-k^2\sin^2\varphi}} = \Pi(\varphi, v, k) \end{aligned} \quad (13)$$

is called the *normal elliptic integral of the third kind in the Legendre form*.

Substitution of the independent variable $z = \operatorname{sn} u$ leads to the formula

$$w(\operatorname{sn} u, v, k) = \int_0^u \frac{d\tau}{1+v\operatorname{sn}^2\tau}. \quad (14)$$

The quantity $\Pi_1(v, k) = w(1, v, k) = \Pi\left(\frac{\pi}{2}, v, k\right)$ is called the *complete elliptic integral of the third kind*.

1263. Find the image of the first quadrant of the z -plane under a mapping by the normal elliptic integral of the third kind (13) for $0 < k < 1$. Consider separately the cases when v belongs to the following intervals:

1) $(-\infty, -1)$; 2) $(-1, -k^2)$; 3) $(-k^2, 0)$; 4) $(0, \infty)$. Also consider the cases $v = -1$ and $v = -k^2$.

Indicate the domains in the w -plane which are obtained by continuation via the symmetry principle over different intervals of the real axis of the z -plane. In each case show the corresponding domains in the u -plane, where $z = \operatorname{sn} u$.

The integral

$$w = \int_z^{\infty} \frac{dz}{\sqrt{4z^3 - g_2z - g_3}} = \int_z^{\infty} \frac{dz}{\sqrt{4(z-e_1)(z-e_2)(z-e_3)}} \quad (15)$$

with the discriminant $\Delta = g_2^3 - 27g_3^2 \neq 0$ (under this condition, e_1, e_2, e_3 are pairwise distinct) is called the *normal elliptic integral of the first kind in the Weierstrass form*, and the function

$$z = \wp(w) \quad (16)$$

is called the *Weierstrassian "p"-function*. This is one of the principal elliptic functions with periods $2\omega, 2\omega'$ ($\operatorname{Im} \frac{\omega'}{\omega} \neq 0$). It sa-

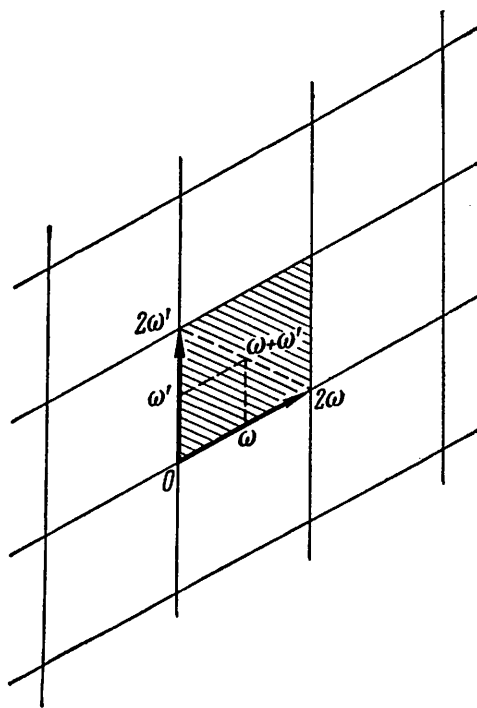


FIG. 55

tifies the differential equation

$$p'^2 = 4p^3 - g_2p - g_3 = 4(p - e_1)(p - e_2)(p - e_3). \quad (17)$$

$p(w)$ is an even, two-sheeted function in the parallelogram of periods (Fig. 55) and has, there, a second-order pole at zero and double points ($p' = 0$), ω , $\omega + \omega'$, ω' :

$$\left. \begin{aligned} e_1 &= p(\omega), \quad e_2 = p(\omega + \omega'), \\ e_3 &= p(\omega'). \end{aligned} \right\} \quad (18)$$

It follows from (17) that

$$\left. \begin{aligned} e_1 + e_2 + e_3 &= 0, \\ e_1e_2 + e_2e_3 + e_3e_1 &= -\frac{1}{4}g_2, \\ e_1e_2e_3 &= \frac{1}{4}g_3, \end{aligned} \right\} \quad (19)$$

1264.* Investigate the mapping of the z -plane with the aid of the normal elliptic integral of the first kind in the Weierstrass form (15) for real g_2 , g_3 , and $\Delta > 0$. Consider the cases $g_3 > 0$, $g_3 < 0$, $g_3 = 0$. Find the periods of $p(w)$.

Hint. Consider the mapping of the upper half-plane $\text{Im } z > 0$ by means of the principle of correspondence of boundaries.

1265.* Investigate the mapping of the z -plane with the aid of the normal elliptic integral of the first kind (15) for real g_2 , g_3 , and $\Delta < 0$. Consider, in particular, the case $g_2 = 0$. Find the periods of $p(w)$.

Hint. Since $\Delta < 0$, two of the quantities e_1 , e_2 , e_3 are complex conjugate and one is real. Let e_2 be real, $e_1 = \alpha + i\beta$, $e_3 = \alpha - i\beta$ ($\beta > 0$). Consider the mapping of the semicircle $|z - e_2| = |e_1 - e_2|$, $\text{Im } z > 0$ with the aid of the principle of correspondence of boundaries and continue the mapping according to the symmetry principle.

1266. Find the mappings of the triangles ABC onto the upper half-plane $\text{Im } w > 0$, given:

$$1) (A = 0, B = \omega > 0, C = \omega(1 + i)) \rightarrow (\infty, -1, 0);$$

$$2) \left(A = 0, B = a > 0, C = \frac{a\sqrt{3}}{2} e^{i\frac{\pi}{3}} \right) \rightarrow (\infty, -1, 1);$$

$$3) \left(A = 0, B = a > 0, C = \frac{a\sqrt{3}}{2} e^{i\frac{\pi}{6}} \right) \rightarrow (\infty, -1, 0).$$

Hint. Use the solutions to Problems 1264 (the case $g_3 = 0$) and 1265 (the case $g_2 = 0$).

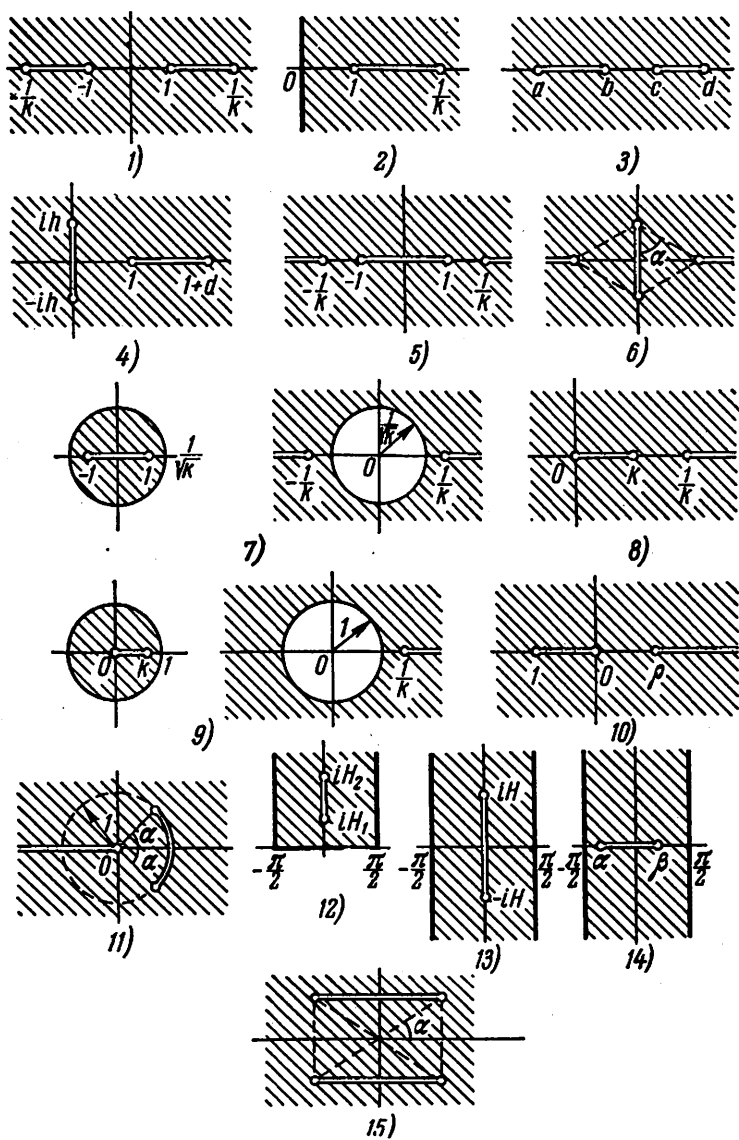


FIG. 56

1267. Map the doubly connected domains 1 to 15 situated in the z -plane (Fig. 56) onto the annulus $\rho_1 < |w| < \rho_2$ and determine the modulus $\mu = \frac{\rho_2}{\rho_1}$ (see page 38).

In Problems 1268 to 1270 map the indicated domains onto the unit circle $|t| < 1$.

1268. The rectangle $|\operatorname{Re} u| < K$, $|\operatorname{Im} u| < K'$ ($0 < K < 1$). Find the positions of vertices under the mapping.

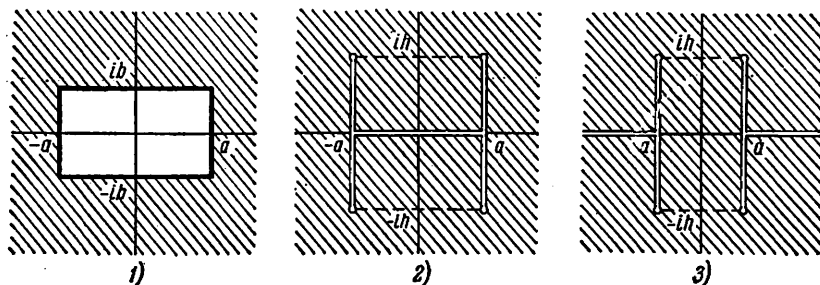


FIG. 57

1269. The interior of the ellipse $|z-1| + |z+1| = 2a$ ($a > 1$) with cuts $[-a, -1]$, $[1, a]$.

1270. The interior of the ellipse $|z-1| + |z+1| = 2a$ ($a > 1$). Find the positions of the foci under the mapping.

1271. Map the exterior of the unit circle $|t| > 1$ onto domains 1 to 3 located in the z -plane and shown in Fig. 57.

CHAPTER X

APPLICATIONS TO MECHANICS AND PHYSICS

§ 1. APPLICATIONS TO HYDRODYNAMICS

A steady-state two-dimensional irrotational flow of an incompressible fluid is characterized by the analytic function

$$w(z) = \varphi(x, y) + i\psi(x, y) \quad (1)$$

called the *complex potential* or *characteristic function* of flow; φ is called the *potential function*, ψ the *stream function*. The lines $\varphi = \text{const}$ are *equipotential lines*, $\psi = \text{const}$ — *streamlines* (or *lines of flow*). The velocity V of flow is connected with $w(z)$ by the following relations:

$$\left. \begin{aligned} V &= Ve^{i\alpha} = V_x + iV_y = \overline{w'(z)}, \\ |V| &= |w'(z)|, \quad \alpha = -\arg w'(z), \\ V &= \text{grad } \varphi. \end{aligned} \right\} \quad (2)$$

Let C be a closed contour traversed in the positive direction (the contour C may also consist of two sides of an arc traced in opposite directions). The quantity

$$\Gamma = \int_C V_s ds = \int_C V_x dx + V_y dy = \int_C d\varphi \quad (3)$$

is called the *circulation* of the vector V around the contour C .

The quantity

$$Q = \int_C V_n ds = \int_C (-V_y dx + V_x dy) = \int_C d\psi \quad (4)$$

(n an external normal to the closed contour C traversed in the positive direction) is called the *flux* of the vector V through the contour C . Similarly, the flux of the vector V through the arc AB is defined as the integral $\int_{AB} V_n ds$ (the direction of the normal n must be indicated).

Combining formulas (3) and (4), we obtain

$$\Gamma + iQ = \int_C w'(z) dz. \quad (5)$$

If $w'(z)$ is defined inside C and has a finite number of singular points there, then

$$\Gamma + iQ = 2\pi i \sum \text{res } w'(z).$$

If a is a pole of the function $w'(z)$, then $w(z)$ has, near a , an expansion of the form

$$w(z) = \frac{c_{-n}}{(z-a)^n} + \dots + \frac{p}{2\pi} \frac{1}{z-a} + \frac{\Gamma+iQ}{2\pi i} \ln(z-a) + \\ + c_0 + c_1(z-a) + \dots$$

The term $\frac{\Gamma+iQ}{2\pi i} \ln(z-a)$ (Γ, Q real numbers) is said to define, at a point a a *vortex source* of strength Q and intensity Γ , which is denoted as $(a; Q, \Gamma)^{1)}$, the term $\frac{p}{2\pi} \frac{1}{z-a}$ is said to be a *doublet* with moment p , it is denoted as $(a; p)$ (p a complex number; the radius vector p determines the direction of the doublet axis which passes through the point a in the direction of the streamline); the remaining terms $\frac{c_{-k}}{(z-a)^k}$ define *multiplets* of order $2k$ at the point a .

Accordingly, if at ∞

$$w(z) = c_n z^n + \dots + \frac{p}{2\pi} z + \frac{\Gamma+iQ}{2\pi i} \ln z + c_0 + \frac{c_{-1}}{z} + \dots,$$

then the term $\frac{\Gamma+iQ}{2\pi i} \ln z$ defines at ∞ a *vortex source* of strength Q and intensity Γ , the term $\frac{p}{2\pi} z$ a *doublet* with moment p (the direction of streamlines at ∞ coincides with the direction of the radius vector p), the remaining terms $c_k z^k$ defining *multiplets* of order $2k$.

Points at which $V=0$ and, hence, $w'(z)=0$, are called *critical points* of flow; the streamlines and equipotential lines flow alternately from these points. If a critical point is a zero of a derivative of order $(n-1)$, then these lines form angles $\frac{\pi}{2n}$. Such branching of lines is also possible at infinity.

In Problems 1272 to 1285, from the given complex potential of flow construct equipotential lines and streamlines, determine V , singular and critical points, strength and intensity of vortex sour-

¹⁾ If $Q=0$, we have a *vortex* $(a; \Gamma)$. If $\Gamma=0$, we have a *source* $(a; Q)$. If the strength of the source $Q < 0$, then we have a *sink*.

ces and moments of doublets, and investigate the behaviour of the flow at infinity.

1272. $w = cz$ ($c = \alpha + i\beta$).

1273. $w = z^n$ (in particular, $n = 2, 3$).

1274. $w = \frac{\Gamma + iQ}{2\pi i} \ln z$. Consider, in particular, the cases $\Gamma = 0$ and $Q = 0$.

1275. $w = \frac{\Gamma + iQ}{2\pi i} \ln \frac{z-a}{z-b}$.

1276. $w = \frac{1}{z}$. Also determine the velocity at the points $2 \pm i$.

1277. 1) $w = z + \frac{R^2}{z}$; 2) $w = z - \frac{R^2}{z}$. 1278. $w = \frac{1}{z^2}$.

1279. $w = \ln(z^2 - a^2)$ ($a > 0$). Also determine the velocity at the points $\pm ia$.

1280. $w = \ln \frac{z^2 - a^2}{z^2 + a^2}$ ($a > 0$). 1281. $w = \frac{Q}{2\pi} \ln \left(z - \frac{1}{z} \right)$.

1282. $w = \ln \left(1 + \frac{1}{z^2} \right)$. 1283. $w = \ln \left(z^2 + \frac{1}{z^2} \right)$.

1284. $w = az + \frac{Q}{2\pi} \ln z$ ($a > 0$, $Q > 0$).

1285. $w = az + \frac{\Gamma}{2\pi i} \ln z$ ($a > 0$, $\Gamma > 0$).

1286. Investigate the character of flow in the domain $|z| \geq R$ if

$$w = a \left(z + \frac{R^2}{z} \right) + \frac{\Gamma}{2\pi i} \ln z \quad (a > 0, \Gamma > 0).$$

Consider the cases: $\Gamma < 4\pi aR$, $\Gamma = 4\pi aR$, $\Gamma > 4\pi aR$.

1287. Find the complex potential $w(z)$ of a flow in the entire plane if the flow is generated by vortex sources $\{(a_k; Q_k, \Gamma_k)\}$ ($k = 1, 2, \dots, n$) and has at infinity a given velocity $V_\infty = Ve^{i\alpha}$.

1288. Can a streamline emanate from a point which is: 1) a vortex, 2) a doublet, 3) both a vortex and a doublet at the same time?

1289. Find the law governing the change of a vortex source, doublet and multiplet found at a point a or at infinity for the following univalent conformal mappings in the neighbourhood of these points ($c_1 \neq 0$, $c_{-1} \neq 0$):

1) $\zeta = \alpha + c_1(z-a) + \dots$; 2) $\zeta = \alpha + \frac{c_{-1}}{z} + \dots$;

3) $\zeta = \frac{c_{-1}}{z-a} + c_0 + \dots$; 4) $\zeta = c_1 z + c_0 + \dots$.

1290. Find the law governing the change of a vortex source under n -valent mappings:

$$\zeta = \alpha + c_n (z - a)^n + \dots, \quad c_n \neq 0;$$

$$\zeta = \alpha + \frac{c_{-n}}{z^n} + \dots, \quad c_{-n} \neq 0.$$

1291. Prove that a flow can be continued by the symmetry principle across a rectilinear or circular portion of a streamline or an equipotential line. In this case a vortex source passes into a vortex source, a doublet into a doublet, and a multiplet, generally speaking, into a set of multiplets up to the same order inclusive. Find the law governing the change of strength and intensity of the vortex source and of the moment of the doublet.

Note. Problem 1291 establishes the *symmetry principle*, which, along with conformal mappings, is widely used in constructing flows (see Problems 1293 to 1301).

It follows from the symmetry principle that if there is a rectilinear or circular section on a streamline or on an equipotential line, the flow must be symmetric with respect to this line. It imposes certain limitations not only on the singularities of the flow outside the indicated lines, but also on these lines themselves or on their end-points (if any).

1292. A flow in the z -plane is formed by a finite number of sources, vortices and doublets.

1) Find a necessary and sufficient condition for the circle $|z| = R$ to be a streamline if the sources, vortices and doublets: a) are not located on the circle; b) are all situated on it; c) are partially located on the circle and partially outside it.

2) Under the same assumptions find conditions for the circle $|z| = R$ to be an equipotential line.

1293. Find the complex potentials of flows in the upper half-plane $\text{Im } z > 0$, given the following singularities and velocity V_∞ :

1) Velocity $V_\infty = V$. 2) Vortex $(a; \Gamma)$ and velocity $V_\infty = 0$. 3) Source $(a; Q)$ and velocity $V_\infty = 0$. 4) Doublet $(a; p)$ and velocity $V_\infty = 0$. 5) Vortex sources $\{(a_k; Q_k, \Gamma_k)\}$ ($k = 1, 2, \dots, n$), doublet $(a; p)$ and velocity $V_\infty = V$. What can be said about the behaviour of the flow at infinity? 6) Vortex source $(0; Q; \Gamma)$ and doublet $(0; p)$; $V_\infty = 0$. What values can the moment of the doublet p take on? Is the flow always possible if $\Gamma \neq 0$?

1294. In the circle $|z| < R$ construct flows that have, respectively:

1) vortex $(a; \Gamma)$; 2) doublet $(a; p)$.

1295. Find the conditions for constructing flows in the circle $|z| < R$, given that:

1) the only sources are $\{(a_k; Q_k)\}$ ($k=1, 2, \dots, n$) located inside the circle;

2) in addition to the sources of item (1) there are sources $\{(a'_k; Q'_k)\}$ ($k=1, 2, \dots, m$) situated on the circle $|z|=R$.

Find the complex potentials of the flows in both cases.

1296. In the domain $|z| > R$ construct flows that have, respectively:

1) vortex $(a; \Gamma)$, velocity $V_\infty = 0$ and circulation at infinity $\Gamma_\infty = 0$;

2) doublet $(a; p)$, velocity $V_\infty = 0$ and circulation $\Gamma_\infty = 0$;

3) velocity $V_\infty = Ve^{i\alpha}$ and circulation $\Gamma_\infty = 0$;

4) velocity $V_\infty = Ve^{i\alpha}$ and circulation Γ around the circle $|z|=R$.

Note. Items (3) and (4) deal with flow around a circle with a given velocity at infinity both with and without circulation.

In Problems 1297 to 1300 construct flows from the given singularities (at infinity and at corner points the velocity is equal to zero), using the symmetry principle.

1297. In the domain $|z| > 1$, $\text{Im } z > 0$, with vortex $(ia; \Gamma)$, $a > 0$.

1298. In the angle $0 < \arg z < \frac{\pi}{3}$, with source $(ae^{i\frac{\pi}{6}}; Q)$, $a > 0$.

1299. In the first quadrant $\text{Re } z > 0$, $\text{Im } z > 0$, with source $(1; Q)$.

1300. 1) In the first quadrant $\text{Re } z > 0$, $\text{Im } z > 0$, with source $(1; Q)$ and sink $(i; -Q)$.

2) In the first quadrant $\text{Re } z > 0$, $\text{Im } z > 0$, with source $(1+i; Q)$ and sink $(0; -Q)$.

1301. Construct a flow in the entire z -plane, given that in the upper half-plane $\text{Im } z > 0$ it has vortex sources $\{(a_k; Q_k, \Gamma_k)\}$ ($k=1, 2, \dots, n$) and a doublet $(a; p)$, the x -axis is an equipotential line and the velocity $V_\infty = Ve^{i\alpha}$. Is such a flow always possible?

1302. Construct a flow in the entire z -plane, given that in the circle $|z| < R$ it has vortex sources $\{(a_k; Q_k, \Gamma_k)\}$ ($k=1, 2, \dots, n$) and a doublet $(a; p)$, the circle $|z|=R$ is an equipotential line and the velocity $V_\infty = Ve^{i\alpha}$. Is such a flow always possible?

1303. In a singly connected domain D bounded by a contour C , construct a flow with a streamline C that has vortex sources $\{(a_k; Q_k, \Gamma_k)\}$ ($k=1, 2, \dots, n$). Is such a flow always possible?

1304. In a domain D bounded by a contour C and containing the point at infinity, construct a flow with a streamline C that has vortex sources $\{(a_k; Q_k, \Gamma_k)\}$ ($k=1, 2, \dots, n$) and a given velocity $V_\infty = Ve^{i\alpha}$. Is such a flow always possible?

Problems 1305 to 1312 consider the streamlining of bounded and unbounded contours. The problems are solved with the aid of conformal mappings onto the exterior of a circle, the upper half-plane and a rectilinear strip.

1305. Construct a streamline flow around a contour C with a given circulation Γ and velocity $V_x = V e^{i\alpha}$. What mapping does the complex potential $w(z)$ effect in the case $\Gamma = 0$?

1306. Construct a streamline flow around the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$:

1) with a given velocity V_x without circulation;

2) with a given velocity V_x and circulation Γ .

1307. Construct a streamline flow around a plate $|x| \leq C, y = 0$:

1) with a given velocity V_x without circulation;

2) with a given velocity V_x and circulation Γ determined from the condition that one of the ends of the plate be the point of departure of the flux (the Zhukovsky-Chaplygin postulate).

1308. Construct a streamline flow around Zhukovsky's profile¹⁾ with a given velocity V_x and circulation Γ determined with the aid of the Zhukovsky-Chaplygin postulate (the acute end of the profile must be the point of departure).

In Problems 1309 to 1312 construct the flow past the given contours.

1309. Parabola $y^2 = 2px$ (outside and inside).

1310. Right-hand branch of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (outside and inside with a velocity $V_x = 0$).

1311. Half-lines: $-\infty < x < -1, y = \pm \pi$.

1312. Half-lines: $1 < |x| < \infty, y = 0$.

Problems 1313 to 1317 involve periodic flows $\{V(z + \omega) = V(z)\}$ and flows in curvilinear strips (channels). To construct these flows, map the curvilinear strips conformally onto rectilinear strips, then continue the flows according to the symmetry principle and use expansions of meromorphic functions into series of partial fractions.

In Problems 1313 and 1314 investigate the singularities, sketch the streamlines and equipotential lines, and determine the velocity at infinity in the strip of periods for periodic flows with the given complex potentials.

1313. 1) $w = \frac{Q}{2\pi} \ln \sin z$; 2) $w = \frac{\Gamma}{2\pi i} \ln \sin z$.

¹⁾ See Problem 313 and the answer to it.

1314. $w = \frac{p}{2\pi} \cot z \quad \left(0 \leq \arg p \leq \frac{\pi}{2}\right).$

1315. In the rectilinear strip $S: 0 < x < \omega$ of the z -plane, construct a flow generated by the vortex source $(a; Q, \Gamma)$, $a \in S$ and having the given velocities $V(x + i\infty) = iV$, $V(x - i\infty) = iV_1$. Is such a flow always possible?

Sketch the streamlines and equipotential lines, given that $\Gamma = 0$ or $Q = 0$.

Hint. Continue the flow by the symmetry principle and use the solution of Problem 1313.

1316. In the rectilinear strip $S: 0 < x < \omega$ of the z -plane, construct a flow generated by the doublet $(a; p)$, $a \in S$ and having a given velocity $V(x \pm i\infty) = iV$. Sketch the streamlines and equipotential lines.

1317. In a curvilinear strip S (of the z -plane) bounded by contours C_1, C_2 , construct the flow streamlining C_1, C_2 that has given vortex sources and doublets in S and given velocities V_1, V_2 at the points at infinity Ω_1, Ω_2 of the strip S . Indicate sufficient conditions for the existence of such a flow.

A flow is called *doubly periodic* if its velocity $\overline{w'(z)}$ is an elliptic function.

An *elliptic* function is a *doubly periodic* meromorphic function with periods 2ω and $2\omega'$, where $\text{Im} \frac{\omega'}{\omega} \neq 0$ (it is assumed henceforward that $\text{Im} \frac{\omega'}{\omega} > 0$). From this definition it follows that $f(z + 2m\omega + 2n\omega') = f(z)$, where m and n are any integers or zeros.

A parallelogram with vertices $z_0, z_0 + 2\omega, z_0 + 2\omega', z_0 + 2\omega + 2\omega'$ (z_0 an arbitrary point) is called a *parallelogram of periods*.

If $f(z)$ is a non-constant elliptic function, it possesses the following properties (Liouville's theorems):

- 1) $f(z)$ has at least one pole in the parallelogram of periods;
- 2) the sum of the residues of the function $f(z)$ with respect to all poles located in the parallelogram of periods equals zero;
- 3) the equation $f(z) = a$ has in the parallelogram of periods an equal number of roots for any complex number a , finite or infinite (this number of roots is called the *order* of the elliptic function);
- 4) the difference between the sum of all zeros and the sum of all poles of the function $f(z)$ situated in the parallelogram of periods is equal to a certain period of the function, i.e.

$$\sum \alpha_k - \sum \beta_k = 2\mu\omega + 2\nu\omega' \quad (\mu \text{ and } \nu \text{ integers}).$$

Weierstrass' Sigma function is the entire function

$$\sigma(z) = z \prod' \left(1 - \frac{z}{\Omega}\right) e^{\frac{z}{\Omega} + \frac{z^2}{2\Omega^2}}, \quad (6)$$

where $\Omega = 2n\omega + 2m\omega'$ and the product is extended over all non-zero Ω . $\sigma(z)$ is an odd function.

Weierstrass' Zeta function is the meromorphic function

$$\zeta(z) = \frac{\sigma'(z)}{\sigma(z)} = \frac{1}{z} + \sum' \left(\frac{1}{z - \Omega} + \frac{1}{\Omega} + \frac{z}{\Omega^2} \right), \quad (7)$$

where the summation is extended over all non-zero Ω . $\zeta(z)$ is an odd function.

The Weierstrass function $\wp(z)$ with periods 2ω and $2\omega'$ (see page 182) is connected with $\zeta(z)$ by the relation $\wp(z) = -\zeta'(z)$.

Since $[\zeta(z+2\omega) - \zeta(z)]' = \wp(z) - \wp(z+2\omega) = 0$, then

$$\zeta(z+2\omega) - \zeta(z) = 2\eta$$

and similarly

$$\zeta(z+2\omega') - \zeta(z) = 2\eta',$$

where η and η' are constants. Taking advantage of the fact that the function $\zeta(z)$ is odd, it is easy to show that $\eta = \zeta(\omega)$ and $\eta' = \zeta(\omega')$.

The quantities η , η' , ω and ω' are connected by the Legendre relation

$$\eta\omega' - \eta'\omega = \frac{\pi i}{2}.$$

Denote

$$\eta = \eta_1, \quad \eta' = \eta_3 \quad \text{and} \quad \eta + \eta' = \eta_2,$$

$$\omega = \omega_1, \quad \omega' = \omega_3 \quad \text{and} \quad \omega + \omega' = \omega_2.$$

The functions $\sigma_k(z)$ are defined by the relations

$$\sigma_k(z) = -e^{\eta_k z} \frac{\sigma(z - \omega_k)}{\sigma(\omega_k)} \quad (k = 1, 2, 3). \quad (8)$$

Accordingly

$$\zeta_k(z) = \frac{\sigma'_k(z)}{\sigma_k(z)}. \quad (9)$$

The functions $\sigma_k(z)$ are related to the Weierstrass function $\wp(z)$ and Jacobi's functions $\operatorname{sn} z$, $\operatorname{cn} z$, $\operatorname{dn} z$ by the following formulas:

$$\sqrt{\wp(z) - e_k} = \frac{\sigma_k(z)}{\sigma(z)}, \quad (10)$$

$$\operatorname{sn} u = \sqrt{e_1 - e_3} \frac{\sigma(z)}{\sigma_3(z)}, \quad \operatorname{cn} u = \frac{\sigma_1(z)}{\sigma_3(z)}, \quad \operatorname{dn} u = \frac{\sigma_2(z)}{\sigma_3(z)}, \quad (11)$$

where $u = z\sqrt{e_1 - e_3}$ and $e_k = \wp(\omega_k)$ (see page 182).

Any elliptic function can be expressed with the aid of $\sigma(z)$ and $\zeta(z)$.

If in the parallelogram of periods $f(z)$ has only simple poles b_k with residues A_k ($k=1, 2, \dots, n$), then

$$f(z) = \sum_{k=1}^n A_k \zeta(z - b_k) + C. \quad (12)$$

If in the parallelogram of periods $f(z)$ has zeros a_k and poles b_k (each one repeated according to its multiplicity; $k=1, 2, \dots, n$), then

$$f(z) = C \frac{\sigma(z - a_1) \sigma(z - a_2) \dots \sigma(z - a_n)}{\sigma(z - b_1^*) \sigma(z - b_2) \dots \sigma(z - b_n)}, \quad (13)$$

where $b_1^* = \sum_{k=1}^n a_k - \sum_{k=2}^n b_k$.

The following functions are called *Jacobi's Theta functions*:

$$\begin{aligned} \vartheta_1(v) &= i \sum_{n=-\infty}^{\infty} (-1)^n q^{\left(n - \frac{1}{2}\right)^2} e^{(2n-1)\pi i v} = \\ &= 2 \sum_{n=0}^{\infty} (-1)^n q^{\left(n + \frac{1}{2}\right)^2} \sin(2n+1)\pi v, \end{aligned} \quad (14)$$

$$\vartheta_2(v) = \vartheta_1\left(v + \frac{1}{2}\right), \quad \vartheta_3(v) = q^{\frac{1}{4}} e^{\pi i v} \vartheta_1\left(v + \frac{1}{2} + \frac{\tau}{2}\right), \quad (15)$$

$$\vartheta_4(v) = -iq^{\frac{1}{4}} e^{\pi i v} \vartheta_1\left(v + \frac{\tau}{2}\right),$$

where $q = e^{\pi i \tau}$, $\tau = \omega'/\omega$.

The Theta functions are connected with the Sigma functions by relations of the form

$$\sigma(z) = 2\omega_1 e^{\frac{\eta_1 z^2}{2\omega_1}} \frac{\vartheta_1(v)}{\vartheta_1(0)}, \quad (16)$$

$$\sigma_i(z) = e^{\frac{\eta_i z^2}{2\omega_i}} \frac{\vartheta_{i+1}(v)}{\vartheta_{i+1}(0)} \quad (i=1, 2, 3), \quad (17)$$

where $z = 2\omega_1 v$.

The advantage of the Theta functions consists in the rapid convergence of the series defining them. Using formula (16) we can write representation (13) in the form

$$f(z) = C \frac{\vartheta_1\left(\frac{z-a_1}{2\omega}\right) \vartheta_1\left(\frac{z-a_2}{2\omega}\right) \dots \vartheta_1\left(\frac{z-a_n}{2\omega}\right)}{\vartheta_1\left(\frac{z-b_1^*}{2\omega}\right) \vartheta_1\left(\frac{z-b_2}{2\omega}\right) \dots \vartheta_1\left(\frac{z-b_n}{2\omega}\right)}. \quad (18)$$

Using formulas (11), (16) and (17), we can write the expressions for the Jacobian functions $\operatorname{sn} z$, $\operatorname{cn} z$, $\operatorname{dn} z$ in terms of the Theta functions.

It is assumed henceforward that ω is a real number and ω' , purely imaginary, i.e. the parallelograms of periods are rectangles.

1318. Show that the function $f(u) = \frac{M}{2\pi} \zeta(u - \alpha) + Cu$ is the complex potential of a doubly periodic flow with one doublet (α ; M) (M is the moment of the doublet) in the parallelogram of periods.

Consider, in particular, the following cases:

1) $\alpha = 0$ and the lines $\text{Im } u = \pm \text{Im } \omega'$ are streamlines; sketch the streamlines and equipotential lines, investigate the conformal mapping effected by the function $t = f(u)$.

2) $f(u + 2\omega) = f(u)$; sketch the streamlines and equipotential lines, investigate the conformal mapping effected by $t = f(u)$.

1319. Show that the flows defined by the complex potentials $\zeta_k(u)$ ($k = 1, 2, 3$) can be reduced to the flows of Problem 1318 (for $C = 0$) with the aid of shifts in the planes u and ζ .

Hint. Use formulas (8) and (9).

1320. Show that the flows defined by the complex potentials $\vartheta'_k\left(\frac{u}{2\omega}\right)$ ($k = 1, 2, 3, 4$) and $Z(u)$ (see page 180) can be reduced to the flows of Problem 1318(2) with the aid of linear transformations.

1321. Show that a flow defined by the complex potential $E(u)$ (see page 180) can be reduced to the flow of Problem 1318(1) with the aid of linear transformations.

Hint. First prove the relations:

$$K^2 = (e_1 - e_3) \omega^2 \quad \text{and} \quad \frac{E}{K} = \frac{e_1}{e_2 - e_3} + \frac{\omega}{K^2} \eta.$$

1322. Find the complex potential $f(u)$ of a doubly periodic flow with two doublets (α ; M), (β ; N) in the parallelogram of periods.

Find out when the function $f(u)$ is elliptic and the lines $\text{Im } u = \pm \text{Im } \omega'$, $\text{Re } u = \pm \omega$ are streamlines and equipotential lines (or vice versa); sketch the streamlines and equipotential lines.

In Problems 1323 to 1325 investigate the doubly periodic flows defined by the given complex potentials $f(u)$.

1323. sn u . 1324. cn u . 1325. dn u .

1326. Find the complex potential $f(u)$ of a doubly periodic flow with two vortex sources (α ; Q , Γ), (β ; $-Q$, $-\Gamma$) in the parallelogram of periods. Consider, in particular, the cases $\alpha = 0$, $\alpha = \omega$, $\alpha = \omega + \omega'$ and $\beta = \omega'$.

Find the form of the function $f(u)$ that satisfies the condition $f(u + 2\omega) = f(u)$.

In Problems 1327 to 1329 investigate the flows defined by the indicated complex potentials $f(u)$.

1327. 1) $\ln \operatorname{sn} u$; 2) $\ln \operatorname{cn} u$; 3) $\ln \operatorname{dn} u$.

1328. $\wp(u)$. 1329. $\ln \vartheta_k(v)$ ($v = \frac{u}{2\omega}$, $k=1, 2, 3, 4$).

To construct the complex potential $f(z)$ in a doubly connected domain D , it is usually first mapped conformally onto an annulus R : $\rho < |t| < 1$ ($\mu = 1/\rho$ is the modulus of D); with the aid of the function $t = e^{\pi ui/\omega}$ the annulus R with a radial cut $[\rho, 1]$ is, in turn, mapped onto a rectangle with vertices $0, 2\omega, 2\omega + \omega', \omega'$ in the u -plane so that the lips of the cut go into the lateral sides and $\tau = \frac{\omega'}{\omega} = \frac{i}{\pi} \ln \frac{1}{\rho}$. The characteristics of flow in the rectangle are determined by the method used in solving Problem 1289. Since the bases of the rectangle are streamlines, the flow continues across them according to the symmetry principle (see Problem 1291). Then determine the complex potential $\Phi(u)$ of the doubly periodic flow thus obtained whose periods are $2\omega, 2\omega'$; then $f(z) = \Phi[u(z)]$.

1330. Find the complex potential of the flow:

1) in the annulus R : $r_1 < |z| < r_2$ with circulations Γ around the boundary circles;

2) in an arbitrary bounded doubly connected domain $D^{1)}$ with circulation Γ around the boundary contours;

3) in the exterior of two circles lying outside one another, with circulations $\pm \Gamma$ on the boundary circles, provided that $V_\infty = 0$;

4) in a doubly connected domain D containing the point at infinity with circulations $\pm \Gamma$ around the boundary contours provided that $V_\infty = 0$.

1331. Construct a flow in the annulus R : $\rho < |z| < 1$ generated by the doublet $(a; p)$ ($\rho < a < 1$) and streamlining the boundary contours without circulation.

Investigate the mapping $t = f(z)$ and sketch the streamlines.

Hint. Use the solutions of Problems 1289(1) and 1322.

1332. Construct a flow in a doubly connected domain D containing the point at infinity and streamlining the boundary contours without circulation with velocity $V_\infty = Ve^{i\alpha}$ at infinity.

1333. Construct a flow in the annulus R : $\rho < |z| < 1$ generated by a doublet and a quadruplet located at the point $z=1$ and

¹⁾ Here and henceforward it is assumed that the function that maps the domain D onto an annulus is known.

streamlining the boundary circles without circulation. Draw the streamlines and investigate the mapping $t=f(z)$. Consider, in particular, the case of a doublet only.

Hint. Write the complex potential $f(z)$ in the form

$$f(z) = \frac{c_{-2}}{(z-1)^2} + \frac{c_{-1}}{(z-1)} + c_0 + c_1(z-1) + \dots$$

and determine the possible values of c_{-2} and c_{-1} .

1334. 1) In the annulus $R: \rho < |z| < 1$ construct a flow generated by the vortex $(a; \Gamma)$ ($\rho < a < 1$) and streamlining the boundary circles with circulations Γ_1 (around the circle $|z|=1$) and Γ_2 (around the circle $|z|=\rho$). Is it possible to specify Γ , Γ_1 , Γ_2 arbitrarily? Consider, in particular, the cases $\Gamma_2=0$ and $\Gamma_2=-\Gamma_1$.

Investigate the mappings effected by the functions $t=f(z)$, $\beta=e^{2\pi it/\Gamma}$ in the first case and by the functions $t=f(z)$, $\beta=e^{-4\pi it/\Gamma}$ and $s=\sqrt{\beta-\beta_0}$ ($\beta_0=-e^{4\pi i\psi_0/\Gamma}$), where ψ_0 is the value of the flow function at a critical point, in the second case. Construct the streamlines and equipotential lines in the u -plane.

2) In a doubly connected domain D containing the point at infinity construct a flow that streamlines the boundary contours with circulations Γ_1 , Γ_2 with velocity $V_\infty=Ve^{i\alpha}$.

§ 2. APPLICATIONS TO ELECTROSTATICS

A plane electrostatic field of intensity $\mathbf{E}=E_x+iE_y=Ee^{i\alpha}$ is characterized by an analytic function $w(z)=u+iv$ called the *complex potential*; v is called the *potential function* (it is always single-valued!) and u is the *force function*. The lines $v=\text{const}$ are *equipotential lines*, and $u=\text{const}$ are *lines of force* of the field. Here,

$$\mathbf{E} = -\text{grad } v = -i\overline{w'(z)}, \quad E = |w'(z)|, \quad \alpha = -\frac{\pi}{2} - \arg w'(z),$$

$$E_x = -\frac{\partial v}{\partial x} = \frac{\partial u}{\partial y}, \quad E_y = -\frac{\partial v}{\partial y} = -\frac{\partial u}{\partial x}.$$

In all problems of this section involving electrostatic fields in domains bounded by one or several boundary contours, it is assumed that the potential function is constant around each simple contour (i.e. each such contour is a conductor).

If a is a pole of $w'(z)$ and w has the following expansion in the neighbourhood of a

$$w(z) = \frac{c_{-n}}{(z-a)^n} + \dots + \frac{pi}{z-a} + 2qi \ln \frac{1}{z-a} + c_0 + c_1(z-a) + \dots,$$

then at the point a the term $2qi \ln \frac{1}{z-a}$ defines a plane point charge of magnitude $p=2q$, which is usually denoted as $(a; 2q)$ (in

space, a unit length of a rectilinear conductor perpendicular to the z -plane at the point a carries a charge q); the term $pi/(z-a)$ determines, at the point a , a *dipole* with a moment p , which is denoted $(a; p)$ (p is a complex number; its argument determines the direction of the dipole axis); the remaining terms $c_{-k}/(z-a)^k$ ($k=2, \dots, n$) determine, at the point a , *multipoles* of order $2k$.

Accordingly, if at infinity

$$w(z) = c_n z^n + \dots + piz + 2qi \ln z + c_0 + \frac{c_{-1}}{z} + \dots,$$

then the term $2qi \ln z$ defines, at infinity, a plane point charge of magnitude $p=2q$ and the term piz , a dipole of moment p .

If the function $w=u+iv$ is considered to be the complex potential of an electrostatic field $\mathbf{E} = -i\overline{w'}(z)$ and simultaneously that of a fluid flow with velocity $\mathbf{V} = \overline{w'}(z)$, then we have the following electrohydrodynamic analogy:

	Fluid flow	Electrostatic field
u	Potential function	Force function
$u = \text{const}$	Equipotential lines	Lines of force
v	Flow function (it may be multi-valued)	Potential function (always single-valued)
$v = \text{const}$	Streamlines	Equipotential lines
$v_2 - v_1$	Flow rate	Potential difference
$\oint du$	Circulation $\Gamma = \oint V_s ds$	Flux $N = \oint E_n ds$
—	Vortex ($a; \Gamma$)	Point charge ($a; 2q$) $q = \Gamma/4\pi$
—	Source	—
—	Doublet with moment p	Dipole with moment $p/2\pi i$
—	Streamlining with given vortices and doublets	Field with given charges, dipoles and equipotential boundary lines

In Problems 1335 to 1342, from the given complex potentials $w(z)$ determine the force function and the potential function, the field strength, the character of the singularities (at infinity as well), and sketch the families of lines of force and equipotential lines (q a real number). Compare with the solutions of Problems 1272 to 1285.

1335. $w = cz$ ($c = \alpha + i\beta$). 1336. $w = 2qi \ln \frac{1}{z}$. 1337. $w = 2qi \times \ln \frac{z-a}{z-b}$. 1338. $w = 2qi \ln(z^2 - a^2)$ ($a > 0$). 1339. $w = \frac{pi}{z}$ ($p = |p|e^{i\varphi}$).

1340. $w = z \pm \frac{R^2}{z}$. 1341. $w = piz + 2qi \ln \frac{1}{z}$ ($p > 0$, $q > 0$).

1342. $w = piz + 2i \sum_{k=1}^n q_k \ln \frac{1}{z-a_k}$ ($p > 0$, $q_k > 0$, $a_1 < a_2 < \dots < a_n$).

1343. Find the law governing the variation of a point charge (a ; $2q$) and a dipole (a ; p):

1) under a univalent conformal mapping;

2) in continuation via the symmetry principle across a rectilinear or circular portion of an equipotential line.

1344. Show that the complex potential of an electrostatic field generated by a point charge (a ; $2q$) in an arbitrary singly connected domain D is defined by the formula

$$w = 2qi \ln \frac{1}{f(z, a)} + c,$$

where $f(z, a)$ is a function that conformally maps the domain D onto the unit circle so that $f(a, a) = 0$, and c is a real constant.

Determine the relationship between the potential function $v(z)$ and Green's function of the domain D (see Problem 1043).

In Problems 1345 to 1351, using the results of Problem 1344 or the symmetry principle, find the complex potentials of electrostatic fields generated by the given point charges in the indicated domains.

1345. In the upper half-plane $\text{Im } z > 0$ by the charge (z_0 ; $2q$).

1346. 1) In a circle $|z| < R$ by the charge (z_0 ; $2q$); 2) in the exterior of the circle $|z| > R$ by the charge (z_0 ; $2q$).

1347. In the exterior of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by the charge (∞ ; $2q$).

1348. In the exterior of the line-segment $|x| < R$, $y = 0$ by the charge (∞ ; $2q$).

1349. In the exterior of the square $|x| < d$, $|y| < d$ by the charge (∞ ; $2q$).

1350. In the rectangle $|x| < a$, $|y| < b$ by the charge (0 ; $2q$).

1351. In the rectangle $0 < x < 2a$, $0 < y < 2b$ by the charge (z_0 ; $2q$).

In Problems 1352 to 1356 construct the electrostatic fields generated by the given dipoles.

1352. In the circle $|z| < R$ by the dipole $(a; p)$.

1353. In the exterior of the circle $|z| > R$ by the dipole $(a; p)$.

1354. In the exterior of the segment $|x| < R, y = 0$ by the dipole $(\infty; p)$.

1355. In the exterior of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by the dipole $(\infty; p)$.

1356. In the rectangle $|x| < a, |y| < b$ by the dipole $(0; p)$ ($p i = \rho e^{i\alpha}$).

1357. Prove that the electrostatic field generated by the dipole $(a; p)$ in an arbitrary singly connected domain D is determined by the complex potential $w = f(z)$, where the function $f(z)$ maps the domain D onto the exterior of a horizontal segment so that $f(a) = \infty$, and the principal part of $f(z)$ at the point a equals $\frac{pi}{z-a}$ if $a \neq \infty$, and $pi z$ if $a = \infty$.

Find $f(z)$, given a function $t(z)$ that maps the domain D :

1) onto the interior of the unit circle, if $a \neq \infty$, where $t(a) = 0$, $t'(a) > 0$;

2) onto the exterior of the circle $|t| > R, a = \infty$, where $t(\infty) = \infty$ and $t'(\infty) = 1$.

1358. In a singly connected domain D , construct an electrostatic field generated by point charges $\{(a_k; 2q_k)\}$ ($k = 1, 2, \dots, n$) and the dipole $(a; p)$.

Let $g(\zeta, z)$ be Green's function of a domain D (see page 140) whose boundary Γ consists of the piecewise-smooth simple contours $\Gamma_1, \Gamma_2, \dots, \Gamma_n$; let n be an internal normal to Γ and let Γ be traversed in the positive direction with respect to D . If $u(z)$ is a function harmonic in the domain D and continuous on Γ , then it follows from Green's formula that

$$u(z) = \frac{1}{2\pi} \int_{\Gamma} u(\zeta) \frac{\partial g(\zeta, z)}{\partial n} ds$$

or

$$u(z) = \frac{1}{2\pi} \int_{\Gamma} \left[u(\zeta) \frac{\partial}{\partial n} \ln \frac{1}{|\zeta - z|} - \ln \frac{1}{|\zeta - z|} \frac{\partial u(\zeta)}{\partial n} \right] ds.$$

If the domain D contains the point at infinity and the function $u(z)$ is harmonic at that point, then we have to add $u(\infty)$ to the right members of the above formulas. In this case, in the neighbourhood of the point at infinity, Green's function $g(z, \infty)$ can be represented in the form

$$g(z, \infty) = \ln |z| + \gamma + o\left(\frac{1}{|z|}\right).$$

The quantity

$$\gamma = \lim_{z \rightarrow \infty} [g(z, \infty) - \ln |z|]$$

is called *Roben's constant* of a closed set, which is the complement of D in the z -plane; the quantity $e^{-\gamma}$ is called the *capacity* of this set.

1359. Prove the following assertions (n is an internal normal):

$$1) \ g(z, a) = \ln \frac{1}{|z-a|} - \frac{1}{2\pi} \int_{\Gamma} \frac{\partial g(\zeta, a)}{\partial n} \ln \frac{1}{|\zeta-z|} ds \text{ if } a \neq \infty;$$

$$2) \ g(z, \infty) = \gamma - \frac{1}{2\pi} \int_{\Gamma} \frac{\partial g(\zeta, \infty)}{\partial n} \ln \frac{1}{|\zeta-z|} ds \text{ if } z \in D \text{ and the domain } D \text{ contains the point at infinity;}$$

$$3) \ \frac{1}{2\pi} \int_{\Gamma} \frac{\partial g(\zeta, \infty)}{\partial n} \ln \frac{1}{|\zeta-z|} ds = \gamma \text{ if } z \notin D \text{ and the domain } D \text{ contains the point at infinity;}$$

$$4) \ \frac{1}{2\pi} \int_{\Gamma} \frac{\partial g(\zeta, a)}{\partial n} ds = 1 \text{ if } a \neq \infty \text{ or if } a = \infty \in D.$$

Hint. In item (1) take advantage of the symmetry property of Green's function $g(\zeta, z) = g(z, \zeta)$ and use the integral representation of a harmonic function with respect to its boundary values. In item (2) make use of the integral representation of the function $\ln |\zeta-z| + g(\zeta, z) - g(\zeta, \infty)$ harmonic in D , the limit process and the symmetry property $g(\infty, z) = g(z, \infty)$. In item (3) do the same, but proceed from the function $\ln |\zeta-z| - g(\zeta, \infty)$.

The function

$$v_0(z) = 2q \ln \frac{1}{|z-a|}$$

is called the *logarithmic potential* of the point charge ($a; 2q$). In the extended z -plane, $v_0(z)$ represents the logarithmic potential of two point charges ($a; 2q$) and ($\infty; -2q$).

Let the contour Γ satisfy the conditions indicated on page 200, and let $\rho(\zeta)$ and $v(\zeta)$ be real and continuous on Γ .

The integral

$$v(z) = \int_{\Gamma} \rho(\zeta) \ln \frac{1}{|\zeta-z|} ds$$

is called the *logarithmic potential of a simple layer of density* $\rho(\zeta)$ (in space it is associated with the potential of a charged cylindrical

surface with base Γ and surface charge density $\frac{\rho}{2}$, i.e. carrying a charge $\frac{\rho}{2} ds^2$ on the square element above ds).

The function $v(z)$ is continuous in a finite portion of the z -plane and harmonic everywhere outside Γ , except at the point $z = \infty$, where it has a logarithmic singularity

$$v(z) = -2q \ln |z| + o\left(\frac{1}{|z|}\right), \quad 2q = \int_{\Gamma} \rho(\zeta) ds;$$

this means that to the potential $v(z)$ there corresponds a charge $(\infty; -2q)$.

The integral

$$v_1(z) = \int_{\Gamma} v(\zeta) \frac{\partial}{\partial n} \ln \frac{1}{|\zeta - z|} ds$$

is called the *logarithmic potential of a double layer of density $v(\zeta)$* (distributed on Γ are dipoles with axes in the given direction of the normal n to Γ , which is internal if Γ bounds a domain; $v(\zeta)$ is the distribution density of dipole moments). If $\theta(\zeta, z)$ is the angle between n and the vector from ζ to z , and $d\varphi(\zeta, z)$, the angle subtended by the element of arc ds at z , then

$$v_1(z) = \int_{\Gamma} v(\zeta) \frac{\cos \theta(\zeta, z)}{|\zeta - z|} ds = \int_{\Gamma} v(\zeta) d\varphi(\zeta, z).$$

In particular, for a closed contour Γ and $v(\zeta) \equiv 1$

$$\int_{\Gamma} \frac{\partial}{\partial n} \ln \frac{1}{|\zeta - z|} ds = \begin{cases} 2\pi & \text{if } z \text{ is inside } \Gamma, \\ \pi & \text{if } z \text{ is on } \Gamma, \\ 0 & \text{if } z \text{ is outside } \Gamma \end{cases}$$

(see also Problem 1041).

Green's function $g(z, a)$ of a domain D can be considered as the potential of an electrostatic field generated by a point charge $(a; 1)$ when the boundary Γ of the domain D is earthed. Problem 1359(1) shows that in the case $a \neq \infty$ the earthing of Γ is equivalent to placing on Γ a charge of linear density $\rho(\zeta) = -\frac{1}{2\pi} \frac{\partial g(\zeta, a)}{\partial n}$. Here, according to Problem 1359(4), the charge is equal to -1 . In the case of $a = \infty$ the point charge $(\infty; 1)$ and the earthing of Γ taken together are equivalent to the placing on Γ of a charge of linear density $\rho(\zeta) = -\frac{1}{2\pi} \frac{\partial g(\zeta, \infty)}{\partial n}$ of the total magnitude -1 and to an additional field with constant potential γ [see Problem 1359(2)]. In these cases the given distributions on Γ are said to be *induced by charge $(a; 1)$* .

In Problems 1360 to 1363 find the linear density $\rho(\zeta, a)$ of a contour Γ induced by the charge $(a; 1)$ and the corresponding potential $v(z, a)$ of this simple layer for the domains bounded by the contour Γ .

1360. Γ is the real axis, $\text{Im } a > 0$.

1361. 1) Γ is the circle $|z| = R$, $|a| < R$;

2) Γ is the circle $|z| = R$, $|a| > R$ (consider, in particular, the case $a = \infty$).

1362. Γ is a segment of the real axis $|x| \leq R$, $y = 0$, $a = \infty$.

1363. Γ is the ellipse $\frac{x^2}{\alpha^2} + \frac{y^2}{\beta^2} = 1$, $a = \infty$.

1364. Assuming Green's function to be known for a domain D containing the point $z = \infty$, solve the *Roben problem*: find the density of the distribution $\rho(\zeta)$ on the boundary Γ of the domain D due to unit charge that establishes a constant potential outside D and on Γ .

Hint. See Problem 1359 (3) and (4).

In Problems 1365 to 1367 solve the Roben problem for the given domains D .

1365. D is the exterior of the circle $|z| > R$.

1366. D is the exterior of the segment $|x| \leq R$, $y = 0$.

1367. D is the exterior of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

In Problems 1368 to 1371 find the capacities (see page 201) of the given closed sets.

1368. $|z| \leq R$. 1369. $|x| \leq R$, $y = 0$. 1370. $\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$.

1371. $|z^2 - a^2| \leq a^2$ ($a > 0$).

1372. Given on a simple closed contour Γ a real function $\varphi(\zeta)$ continuous and differentiable along the contour.

Prove that the real part of the Cauchy-type integral $\frac{1}{2\pi i} \int_{\Gamma} \frac{\varphi(\zeta)}{\zeta - z} d\zeta$

is the logarithmic potential of a double layer of density $\varphi(\zeta)$, and its imaginary part is the logarithmic potential of a single layer of density $-\frac{1}{2\pi} \frac{d\varphi}{ds}$.

1373. Prove that the function $v(z)$ bounded and harmonic in the upper half-plane $\text{Im } z > 0$ can be represented in the form of the logarithmic potential of a double layer:

$$v(z) = \frac{1}{\pi} \int_{-\infty}^{\infty} v(t) \frac{\partial}{\partial n} \ln \frac{1}{|t-z|} dt.$$

However, if $v(z)$ is regular at ∞ , it can also be represented in the form of the logarithmic potential of a simple layer:

$$v(z) = v(\infty) - \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\partial v(t)}{\partial n} \ln \left| \frac{1}{t-z} \right| dt.$$

1374. In the upper half-plane $\text{Im } z > 0$ find the complex potential of an electrostatic field if its potential $v(z)$ takes on the given piecewise continuous values on the real axis. Write the potential function with the aid of harmonic measures of the corresponding segments of the real axis (see page 144):

- 1) φ on the interval $(-\infty, a)$, 0 on the interval (a, ∞) ;
- 2) φ on the interval (a, b) , 0 on the intervals $(-\infty, a)$, (b, ∞) ;
- 3) $\varphi_1, \varphi_2, \dots, \varphi_n$ on the intervals $(-\infty, a_1)$, (a_1, a_2) , $\dots$, (a_{n-1}, a_n) , respectively, and 0 on the interval (a_n, ∞) (here $a_1 < a_2 < \dots < a_n$);
- 4) φ_0 on the interval (a_n, ∞) , $\varphi_1, \varphi_2, \dots, \varphi_n$ on the intervals $(-\infty, a_1)$, (a_1, a_2) , and so on.

Hint. In item (1) use conformal mapping on a strip; in the other items apply the superposition method or the integral formula of Schwarz for a half-plane (see pages 143, 144).

1375. Find the complex potentials $w(z)$ and the potentials $v(z)$ in doubly connected domains with a given difference $d = v_2 - v_1$ of potentials v_1 and v_2 on the boundary contours:

- 1) in the annulus $r_1 < |z| < r_2$;
- 2) in an arbitrary doubly connected domain D .

1376. Prove that if D is an arbitrary doubly connected domain and the potential function takes on constant values (v_1 and v_2) on each of the contours bounding this domain, then

$$w(z) = \frac{i(v_2 - v_1)}{\ln \mu} \ln t(z) + c + iv_1, \quad v(z) = \frac{v_2 - v_1}{\ln \mu} \ln |t(z)| + v_1,$$

where $t(z)$ maps D conformally onto the annulus $1 < |t| < \mu$ (μ is the modulus of D), and the boundary contour of potential v_1 goes into the circle $|t| = 1$ (c a real number).

1377. Find the complex potentials in the indicated doubly connected domains (the potentials v_1 and v_2 on the boundary contours are constant).

- 1) In the exterior of the circles $|z \pm a| = R$ ($a > R$) (v_1 is the potential of the circle on the left).
- 2) In the exterior of the circles $|z| = r_1$ (potential v_1) and $|z - a| = r_2$ ($a > r_1 + r_2$).

3) In the annular domain between two non-concentric circles $|z|=R$ (potential v_1) and $|z-a|=r$ ($0 < a < R-r$).

4) In the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} < 1$ with a cut along the line segment joining the foci (the potential on the ellipse is v_1).

5) In the exterior of the line segments $1 < |x| < \frac{1}{k}$, $y=0$ ($0 < k < 1$). On the left-hand segment the potential is v_1 .

6) In the exterior of the segments $|x| < 1$, $y = \pm \pi$. On the upper segment the potential is v_1 .

1378. Let D be a multiply connected domain with boundary Γ consisting of n piecewise smooth contours Γ_k ($k=1, 2, \dots, n$), and let $\omega_k(z)$ be a harmonic measure of Γ_k (page 144). If D is bounded, then Γ_n is considered to be its exterior contour. Prove the following assertions:

1) If the domain D is bounded, then

$$\omega_k(z) = -\frac{1}{2\pi} \int_{\Gamma} \frac{\partial \omega_k(\zeta)}{\partial n} \ln \frac{1}{|\zeta - z|} ds \quad (k=1, 2, \dots, n-1),$$

$$\omega_n(z) = 1 - \frac{1}{2\pi} \int_{\Gamma} \frac{\partial \omega_n(\zeta)}{\partial n} \ln \frac{1}{|\zeta - z|} ds.$$

If the domain D contains the point at infinity, then

$$\omega_k(z) = \omega_k(\infty) - \frac{1}{2\pi} \int_{\Gamma} \frac{\partial \omega_k(\zeta)}{\partial n} \ln \frac{1}{|\zeta - z|} ds \quad (k=1, 2, \dots, n).$$

2) For the points z not belonging to the domain D the right members of the equalities indicated in item (1) take the value of unity in the domain complementary to D and bounded by Γ_k (or Γ_n), and zero in the domain complementary to D and bounded by Γ_i , $i \neq k$.

Note. According to item (1) the functions $\omega_k(z)$ represent in D potentials set up by induced plate charges on Γ , with densities of the layer $\rho_k(\zeta) = -\frac{1}{2\pi} \frac{\partial \omega_k(\zeta)}{\partial n}$. In the case of a bounded domain D the quantities $\omega_1(z), \dots, \omega_{n-1}(z)$ coincide exactly with the logarithmic potentials of the indicated induced layers on Γ .

The magnitudes of the charges of the layer ρ_{ik} induced by the potential $\omega_k(z)$ on the contour Γ_i ($i, k=1, 2, \dots, n$), i.e.

$$\rho_{ik} = -\frac{1}{2\pi} \int_{\Gamma_i} \frac{\partial \omega_k(\zeta)}{\partial n} ds = -\frac{1}{2\pi} \int_{\Gamma} \omega_i \frac{\partial \omega_k(\zeta)}{\partial n} ds$$

are called the *mutual capacities* of the boundary contours (some properties of the numbers p_{ik} are considered in Problems 1075 to 1078).

1379. Find the harmonic measures $\omega_k(z)$ and also the quantities $\rho_k(\zeta)$ and p_{ik} , defined in the note to the preceding problem, for:

1) the annulus $1 < |z| < \mu$;

2) an arbitrary doubly connected domain D , assuming that the function mapping this domain onto the annulus is known.

1380. Let D be the domain of Problem 1378 and $v(z)$ a bounded potential of an electrostatic field which assumes the constant values α_k ($k=1, 2, \dots, n$) on the boundary contours (conductors) Γ_k . Prove the following assertions:

$$1) v(z) = \sum_{k=1}^n \alpha_k \omega_k(z).$$

2) If the domain D is bounded and Γ_n is an external contour, then

$$v(z) = \alpha_n - \frac{1}{2\pi} \int_{\Gamma} \frac{\partial v(\zeta)}{\partial n} \ln \frac{1}{|\zeta - z|} ds;$$

but if D contains the point at infinity, then

$$v(z) = v(\infty) - \frac{1}{2\pi} \int_{\Gamma} \frac{\partial v(\zeta)}{\partial n} \ln \frac{1}{|\zeta - z|} ds.$$

Prove that the right members of the above formulas equal α_k ($k=1, 2, \dots, n$) in domains complementary to D and bounded by Γ_k .

3) The magnitudes of the charges of the layer induced on Γ_i are equal to

$$2q_i = -\frac{1}{2\pi} \int_{\Gamma_i} \frac{\partial v(\zeta)}{\partial n} ds = \sum_{k=1}^n p_{ik} \alpha_k.$$

where $\sum_{i=1}^n q_i = 0$.

Hint. See Problem 1075(1).

$$4) \frac{1}{2\pi} \iint_D (\text{grad } v)^2 dx dy = \sum_{i,k=1}^n p_{ik} \alpha_i \alpha_k.$$

Hint. See Problem 1076.

5) If $w(z)$ is the complex potential of the field, then the density of the induced layer is $\rho(\zeta) = -\frac{1}{2\pi} \frac{\partial v(\zeta)}{\partial n} = \pm \frac{1}{2\pi} |w'(\zeta)|$.

Let D be an arbitrary multiply connected domain with a boundary Γ consisting of Jordan contours $\Gamma_1, \dots, \Gamma_n$. There exist conformal mappings of the domain D onto each of the following canonical domains with the indicated uniqueness conditions (a and b are arbitrary points of the domain D , and A is an arbitrary complex number):

1) Onto a *plane with parallel cuts*. The mapping function $f(z)$ is defined uniquely by specifying its pole b and the coefficient A of the expansion

$$f(z) = \begin{cases} \frac{A}{z-b} + c_1(z-b) + \dots & (b \neq \infty), \\ Az + \frac{c_{-1}}{z} + \dots & (b = \infty). \end{cases}$$

2) Onto a *plane with radial cuts* (we will use this term for a plane with cuts along rays emanating from the origin), or with *cuts along concentric circular arcs* with centre at the origin of coordinates. The function $f(z)$ is defined by the zero a , the pole b and the coefficient A of the expansion

$$f(z) = \begin{cases} \frac{A}{z-b} + c_1(z-b) + \dots & (b \neq \infty), \\ Az + \frac{c_{-1}}{z} + \dots & (b = \infty). \end{cases}$$

3) Onto a *circle with radial cuts* or *with cuts along concentric circular arcs* (centred at the origin of coordinates). The function $f(z)$ is defined by the conditions $f(a) = 0$, $f'(a) = 1$ and by specifying the contour Γ_k which passes into a circle.

4) Onto an *annulus with radial cuts* or *with cuts along concentric circular arcs* (centred at the origin). By specifying contours that go into the internal and into the external boundary circles, the mapping is defined to within transformations of similarity and rotation.

1381. Assuming as known the function mapping the domain D conformally onto a plane with parallel cuts and the harmonic measures $\omega_k(z)$ of the boundary contours Γ_k ($k = 1, 2, \dots, n$)¹⁾, find the potential of the electrostatic field set up by the dipole $(a; p)$ in the domain D when the boundary Γ of the domain is earthed.

1382. Assuming Green's function of a domain D to be known, find the potential of the electrostatic field in this domain generated by the point charge $(a; 2q)$ ($a \in D$) which has given potentials α_k on the boundary contours Γ_k . ($k = 1, 2, \dots, n$). Given that D is a singly connected domain, obtain the formula of Problem 1344.

¹⁾ Both are defined with the aid of Green's function.

1383. Determine the character of an electrostatic field defined in a multiply connected domain D by the complex potential $w=f(z)$, where $f(z)$ is a function that maps this domain onto a plane with cuts parallel to the real axis.

1384. Determine the character of an electrostatic field defined in the domain D by the complex potential $w=2qi \ln \frac{1}{f(z)}$ given that the function $f(z)$ maps the domain D onto:

- 1) a plane with cuts along concentric circular arcs centred at the origin of coordinates;
- 2) a circle with cuts along concentric circular arcs centred at the origin;
- 3) an annulus with cuts along concentric circular arcs centred at the origin.

In all cases find the fluxes of the field intensity vector across the boundary contours.

1385. Construct a diagram of equipotential lines and lines of force of electrostatic fields:

- 1) set up in an infinite doubly connected domain by a dipole at ∞ .
- 2) set up in a bounded doubly connected domain D by a point charge.

In both cases the potentials on the boundary contours are constant.

1386. 1) Express the potential $v(z)$ of an electrostatic field set up in a multiply connected domain D by the charge distribution

$2q_k$ on $\Gamma_k \left(\sum_{k=1}^n q_k = 0 \right)$ in terms of the harmonic measures $\omega_k(z)$ of its boundary contours. (On each contour Γ_k the potential is constant.)

Hint. Use the results of Problem 1380.

2) Express the potential $v(z)$ if $\sum_{k=1}^n q_k = q \neq 0$ and there is a point charge ($a; -2q$).

Hint. The determination of the potential $v(z) + 2qg(z, a)$ is reduced to case (1).

1387. Find the potential $v(z)$ in the annulus $r_1 < |z| < r_2$, given that the charge distribution on its contours is $2q_1$ and $2q_2$, where in the case $q_1 + q_2 = q \neq 0$ there is also a point charge ($a; -2q$).

Hint. Green's function of an annulus can be determined via a solution of Problem 1334 by choosing suitable circulations. They also serve to determine the induced charges on the boundary contours which are connected with Green's function.

§ 3. APPLICATIONS TO THE PLANE PROBLEM OF HEAT CONDUCTION

The plane problem on the stationary distribution of temperature inside a field is characterized by the analytic function $w(z) = u + iv$ (where u is the temperature) called the *complex potential of a thermal field*. The vector $\mathbf{Q} = -k \text{grad } u = -k\overline{w'(z)}$ (k is the thermal conductivity, which is henceforward assumed constant) is called the *vector of heat flux*. The heat flux across a contour C is equal to

$$\int_C Q_n ds = -k \int_C \frac{\partial u}{\partial n} ds = -k \int_C dv$$

(n is an external normal to the contour C traversed in the positive direction). Since u is a single-valued function, the heat flux for a closed contour C also equals $ik \int_C w'(z) dz$. If, near a point a , we have

$$w(z) = \left[\dots + \frac{c_{-1}}{z-a} + c_0 + c_1(z-a) + \dots \right] + \frac{q}{2\pi k} \ln \frac{1}{z-a},$$

then the term $\frac{q}{2\pi k} \ln \frac{1}{z-a}$ defines at the point a a *source* (a ; q) of strength q and the term $\frac{c_{-1}}{z-a}$, a *doublet* at the point a .

We have the following analogy with a fluid flow and an electrostatic field:

	Thermal field	Fluid flow	Electrostatic field
Complex potential	$w(z) = u + iv$	$w(z) = u + iv$	$iw(z) = -v + iu$
Field vector	$\mathbf{Q} = -k \text{grad } u = -k\overline{w'(z)}$	$\mathbf{V} = \text{grad } u = \overline{w'(z)}$	$\mathbf{E} = -\text{grad } u = -\overline{w'(z)}$
u	Temperature	Potential function	Potential function
$u = \text{const}$	Isotherms	Equipotential lines	Equipotential lines
v	Flow function	Flow function	$-v$ is the force function
$v = \text{const}$	Streamlines	Streamlines	Lines of force
	Source (a ; q)	Source (a ; $-q/k$)	Point charge (a ; $q/2\pi k$)
	Doublet	Doublet	Dipole
	Thermal field with given sources, doublets and isothermal boundary contours	Flow determined by complex potential $iw(z)$ with given vortices and doublets streamlining the boundary contours	Field with given charges, dipoles and equipotential boundary contours

1388. Formulate the symmetry principle for the continuation of a heat source across a rectilinear or circular section of the boundary of the domain. Find the temperature distribution in an arbitrary singly connected domain D , given that there is a source $(a; q)$ inside this domain and that the temperature on the boundary has a constant value C .

In Problems 1389 to 1392 find the temperature distribution in the indicated domains from the given sources, assuming that the temperature on the boundary of the domain is constant.

1389. In the upper half-plane $\text{Im } z > 0$; source $(a; q)$.

1390. In the circle $|z| < R$; source $(a; q)$.

1391. In the semi-strip $|x| < a, y > 0$ with source $(ih; q)$ ($h > 0$).

1392. In the rectangle $|x| < a, |y| < b$ with source $(0; q)$.

1393. 1) Interpret Green's function $g(z, a)$ of a plane domain D in terms of heat conduction theory.

2) Assuming Green's function of the domain D to be known, find the temperature distribution in this domain if it is known that there is a source $(a; q)$ in D and the temperature on the boundary contours Γ_k ($k=1, 2, \dots, n$) has constant values u_k . Write down the answer with the aid of harmonic measures $\omega_k(z)$ of the boundary contours.

1394. Find the temperature distribution inside the annulus $r_1 < |z| < r_2$, given that there is a source $(a; q)$ inside the annulus and that the temperature on the boundary circles has constant values: u_1 on the circle $|z|=r_1$ and u_2 on the circle $|z|=r_2$.

Hint. See similar problems 1334 and 1387.

CHAPTER XI

GENERALIZATION OF ANALYTIC FUNCTIONS

In this chapter we use the notation introduced in § 5 of Chapter I for formal Cauchy derivatives. Differentials are denoted accordingly:

$$d\bar{z} = dx - i dy = \overline{dz}, \quad d\varphi = \frac{\partial \varphi}{\partial z} dz + \frac{\partial \varphi}{\partial \bar{z}} d\bar{z}.$$

§ 1. QUASI-CONFORMAL MAPPINGS

By the *characteristics* of an ellipse are meant the ratio p of its semi-axes ($p \geq 1$) and, if $p \neq 1$, the angle θ ($0 \leq \theta < \pi$) formed by the major axis of the ellipse and the x -axis.

1395. Show that the equation of an ellipse with centre at the origin, semi-minor axis h and characteristics p and θ can be written in the form

$$\gamma x^2 - 2\beta xy + \alpha y^2 = ph^2,$$

where $\alpha = p \cos^2 \theta + \frac{1}{p} \sin^2 \theta$, $\beta = \left(p - \frac{1}{p}\right) \cos \theta \sin \theta$, $\gamma = p \sin^2 \theta + \frac{1}{p} \cos^2 \theta$, or in the form

$$|z + \mu \bar{z}| = \lambda,$$

where

$$\mu = -\frac{p-1}{p+1} e^{2i\theta}, \quad \lambda = \frac{2ph}{p+1}.$$

The quantities α , β , γ in Problem 1395 are also called the *characteristics* of an ellipse. They are connected by the relation $\alpha\gamma - \beta^2 = 1$. The quantity μ is called the *complex characteristic* of an ellipse. Note that $|\mu| < 1$. In the case of the circle $\mu = 0$, $p = 1$, $\alpha = \gamma = 1$, $\beta = 0$.

1396. Prove the following relations between the different characteristics of the ellipse:

$$1) \rho = \frac{1+|\mu|}{1-|\mu|} = \frac{\alpha+\gamma}{2} + \sqrt{\left(\frac{\alpha-\gamma}{2}\right)^2 + \beta^2}, \quad \rho + \frac{1}{\rho} = \alpha + \beta;$$

$$2) 2\theta = \arg \mu + \pi \quad (-\pi \leq \arg \mu < \pi);$$

$$3) \begin{cases} \tan 2\theta = \frac{2\beta}{\alpha-\gamma}, \\ \tan \theta = \frac{\gamma-\alpha + \sqrt{(\gamma-\alpha)^2 + 4\beta^2}}{2\beta}; \end{cases}$$

$$4) \begin{cases} (1-|\mu|^2)\alpha = 1 + 2|\mu|\cos 2\theta + |\mu|^2, \\ (1-|\mu|^2)\beta = 2|\mu|\sin 2\theta, \\ (1-|\mu|^2)\gamma = 1 - 2|\mu|\cos 2\theta + |\mu|^2; \end{cases}$$

$$5) \frac{1}{\rho} \leq \alpha, \quad \gamma \leq \rho; \quad |\beta| \leq \frac{1}{2} \left(\rho - \frac{1}{\rho} \right).$$

Prove the following relations between the characteristics of two ellipses:

$$6) |p_2 - p_1| \leq |\alpha_2 - \alpha_1| + |\beta_2 - \beta_1| + |\gamma_2 - \gamma_1|;$$

$$7) \begin{cases} |\alpha_2 - \alpha_1| \leq |p_2 - p_1| + \sqrt{\left(p_2 - \frac{1}{p_2}\right)\left(p_1 - \frac{1}{p_1}\right)} \sin |\theta_2 - \theta_1|, \\ |\beta_2 - \beta_1| \leq |p_2 - p_1| \left(1 + \frac{1}{p_2 p_1}\right) + \\ \quad + 2 \sqrt{\left(p_2 - \frac{1}{p_2}\right)\left(p_1 - \frac{1}{p_1}\right)} \sin |\theta_2 - \theta_1|, \\ |\gamma_2 - \gamma_1| \leq |p_2 - p_1| + \sqrt{\left(p_2 - \frac{1}{p_2}\right)\left(p_1 - \frac{1}{p_1}\right)} \sin |\theta_2 - \theta_1|; \end{cases}$$

$$8) |\mu_2 - \mu_1| \leq \|\mu_2\| - \|\mu_1\| + 2 \sqrt{|\mu_2 \mu_1|} \sin |\theta_2 - \theta_1|;$$

$$\left| \frac{\mu_2 - \mu_1}{1 - \mu_1 \mu_2} \right| \leq \frac{|p_2 - p_1|}{p_2 + p_1} + \sqrt{(p_2 - 1)(p_1 - 1)} \sin |\theta_2 - \theta_1|.$$

Hint. To prove inequalities (6), (7), (8) use inequalities of the form

$$|r_2 \cos \lambda_2 - r_1 \cos \lambda_1| \leq |r_2 e^{i\lambda_2} - r_1 e^{i\lambda_1}| \leq |r_2 - r_1| + \\ + 2 \sqrt{r_2 r_1} \sin |\lambda_2 - \lambda_1|,$$

$$\|a_2 + ib_2\| - \|a_1 + ib_1\| \leq |a_2 - a_1| + |b_2 - b_1|.$$

The mapping (transformation) $w = f(z) = u(x, y) + iv(x, y)$ is called *affine* if

$$u = a_1 x + b_1 y + c_1, \quad v = a_2 x + b_2 y + c_2. \quad (1)$$

The Jacobian of this mapping is $\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$. If $\Delta = 0$, then the mapping is degenerate.

1397. Prove the following properties of affine mappings:

1) An affine mapping can be represented in the form

$$w = Az + B\bar{z} + C.$$

Express the coefficients A , B and C in terms of the transformation coefficients (1) and show that $\Delta = |A|^2 - |B|^2$.

2) If $\Delta \neq 0$, then there exists an inverse mapping

$$z = A_1 w + B_1 \bar{w} + C_1.$$

Express its coefficients in terms of the coefficients of transformation (1) and show that $\Delta_1 = |A_1|^2 - |B_1|^2 = 1/\Delta$.

3) If $\Delta \neq 0$, then the mapping preserves parallelism of straight lines and transforms ellipses into ellipses. Ellipses with a complex characteristic $\mu = B/A$ are transformed into circles if $\Delta > 0$, and so are ellipses with $\mu = \bar{A}/\bar{B}$ if $\Delta < 0$.

Circles are transformed into circles only under the orthogonal transformations $w = Az + C$ or $w = B\bar{z} + C$.

4) If $\Delta > 0$, the mapping preserves its direction of traversal; if $\Delta < 0$, the direction is reversed.

5) If $\Delta = 0$, but not all of the coefficients a_1, b_1, a_2, b_2 are equal to zero, then the mapping can be represented in the form

$$w = 2|A|e^{i\frac{\alpha+\beta}{2}}|z|\cos\left(\varphi + \frac{\alpha-\beta}{2}\right) + C,$$

where $\varphi = \arg z$, $\alpha = \arg A$, $\beta = \arg B$. Give the geometric meaning of this mapping.

By the characteristics of an affine mapping are meant the characteristics (p, θ) , (α, β, γ) and the complex characteristic μ of ellipses transformed into circles [see Problem 1397(3)].

The characteristics $p(z)$, $\theta(z)$; $\alpha(z)$, $\beta(z)$, $\gamma(z)$ and the complex characteristic $\mu(z)$ of the affine mapping

$$du = u_x dx + u_y dy, dv = v_x dx + v_y dy \text{ or } dw = w_z dz + w_{\bar{z}} d\bar{z}$$

are called the characteristics of the continuously differentiable mapping

$$w = u(x, y) + iv(x, y)$$

with a Jacobian $J > 0$.

Under such transformation, to an infinitely small circle

$$du^2 + dv^2 = d\rho^2$$

there corresponds an infinitely small ellipse

$$\gamma dx^2 - 2\beta dx dy + \alpha dy^2 = p dh^2$$

(dh is the semi-minor axis) or, in different notation,

$$|dz + \mu d\bar{z}| = \frac{2p}{p+1} dh.$$

1398. Prove that for a non-degenerate affine mapping the following relations hold:

$$\frac{\gamma}{a_1^2 + a_2^2} = \frac{-\beta}{a_1 b_1 + a_2 b_2} = \frac{\alpha}{b_1^2 + b_2^2} = \frac{1}{|\Delta|}.$$

1399. Prove that for the characteristics of a continuously differentiable mapping with a positive Jacobian the following relations hold true:

$$1) \frac{\gamma}{u_x^2 + v_x^2} = \frac{-\beta}{u_x u_y + v_x v_y} = \frac{\alpha}{u_y^2 + v_y^2} = \frac{p dh^2}{d\rho^2} = \frac{1}{J};$$

$$2) u_x^2 + u_y^2 + v_x^2 + v_y^2 = J \left(p + \frac{1}{p} \right);$$

$$3) p = \frac{\max \left| \frac{dw}{dz} \right|}{\min \left| \frac{dw}{dz} \right|} = \frac{|w_z| + |w_{\bar{z}}|}{||w_z| - |w_{\bar{z}}||};$$

$$4) \mu = \frac{w_{\bar{z}}}{w_z} = -\frac{p-1}{p+1} e^{2i\theta};$$

$$5) \sqrt{\frac{J}{p}} \leq \left| \frac{dw}{dz} \right| \leq \sqrt{pJ}.$$

A univalent continuously differentiable mapping $w = u + iv$ with a positive Jacobian is called a *quasi-conformal mapping with characteristics* $p(z)$, $\theta(z)$ or $\alpha(z)$, $\beta(z)$, $\gamma(z)$ or $\mu(z)$ if it transforms infinitesimal ellipses with these characteristics into infinitesimal circles.

1400. Prove that a quasi-conformal mapping satisfies the system of equations

$$\alpha u_x + \beta u_y = v_y, \quad \beta u_x + \gamma u_y = -v_x$$

or, in complex notation, the equation

$$w_{\bar{z}} = \mu w_z.$$

Hint. To the equation $|dz + \mu d\bar{z}| = \text{const}$ corresponds the equation $|dw| = \text{const}$.

Note. The above equations are called the *Beltrami equations*, and the characteristic μ , the *Beltrami coefficient*.

1401. Show that the univalent continuously differentiable mapping $w = u + iv$ with a positive Jacobian which transforms infinitesimal

tely small circles into infinitely small ellipses with characteristics $\rho(z)$, $\theta(z)$, or $\alpha(z)$, $\beta(z)$, $\gamma(z)$, or $\mu(z)$ satisfies the system of equations

$$u_x + \beta u_y = \alpha v_y, \quad -\beta u_x + u_y = -\alpha v_x$$

or, in complex notation, the equation

$$w_z = \nu \bar{w}_z, \quad \nu = -\mu = \frac{p-1}{p+1} e^{2i\theta}.$$

Note. If the indicated system is written in the form

$$pu_x + qu_y = v_y, \quad -qu_x + pu_y = -v_x$$

(here $p = 1/\alpha$ and $q = \beta/\alpha$), then we arrive at (p, q) -analytic functions, to use Polozhy's terminology.

1402. Prove that the Beltrami equation $w_z = \mu(z) w_{\bar{z}}$ is invariant with respect to analytic transformations of the function w , and the equation $w_z = \nu(z) \bar{w}_z$ is invariant with respect to conformal transformations of the variable z .

1403. Let the univalent continuously differentiable mapping $w = u + iv$ with a positive Jacobian transform infinitely small ellipses with the characteristics ρ, θ into infinitely small ellipses with the characteristics ρ_1 and θ_1 (such a mapping is called a quasi-conformal mapping with two pairs of characteristics $\rho, \theta; \rho_1, \theta_1$ or with two triples of characteristics $\alpha, \beta, \gamma; \alpha_1, \beta_1, \gamma_1$). Prove that the mapping $w = u + iv$ satisfies the system of equations

$$\alpha u_x + (\beta + \beta_1) u_y = \alpha v_y, \quad (\beta - \beta_1) u_x + \gamma u_y = -\alpha_1 v_x,$$

which can also be written in the form

$$w_z = q_1 w_z + q_2 \bar{w}_z,$$

where

$$q_1 = -\frac{p_1(p^2-1)}{(pp_1+1)(p+p_1)} e^{2i\theta}, \quad q_2 = \frac{p(p_1^2-1)}{(pp_1+1)(p+p_1)} e^{2i\theta_1}.$$

Hint. The equation $|dz + \mu d\bar{z}| = \text{const}$ corresponds to the equation $|dw + \mu_1 d\bar{w}| = \text{const}$ with known μ, μ_1 .

1404. Let $\zeta = f(z)$ be a quasi-conformal mapping with characteristic μ_f :

$$\bar{f}_z = \mu_f f_z$$

and $w = g(\zeta)$ a quasi-conformal mapping with characteristic μ_g :

$$g_{\bar{\zeta}} = \mu_g g_{\zeta}.$$

Prove that their superposition $F(z) = g[f(z)] = g \circ f$ is a quasi-conformal mapping with characteristic

$$\mu_F = \frac{\mu_f + \mu_g \bar{f}_z / f_z}{1 + \mu_g \mu_f \bar{f}_z / f_z}.$$

Also prove the relations

$$\begin{aligned}\mu_g &= \frac{f_z}{\bar{f}_z} \frac{\mu_F - \mu_f}{1 - \bar{\mu}_f \mu_F}, \\ \mu_{f^{-1}(z)} &= -\mu_{f(z)} \frac{f_z}{\bar{f}_z}, \\ \mu_F &= \frac{\bar{f}_z}{f_z} \frac{\mu_g - \mu_{f^{-1}}}{1 - \bar{\mu}_{f^{-1}} \mu_g}.\end{aligned}$$

Note. If μ_F is assumed to be known, then the quantity

$$\sup \ln \frac{1 + |\mu_g|}{1 - |\mu_g|}$$

may be taken as the distance between μ_f and μ_F . Similarly, the distance between μ_g and $\mu_{f^{-1}}$ is determined via μ_F .

1405. Let $p_{w/z}$ and $\theta_{w/z}$ be the characteristics ρ and θ of the quasi-conformal mapping $w = f(z)$. Show that

$$\begin{aligned}\rho_{z/w} &= \rho_{w/z}, \\ \theta_{z/w} &= \theta_{w/z} \pm \frac{\pi}{2} + \arg f_z\end{aligned}$$

and that for the composite quasi-conformal mapping $w[z(t)]$

$$\rho_{w/t} \leq \rho_{w/z} \rho_{z/t}.$$

1406. Show that for quasi-conformal mappings

$$\begin{aligned}u &= f(x), \quad v = y \text{ (longitudinal expansion-contraction),} \\ u &= x, \quad v = f(y) \text{ (transverse expansion-contraction),} \\ \rho &= r, \quad \theta = f(\varphi) \text{ (angular expansion-contraction)}\end{aligned}$$

the characteristic

$$\rho = \max \left(f', \frac{1}{\bar{f}'} \right),$$

and for the mapping $\rho = f(r)$, $\theta = \varphi$ (radial expansion-contraction) the characteristic

$$\rho = \max \left(\frac{rf'}{f}, \frac{f}{r\bar{f}'} \right).$$

1407. Construct a quasi-conformal mapping of the circle $|z| < R$ onto itself such that carries the point $z = a$ ($|a| < R$) to the origin of coordinates and leaves the points of the circle $|z| = R$ fixed. Estimate the characteristic ρ .

1408. Construct a quasi-conformal mapping of a skew semi-strip $x > 0$, $x \tan \alpha < y < x \tan \alpha + h$ onto a rectangular semi-strip $u > 0$, $0 < v < h$ without stretching at the base and with constant stretching on the lateral side. Estimate the characteristic ρ .

1409.* Construct a quasi-conformal mapping of a lune consisting of a half-plane and a circular segment with a central angle $2\theta_0$ (Fig. 58) onto a half-plane with preservation of lengths on the boundary. Estimate the characteristic ρ .

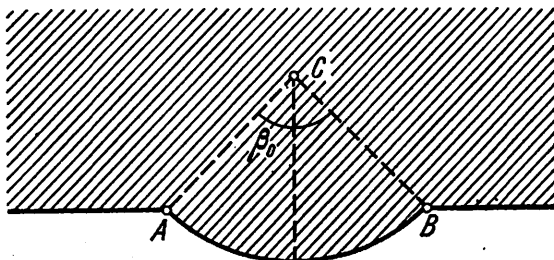


FIG. 58

1410. By means of the univalent mapping $\zeta = \zeta(z) = \xi + i\eta$ reduce the quasi-linear equation

$$A \frac{\partial^2 u}{\partial x^2} + 2B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} = F\left(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right)$$

of the elliptic type ($AC - B^2 > 0$) to the canonical form

$$\frac{\partial^2 u}{\partial \xi^2} + \frac{\partial^2 u}{\partial \eta^2} = F_1\left(\xi, \eta, u, \frac{\partial u}{\partial \xi}, \frac{\partial u}{\partial \eta}\right).$$

Prove that the mapping $\zeta(z)$ satisfies the system of Beltrami equations

$$\frac{A \frac{\partial \xi}{\partial x} + B \frac{\partial \xi}{\partial y}}{\sqrt{AC - B^2}} = \frac{\partial \eta}{\partial y}, \quad \frac{B \frac{\partial \xi}{\partial x} + C \frac{\partial \xi}{\partial y}}{\sqrt{AC - B^2}} = -\frac{\partial \eta}{\partial x}$$

and is a quasi-conformal mapping with characteristics α , β and γ determined from the relations

$$\frac{\gamma}{C} = \frac{\beta}{B} = \frac{\alpha}{A} = \frac{1}{\sqrt{AC - B^2}}$$

(assuming $A > 0$).

§ 2. GENERALIZED ANALYTIC FUNCTIONS

A function $w = u + iv$ satisfying the equation

$$w_{\bar{z}} + Aw + B\bar{w} = F, \quad (1)$$

where A , B and F are functions of z , is called a *generalized analytic function*.

The problems of this section deal with equations and systems of equations reducible to form (1), and also with some properties of their solutions.

1411. Show that the system of Carleman's equations

$$\begin{cases} u_x - v_y + au + bv = f, \\ u_y + v_x + cu + dv = g, \end{cases}$$

where a, b, c, d, f and g are functions of the variables x and y , can be written in the form (1) $\omega_z - q_1(z)\bar{\omega}_z + A\omega + B\bar{\omega} = F$.

Express A, B and F in terms of the coefficients of this system.

1412. Show that the equation $\omega_z - q_1(z)\bar{\omega}_z + A\omega + B\bar{\omega} = F$ can be reduced to the form (1) with the aid of the "affine" transformation

$$\omega = a(z)\omega + b(z)\bar{\omega}. \quad (2)$$

Find the general form of transformation (2) and find out when it is nonsingular.

1413. Show that the equation

$$\omega_z - q_1(z)\omega_z - q_2(z)\bar{\omega}_z + A\omega + B\bar{\omega} = F$$

can be reduced to the form $\omega_z - q'_1\omega_z + A'\omega + B'\bar{\omega} = F'$ by the transformation of the preceding problem. Find the general form of the transformation and find out when it is nonsingular.

Hint. Apply the transformation in question to this equation and to the equation $\omega_z - \bar{q}_1(z)\bar{\omega}_z - \bar{q}_2(z)\omega_z + \bar{A}\omega + \bar{B}\bar{\omega} = \bar{F}$, eliminate $\bar{\omega}_z$ and then select the coefficients $a(z)$ and $b(z)$.

1414. Show that the equation

$$\begin{aligned} &\omega_z - q_1(z)\omega_z - q_2(z)\bar{\omega}_z + A\omega + B\bar{\omega} = F \\ &(|q_1(z)| + |q_2(z)| < 1) \text{ can be reduced to the form} \end{aligned}$$

$$\omega_\zeta - q_1^*\bar{\omega}_\zeta + A\omega + B\bar{\omega} = F$$

by replacing the independent variable z by the variable ζ connected with z by the relation $\zeta_z = q_1^*\bar{\zeta}_z$.

Find q_1^* and q_2^* and elucidate the geometric meaning of the transformation $\zeta(z)$.

1415. Prove that an elliptic system of differential equations of the form

$$\begin{cases} \alpha_1 v_y = \alpha u_x + (\beta + \beta_1) u_y + au + bv + f, \\ -\alpha_1 v_x = (\beta - \beta_1) u_x + \gamma u_y + cu + dv + g \end{cases}$$

(here the ellipticity condition is $\alpha\gamma - \beta^2 > 0$; besides, $\alpha > 0$) can be reduced to the form $\omega_z - q_1(z)\omega_z - q_2(z)\bar{\omega}_z + A\omega + B\bar{\omega} = F$; here $|q_1(z)| + |q_2(z)| < 1$ if $\alpha_1 > 0$, and $\|q_1(z) - |q_2(z)\| > 1$ if $\alpha_1 < 0$.

Hint. See Problem 1403. The case $\alpha_1 < 0$ is reduced to the case $\alpha_1 > 0$ by substituting $\omega = u - iv$ for $\omega = u + iv$.

1416. Prove that if $w(z)$ is a continuously differentiable solution of the equation $w_z - q_2(z) \bar{w}_z = 0$, where $q_2(z)$ is an analytic function of z and $|q_2(z)| \neq 1$, then

$$w(z) = \frac{\varphi(z) + q_2(z) \bar{\varphi}(z)}{1 - |q_2(z)|^2},$$

where $\varphi(z)$ is an arbitrary analytic function.

1417. Prove that if $w(z)$ is a twice continuously differentiable solution of the equation $w_z - q_2(\bar{z}) \bar{w}_z = 0$, where $q_2(\bar{z})$ is an analytic function of $\bar{z}$ and $|q_2(\bar{z})| \neq 1$, then

$$w(z) = \varphi(z) + \int q_2(\bar{z}) \bar{\varphi}_z d\bar{z},$$

where $\varphi(z)$ is an arbitrary analytic function of z .

Hint. First prove that $w(z)$ is the sum of an analytic function of z and an analytic function of $\bar{z}$.

§ 3. CERTAIN INTEGRAL RELATIONS AND DOUBLE INTEGRALS

In the problems of this section G is a domain bounded by a contour C .

1418. Using Green's formula, prove the following relations [$f(x, y)$ and $g(x, y)$ are continuously differentiable in G]:

$$1) \iint_G \frac{\partial f}{\partial z} dx dy = \frac{1}{2i} \int_C f d\zeta;$$

$$2) \int_C f d\zeta + g d\bar{\zeta} = 2i \iint_G \left(\frac{\partial f}{\partial z} - \frac{\partial g}{\partial \bar{z}} \right) dx dy;$$

$$3) f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta) d\zeta}{\zeta - z} - \frac{1}{\pi} \iint_G \frac{\partial f(\zeta)}{\partial \bar{\zeta}} \frac{\partial \bar{\zeta}}{\zeta - z} \quad (\zeta = \xi + i\eta);$$

Hint. Take advantage of the fact that

$$\frac{\partial f(\zeta)}{\partial \bar{\zeta}} \frac{1}{\zeta - z} = \frac{\partial}{\partial \bar{\zeta}} \left(\frac{f(\zeta)}{\zeta - z} \right),$$

and apply the formula in item (1) to the domain G without the small circle $|\zeta - z| < \rho$ ($\rho \rightarrow 0$).

1419. Assuming that the functions f and g are continuously differentiable in a closed domain $\bar{G}$, prove that the expression $f dz + g d\bar{z}$ is the total differential of a certain function if and only if

$$\frac{\partial f}{\partial \bar{z}} = \frac{\partial g}{\partial z}.$$

1420. Prove that if f and g are functions analytic in the domain G and continuously differentiable on C , then

$$\iint_G f' \bar{g} \, dx \, dy = \frac{i}{2} \int_C f \bar{g} \, d\bar{z},$$

in particular

$$\iint_G f' \bar{g}' \, dx \, dy = \frac{i}{2} \int_C f \, d\bar{g}.$$

1421. Prove that if the function $f(z)$ maps a domain G conformally onto a domain G' bounded by the contour $C' = f(C)$, then the integral

$$I = \frac{i}{2} \int_C f \, d\bar{f}$$

is equal to the area S of the domain G' , but if $f(z)$ maps the domain G conformally onto the exterior of the contour C' , then $I = -S$.

1422. Let G be the circle $|z| < R < 1$ and

$$\varphi(\xi) = \frac{e^{2i\eta}}{\ln \frac{1}{\rho}} \quad (\xi = \rho e^{i\theta} = \xi + i\eta, \quad z = re^{i\varphi}).$$

Find the function

$$F(z) = \frac{1}{\pi} \iint_G \frac{\varphi(\xi) \, d\xi \, d\eta}{\xi - z} \quad (z \in G)$$

and its partial derivatives $\frac{\partial F}{\partial x}$, $\frac{\partial F}{\partial y}$, $\frac{\partial F}{\partial z}$, $\frac{\partial F}{\partial \bar{z}}$ for $z \neq 0$. Show that of these derivatives only $\frac{\partial F}{\partial \bar{z}}$ exists and is continuous at the origin of coordinates.

1423. Prove that if the function $f(z)$ is continuously differentiable, then

$$\frac{\partial f}{\partial \bar{z}} = \lim_{G \rightarrow z} \frac{\frac{1}{2i} \int_C f \, d\bar{z}}{\iint_G d\xi \, d\eta}$$

when the domain G shrinks to the point z .

Hint. Use the relation of Problem 1418(1).

1424. Let the function $\varphi(\xi)$ be continuous in a closed domain $\bar{G}$. Prove that the function

$$F(z) = -\frac{1}{\pi} \iint_G \frac{\varphi(\xi) \, d\xi \, d\eta}{\xi - z}$$

in the domain G satisfies the equation $\frac{\partial F}{\partial \bar{z}} = \varphi(z)$ if the derivative $\frac{\partial F}{\partial \bar{z}}$ is determined by means of the formula of Problem 1423.

1425. Prove that under the conditions of Problem 1424 the general solution of the equation

$$\frac{\partial f}{\partial \bar{z}} = \varphi(z)$$

can be represented (in the domain G) in the form

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\xi) d\xi}{\xi - z} - \frac{1}{\pi} \iint_G \frac{\varphi(\xi) d\xi d\eta}{\xi - z}.$$

Hint. Use the relation of Problem 1418(3).

ANSWERS AND SOLUTIONS

Chapter I

1. 1) $-i$; 2) $-i$; 3) $\frac{1}{5}(1+3i)$; 4) -8 . 2. 1) 3 , $\frac{\pi}{2}$ (here and henceforward only values of $\arg z$ are given); 2) 2 , π ; 3) $\sqrt{2}$, $\frac{\pi}{4}$; 4) $\sqrt{2}$, $-\frac{3\pi}{4}$; 5) $\sqrt{29}$, $\arctan \frac{5}{2}$; 6) $\sqrt{29}$, $-\arctan \frac{5}{2}$; 7) $\sqrt{29}$, $\pi - \arctan \frac{5}{2}$; 8) $\sqrt{29}$, $\arctan \frac{5}{2} - \pi$; 9) $|b|$, $\frac{\pi}{2} \frac{|b|}{b} = \frac{\pi}{2} \operatorname{sgn} b^{1)}$; 10) $\sqrt{a^2+b^2}$, $\arctan \frac{b}{a}$ for $a > 0$, $\arctan \frac{b}{a} + \pi$ for $a < 0$ and $b \geq 0$, $\arctan \frac{b}{a} - \pi$ for $a < 0$ and $b < 0$.

3. $z = \cos \varphi_k + i \sin \varphi_k$, where $\varphi_k = \frac{2k\pi}{n}$ $k=0, 1, \dots, n-1$; $z=0$.

4. 1) 1 , $-\frac{1}{2} \pm \frac{i\sqrt{3}}{2}$; 2) $\pm \frac{\sqrt{3}}{2} + \frac{i}{2}$, $-i$; 3) $\pm \frac{\sqrt{2}}{2}(1+i)$, $\pm \frac{\sqrt{2}}{2}(1-i)$; 4) $\pm \frac{\sqrt{2}}{2}(\sqrt{3}+i)$, $\pm \frac{\sqrt{2}}{2}(\sqrt{3}-i)$, $\pm \sqrt{2}i$; 5) $\pm 1 \pm i$, $\pm \frac{\sqrt{2}}{2}(1+i)$, $\pm \frac{\sqrt{2}}{2}(1-i)$; 6) $\pm \frac{\sqrt{2}}{2}(\sqrt{\sqrt{2}+1} - i\sqrt{\sqrt{2}-1})$;

7) $\pm(2+i)$; 8) $\sqrt{2} \left[\cos \frac{(2k+\frac{3}{4})\pi}{3} + i \sin \frac{(2k+\frac{3}{4})\pi}{3} \right]$ ($k=0, 1, 2$);

9) $\sqrt[5]{5} \left[\cos \frac{(2k+1)\pi - \arctan \frac{3}{4}}{5} + i \sin \frac{(2k+1)\pi - \arctan \frac{3}{4}}{5} \right]$ ($k=0, 1, 2, 3, 4$).

15. $z_k = z_1 \left(\cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n} \right)$ ($k=0, 1, 2, \dots, n-1$).

¹⁾ $\operatorname{sgn} b$ denotes:

$$\begin{aligned} \operatorname{sgn} b &= 1 \text{ for } b > 0, \\ \operatorname{sgn} b &= -1 \text{ for } b < 0. \end{aligned}$$

$$16. z_3 = z_2 + (z_2 - z_1) \left(\cos \frac{2\pi}{n} \pm i \sin \frac{2\pi}{n} \right). \quad 17. z_4 = z_1 + z_3 - z_2.$$

18. The ratio $\frac{z_3 - z_1}{z_2 - z_1}$ must be a real number (the condition is necessary

and sufficient). 19. The anharmonic ratio $(z_1, z_2, z_3, z_4) = \frac{z_1 - z_3}{z_1 - z_4} \cdot \frac{z_2 - z_4}{z_2 - z_3}$ must be a real number (the condition is necessary and sufficient).

20. *Solution.* Here we may assume (without loss of generality) that the straight line in question is the imaginary axis and that all the points under consideration are situated to the right of it (otherwise all z_k should be multiplied by a certain number of the form $\cos \alpha + i \sin \alpha$). Then it is obvious that

$\operatorname{Re} z_k > 0$ and $\operatorname{Re} \frac{1}{z_k} > 0$ for any k . 23. The interior of a circle of radius R with centre at point $z = z_0$; the exterior of the same circle; the circumference of the same circle. 24. An ellipse with foci at points $z = \pm 2$ and semi-major axis $\frac{5}{2}$. 25. The interior of the left-hand branch of a hyperbola with foci at

points $z = \pm 2$ and real semi-axis $\frac{3}{2}$. 26. A straight line perpendicular to the

segment joining points z_1 and z_2 and passing through the mid-point of this segment. 27. 1) A straight line $x = C$ and the half-plane located to the right of it; 2) the half-plane located below straight line $y = C$. 28. The strip $-1 < y < 0$. 29. The interior of an angle (containing the positive real axis)

with vertex at the origin and with sides forming angles α and β with the real axis; the interior of a similar angle with vertex at point z_0 . 30. A parabola $y^2 = 2x + 1$. 31. The half-plane bounded by the straight line $x + y = 1$ and containing the origin. 32. The straight line passing through points z_1 and z_2

(with point z_2 excluded); a circle whose diameter is a segment joining points z_1 and z_2 (with point z_2 removed). 33. The interior of the circles $|z - i| = \sqrt{2}$

and $|z + i| = \sqrt{2}$ with the exception of their common part. 34. 1) The interior of the domain bounded by the segment $0 \leq x \leq 2\pi$ of the real axis and by one turn of Archimedean spiral $r = \varphi$; 2) the set of points determined in item (1)

and complemented to the interval $(0, 2\pi)$ of the real axis. 35. 1) A family of circles tangent to the imaginary axis at the origin, and the imaginary axis itself (the equation of the family is $C(x^2 + y^2) = x$); 2) a family of circles tangent

to the real axis at the origin, and the real axis itself. 36. 1) A family of hyperbolas $x^2 - y^2 = C$; 2) a family of hyperbolas $xy = \frac{C}{2}$.

37. Every line is a circle which is the locus of points for which the ratio

of distances from points z_1 and z_2 is a constant (the circle of Apollonius with respect to the points z_1 and z_2). 38. A family of circular arcs with end-points

at z_1 and z_2 (this family also includes the two rectilinear segments with end-points at z_1 and z_2 ; one of these segments contains the point at infinity).

39. 1) Each line is the locus of points for which the product of the distances from points $z = -1$ and $z = 1$ is a constant (a lemniscate with foci $z = \pm 1$). For $\lambda > 1$ the lines of the family are simple closed curves, for $\lambda < 1$

they decompose into two simple closed curves, which shrink to the points ± 1 as $\lambda \rightarrow 0$. For $\lambda = 1$ we have Bernoulli's lemniscate; its equation in polar coordinates is $r^2 = 2 \cos 2\varphi$. 2) Lemniscates with foci at points z_1 and z_2 , where

z_1 and z_2 are the roots of the equation $z^2 + az + b = 0$. The lemniscates consist

of one line if $\lambda > \sqrt{\frac{|z_2 - z_1|}{2}}$ and of two lines if $\lambda < \sqrt{\frac{|z_2 - z_1|}{2}}$.

When $\lambda = \sqrt{\frac{|z_2 - z_1|}{2}}$ we have Bernoulli's lemniscate with the double

point $\frac{z_1 + z_2}{2}$.

40. 1) $|z|_{\max} = \frac{1}{2}(\sqrt{a^2 + 4} + a)$,

$|z|_{\min} = \frac{1}{2}(\sqrt{a^2 + 4} - a)$;

2) $|z|_{\max} = \frac{1}{2}(\sqrt{a^2 + 4|b|} + a)$,

$|z|_{\min} = \frac{1}{2}(\sqrt{a^2 + 4|b|} - a)$. 41. Spiral of Archimedes $r = \varphi$. 42. Logarithmic

spiral $r = e^{\varphi}$.

43. 1) π ; 2) $\frac{2\pi}{3}$; 3) 2π ; 4) π ; 5) 0.

44. $\xi = \frac{x}{x^2 + y^2 + 1}$,

$\eta = \frac{y}{x^2 + y^2 + 1}$,

$\zeta = \frac{x^2 + y^2}{x^2 + y^2 + 1}$,

$z = x + iy = \frac{\xi + i\eta}{1 - \zeta}$.

45. $\left(\frac{1}{2}, 0, \frac{1}{2}\right)$,

$\left(-\frac{1}{2}, 0, \frac{1}{2}\right)$, $\left(0, \frac{1}{2}, \frac{1}{2}\right)$, $\left(\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{1}{2}\right)$. All four points lie

on the equator, their respective longitudes being equal to $0, \pi, \frac{\pi}{2}, -\frac{\pi}{4}$ (a longitude is measured from the initial meridian contained in the plane ξ, ζ).

46. Circle of radius $\tan\left(\frac{\beta}{2} + \frac{\pi}{4}\right)$ with centre at $z = 0$. The "South" pole corresponds to the origin, the "North" pole, to the point at infinity.

47. 1) Semi-meridians with longitude α ; 2) parallels with latitude $\beta = 2 \arctan r - \frac{\pi}{2}$.

48. 1) Diametrically opposite points on one parallel; 2) points mutually symmetric about the initial meridian (i.e. with longitudes differing in sign); 3) points mutually symmetric about the plane of the equator (i.e. with the same longitude and with latitudes differing in sign). 49. $z_1 \times z_2 = -1$. 50. Points obtainable one from the other by a rotation of the sphere through 180° about the diameter parallel to the real axis of the z -plane. 51. 1) The Eastern hemisphere;

2) the Western hemisphere; 3) the hemisphere $-\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ (α the longitude);

4) the hemisphere $\frac{\pi}{2} < |\alpha| < \pi$; 5) the Southern hemisphere; 6) the Northern

hemisphere. 52. A family of circles tangent to one another at the "North pole" (pole of projections); to a straight line passing through the origin corresponds a great circle, and to a straight line parallel to it and distant d from the origin corresponds a circle contained in a plane inclined at an angle $\arctan d$ to the meridional plane. 53. To straight lines there correspond circles passing through

the "North" pole. 56. $k(z, a) = \frac{|z - a|}{\sqrt{1 + |z|^2} \sqrt{1 + |a|^2}}$; $k(z, \infty) = \frac{1}{\sqrt{1 + |z|^2}}$.

57. The circle of Apollonius $\left|\frac{z - z_1}{z - z_2}\right| = \frac{\sqrt{1 + |z_1|^2}}{\sqrt{1 + |z_2|^2}}$, in particular the straight

line $|z - z_1| = |z - z_2|$ if $|z_1| = |z_2|$, and the circle $|z - z_1| = \sqrt{1 + |z_1|^2}$ if $z_2 = \infty$.

59. 1, $e^{n!}$, $e^{\frac{\pi}{2}i}$, $e^{-\frac{\pi}{2}i}$, $\sqrt{2}e^{\frac{\pi}{4}i}$, $\sqrt{2}e^{-\frac{\pi}{4}i}$, $\sqrt{2}e^{\frac{3\pi}{4}i}$, $\sqrt{2}e^{-\frac{3\pi}{4}i}$. 60. $\pm i$; $(-1)^k$. 61. $e^2, 1; e^2, -3; e^3, 4 - 2\pi; e^{-3}, 2\pi - 4; a, \varphi - \pi$ if $\varphi > 0$, and $\varphi + \pi$ if $\varphi \leq 0$; 1, $-\varphi$ if $|\varphi| < \pi$, and π if $|\varphi| = \pi$; $2 \sin \frac{\alpha - \beta}{2}$;

$\frac{\alpha + \beta + \pi}{2}$ if $\alpha + \beta \leq \pi$, and $\frac{\alpha + \beta - 3\pi}{2}$ if $\alpha + \beta > \pi$. 62. 1) $\frac{\sin \frac{(n+1)x}{2} \cos \frac{nx}{2}}{\sin \frac{x}{2}}$;

$$2) \frac{\sin \frac{(n+1)x}{2} \sin \frac{nx}{2}}{\sin \frac{x}{2}}; \quad 3) \frac{\sin 2nx}{2 \sin x}; \quad 4) \frac{\sin^2 nx}{\sin x}; \quad 5) \frac{\sin \frac{(n+1)x}{2} \cos \frac{nx}{2}}{\cos \frac{x}{2}};$$

if n is an odd number; $-\frac{\cos \frac{(n+1)x}{2} \sin \frac{nx}{2}}{\cos \frac{x}{2}}$ if n is an even number.

$$63. \quad 1) \frac{\sin \frac{n+1}{2} \beta}{\sin \frac{\beta}{2}} \cos \left(\alpha + \frac{n\beta}{2} \right); \quad 2) \frac{\sin \frac{n+1}{2} \beta}{\sin \frac{\beta}{2}} \sin \left(\alpha + \frac{n\beta}{2} \right).$$

$$67. \quad 1) \sin z = \sin(x+iy) = \sin x \cosh y + i \cos x \sinh y, \quad |\sin z| = \sqrt{\sinh^2 y + \sin^2 x};$$

$$2) \cos z = \cos x \cosh y - i \sin x \sinh y, \quad |\cos z| = \sqrt{\sinh^2 y + \cos^2 x};$$

$$3) \tan z = \frac{\sin 2x + i \sinh 2y}{\cos 2x + \cosh 2y}, \quad |\tan z| = \frac{\sqrt{\sin^2 2x + \sinh^2 2y}}{\cos 2x + \cosh 2y};$$

$$4) \sinh z = \sinh x \cos y + i \cosh x \sin y, \quad |\sinh z| = \sqrt{\sinh^2 x + \sin^2 y};$$

$$5) \cosh z = \cosh x \cos y + i \sinh x \sin y, \quad |\cosh z| = \sqrt{\sinh^2 x + \cos^2 y};$$

$$6) \tanh z = \frac{\sinh 2x + i \sin 2y}{\cosh 2x + \cos 2y}, \quad |\tanh z| = \frac{\sqrt{\sinh^2 2x + \sin^2 2y}}{\cosh 2x + \cos 2y}.$$

$$68. \quad 1) \cos 2 \cosh 1 - i \sin 2 \sinh 1; \quad 2) i \sinh 2; \quad 3) \frac{\sin 4 - i \sinh 2}{2(\cos^2 2 + \sinh^2 1)};$$

$$4) \frac{8+15i}{17}; \quad 5) \frac{\sinh 4 - i \sin 2}{\cosh 4 - \cos 2}; \quad 6) \frac{40+9i}{41}. \quad 69. \quad \operatorname{Im} e^z = 0 \quad \text{if} \quad \operatorname{Im} z = k\pi;$$

$$\operatorname{Re} e^z = 0 \quad \text{if} \quad \operatorname{Im} z = (2k+1)\frac{\pi}{2}; \quad \operatorname{Im} \cos z = 0 \quad \text{if} \quad \operatorname{Re} z = k\pi \quad \text{or} \quad \operatorname{Im} z = 0;$$

$$\operatorname{Re} \cos z = 0 \quad \text{if} \quad \operatorname{Re} z = (2k+1)\frac{\pi}{2}; \quad \operatorname{Im} \sin z = 0 \quad \text{if} \quad \operatorname{Re} z = (2k+1)\frac{\pi}{2}$$

$$\text{or } \operatorname{Im} z = 0; \quad \operatorname{Re} \sin z = 0 \quad \text{if} \quad \operatorname{Re} z = k\pi; \quad \operatorname{Im} \tan z = 0 \quad \text{if} \quad \operatorname{Im} z = 0; \quad \operatorname{Re} \tan z = 0$$

$$\text{if } \operatorname{Re} z = \frac{k\pi}{2}; \quad \operatorname{Im} \cosh z = 0 \quad \text{if} \quad \operatorname{Im} z = k\pi \quad \text{or} \quad \operatorname{Re} z = 0; \quad \operatorname{Re} \cosh z = 0 \quad \text{if}$$

$$\operatorname{Im} z = (2k+1)\frac{\pi}{2}; \quad \operatorname{Im} \coth z = 0 \quad \text{if} \quad \operatorname{Im} z = \frac{k\pi}{2}; \quad \operatorname{Re} \coth z = 0 \quad \text{if} \quad \operatorname{Re} z = 0;$$

$$k \text{ is an integer throughout } (k=0, \pm 1, \pm 2, \dots). \quad 70. \quad 1) \operatorname{Re} z = \frac{\pi}{4} + \frac{k\pi}{2};$$

$$2) \operatorname{Im} z = \frac{\pi}{4} + \frac{k\pi}{2}. \quad 71. \quad 1) \ln 4 + 2k\pi i, (2k+1)\pi i, \pi i; \quad 2) \left(2k + \frac{1}{2}\right)\pi i,$$

$$\frac{\pi i}{2}; \quad 3) \left(2k \pm \frac{1}{4}\right)\pi i; \quad 4) \frac{1}{2} \ln 13 + \left(2k\pi - \arctan \frac{3}{2}\right)i;$$

$$\frac{1}{2} \ln 13 + \left[(2k+1)\pi - \arctan \frac{3}{2}\right]i. \quad 72. \quad \text{The set of values of } 2 \operatorname{Ln} z \text{ constitutes only a part of the set of values of } \operatorname{Ln}(z^2). \quad 73. \quad 1) 4\pi; \quad 2) -2\pi; \quad 3) 0; \quad 4) 4\pi.$$

$$74. \quad 1) \cos(2k\sqrt{2}\pi) + i \sin(2k\sqrt{2}\pi); \quad 2) 2^{1/2} [\cos(2k+1)\pi\sqrt{2} + i \sin(2k+1)\pi\sqrt{2}];$$

$$3) e^{2k\pi} (\cos \ln 2 + i \sin \ln 2); \quad 4) e^{2k\pi}; \quad 5) e^{\left(2k - \frac{1}{2}\right)\pi}; \quad 6) \frac{1-i}{\sqrt{2}} e^{\left(2k + \frac{1}{4}\right)\pi};$$

$$7) 5e^{\arctan \frac{4}{3} + 2k\pi} \left[\cos \left(\ln 5 - \arctan \frac{4}{3} \right) + i \sin \left(\ln 5 - \arctan \frac{4}{3} \right) \right];$$

$$8) -5e^{\arctan \frac{4}{3} + (2k+1)\pi} \left[\cos \left(\ln 5 - \arctan \frac{4}{3} \right) + i \sin \left(\ln 5 - \arctan \frac{4}{3} \right) \right].$$

Everywhere k is an integer ($k=0, \pm 1, \pm 2, \dots$).

76. The sets of values of $a^{2\pi}$ and $(a^2)^{\pi}$ coincide with one another, but do not coincide, generally speaking, with the set of values of $(a^2)^{\pi}$.

80. 1) $\operatorname{Im} \operatorname{Arc} \cos z = \operatorname{Im} \operatorname{Arc} \sin z = 0$ if z is a real number and $|z| \leq 1$. $\operatorname{Im} \operatorname{Arc} \tan z = 0$ if z is a real number;

2) $\operatorname{Re} \operatorname{Arsinh} z = 0$ if z is a purely imaginary number and $|z| \leq 1$.

$$81. 1) \frac{\pi}{6} + 2k\pi, \frac{5\pi}{6} + 2k\pi; 2) 2k\pi \pm \frac{\pi}{3}; 3) 2k\pi \pm i \ln(2 + \sqrt{3}); 4) 2k\pi - i \ln(\sqrt{2} - 1), (2k+1)\pi - i \ln(\sqrt{2} + 1); 5) \frac{1}{2} \left[\arctan \frac{1}{2} + (2k+1)\pi \right] + \frac{i}{4} \ln 5;$$

$$6) \ln(\sqrt{5} \pm 2) + \left(2k \pm \frac{1}{2} \right) \pi i; 7) \frac{1}{4} \ln 5 + \left[\frac{1}{2} \arctan 2 + \left(k + \frac{1}{2} \right) \pi \right] i.$$

Everywhere k is an integer ($k=0, \pm 1, \pm 2, \dots$).

$$82. 1) z = \frac{\pi}{4} + 2k\pi - i \ln(\sqrt{2} \pm 1); 2) z = \frac{3\pi}{4} + 2k\pi - i \ln \frac{3 \pm \sqrt{7}}{\sqrt{2}};$$

$$3) z = \frac{\pi}{4} + 2k\pi - i \ln \frac{\sqrt{3}-1}{\sqrt{2}} \text{ and } z = -\frac{3\pi}{4} + 2k\pi - i \ln \frac{\sqrt{3}+1}{\sqrt{2}}; 4) z = 2k\pi i;$$

$$5) z = -\ln 2 + (2k+1)\pi i; 6) z = \left(2k + \frac{1}{2} \right) \pi i \text{ and } z = -\ln 3 + \left(2k - \frac{1}{2} \right) \pi i.$$

Everywhere k is an integer.

$$83. 1) z = k\pi(1 \pm i); 2) z = k\pi(1+i) \text{ and } z = \frac{(2k+1)\pi}{1+i}; 3) z = \frac{(4k+1)\pi}{2(1+2i)} \text{ and } z = \frac{(4k-1)\pi}{2(1-2i)}.$$

Everywhere k is an integer.

95. Converges absolutely. 96. Converges absolutely. 97. Diverges.

98. Converges non-absolutely. 99. Converges non-absolutely for $\varphi \neq 2k\pi$ ($k=0, \pm 1, \pm 2, \dots$), diverges for $\varphi = 2k\pi$. 100. Converges absolutely.

101. Diverges. 102. Converges absolutely. 103. Diverges. 104. Converges absolutely.

105. 1) $z=0$ and $z=2$; 2) $z=0$, $z=\frac{1}{m}$, $z=\frac{i}{n}$ (m and n any integers); 3) all points of the plane; 4) all points of the circle $|z| \leq 1$. 107. 2) When $\lim z_n$

equals 0 or ∞ . 109. A segment of a straight line: $x=1$, $-2 \leq y \leq 0$.

110. Parabola $y=x^2$. 111. The right-hand half of the parabola $y=x^2$, traversed twice. 112. The left semicircle of radius a with centre at $z=0$.

113. A branch of the hyperbola $y=\frac{1}{x}$, lying in the third quadrant.

114. 1) The upper semicircle of radius 1 with centre at $z=0$; 2) A quarter circumference of radius 1 with centre at $z=0$ and located in the first quadrant.

115. 1) Cycloid: $x=a(t-\sin t)$, $y=a(1-\cos t)$; 2) the first (counting from the origin) arc of a prolate ($a < b$), curtate ($a > b$) or ordinary ($a=b$) cycloid: $x=at-b\sin t$, $y=a-b\cos t$. 116. 1) Images of straight lines $x=C$ are

parabolas $u=C^2-\frac{v^2}{4C^2}$ for $C \neq 0$, semi-axis $v=0$, $u \leq 0$ for $C=0$; images of

straight lines $y=C$ are parabolas $u=\frac{v^2}{4C^2}-C^2$ for $C \neq 0$, semi-axis $v=0$, $u \geq 0$ for $C=0$; the image of the straight line $y=x$ is the semi-axis $u=0$, $v \geq 0$; the images of circles $|z|=R$ are circles $|w|=R^2$; the images of rays

$\arg z = \alpha$ are rays $\arg w = 2\alpha$; one-to-one mapping in case of straight lines $x=C$, $y=C$ for $C \neq 0$ and rays $\arg z = \alpha$; 2) preimages of straight lines $u=C$ are hyperbolas $x^2 - y^2 = C$ (for $C=0$, a pair of straight lines), preimages of straight lines $v=C$ are hyperbolas $xy = \frac{C}{2}$ (for $C=0$, a pair of straight lines).

117. 1) Images of straight lines $x=C$ are circles $u^2 + v^2 - \frac{u}{C} = 0$ (for $C=0$,

axis $u=0$); images of straight lines $y=C$ are circles $u^2 + v^2 + \frac{v}{C} = 0$ (for $C=0$,

axis $v=0$); images of circles $|z|=R$ are circles $|w| = \frac{1}{R}$; images of rays $\arg z = \alpha$ are rays $\arg w = -\alpha$; the image of the circle $|z-1|=1$ is the straight line $u = \frac{1}{2}$; 2) preimages of straight lines $u=C$ are circles $x^2 + y^2 - \frac{x}{C} = 0$

(for $C=0$, axis $x=0$); preimages of straight lines $v=C$ are circles $x^2 + y^2 + \frac{y}{C} = 0$

(for $C=0$, axis $y=0$). 118. The function $w = z + \frac{1}{z}$ maps circles $|z|=R \neq 1$

onto ellipses $\frac{u^2}{\left(R + \frac{1}{R}\right)^2} + \frac{v^2}{\left(R - \frac{1}{R}\right)^2} = 1$, and the circle $|z|=1$ onto the

segment $v=0$, $-2 \leq u \leq 2$; the function $w = z - \frac{1}{z}$ maps circles $|z|=R \neq 1$

onto ellipses $\frac{u^2}{\left(R - \frac{1}{R}\right)^2} + \frac{v^2}{\left(R + \frac{1}{R}\right)^2} = 1$, and the circle $|z|=1$ onto the

segment $u=0$, $-2 \leq v \leq 2$. 119. The preimage of the family $u=C$ is the family $x(x^2 + y^2 + 1) = C(x^2 + y^2)$; the preimage of the family $v=C$ is the family $y(x^2 + y^2 - 1) = C(x^2 + y^2)$ (Fig. 59). 120. Into a ray passing along the negative

real axis from $w = -\frac{1}{4}$ to $w = \infty$. 121. 1) Circles $\rho = e^C$, rays $\theta = C$, spiral

$\rho = e^{\theta}$; 2) lines $y = e^x + 2k\pi$. 122. 1) The family of straight lines $x=C$ is trans-

formed into the family $v^2 = -4a^2 \left(u - a^2 + \frac{1}{4}\right)$ (parabolas with focus at point

$w = -\frac{1}{4}$, $a = C + \frac{1}{2}$), containing also a ray passing along the negative real

axis ($a=0$) from the point $w = -\frac{1}{4}$; the family $y=C$ is transformed into a fa-

family of confocal parabolas: $v^2 = 4C^2 \left(u + C^2 + \frac{1}{4}\right)$, containing also a ray ema-

nating from point $w = -\frac{1}{4}$ and passing along the real axis in the positive direc-

tion; 2) the family $x=C$ is transformed into a family of circles of Apollonius

with respect to the points $w=-1$ and $w=1$ (the latter family includes the

imaginary axis as well); the equation of a family of circles of Apollonius:

$(u-a)^2 + v^2 = a^2 - 1$; $|a| > 1$ ($a = \coth 2C$); the family $y=C$ is transformed into

a pencil of circular arcs with end-points at $w=-1$ and $w=1$; the pencil also

includes appropriate parts of the real axis; the equation of the pencil of circles:

$u^2 + (v+b)^2 = 1 + b^2$ ($b = \cot 2C$); 3) the family $x=C$ is transformed into a family

of spirals $\rho = e^{C - \frac{\theta^2}{4C^2}}$; to the axis $x=0$ there corresponds a segment $\theta=0$,

$0 < \rho \leq 1$; the family $y=C$ is transformed into a family of spirals $\rho = e^{\frac{\theta^2}{4C^2} - C^2}$; to the axis $y=0$ there corresponds the ray $\theta=0$, $1 \leq \rho < \infty$. 123. To the lines $y=C$ there correspond the lines $u=x+e^x \cos C$, $v=C+e^x \sin C$; to segments of straight lines $x=C$, arcs of the lines $u=C+e^C \cos y$, $v=y+e^C \sin y$.

124. 1) To the family $|w|=R$ there correspond circles $r = \frac{\cos \varphi}{\ln R}$ ($R \neq 1$) and the imaginary axis ($R=1$); to each ray $\arg w = \alpha$ there corresponds a family of

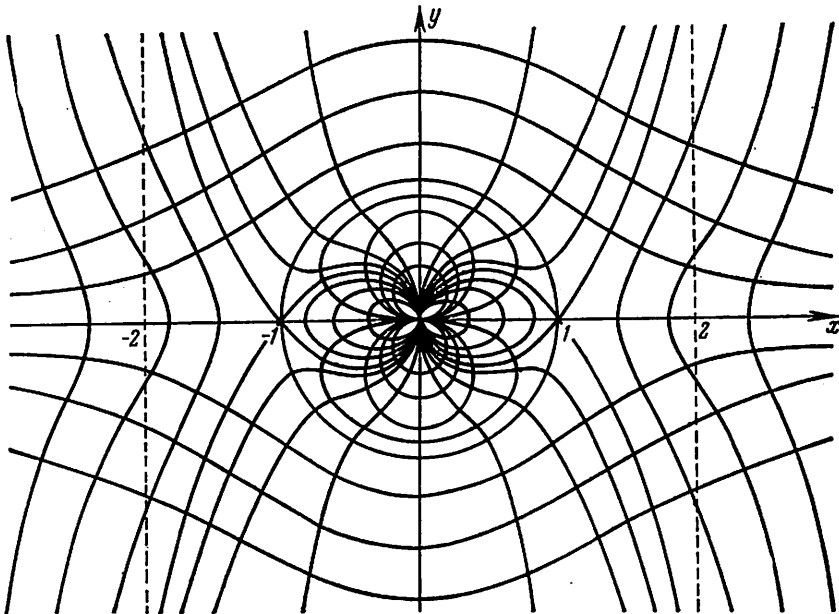


FIG. 59

circles $r = \frac{\sin \varphi}{2k\pi - \alpha}$; for $\alpha=0$, this family includes the real axis (for $k=0$);

2) hyperbolas $x^2 - y^2 = \ln R$ and $2xy = \alpha + 2k\pi$. 126. Only $f(z) = \frac{z \operatorname{Re} z}{|z|}$ ($f(0)=0$).

127. 1) and 2) Continuous but not uniformly. 128. 2) No; 3) yes. 132. 1) $c=1$, $b=-a$; $f(z) = (1-ai)z$; 2) $a=b=-1$; $f(z) = e^{iz}$. 133. The function is analytic for $0 < \arg z < \frac{\pi}{4}$, $\pi < \arg z < \frac{5\pi}{4}$ ($f(z)=z^2$) and for $\frac{\pi}{2} < \arg z < \frac{3\pi}{4}$, $\frac{3\pi}{2} < \arg z < \frac{7\pi}{4}$ ($f(z)=-z^2$).

135. $r \frac{\partial u}{\partial r} = \frac{\partial v}{\partial \varphi}$, $\frac{\partial u}{\partial \varphi} = -r \frac{\partial v}{\partial r}$. 138. 0. 155. 1) No if $u \neq \text{const}$; 2) $f(u) = au + b$. 156. $|f(z)|$ is not a harmonic function, $\arg f(z)$ and $\ln |f(z)|$ are harmonic functions. 157. $\Delta u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \varphi^2}$;

$u = C_1 \ln r + C_2$. 158. $p_1 = x$, $q_1 = y$; $p_2 = x^2 - y^2$, $q_2 = 2xy$, $p_3 = x^3 - 3xy^2$, $q_3 = 3x^2y - y^3$, $p_4 = x^4 - 6x^2y^2 + y^4$, $q_4 = 4x^3y - 4xy^3$, $p_n = r^n \cos n\varphi$, $q_n = r^n \sin n\varphi$.

159. $v(x, y) = 2xy + y + C$. 160. $v(x, y) = -\frac{y}{x^2 + y^2} + C$. 161. a) $v(x, y) = \arg z + C$;

b) $v(x, y) = \arg z + 2m\pi + C$. 162. a) $v(x, y) = \arg z - \arg(z-1) + 2m\pi + C$;
 b) $v(x, y) = \arg z - \arg(z-1) + C$; c) $v(x, y) = \arg z - \arg(z-1) + 2m\pi + C$.

163. a) $v(x, y) = \sum_{k=1}^n \alpha_k \arg(z-z_k) + 2\pi \sum_{k=1}^n m_k \alpha_k + C$; b) $v(x, y) =$
 $= \sum_{k=1}^n \alpha_k \arg(z-z_k) + 2\pi m \sum_{k=1}^n \alpha_k + C$ (if $\sum_{k=1}^n \alpha_k = 0$, then the function $v(x, y)$
 is single-valued in the domain under consideration!). 164. 1) Exists; 2) exists;
 3) does not exist. 165. $f(z) = z^2 + (5-i)z - \frac{i}{z} + Ci$. 166. $f(z) = ze^z + 2i \cos z +$

$+ z^3 - iz + Ci$. 167. $f(z) = \frac{1}{2z} + iz^3 + 3i + C$. 168. $f(z) = 2i \ln z - (2-i)z + C$.
 Everywhere C is an arbitrary real constant. 169. $u = C_1 x + C_2$. 170. $u =$
 $= C_1(ax + by) + C_2$. 171. $u = C_1 \arctan \frac{y}{x} + C_2$. 172. $u = C_1 xy + C_2$. 173. $u =$
 $= C_1 \ln(x^2 + y^2) + C_2$. 174. $u = \frac{C_1 x}{x^2 + y^2} + C_2$. 175. $u = C_1 \sqrt{x + \sqrt{x^2 + y^2}} + C_2$.

176. Does not exist. 177. $f(z) = e^{iaz^2} e^z$. 178. $f(z) = e^{iaz^2}$. 179. $f(z) = Ae^{\frac{z^2}{2}}$.
 180. $f(z) = Aze^z$ (α is an arbitrary real constant, A is an arbitrary positive
 constant). 182. $az + \lambda$, $iaz + \lambda$, λe^{az} , λe^{iaz} . 183. $iaz + \lambda$, $az + \lambda$, λe^{iaz} , λe^{az} .
 184. $ai \ln z + \lambda$, $a \ln z + \lambda$, $\lambda e^{ai \ln z}$, $\lambda e^{a \ln z}$. 185. $a \ln z + \lambda$, $ai \ln z + \lambda$; $\lambda e^{a \ln z}$,
 $\lambda e^{ai \ln z}$. 186. $\frac{a}{z} + \lambda$, $\frac{ai}{z} + \lambda$, $\lambda e^{\frac{a}{z}}$, $\lambda e^{\frac{ai}{z}}$ (a is an arbitrary real constant, λ an

arbitrary complex constant). 187. For $w = z^2$: 1) $\vartheta = 0$, $k = 2$; 2) $\vartheta = \pi$, $k = \frac{1}{2}$;
 3) $\vartheta = \frac{\pi}{4}$, $k = 2\sqrt{2}$; 4) $\vartheta = \pi - \arctan \frac{4}{3}$, $k = 10$. For $w = z^3$: 1) $\vartheta = 0$, $k = 3$;
 2) $\vartheta = 0$, $k = \frac{3}{16}$; 3) $\vartheta = \frac{\pi}{2}$, $k = 6$; 4) $\vartheta = -2 \arctan \frac{4}{3}$, $k = 75$. 188. 1) Con-
 traction for $|z| < \frac{1}{2}$, stretching for $|z| > \frac{1}{2}$; 2) contraction for $|z+1| < \frac{1}{2}$,
 stretching for $|z+1| > \frac{1}{2}$; 3) contraction for $|z| > 1$, stretching for $|z| < 1$;
 4) contraction for $\operatorname{Re} z < 0$, stretching for $\operatorname{Re} z > 0$; 5) contraction for $|z-1| > 1$,
 stretching for $|z-1| < 1$. 189. $S = \int_G |f'(z)|^2 dx dy$, $L = \int_l |f'(z)| ds$.

190. $\sqrt{2}(e^{2\pi} - 1)$. 191. $2e^2(e^2 - 1)$. 192. Domain D is the annulus $e \leq |w| \leq e^2$.
 The formula of Problem 189 cannot be used, since the mapping is not one-to-one.

Chapter II

193. $w = (1+i)(1-z)$. 194. $w = (2+i)z + 1 - 3i$. 195. 1) $z_0 = -1 + 3i$, $\vartheta = 0$,
 $k = 2$, $w + 1 - 3i = 2(z + 1 - 3i)$; 2) $z_0 = 2 + 2i$, $\vartheta = \frac{\pi}{2}$, $k = 1$, $w - 2 - 2i =$
 $= i(z - 2 - 2i)$; 3) there is no finite fixed point; 4) if $a = 1$, then there is no
 finite fixed point; if $a \neq 1$, then $z_0 = \frac{w_1 - az_1}{1-a}$, $\vartheta = \arg a$, $k = |a|$, $w - \frac{w_1 - az_1}{1-a} =$
 $= a \left(z - \frac{w_1 - az_1}{1-a} \right)$; 5) if $a = 1$, then there is no finite fixed point; if $a \neq 1$,

then $z_0 = \frac{b}{1-a}$, $\varphi = \arg a$, $k = |a|$, $w - \frac{b}{1-a} = a \left(z - \frac{b}{1-a} \right)$. 196. 1) $w = az + b$; 2) $w = -az + b$; 3) $w = -i(az + b)$; 4) $w = az + bi$. Everywhere a and b are real numbers and $a > 0$. 197. 1) $w = z + bi$ or $w = -z + 1 + bi$; 2) $w = z + b$ or $w = -z - i + b$; 3) $w = z + b(1+i)$ or $w = -z + 1 + b(1+i)$. Everywhere b is a real number. Points can correspond to each other if they lie either on a straight line parallel to the boundaries of the strip or on parallel lines symmetric about the centre-line of the strip. The mapping is not defined uniquely if the corresponding points lie on the centre-line of the strip. 198. 1) $w = \frac{z-a}{h}$;

$$2) \quad w = \frac{-z+a+h}{h} + i; \quad 3) \quad w = \frac{\sqrt{1+k^2}}{b} e^{-i\left(\frac{\pi}{2} + \arctan k\right)} z;$$

$$4) \quad w = \frac{\sqrt{1+k^2}}{b_2-b_1} e^{-i\left(\frac{\pi}{2} + \arctan k\right)} (z - ib_1). \quad 199. \quad w = e^{i\alpha} R z + w_0. \quad 200. \quad 1) \quad \text{The}$$

family of straight lines $u = \frac{1}{a}$ parallel to the imaginary axis (the family does not include the imaginary axis itself); 2) the family of straight lines $v = -\frac{1}{b}$ parallel to the real axis (the family does not include the real axis itself); 3) the family of circles $b(u^2 + v^2) + u + v = 0$ tangent to the straight line $v = -u$ at the origin (the family includes the line itself); 4) the pencil of straight lines $v = -ku$; 5) the pencil of circles passing through the origin and through point $w_0 = \frac{1}{z_0}$ (this pencil includes the straight line passing through points $w=0$ and $w=w_0$); 6) cissoid $u^2 = -\frac{v^3}{v+1}$. 201. Into families of circles tangent at point $w=h$ to straight lines parallel to the imaginary and real axes respectively (including the straight lines themselves); the equations of these families are:

$$\begin{aligned} (C-x_0)[(u-h_1)^2 + (v-h_2)^2] - (u-h_1) &= 0; \\ (C-y_0)[(u-h_1)^2 + (v-h_2)^2] + (v-h_2) &= 0, \end{aligned}$$

where $z_0 = x_0 + iy_0$, $h = h_1 + ih_2$; 2) into a family of circles with centre at point $w = h \left(|w-h| = \frac{1}{R} \right)$ and a family of rays emanating from point $w = h \left(\arg(w-h) = -\alpha \right)$.

202. 1) The equation of a family of circles of Apollonius with respect to points z_1 and z_2 : $\left| \frac{z-z_1}{z-z_2} \right| = \lambda$. The end-points A and B of a diameter lying on a straight line that passes through points z_1 and z_2 (Fig. 60) divide the segment $z_1 z_2$ internally and externally in the ratio λ .

Using the notation in the figure (C is the centre of a circle of diameter AB) and denoting $|z_1 - z_2| = d$, we have the following relations for $\lambda < 1$: $\lambda = \cos \alpha = \frac{r_1}{R} = \frac{R}{r_2}$, $R = \frac{\lambda d}{1-\lambda^2}$, $r_1 = \frac{\lambda^2 d}{1-\lambda^2}$, $r_2 = \frac{d}{1-\lambda^2}$; 2) arcs of the circles passing

through points z_1 and z_2 : $\arg \frac{z-z_1}{z-z_2} = \theta$. The arcs corresponding to the values $\theta = \alpha$ and $\theta = \pi - \alpha$ are complements of one another in forming a complete circle; 3) to the polar grid there corresponds (Fig. 61) a grid made up of the circles of Apollonius $\left| \frac{z-z_1}{z-z_2} \right| = R$ and of arcs $\arg \frac{z-z_1}{z-z_2} = \theta$ orthogonal to them

(if $\theta > 0$, then the arc is situated to the right of the direction $\overrightarrow{z_1 z_2}$; if $\theta < 0$,

then to the left); 4) to the upper semicircle there corresponds the right angle shown in the figure. 203. Into the semicircle $|w| < 1$, $\text{Im } w < 0$. 204. Into a domain containing the point $w=0$ and bounded by two arcs of the circles $|w|=1$

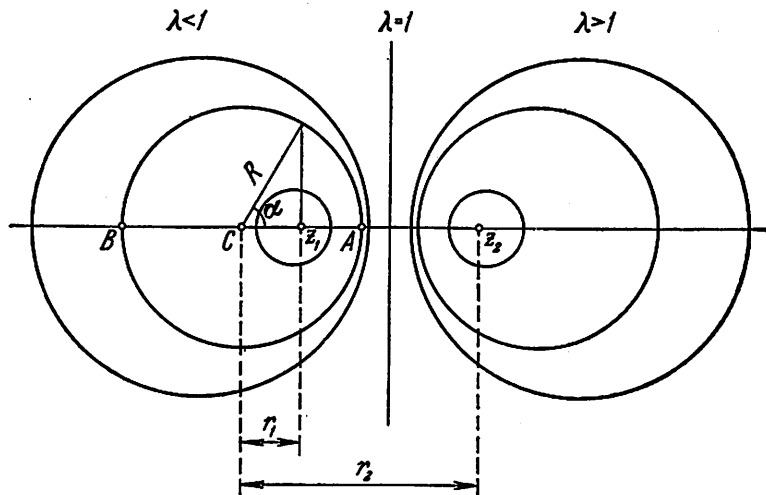


FIG. 60

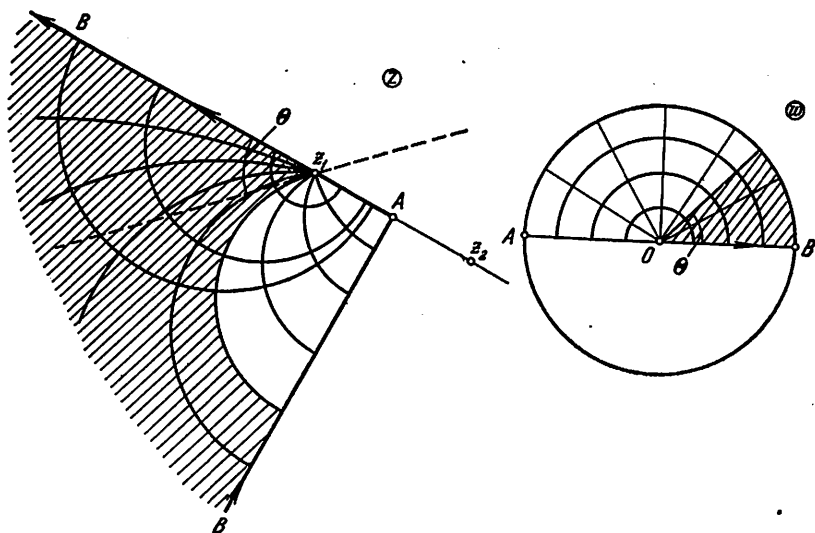


FIG. 61

and $\left|w + \frac{5i}{4}\right| - \frac{3}{4}$. 205. Into the domain obtained from the lower half-plane ($\text{Im } w < 0$) by removing a part of the circle $\left|w - \frac{1}{2} + \frac{i}{2}\right| < \frac{\sqrt{2}}{2}$ contained in that half-plane. 206. 1) Into a domain bounded by the straight line $\text{Re } w = 1$

- and a circle $\left|w - \frac{1}{2}\right| = \frac{1}{2}$ tangent to it; 2) into a domain bounded by the circles $\left|w - \frac{1}{2}\right| = \frac{1}{2}$ and $\left|w - \frac{3}{4}\right| = \frac{1}{4}$ tangent to one another. 207. Into a doubly connected domain whose boundary consists of the straight line $\operatorname{Re} w = \frac{1}{2}$ and the circle $\left|w - \frac{4}{3}\right| = \frac{2}{3}$. 208. 1) $w = -\frac{d}{z} + 1 + hi$ or $w = \frac{d}{z} + hi$; 2) $w = \frac{d_1}{d_2 - d_1} \times \left(\frac{d_2}{z} - 1\right) + hi$ or $w = \frac{d_2}{d_1 - d_2} \left(\frac{d_2}{z} - 1\right) + 1 + hi$; 3) $w = \frac{d_1(z - d_2)}{z(d_1 + d_2)}$.
209. 1) $w = -\frac{2i(z+1)}{4z-1-5i}$; 2) $w = \frac{(1+2i)z+6-3i}{5(z-i)}$. 210. 1) $w = \frac{(1+i)z+1+3i}{(1+i)z+3+i}$; 2) $w = \frac{iz+2+i}{z+1}$; 3) $w = \frac{1-i}{2}(z+1)$. 211. 1) $w = \frac{(-1+3i)z+1-i}{(1+i)z-1+i}$; 2) $w = \frac{z(1-4i)-2(1-i)}{2z(1-i)-(4-i)}$; 3) $w = \frac{z(3-i)-(1+i)}{(1+i)(1-z)}$. 212. $w = \frac{z-i}{iz-1}$; the upper half-plane goes into the unit circle. 213. 1) $w = \frac{az+b}{cz+d}$, where a, b, c, d are real numbers and $ad-bc > 0$; 2) $w = \frac{az+b}{cz+d}$, where a, b, c, d are real numbers and $ad-bc < 0$; 3) $w = i \frac{az+b}{cz+d}$, where a, b, c, d are real numbers and $ad-bc < 0$. 214. 1) $w = \frac{2}{2-z}$, 2) $w = -2 \left(\frac{2z+1}{z-2}\right)$. 215. $w = \frac{R-z}{R+z}$; the image of the upper semicircle is the angle $u > 0, v < 0$. 216. 1) $\frac{2+i}{5}$; 2) $\frac{9}{2} + i$.
217. 1) $|z|=2$; 2) straight line $x = \frac{1}{2}$; 3) $\left|z - \frac{i}{4}\right| = \frac{1}{4}$; 4) $xx_0 + yy_0 = \frac{1}{2}$; $|z - z_0| = \sqrt{|z_0|^2 - 1}$ (i.e. this circle is symmetric to itself about the unit circle); 6) $(x^2 + y^2)^2 - (x^2 - y^2) = 0$ (lemniscate); 7) curvilinear triangle with vertices at points $\frac{1}{z_1}, \frac{1}{z_2}, \frac{1}{z_3}$, whose sides are arcs of circles passing through two vertices and point $z=0$ (one of the arcs may be a segment of a straight line). 219. 1) $\theta(x) = \alpha + 2\arg(x - \beta)$; 2) $w'(\beta) = \frac{e^{\left(\alpha - \frac{\pi}{2}\right)i}}{2b}$; 3) if $b \geq 2$, the entire half-plane contracts; if $b < 2$, the domain lying inside the circle $|z - \beta| < \sqrt{2b}$ stretches. (The circle $|z - \beta| = \sqrt{2b}$ is called *isometric*.) 220. 1) $w = \frac{z-i}{z+i}$, 2) $w = i \frac{z-2i}{z+2i}$; 3) $w = e^{i\left(\frac{\pi}{2} + \theta\right)} \frac{z-(a+bi)}{z-(a-bi)}$ 221 $w = R_1 \frac{z-i}{z+i} + w_0$.
222. $w = -\frac{z-2i}{z+2i}$. 223 $w = -4 \frac{zi+2}{z-2-4i}$. 224 $w = ke^{\frac{1}{2}\left(\pi + \arg \frac{z_2}{z_1}\right)i} \frac{z-z_1}{z-z_2}$, where $k > 0$. To the rays emanating from the point $w=0$ in the half-plane $\operatorname{Re} w > 0$ there correspond circular arcs lying inside the disk $|z| < 1$ and passing through points z_1 and z_2 ; to the semicircles with centre at $w=0$ lying in the half-plane $\operatorname{Re} w > 0$ there correspond arcs of the circles of Apollonius with respect to points z_1 and z_2 situated within the disk $|z| < 1$. 225. $w_0 =$

$$= \frac{\bar{z}_2 - z_1}{z_2 - z_1}, \quad R = \frac{|z_2 - z_1|}{2|\operatorname{Im} z_2|}. \quad 226. \quad \frac{w-b}{w-\bar{b}} = e^{i\alpha} \frac{z-a}{z-\bar{a}}. \quad 227. \quad \frac{w-\bar{a}}{w-a} = i \frac{z-\bar{a}}{z-a}.$$

$$228. \quad 1) \theta(\varphi) = \alpha - \varphi + 2 \arg(e^{i\varphi} - a) = \alpha - \varphi + 2 \arctan \frac{\sin \varphi - \lambda \sin \vartheta}{\cos \varphi - \lambda \cos \vartheta}, \quad \text{where } a = \lambda e^{i\vartheta}. \quad 2) w'(0) = (1 - |a|^2) e^{i\alpha}, \quad w'(a) = \frac{e^{i\alpha}}{1 - |a|^2}. \quad 3) \text{ If } a \neq 0, \text{ the domain}$$

lying inside the circle $\left| z - \frac{1}{a} \right| < \sqrt{\frac{1}{|a|^2} - 1}$ stretches, and the domain located outside this circle contracts. (The circle $\left| z - \frac{1}{a} \right| = \sqrt{\frac{1}{|a|^2} - 1}$ is isometric.) If $a=0$, then $|w'(z)| \equiv 1$.

$$4) \max \left| \frac{dw}{dz} \right| = \frac{1+|a|}{1-|a|}, \quad \min \left| \frac{dw}{dz} \right| = \frac{1-|a|}{1+|a|}. \quad 229. \quad 1) w = \frac{2z-1}{2-z}; \quad 2) w = \frac{2iz+1}{2+iz}; \quad 3) w = -iz; \quad 4) \frac{w-a}{1-\bar{a}w} = e^{i\alpha} \frac{z-a}{1-\bar{a}z}.$$

$$230. \quad R_2 \frac{w-b}{R_2^2 - \bar{b}w} = e^{i\alpha} R_1 \frac{z-a}{R_1^2 - \bar{a}z}. \quad 231. \quad w = \frac{1-z}{z+2}. \quad 232. \quad w = 2 \frac{z-2+i}{iz+2-2i}.$$

$$233. \quad 1) w = R^2 e^{i\alpha} \frac{z-a}{R^2 - \bar{a}z}; \quad 2) \frac{w-b}{R^2 - \bar{b}w} = e^{i\alpha} \frac{z-a}{R^2 - \bar{a}z}; \quad 3) w = R^2 \frac{z-a}{R^2 - \bar{a}z},$$

where a is a real number and $|a| < R$. $234. \quad \frac{w-a}{1-\bar{a}w} =$

$$= e^{i\varphi} \frac{z-z_1}{1-\bar{z}_1 z}, \quad \text{where } \varphi = \pi - \arg \frac{z_2 - z_1}{1 - \bar{z}_1 z_2}, \quad a = \frac{|z_2 - z_1|}{|1 - \bar{z}_1 z_2| + \sqrt{(1 - |z_1|^2)(1 - |z_2|^2)}}.$$

$$235. \quad w = \pm \frac{az - 1 + \sqrt{1-a^2}}{(1 - \sqrt{1-a^2})z - a}, \quad \rho = 2 \frac{1 - \sqrt{1-a^2}}{a}. \quad 238. \quad 1) w = z;$$

$$2) w = \frac{z+2-\sqrt{3}}{1+(2-\sqrt{3})z}; \quad w = i \frac{z-2+\sqrt{3}}{1-(2-\sqrt{3})z}. \quad 241. \text{ Circles passing through point } z_0 \text{ and having at this point a tangent defined by the vector } \bar{h}.$$

$$242. \quad w = \frac{(R-ki)z - R^2}{z - (R+ki)}, \quad \text{where } k \text{ is a real number, } k \neq 0 \text{ (for } k = \infty \text{ we put } w = z).$$

247. If $|a| < \sin \frac{\lambda}{2}$, the transformation is elliptic. Denoting $|a| = \sin \frac{\lambda}{2} \sin \beta$,

$$z_1 = e^{i\left(\alpha + \frac{\lambda}{2} - \frac{\pi}{2}\right)} \tan \frac{\beta}{2}, \quad z_2 = e^{i\left(\alpha + \frac{\lambda}{2} - \frac{\pi}{2}\right)} \cot \frac{\beta}{2}, \quad \tan \frac{\lambda_0}{2} = \tan \frac{\lambda}{2} \cos \beta,$$

we can write the transformation in the form $\frac{w-z_1}{w-z_2} = e^{i\lambda_0} \frac{z-z_1}{z-z_2}$. If $|a| = \sin \frac{\lambda}{2}$,

we have a parabolic transformation in the form $\frac{1}{w-z_0} = \frac{1}{z-z_0} + i h \bar{z}_0$, where

$$z_0 = e^{i\left(\alpha + \frac{\lambda}{2} - \frac{\pi}{2}\right)}, \quad h = \tan \frac{\lambda}{2}. \quad \text{If } |a| > \sin \frac{\lambda}{2}, \text{ the transformation is hyper-}$$

bolic. Denoting $\sin \frac{\lambda}{2} = |a| \sin \beta$, $z_1 = e^{i\left(\alpha + \frac{\lambda}{2} - \beta\right)}$, $z_2 = e^{i\left(\alpha + \frac{\lambda}{2} - \pi + \beta\right)}$,

$$K = \frac{\cos \frac{\lambda}{2} + \sqrt{|a|^2 - \sin^2 \frac{\lambda}{2}}}{\cos \frac{\lambda}{2} - \sqrt{|a|^2 - \sin^2 \frac{\lambda}{2}}}, \quad \text{we can write the transformation in the form}$$

$$\frac{w-z_1}{w-z_2} = K \frac{z-z_1}{z-z_2}. \quad 248. \quad \Gamma = 2 \arctan \frac{a}{h}; \quad \Gamma = \frac{2a}{h} + O\left[\left(\frac{a}{h}\right)^3\right] \text{ for small } \frac{a}{h};$$

$$\Gamma = \pi - \frac{2h}{a} + O\left[\left(\frac{h}{a}\right)^3\right] \text{ for small } \frac{h}{a}. \quad 249. \quad \Gamma = \gamma + 2 \arctan \frac{x_0 \sin \gamma}{1 - x_0 \cos \gamma} = 2 \arctan \left(\frac{1+x_0}{1-x_0} \tan \frac{\gamma}{2} \right); \quad \Gamma < \gamma \text{ if } x_0 < 0, \text{ and } \Gamma > \gamma \text{ if } x_0 > 0.$$

$$253. \quad 1) \quad w = -\frac{20}{z}; \quad 2) \quad w = -(2iz + 1 + 2i). \quad 254. \quad w = \frac{z - \sqrt{h^2 - R^2}}{z + \sqrt{h^2 - R^2}} e^{i\alpha}; \quad \rho =$$

$$= \frac{h}{R} + \sqrt{\left(\frac{h}{R}\right)^2 - 1}. \quad 255. \quad h = \frac{5}{4}; \quad w = 2e^{i\alpha} \frac{4z-3}{4z+3} \text{ or } w = e^{i\alpha} \frac{4z+3}{4z-3}. \quad 256. \quad w =$$

$$= \frac{2z}{z+24} e^{i\alpha} \text{ or } w = \frac{z+24}{3z} e^{i\alpha}, \quad \rho = \frac{2}{3}. \quad 257. \quad w = \lambda \frac{z-z'_1}{z-z'_2} \text{ or } w = \lambda \frac{z-z'_2}{z-z'_1},$$

$$\text{where } \lambda \text{ is an arbitrary complex number, } z'_1 = z_1 + \frac{u_1(z_2 - z_1)}{d},$$

$$z'_2 = z_1 + \frac{u_2(z_2 - z_1)}{d}, \quad d = |z_2 - z_1|,$$

$$u_1 = \frac{1}{2d} (r_1^2 + d^2 - r_2^2 - \sqrt{[d^2 - (r_1 + r_2)^2][d^2 - (r_1 - r_2)^2]}),$$

$$u_2 = \frac{1}{2d} (r_1^2 + d^2 - r_2^2 + \sqrt{[d^2 - (r_1 + r_2)^2][d^2 - (r_1 - r_2)^2]});$$

$$\mu \left(\text{or } \frac{1}{\mu} \right) = \left| \frac{(d + r_2 - u_1)(r_1 - u_2)}{(d + r_2 - u_2)(r_1 - u_1)} \right|.$$

258. 1) $\mu = 2$; 2) $\mu = 5 + 2\sqrt{6}$. 259. The group is finite if α is commensurable with π . 263. Fundamental domains (one of the possible kinds of them) are shown hatched in Fig. 62. Equivalent boundary sides are connected by arrows. The numbered points are fixed points of rotations leading to the groups (the figures show the number of rotations). Examples 5 to 9 are illustrated by the parallelogram of a doubly periodic subgroup; in example 7 it is a square, in examples 8 and 9, a rhombus with angles of 120° and 60° . Note. It can be shown that, to within a linear transformation, groups 3 to 9 exhaust the groups of linear transformations with one limit point (this is the term for the limit point of a set of equivalent points). 264. 1) $w = e^{i\alpha} z$;

$$2) \quad \frac{w-1}{w+1} = e^{i\alpha} \frac{z-1}{z+1}; \quad 3) \quad \frac{w-i}{w+i} = e^{i\alpha} \frac{z-i}{z+i}; \quad 4) \quad \frac{w-a}{1+\bar{a}w} = e^{i\alpha} \frac{z-a}{1+\bar{a}z}. \quad 270. \quad 1) \text{ and}$$

2) The construction is obvious; 3) equidistant curves of the "straight line" $\alpha\beta$ (α and β are the "points at infinity" of this straight line) are circular arcs with end-points α and β (they are called *hypercycles*); 4) limit lines of a pencil of "parallel lines" with a common point α "at infinity" are circles tangent (internally) to the unit circle at the point α (they are called *oricycles*). 271. 2) To construct a "rectilinear" triangle with angles $\varphi_1, \varphi_2, \varphi_3$, first construct a circular sector OAB with a central angle $\Delta = \pi - (\varphi_1 + \varphi_2 + \varphi_3)$, then draw a "straight line" AB , and, through the points A and B , draw "straight lines" at angles φ_2 and φ_3 to AB to intersect at point C . $\triangle ABC$ is the desired triangle.

$$272. \quad 1) \quad w = i(z^2 - a^2); \quad 2) \quad w = \sqrt{z - \frac{p}{2}} - i \sqrt{\frac{p}{2}}. \quad 273. \quad 1) \quad w = \frac{z^2}{a^2}; \quad 2) \quad w =$$

$$= \sqrt{\frac{z}{a}}; \quad 3) \quad w = \frac{A}{2} \left(\frac{z+1}{z-2} \right)^3. \quad 274. \text{ The domain is bounded by the limaçon of Pascal: } u = R(\cos \varphi + m \cos 2\varphi), \quad v = R(\sin \varphi + m \sin 2\varphi). \text{ Transferring the}$$

origin in the w -plane to the point $w = -Rm$, we obtain the equation of the

limaçon in its ordinary form (in polar coordinates): $\rho = R(1 + 2m \sin \theta)$. For $m=0$ the limaçon of Pascal becomes a circle, for $m=\frac{1}{2}$, a cardioid with a cusp $w = -\frac{R}{2}$. The images of circles $|z|=r < 1$ are also limaçons of Pascal whose polar equations are readily obtained by transferring the origin to the point $w = -Rmr^2$: $\rho = Rr(1 + 2mr \cos \theta)$. The images of the radii of the circle

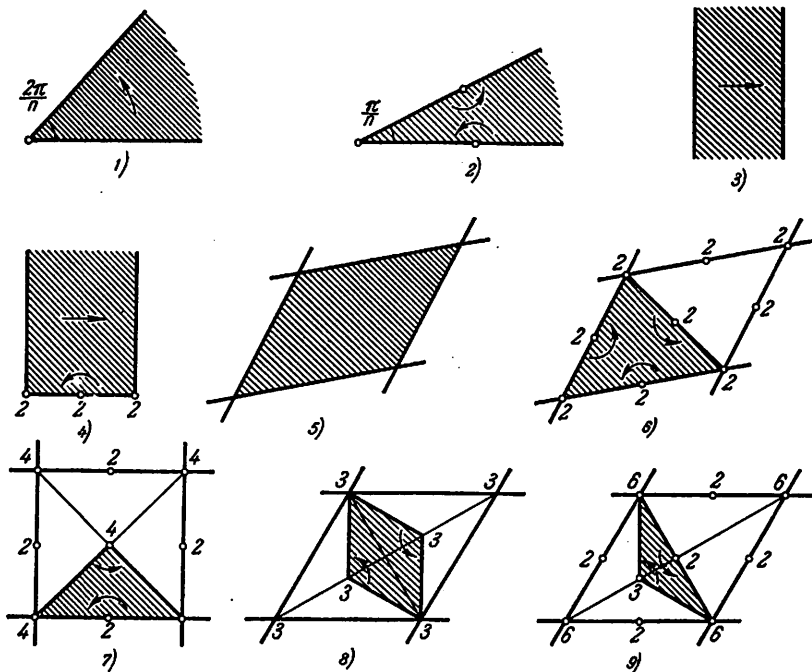


FIG 62.

$\arg z = \alpha$ are parabolas passing through the origin: $m(u \sin 2\alpha - v \cos 2\alpha)^2 + R \sin \alpha (u \sin \alpha - v \cos \alpha) = 0$. To the radii $\alpha=0$ and $\alpha=\pi$ there correspond the segments of the real axis $0 \leq u \leq R(1+m)$ and $R(m-1) \leq u \leq 0$.

275. The domain is bounded by a parabola $u = -v^2$ and a curve $\rho = 2 \cos \frac{\theta}{3}$,

$|\theta| < \frac{3\pi}{4}$ (Fig. 63). 276. 1) The domain is bounded by an epicycloid:

$u = R \left(\cos \varphi + \frac{\cos n\varphi}{n} \right)$, $v = R \left(\sin \varphi + \frac{\sin n\varphi}{n} \right)$ having $(n-1)$ cusps, which

are images of the points $z = \sqrt[n-1]{-1}$; 2) the exterior of the hypocycloid

$u = R \left(\cos \varphi + \frac{\cos n\varphi}{n} \right)$, $v = R \left(\sin \varphi - \frac{\sin n\varphi}{n} \right)$ having $(n+1)$ cusps (images of

the points $z = \sqrt[n+1]{1}$). 277. 1) $|m| \leq \frac{1}{n}$. The domain is bounded by an

"elongated" epicycloid (epitrochoid), i.e. by the trajectory of a point situated

at a distance mR from the centre of a circle of radius $\frac{R}{n}$ rolling along the outside of a circle of radius $\frac{R(n-1)}{n}$ 2) $|m| \leq \frac{1}{n}$. In the first case, the exterior of the unit circle, and in the second case its interior is mapped onto the exterior of a "shortened" hypocycloid (hypotrochoid). 278. 1) $w = z^{\frac{1}{\alpha}}$; 2) $w = \frac{2(\sqrt[3]{4}+1)e^{\frac{\pi i}{3}}z^{\frac{4}{3}}}{(\sqrt[3]{4}-2)e^{\frac{\pi i}{3}}z^{\frac{4}{3}}+3\sqrt[3]{4}}$. 279. 1) $w = \left(\frac{1+z}{1-z}\right)^2$; 2) $w = -\frac{z^2+1}{2z}$;

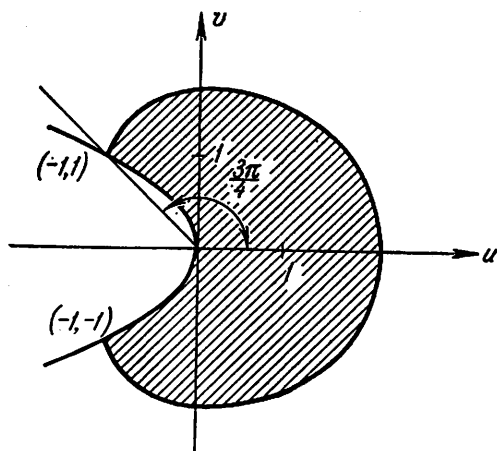


FIG. 63

- 3) $w = \frac{-2z^2+3z-2}{2z^2+3z+2}$. 280. 1) $w = \frac{z^2+2iz+1}{iz^2+2z+i}$; 2) $w = \frac{2z^2+3iz+2}{2z^2-3iz+2}$.
 281. 1) $w = \left(\frac{z-1}{z+1}\right)^2$. 282. 1) $w = \left(\frac{\frac{1}{z^\alpha} + R\frac{1}{\alpha}}{\frac{1}{z^\alpha} - R\frac{1}{\alpha}}\right)^2$. 2) $w = \left(\frac{\frac{1}{z^\alpha} - R\frac{1}{\alpha}}{\frac{1}{z^\alpha} + R\frac{1}{\alpha}}\right)^2$.
 283. 1) $w = -\left(\frac{2z+\sqrt{3}-i}{2z-\sqrt{3}-i}\right)^{\frac{3}{2}}$; 2) $w = -\left(\frac{2z+\sqrt{3}-i}{2z-\sqrt{3}-i}\right)^3$; 3) $w = \left(\frac{2z+\sqrt{3}-i}{2z-\sqrt{3}-i}\right)^3$; 4) $w = i\left(\frac{2z+\sqrt{3}-i}{2z-\sqrt{3}-i}\right)^{\frac{3}{2}}$; 5) $w = \left[\frac{z-\sqrt{2}(1-i)}{z-\sqrt{2}(1+i)}\right]^4$.
 284. $w = e^{\frac{\pi i}{3}}\left(\frac{z+1}{z-1}\right)^{\frac{2}{3}}$. 285. $w = \sqrt{\frac{z+1}{1-z}}$. 286. $w = \sqrt{\frac{z+i}{i-z}}$. 287. $w = \sqrt{\frac{z-z_1}{z_2-z}}$.
 288. $w = \sqrt{\frac{z+R}{z-R}}$. 289. $w = e^{-\frac{\pi i}{8}}\sqrt{z-i}$. 290. $w = e^{\frac{i}{3}\arctan \frac{2h}{1-h^2}}\sqrt{\frac{z+1}{z-1}}$.
 291. $w = \sqrt{z^2+h^2}$. 292. $w = \frac{\sqrt{z^2+h^2}}{z}$. 293. $w = \sqrt{\left(\frac{z-1}{z+1}\right)^2 + \tan^2 \frac{\alpha}{2}}$.

$$294. w = \sqrt{\left(\frac{\frac{1}{z^{\frac{1}{\beta}}}-1}{\frac{1}{z^{\frac{1}{\beta}}}+1}\right)^2 + \tan^2 \frac{\alpha}{2\beta}}. \quad 295. w = \sqrt{\left[\frac{(1-z)^{\frac{2}{3}}-(1+z)^{\frac{2}{3}}}{(1-z)^{\frac{2}{3}}+(1+z)^{\frac{2}{3}}}\right]^2} + \frac{1}{3}.$$

296. $w = \left(\frac{\sqrt{z}+1}{\sqrt{z}-1}\right)^2$. For one choice of the branch of $\sqrt{z}$ this function yields the solution to item (1), for the other, to item (2). 297. $w = \frac{z}{(1-z)^2}$.

298. To the circles $|z|=R$ (Fig. 64) there correspond confocal ellipses $\frac{4u^2}{\left(R+\frac{1}{R}\right)^2} + \frac{4v^2}{\left(R-\frac{1}{R}\right)^2} = 1$ (to the circle $|z|=1$, the segment $v=0$, $-1 \leq u \leq 1$); to the rays $\arg z = \alpha$ there correspond the branches of confocal hyperbolas $\frac{u^2}{\cos^2 \alpha} - \frac{v^2}{\sin^2 \alpha} = 1$ (to the ray $\arg z = 0$, the ray $v=0$, $u \geq 1$; to the ray $\arg z = \pi$, the ray $v=0$, $u \leq -1$; to the rays $\arg z = \pm \frac{\pi}{2}$, the axis $u=0$).

299. 1) and 2) The exterior of the ellipse $\frac{4u^2}{\left(R+\frac{1}{R}\right)^2} + \frac{4v^2}{\left(R-\frac{1}{R}\right)^2} = 1$

(Fig. 65, 1, 2); 3) and 4). The entire plane with a cut along the segment $[-1, 1]$ (Fig. 65, 3, 4); 5) and 6) the entire plane with cuts along the rays $(-\infty, -1]$ and $[1, \infty)$ lying on the real axis; 7) the lower half-plane; 8) the upper half-plane; 9) the upper half-plane; 10) the upper half of the interior of the ellipse $\frac{4u^2}{\left(R+\frac{1}{R}\right)^2} + \frac{4v^2}{\left(R-\frac{1}{R}\right)^2} = 1$; 11) the lower half of the interior

of the ellipse $\frac{4u^2}{\left(R+\frac{1}{R}\right)^2} + \frac{4v^2}{\left(R-\frac{1}{R}\right)^2} = 1$; 12) the right half of the interior

of the ellipse $\frac{4u^2}{\left(R+\frac{1}{R}\right)^2} + \frac{4v^2}{\left(R-\frac{1}{R}\right)^2} = 1$ with a cut along the segment

$\left[1, \frac{1}{2}\left(R+\frac{1}{R}\right)\right]$; 13) the domain between the branches of the hyperbola

$\frac{u^2}{\sin^2 \alpha} - \frac{v^2}{\cos^2 \alpha} = 1$. 300. 1) To the circles $|z|=R$ there correspond the ellipses

$\frac{4u^2}{\left(R-\frac{1}{R}\right)^2} + \frac{4v^2}{\left(R+\frac{1}{R}\right)^2} = 1$ (to the circle $|z|=1$, the segment $u=0$, $-1 \leq v \leq 1$),

to the rays $\arg z = \alpha$, branches of the hyperbolas $\frac{v^2}{\sin^2 \alpha} - \frac{u^2}{\cos^2 \alpha} = 1$ (to the rays $\varphi=0$ and $\varphi=\pi$, the axis $v=0$, to the ray $\varphi=\frac{\pi}{2}$, the ray $u=0$,

$v \geq 1$, to the ray $\varphi = -\frac{\pi}{2}$, the ray $u=0$, $v \leq -1$); 2) to the

circles $|z| = R$ there correspond the ellipses $\frac{4u^2}{\left(R + \frac{a^2}{R}\right)^2} + \frac{4v^2}{\left(R - \frac{a^2}{R}\right)^2} = 1$ (to the circle $|z| = a$, the segment $v=0$, $-a \leq u \leq a$), to the rays $\arg z = \alpha$,

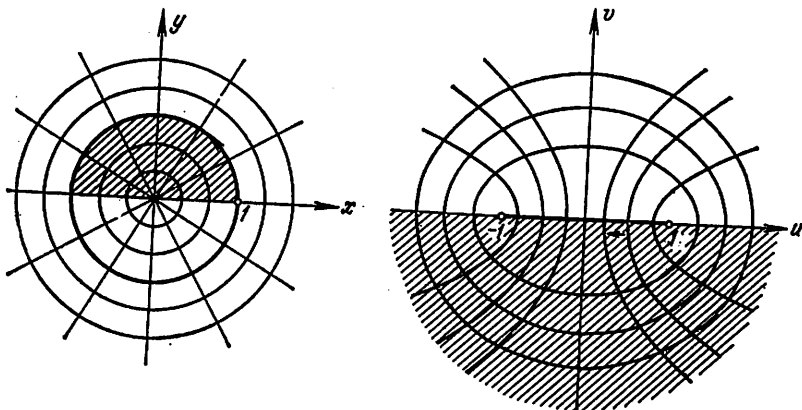


FIG. 64

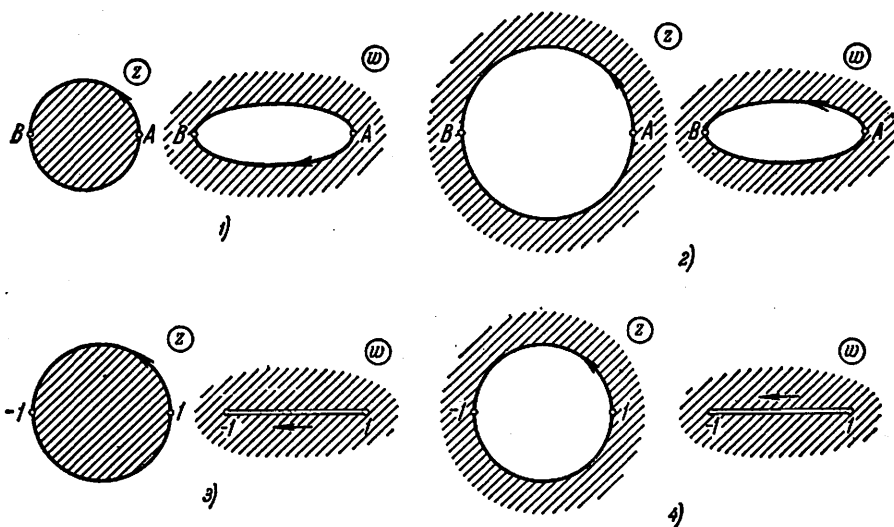


FIG. 65

branches of the hyperbolas $\frac{u^2}{a^2 \cos^2 \alpha} - \frac{v^2}{a^2 \sin^2 \alpha} = 1$ (to the ray $\arg z = 0$, the ray $v=0$, $u \geq a$, to the ray $\arg z = \pi$, the ray $v=0$, $u \leq -a$, to the rays $\arg z = \pm \frac{\pi}{2}$, the axis $u=0$); 3) families of confocal ellipses and hyperbolas

obtained from the appropriate families for Zhukovsky's function (see Problem 298) by rotating through an angle γ and by a similarity transformation with the expansion factor $|c|$ (the centre of similarity is at the origin). 301. 1) $w = \frac{e^{i\alpha}}{c} (z + \sqrt{z^2 - c^2})$; 2) $w = \frac{1}{a+b} (z + \sqrt{z^2 - (a^2 - b^2)})$, in both cases, with one choice of the root branch, we get a mapping onto the exterior of the unit

circle, and with the other, onto the interior. 302. $w = \frac{az - b\sqrt{z^2 - (a^2 - b^2)}}{a^2 - b^2}$.

303. $w = A (z + \sqrt{z^2 - (a^2 - b^2)})$, where A is an arbitrary complex number;

$\mu = \frac{1}{\sqrt{a^2 + k^2} - \sqrt{b^2 + k^2}}$. 304. The entire plane with a cut along the segment $[-1, \frac{1}{2}(a + \frac{1}{a})]$ if $a > 0$; the entire plane with cuts along the rays

$(-\infty; \frac{1}{2}(a + \frac{1}{a})]$ and $[-1, +\infty)$ if $a < 0$. 305. $w = \sqrt{\frac{1 + \frac{1}{2}(z + \frac{1}{z})}{\frac{5}{4} - \frac{1}{2}(z + \frac{1}{z})}}$.

306. $w = \sqrt{\frac{1}{2}[(a + \frac{1}{a}) - (z + \frac{1}{z})]}$. 307. $w = \sqrt{\frac{1}{2}(z + \frac{1}{z} + a + \frac{1}{a})}$.

308. $w = \frac{\sqrt{(z^2 + \frac{1}{z^2}) + (\alpha^2 + \frac{1}{\alpha^2})}}{z + \frac{1}{z}}$. 309. $w = \sqrt{\frac{1}{2}[(\alpha^2 + \frac{1}{\alpha^2}) + (z^2 + \frac{1}{z^2})]}$.

310. $\left[1 + \frac{h^2}{4(1-h)}\right] \left(w + \frac{1}{w}\right) = \left(\frac{z}{e^{i\alpha}} + \frac{z}{z}\right) - \frac{h^2}{2(1-h)}$. 311. $w + \frac{1}{w} = \frac{4a}{(1+a)^2} \left(z + \frac{1}{z}\right) - 2 \frac{(1-a)^2}{(1+a)^2}$; $w'(0) = \frac{(1+a)^3}{4a}$; the arc length corresponding

to the cut is $2 \arccos \frac{6a - 1 - a^2}{(1+a)^2}$; it equals π for $a = 3 - \sqrt{8}$. 312. $w + \frac{1}{w} =$

$= 2 \frac{2(z + \frac{1}{z}) - (a + \frac{1}{a}) + (b + \frac{1}{b})}{(a + \frac{1}{a}) + (b + \frac{1}{b})}$; $w'(0) = \frac{a + \frac{1}{a} + b + \frac{1}{b}}{4}$; the arc lengths

corresponding to the cuts are: $2 \arccos \frac{4 - (a + \frac{1}{a}) + (b + \frac{1}{b})}{(a + \frac{1}{a}) + (b + \frac{1}{b})}$,

$2\pi - 2 \arccos \frac{-4 - (a + \frac{1}{a}) + (b + \frac{1}{b})}{(a + \frac{1}{a}) + (b + \frac{1}{b})}$. 313. 1) The image of the circle C is

a circular arc with end-points at points ± 1 inclined at an angle of 2α to the real axis at point 1; the exterior of the given circle is mapped onto the entire plane with a cut along the indicated arc; 2) the image of the circle C (Fig. 66) is a closed curve (Zhukovsky airfoil) with cusp $w = 1$; a line tangent at this point makes an angle of 2α with the real axis; the arc of a circle with end-points ± 1 which is considered in (1) is contained in the domain bounded by the Zhukovsky airfoil; the exterior of the circle C is map-

ped onto the exterior of the Zhukovsky airfoil. 314. 1) The image of the circle C is a closed curve consisting of two circular arcs with common end-points at ± 1 ; the lines tangent to these arcs at point 1 form angles of $2\alpha - \alpha\delta$ and $2\alpha + (\pi - \alpha)\delta$, respectively, with the real axis; the exterior of the circle is

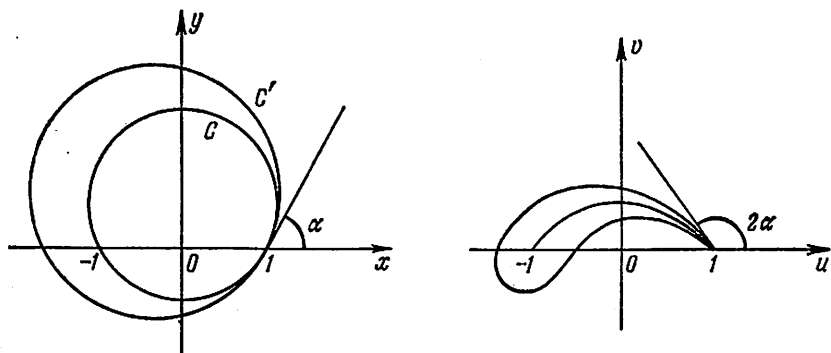


FIG. 66

mapped onto the exterior of the domain bounded by the above-mentioned arcs; the image of the circle C' is (Fig. 67) a closed curve with corner point $w=1$, the lines tangent at this point being inclined at angles of $2\alpha - \alpha\delta$ and $2\alpha + (\pi - \alpha)\delta$, respectively, to the real axis; the image of the circle C is contained in the domain bounded by the image of the circle C' ;

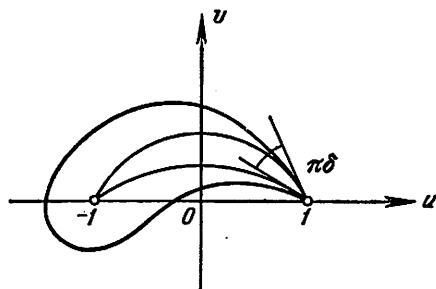


FIG. 67

the exterior of the circle C' is mapped onto the exterior of the image of this circle. 2) The interior of the circle C is mapped onto the exterior of the domain bounded by circular arcs passing through points $-1, 1$, the lines tangent to these circles at point 1 making angles equal to: a) $2\alpha + (\pi - \alpha)\delta$, $2\alpha + (2\pi - \alpha)\delta$, respectively, with the real axis if the function $w(z)$ is defined in the z -plane with a cut along an arc of the circle C lying in the lower half-plane; b) $2\alpha - \alpha\delta$, $2\alpha - (\pi + \alpha)\delta$ if the cut defining the function $w(z)$ is made along an arc of the circle C lying in the upper half-plane. 315. $\frac{w-1}{w+1} =$

$$= \left(\frac{ze^{i\gamma} + ie^{\frac{i\beta}{2}}}{ze^{i\gamma} - ie^{-\frac{i\beta}{2}}} \right)^2, \text{ where } \gamma = \alpha \text{ if } \beta > 0, \text{ and } \gamma = \alpha + \pi \text{ if } \beta < 0. \text{ 316. The}$$

entire plane with cuts along the rays $y=0, x \leq -\frac{1}{2}$ and $y=0, x \geq \frac{1}{2}$. 317. The half-plane $x > \frac{1}{2}$ with a cut along the line segment $y=0, \frac{1}{2} \leq x \leq 1$. 318. The entire plane with cuts along the rays $y=0, 1 \leq x < \infty$ and $y=0, -\infty < x \leq -1$. 319. An angle $-\frac{\pi}{n} < \arg z < \frac{\pi}{n}$ with a cut along the ray $y=0, \sqrt[n]{\frac{1}{4}} \leq x < \infty$. 320. 1) The entire plane with cuts along the rays $|w| \geq \frac{1}{\sqrt[n]{4}}, \arg w = \frac{2k\pi}{n}$

$$(k=0, 1, \dots, n-1); 2) w = \frac{(1+z^n)^{\frac{2}{n}}}{\sqrt[n]{4z}}.$$

$$321. 1) w = \sqrt{z^2 + \sqrt{z^4 - 1}} = \frac{1}{\sqrt{2}} (\sqrt{z^2 + 1} + \sqrt{z^2 - 1}); 2) w = \frac{\sqrt{1 + \sqrt{1 - z^4}}}{z} = \frac{1}{z\sqrt{2}} (\sqrt{1 + z^2} + \sqrt{1 - z^2}).$$

$$322. 1) w = (\alpha^n + \alpha^{-n})^{\frac{1}{n}} \sqrt[n]{z^n + z^{-n} + \sqrt{(z^n + z^{-n})^2 - (\alpha^n + \alpha^{-n})^2}}.$$

Solution. The function $\xi = \frac{1}{2} \left(z^n + \frac{1}{z^n} \right)$ maps the sector onto the lower half-plane, the points α and $\alpha e^{\frac{\pi i}{n}}$ go into the points $\pm \frac{1}{2} \left(\alpha^n + \frac{1}{\alpha^n} \right)$. Furthermore, the half-plane $\left(p = \frac{\xi}{\frac{1}{2} (\alpha^n + \alpha^{-n})} \right)$ is to be contracted and mapped

onto the unit semicircle ($\tau = p + \sqrt{p^2 - 1}$). The function $w = \sqrt[n]{\tau}$ will be the required function. 2) $w = \left(\alpha^{\frac{n}{2}} + \alpha^{-\frac{n}{2}} \right) - \frac{2}{n} \left(\alpha^{\frac{n}{2}} + z^{-\frac{n}{2}} + \sqrt{\left(\alpha^{\frac{n}{2}} + z^{-\frac{n}{2}} \right)^2 - \left(\alpha^{\frac{n}{2}} + \alpha^{-\frac{n}{2}} \right)^2} \right)^{\frac{2}{n}}$. 323. $w = \sqrt{\frac{\sqrt{z^2 + c^2} + \sqrt{a^2 + c^2}}{\sqrt{b^2 + c^2} - \sqrt{z^2 + c^2}}}$

and $w = \frac{1}{\beta} (\sqrt{z^2 + c^2} + \alpha + \sqrt{(\sqrt{z^2 + c^2} + \alpha)^2 - \beta^2})$, where $\alpha = \frac{1}{2} (\sqrt{a^2 + c^2} - \sqrt{b^2 + c^2})$, $\beta = \frac{1}{2} (\sqrt{a^2 + c^2} + \sqrt{b^2 + c^2})$. 324. $w = \sqrt{\sqrt{z^2 + c^2} + \sqrt{a^2 + c^2}}$.

$$325. w = \frac{1}{\sqrt{2}} \left(\sqrt{\left(\frac{z+i}{z-i} \right)^2 + 1} + \sqrt{\left(\frac{z+i}{z-i} \right)^2 - 1} \right). \quad 326. w =$$

$$= \sqrt{\frac{\sqrt{z^2 - 1} + z - i}{\frac{\sqrt{a^2 + 1}}{\alpha + 1} (z - i) - \sqrt{z^2 - 1}}}. \quad 327. \text{ When } b < 1,$$

$$w = \sqrt{\frac{\sqrt{z_1^2 + \cot^2 \frac{\alpha}{2}} + \cot \frac{\alpha}{2}}{\sqrt{b_1^2 + \cot^2 \frac{\alpha}{2}} - \sqrt{z_1^2 + \cot^2 \frac{\alpha}{2}}}}; \text{ for } b = 1,$$

$$w = \sqrt{\sqrt{z_1^2 + \cot^2 \frac{\alpha}{2}} + \cot \frac{\alpha}{2}}; \text{ for } b > 1,$$

$$w = \sqrt{-\frac{\sqrt{z_1^2 + \cot^2 \frac{\alpha}{2}} + \cot \frac{\alpha}{2}}{\sqrt{z_1^2 + \cot^2 \frac{\alpha}{2}} + \sqrt{b_1^2 + \cot^2 \frac{\alpha}{2}}}}; \quad z_1 = \frac{1+z}{1-z}, \quad b_1 = \frac{1+b}{1-b}.$$

$$328. w = \frac{\sqrt{\left(z^2 + \frac{1}{z^2}\right) + \left(b^2 + \frac{1}{b^2}\right)} + \sqrt{\left(a^2 + \frac{1}{a^2}\right) + \left(b^2 + \frac{1}{b^2}\right)}}{\sqrt{\left(a^2 + \frac{1}{a^2}\right) - \left(z^2 + \frac{1}{z^2}\right)}}.$$

$$329. w = \frac{1}{\sqrt{2 + \sqrt{5}}} \left(\sqrt{\sqrt{z^4 + 4} + 2} + \sqrt{\sqrt{z^4 + 4} - \sqrt{5}} \right).$$

Solution. The function $\zeta = z^2$ maps the upper half-plane with cuts along the line segments $[0, 1+i]$, $[0, -1+i]$ onto the domain of Problem 324, which we map onto the upper half-plane. By virtue of the symmetry principle, the function obtained maps the given domain onto a plane with a cut along a certain segment. It now remains to map the exterior of this segment onto the exterior

of the unit circle. 330. $w = \frac{i}{\sqrt{2}} [(z + \sqrt{z^2 - 1})^{\frac{\pi}{\alpha}} + (z + \sqrt{z^2 - 1})^{-\frac{\pi}{\alpha}} + 2]^{\frac{1}{2}} =$

$$= \frac{i}{\sqrt{2}} \left[(z + \sqrt{z^2 - 1})^{\frac{\pi}{2\alpha}} + (z + \sqrt{z^2 - 1})^{-\frac{\pi}{2\alpha}} \right]. \text{ Solution. With the aid of the}$$

function $\tau = \zeta^{\frac{\pi}{\alpha}}$, where $\zeta = z + \sqrt{z^2 - 1}$ is the inverse of Zhukovsky's function, the upper half of the given domain is mapped onto the domain $|\tau| > 1$, $\operatorname{Im} \tau > 0$, which is mapped onto the upper half-plane by Zhukovsky's function. Applying the symmetry principle, we obtain the mapping of the interior of the right-hand branch of the hyperbola onto the entire plane with a cut along the ray $(-\infty, -1]$; this latter domain is easily mapped onto the upper half-plane.

Note. The factor $\frac{1}{\sqrt{2}}$ plays no role, since the transformation $w' = kw$ ($k > 0$)

maps the half-plane onto itself. 331. $w = [e^{-i\alpha}(z + \sqrt{z^2 - 1})]^{\frac{\pi}{2\beta}} - [e^{-i\alpha}(z + \sqrt{z^2 - 1})]^{-\frac{\pi}{2\beta}}$, where $\beta = \pi - \alpha$. 332. $w = \left[e^{-i\alpha} \frac{z + \sqrt{z^2 - c^2}}{c} \right]^p$,

where $c = \sqrt{a^2 + b^2}$, $\alpha = \arctan \frac{b}{a}$, $p = \frac{\pi}{2 \arctan \frac{a}{b}}$. 333. 1) The domain is

constructed in the following way: the annulus $r_1^2 < |w| < r_2^2$ is cut along the segment $r_1^2 \leq u \leq r_2^2$ of the real axis, and a portion of a similar annulus ($r_1^2 < |w| < r_2^2$, $0 \leq \arg w < 2\alpha$) is glued to the lower lip of the cut; if $\alpha = \pi$, the second annulus is complete and its free lip should be glued to the free lip of the first annulus (in this case we obtain a two-sheeted annulus $r_1^2 < |w| < r_2^2$); 2) if $\alpha \leq 1$, the inequality $|z^2 - 1| < a$ defines two domains (see Problem 39), each of which is mapped onto a univalent circle $|w - 1| < a$; if $a > 1$, the inequality $|z^2 - 1| < a$ defines one domain, which is mapped onto a two-sheeted circle $|w - 1| < a$ (to construct this two-sheeted circle it is sufficient to cut

two identical sheets of the circle $|w-1| < a$ along any radius and to glue the lower lip of the cut of the first sheet to the upper lip of the cut of the second sheet, and the upper lip of the cut of the first sheet to the lower lip of the cut of the second sheet). 334. 1) The domain is constructed in the following way: to the w -plane cut along the segment $[-1, 1]$ is glued the interior of the

ellipse $\frac{4u^2}{\left(R+\frac{1}{R}\right)^2} + \frac{4v^2}{\left(R-\frac{1}{R}\right)^2} = 1$, which is also cut along the segment $[-1, 1]$,

the upper lip of the cut of the ellipse being glued to the lower lip of the cut of the plane, and the lower lip of the cut of the ellipse to the upper lip of the

cut of the plane; 2) the two-sheeted domain $\left| \frac{w-1}{w+1} \right| < R^2$ (the inequality

$\left| \frac{w-1}{w+1} \right| < R^2$ defines the interior of a circle if $R < 1$, a half-plane when $R = 1$,

and the exterior of a circle when $R > 1$). The cut and the corresponding attachments go along the line joining the point $w=1$ with some boundary point

of the domain $\left| \frac{w-1}{w+1} \right| < R^2$. 335. 1) and 2) The surface consists of two sheets

of the z -plane which are cut along the segment $[-1, 1]$, the lower lip of the cut of the first sheet being glued to the upper lip of the cut of the second,

and the upper lip of the cut of the first sheet to the lower lip of the cut of the second. 336. 1) and 2) The surface consists of two sheets cut along the rays

going from points $-i, 0, i$, respectively, to infinity, say, parallel to the real axis in the positive direction. The lower lips of the cuts of the first sheet are

glued to the upper lips of the appropriate cuts of the second sheet, and vice versa. 337. The surface consists of three sheets of the z -plane cut along the rays

$(-\infty, -1]$ and $[1, \infty)$. Joining along the ray $(-\infty, -1]$ is effected in the following manner: the upper lip of the cut of the first sheet is joined to the

lower lip of the cut of the second sheet, the upper lip of the cut of the second sheet to the lower lip of the cut of the third sheet, and the upper lip of the

cut of the third sheet to the lower lip of the cut of the first sheet. Along the ray $[1, \infty)$ join the following: the lower lip of the cut of the first sheet to the

upper lip of the cut of the second sheet, the lower lip of the cut of the second sheet to the upper lip of the cut of the third sheet, and the lower lip of the

cut of the third sheet to the upper lip of the cut of the first sheet. 338. 1) Into

the polar grid $\rho = \text{const}$, $\theta = \text{const}$; 2) into spirals $\rho = e^{\frac{\theta-b}{k}}$ (for $k=0$, into rays $\theta=b$); 3) into an angle $\alpha < \theta < \beta$ (for $\alpha=0$ and $\beta=2\pi$, into a plane with a

cut along the positive real axis); 4) into the entire plane with a cut along the spiral $\rho=e^{\theta}$; 5) into the sector $\rho < 1$, $0 < \theta < \alpha$ (for $\alpha=2\pi$, into the unit

circle with a cut along the radius $v=0$, $0 \leq u \leq 1$); 6) into the domain $\rho > 1$, $0 < \theta < \alpha$ (for $\alpha=2\pi$, into the exterior of the unit circle with a cut along the

ray $v=0$, $1 \leq u < \infty$); 7) into the domain $e^\alpha < \rho < e^\beta$, $\gamma < \theta < \delta$ (for $\delta-\gamma=2\pi$, this domain is a concentric annulus with a cut along the segment $\theta=\gamma$,

$e^\alpha \leq \rho \leq e^\beta$). 339. Angle $0 < \arg(z+n) < \frac{\pi}{n}$; strip $0 < y < \pi$. 340. 1) Into an

orthographic Cartesian grid $u=C$, $v=C$; 2) into straight lines; 3) into the strip $0 < v < \alpha$; 4) into the semi-strip $u < 0$, $0 < v < \alpha$; 5) into the rectangle

$\ln r_1 < u < \ln r_2$, $0 < v < 2\pi$. 341. 5) $b=a \left| \tanh \frac{\xi_0}{2} \right|$, $l=a \left| \tan \frac{\eta_0}{2} \right|$. 342. 1) The

family $x=C$ is transformed into a family of confocal hyperbolas with foci at

points $\pm 1 \left(\frac{u^2}{\cos^2 C} - \frac{v^2}{\sin^2 C} = 1 \right)$; the family $y=C$, into a family of confocal

ellipses with the same foci $\left(\frac{u^2}{\cosh^2 C} + \frac{v^2}{\sinh^2 C} = 1\right)$; 2) into the upper half-plane; 3) into the fourth quadrant; 4) into the right half-plane with a cut along the segment $[0, 1]$; 5) into the entire plane with cuts on the real axis along the rays $(-\infty, -1]$ and $[1, \infty)$; 6) into the interior of the ellipse $\frac{u^2}{\cosh^2 h} + \frac{v^2}{\sinh^2 h} = 1$ with cuts along the segments $[-\cosh h, -1]$ and $[1, \cosh h]$.

343. 1) Into the semi-strip $-\frac{\pi}{2} < u < \frac{\pi}{2}$, $v > 0$; 2) into the strip

$-\frac{\pi}{2} < u < \frac{\pi}{2}$; 3) into the semi-strip $0 < u < \frac{\pi}{2}$, $v > 0$; 4) into the strip

$-\frac{\pi}{2} < u < 0$. 344. 1) The family $x=C$ is transformed into a family of confocal

ellipses with foci at points ± 1 $\left(\frac{u^2}{\cosh^2 C} + \frac{v^2}{\sinh^2 C} = 1\right)$; the family $y=C$, into

a family of confocal hyperbolas with the same foci $\left(\frac{u^2}{\cosh^2 C} - \frac{v^2}{\sinh^2 C} = 1\right)$;

2) into the entire plane with cuts on the real axis along the rays $(-\infty, -1]$

and $[1, \infty)$; 3) into the upper half-plane. 345. 1) Into the strip $-\frac{\pi}{2} < v < \frac{\pi}{2}$;

2) into the semi-strip $0 < v < \frac{\pi}{2}$, $u > 0$. 346. 1) The family $x=C$ is transformed

into a pencil of circular arcs with end-points at $w = \pm i$; the pencil includes the corresponding parts of the imaginary axis; the equation of the pencil of circles is $(u-a)^2 + v^2 = 1 + a^2$ ($a = \cot 2C$); the family $y=C$ is transformed into a family of circles of Apollonius with respect to points $w = \pm i$ (the family includes the real axis as well); the equation of the family of circles of Apollonius is $u^2 + (v-b)^2 = b^2 - 1$, $|b| > 1$ ($b = \coth 2C$); 2) into the upper half-plane with a cut on the imaginary axis along the segment $0 \leq v \leq 1$;

3) into the entire plane with a cut on the imaginary axis (along the segment $-1 \leq v \leq 1$); 4) into the semicircle $|w| < 1$, $\operatorname{Re} w > 0$; 5) into the unit circle.

347. 1) Into the entire plane with cuts on the real axis (along the rays $(-\infty, -1]$ and $[1, \infty)$); 2) into the right half-plane with a cut on the real

axis (along the ray $[1, \infty)$). 348. $w = e^{\frac{\pi(1-i)z}{h}}$. 349. $w = -\cosh \frac{(z-1)\pi}{h}$.

350. $w = e^{\frac{2\pi iz}{z-2}}$. 351. $w = e^{\frac{\pi i(z+2)}{3(z-2)}}$. 352. $w = \cos \frac{\pi(z+2)}{2z}$. 353. $w = e^{\frac{\pi}{\sqrt{z-1}}}$.

354. 1) $w = -\frac{1+i\sqrt{3}}{2} \tanh \frac{\pi i(z+3i)}{4(z-i)}$; 2) $w = -\frac{e^{\frac{\pi i}{z-i}} + 2-i}{e^{\frac{\pi i}{z-i}} + 2+i}$;

3) $w = \frac{2e^{\frac{\pi i}{z-i}}}{1 + e^{\frac{\pi i}{z-i}}}$. 355. 1) $w = \sqrt{e^{2\pi iz} + e^{-2\pi h}}$; 2) $w = \sqrt{\frac{e^{2\pi iz} + e^{-2\pi h_1}}{e^{2\pi iz} + e^{-2\pi h_2}}}$.

356. $w = \sqrt{\frac{\cos \pi z - \cos \pi h_1}{1 + \cos \pi z}}$. 357. $w = \sqrt{\frac{\cos \pi z - \cos \pi h_1}{\cos \pi z + \cos \pi h_2}}$.

358. $w = \sqrt{\cos 2z + \cosh 2h}$. 359. $w = \sqrt{\frac{\cos 2z + \cosh 2h}{\cos 2z + 1}}$.

$$360. w = \sqrt{\frac{\cos 2z + \cosh 2h_1}{\cos 2z + \cosh 2h_2}}$$

$$362. w = \sqrt{\frac{1 + \cos \frac{4\pi}{z}}{\cos \frac{4\pi}{z} - \cos \frac{4\pi}{a}}}$$

$$364. w = \sqrt{\frac{\cos \frac{2\pi}{z} - \cos \frac{2\pi}{b}}{\cos \frac{2\pi}{z} - \cos \frac{2\pi}{a}}}$$

$$366. w = \sqrt{\frac{\cosh \frac{\pi}{h} - \cos \frac{\pi}{z}}{1 - \cos \frac{\pi}{z}}}$$

$$361. w = \sqrt{\frac{\sin \frac{\pi}{z}}{1 + \sin \frac{\pi}{z}}}$$

$$363. w = \sqrt{\frac{\cos \frac{4\pi}{z} - \cos \frac{4\pi}{b}}{\cos \frac{4\pi}{z} - \cos \frac{4\pi}{a}}}$$

$$365. w = \sqrt{\frac{e^{-\frac{2\pi}{\beta}} - e^{\frac{2\pi i}{z}}}{e^{\frac{2\pi}{\alpha}} - e^{\frac{2\pi i}{z}}}}$$

$$367. w = i \cosh \frac{\pi \sqrt{z}}{2\alpha}, \quad w = \tanh^2 \frac{\pi \sqrt{z}}{4\alpha}$$

(see the note in the answer to Problem 330). 368. $w = \arcsin \frac{\sin z}{\cosh a}$. *Solution.*

The function $\sin z$ maps the semi-strip $y > 0$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$ onto the upper half-plane, the points $\pm \frac{\pi}{2} + ai$ going into the points $\pm \cosh a$. Hence, we readily find that the function $w = \arcsin \frac{\sin z}{\cosh a}$ will map the above-mentioned semi-strip onto itself so that the rays $x = \pm \frac{\pi}{2}$, $a \leq y < \infty$ will correspond to the rays $u = \pm \frac{\pi}{2}$, $0 \leq v < \infty$. Applying the symmetry principle an unlimited number of times we are convinced that the function found is the desired

function. 369. $w = \frac{b \arcsin \frac{i \sinh z}{\cosh a}}{\arcsin \frac{1}{\cosh a}}$.

$$370. w = \frac{\arcsin \frac{1}{\cosh a} + \sqrt{\left(\arcsin \frac{1}{\cosh a}\right)^2 - \left(\arcsin \frac{\sin z}{\cosh a}\right)^2}}{\arcsin \frac{\sin z}{\cosh a}}$$

$$371. w = \sqrt{\frac{\arcsin \frac{\sin z}{\cosh a} - \arcsin \frac{\sin p}{\cosh a}}{\arcsin \frac{\sin q}{\cosh a} - \arcsin \frac{\sin z}{\cosh a}}}. \quad 372. w = \arcsin e^{2iz}. \quad \text{Solution.}$$

The function $\zeta = e^{2iz}$ maps the strip $0 < x < \frac{\pi}{2}$ onto the upper half-plane, and the function $w = \arcsin \zeta$ maps the upper half-plane onto the semi-strip $-\frac{\pi}{2} < u < \frac{\pi}{2}$, $v > 0$. The function $w = \arcsin e^{2iz}$ that maps the strip onto

the semi-strip transforms the rays $x=0$, $-\infty < y < 0$; $x=\frac{\pi}{2}$, $-\infty < y < 0$ into the rays $x=\frac{\pi}{2}$, $0 < y < \infty$; $x=-\frac{\pi}{2}$, $0 < y < \infty$, respectively. Applying the symmetry principle an unlimited number of times, we are convinced that the function thus found executes the desired mapping. 373. 1) If $b < 2\pi$, the curvilinear rectangle $1 < \rho < e^a$, $0 < \theta < b$; if $b=2\pi$, the annulus $1 < \rho < e^a$ with a cut along the segment $[1, e^a]$; if $b=2k\pi$ ($k=2, 3, \dots$), a multivalent domain made up of k annuli $1 < \rho < e^a$ cut along the segment $[1, e^a]$ and glued together so that the lower lip of the cut of the first annulus is joined to the upper lip of the cut of the second annulus, the lower lip of the cut of the second annulus to the upper lip of the cut of the third annulus and so on; if $b=2k\pi+\beta$ ($k=2, 3, \dots$, $0 < \beta < 2\pi$), a curvilinear rectangle $1 < \rho < e^a$, $0 < \theta < \beta$ has to be joined to the free lower lip of the last annulus of the constructed surface along the segment $[1, e^a]$; 2) a domain of infinitely many sheets consisting of annuli $1 < \rho < e^a$ cut along the segment $[1, e^a]$ and joined in the above-mentioned manner; 3) a domain of infinitely many sheets made up of annuli $1 < \rho < e^a$, cut along the segment $[1, e^a]$, numbered with the integers $(\dots, -2, -1, 0, 1, 2, \dots)$ and joined so that the lower lip of the cut of each annulus is joined to the upper lip of the cut of the annulus with the next higher number. This domain is part of the Riemann surface of the function $\text{Ln } w$ lying above the annulus $1 < \rho < e^a$. 374. 1) A two-sheeted domain obtained by joining two right half-planes, each being cut along the ray $v=0$, $1 \leq u < \infty$; the edges of the cuts are joined crosswise, i.e. so that the lower lip of the first sheet is glued to the upper lip of the second sheet, and vice versa; 2) a two-sheeted domain consisting of two planes with the cuts on the real axis along the rays $-\infty < u \leq -1$ and $-1 \leq u < \infty$ and glued crosswise along the cuts $-\infty < u \leq -1$. The lips of the cuts $-1 \leq u < \infty$ remain free. 375. A two-sheeted domain consisting of two planes cut along the imaginary axis (along the segment $-1 \leq v \leq 1$) and joined so that the left-hand lip of the cut of the first sheet is joined to the right-hand lip of the cut on the second sheet. The remaining lips are free. 376. The Riemann surface has infinitely many sheets and has two logarithmic branch-points located above the points $w=0$ and $w=\infty$. The domains of univalence in the z -plane which correspond to sheets of the w -plane with cuts along the positive real axis are bounded by the circles $2k\pi(x^2+y^2)+y=0$ ($k=0, \pm 1, \pm 2, \dots$). 377. $r_1=\infty$, $r_2=r_3=1$. The image of the circle $|z| < r_1$ is the entire plane with the point $w=-1$ removed; the image of the circle $|z| < r_2$ (and of the circle $|z| < r_3$) is the half-plane $\text{Re } w > -\frac{1}{2}$. 378. 1) $r_1=\frac{1}{2}$; 2) $r_1=\frac{1}{2|a|}$; 3) $r_1=1$. 383. 1) $r_2=\frac{1}{4}$; 2) $r_2=\frac{1}{4|a|}$; 3) $r_2=2-\sqrt{3}$. 384. $r_1=\pi$, $r_2=1$. 385. 1) $r_3=\frac{1}{2}$; 2) $r_3=\frac{1}{2|a|}$; 3) $r_3=1$.

Chapter III

388. 1) $I_1=2+i$, $I_2=1+\frac{i}{2}$; 2) $I_1=\frac{i\pi}{2}$, $I_2=-\frac{\pi}{2}$; 3) $I_1=i\pi R^2$, $I_2=-\pi R^2$.
 389. 1) $\sqrt{5}\left(1-\frac{i}{2}\right)$; 2) 2; 3) $2i$; 4) 0. 390. πi . 391. $\frac{4}{3}$. 392. 1) $\frac{R^{n+1}}{n+1} \times$
 $\times [(-1)^{n+1}-1]$ if $n \neq -1$; πi if $n=-1$; 2) and 3) 0 if $n \neq -1$; $2\pi i$ if $n=-1$.
 393. 1) $-2(1-i)$; 2) $2(1-i)$; 3) $-2(1+i)$; 4) -4 ; 5) $4i$. 394. 1) $2\pi i$; 2) $-2\pi i$;
 3) $2\pi Ri$; 4) $2\pi Ri$. 395. 1) $\frac{2\pi i}{n+1}$ if $n \neq -1$; $-2\pi^2$ if $n=-1$; 2) $(-1)^{n+1} \frac{2\pi i}{n+1}$

if $n \neq -1$, $-2\pi^2$ if $n = -1$. 396. $\frac{e^{2\pi i} - 1}{1 + \alpha}$ if $\alpha \neq -1$; $2\pi i$ if $\alpha = -1$.

398. 1) $|\alpha| \leq \frac{\pi}{2}$; 2) $\sin p\alpha \geq 0$. 412. 1) $\frac{\pi}{3}$; 2) $-\frac{\pi}{3}$; 3) 0. 413. If the contour C contains within it the point 0 and does not contain 1 and -1 , then $I = -2\pi i$; if it contains only one of the points -1 or 1 and does not contain the point 0, then $I = \pi i$. Hence, it is clear that the integral can have five different values $(-2\pi i; -\pi i; 0; \pi i; 2\pi i)$. 414. $2^n - 1$ if $n > 1$; 2 if $n = 1$. 415. $\frac{\pi i}{2}$.

416. $\frac{\sin a}{a}$. 417. $e^a \left(1 + \frac{a}{2}\right)$. 418. 1) 1; 2) $-\frac{e}{2}$; 3) $1 - \frac{e}{2}$. 420. 1) $\frac{2}{3}$;

2) $1 - \frac{2i}{3}$. 425. $R = 1$. 426. ∞ . 427. 0. 428. 2. 429. e . 430. 1. 431. 1. 432. 1.

433. $\frac{1}{4}$. 434. $\frac{1}{e}$. 435. 1 if $|a| \leq 1$; $\frac{1}{|a|}$ if $|a| > 1$. 436. 1. 437. 1) R ;

2) $\frac{R}{2}$, 3) ∞ ; 4) 0; 5) R^k ; 6) R if $|z_0| \leq 1$, and $\frac{R}{|z_0|}$ if $|z_0| > 1$. 438. 1) $R \geq$

$\geq \min(r_1, r_2)$; 2) $R \geq r_1 r_2$; 3) $R \leq \frac{r_1}{r_2}$. 439. 1) $\frac{z}{(1-z)^2}$; 2) $-\ln(1-z)$;

3) $\frac{1}{2} \ln \frac{1+z}{1-z}$; 4) $\ln(1+z)$. 440. Diverges at all points. 441. Converges (non-

absolutely) at all points except $z = 1$. 442. Converges absolutely. 443. Converges (non-absolutely) at all points except $z = -1$. 444. Converges (non-abs-

olutely) at all points except $z = e^{\frac{2k\pi i}{p}}$ ($k = 0, 1, \dots, p-1$). 445. Converges (non-

absolutely) at all points except $z = \frac{1 \pm i\sqrt{3}}{2}$ and $z = -1$. 446. Converges abso-

lutely. 447. For example, $c_n = (-1)^n$. 449. Solution. First investigate conver-

gence at point $z = 1$. The terms of the series $\sum_{n=1}^{\infty} \frac{(-1)^{[Vn]}}{n}$ with denominators

$k^2, k^2+1, \dots, (k+1)^2-1$ have the sign $(-1)^k$; let us denote the sum of these terms by $(-1)^k \sigma_k$ and prove that $\sigma_k \rightarrow 0$ monotonically. We have $0 < \sigma_k <$

$< \frac{(k+1)^2 - k^2}{k^2} = \frac{2k+1}{k^2} \rightarrow 0$ as $k \rightarrow \infty$. Further, the difference $\sigma_k - \sigma_{k+1} =$

$= \left[\frac{1}{k^2} - \frac{1}{(k+1)^2} \right] + \left[\frac{1}{k^2+1} - \frac{1}{(k+1)^2+1} \right] + \dots + \left[\frac{1}{(k+1)^2-1} - \frac{1}{(k+2)^2-3} \right] -$

$-\frac{1}{(k+2)^2-2} - \frac{1}{(k+2)^2-1} > (2k+1) \left[\frac{1}{(k+1)^2-1} - \frac{1}{(k+2)^2-3} \right] - \frac{1}{(k+2)^2-2} -$

$-\frac{1}{(k+2)^2-1} > 0$. This proves that the given series converges for $z = 1$. If

$|z| = 1$, but $z \neq 1$, then, according to the hint, we use the convergence test of Problem 90, putting $a_n = (-1)^{[Vn]} z^n$, $b_n = \frac{1}{n}$. The first two conditions are

obviously true; to prove the third one let us estimate S_n . $S_n = -z - z^2 - z^3 +$

$+ z^4 + \dots + (-1)^{[Vn]} z^n = -z \frac{1-z^3}{1-z} + z^4 \frac{1-z^5}{1-z} - z^9 \frac{1-z^7}{1-z} + \dots \pm z^{p^2} \frac{1-z^{2p+1}}{1-z} \mp$

$\mp (z^{(p+1)} + \dots + z^n)$, where $p = [Vn] - 1$. Whence $|S_n| < \frac{2p}{|1-z|} + 2p + 3$;

hence, for each z there exists a k such that $|S_n| < kp < k\sqrt{n}$, and thus our

- proof is complete.
452. $\sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!}$, $R = \infty$. 453. $\sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!}$, $R = \infty$.
454. $\sum_{n=0}^{\infty} (-1)^{n+1} \frac{z^{2n-1} z^{2n}}{(2n)!}$, $R = \infty$. 455. $\frac{1}{2} + \sum_{n=0}^{\infty} \frac{2^{2n-1} z^{2n}}{(2n)!}$, $R = \infty$.
456. $a^2 \sum_{n=0}^{\infty} \binom{\alpha}{n} \left(\frac{z}{a}\right)^n$, where $\binom{\alpha}{0} = 1$, $\binom{\alpha}{n} = \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!}$ ($n=1, 2, \dots$); $R = |a|$. 457. $\frac{1+i}{\sqrt{2}} \left[1 + \frac{1}{2} \frac{z}{i} + \sum_{n=2}^{\infty} (-1)^{n-1} \frac{1 \cdot 3 \dots (2n-3)}{2 \cdot 4 \dots 2n} \left(\frac{z}{i}\right)^n \right]$, $R = 1$.
458. $\sum_{n=0}^{\infty} (-1)^n \frac{a^n z^n}{b^{n+1}}$, $R = \left| \frac{b}{a} \right|$. 459. $\frac{i}{6} \sum_{n=1}^{\infty} \frac{(2-3i)^n - (2+3i)^n}{13^n} z^n =$
 $= \sum_{n=1}^{\infty} c_n z^n$, where $c_n = \sum_{m=0}^{\left[\frac{n-1}{2}\right]} (-1)^m \binom{n}{2m+1} 2^{n-2m-1} 3^{2m}$, $R = \sqrt{13}$.
460. $\sum_{n=2}^{\infty} (-1)^n (n-1) z^n$, $R = 1$. 461. $2 \sum_{n=0}^{\infty} \frac{z^{2n+1}}{2n+1}$, $R = 1$. 462. $\sum_{n=0}^{\infty} (-1)^n \times$
 $\times \frac{z^{2n+1}}{2n+1}$, $R = 1$. 463. $z + \sum_{n=1}^{\infty} (-1)^n \frac{1 \times 3 \dots (2n-1)}{2^n \times n! (2n+1)} z^{2n+1}$, $R = 1$. 464. $\ln 2 -$
 $-\sum_{n=1}^{\infty} \left(1 + \frac{1}{2^n}\right) \frac{z^n}{n}$, $R = 1$. 465. $\sum_{n=0}^{\infty} \frac{z^{2n+1}}{n! (2n+1)}$, $R = \infty$. 466. $\sum_{n=0}^{\infty} (-1)^n \times$
 $\times \frac{z^{2n+1}}{(2n+1)! (2n+1)}$, $R = \infty$. 467. $\frac{1}{3} + 2 \sum_{n=0}^{\infty} (-1)^{n+1} \frac{(z-1)^n}{3^{n+1}}$, $R = 3$.
468. $\frac{1}{4} \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{2n}} [(z-1)^{2n} + (z-1)^{2n+1}]$, $R = 2$. 469. $\frac{1}{4} + \sum_{n=1}^{\infty} (-1)^n \times$
 $\times \frac{(n-3)(z-1)^n}{2^{n+2}}$, $R = 2$. 470. $\frac{-1+i\sqrt{3}}{2} \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n (z-1)^n$, $R = 1$.
471. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{(z-1)^n}{n}$, $R = 1$. 472. $\sum_{n=0}^{\infty} \frac{\sin\left(1 + \frac{n\pi}{2}\right)}{n!} (z-1)^{2n}$, $R = \infty$. 473. $1 +$
 $+ z^2 + \frac{z^4}{3} + \dots$. 474. $1 - \frac{z^3}{4} - \frac{z^4}{96} + \dots$. 475. $1 + z^2 - \frac{1}{2} z^3 + \frac{5}{6} z^4 - \frac{3}{4} z^5 + \dots$.
476. $1 + z + z^2 + \frac{5}{6} z^3 - \frac{5}{8} z^4 + \frac{13}{30} z^5 + \dots$. 477. $z + \frac{z^2}{2!} + \frac{2z^3}{3!} + \frac{9z^5}{5!} + \dots$.
478. $\ln 2 + \frac{z}{2} + \frac{z^2}{8} - \frac{z^4}{192} + \dots = \ln 2 + \sum_{n=0}^{\infty} \frac{c_n}{n+1} z^{n+1}$; $c_0 = \frac{1}{2}$, $c_0 + nc_1 +$

$$+ n(n-1)c_2 + \dots + n!c_{n-1} + 2n!c_n = 1, \quad n \geq 1. \quad 479. \quad 2 \left[\frac{z^2}{2} + \left(1 + \frac{1}{2}\right) \frac{z^3}{3} + \left(1 + \frac{1}{2} + \frac{1}{3}\right) \frac{z^4}{4} + \dots + \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n-1}\right) \frac{z^n}{n} + \dots \right]. \quad 480. \quad -4\pi^2 + 2 \left[-2\pi iz + (1-2\pi i) \frac{z^2}{2} + \left(1 + \frac{1}{2} - 2\pi i\right) \frac{z^3}{3} + \dots + \left(1 + \frac{1}{2} + \dots + \frac{1}{n-1} - 2\pi i\right) \frac{z^n}{n} + \dots \right]. \quad 481. \quad z^3 - \left(1 + \frac{1}{3}\right) \frac{z^4}{2} + \left(1 + \frac{1}{3} + \frac{1}{5}\right) \frac{z^5}{3} - \dots + (-1)^{n+1} \times \left(1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2n-1}\right) \frac{z^{2n}}{n} + \dots. \quad 482. \quad 2 \left[\left(1 + \frac{1}{2}\right) \frac{z^3}{3} - \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}\right) \frac{z^5}{5} + \dots + (-1)^{n+1} \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{2n}\right) \frac{z^{2n+1}}{2n+1} + \dots \right].$$

$$483. \quad 1 + \sum_{n=1}^{\infty} \left[\sum_{k=1}^n \frac{1}{k!} \binom{n-1}{k-1} \right] z^n. \quad 487. \quad 1 + \sum_{k=1}^{\infty} (-1)^k \frac{2^{2k} B_{2k} z^{2k}}{(2k)!}, \quad R = \pi.$$

$$488. \quad 1) \sum_{k=1}^{\infty} (-1)^k \frac{2^{2k} B_{2k}}{2k (2k)!} z^{2k}, \quad R = \pi; \quad 2) \sum_{k=1}^{\infty} (-1)^{k-1} \frac{2^{2k} (2^{2k}-1) B_{2k}}{(2k)!} z^{2k-1}, \quad R = \frac{\pi}{2}; \quad 3) \sum_{k=1}^{\infty} (-1)^k \frac{2^{2k} (2^{2k}-1) B_{2k}}{2k (2k)!} z^{2k}, \quad R = \frac{\pi}{2}; \quad 4) \quad 1 + \sum_{k=1}^{\infty} (-1)^{k-1} \times$$

$$\times \frac{(2^{2k}-2) B_{2k}}{(2k)!} z^{2k}, \quad R = \pi. \quad 489. \quad c_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^{n+1} - \left(\frac{1-\sqrt{5}}{2} \right)^{n+1} \right], \quad n \geq 0, \quad R = \frac{\sqrt{5}-1}{2}. \quad 490. \quad c_0 = \frac{A}{\alpha}, \quad c_1 = \frac{\alpha B - \beta A}{\alpha^2}, \quad c_n = \frac{\alpha^2 C - \alpha \beta B + (\beta^2 - \alpha \gamma) A}{\alpha^3};$$

$$\alpha c_n + \beta c_{n-1} + \gamma c_{n-2} + \delta c_{n-3} = 0 \quad (n=3, 4, \dots). \quad 496. \quad \frac{e^{\frac{-z}{1-t}}}{1-t}; \quad L_{n+1}(z) - (2n+1-2)L_n(z) + n^2 L_{n-1}(z) = 0 \quad (n \geq 1). \quad 497. \quad z + z^3 + \frac{z^5}{4 \times 5} + \frac{z^9}{4 \times 5 \times 8 \times 9} + \dots + \frac{z^{4n+1}}{4 \times 5 \times 8 \times 9 \dots 4n(4n+1)} + \dots; \quad |z| < \infty. \quad 498. \quad z - \frac{n(n+1)-1 \times 2}{3!} z^3 + \frac{[n(n+1)-1 \times 2][n(n+1)-3 \times 4]}{5!} z^5 + \dots + \frac{[n(n+1)-1 \times 2][n(n+1)-3 \times 4] \dots [n(n+1)-(2k-1)2k]}{(2k+1)!} z^{2k+1} + \dots;$$

$$|z| < 1. \quad 499. \quad \frac{z}{1-z^2}. \quad 500. \quad 1 - \frac{m^2}{2!} z^2 + \frac{m^2(m^2-2^2)}{4!} z^4 - \dots + (-1)^m \times \frac{m^2(m^2-2^2) \dots [m^2-(2n-2)^2]}{(2n)!} z^{2n} + \dots. \quad 501. \quad w = F(a, b, c, z) = 1 +$$

$$+ \frac{ab}{1 \times c} z + \frac{a(a+1)b(b+1)}{2!c(c+1)} z^2 + \dots + \frac{a(a+1) \dots (a+n-1)b(b+1) \dots (b+n-1)}{n!c(c+1) \dots (c+n-1)} z^n; \quad |z| < 1. \quad 503. \text{ Solution. Differentiating the hypergeometric equation we obtain an equation satisfied by the function } \xi = \frac{d}{dz} F(a, b, c, z):$$

$$z(1-z) \frac{d^2 \xi}{dz^2} + [(c+1) - (a+b+3)z] \frac{d \xi}{dz} - (a+1)(b+1)\xi = 0. \quad (1)$$

Since the function $\frac{d}{dz} F(a, b, c, z)$, being a derivative of the function

$F(a, b, c, z)$, which is analytic at $z=0$, is also analytic at the point $z=0$, and since any solution of equation (1) which is analytic at $z=0$ must have the form $kF(a+1, b+1, c+1, z)$ (see Problem 501), where k is a constant, it follows that $\frac{d}{dz}F(a, b, c, z) = kF(a+1, b+1, c+1, z)$. Putting $z=0$, we find

that $k = \frac{ab}{c}$. 505. 1) 4; 2) 15; 3) 3. 506. 1) A zero of order $k+l$; 2) a zero

whose order is not lower than $\min(k, l)$; 3) a zero of order $k-l$ if $k > l$; a regular nonzero point if $k=l$, and a singular point if $k < l$. 507. The points $z = \pm 3i$ are zeros of the first order. 508. The points $z = \pm 3i$ are zeros of the first order; the point at infinity is a second-order zero. 509. $z=0$ is a zero of the second order; $z = k\pi$ ($k = \pm 1, \pm 2, \dots$) are zeros of the first order.

510. $z = \pm 2$ are zeros of the third order; $z = 2k\pi i$ ($k = 0, \pm 1, \pm 2, \dots$) are zeros of the first order. 511. $z = 2k\pi$ ($k = 0, \pm 1, \pm 2, \dots$) are zeros of the second order. 512. $z = \pm \pi$ are zeros of the third order; all remaining points of the form

$z = k\pi$ ($k = 0, \pm 2, \pm 3, \dots$) are zeros of the first order. 513. $z = \frac{\pi}{4} + k\pi$ ($k = 0,$

$\pm 1, \pm 2, \dots$) are zeros of the first order. 514. No zeros. 515. $z = k\pi$ ($k = 0, \pm 1, \pm 2, \dots$) are zeros of the third order. 516. $z=0$ is a zero of the second order; $z = k\pi$ ($k = \pm 1, \pm 2, \dots$) are zeros of the third order. 517. $z=0$ is a

zero of the third order; $z = \sqrt[3]{k\pi}$ and $z = \frac{1}{2}\sqrt[3]{k\pi}(1 \pm i\sqrt{3})$ ($k = \pm 1, \pm 2, \dots$)

are zeros of the first order. 518. $z = (2k+1)\frac{\pi}{2}$ ($k = 0, \pm 1, \pm 2, \dots$) are zeros

of the third order. 519. $z = \sqrt[3]{(2k+1)\frac{\pi}{2}}$ and $z = \frac{1}{2}\sqrt{(2k+1)\frac{\pi}{2}}(1 \pm i\sqrt{3})$

($k = 0, \pm 1, \pm 2, \dots$) are zeros of the first order. 520. $z=4$ is a zero of the third order for one of the branches. 521. Two functions are specified

here: one of them has zeros of the second order at the points $z = 2k\pi \pm \frac{\pi}{6}$, the

other, zeros of the second order at the points $z = (2k+1)\pi \pm \frac{\pi}{6}$ ($k = 0, \pm 1, \pm 2, \dots$). 522. Only the point at infinity can be a limit point. 523. 1), 2) and

3) Does not exist; 4) exists $\left(f(z) = \frac{1}{z+1}\right)$. 524. 1) Exists ($f(z) = z^2$); 2) does

not exist. 525. It does not contradict the uniqueness theorem since the point $z=1$ does not belong to the domain of analyticity of the function. 526. *Solu-*

tion. From the expansion $f(z) = \sum_{n=0}^{\infty} c_n(z-z_0)^n$ it follows that $u(x, y) = \frac{c_0 + \bar{c}_0}{2} +$

$+ \sum_{n=1}^{\infty} \{c_n[(x-x_0) + i(y-y_0)]^n + \bar{c}_n[(x-x_0) - i(y-y_0)]^n\}$. Putting $x = x_0 +$

$+\frac{\zeta - z_0}{2}$, $y = y_0 + \frac{\zeta - z_0}{2i}$, where ζ is sufficiently close to z , we obtain (estab-

lish this!) $u\left(x_0 + \frac{\zeta - z_0}{2}, y_0 + \frac{\zeta - z_0}{2i}\right) = \frac{1}{2}[\bar{c}_0 + f(\zeta)]$, and, substituting z for

ζ , we arrive at the desired equality. 528. $z^2 + 2 + Ci$. 529. $ze^z - \frac{i}{z} + Ci$.

530. $(1+i)z - 3i + C$. 531. $\sin z - \cosh z + C$ (C an arbitrary real constant).

538. 2) $2\pi^2 M$.

Chapter IV

543. $-\frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z}{2}\right)^n$ for $|z| < 2$; $\sum_{n=0}^{\infty} \frac{2^n}{z^{n+1}}$ for $|z| > 2$.
544. $\frac{(-1)^k}{a^k} \sum_{n=0}^{\infty} \binom{n+k-1}{k-1} \left(\frac{z}{a}\right)^n$ for $|z| < |a|$; $\frac{1}{z^k} \sum_{n=0}^{\infty} \binom{n+k-1}{k-1} \left(\frac{a}{z}\right)^n$ for $|z| > |a|$.
545. $\frac{1}{z} + \sum_{n=0}^{\infty} z^n$ for $|z| < 1$; $-\frac{1}{z-1} + \sum_{n=0}^{\infty} (-1)^n (z-1)^n$ for $0 < |z-1| < 1$; $-\sum_{n=2}^{\infty} \frac{1}{z^n}$ for $|z| > 1$.
546. $\frac{1}{b-a} \sum_{n=0}^{\infty} \frac{b^{n+1}-a^{n+1}}{a^{n+1}b^{n+1}} z^n$ for $|z| < |a|$; $\frac{1}{a-b} \left[\frac{1}{z-a} + \sum_{n=0}^{\infty} \frac{(z-a)^n}{(b-a)^{n+1}} \right]$ for $0 < |z-a| < |b-a|$; $\frac{1}{b-a} \sum_{n=2}^{\infty} \frac{b^{n-1}-a^{n-1}}{z^n}$ for $|z| > |b|$; $\frac{1}{a-b} \sum_{n=0}^{\infty} \left(\frac{z^n}{b^{n+1}} + \frac{a^n}{z^{n+1}} \right)$ for $|a| < |z| < |b|$.
547. $\frac{1}{z-2} + i \sum_{n=0}^{\infty} (-1)^n \frac{(2+i)^{n+1} - (2-i)^{n+1}}{5^{n+1}} (z-2)^n$ for $0 < |z-2| < \sqrt{5}$;
548. $-\frac{i}{4(z-i)} - \frac{1}{4(z-i)^2} + \sum_{n=0}^{\infty} \frac{(n+3)i^n (z-i)^n}{2^{n+4}}$ for $0 < |z-i| < 2$; $\sum_{n=1}^{\infty} (-1)^n \frac{n}{z^{2n+2}}$ for $|z| > 1$.
549. $\pm \left[z - \frac{1}{2}(a+b) + \sum_{n=1}^{\infty} \frac{c_{-n}}{z^n} \right]$ for $|z| > |b|$, where $c_{-(n-1)} = (-1)^n \left[\binom{1}{2} b^n + \binom{1}{n-1} \binom{1}{2} b^{n-1} a + \binom{1}{n-2} \binom{1}{2} b^{n-2} a^2 + \dots + \binom{1}{n} a^n \right]$.
550. $\sum_{n=1}^{\infty} \frac{c_{-n}}{z^n} + \sum_{n=0}^{\infty} c_n z^n$, where $c_{-n} = \frac{(-1)^n i}{\sqrt{2}} \left[\binom{-1}{n} + \sum_{m=1}^{\infty} \binom{-1}{n+m} \left(\frac{-1}{2}\right)^{2-m} \right]$, $c_n = 2^{-n} c_{-n}$ ($n = 1, 2, \dots$), $c_0 = \frac{i}{\sqrt{2}} \left[1 + \sum_{m=1}^{\infty} \binom{-1}{m} \left(\frac{-1}{2}\right)^2 2^{-m} \right]$.
551. $\frac{1}{2} + z + z^2 + \sum_{n=1}^{\infty} \frac{1}{(n+2)! z^n}$ for $0 < |z| < \infty$.
552. $\sum_{n=0}^{\infty} (-1)^n \frac{1}{n!(z-1)^n}$ for $0 < |z-1| < \infty$;
553. $1 - \frac{1}{z} + \sum_{n=2}^{\infty} c_{-n} z^{-n}$ for $|z| > 1$, where $c_{-n} = -1 + \sum_{k=1}^{n-1} \frac{(-1)^{k+1}}{(k+1)!} \binom{n-1}{k}$.

$$(n=2, 3, \dots). \quad 553. \cos 1 + \sum_{k=1}^{\infty} \left[\frac{(-1)^{k-1} 4^{2k-1} \sin 1}{(2k-1)! (z-2)^{4k-2}} + \frac{(-1)^k 4^{2k} \cos 1}{(2k)! (z-2)^{4k}} \right]$$

$$\text{for } 0 < |z-2| < \infty. \quad 554. (z-1) + 2 + \sum_{k=1}^{\infty} \left[\frac{(-1)^{k-1} \left[\frac{1}{(2k-1)!} - \frac{1}{(2k+1)!} \right]}{(z-1)^{2k-1}} + \right.$$

$$\left. + \frac{2(-1)^k}{(2k+1)! (z-1)^{2k}} \right] \text{ for } 0 < |z-1| < \infty. \quad 555. \sum_{n=0}^{\infty} c_n z^n + \sum_{n=1}^{\infty} c_{-n} z^{-n}, \text{ where}$$

$$c_n = c_{-n} = \sum_{k=0}^{\infty} \frac{1}{k! (n+k)!} \quad (n=0, 1, 2, \dots). \quad 556. \sum_{n=0}^{\infty} c_{2n} z^{2n} + \sum_{n=0}^{\infty} c_{-2n} z^{-2n},$$

$$\text{where } c_{2n} = c_{-2n} = (-1)^n \sum_{k=0}^{\infty} \frac{1}{(2k+1)! (2n+2k+1)!} \quad (n=0, 1, 2, \dots).$$

$$557. - \sum_{n=0}^{\infty} \frac{\sin \left(1 + \frac{n\pi}{2} \right)}{n! (z-1)^n} \text{ for } 0 < |z-1| < \infty; -\sin 1 - \frac{\cos 1}{z} + \frac{\sin 1 - 2\cos 1}{2! z^2} +$$

$$+ \frac{6 \sin 1 - 5 \cos 1}{3! z^3} + \dots \text{ for } |z| > 1. \quad 558. \frac{1}{z} + \sum_{n=1}^{\infty} (-1)^n \frac{2^{2n} B_{2n}}{(2n)!} z^{2n-1}, \text{ where}$$

$$B_{2n} \text{ are Bernoulli's numbers (see Problem 485), for } 0 < |z| < 2\pi; \frac{3}{z} + 2 \sum_{n=1}^{\infty} \frac{\pi^{2n}}{z^{2n+1}} +$$

$$+ \sum_{n=1}^{\infty} \left[(-1)^n \frac{2^{2n} B_{2n}}{(2n)!} + \frac{2}{\pi^{2n}} \right] z^{2n-1} \text{ for } \pi < |z| < 2\pi. \quad 559. \sum_{n=1}^{\infty} \frac{b^n - a^n}{n z^n} \text{ for } |z| >$$

$$> \max(|a|, |b|). \quad 560. \sum_{n=2}^{\infty} \frac{c_{-n}}{z^n}, \text{ where } c_{-2k} = -i \left(2^{2k-1} - \frac{2^{2k-3}}{3} + \frac{2^{2k-5}}{5} - \dots + \right.$$

$$\left. + (-1)^{k+1} \frac{2}{2k-1} \right), \quad c_{-(2k+1)} = 2c_{-2k} \quad (k=1, 2, \dots), \quad \text{for } |z| > 2,$$

$$i \arctan \frac{1}{2} \times \sum_{n=0}^{\infty} \frac{z^n}{2^n} + \sum_{n=1}^{\infty} \frac{c_{-n}}{z^n}, \text{ where } c_{-1} = 2i \arctan \frac{1}{2}, \quad c_{-2k} = 2^{2k} i \left(\arctan \frac{1}{2} - \right.$$

$$\left. - \sum_{m=0}^{k-1} \frac{(-1)^m}{(2m+1) 2^{2m+1}} \right), \quad c_{-(2k+1)} = 2c_{-2k} \quad (k=1, 2, \dots), \quad \text{for } 1 < |z| < 2.$$

561. 1) Yes; 2) yes; 3) no (the point $z=1$ is not an isolated singular point); 4) no; 5) no; 6) no; 7) no; 8) no (in any annulus enclosing the point $z=0$, the function is not continuous); 9) no; 10) yes; 11) yes, if α is an integer or zero; no in all other cases. 562. 1) No; 2) yes, both branches allow expansion; 3) no; 4) yes, all three branches allow expansion; 5) no; 6) two branches out of the

four determined by the conditions $\sqrt{1+\sqrt{1}} = \pm \sqrt{2}$ have expansions; 7) no; 8) no; 9) no; 10) yes, all six branches permit expansion; 11) no; 12), 13) and 14) yes, any branch can be expanded; 15) no; 16) no; 17) expansion is possible

for all branches except two determined by the value of $\text{Arc sin } \frac{\sqrt{2}}{2} = \frac{\pi}{4}$.

565. $z=0$, $z=\pm 1$ are poles of the first order, $z=\infty$ is a regular point (zero of the third order). 566. $z=\frac{1\pm i}{\sqrt{2}}$, $z=\frac{-1\pm i}{\sqrt{2}}$ are poles of the first order;

$z=\infty$ is a regular point. 567. $z=1$ is a pole of the second order; $z=\infty$ is a pole of the third order. 568. $z=0$ is a pole of the first order; $z=\pm 2i$ are poles of the second order; $z=\infty$ is a regular point (zero of the fifth order). 569. $z=\pm i$ are poles of the first order; $z=\infty$ is an essential singularity. 570. $z=\infty$ is an essential singularity. 571. $z=\infty$ is an essential singularity. 572. $z=2k\pi i$ ($k=\pm 1, \pm 2, \dots$) are poles of the first order; $z=\infty$ is a limit point for poles. 573. $z=0$ is a pole of the second order; $z=2k\pi i$ ($k=\pm 1, \pm 2, \dots$) are poles of the first order; $z=\infty$ is a limit point for poles. 574. $z=(2k+1)\pi i$ ($k=0, \pm 1, \pm 2, \dots$) are poles of the first order; $z=\infty$ is a limit point for poles. 575. $z=0$ is a pole of the third order; $z=2k\pi \pm \pm i\ln(2+\sqrt{3})$ ($k=0, \pm 1, \pm 2, \dots$) are poles of the first order; $z=\infty$ is a limit point for poles. 576. $z=k\pi i$ ($k=0, \pm 1, \pm 2, \dots$) are poles of the first order; $z=\infty$ is a limit point for poles. 577. $z=0$ is an essential singularity; $z=\infty$ is a regular point. 578. $z=0$ is an essential singularity; $z=\infty$ is a pole of the first order. 579. $z=1$ is an essential singularity; $z=\infty$ is a regular point. 580. $z=0$ is an essential singularity; $z=\infty$ is an essential singularity. 581. $z=1$ is an essential singularity; $z=2k\pi i$ ($k=0, \pm 1, \pm 2, \dots$) are poles of the first order; $z=\infty$ is a limit point for poles. 582. $z=k\pi$ ($k=0, \pm 1, \pm 2, \dots$) are poles of the first order; $z=\infty$ is a limit point for poles. 583. $z=0$ is a pole of the second order; $z=\infty$ is an essential singularity. 584. $z=(2k+1)\frac{\pi}{2}$ ($k=0, \pm 1, \pm 2, \dots$) are poles of the first order; $z=\infty$ is a limit

point for poles. 585. $z=(2k+1)\frac{\pi}{2}$ ($k=0, \pm 1, \pm 2, \dots$) are poles of the second order; $z=\infty$ is a limit point for poles. 586. $z=0$ is a pole of the third order; $z=k\pi$ ($k=\pm 1, \pm 2, \dots$) are poles of the first order; $z=\infty$ is a limit point for poles. 587. $z=k\pi$ ($k=\pm 1, \pm 2, \dots$) are poles of the first order; $z=\infty$ is a limit point for poles. 588. $z=k\pi$ ($k=0, \pm 1, \pm 2, \dots$) are poles of the first order; $z=\infty$ is a limit point for poles. 589. If $a \neq m\pi + \frac{\pi}{2}$ ($m=0, \pm 1, \pm 2, \dots$), then $z=2k\pi + a$ and $z=(2k+1)\pi - a$ ($k=0, \pm 1, \pm 2, \dots$) are simple poles; if $a = m\pi + \frac{\pi}{2}$, then, for even m , $z=2k\pi + \frac{\pi}{2}$ and, for odd m , $z=(2k+1)\pi + \frac{\pi}{2}$

are poles of the second order; $z=\infty$ is in all cases a limit point for poles. 590. If $a \neq m\pi$ ($m=0, \pm 1, \pm 2, \dots$), then $z=(2k+1)\pi \pm a$ ($k=0, \pm 1, \pm 2, \dots$) are poles of the first order; if $a = m\pi$, then, for odd m , $z=2k\pi$, and, for even m , $z=(2k+1)\pi$ are poles of the second order; $z=\infty$ is in all cases a limit point for poles. 591. $z=1$ is an essential singularity; $z=\infty$ is a regular point (zero of the first order). 592. $z=-2$ is a pole of the second order; $z=2$ is an essential singularity; $z=\infty$ is a pole of the third order. 593 and 594. $z=\frac{1}{k\pi}$

($k=\pm 1, \pm 2, \dots$) are poles of the first order; $z=0$ is a limit point for poles; $z=\infty$ is a pole of the first order. 595. $z=0$ is an essential singularity; $z=\infty$ is a regular point (zero of the first order). 596. $z=0$ is an essential singularity;

$z=\infty$ is an essential singularity. 597. $z=\frac{1}{k\pi}$ ($k=\pm 1, \pm 2, \dots$) are essential singularities; $z=0$ is a limit point for essential singularities; $z=\infty$ is an essential singularity. 598. $z=\frac{2}{(2k+1)\pi}$ ($k=0, \pm 1, \pm 2, \dots$) are essential singularities; $z=0$ is a limit point for essential singularities; $z=\infty$ is a regular

point. 599. $z = \frac{1}{k\pi}$ ($k = \pm 1, \pm 2, \dots$) are essential singularities; $z = 0$ is a limit

point for essential singularities; $z = \infty$ is an essential singularity. 600. $z = \frac{2}{(2k+1)\pi}$

($k = 0, \pm 1, \pm 2, \dots$) are essential singularities; $z = 0$ is a limit point for essential singularities; $z = \infty$ is a regular point. 601. For one branch, a regular point, for the other, a pole of the first order. 602. For one branch, a pole of the first order, for the remaining five branches, a regular point. 603. For one branch, a regular point, for the other, a pole of the second order. 604. For one branch, a regular point, for the other, an essential singularity. 605. For both branches, a pole of the first order. 606. $z = \left(1 + \frac{1}{k\pi}\right)^2$ is a regular point for one branch

and a pole of the first order for the other; $z = 1$ is a regular point for one branch and a nonisolated singularity (a limit point for poles), for the other. 607. Each of the given points is a regular point for one branch and a pole of the first order for the other. 608. A regular point for one branch and an essential singularity for the other. 609. 1) For all the branches, a pole of the first order; 2) for one infinite set of branches, a regular point, for the remaining infinite set of branches, a pole of the first order. 610. For one infinite set of branches, a regular point, for the remaining infinite set of branches, an essential singularity. 611. 1) The point $z = \infty$ is a pole of order

$k = \max(n, m)$ if $n \neq m$; if $n = m$, $z = \infty$ is either a pole of order $k \leq n$, or a regular point; 2) a pole of order $n - m$ if $n > m$, and a regular point if $n \leq m$; if $n < m$, then $z = \infty$ is a zero of order $m - n$; 3) a pole of order $n + m$. 613. Examples: 1) z^2 ; 2) $\frac{1}{z^2} + z$; 3) $\frac{1}{z^2 - 1}$. 614. 1) $\frac{a}{z - \alpha}$ ($a \neq 0$) or $az + b$ ($a \neq 0$); 2) $\frac{a}{(z - \alpha)^n}$ ($a \neq 0$) or $a_0 + a_1 z + \dots + a_n z^n$ ($a_n \neq 0$); 3) $\frac{1}{z^2} + C$;

4) $\frac{a_0 + a_1 z + \dots + a_n z^n + a_{n+m} z^{n+m}}{z^n}$ ($a_0 \neq 0, a_{n+m} \neq 0$); 5) $\frac{a_0 + a_1 z + \dots + a_n z^n}{(z - \alpha_1)(z - \alpha_2) \dots (z - \alpha_n)}$

($\alpha_k \neq \alpha_l$ for $k \neq l$ and at least one of the numbers a_m is nonzero) or

$\frac{a_0 + a_1 z + \dots + a_n z^n}{(z - \alpha_1)(z - \alpha_2) \dots (z - \alpha_{n-1})}$ ($a_n \neq 0, \alpha_k \neq \alpha_l$ for $k \neq l$). 616. 1) z_0 is a removable singularity; 2) z_0 is a pole of order n if $\varphi(z)$ is univalent in the neighbourhood of the point z_0 , and a pole of order nm if $\varphi(z)$ is m -valent in the neighbourhood of this point; 3) z_0 is an essential singularity. 617. 1) The point z_0^* is a pole of order n if γ' is a rectilinear segment, and a regular point of multiplicity n if γ' is a circular arc, i.e. $f(z) - f(z_0^*) = (z - z_0^*)^n \varphi(z)$, where $\varphi(z)$ is analytic in the neighbourhood of the point z_0^* and $\varphi(z_0^*) \neq 0$. If $z_0^* = \infty$, this condition is written in the form $f(z) - f(\infty) = z^{-n} \varphi(z)$, where $\varphi(z)$ is analytic at ∞ and $\varphi(\infty) \neq 0$; 2) z_0^* is an essential singularity. 619. 1) -1 ; 2) 0; 3) 0; 4) 0. 620. 1) An essential singularity $z = \infty$; an exceptional value is 0 (and ∞); $e^z \rightarrow 0$ if $x \rightarrow -\infty$ ($e^z \rightarrow \infty$ if $x \rightarrow +\infty$); 2) an essential singularity $z = 0$; the

exceptional value 0 (and ∞); $e^{\frac{1}{z}} \rightarrow 0$ if $z \rightarrow 0$, for instance, along the path $y = 0$, $x < 0$ ($e^{\frac{1}{z}} \rightarrow \infty$ as $z \rightarrow 0$ along the path $y = 0, x > 0$); 3) an essential singularity $z = 0$; there are no exceptional values (except ∞); $\cos \frac{1}{z} \rightarrow \infty$ for $x = 0, y \rightarrow 0$; 4) an essential singularity $z = \infty$; the exceptional values are i and $-i$. 621. $\operatorname{res} [f(z)]_{z=\pm 1} = -\frac{1}{2}$; $\operatorname{res} [f(z)]_{z=0} = 1$; $\operatorname{res} [f(z)]_{z=\infty} = 0$. 622. $\operatorname{res} [(f)]_{z=i} =$

$$= -\frac{i}{4}; \text{ res } [f(z)]_{z=-i} = \frac{i}{4}; \text{ res } [f(z)]_{z=\infty} = 0. \quad 623. \text{ res } [f(z)]_{z=-1} =$$

$$= (-1)^{n+1} \frac{(2n)!}{(n-1)!(n+1)!}; \text{ res } [f(z)]_{z=\infty} = (-1)^n \frac{(2n)!}{(n-1)!(n+1)!}.$$

$$624. \text{ res } [f(z)]_{z=0} = 1; \text{ res } [f(z)]_{z=\pm 1} = -\frac{1}{2}; \text{ res } [f(z)]_{z=\infty} = 0.$$

$$625. \text{ res } [f(z)]_{z=0}=0; \text{ res } [f(z)]_{z=1}=1; \text{ res } [f(z)]_{z=\infty}=-1. \quad 626. \text{ res } [f(z)]_{z=-1}=2 \sin 2;$$

$$\text{ res } [f(z)]_{z=\infty} = -2 \sin 2. \quad 627. \text{ res } [f(z)]_{z=0} = \frac{1}{9}; \text{ res } [f(z)]_{z=3i} = -\frac{1}{54} (\sin 3 -$$

$$-i \cos 3); \text{ res } [f(z)]_{z=-3i} = -\frac{1}{54} (\sin 3 + i \cos 3); \text{ res } [f(z)]_{z=\infty} = \frac{1}{27} (\sin 3 - 3).$$

$$628. \text{ res } [f(z)]_{z=\frac{2k+1}{2}\pi} = -1 \quad (k=0, \pm 1, \pm 2, \dots). \quad 629. \text{ res } [f(z)]_{z=k\pi} =$$

$$= (-1)^k \quad (k=0, \pm 1, \pm 2, \dots). \quad 630. \text{ res } [f(z)]_{z=k\pi} = 0 \quad (k=0, \pm 1, \pm 2, \dots).$$

$$631. \text{ res } [f(z)]_{z=k\pi} = -1 \quad (k=0, \pm 1, \pm 2, \dots). \quad 632. 1) \text{ res } [f(z)]_{z=2} =$$

$$= \text{ res } [f(z)]_{z=\infty} = 0; 2) \text{ res } [f(z)]_{z=2} = -\text{ res } [f(z)]_{z=\infty} = -\frac{143}{24}.$$

$$633. \text{ res } [f(z)]_{z=0} = -\text{ res } [f(z)]_{z=\infty} = \sum_{n=0}^{\infty} \frac{1}{n!(n+1)!}. \quad 634. \text{ res } [f(z)]_{z=0} =$$

$$= \text{ res } [f(z)]_{z=\infty} = 0. \quad 635. \text{ res } [f(z)]_{z=-1} = -\text{ res } [f(z)]_{z=\infty} = -\cos 1.$$

$$636. \text{ res } [f(z)]_{z=-3} = -\text{ res } [f(z)]_{z=\infty} = -\sin 2 \left[\sum_{n=1}^{\infty} \frac{4^{2n}}{(2n-1)!(2n)!} + \right.$$

$$\left. + \sum_{n=0}^{\infty} \frac{4^{2n+1}}{(2n)!(2n+1)!} \right]. \quad 637. \text{ res } [f(z)]_{z=0} = \frac{1}{2}; \text{ res } [f(z)]_{z=\frac{2k\pi i}{h}} = \frac{1}{2k\pi i}$$

$$(k=\pm 1, \pm 2, \dots). \quad 638. \text{ res } [f(z)]_{z=0}=0 \text{ if } n < 0 \text{ and also if } n > 0 \text{ is odd;}$$

$$\text{ res } [f(z)]_{z=0} = \frac{(-1)^{\frac{n}{2}}}{(n+1)!} \text{ if } n=0 \text{ or } n > 0 \text{ is even; res } [f(z)]_{z=\infty} = -\text{ res } [f(z)]_{z=0}.$$

$$639. \text{ res } [f(z)]_{z=\frac{1}{k\pi}} = (-1)^{k+1} \frac{1}{k^3\pi^3} \quad (k=\pm 1, \pm 2, \dots); \text{ res } [f(z)]_{z=\infty} =$$

$$= \frac{2}{\pi^2} \sum_{k=1}^{\infty} \frac{(-1)^k}{k^2} = -\frac{1}{6}. \quad 640. \text{ res } [f(z)]_{z=k^2\pi^2} = (-1)^k 2k^2\pi^2 \quad (k=1, 2, \dots).$$

$$641. \text{ res } [f(z)]_{z=0}=0 \text{ if } n \text{ is odd, res } [f(z)]_{z=0} = (-1)^{k+1} \frac{2^{2k}(2^{2k}-1)}{(2k)!} B_{2k} \text{ if } n=2k$$

$$(k=0, 1, 2, \dots), \text{ where } B_{2k} \text{ are Bernoulli's numbers (see Problem 485);}$$

$$\text{ res } [f(z)]_z = \left(k + \frac{1}{2}\right)\pi = -\frac{1}{\left(k + \frac{1}{2}\right)^n \pi^n} \quad (k=0, \pm 1, \pm 2, \dots). \quad 642. 1; -1.$$

$$643. 0; 2. \quad 644. -2e^{2k\pi i} \text{ if } \sqrt{1}=1 \text{ and } \text{Ln } 1=2k\pi i; 0 \text{ for branches defined by}$$

$$\text{ the value } \sqrt{1}=-1. \quad 645. \pm \frac{(a-b)^2}{8}. \quad 646. 1) \alpha-\beta \text{ (for all branches); 2) } e^\alpha - e^\beta$$

$$\text{ (for all branches). } \quad 647. 1) 2k\pi i + \frac{1}{2 \times 3!} - \frac{1}{4 \times 5!} + \dots \text{ if } \text{Ln } 1=2k\pi i; 2) -\frac{1}{2!} +$$

$$+ \frac{1}{3 \times 4!} - \frac{1}{5 \times 6!} + \dots \text{ (for all branches). } \quad 648. \text{ res } [f(z)]_{z=0}=k\pi \text{ if } \text{Arc tan } 0=k\pi;$$

- $\operatorname{res} [f(z)]_{z=\infty} = -\frac{(2k+1)\pi}{2}$ if $\operatorname{Arc} \tan \infty = \frac{(2k+1)\pi}{2}$. 649. $\operatorname{res} [f(z)]_{z=0} = 0$ if $n \geq 0$; $\operatorname{res} [f(z)]_{z=0} = \operatorname{Ln} \frac{\alpha}{\beta}$ if $n = -1$, and $\operatorname{res} [f(z)]_{z=0} = \frac{1}{n+1} (\alpha^{n+1} - \beta^{n+1})$ if $n \leq -2$; $\operatorname{res} [f(z)]_{z=\infty} = \frac{1}{n+1} (\alpha^{n+1} - \beta^{n+1})$ if $n \geq 0$; $\operatorname{res} [f(z)]_{z=\infty} = -2k\pi i$ if $n = -1$ and $\operatorname{Ln} 1 = 2k\pi i$, and $\operatorname{res} [f(z)]_{z=\infty} = 0$ if $n \leq -2$. 650. $-2c_0c_1$.
 651. $A\varphi(a)$; 2) $c_{-1}\varphi(a) + \frac{c_{-2}\varphi'(a)}{1!} + \dots + \frac{c_{-k}\varphi^{(k-1)}(a)}{(k-1)!}$. 652. 1) n ; 2) $-n$.
 653. 1) $n\varphi(a)$; 2) $-n\varphi(a)$. 654. $\frac{A}{\varphi'(a)}$. 655. AB . 656. $\sum_{n=1}^k (-1)^n \times \frac{\bar{n}c_{-n}(\beta^* - a)^{n+1}}{R^{2n}}$. 657. $-\frac{\pi i}{\sqrt{2}}$. 658. $-2\pi i$. 659. $-\frac{\pi i}{121}$. 660. πi .
 661. $-\frac{2\pi i}{9}$. 662. 1. 663. 0. 664. $\frac{2^{n+1}}{(n+1)!}$ if $n \geq -1$, and 0 if $n < -1$.
 665. $32\pi i$. 666. 0. 667. $\frac{f(0)}{g(0)} + \sum_{k=1}^n \frac{f(a_k)}{a_k g'(a_k)}$. 669. 0 if $r < 1$; ± 1 if $r > 1$ (the sign depends on the branch of the integrand). 670. $\frac{1}{4} \sqrt{1 + \sqrt{2}}$.
 671. $\frac{1}{\pi(1+a)}$. 672. $\frac{e^{\frac{\pi}{2}}}{1+e^{\pi}}$. 673. $\frac{2\pi}{\sqrt{a^2-1}}$. 674. $\frac{2\pi a}{(a^2-b^2)^{\frac{3}{2}}}$.
 675. $\frac{(2a+b)\pi}{3}$. 676. $\frac{2\pi}{1-a^2}$ if $|a| < 1$; $\frac{2\pi}{a^2-1}$ if $|a| > 1$; 0 (principal value) if $|a|=1$, $a \neq \pm 1$ (for $a = \pm 1$ there is no principal value).
 677. $\frac{\pi(a^6+1)}{1-a^2}$ if $|a| < 1$; $\frac{\pi(a^6+1)}{a^6(a^2-1)}$ if $|a| > 1$; $\frac{\pi}{2} \frac{1-a^{12}}{a^6(a^2-1)}$ (principal value) if $|a|=1$, $a \neq \pm 1$ (for $a = \pm 1$ there is no principal value). 678. $\frac{2\pi}{n!}$ if $n \geq 0$, 0 if $n < 0$. 679. $\pi i \operatorname{sign} a$ (for $a=0$ the principal value of the integral is 0).
 680. $-2\pi i \operatorname{sign} \operatorname{Im} a$. 682. $-\frac{\pi}{27}$. 683. $\frac{\pi}{4a}$. 684. $\frac{1 \times 3 \times 5 \dots (2n-3)}{2 \times 4 \times 6 \dots (2n-2)} \frac{\pi}{2}$ if $n > 1$; $\frac{\pi}{2}$ if $n=1$. 685. $\frac{\pi}{ab(a+b)}$. 686. $\frac{\pi\sqrt{2}}{2}$. 687. $\frac{\pi}{n \sin \frac{\pi}{n}}$. 688. $\frac{\pi}{n} \frac{\sin \frac{\pi}{2n}}{\sin \frac{\pi}{n}}$.
 690. $\frac{(-1)^{k-1}}{(2ih-z)^k}$. 691. 1) $\frac{\pi}{3e^3} (\cos 1 - 3 \sin 1)$; 2) $\frac{\pi}{3e^3} (3 \cos 1 + \sin 1)$.
 692. $\frac{\pi}{2e^3} (2 \cos 2 + \sin 2)$. 693. $\frac{\pi e^{-ab}}{2b}$. 694. $\frac{\pi}{2} e^{-ab}$. 696. πi if $t > 0$; 0 if $t=0$; $-\pi i$ if $t < 0$. 697. $\pi (2 \sin 2 - 3 \sin 3)$. 698. $\frac{\pi}{5} \left(\cos 1 - \frac{1}{e^2} \right)$.
 699. $\frac{\pi}{3} \left[\sin |t| + e^{-\frac{|t| \sqrt{3}}{2}} \left(\sin \left| \frac{t}{2} \right| + \sqrt{3} \cos \frac{t}{2} \right) \right]$. 700. $\frac{\pi}{4} [e^{-|t|} - \sin |t|]$.

$$701. \pi \left(e^{-ab} - \frac{1}{2} \right). \quad 702. \frac{\pi}{2b^2} (1 - e^{-ab}). \quad 703. \frac{\pi}{4b^4} [2 - (2 + ab)e^{-ab}].$$

$$704. \pi(b-a). \quad 705. \frac{\pi}{2}. \quad 706. \frac{3\pi}{8}. \quad 707. 1) \frac{\Gamma(p) \cos \frac{\pi p}{2}}{a^p}; \quad 2) \frac{\Gamma(p) \sin \frac{\pi p}{2}}{a^p}.$$

to see that the answer for $-1 < p < 0$ is correct, it suffices to note that for these values of p the integral converges, and the function given in the answer is analytic.

$$708. \frac{1}{p} \Gamma\left(\frac{1}{p}\right) \cos \frac{\pi}{2p}. \quad 709. \frac{1}{p} \Gamma\left(\frac{1}{p}\right) \sin \frac{\pi}{2p}. \quad 710. \frac{1}{p-1} \Gamma\left(\frac{1}{p}\right) \cos \frac{\pi}{2p}$$

(for $p=1$ the integral equals $\frac{\pi}{2}$). $712. \frac{\pi}{\sin p\pi}. \quad 714. \frac{\pi}{2 \cos \frac{\pi p}{2}}. \quad 715. \frac{\pi(1-p)}{4 \cos \frac{\pi p}{2}}$

(for $p=1$ the integral equals $\frac{1}{2}$). $716. \frac{\pi}{\sin p\pi} \frac{\sin p\lambda}{\sin \lambda}$ if $\lambda \neq 0$, and $\frac{p\pi}{\sin p\pi}$

if $\lambda=0$. $718. 0. \quad 719. \frac{\pi}{4}. \quad 720. \pi \cot p\pi. \quad 721. \pi \cot p\pi. \quad 722. \frac{\pi p(1-p)}{2^{2p-1} \sin p\pi}.$

$$723. \frac{\pi}{\sin p\pi} \left(2^{\frac{p}{2}} \cos \frac{p\pi}{4} - 1 \right). \quad 724. \frac{\pi}{\sin p\pi} \left[2^p \left(1 - \frac{p}{2} \right) - 1 \right]. \quad 725. \frac{\pi}{\sin p\pi} \times$$

$$\times \left[1 - \left(\frac{a}{1+a} \right)^p \right]. \quad 726. \frac{p\pi}{\sin p\pi} \frac{a^{p-1}}{(1+a)^{p+1}}. \quad 727. \frac{\pi}{\sin p\pi} \left(\sin \frac{p\pi}{2} + \cos \frac{p\pi}{2} - 1 \right).$$

$$728. \frac{3\sqrt[3]{4}}{\sqrt{3}}. \quad 729. \text{If } a \text{ does not belong to the interval } (-1, 1), \text{ then}$$

$I = -\frac{\pi}{\sqrt{a^2-1}}$, where $\sqrt{a^2-1} > 0$ for $a > 1$ (in the plane with a cut along the segment $[-1, 1]$ the quantity $\sqrt{a^2-1}$ is single-valued); for $a = \pm e^{i\alpha}$,

$$I = \pm \frac{\pi}{\sqrt{2} \sin \alpha} e^{i\left(\frac{3\pi}{4} - \frac{\alpha}{2}\right)}; \text{ for } a = iy, I = \frac{\pi i}{\sqrt{1+y^2}} \operatorname{sign} y; \text{ for } -1 < a < 1$$

$I=0$ (principal value). $730.$ If b does not belong to the interval $(0, 1)$, then

$$I = \frac{\pi}{\sin p\pi} b^{p-1} (b-1)^{-p}, \text{ where } (b-1)^{-p} > 0, b^{p-1} > 0 \text{ for } b > 1; \text{ if } 0 < b < 1,$$

then $I = -\pi b^{p-1} (1-b)^{-a} \cot p\pi$ (principal value). $731. \frac{\pi}{n \sin \frac{\pi}{n}}. \quad 732. \frac{\pi}{2a} \ln a.$

$$733. \frac{\pi}{8a} (\pi^2 + 4 \ln^2 a). \quad 734. \frac{\pi}{2a^2 \sqrt{2a}} \left(\frac{3}{2} \ln a - 1 - \frac{3\pi}{4} \right). \quad 735. -\pi. \quad 736. \frac{\pi}{4} \ln 2.$$

$$737. \frac{1}{2} \ln^2 \frac{a}{1+a}. \quad 739. 1) \frac{1}{1-a} + \frac{1}{\ln a} \left(\text{for } a=1 \quad I = \frac{1}{2} \right); \quad 2) \frac{\pi}{2a \left(\ln^2 a + \frac{\pi^2}{4} \right)} -$$

$$-\frac{1}{1+a^2}, \quad 740. \frac{1}{2n+1} \left\{ \frac{\pi}{2a} \sum_{k=0}^{2n} (-1)^k \frac{2k+1}{\ln^2 a + \left(k + \frac{1}{2} \right)^2 \pi^2} - \frac{1}{1+a^2} \right\}.$$

$$741. \frac{1}{2n} \left\{ \frac{1}{1+a^2} + \frac{\pi}{2a} \sum_{k=0}^{2n-1} (-1)^{k+1} \frac{2k+1}{\ln^2 a + \left(k + \frac{1}{2} \right)^2 \pi^2} \right\}. \quad 743. \frac{\pi(1-2^{a-1})}{\sin a\pi}.$$

- (for $a=1$, $l=\ln 2$). 744. $\frac{\pi}{2} \tanh \frac{\pi a}{2}$. 745. $\frac{\pi}{2 \cosh \frac{\pi a}{2}}$. 746. $\frac{\pi^2 e^{\pi a}}{(e^{\pi a} + 1)^2}$.
747. $\frac{1}{2 \cos \frac{a}{2}}$. 748. $\frac{\pi}{a} \ln(1+a)$ if $0 < a < 1$; $\frac{\pi}{a} \ln \frac{1+a}{a}$ if $a > 1$. 750. 1) $\frac{t^n}{n!}$;
 2) $\frac{\ln^n t}{n!}$ if $t > 1$; 0 if $t < 1$; if $t=1$, then $l=0$ for $n > 1$ and $l=\frac{1}{2}$ (principal value) for $n=1$. 751. $\frac{e^{at} t^n}{n!}$. 752. $\sin t$. 753. 1) $\cos t$; 2) $t - \sin t$.
754. $\frac{e^{at}}{(b-a)(c-a)} + \frac{e^{bt}}{(a-b)(c-b)} + \frac{e^{ct}}{(a-c)(b-c)}$. 755. $\frac{1}{n!} \left(1 - \frac{1}{t}\right)^n$ if $t \geq 1$;
 0 if $t \leq 1$. 758. 1) $\frac{e^{-t}}{\sqrt{\pi t}}$; 2) $\frac{e^{-it}}{\sqrt{\pi t}}$. 759. $\operatorname{erf} \sqrt{t}$, where $\operatorname{erf} u = \frac{2}{\sqrt{\pi}} \int_0^u e^{-x^2} dx$.
760. $e^{-t} \operatorname{erf} \sqrt{t}$. 761. $\sin t$ if $t < \pi$; 0 if $t \geq \pi$. 762. 1 if $0 < t < a$; 0 if $t=a$;
 -1 , if $a < t < 2a$; $-\frac{1}{2}$ if $t=2a$; 0 if $t > 2a$. 763. $n+1$ if $na < t < (n+1)a$;
 $n + \frac{1}{2}$ if $t=na$ ($n=0, 1, 2, \dots$). 764. $1 - \operatorname{erf} \frac{x}{2\sqrt{t}}$ (see the answer to Problem 759). 765. $\frac{1}{a} + 2 \sum_{n=1}^{\infty} (-1)^n e^{-\frac{n^2 \pi^2 t}{a^2}} \frac{\sin \frac{n \pi x}{a}}{n \pi x}$. 766. $-Ei(-t)$, where
 $Ei(t) = \int_{-\infty}^t \frac{e^u}{u} du$. 767. $\frac{2 \sinh t}{t}$. 768. $\frac{2 \pi i}{a}$. 769. 0 if $b < 0$; $\operatorname{arcosh} \frac{b}{a}$ if $b > 0$.
772. $\frac{1}{\sqrt{z^2+1} (z + \sqrt{z^2+1})^n}$ ($\sqrt{z^2+1} > 0$ for $z > 0$). 773. 1) $\frac{1}{\sqrt{a^2-b^2}}$
 if $a > b$; 0 if $a < b$; 2) 0 if $a > b$; $\frac{1}{\sqrt{b^2-a^2}}$ if $a < b$. 774. $\frac{\pi}{2} J_0(i|a|\sqrt{t^2-b^2})$.
775. If the real part of at least one pole is positive, then $\lim_{t \rightarrow \infty} f(t) = \infty$; if the real parts of all poles are negative, then $\lim_{t \rightarrow \infty} f(t) = 0$; if some poles are situated on the imaginary axis and all the others have a negative real part, then $f(t)$ oscillates as $t \rightarrow \infty$, the amplitude increasing without bound if at least one pole on the imaginary axis is of order higher than first, and it remains bounded if all the poles located on the imaginary axis are simple. If there is only one pole on the imaginary axis (at the origin), then $f(t) \rightarrow \infty$ if the pole is multiple, and $f(t) \rightarrow \operatorname{res}[e^{zt}\varphi(z)]_{z=0}$ if the pole is simple. 777. $f(t) \sim \frac{at-3}{a^4}$.
778. $f(t) \sim \frac{i\omega}{\sqrt{2a i \omega - \omega^2}} e^{i\omega t - \sqrt{2a i \omega - \omega^2} t}$. 781. Solution. Rewrite the integral in the form $\int_{-\frac{x}{t}}^x \frac{e^{-x-t}}{t} dt + \int_x^{\infty} \frac{e^{-x-t}}{t} dt$. Since in an asymptotic expansion in negative powers of x , $e^{-x} \sim 0$, it follows from the solution of Problem 780 that the second integral is asymptotically equal to zero. By the definition of the principal

pal value of the integral

$$\begin{aligned} \int_{-x}^x \frac{e^{-x-t}}{t} dt &= e^{-x} \lim_{\varepsilon \rightarrow 0} \left[\int_{-x}^{-\varepsilon} \frac{e^{-t}}{t} dt + \int_{\varepsilon}^x \frac{e^{-t}}{t} dt \right] = \\ &= e^{-x} \lim_{\varepsilon \rightarrow 0} \int_{\varepsilon}^x \frac{e^{-t} - e^t}{t} dt = e^{-x} \int_0^x \frac{e^{-t} - e^t}{t} dt \end{aligned}$$

(in the latter integral the integrand is continuous). Further

$$\int_0^x \frac{e^{-t} - e^t}{t} dt = \int_0^1 \frac{e^{-t} - e^t}{t} dt + \int_1^x \frac{e^{-t}}{t} dt + \int_1^x \frac{e^t}{t} dt \sim O(1) + \int_1^x \frac{e^t}{t} dt.$$

Integrating by parts, we obtain

$$\int_1^x \frac{e^t}{t} dt = C + e^x \left(\frac{1}{x} + \frac{1}{x^2} + \dots + \frac{(n-1)!}{x^n} \right) + n! \int_1^x \frac{e^t}{t^{n+1}} dt,$$

where $C = C(n)$ is a constant. It remains to establish that

$$\lim_{x \rightarrow \infty} x^n e^{-x} \int_1^x \frac{e^t}{t^{n+1}} dt = 0,$$

which is easy to prove with the aid of l'Hospital's rule. **784. Solution.** Consider $\int_C e^{z^2-t^2} dt$, where the contour C is shown in Fig. 68 (for $\operatorname{Re} z > 0$). This

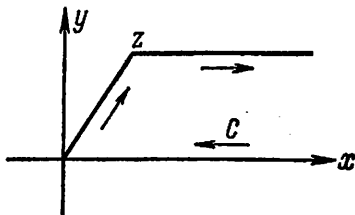


FIG. 68

integral is equal to zero and hence ($z = x + iy$)

$$\int_0^z e^{z^2-t^2} dt = \int_0^\infty e^{z^2-t^2} dt - \int_x^\infty e^{z^2-(t+iy)^2} dt.$$

The first integral in the right member is equal to $\frac{\sqrt{\pi}}{2} e^{z^2}$, rewrite the second one in the form

$$-\frac{1}{2} \int_x^\infty \frac{1}{t+iy} de^{z^2-(t+iy)^2} = \frac{1}{2z} - \frac{1}{2} \int_x^\infty \frac{e^{z^2-(t+iy)^2}}{(t+iy)^2} dt.$$

Repeating integration by parts, we obtain the required expansion. For the re-

mainder we have the estimate

$$\left| \frac{1 \times 3 \dots (2n-1)}{2^n} \int_x^\infty \frac{e^{z^2 - (t+iy)^2}}{(t+iy)^{2n}} dt \right| < \frac{1 \times 3 \dots (2n-1)}{2^n} \int_x^\infty \frac{e^{x^2 - t^2}}{|t+iy|^{2n}} dt.$$

Integrating by parts once again, we obtain $\int_x^\infty \frac{e^{x^2 - t^2}}{|t+iy|^{2n}} dt < \frac{1}{2|z|^{2n}x}$, whence it follows that the expansion is asymptotic. The case $\operatorname{Re} z < 0$ is considered in a similar way. If $\operatorname{Re} z = 0$, then

$$\left| \int_0^\infty \frac{e^{z^2 - (t+iy)^2}}{(t+iy)^{2n}} dt \right| < \int_0^\infty \frac{e^{-t^2}}{y^{2n}} dt = \frac{\sqrt{\pi}}{2y^{2n}}.$$

$$786. f(t) \sim \frac{1}{\omega} \sin\left(\omega t + \frac{\pi}{4}\right) - \frac{1}{\omega^2 \pi} \left[\frac{\Gamma\left(\frac{3}{2}\right)}{\frac{3}{t^{\frac{3}{2}}}} - \frac{\Gamma\left(\frac{7}{2}\right)}{\omega^2 t^{\frac{7}{2}}} + \dots + (-1)^n \frac{\Gamma\left(2n + \frac{3}{2}\right)}{\omega^{2n} t^{2n + \frac{3}{2}}} + \dots \right];$$

for small t

$$f(t) = 2 \sqrt{\frac{t}{\pi}} \left[1 - \frac{(2\omega t)^2}{1 \times 3 \times 5} + \frac{(2\omega t)^4}{1 \times 3 \times 5 \times 7 \times 9} - \dots \right] \approx 2 \sqrt{\frac{t}{\pi}}.$$

$$787. f(t) \sim 1 + \frac{1}{\pi} \sum_0^\infty (-1)^n \frac{\Gamma\left(3n + \frac{3}{2}\right)}{t^{3n + \frac{3}{2}}}; \quad \text{for small } t \quad f(t) = \frac{t^{\frac{3}{2}}}{\Gamma\left(\frac{5}{2}\right)} - \frac{t^{\frac{9}{2}}}{\Gamma(4)} + \frac{t^{\frac{9}{2}}}{\Gamma\left(\frac{11}{2}\right)} - \dots \approx \frac{4t^{\frac{3}{2}}}{3\sqrt{\pi}}.$$

788. 1. 789. 0. 790. 4. 792. 1; 3. 793. 0; 4. 794. 2. 795. 1. 796. n . 797. n . 799. *Solution.* Since the sequence of functions $f_n(z)$ converges to the function $e^{\frac{1}{z}}$ everywhere except at the point $z=0$, for any circle K with centre at $z \neq 0$, which does not enclose the origin or contain it on the boundary, for a given $\varepsilon > 0$ there exists an N such that the inequality $\left| f_n(z) - e^{\frac{1}{z}} \right| < \varepsilon$ will be satisfied for all points of the circle K when $n > N$. It remains to choose $\varepsilon < \min_{z \in C} \left| e^{\frac{1}{z}} \right|$, where C is the circumference of the circle K , and to use Rouché's theorem. 803. 0.

804. 2; 1. 805. One root in each quadrant. 806. There are two roots each in the second and third quadrants. 808. In the domain $\beta > 0$, $\alpha > +\sqrt{\beta}$ (domain I in Fig. 69), $m=0$; in the domain $\beta > 0$, $\alpha < +\sqrt{\beta}$ (domain II), $m=2$; in the domain $\beta < 0$ (domain III), $m=1$. 809. In the domain $\alpha > 0$, $\beta > \frac{1}{\alpha}$ (domain I in Fig. 70), $m=0$; in the domain where either $\alpha \leq 0$ or $\alpha > 0$, $\beta < \frac{1}{\alpha}$ (domain II),

$m=2$. 810. In the domain $\alpha > \frac{1}{2} + \sqrt{\beta^2 + \frac{1}{4}}$ (domain I in Fig. 71), $m=0$; in the domain $0 < \alpha < \frac{1}{2} + \sqrt{\beta^2 + \frac{1}{4}}$ (domain II), $m=2$; in the domain

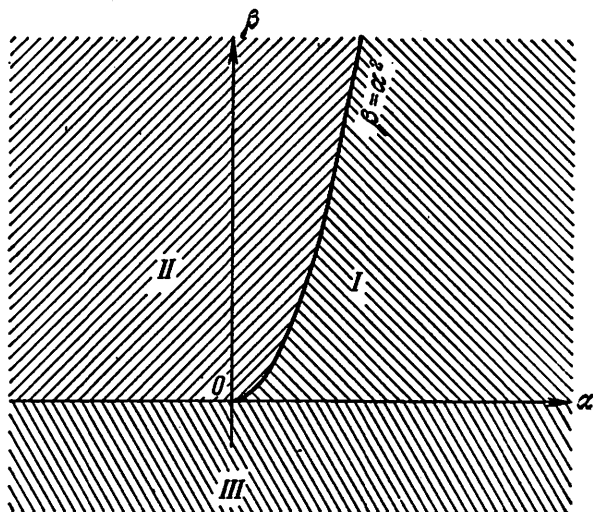


FIG. 69

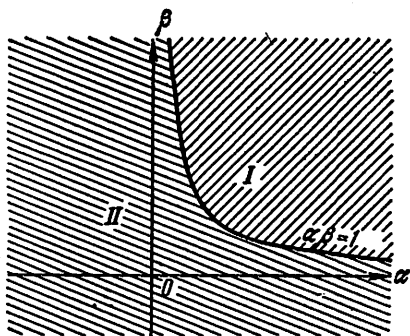


FIG. 70

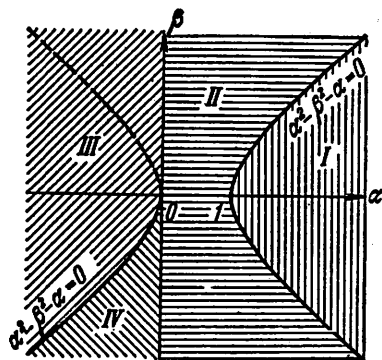


FIG. 71

where either $\alpha < \frac{1}{2} - \sqrt{\beta^2 + \frac{1}{4}}$ or $\frac{1}{2} - \sqrt{\beta^2 + \frac{1}{4}} \leq \alpha < 0$, $\beta > 0$ (domain III), $m=1$; in the domain $\frac{1}{2} - \sqrt{\beta^2 + \frac{1}{4}} < \alpha < 0$, $\beta < 0$ (domain IV), $m=3$. 812. The domain containing the positive semi-axis α and bounded by

the lines $a+b=0$

$$\text{and } \begin{cases} a = -\frac{t \cos \tau t}{\sin \tau t}, \\ b = \frac{t}{\sin \tau t}, \end{cases} \quad 0 \leq t \leq \frac{\pi}{2}.$$

813. The domain lying in the first quadrant and bounded

$$\text{by the lines } b=0 \text{ and } \begin{cases} a = \tan \tau t, \\ b = \frac{t^2}{\cos \tau t}, \end{cases} \quad 0 \leq t \leq \frac{\pi}{2\tau}.$$

814. A finite domain bounded by the segment $b=0$, $0 \leq a \leq \frac{\pi}{2\tau}$ and the arc

$$\begin{cases} a = t \sin \tau t, \\ b = t^2 \cos \tau t, \end{cases} \quad 0 \leq t \leq \frac{\pi}{2\tau}.$$

$$824. z = a + \frac{w}{1!} f(a) + \frac{w^2}{2!} \frac{d}{da} \{[f(a)]^2\} + \dots + \frac{w^n}{n!} \frac{d^{n-1}}{da^{n-1}} \{[f(a)]^n\} + \dots \quad 825. z =$$

$$= \frac{1}{2} w + \sum_{n=2}^{\infty} (-1)^{n-1} \frac{n(n+1) \dots (2n-2)}{2^{2n-1} \times n!} w^n,$$

$$z = -2 - \frac{1}{2} w + \sum_{n=2}^{\infty} (-1)^n \frac{n(n+1) \dots (2n-2)}{2^{2n-1} \times n!} w^n.$$

$$826. z = a + \frac{w}{1!} \frac{1}{2} (a^2 - 1) + \frac{w^2}{2!} \frac{1}{2^2} \frac{d}{da} [(a^2 - 1)^2] + \dots + \frac{w^n}{n!} \frac{1}{2^n} \frac{d^{n-1}}{da^{n-1}} [(a^2 - 1)^n] + \dots$$

$$828. 1) z = \sum_{n=1}^{\infty} \frac{n^{n-1} a^{n-1}}{n!} w^n, \quad 2) e^{bz} = b \sum_{n=0}^{\infty} \frac{(an+b)^{n-1}}{n!} w^n. \quad 829. z = a +$$

$$+ \sum_{n=1}^{\infty} \frac{w^n}{n!} \frac{d^{n-1}}{da^{n-1}} (\sin^n a). \quad 830. \text{Solution. The function } w = \frac{z - \frac{\pi}{2}}{\sin z} \text{ is analytic in the}$$

circle $\left| z - \frac{\pi}{2} \right| < r$ if $r < \frac{\pi}{2}$, and has no other zeros in this circle except

$z = \frac{\pi}{2}$. On the circumference of this circle $|w| \geq \frac{2r}{e^r + e^{-r}}$, that is why the circle

$|w| < \rho$ of radius $\rho < \frac{2r}{e^r + e^{-r}}$ is mapped one-to-one onto the corresponding neighbourhood of the point $z = \frac{\pi}{2}$, and the expansion $z(w)$ converges in this

circle. The function $\frac{2r}{e^r + e^{-r}}$ has a maximum at $r = r^*$, where $e^{2r^*} = \frac{r^* + 1}{r^* - 1}$ ($r^* = 1.19 \dots < \frac{\pi}{2}$), and this maximum is equal to $\frac{2r^*}{e^{r^*} + e^{-r^*}} =$

$= \sqrt{r^{*2} - 1} = 0.6627 \dots$. Thus, the desired radius of convergence is not less than 0.6627. At the same time, points w at which $\frac{dw}{dz} = 0$, i.e. $\tan z =$

$= z - \frac{\pi}{2}$, cannot be situated inside the circle of convergence of the expansion

of $z(w)$. Denoting $z - \frac{\pi}{2} = t$, transform the equation $\frac{dw}{dz} = 0$ to the form $\cot t = -t$ or $e^{2it} = \frac{it+1}{it-1}$. Hence, $it = r^*$ is a root of the equation $\frac{dw}{dz} = 0$. The appropriate value of $w = \frac{t}{\cos t} = -\frac{2ir^*}{e^{r^*} + e^{-r^*}}$, whence $|w| = \frac{2r^*}{e^{r^*} + e^{-r^*}} = \sqrt{r^{*2} - 1} = 0.6627 \dots$. This point is thus found on the circumference of the circle of convergence, and hence the radius of convergence is equal to $\sqrt{r^{*2} - 1} = 0.6627 \dots$.

Chapter V

832. The annulus $\frac{1}{2} < |z| < 1$. 833. The exterior of the unit circle ($|z| > 1$). 834. $|z| < 1$. 835. The half-plane $\operatorname{Re} z < -1$. 836. The real axis. 837. The entire plane except the points $z=0, \pm 1, \pm 2, \dots$. 838. $|z| > 1$. 839. $|z| < 1$. 840. The entire plane except the unit circle ($|z| \neq 1$). 841. The entire plane except the points $z = 4^n e^{\frac{(2k+1)\pi i}{n}}$ ($k, n = 1, 2, \dots$). 842. *Solution.* If the series

$\sum_{n=1}^{\infty} a_n$ converges, then for $|z| < 1$ the convergence of the series $\sum_{n=1}^{\infty} \frac{a_n z^n}{1-z^n}$ is

obvious, and from the identity $\sum_{n=1}^{\infty} \frac{a_n z^n}{1-z^n} = -\sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} \frac{a_n \left(\frac{1}{z}\right)^n}{1 - \left(\frac{1}{z}\right)^n}$ it fol-

lows that the series converges for $|z| > 1$ as well. If the series $\sum_{n=1}^{\infty} a_n$ diverges,

then the series $\sum_{n=1}^{\infty} a_n z^n$ has the radius of convergence $R \leq 1$. For $|z| > 1$ the

divergence of the series $\sum_{n=1}^{\infty} \frac{a_n z^n}{1-z^n}$ follows from the fact that otherwise

$\sum_{n=1}^{\infty} \frac{a_n}{1-z^n}$ would be a convergent series, and hence also the series

$\sum_{n=1}^{\infty} \left(\frac{a_n}{1-z^n} - \frac{a_n z^n}{1-z^n} \right) = \sum_{n=1}^{\infty} a_n$. But if $|z| < 1$, then the modulus q of the

ratio of the general terms of the series $\sum_{n=1}^{\infty} a_n z^n$ and $\sum_{n=1}^{\infty} \frac{a_n z^n}{1-z^n}$ lies within the range $1 - |z| \leq q \leq 2$ and, hence, both series converge or diverge simultaneously.

843. 1) $\sum_{n=1}^{\infty} b_n z^n$, where $b_n = \sum a_p$, the summation extending over those p 's which are divisors of the number n , including 1 and n . The radius of convergence $R = \min \{r, 1\}$, where r is the radius of convergence of the series

$\sum_{n=1}^{\infty} a_n z^n$. 844. $\sum_{n=0}^{\infty} a_n (z-2)^n$, where $a_n = \frac{(-1)^n}{n!} \sum_{k=1}^{\infty} \frac{(\ln k)^n}{k!} \left(a_0 = \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6} \right)$;

$R=1$, 845. $\frac{1}{2}$ if $|z| < 1$, and $-\frac{1}{2}$ if $|z| > 1$. 846. $\frac{z}{(1-z)^2}$ if $|z| < 1$; and $\frac{1}{(1-z)^2}$ if $|z| > 1$. 847. z if $|z| < 1$, and 1 if $|z| > 1$. 848. $\frac{z}{z-1}$ if $|z| < 1$, and $\frac{1}{z-1}$ if $|z| > 1$. 850. 1) and 2) Converge uniformly in any circle $|z| \leq r < 1$ and in any domain $|z| \geq R > 1$; 3) converges uniformly along the whole real axis; at all other points it diverges.

852. Converges uniformly on the circle $|z|=1$, at all other points it diverges.

853. Converges uniformly in any half-plane $\operatorname{Re} z \geq \delta$, where $\delta > 0$.

854. Converges uniformly in any half-plane $\operatorname{Re} z \geq 1 + \delta$, where $\delta > 0$.

855. Converges uniformly along the real axis; at all other points it diverges.

856. Converges uniformly along any segment $[2k\pi + e, 2(k+1)\pi - e]$ of the real axis ($k=0, 1, 2, \dots$). 857. No. 864. $x_c = x_a = -\infty$. 865. $x_c = -\infty$, $x_a = 1$. 866. $x_c = -\infty$, $x_a = +\infty$. 867. $x_c = 0$, $x_a = +\infty$. 868. $x_c = 0$, $x_a = 1$.

869. $x_c = x_a = -1$. 870. $x_c = x_a = +\infty$. 873. $x_c = 0$; diverges at all points of the boundary. 874. $x_c = x_a = -2$; diverges at all points of the boundary.

875. $x_c = x_a = 0$; converges (non-absolutely) at the points $z = (2k+1)\pi i$ ($k=0, \pm 1, \pm 2, \dots$), and diverges at all other points of the boundary.

876. $x_c = x_a = 0$; converges absolutely at all points of the boundary.

877. $x_c = x_a = 0$; converges non-absolutely at all points of the boundary.

885. The integral converges uniformly in any strip $0 < \alpha \leq \operatorname{Re} z \leq A < \infty$.

886. The integral converges uniformly in any half-plane $\operatorname{Re} z \geq \alpha > 0$. 887. The integral converges uniformly in any strip $\alpha \leq \operatorname{Re} z \leq 2 - \alpha$, where $\alpha > 0$.

888. The integral converges uniformly in any strip $\alpha \leq \operatorname{Re} z \leq 1 - \alpha$, where $\alpha > 0$. 889 and 890. The integral converges uniformly in any closed interval of the real axis which does not contain the origin. 891. The integral converges uniformly in a half-plane $\operatorname{Im} z \geq 0$ with the $|z| < r$ semicircle removed, where r is an arbitrarily small positive number.

892. The integral converges uniformly in the intervals $0 < \alpha \leq z \leq \beta < 1$ and $1 < \gamma \leq z < \infty$. 893. Example: $f(t) = e^{t^2}$ for $n < t < n + e^{-n^3}$ ($n=1, 2, \dots$) and $f(t) = 0$ for all the remaining t 's. 894. $x_c = x_a = 0$. 895. $x_c = x_a = -\infty$.

896. $x_c = x_a = +\infty$. 897. $x_c = \infty$; $x_a = 1$. 898. $x_c = -\infty$; $x_a = +\infty$. 899. $x_c = -1$; $x_a = +\infty$. 900. $x_c = 0$; $x_a = 1$. 901. Diverges at all points of the boundary.

902. Converges absolutely. 903. Diverges at the point $z=0$, converges (non-absolutely) at all the remaining points of the boundary. 904. Converges (non-absolutely) at all points of the boundary.

Chapter VI

914. *Solution.* Evaluating the integrals entering into the obvious inequalities

$$\int_0^{\pi/2} \sin^{2n+1} x \, dx < \int_0^{\pi/2} \sin^{2n} x \, dx < \int_0^{\pi/2} \sin^{2n-1} x \, dx,$$

we obtain

$$\left[\frac{(2n)!!}{(2n-1)!!} \right]^2 \frac{1}{2n+1} < \frac{\pi}{2} < \left[\frac{(2n)!!}{(2n-1)!!} \right]^2 \times \frac{1}{2n}.$$

To prove the Wallis formula it remains to establish that the difference between the extreme terms in this system of inequalities tends to zero as $n \rightarrow \infty$.

916. 1) It will not hold; 2) it will hold. 918. 1) Diverges; 2) diverges (to zero); 3) converges; 4) converges. 919. Converges non-absolutely. 920. Diverges. 921. Converges if

$p > \frac{1}{2}$, converges absolutely if $p > 1$; diverges if $p \leq \frac{1}{2}$. 922. Converges absolutely if $p > 1$; diverges if $p \leq 1$. 923. Converges absolutely. 925. $|z| < 1$.

926. $|z| < 2$. 927. $|z| < \infty$. 928. $|z| > 1$. 929. $|z| < \frac{1}{e}$. 930. $|z| < \infty$.
 931. $|z| < \infty$. 932. $|z| < \infty$. 933. $|z| < \infty$. 940. *Solution.* 1) Deducting the series for $2^{-s}\zeta(s)$ from the series $\zeta(s) = 1 + \frac{1}{2^s} + \frac{1}{3^s} + \dots$, we obtain

$(1 - 2^{-s})\zeta(s) = 1 + \frac{1}{3^s} + \frac{1}{5^s} + \frac{1}{7^s} + \dots$; the right member of this equality lacks those terms $1/n^s$ for which n is divisible by 2. Similarly, $(1 - 2^{-s})(1 - 3^{-s})\zeta(s) = 1 + \frac{1}{5^s} + \frac{1}{7^s} + \dots$, the right member lacking those terms $\frac{1}{n^s}$ for which n is divisible by 2 or by 3. In general,

$$(1 - p_1^{-s})(1 - p_2^{-s}) \dots (1 - p_m^{-s})\zeta(s) = 1 + \sum \frac{1}{n^s}, \quad (1)$$

the summation in the right member of (1) extending to those n 's (greater than unity) which are not divisible by any of the numbers $p_1, p_2, \dots, p_m$. It is easy to prove that for $\operatorname{Re} s \geq 1 + \delta$ ($\delta > 0$) the sum of the series in the right mem-

ber of (1) tends to zero as $m \rightarrow \infty$ and, hence, $\zeta(s) \prod_{m=1}^{\infty} (1 - p_m^{-s}) = 1$. 2) Since

it follows from the absolute convergence test (see Problem 917) that the product $\prod_{m=1}^{\infty} (1 - p_m^{-s})$ converges for $\operatorname{Re} s \geq 1 + \delta$, the function $\zeta(s)$ has no zeros for $\operatorname{Re} s > 1$. 941. *Solution.* From what was proven in the preceding problem

it follows that for any $\delta > 0$ we have $\prod_{n=1}^{\infty} (1 - p_n^{-(1+\delta)}) = \frac{1}{\zeta(1+\delta)}$, whence it

is easy to conclude that $\lim_{\delta \rightarrow 0} \prod_{n=1}^{\infty} (1 - p_n^{-(1+\delta)}) = 0$. Since $(1 - p_n^{-1}) <$

$< (1 - p_n^{-(1+\delta)})$, it is obvious that the product $\prod_{n=1}^{\infty} (1 - p_n^{-1})$ diverges (to

zero) and, consequently, $\sum_{n=1}^{\infty} p_n^{-1}$ is also a divergent series.

961. $\frac{\pi^2}{\sin^2 \pi a}$. 962. $\frac{\pi^2 \cot \pi a}{\sin \pi a}$. 963. $\frac{\pi^2}{8}$. 964. $\frac{1}{2a^2}(1 + \pi a \coth \pi a)$.
 965. $\frac{1}{2a^2} \left(1 + \frac{\pi a}{\sinh \pi a}\right)$. 966. $\frac{\pi^3}{32}$. 967. $\frac{(-1)^{k+1} 2^{2k-1} \pi^{2k}}{(2k)!} B_{2k}$, (B_{2k} is Bernoulli number (see Problem 485)). 968. $\frac{\pi \sin ab}{2 \sin \pi a}$. 971. $\rho = 0$. 972. $\rho = n$, $\sigma = a$.
 973. $\rho = 1$, $\sigma = 3$. 974. $\rho = 1$, $\sigma = 3$. 975. $\rho = 3$, $\sigma = 2$. 976. $\rho = 2$, $\sigma = \sqrt{5}$.
 977. $\rho = 1$, $\sigma = 1$. 978. $\rho = 1$, $\sigma = 1$. 979. $\rho = 1$, $\sigma = \sqrt{2}$. 980. $\rho = \frac{1}{2}$, $\sigma = 1$.

981. $\rho = \frac{1}{2m}$, $\sigma = 1$. *Solution* $\sum_{n=0}^{\infty} \frac{z^{2n}}{(2^2 \times n)!} = \frac{1}{2} (\cos \sqrt{z} + \cos i \sqrt{z})$, whence

$$\sum_{n=0}^{\infty} \frac{z^n}{(2^2 \times n)!} = \frac{1}{2} \left(\cos \sqrt[4]{z} + \cos i \sqrt[4]{z} \right). \quad \text{Similarly} \quad \sum_{n=0}^{\infty} \frac{z^{2n}}{(2^3 \times n)!} =$$

$$= \frac{1}{4} \left(\cos \alpha \sqrt[4]{z} + \cos \alpha^2 \sqrt[4]{z} + \cos \alpha^3 \sqrt[4]{z} + \cos \alpha^4 \sqrt[4]{z} \right), \text{ where } \alpha = \sqrt[4]{-1},$$

whence $\sum_{n=0}^{\infty} \frac{z^n}{(2^3 \times n)!} = \frac{1}{4} \sum_{k=1}^4 \cos \alpha^k \sqrt[8]{z}$, and so on. 982. $\rho = \infty$. 983. *Solution.*

As is easy to see, it is sufficient to consider the values $z > 0$. Then $\frac{\int_0^1 e^{zt^2} dt}{e^z} < 1$;

on the other hand, if $0 < \alpha < 1$, then $\frac{\int_0^1 e^{zt^2} dt}{e^{\alpha z}} = \int_0^1 e^{z(t^2 - \alpha)} dt \geq \int_{\sqrt{\alpha}}^1 e^{z(t^2 - \alpha)} dt \rightarrow$

$\rightarrow \infty$ as $z \rightarrow \infty$. Hence, $\rho = 1$, $\sigma = 1$. 985. $\rho^* = \max(\rho_1, \rho_2)$. 986. 1) $\rho^* = \rho$, $\sigma^* \leq \sigma_1 + \sigma_2$; 2) $\rho^* = \rho$, $\sigma^* = \max(\sigma_1, \sigma_2)$. 987. 1) $\rho^* \leq \rho$, $\sigma^* \leq 2\sigma$; 2) $\rho^* \leq \rho$, $\sigma^* \leq \sigma$. 988. *Solution.* If $z = re^{i\varphi}$, then for any given r the quantity $\left| 1 - \frac{z^2}{\lambda_n^2} \right|$

attains the maximum value when $\varphi = \pm \frac{\pi}{2}$; it equals $1 + \frac{r^2}{\lambda_n^2}$. On the other

hand, for any $\varepsilon > 0$ ($\varepsilon < \alpha$) and $n > n_0(\varepsilon)$ we have $\frac{n}{\beta + \varepsilon} < \lambda_n < \frac{n}{\alpha - \varepsilon}$ and

hence, $1 + \frac{(\alpha - \varepsilon)^2 r^2}{n^2} < 1 + \frac{r^2}{\lambda_n^2} < 1 + \frac{(\beta + \varepsilon)^2 r^2}{n^2}$ (*). And since $\prod_{n=1}^{\infty} \left(1 + \frac{\gamma^2 r^2}{n^2} \right) =$

$= \frac{\sin i\pi\gamma r}{i\pi\gamma r} = \frac{e^{\pi\gamma r} - e^{-\pi\gamma r}}{2\pi\gamma r}$ (see Problem 952 or 957), it follows from the inequalities (*) (because of the arbitrary nature of ε and the solution of Problem 984) that the order ρ of the function $f(z)$ equals 1, and the type σ is not less than the type of the function $e^{\pi\alpha z}$ and not greater than the type of the function $e^{\pi\beta z}$, i.e. $\pi\alpha \leq \sigma \leq \pi\beta$.

990. *Solution.* Let $M(r)$, ρ , σ be the maximum of $|f(z)|$ on the circle $|z| = r$, the order and the type of $f(z)$ and let $M_1(r)$, ρ_1 and σ_1 be the respective

characteristics for the function $f'(z)$. From the equality $f(z) = \int_0^z f'(t) dt + f(0)$

it follows that $M(r) \leq rM_1(r) + |f(0)|$ and hence $\rho \leq \rho_1$. Since $f'(z) = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta) d\zeta}{(\zeta - z)^2}$, where Γ may be a circle of radius δ (δ any positive number)

with centre at the point z , then $M_1(r) \leq \frac{M(r + \delta)}{\delta}$, i.e. $\rho_1 \leq \rho$ and thus

$\rho_1 = \rho$. From this (and from the above-mentioned inequalities) we come to a conclusion that $\sigma_1 = \sigma$. An alternative solution of the problem is based on the

theorem given on page 130. 992. $\rho = 1$, $\sigma = \frac{1}{e}$. 993. $\rho = a$, $\sigma = \infty$. 994. $\rho = a$,

$\sigma = 0$. 995. $\rho = 0$. 996. $\rho = 0$. 997. $\rho = 1$, $\sigma = 1$. 998. $\rho = 1$, $\sigma = 2$. 999. $\rho = 1$;

$h(\varphi) = \cos \varphi$. 1000. $\rho = 1$; $h(\varphi) = \cos \varphi$ if $\cos \varphi \geq 0$; $h(\varphi) = 0$ if $\cos \varphi < 0$.

1001. $\rho = 1$; $h(\varphi) = |\sin \varphi|$. 1002. $\rho = 1$; $h(\varphi) = |\sin \varphi|$. 1003. $\rho = 1$; $h(\varphi) =$

$|\cos \varphi|$. 1004. $\rho = n$; $h(\varphi) = \cos n\varphi$. 1005. $\rho = \frac{1}{2}$, $h(\varphi) = \left| \sin \frac{\varphi}{2} \right|$.

1006. 1) $h^*(\varphi) = h(\varphi)$ if $h(\varphi) > 0$; $h^*(\varphi) = 0$ if $h(\varphi) < 0$, $h^*(\varphi) \leq 0$ if $h(\varphi) = 0$;

2) always $h^*(\varphi) = h(\varphi)$. 1007. Example: $f(z) = e^z - P(z)$, $\frac{\pi}{2} < \varphi < \frac{3\pi}{2}$.

Chapter VII

1009. 1) $F^+(z) = (z-a)^n$, $F^-(z) \equiv 0$; 2) $F^+(z) \equiv 0$, $F^-(z) = -\frac{1}{(z-a)^n}$;

3) $F^+(z) = \frac{1}{(z-a)^n}$, $F^-(z) \equiv 0$. In all three cases the Cauchy-type integral turns into Cauchy's integral. 1010. 1) $F^+(z) = \varphi(z) - \sum g_k \left(\frac{1}{z-a_k} \right)$, $F^-(z) = -\sum g_k \left(\frac{1}{z-a_k} \right)$; 2) $F^+(z) = \sum g_k \left(\frac{1}{z-a_k} \right) + g(z)$, $F^-(z) = -\varphi(z) + \sum g_k \left(\frac{1}{z-a_k} \right) + g(z)$. $g_k \left(\frac{1}{z-a_k} \right)$ is the principal part of the expansion of the function $\varphi(z)$ in the neighbourhood of the pole a_k ; $g(z)$ is the principal part of the expansion of $\varphi(z)$ in the neighbourhood of the point at infinity

with the constant term included. 1011. $F^+(z) = \frac{z + \ln \frac{z-2}{z-3}}{z^2-4}$, $F^-(z) = -\frac{i}{z^2+1}$.

1012. $F^+(z) = \cot z - \frac{1}{z} - \frac{2z}{z^2-\pi^2}$, in particular, $F^+(0) = 0$, $F^+(\pi) = -\frac{3}{2\pi}$,

$F^+(-\pi) = \frac{3}{2\pi}$; $F^-(z) = -\frac{1}{z} - \frac{2z}{z^2-\pi^2}$. 1013. $F^+(z) = \frac{z}{2(z+i)}$, $F^-(z) =$

$-\frac{z}{2(z-i)}$. 1014. $F^+(z) = \frac{1}{2} \sum_{n=0}^{\infty} \frac{(a_n - ib_n) z^n}{R^n}$, $F^-(z) = -\frac{1}{2} \sum_{n=1}^{\infty} \frac{R^n (a_n + ib_n)}{z^n}$;

limiting values:

$$F^+(Re^{i\theta}) = \frac{1}{2} \sum_{n=0}^{\infty} (a_n - ib_n) e^{in\theta} =$$

$$= \frac{a_0}{4} + \frac{1}{2} \varphi(Re^{i\theta}) + \frac{i}{2} \sum_{n=1}^{\infty} (-b_n \cos n\theta + a_n \sin n\theta);$$

$$F^-(Re^{i\theta}) = -\frac{1}{2} \sum_{n=1}^{\infty} (a_n + ib_n) e^{in\theta} =$$

$$= \frac{a_0}{4} - \frac{1}{2} \varphi(Re^{i\theta}) + \frac{i}{2} \sum_{n=1}^{\infty} (-b_n \cos n\theta + a_n \sin n\theta).$$

1015. 1) If point z lies in circle Q_k : $|z - k\pi| < \frac{\pi}{2}$, then $I_1(z) = f(z - k\pi)$,

$I_2(z) = (-1)^k f(z - k\pi)$. In particular, if $|z| < \frac{\pi}{2}$, then $I_1(z) = I_2(z) = f(z)$.

If point z lies outside all closed $\bar{Q}_k$, then $I_1(z) = I_2(z) = 0$. 2) Let Q_k be a circle $|z - k\pi| < \pi$. Then $I_1(z) = f(z - k\pi) + f[z - (k-1)\pi]$, $I_2(z) = (-1)^k [f(z - k\pi) - f(z - (k-1)\pi)]$ if $z \in Q_{k-1}Q_k$; $I_1(z) = f(z - k\pi) + f[z - (k+1)\pi]$, $I_2(z) = (-1)^k [f(z - k\pi) - f(z - (k+1)\pi)]$ if $z \in Q_kQ_{k+1}$; $I_1 = f(z - k\pi)$, $I_2(z) = (-1)^k f(z - k\pi)$ if z is inside the domain $Q_k - Q_{k-1}Q_k - Q_kQ_{k+1}$; $I_1(z) = 0$, $I_2(z) = 0$ if z lies inside the complement to all Q_k .

1016. $F(z) = \frac{1}{2\pi i} \ln \frac{z-1}{z+1}$; $F^\pm(\zeta) = \frac{1}{2\pi i} \ln \frac{1-\zeta}{1+\zeta} \pm \frac{1}{2}$, $F(\zeta) = \frac{1}{2\pi i} \ln \frac{1-\zeta}{1+\zeta}$ ($-1 < \zeta < 1$); $F(\pm i) = \pm \frac{1}{4}$, $F^\pm(0) = \pm \frac{1}{2}$, $F(0) = 0$. 1017. $F(z) = \frac{1}{2\pi i} \ln \frac{z+R}{z-R}$ is a single-valued branch in the z -plane with a cut along C determined by the value $\ln 1 = 0$. $\ln \frac{z+R}{z-R} = \ln \left| \frac{z+R}{z-R} \right| + i\Delta_C \arg(\zeta - z)$, where $\Delta_C \arg(\zeta - z)$ is the increment of $\arg(\zeta - z)$ along C .

$$F^+(\zeta) = \frac{1}{2\pi i} \ln \left| \frac{\zeta+R}{\zeta-R} \right| + \frac{3}{4}, \quad F^-(\zeta) = \frac{1}{2\pi i} \ln \left| \frac{\zeta+R}{\zeta-R} \right| - \frac{1}{4}.$$

$$F(\zeta) = \frac{1}{2\pi i} \ln \left| \frac{\zeta+R}{\zeta-R} \right| + \frac{1}{4}; \quad F(0) = \frac{1}{2},$$

$$F^+(iR) = \frac{3}{4}, \quad F^-(iR) = -\frac{1}{4}, \quad F(iR) = \frac{1}{4}; \quad F'(0) = -\frac{i}{\pi R}.$$

1018. $F(z) = \frac{1}{2\pi i} \ln \frac{z+R}{z-R}$ (a single-valued branch in the z -plane with a cut along C determined by the value $\ln 1 = 0$; for $|z| > R$ it coincides with the similar branch in Problem 1017); $F^+(\zeta) = \frac{1}{2\pi i} \ln \left| \frac{\zeta+R}{\zeta-R} \right| + \frac{1}{4}$, $F^-(\zeta) = \frac{1}{2\pi i} \ln \left| \frac{\zeta+R}{\zeta-R} \right| - \frac{3}{4}$; $F(0) = -\frac{1}{2}$, $F'(0) = -\frac{i}{\pi R}$. 1019. 1) 0 if $|z| < r$ or $|z| > R$, and $\frac{1}{2^n}$ if $r < |z| < R$; 2) $\frac{1}{2^n}$ if $\operatorname{Im} z > \pi$, and 0 if $\operatorname{Im} z < \pi$; 3) 0 if $|\operatorname{Im} z| < \pi$, and $-\frac{1}{2^n}$ if $|\operatorname{Im} z| > \pi$; 4) $\frac{1}{2\pi i z^n} \left[\ln \frac{R+z}{R-z} - 2 \sum_{k=1}^m \frac{z^{2k-1}}{(2k-1) R^{2k-1}} \right]$ if $m = \left[\frac{n}{2} \right]$,

$\ln \frac{R+z}{R-z} = \ln \left| \frac{R+z}{R-z} \right| + i\Delta_C \{ \arg(\zeta - z) - \arg \zeta \}$ is a single-valued branch in the z -plane with a cut along C determined by the value $\ln 1 = 0$. The limiting value of this branch on C has an imaginary part $\frac{\pi}{2}$ on the left and $\left(-\frac{3\pi}{2} \right)$ on the right. This determines the limiting values $F^\pm(\zeta)$ on C ; $F(0) = \frac{1}{2\pi i} \frac{1 + (-1)^{n-1}}{nR^n}$; 5) the function $F(z)$ is the same as in the preceding item,

the only difference being that $\ln \frac{R+z}{R-z}$ is a single-valued branch in the z -plane with a cut along another semicircle C with the same value $\ln 1 = 0$. The limiting value of this branch on C has an imaginary part $\frac{3\pi}{2}$ on the left and

¹⁾ At first glance the solution seems to be obvious:

$$\frac{1}{2\pi i} \int_{-1}^1 \frac{d\zeta}{\zeta-z} = \frac{1}{2\pi i} [\ln(1-z) - \ln(-1-z)] = \frac{1}{2\pi i} \ln \frac{z-1}{z+1}.$$

But it is necessary to check whether the final transformation actually leads to the indicated branch of the logarithm since the equality $\ln z_2 - \ln z_1 = \ln \frac{z_2}{z_1}$ is, generally speaking, invalid. This should also be borne in mind in the sequel.

$\left(-\frac{\pi}{2}\right)$ on the right. This determines the limiting values $F^\pm(\zeta)$ on C ;

$F(0) = \frac{1}{2\pi i} \frac{1 + (-1)^{n-1}}{nR^n}$. Note. In items (4) and (5) the branches of $\text{Ln} \frac{R+z}{R-z}$ coincide in the circle $|z| < R$, but in general they are different; for instance, at ∞ in the first case the branch has the value $-\pi i$, and in the second $+\pi i$.

But $F(\infty) = 0$. 1020. $\text{Ln} \frac{b-z}{a-z} = \ln \left| \frac{b-z}{a-z} \right| + i\Delta_C \arg(\zeta - z)$ is a single-valued branch in the z -plane with a cut along C determined by the value $\text{Ln} 1 = 0$ (the branches in Problems 1021 to 1024 are defined in the same manner).

$$1021. b-a+z \text{Ln} \frac{b-z}{a-z}; \quad 1022. 1) \sum_{k=1}^n A_{nk} z^{k-1} + z^n \text{Ln} \frac{b-z}{a-z};$$

$$2) \sum_{n=1}^{\infty} \sum_{k=1}^n c_n A_{nk} z^{k-1} + \varphi(z) \text{Ln} \frac{b-z}{a-z}, \text{ where } A_{nk} = \frac{b^{n-k+1} - a^{n-k+1}}{n-k+1}.$$

$$1023. \frac{1}{z-z_0} \left(\text{Ln} \frac{b-z}{a-z} - \text{Ln} \frac{b-z_0}{a-z_0} \right). \quad 1024. \frac{1}{(z-z_0)^n} \left[\text{Ln} \frac{b-z}{a-z} - \text{Ln} \frac{b-z_0}{a-z_0} + \right. \\ \left. + \sum_{k=2}^n A_k (z-z_0)^{k-1} \right],$$

$$A_k = \frac{1}{k-1} \left[\frac{1}{(b-z_0)^{k-1}} - \frac{1}{(a-z_0)^{k-1}} \right], \quad F(z_0) = -A_{n+1}.$$

$$1025. 1) F^+(z) = \ln(z+R), \quad F^-(z) = \ln\left(1 + \frac{R}{z}\right); \quad F^+(\zeta) = \ln 2R \cos \frac{\varphi}{2} + i \frac{\varphi}{2},$$

$$F^-(\zeta) = \ln 2 \cos \frac{\varphi}{2} - i \frac{\varphi}{2}; \quad 2) F^+(z) = \ln(R-z) + \pi i, \quad F^-(z) = \ln\left(1 - \frac{R}{z}\right);$$

$$F^+(\zeta) = \ln 2R \left| \sin \frac{\varphi}{2} \right| + i \frac{\pi + \varphi}{2}, \quad F^-(\zeta) = \ln 2 \left| \sin \frac{\varphi}{2} \right| + i \frac{\pi - \varphi}{2}. \quad 1026. 1) F^+(z) = 0,$$

$$F^-(z) = \ln\left(1 - \frac{1}{z}\right); \quad F^+(\zeta) = 0, \quad F^-(\zeta) = \ln\left(1 - \frac{1}{\zeta}\right); \quad 2) F^+(z) = \ln \frac{z}{z-1},$$

$$F^-(z) = 0, \quad F^+(\zeta) = \ln \frac{\zeta}{\zeta-1}; \quad F^-(\zeta) = 0. \quad 1027. F(z) = \frac{1}{2\pi i} \sum_{n=1}^{\infty} \frac{1}{n z^n} \left[\ln \frac{R+z}{R-z} - \right.$$

$$\left. - 2 \sum_{k=1}^m \frac{z^{2k-1}}{(2k-1) R^{2k-1}} \right], \text{ where } m = \left[\frac{n}{2} \right], \text{ and } \text{Ln} \frac{R+z}{R-z} \text{ has the same value}$$

as in Problem 1019(4). This also determines the limiting values $F^\pm(\zeta)$ on C . If

$$|z| < R, \text{ then } F(z) = \sum_{n=0}^{\infty} c_n z^n, \text{ where } c_n = \frac{1}{2\pi i} \int_C \ln \frac{\zeta}{\zeta-1} \frac{d\zeta}{\zeta^{n+1}}. \quad 1028. F^+(z) =$$

$$= -\frac{1}{\pi i} \left[2 + \sqrt{z} \ln \frac{1-\sqrt{z}}{1+\sqrt{z}} \right], \quad F^-(z) = -\frac{1}{\pi i} \left[2 + \sqrt{z} \ln \frac{1-\sqrt{z}}{1+\sqrt{z}} \right] - \sqrt{z}.$$

$$1029. F^+(z) = 0, \quad F^-(z) = \text{Ln} \frac{z-b}{z-a} \text{ (the branch of the logarithm is determined$$

¹⁾ If $\text{Ln} z$ and $\ln z$ are the branches given in the problem, $\text{Ln} z = \ln(-z) + \pi i$.

by the conditions of the problem). 1030. $F^+(z) \equiv 1$, $F^-(z) = 1 - \sqrt{\frac{z-a}{z-b}}$.

1031. $F^+(z) = z - \lambda a - (1-\lambda)b$, $F^-(z) = z - \lambda a - (1-\lambda)b - (z-a)^\lambda (z-b)^{1-\lambda}$.

1032. $F^+(z) = \operatorname{Ln} \frac{z-\zeta_0}{z-b}$, $F^-(z) = \operatorname{Ln} \frac{z-\zeta_0}{z-a}$. 1034. $-x^2 + x - 3.5$.

1035. 1) $\frac{1}{2} \left(\ln \frac{z}{z-1} \right)^2$; 2) $\frac{1}{2} \left(\ln \frac{\tau}{1-\tau} \right)^2 - \frac{\pi^2}{2}$. 1037. If $|z| > 1$ and $z \notin C$,

then $F(z) = \frac{1}{2\pi i} \operatorname{Ln} \frac{z}{z-1} \operatorname{Ln} \frac{R+z}{R-z} + F_1(z)$, where $F_1(z)$ is an analytic function for

$|z| > 1$, and the branch $\operatorname{Ln} \frac{R+z}{R-z}$ is chosen as in Problem 1019(4). From this

one can see the behaviour of $F(z)$ at the points $\pm R$, i.e. the ends of the arc C . 1038. $D(f)$ is the area of the domain G' onto which $f(z)$ maps the domain G .

1044. 1) $\ln \left| \frac{R^2 - \bar{\zeta} z}{R(z - \bar{\zeta})} \right|$; 2) $\ln \left| \frac{z - \bar{\zeta}}{z - \zeta} \right|$; 3) $\ln \left| \frac{e^{\pi z} - e^{\pi \bar{\zeta}}}{e^{\pi z} - e^{\pi \zeta}} \right|$. 1050. $u(z) =$

$= \frac{1}{2\pi} \int_0^{2\pi} u(\zeta) \frac{r^2 - R^2}{R^2 - 2Rr \cos(\theta - \varphi) + r^2} d\theta$, $u(\infty) = \frac{1}{2\pi} \int_0^{2\pi} u(\zeta) d\theta$. 1052. 1) $f(z) =$

$= \varphi(z) + \psi\left(\frac{R^2}{z}\right)$, $f_1(z) = -\varphi\left(\frac{R^2}{z}\right) - \psi(z)$, where $v(0) = \operatorname{Im} f(0) =$

$= \operatorname{Im} [\varphi(0) + \overline{\psi(\infty)}]$; 2) $f(z) = -i\varphi(z) + i\psi\left(\frac{R^2}{z}\right)$, $f_1(z) = i\varphi\left(\frac{R^2}{z}\right) + i\psi(z)$,

$v(0) = \operatorname{Im} f(0) = \operatorname{Im} [-i\varphi(0) + i\overline{\psi(\infty)}]$. 1053. 1) $f(z) = z^n$, $f_1(z) = -\frac{R^{2n}}{z^n}$ (here

and in the answers to Problems 1054–1057 the value of $v(0)$ is taken to be

zero). 1054. $f(z) = \frac{z^n}{R^{2n}}$, $f_1(z) = -\frac{1}{z^n}$. 1055. $f(z) = -\ln\left(1 - \frac{z}{R^2}\right)$ ($\ln 1 = 0$),

$f_1(z) = \ln\left(1 - \frac{1}{z}\right)$. 1056. $f(z) = \frac{R}{\sqrt{R^2 - z}}$ ($f(0) = \frac{1}{R}$), $f_1(z) = -\sqrt{\frac{z}{z-1}}$.

1057. $f(z) \equiv \ln R$, $f_1(z) \equiv -\ln R$. 1059. $u(z) = \frac{1}{\pi} \int_{-\infty}^{\infty} u(t) \frac{y dt}{(t-x)^2 + y^2}$, $f(z) =$

$= \frac{1}{\pi i} \int_{-\infty}^{\infty} \frac{u(t) dt}{t-z} + iC$ (for the first integral to exist it will suffice to ensure

piecewise continuity and boundedness of $u(t)$ over the entire interval $(-\infty, \infty)$, for the second, in addition, for instance, the function $u(t)$ must have order

$\frac{1}{|t|^\alpha}$ ($\alpha > 0$) at infinity). 1060. $f(z) = u(z) + iv(z) = \frac{1}{2i} \int_{-\infty}^{\infty} u(t) \coth \frac{\pi(t-z)}{2} dt -$

$-\frac{1}{2i} \int_{-\infty}^{\infty} u_1(t) \tanh \frac{\pi(t-z)}{2} dt$, where $u_1(t) = u(t+i)$ (for the integrals to exist

it is sufficient, for instance, that $u(t)$ decrease at infinity as $\frac{1}{|t|^{1+\alpha}}$, $\alpha > 0$).

1062. Circles in the disk $|z| < 1$ and tangential to the circle $|z| = 1$ at the point $e^{i\theta}$. 1064. Circular arcs joining points $e^{i\theta_1}$, $e^{i\theta_2}$ inside the circle $|z| < 1$.

1065. $\omega(z; a, b) = \frac{1}{\pi} \arg \frac{z-b}{z-a}$, $\omega(z; -\infty, b) = \frac{1}{\pi} \arg(z-b)$; $\omega(z; a, \infty) = 1 - \frac{1}{\pi} \arg(z-a)$. The geometric meaning of these harmonic measures is the angle (divided by π) subtended by the segment or ray at the point z .

1066. $1 - \frac{1}{\gamma} \arg z$ for the ray $\arg z = 0$ and $\frac{1}{\gamma} \arg z$ for the ray $\arg z = \gamma$.

1067. $\omega(z, \Delta) = \frac{2}{\pi} \arg \frac{z-R}{z+R} - 1 = 1 - \frac{2}{\pi} \arctan \frac{2Ry}{R^2 - |z|^2}$, $\omega(z, \Gamma) = 1 - \omega(z, \Delta)$.

The level lines are circular arcs (joining points $\pm R$) from whose points the diameter Δ subtends an angle $\frac{\pi}{2}(1+\omega)$ in the case of $\omega(z, \Delta)$ and $\pi\left(1-\frac{\omega}{2}\right)$

in the case of $\omega(z, \Gamma)$. 1068. $\omega(z, \Gamma) = \frac{2}{\pi} \arg \frac{z-R}{z+R}$. 1069. $\omega(z, \Gamma) =$

$\frac{2}{\pi} \arg \frac{\sqrt{z}-\sqrt{R}}{\sqrt{z}+\sqrt{R}}$. 1070. $\frac{\ln|z| - \ln r}{\ln R - \ln r}$ for $|z| = R$ and $\frac{\ln|z| - \ln R}{\ln r - \ln R}$ for

$|z| = r$. 1073. $u(z) = \sum_{v=1}^n c_v \omega_v(z)$.

Chapter VIII

1078. $f(z) = \sum_{n=0}^{\infty} \frac{(z-a)^n}{(1-a)^{n+1}}$; by this expansion $f(z)$ is analytically continued inside the circle $|z-a| < |1-a|$, which does not lie entirely inside the circle $|z| < 1$ if a does not belong to the interval $[0, 1)$. 1079. $f(z) = \ln \frac{2}{3} +$

$+\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n \frac{\left(z+\frac{1}{2}\right)^n}{n}$; the circle of convergence of this series is $\left|z+\frac{1}{2}\right| < \frac{3}{2}$.

1093. Solution. The substitution $e^t = x$ reduces the integral to the form

$f(s) = \int_1^{\infty} \frac{\sin x \, dx}{x^s} \quad (x^s = e^{s \ln x})$. Integrating by parts, we obtain $f(s) = \cos 1 -$

$-\int_1^{\infty} \frac{\cos x}{x^{s+1}} dx$. The latter integral converges in the half-plane $\operatorname{Re} s > -1$.

1096. $0 < \operatorname{Re} z < 1$, $-1 < \operatorname{Re} z < 1$. 1098. The point $z=1$ is a simple pole

with residue unity. 1099. Solution. Denote $f_1(z) = \int_0^{\infty} e^{-t} \varphi(zt) \, dt$. Analyticity of

the function $f_1(z)$ in the circle $|z| < 1$ follows from the result of Problem 535 and from the general properties of the Laplace integral (see page 119). Integrating by parts $(n+1)$ times and using the inequalities of Problem 535,

we obtain for $|z| < 1$

$$\begin{aligned} f_1(z) &= - \sum_{k=0}^n z^k [e^{-t} \varphi^k(zt)]_0^\infty + z^{n+1} \int_0^\infty e^{-t} \varphi^{(n+1)}(zt) dt = \\ &= \sum_{k=0}^n a_n z^n + z^{n+1} \int_0^\infty e^{-t} \varphi^{(n+1)}(zt) dt. \end{aligned}$$

From the estimate of $|\varphi^{(n+1)}|$ it follows that the second term in the right member of the latter equality tends to zero as $n \rightarrow \infty$ ($|z| < r$). To prove the second assertion, take any point $z \in G$. Then it is easy to prove that there will be no singular point of the function $f(z)$ inside and on the boundary of a circle for which Oz is the diameter. That is why for a sufficiently small $\delta > 0$ the function $f(z)$ will also be analytic inside (and on the boundary C of) a circle of radius $\frac{|z|}{2} + \delta$ which is concentric with the one already constructed.

Thus, for the coefficient c_n of the expansion $f(z) = \sum_{n=0}^\infty c_n z^n$ the equalities $c_n = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta^{n+1}} d\zeta$ are valid and, hence,

$$\varphi(zt) = \frac{1}{2\pi i} \sum_{n=0}^\infty \frac{z^n t^n}{n!} \int_C \frac{f(\zeta)}{\zeta^{n+1}} dz.$$

Since the series $\sum_{n=0}^\infty \frac{z^n t^n}{n!} \int_C \frac{f(\zeta)}{\zeta^{n+1}} dz$ converges uniformly on C , $\varphi(zt) = \frac{1}{2\pi i} \int_C f(\zeta) e^{zt} \frac{d\zeta}{\zeta}$. The maximum of the quantity $\operatorname{Re}\left(\frac{z}{\zeta}\right)$ on C is equal

to $\frac{|z|}{|z| + \delta} = q < 1$ (in proving this assertion it will suffice to consider the case when z is real and positive, since rotation about the origin does not change $\operatorname{Re}(z/\zeta)$, and, hence, $|\varphi(zt)| < A e^{qt}$ ($A = \text{const}$)); it follows herefrom that the integral $\int_0^\infty e^{-t} \varphi(zt) dt$ converges. 1103. If $\operatorname{Ln}^{(1)} \frac{z-z_2}{z-z_1}$ is a branch of the function

$\operatorname{Ln} \frac{z-z_2}{z-z_1}$ analytic in the z -plane with a cut along the arc γ_1 and vanishing at $z = \infty$, then $F^-(z) = (a-b) \operatorname{Ln}^{(1)} \frac{z-z_2}{z-z_1}$, $F^+(z) = (a-b) \operatorname{Ln}^{(1)} \frac{z-z_2}{z-z_1} + 2\pi bi$;

inside G^+ : $F^-(z) = (a-b) \operatorname{Ln}^{(1)} \frac{z-z_2}{z-z_1} - 2\pi i(a-b)$ in analytic continuation across γ_1 and $F^-(z) = (a-b) \operatorname{Ln}^{(1)} \frac{z-z_2}{z-z_1}$ in analytic continuation across γ_2 .

1105. 1) $z = \pm 3$; 2) $z = \pm 2i$; 3) $z = \pm 2i$. In all three cases $w'(z)$ has different values. 1107. Above $z=1$ there are two elements: $z=1+t$,

$$w_1 = \frac{1-(1-t)^{1/2}}{t} = \frac{1}{2} + \frac{1}{8}t + \dots, \quad |t| < 1,$$

$$w_2 = \frac{1+(1-t)^{1/2}}{t} = \frac{2}{t} - \frac{1}{2} + \dots, \quad 0 < |t| < 1;$$

above $z=2$ there is one algebraic element: $z=2+t^2$, $w=\frac{1}{1+it}=1-it-t^2+\dots$, $|t|<1$. 1108. Above $z=1$ there are two algebraic elements: $z=1+t^6$, $w=\pm i\sqrt[6]{2}\left(1-\frac{t}{2}\right)^{1/2}=\pm i\sqrt[6]{2}\left(1-\frac{t}{4}+\dots\right)$, $|t|<2$; above $z=5$ there is one algebraic element: $z=5+t^2$, $w=\frac{t}{2}\left(1-\frac{t^2}{32}+\dots\right)$, $|t|<2$, and two regular elements: $z=5+t$, $w=\pm 2i\left(1+\frac{t}{32}+\dots\right)$, $|t|<4$; above $z=\infty$ there is one algebraic element: $z=t^{-4}$, $w=\frac{1}{t}-t+\dots$, $0<|t|<\sqrt[4]{\frac{1}{5}}$.

1109. Above $z=1$ there are two algebraic elements: $z=1+t^2$, $w=\pm(1+t)^{1/2}=\pm\left(1+\frac{1}{2}t-\dots\right)$, $|t|<1$. 1110. Above $z=1$ there are three algebraic elements: $z=1+t^2$, $w=\omega_k\left(1+\frac{t}{3}+\dots\right)$, $|t|<1$, where $\omega_1=1$,

$\omega_{2,3}=e^{\pm\frac{2\pi i}{3}}$. Above $z=2$ there is one algebraic element: $z=2+t^3$, $w=-\frac{t}{\sqrt[3]{2}}\left(1-\frac{t^3}{12}+\dots\right)$, $|t|<1$ and three regular elements: $z=2+t$,

$$w=\omega_k\sqrt[3]{2}\left(1+\frac{t}{12}+\dots\right), \quad |t|<1.$$

Above $z=\infty$ there is one algebraic element: $z=t^{-6}$, $w=\frac{1}{t}+\frac{t^2}{3}+\dots$, $0<|t|<\sqrt[6]{\frac{1}{2}}$. 1111. Above $z=\infty$ there are two algebraic elements: $z=t^{-2}$,

$$w=\pm\frac{1}{t}(1-at)^{\frac{1}{2}}(1-bt)^{\frac{1}{2}}=\pm\frac{1}{t}\left(1-\frac{a+b}{2}t+\dots\right),$$

$0<|t|<\min\left(\frac{1}{|a|}, \frac{1}{|b|}\right)$. 1112. $z=0$ is a double-valued essential singularity: $z=t^2$, $w=e^{\frac{1}{t}}=1+\frac{1}{t}+\frac{1}{2!t^2}+\dots$, $0<t<\infty$. 1113. Above $z=0$ there

is an algebraic element: $z=t^2$, $w=\frac{\sin t}{t^4}=\frac{1}{t^3}-\frac{1}{3!t}+\frac{1}{5!}t-\dots$, $0<|t|<\infty$.

1114. Above $z=0$ there is an algebraic element: $z=t^3$, $w=\cot t=\frac{1}{t}-\frac{t}{3}+$

$+\dots$, $0<|t|<\pi$. 1115. Above $z=0$ there is an algebraic element: $z=t^2$, $w=t\left(1-\frac{t^4}{12}+\dots\right)$, $|t|<\sqrt[4]{\pi}$. 1116. Above $z=1$ there is one transcendent

branch-point of the first order; above $z=\infty$ there is one algebraic branch-point of the first order. 1117. Above $z=0$ and $z=\infty$ each there is one algebraic branch-point of the first order; above $z=1$ there is one regular point and one essential singularity (isolated singularity). 1118. Above $z=0$ and $z=\infty$

each there is one algebraic branch-point of the first order; above each of the points $z=\left(1+\frac{1}{k\pi}\right)^2$ ($k=\pm 1, \pm 2, \dots$) there is one regular point and one

pole of the first order; above $z=1$ there is one regular point and one non-isolated singularity of a single-valued nature (a limit point for poles).

1119. Above $z=0$ there is one algebraic branch-point of the first order; above each of the points $z=k^2\pi^2$ ($k=\pm 1, \pm 2, \dots$) there are two poles of the first order; above $z=\infty$ there is one non-isolated branch-point (a limit point for poles). 1120. Above points $z=0$, $z=2$ each there are six algebraic branch-points of the first order; above $z=1$ there are two algebraic branch-points of the second order and six regular points; above $z=\infty$ there are two algebraic branch-points of the fifth order. 1121. Above points $z=i$; $z=-i$ each there is an infinite set of poles of the first order; above points $z=0$ and $z=\infty$ each there is one logarithmic branch-point. 1122. Above $z=1$ there are infinitely many regular points and infinitely many poles of the first order; above points $z=0$ and $z=\infty$ each there is one logarithmic branch-point. 1123. $w=\frac{1}{4}$,

$w=\infty$ are algebraic branch-points of the first order. 1124. $w=\pm 2$ is an algebraic branch-point of the first order; $w=\infty$ is an algebraic branch-point of the second order. 1125. $w=\frac{1}{4}$, $w=\infty$ are algebraic branch-

points of the first order. 1126. $w=0$, $w=\infty$ are algebraic branch-points of the first order. 1127. $w=\frac{1}{2}e^{\pm i\alpha}$ ($\alpha=\cos \alpha$) is an algebraic branch-point of the first order. 1128. $w=\infty$ is an algebraic branch-point of the $(n-1)$ th order. To each zero of the derivative $w'(z)$ of order k there corresponds an algebraic branch-point of the function $z(w)$ of the same order. 1129. To zeros of the derivative $w'(z)$ there correspond algebraic branch-points, just as in the preceding case. To the poles of the function $w(z)$ whose order is greater than unity there correspond algebraic branch-points of an order lower by unity than that of the pole. If, at ∞ , the function $w(z)=w_0+\frac{c-k}{z^k}+\dots$ ($k>1$), then $z=\infty$

corresponds to an algebraic branch-point of the $(k-1)$ th order above $w=w_0$.

1130. The surface for $z(w)$ is the same as for $\sqrt[n]{w}$, its branch-points lie above $w=0$, $w=\infty$ and correspond to $z=-n$, $z=\infty$. To sheets of the w -plane with cuts $0 < u < \infty$, $v=0$ there correspond angles $\frac{2\pi}{n}$ with vertex at point $z=-n$.

As $n \rightarrow \infty$ these angles are transformed into horizontal strips 2π wide, the function $w(z)$, into e^z , the surface for $z(w)$, into the surface for $\ln w$.

1131. The surface for $z(w)$ is the same as for $\sqrt[n]{w}$. To sheets of the w -plane with cuts $-\infty < u < 0$ there correspond lunes with angles $2\pi/n$ at points $z=a$, $z=b$ (Fig. 72, where $n=3$)¹⁾. 1132. The surface for $z(w)$ is obtained by joining two sheets of the w -plane with cuts $|u| < 1$, $v=0$ which sheets correspond to the domains $|z| < 1$ and $|z| > 1$. The branch-points lie above $w=\pm 1$ and correspond to $z=\pm 1$. To the polar grid $|z|=r$, $\arg z=\varphi$ there correspond ellipses and hyperbolas with foci ± 1 :

$$\frac{x^2}{\left[\frac{1}{2}\left(r+\frac{1}{r}\right)\right]^2} + \frac{y^2}{\left[\frac{1}{2}\left(r-\frac{1}{r}\right)\right]^2} = 1; \quad \frac{x^2}{\cos^2 \varphi} - \frac{y^2}{\sin^2 \varphi} = 1. \quad 1133. w(z) = \frac{z}{(1-z)^3} =$$

$= z + 2z^2 + 3z^3 + \dots$ is a well-known extremal function in the theory of univalent conformal mappings. It maps the unit circle $|z| < 1$ onto the w -plane with a cut $-\infty < u < -\frac{1}{4}$, $v=0$. The surface $z(w)$ is obtained by joining two

such sheets. 1134. The surface for $z(w)$ is obtained by successive joining of $2n$ sheets of the w -plane with cuts $1 < |u| < \infty$, $v=0$. It has $2n$ branch-

¹⁾ In the figures to the problems of this chapter, corresponding points are denoted by the same letters. Points at infinity are, as a rule, denoted by Ω and Ω' .

points of the first order above $w = \pm 1$ and 2 branch-points of the $(n-1)$ th order above $w = \infty$. The main mapping is shown in Fig. 73. The function $w(z)$ is automorphic (invariant) with respect to a group of linear transformations generated by the transformations $T = \omega z$ ($\omega = e^{\frac{2\pi i}{n}}$), $S = \frac{1}{z}$. To these transformations there correspond transformations of the surface for $z(w)$ into itself [under which the sheets go into each other cyclically with projections of points onto the w -plane preserved. 1135. The surface for $z(w)$ is $2n$ -valent with a branch-point of the $(2n-1)$ th order above $w = 0$ corresponding to $z = \infty$,

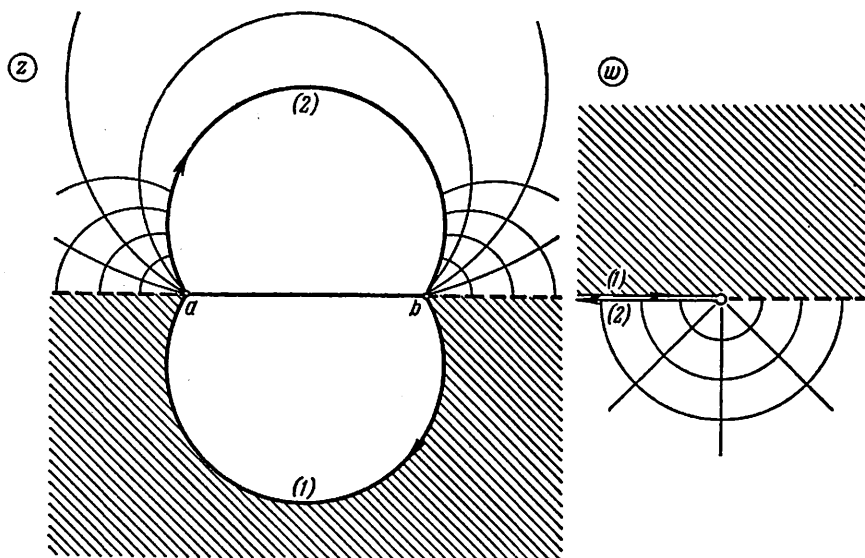


FIG. 72

and with $2n$ branch-points of the first order, of which n are situated above $w = \infty$ and correspond to the points $z_k = e^{\frac{\pi i}{n}} \omega^k$ ($\omega = e^{\frac{2\pi i}{n}}$, $k = 0, 1, 2, \dots, n-1$), and n are situated above the points $w'_k = \left(\frac{2n-1}{2n}\right)^2 z'_k$ and correspond to the points $z'_k = \sqrt[n]{\frac{1}{2n-1}} \omega^k$. The main mapping is shown in Fig. 74. The rest is obtained by continuation according to the symmetry principle. 1136. The surface for $z(w)$ is n -valent, with a branch-point of the $(n-1)$ th order above $w = \infty$ corresponding to $z = \infty$ and with $n-1$ branch-points of the first order situated above $w_k = z_k \left(1 - \frac{1}{n}\right)$ and corresponding to $z_k = \omega^k$ ($\omega = e^{\frac{2\pi i}{n-1}}$; $k = 0, 1, \dots, n-2$). To construct the surface, take its zero sheet (w -plane with $n-1$ radial cuts emanating from w_k) and along each cut join one sheet at a time criss-cross (w -plane with one radial cut). The main mapping is shown in Fig. 75. To the circle $|z| < 1$ there corresponds the interior of an epicycloid (cardioid for $n=2$). 1137. The surface for $z(w)$ is $(n+1)$ -valent,

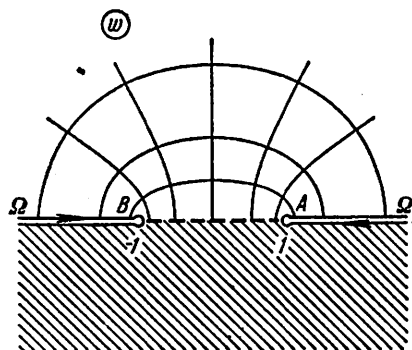
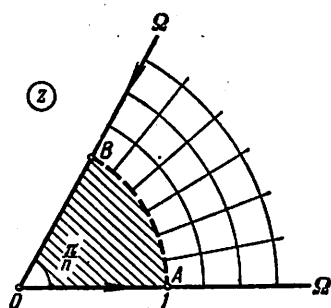


FIG. 73

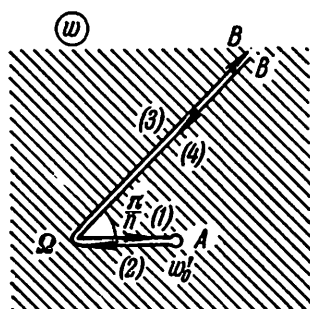
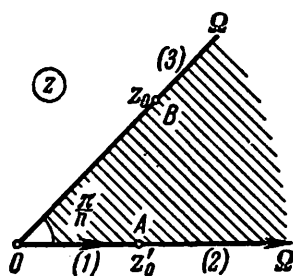


FIG. 74

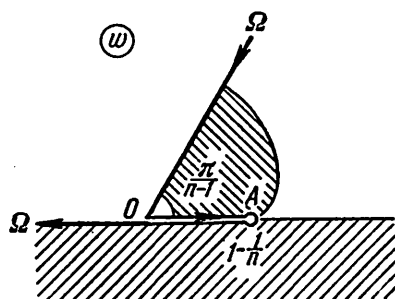
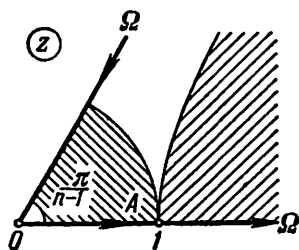


FIG. 75

with a branch-point of the $(n-1)$ th order above $w=\infty$ corresponding to $z=\infty$, and with $n+1$ branch-points of the first order situated above $w_k = \bar{z}_k \left(1 + \frac{1}{n}\right)$ and corresponding to $z_k = \omega^k \left(\omega = e^{\frac{2\pi i}{n+1}}, k=0, 1, \dots, n\right)$. The main mapping is shown in Fig. 76. To the domain $|z| < 1$ corresponds the exterior of a hypocycloid (a segment for $n=1$). 1138. The surface for $z(w)$

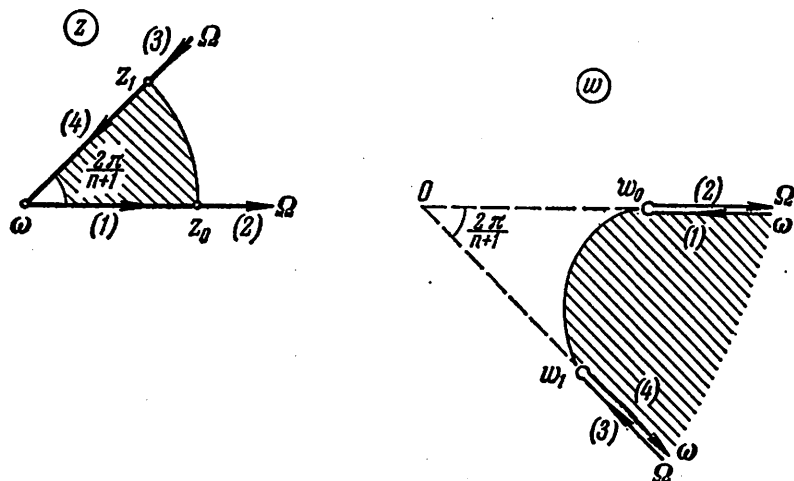


FIG. 76

is n -valent, with a branch-point of the $(n-1)$ th order above $w=\infty$ corresponding to $z=\infty$ and with $n-1$ branch-points of the first order situated above $w_k = \frac{(-1)^k}{2^{n-1}}$ and corresponding to $z_k = \cos \frac{k\pi}{n}$ ($k=1, 2, \dots, n-1$). To construct the surface, join to its zero sheet (the w -plane with cut $-\infty < u \leq -\frac{1}{2^{n-1}}, v=0$) successively $n-2$ sheets with two cuts $\frac{1}{2^{n-1}} \leq u < \infty, v=0$, then to the last of them join one more sheet having the cut $-\infty < u \leq -\frac{1}{2^{n-1}}, v=0$ if n is an even number, and the cut $\frac{1}{2^{n-1}} \leq u < \infty, v=0$ if n is an odd number. Under the mapping $w(z)$, ellipses and hyperbolas with foci ± 1 transform into ellipses and hyperbolas with foci $\pm \frac{1}{2^{n-1}}$. The change in the semi-axes is obtained from the relations $z=\text{const}, w=\text{const}$. Fig. 77 shows a partition of the z -plane into domains corresponding to the half-planes $v > 0$ and $v < 0$ (the former are hatched) for $n=5$. 1139. $w=0, w=\infty$ are logarithmic branch-points, $w=1$ is a pole of the first order for one of the branches of the function $z(w)$. 1140. The function $z(w)$ has one algebraic branch-point of the first order each above the points $w=e^{\pm 2\pi i}$ and two logarithmic branch-points each above the points $w=0$ and $w=\infty$. Its Riemann surface is obtained by joining two sheets of the Riemann surface of the function $\text{Ln } w$ along a cut made on them above the arc joining the points $w=e^{\pm 2\pi i}$ in the w -plane. 1141. The function $z(w)$ has an infinity of algebraic

branch-points of the first order above $w = \pm\sqrt{2}$ and two logarithmic branch-points above $w = \infty$. 1142. The function $z(w)$ has one algebraic branch-point of the first order each above the points $w_k = \frac{\sin z_k}{z_k}$ (z_k are roots of the equation $\tan z = z$; all of them are real), two logarithmic branch-points above $w = \infty$ and an indirectly critical singularity above $w = 0$, which is the limit point of the indicated algebraic branch-points.

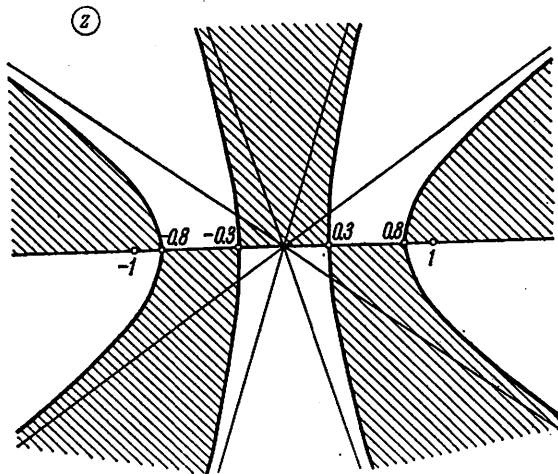


FIG. 77

1143. $z = \text{Arc cos } w = \frac{1}{i} \text{Ln}(w + \sqrt{w^2 - 1})$. The surface for $z(w)$ has infinitely many sheets with two logarithmic branch-points above $w = \infty$ and an algebraic branch-point of the first order above $w = \pm 1$ corresponding to $z = k\pi$ ($k = 0, \pm 1, \pm 2, \dots$); it is obtained by joining an infinite number of w -planes with cuts $1 \leq |u| < \infty, v = 0$, to which there correspond vertical strips $k\pi < x < (k+1)\pi$ (Fig. 78). 1144. $z = \text{Arc sin } w = \frac{\pi}{2} - \text{Arc cos } w$. The surface

for $z(w)$ is the same as for $\text{Arc cos } w$. 1145. $z = \text{Arc tan } w = \frac{1}{2i} \text{Ln} \frac{i-w}{i+w}$. The surface for $z(w)$ has infinitely many sheets with two logarithmic branch-points above $w = \pm i$; it is obtained by joining an infinite number of w -planes with a cut $u = 0, |v| \leq 1$, corresponding to the vertical strips $k\pi < x < (k+1)\pi$ (Fig. 79). 1146. $z = \text{Arc cot } w = \frac{\pi}{2} - \text{Arc tan } w$. The surface for $z(w)$ is the same

as for $\text{Arc tan } w$. 1147. $z = \text{Arcosh } w = \text{Ln}(w + \sqrt{w^2 - 1})$; $\cosh z = \cosh iz$. The surface for $z(w)$ is the same as that for $\text{Arc cos } w$.

1148. $z = \text{Arsinh } w = \text{Ln}(w + \sqrt{w^2 + 1})$; $\sinh z = -i \sin iz$. The surface for $z(w)$ is obtained by rotating the preceding one through $\frac{\pi}{2}$ about the origin.

1149. $z = \text{Artanh } w = \frac{1}{2} \text{Ln} \frac{1+w}{1-w}$; $\tanh z = -i \tan iz$. The surface for $z(w)$ is obtained by rotating the surface for $\text{Arc tan } w$ through $\frac{\pi}{2}$ about the origin.

1150. $z = \operatorname{Arcoth} w = \frac{1}{2} \operatorname{Ln} \frac{w+1}{w-1}$, $\coth z = i \cot iz$. The surface for $z(w)$ is the same as that for $\operatorname{Artanh} w$. 1151. The surface for $z(w)$ is constructed as follows: make horizontal cuts $-\infty < u \leq 1$, $v = (2k+1)\pi$ ($k=0, \pm 1, \pm 2, \dots$) in the w -plane and along each of them attach a sheet of the w -plane with an identical cut (one cut). The construction of the surface for $z(w)$ is based on the fact that $w(z)$ maps each strip $2k\pi < y < (2k+1)\pi$ onto the strip

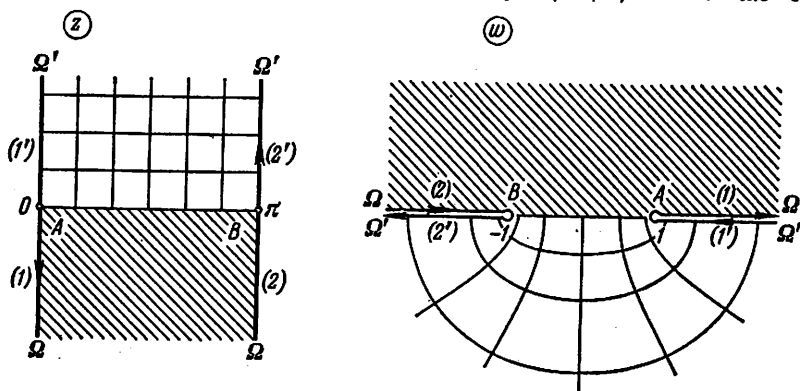


FIG. 78

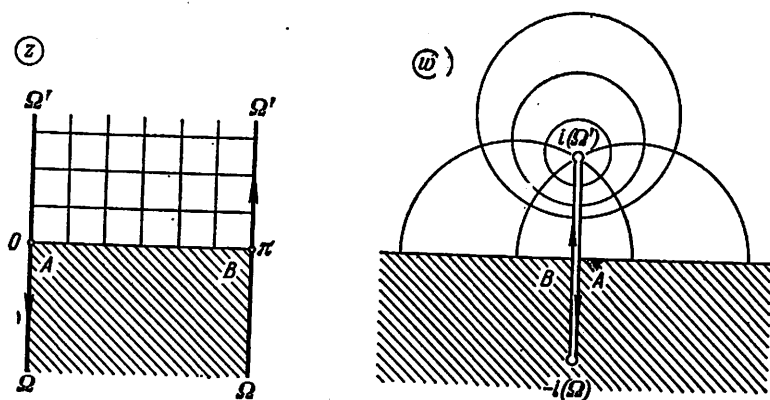


FIG. 79

$2k\pi < v < (2k+2)\pi$ bearing the w -plane joined along the cut $-1 \leq u < \infty$, $v = (2k+1)\pi$ (see Fig. 80; the $+$ sign means that the domains must be joined). 1152. 1) Let F be a Riemann surface onto which the function $w = R(\xi)$ maps the ξ -plane. To construct the Riemann surface of the function $z(w)$, join an infinite number of sheets of the surface F with a cut along the arc connecting the points $w(0)$ and $w(\infty)$ on F (this is similar to the construction of the Riemann surface of the function $\operatorname{Ln} w$). The Riemann surface thus obtained has two logarithmic branch-points at the end-points of the joining arc and an infinite number of algebraic branch-points belonging to the F -surfaces. 2) To construct a Riemann surface of the function $z(w)$, join an infinite number of

sheets of the surface F alternately along the cuts running from point $w(\pm 1)$ to point $w(\infty)$ (this is similar to the construction of the Riemann surface of the function $\text{Arc sin } w$). The Riemann surface thus obtained has two logarithmic branch-points above $w(\infty)$ and, in addition to the algebraic branch-points mentioned in item (1), it also has an infinite number of algebraic branch-points of the first order at points $w(\pm 1)$ (if $w(+1)$ or $w(-1)$ is an algebraic branch-point of order k of the F -surface, then, for $z(w)$, it will be an algebraic branch-point of order $2k+1$). Investigation of the function $z(w)$ can be reduced to the preceding case by substituting $z_1 = iz$. 1153. All Riemann surfaces of

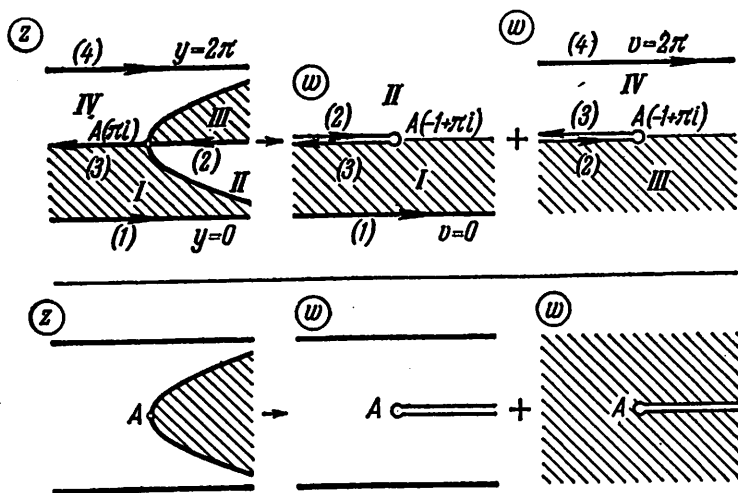


FIG. 80

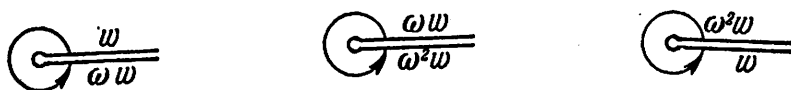


FIG. 81

$w(z)$ are double-sheeted and have algebraic branch-points of the first order above the points: 1) $z=a$, $z=b$; 2) $z=a$, $z=b$, $z=c$, $z=\infty$; 3) $z=a_k$ and $z=\infty$ if n is odd. To construct the surfaces, take two sheets of the z -plane with cuts running from the above-mentioned points to ∞ and join them along identical cuts. 1154. All Riemann surfaces of $w(z)$ are three-sheeted and have algebraic branch-points of the second order above the points: 1) $z=a$, $z=\infty$; 2) $z=a$, $z=b$, $z=\infty$; 3) $z=a$, $z=b$, $z=c$; 4) $z=a_k$ and $z=\infty$ if n is not a multiple of 3. To construct the surfaces, take three sheets of the z -plane with cuts running from the indicated points to ∞ , on which sheets we define three

single-valued branches w , ωw , $\omega^2 w$ ($\omega = e^{\frac{2\pi i}{3}}$). In traversing the branch-points, $w(z)$ acquires a factor ω from among the factors of the root; that is why the

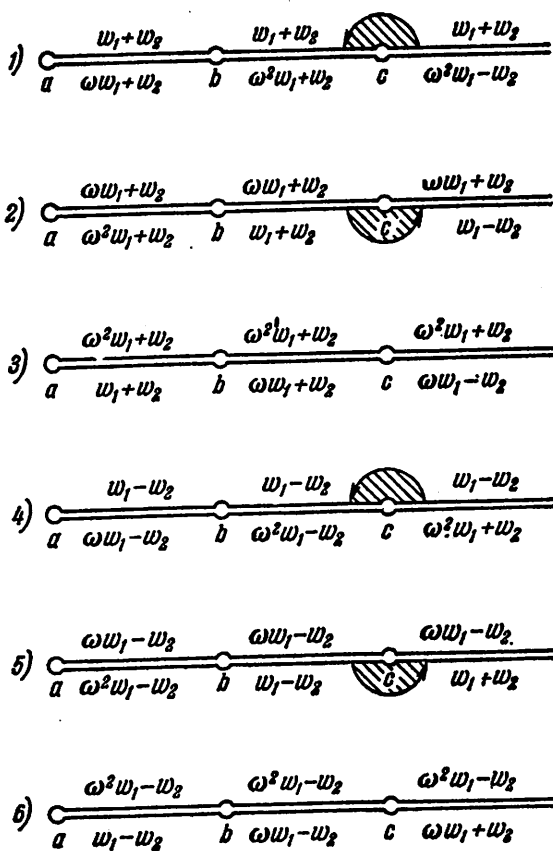


FIG. 82

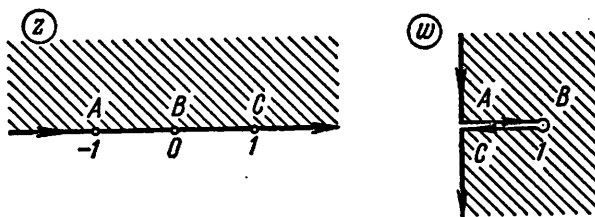


FIG. 83

order of joining the sheets along all cuts is the same, i.e. cyclic (see the diagram in Fig. 81). 1155. The Riemann surface of $w(z)$ is n -sheeted with algebraic branch-points of $(n-1)$ th order above $z=a$, $z=b$, $z=c$, $z=\infty$. The surface is obtained by joining n sheets of the z -plane with cuts running from points $z=a$, $z=b$, $z=c$ to ∞ . The joining is cyclic and simultaneous along all cuts. The sheets correspond to single-valued branches of the function $\omega^k w$ ($\omega = e^{2\pi i/n}$, $k=0, 1, \dots, n-1$). 1156. The Riemann surface of $w(z)$ is 6-sheeted, with algebraic branch-points of the fifth order above $z=\infty$, two algebraic branch-points of the second order above each of the points $z=a$, $z=b$, and three algebraic branch-points of the first order above $z=c$. The surface is obtained by joining six sheets of the z -plane with a cut along a curve running from a to b , from b to c , and from c to ∞ . These sheets correspond to single-valued branches: $\omega_1 + w_2$, $\omega w_1 + w_2$, $\omega^2 w_1 + w_2$, $w_1 - w_2$, $\omega w_1 - w_2$, $\omega^2 w_1 - w_2$,

where $\omega = e^{\frac{2\pi i}{3}}$, and w_1, w_2 are single-valued branches $\sqrt[3]{\frac{z-a}{(z-b)^2}}$ and $\sqrt{z-c}$.

The detour around a cyclically joins the sheets 1, 2, 3 and 4, 5, 6; that around b , the sheets 1, 3, 2 and 4, 6, 5 (joined twice along half-sheets); that around c , the half-sheets 1, 2, 4, 5 (as shown in Fig. 82); 2, 3, 5, 6; 3, 1, 6, 4; that around ∞ , cyclically the sheets 1, 6, 2; 4, 3, 5 (see Fig. 82). 1157. The Riemann surface of $w(z)$ is 6-sheeted, with two algebraic branch-points of the second order above $z=0$, one algebraic branch-point of the first order above $z=1$ and an algebraic branch-point of the fifth order above $z=\infty$. To construct it, join together two sheets of the surface of the function $\sqrt[3]{z}$, each of which has a cut along the ray $y=0$, $1 \leq x < \infty$ on one of the sheets. 1158. The Riemann surface of $w(z)$ is two-sheeted, with algebraic branch-points of the first order above the points $z=k\pi$ ($k=0, \pm 1, \pm 2, \dots$); above $z=\infty$ the surface has a transcendental singularity, namely the limit of algebraic branch-points. To construct the surface, take two sheets of the z -plane with cuts running from algebraic branch-points to ∞ (say, along rays parallel to the imaginary axis), and join the sheets along identical cuts. 1159. $^1) F_z$ and F_w are each obtained by joining two planes with cuts $[-1, 1]$ (Fig. 83). 1160. F_z is obtained by joining two z -planes with cuts $(-\infty < x \leq 0, y=0)$, F_w , by joining two w -planes with cuts $[-\frac{i}{2}, \frac{i}{2}]$ (Fig. 84). 1161. F_z and F_w are each obtained by joining three planes having (in cyclic order) cuts along two of the line segments $[0, \sqrt[3]{4}]$,

$[0, \omega \sqrt[3]{4}]$, $[0, \omega^2 \sqrt[3]{4}]$, where $\omega = e^{\frac{2\pi i}{3}}$ (Fig. 85). 1162. F_z is obtained by joining two z -planes with cuts $[0, \frac{3}{2}]$, F_w , by joining to the w -plane with

cuts $\gamma_1: [i \frac{3\sqrt{3}}{2}, +i\infty]$, $\gamma_2: [-i \frac{3\sqrt{3}}{2}, -i\infty]$ two other w -planes with cuts γ_1 and γ_2 , respectively (Fig. 86). 1163. F_z is obtained by joining two z -planes with cuts $[-1, 1]$, F_w , by joining three w -planes having one cut $[-ib, ib]$, two cuts $[-a, a]$, $[0, ib]$, and two cuts $[-a, a]$, $[0, -ib]$, respectively, where $a = \frac{\sqrt{5}-1}{2} \sqrt{\sqrt{5}-2}$, $b = \frac{\sqrt{5}+1}{2} \sqrt{\sqrt{5}+2}$. The mapping

and the three w -planes with cuts are shown in Fig. 87 (not to scale). 1164. a) n is odd: F_z is n -sheeted, has n algebraic branch-points of the $(n-1)$ th

¹⁾ In the answers to problems 1159–1164, F_w denotes the surfaces of $z(w)$, i.e. those above the w -plane, and F_z , the surfaces of the function $w(z)$ above the z -plane.

order above points $z = \omega^k$ ($\omega = e^{\frac{\pi i}{n}}$, $k = 1, 2, \dots, n$); F_w is $2n$ -sheeted, has n^2 algebraic branch-points of the first order; n above each of the points $w = \eta^v$ $\sqrt[n]{\frac{1}{4}}$

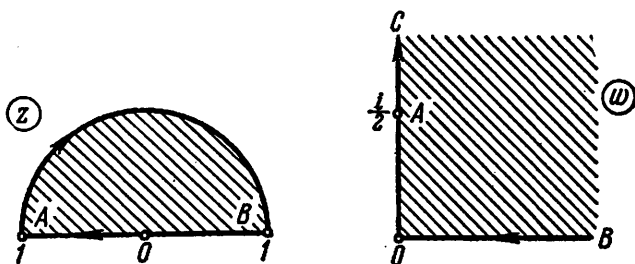


FIG. 84

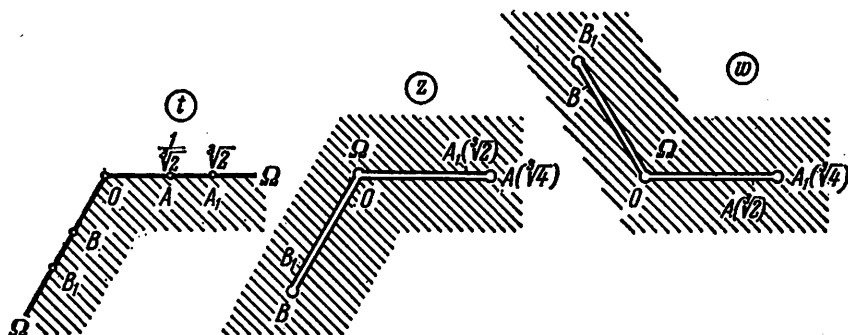


FIG. 85

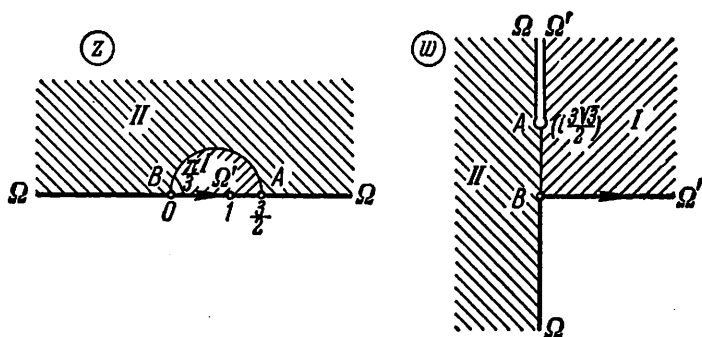


FIG. 86

$\left(\eta = e^{\frac{2\pi i}{n}}, v = 0, 1, 2, \dots, n-1\right)$, corresponding to points of F_z above $z = \eta^v$, and n algebraic branch-points of the first order above $w = \infty$, corresponding to the algebraic branch-points of F_z above $z = \omega^k$. The mapping is shown in Fig. 88

(investigation involves use of the substitution $\zeta = z^n$, $W = w^n$). The function $w(z)$ maps the circle $|z| < 1$ onto the w -plane with n radial cuts emanating from the points $\eta^k \sqrt[n]{\frac{1}{4}}$. b) n is even. If $n = 2m$, then $w(z)$ breaks up into

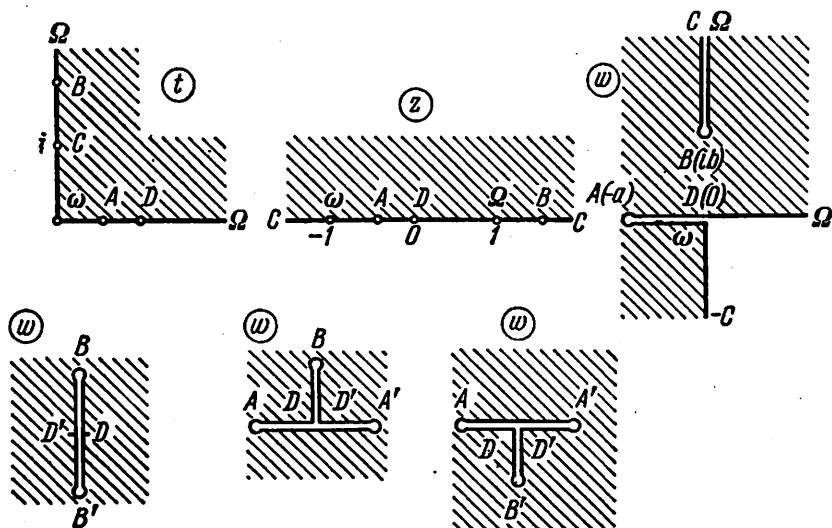


FIG. 87

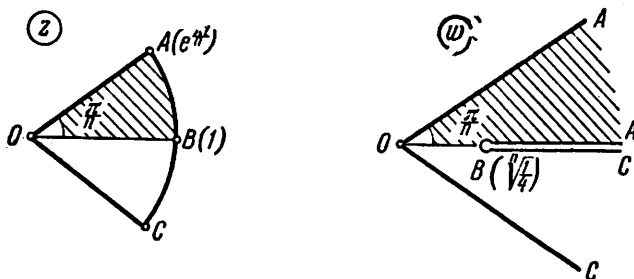


FIG. 88

two functions: $w_{1,2}^m = \pm \frac{z^m}{1+z^{2m}}$. To construct F_z , take n "half-sheets" $|z| < 1$ and n "half-sheets" $|z| > 1$. Denote them by H_k^0 and H_k^+ , respectively, and denote by γ_k the boundary arcs defined by points ω^k . Then join to H_k^+ the half-sheets H_1^0 along γ_k , the half-sheets H_2^0 along γ_{k+1} , the half-sheets H_3^0 along γ_{k+2} , and so on, cyclically. This also determines the order of building up F_w from $2n$ sheets representing w -planes with the n radial cuts indicated above.

1165. $w = z^{\frac{2}{3}}$. $z = 0$ and $z = \infty$ are algebraic branch-points of the second order.

1166. $w = z^{\frac{n_1}{m_1}}$ ($\frac{n_1}{m_1}$ is an irreducible fraction equal to $\frac{n}{m}$), $z=0$ and $z=\infty$ are algebraic branch-points of the (m_1-1) th order. 1167. $z=0$ and $z=\infty$ are algebraic branch-points of the first order; $z=1$ is an essential singularity for one of the two branches of the function. 1168. $w = 1 - \frac{z}{3} + \frac{z^2}{5} - \dots$ is an entire

function; $z=\infty$ is an essential singularity. 1169. $z=0$ and $z=\infty$ are algebraic branch-points of the first order; $z=1$ is an essential singularity for one of the two branches of the function and is a limit point for its poles of the first order

at points $z_k = \left(\frac{1-ak}{1+ak}\right)^2$, $ak = \frac{\pi}{2} + \pi k$ ($k=0, \pm 1, \dots$). According to Problem

618, the domain of indeterminacy at point $z=1$ coincides with the entire plane.

1170. If $n=0$, then $z=0$ and $z=\infty$ are removable singularities and $w=1$; if $n < 0$, then $z=0$ and $z=\infty$ are logarithmic branch-points, where $\lim_{z \rightarrow 0} w =$

$= \lim_{z \rightarrow 0} w = 1$ and $z=1$ is an essential singularity for one of the branches of the

function; if $n=1$, then $w=z$; if $n > 0$, then $z=0$ and $z=\infty$ are logarithmic branch-points; the domain of indeterminacy of $w(z)$ at these points coincides with the entire extended w -plane. 1171 and 1172. $z=0$ and $z=\infty$ are logarithmic branch-points; the domain of indeterminacy of $w(z)$ at these points coincides with the entire extended w -plane. 1173. $z=1$ and $z=\infty$ are logarithmic branch-points, where $\lim_{z \rightarrow 1} w = \lim_{z \rightarrow \infty} w = \infty$ and $z=0$ is a pole of the first order

for all the branches of $w(z)$ except one. 1174. $z=0$ and $z=\infty$ are logarithmic branch-points, and $\lim_{z \rightarrow 0} w = \lim_{z \rightarrow \infty} w = \infty$. 1175. The branch-points are the same

as for Arc sin z (i.e. infinitely many algebraic branch-points of the first order above $z=\pm 1$ and two logarithmic branch-points above $z=\infty$, $\lim_{z \rightarrow \infty} w=0$;

$z=0$ is a pole of the first order for all the branches except one). 1176. $z=\pm i$ is a logarithmic branch-point, $\lim_{z \rightarrow \pm i} w = \infty$; $z=0$ is a pole of the second order

for all branches of the function. 1177. The surfaces of $w(z)$ and $z(w)$ are the same as for the logarithmic function (logarithmic branch-points above 0 and ∞). The mapping is readily obtained with the aid of the parametric representation $z=e^{\zeta}$, $w=e^{a\zeta}$. 1178. The Riemann surface of $w(z)$ has infinitely many sheets with one logarithmic branch-point above each of the points $z=a$, $z=b$ and with two logarithmic branch-points above $z=\infty$. The surface is obtained by joining an infinite number of sheets of the z -plane with cuts running from points $z=a$, $z=b$ to ∞ . These sheets correspond to the single-valued branches of the function $w+2\pi in$ ($n=0, \pm 1, \pm 2, \dots$). In going around $z=a$ and $z=b$ these branches turn into one another successively, and this determines the manner of joining the sheets. 1179. The Riemann surface of $w(z)$ has infinitely many sheets with one logarithmic branch-point above each of the points $z=a$, $z=b$, $z=c$ and with three logarithmic branch-points above $z=\infty$. The construction is similar to that of the preceding problem. 1180. The Riemann surface of $w(z)$ has infinitely many sheets, with one logarithmic branch-point above each of the points $z=k\pi$ ($k=0, \pm 1, \pm 2, \dots$). We may take, as sheets, z -planes with cuts running from points $z=k\pi$ to ∞ (for instance, along vertical half-lines). Two sheets are joined simultaneously along all cuts, on one side, as in the construction of the surface of a logarithmic function. At ∞ there is a transcendental singularity, which is a limit point for the logarithmic branch-points. 1181. 1) In each connected part of the Riemann surface of the function $\zeta=\varphi(z)$ above the z -plane, to which there corresponds a connected part of the Riemann surface of the inverse function $z=\varphi^{-1}(\zeta)$ above G ; $w(z)$ is a single analytic function; 2) in each connected part of the Riemann surface of the func-

tion $\zeta = \varphi(z)$ situated above G_z (this is the domain G , transferred to the z -plane), $w(z)$ is a single analytic function; 3) the same as in item (2). In the particular case indicated in the statement of the problem, $w(z)$ is always one analytic function. 1182. $w(z)$ consists of two analytic functions $\pm z^{\frac{2}{3}}$. 1183. $w(z)$ consists

of two analytic functions $\pm z^{\frac{2}{p}}$. 1184. $w(z)$ consists of p analytic functions $\omega^k z^{\frac{2\pi i}{n}}$ ($\omega = e^{\frac{2\pi i}{n}}$; p is the greatest common divisor of (m, n) ; $k=0, 1, \dots, p-1$; $m_1 = \frac{m}{p}$, $n_1 = \frac{n}{p}$). 1185. $w(z)$ consists of n entire functions $\omega^k e^{\frac{z}{n}}$

($\omega = e^{\frac{2\pi i}{n}}$, $k=0, 1, \dots, n-1$). 1186. $w(z)$ is one n -valued function with algebraic branch-points of the $(n-1)$ th order above $z=k\pi$ ($k=0, \pm 1, \dots$). At ∞ it has a non-isolated singularity which is a limit point for the algebraic branch-points. 1187. $w = n \operatorname{Ln} z + 2\pi i k$ ($k=0, 1, \dots, n-1$) are n distinct analytic functions. 1188. $w(z)$ consists of functions $z + 2\pi i k$ ($k=0, \pm 1, \dots$). 1189. One infinitely-valued function with a logarithmic branch-point above each of the points $z=0, \pm 1, \infty$. 1190. $w(z)$ is one infinitely multi-valued analytic function with one logarithmic branch-point above $z_k = 2\pi i k$ ($k=0, \pm 1, \dots$). At ∞ it has a non-isolated singularity, which is a limit point for the logarithmic branch-points. The Riemann surface of the function $w(z)$ is singly connected and is obtained by joining an infinite number of sheets of w -planes with cuts (having no common points pairwise) running from z_k to ∞ (two sheets are joined simultaneously on all cuts, but only along a definite side). 1191. The Riemann surface is the same as in Problem 1190, but with logarithmic branch-points above $z=k\pi$ ($k=0, \pm 1, \dots$). 1192. The Riemann surface is the same as in

Problem 1190, but with logarithmic branch-points above $z = \frac{\pi k}{2}$ ($k=0, \pm 1, \dots$).

1193. $w(z)$ consists of functions $\pm z + 2k\pi$ ($k=0, \pm 1, \dots$). 1194. $w(z)$ consists of functions $z + k\pi$ ($k=0, \pm 1, \dots$). 1195. 1) Let $r_1 = \frac{m_1}{n_1}$, $r_2 = \frac{m_2}{n_2}$,

$r = r_1 r_2 = \frac{m}{n}$ be irreducible fractions, p the greatest common divisor of (m_1, n_2) , and q the greatest common divisor of (m_2, n_1) . Then $(z^{r_1})^{r_2}$ consists of p different

n -valued analytic functions equal to $\omega^k z^r$, $\omega = e^{\frac{2\pi i}{np}}$, $k=0, 1, \dots, p-1$,

and $(z^{r_2})^{r_1}$ consists of q functions $\omega_1^k z^r$, $\omega_1 = e^{\frac{2\pi i}{nq}}$, $k_1=0, 1, \dots, q-1$. One

of them, namely z^r , always belongs to both cases. In particular, $\left(z^{\frac{2}{3}}\right)^{\frac{3}{2}} = \pm z$,

$\left(z^{\frac{3}{2}}\right)^{\frac{2}{3}} = \sqrt[3]{1} z$. 2) Let $r_1 = \frac{m_1}{n_1}$, $r_2 = \frac{m_2}{n_2}$, $r = r_1 + r_2 = \frac{m}{n}$ be irreducible fractions and p , the greatest common divisor of (n_1, n_2) . Then $z^{r_1} z^{r_2}$ consists

of p different n -valued analytic functions equal to $\omega^k z^r$, $\omega = e^{\frac{2\pi i}{np}}$, p_1 being the greatest common divisor of $(m_1 n_2' + m_2 n_1', p)$, $n_1' = \frac{n_1}{p}$, $n_2' = \frac{n_2}{p}$, $k=0, 1, \dots, p_1-1$. 3) Let p be the greatest common divisor of (n_1, n_2) and N , the least common multiple of $(n_1, n_2) = \frac{n_1 n_2}{p}$. Then $w(z)$ consists of p different N -valued

analytic functions. 1196. Let N be the least common multiple of (m, n) , p the greatest common divisor of (m, n) ; q the greatest common divisor of $\left(\frac{m+n}{p}, \frac{mn}{p}\right)$ and $\frac{1}{m} + \frac{1}{n} = \frac{\mu}{v}$. Then $w(z)$ is a single N -valued analytic function with $\frac{N}{n}$ algebraic branch-points of the $(n-1)$ th order above $z=0$; $\frac{N}{m}$, algebraic branch-points of the $(m-1)$ th order above $z=1$; and $\frac{N}{q}$, algebraic branch-points of the $(v-1)$ th order above $z=\infty$. 1197. One nm -valued analytic function with one algebraic branch-point of the $(n-1)$ th order above $z=1$, n algebraic branch-points of the $(m-1)$ th order above $z=0$ and one algebraic branch-point of the $(mn-1)$ th order above $z=\infty$. 1198. Two distinct four-valued functions differing in sign with the same Riemann surface as that of the function $\sqrt[4]{z}$. Each of these functions has one branch for which the point $z=1$ is a pole of the first order. 1199. An infinitely multi-valued function with one algebraic branch-point of the $(n-1)$ th order above $z=1$ and n logarithmic branch-points above each of the points $z=0$, $z=\infty$. To construct the surface, join n surfaces of $\text{Ln } z$ with the cut $[1, \infty)$ on one of the sheets of each surface. The lines $\rho^n \sin n\theta = k\pi$ ($k=0, \pm 1, \pm 2, \dots$) divide the w -plane into domains corresponding to the half-planes $y \geq 0$ (Fig. 89, for $n=2$, $\zeta=w^2$

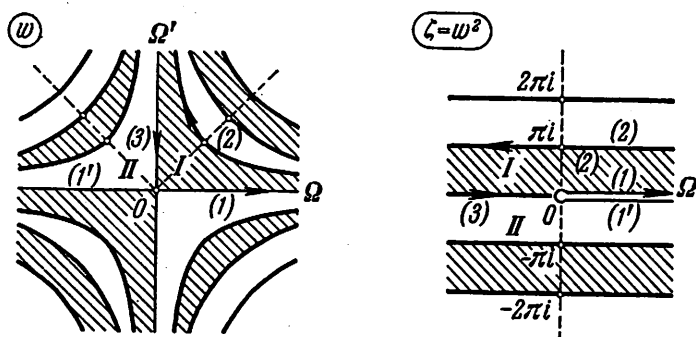


FIG. 89

is an auxiliary plane). 1200. An infinitely multi-valued function with one logarithmic branch-point above $z=1$ and an infinite number of solely logarithmic branch-points above $z=0$ and $z=\infty$. The Riemann surface is obtained by joining an infinite number of surfaces of $\text{Ln } z$ with the cut $[1, \infty)$ on one of the sheets. The surface has an infinite number of solely logarithmic branch-points above $z=0$, $z=\infty$, and regular points and one logarithmic branch-point above $z=1$. The lines $e^{u} \sin v = (2k+1)\pi$ and $v = 2k\pi$ ($k=0, \pm 1, \pm 2, \dots$) divide the w -plane into domains, each corresponding to the z -plane with the cut $-\infty < x \leq 0$, $y=0$ and with an additional cut $1 \leq x < \infty$, $y=0$ for domains whose boundary contains one of the straight lines $v = 2k\pi$ (Fig. 90, $\zeta = e^w$ is an auxiliary plane). 1201. An infinitely multi-valued function with the same Riemann surface as that of $\text{Ln Ln } z$. 1202. The Riemann surface has infinitely many sheets, an algebraic branch-point of the $(n-1)$ th order above $z=0$, solely algebraic branch-points of the first order above $z=\pm 1$ and $2n$ logarithmic branch-points above $z=\infty$. To construct the surface, join n surfaces for $\text{Arc sin } z$ with the cut $[0, 1]$ on one of the sheets for each surface. The lines $n\theta = \frac{k\pi}{n}$ and

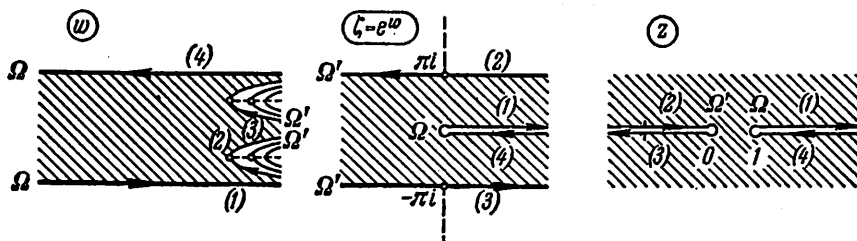


FIG. 90

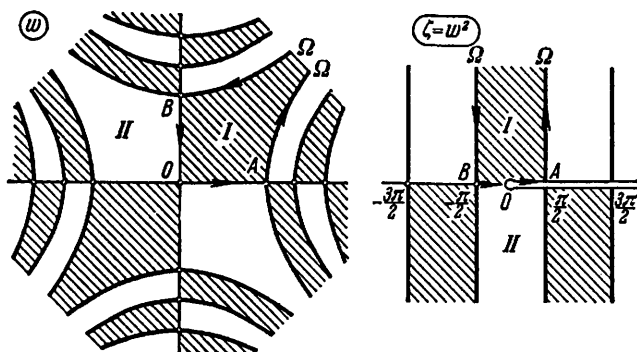


FIG. 91

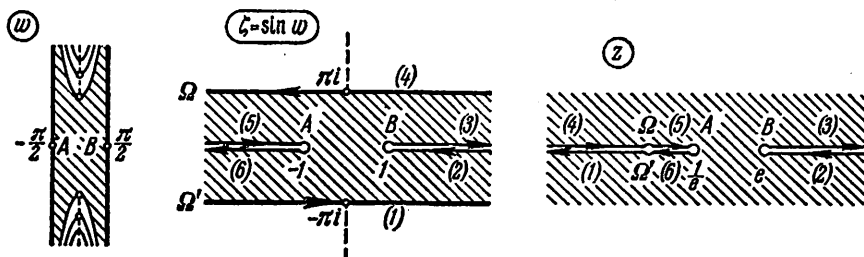


FIG. 92

$\rho^n \cos n\theta = k\pi$ ($k=0, \pm 1, \dots$) divide the w -plane into domains corresponding to half-planes $y \geq 0$ (Fig. 91, where $n=2$, $\zeta = w^2$ is an auxiliary surface). 1203. The Riemann surface is obtained by joining an infinite number of surfaces for $\text{Ln } z$ with cuts $[0, \frac{1}{e}]$, $[e, \infty)$ on one of the sheets (the surfaces for $\text{Ln } z$ are joined in pairs, alternately along one cut and then the other). The surface has an infinite number of logarithmic branch-points above $z=0$, $z=\infty$, and regular points and infinitely many algebraic branch-points of the first order above $z = \frac{1}{e}$, $z=e$. The lines $\text{Im } \sin w = \cos u \sinh v = (2k+1)\pi$ and $u = \frac{\pi}{2} + k\pi$ ($k=0, \pm 1, \pm 2, \dots$) divide the w -plane into domains, each corresponding to the z -plane with the cut $-\infty < x < 0$, $y=0$ and with two additional cuts,

$0 \leq x \leq \frac{1}{e}$, $y=0$ and $e \leq x < \infty$, $y=0$ for domains whose boundary contains

one of the straight lines $u = \frac{\pi}{2} + k\pi$ (Fig. 92; $\xi = \sin w$ is an auxiliary plane).

1204. An infinitely multi-valued function. Its Riemann surface is obtained by joining an infinite number of sheets of the Riemann surface of the function

$\sqrt[n]{z-1}$ with a cut along the ray $1 \leq x < \infty$, $y=0$ on one of the sheets. The joining is performed as in constructing the Riemann surface of a logarithmic function. 1205. An infinitely multivalued function with only algebraic branch-points of the first order above $z=0$ and $z=\infty$ and with two logarithmic branch-points above $z=1$. 1206. An infinitely multi-valued function with two logarithmic branch-points above $z=-1$, and with an infinite number of algebraic

branch-points alone of the first order above $z=0$ and $z=\infty$; 1207. If $\alpha = \frac{m}{n}$,

then $w = \frac{m}{n} \operatorname{Ln} z + \frac{2\pi i}{n} k$ ($k=0, 1, \dots, m-1$) are m distinct analytic functions;

if α is irrational, then $w = \alpha \operatorname{Ln} z + 2\pi i k$ ($k=0, \pm 1, \dots$) are an infinite number of different functions. 1208. Two different infinitely multi-valued analytic functions with the same Riemann surface as that of $\operatorname{Ln} z$. 1209. Two different analytic functions equal to $2 \operatorname{Ln} z$ and $2 \operatorname{Ln}(-z)$, respectively. 1210. $w = \frac{\pi}{2} + 2\pi k$,

$k=0, \pm 1, \dots$, $w = \frac{\pi}{2} + 2 \operatorname{Arc} \cos z$; $w = -\frac{3\pi}{2} + 2 \operatorname{Arc} \cos z$. 1211. $w = \frac{\pi}{2} + \pi k$

($k=0, \pm 1, \dots$). 1212. $w = i \left(\frac{\pi}{2} + \pi k \right)$ ($k=0, \pm 1, \dots$). 1213. w is an infinitely multi-valued function with the same Riemann surface as that of $\operatorname{Ln} \operatorname{Ln} z$

(see Problem 1200). If $\xi = \rho e^{i\varphi} = \operatorname{Ln} z = \ln r + i\varphi$ ($\varphi = \operatorname{Arg} z$), then $w = e^{i \ln \xi} = e^{i(\ln \rho + i\varphi + 2\pi i k)}$. For $f_0(z) = e^{-\theta + i \ln \rho}$ ($0 \leq \theta \leq 2\pi$) the sets of limiting values indicated in the problem represent, respectively: 1) and 2) the circle $|w| = e^{-\pi}$;

3) circles $|w| = e^{-\frac{\pi}{2}}$ as $\varphi \rightarrow +\infty$ and $|w| = e^{-\frac{3\pi}{2}}$ as $\varphi \rightarrow -\infty$; 4) the annulus

$e^{-\frac{3\pi}{2}} \leq |w| \leq e^{-\frac{\pi}{2}}$. For other groups of branches the factor $e^{-2\pi k}$ ($k = \pm 1, \pm 2, \dots$) is added. 1214. In all cases $w(z)$ is one analytic function if $|a| < 1$; if $|a| \geq 1$, then in cases (1) and (3) the function decomposes into n , and in cases (2) and (4), into an infinite number of distinct analytic functions.

1215. If $|a| < 1$, then in both cases $w(z)$ is one analytic function; if $|a| \geq 1$, then in case (1) there will be n , and in case (2) an infinite number of different analytic functions. 1216. 1) Let $z = re^{i\varphi}$ and $\xi = \rho e^{i\theta} = \operatorname{Ln} z$. Then $w(z)$ consists

of multi-valued analytic functions equal, respectively, to $\chi(z) e^{i \ln \rho - \theta + 2\pi k}$ ($|z| < 1$; $\rho = \sqrt{\ln^2 r + \varphi^2}$, $\theta = \arg(\ln r + i\varphi)$; here $-\infty < \varphi < \infty$ and $\frac{\pi}{2} < \theta < \frac{3\pi}{2}$; ($k=0, \pm 1, \dots$). Each of them has one logarithmic branch-point

above $z=0$ in the neighbourhood of which the domain of indeterminacy is an

annulus (in particular, for $k=0$ the annulus is $\chi(0) e^{-\frac{3\pi}{2}} \leq |w| \leq \chi(0) e^{-\frac{\pi}{2}}$).

2) Let $z-1 = re^{i\varphi}$ and $\xi = \rho e^{i\theta} = \operatorname{Ln}(z-1)$. Then $w(z)$ consists of single-valued analytic functions equal, respectively, to $\chi(z) e^{i \ln \rho - \theta + 2\pi k}$ ($|z| < 1$; $\rho =$

$\sqrt{\ln^2 r + (\varphi + 2\pi n)^2}$, $\theta = \arg(\ln r + i(\varphi + 2\pi n))$ here $\frac{\pi}{2} < \varphi < \frac{3\pi}{2}$, $0 < \theta < 2\pi$; $k, n=0, \pm 1, \dots$). For different k, n , these are distinct analytic functions.

If there exists $\chi(1) = \lim_{z \rightarrow 1} \chi(z)$, then, as $z \rightarrow 1$, the domain of indeterminacy

is a circle (in particular, for $k=n=0$ the circle $|w|=\chi(1)e^{-\pi}$). 1217. 1) If $f(z)$ has no odd-order zeros, then $\sqrt{f(z)}$ decomposes into two entire functions. If $a_1, a_2, \dots$ are odd-order zeros of the function $f(z)$, then the Riemann surface of $\sqrt{f(z)}$ is two-sheeted with algebraic branch-points above $a_1, a_2, \dots$ and above ∞ provided $f(z)$ has an odd-order pole there. If $f(z)$ is a transcendental function, then there are two essential singularities of single-valued character above $z=\infty$ (if $f(z)$ has an even number of odd-order zeros) and one essential singularity of a double-valued character (if the number of such zeros is odd or infinitely large). 2) If $f(z)$ has zeros $a_1, a_2, \dots$, then the Riemann surface of $\text{Ln } f(z)$ has one logarithmic branch-point and no other points above each of the points $a_1, a_2, \dots$. If $f(z)$ has no zeros, then $\text{Ln } f(z)$ decomposes into infinitely many entire functions which differ from one another by summands of the form $2\pi ki$ ($k=0, \pm 1, \pm 2, \dots$). 3) If $f(z)$ has zeros, then the Riemann surface of $[f(z)]^a$ is the same as that of $\text{Ln } f(z)$ [see item (2)]. If $f(z)$ has no zeros, then $[f(z)]^a$ decomposes into infinitely many entire functions, which differ from one another by factors of the form $e^{2\pi k a i}$ ($k=0, \pm 1, \pm 2, \dots$). 1218. A double-sheeted

circle $|z| < 1$ with algebraic branch-points at zeros of the function $\sum_{n=1}^{\infty} z^{n!}$, in

particular, at the point $z=0$. 1219. 1) A double-sheeted circle $|z| < 1$ with a sole branch-point at $z=0$ (the part of the Riemann surface of the function $\sqrt{z}$ situated above the circle $|z| < 1$); 2) the part of the Riemann surface of the function $\text{Ln } z$ situated above the circle $|z| < 1$; 3) the part of the Riemann surface of the function $\text{Ln } z$ situated above the annulus $\frac{1}{2} < |z| < 2$.

Chapter IX

1222. 1) At the vertex A_2 , $\alpha_2=0$; 2) at the vertices A_1 and A_2 , $\alpha_1=\alpha_2=0$; 3) at the vertex A_3 , $\alpha_3=0$; 4) at the vertices A_2 and A_4 , $\alpha_2=\alpha_4=-1$; 5) at the vertex A_2 , $\alpha_2=-2$; at the vertex A_4 , $\alpha_4=0$; 6) at the vertices A_2 , A_3 and A_4 , $\alpha_2=\alpha_3=\alpha_4=0$; 7) at the vertex A_2 , $\alpha_2=-2$; at the vertices A_4 and A_6 , $\alpha_4=\alpha_6=0$; 8) at the vertex A_2 , $\alpha_2=-2$; at the vertex A_4 , $\alpha_4=\alpha-2$.

1224. It is necessary and sufficient that $\alpha_k = \frac{1}{n_k}$ (n_k natural numbers or ∞) and

$\sum_{k=1}^n \left(1 - \frac{1}{n_k}\right) = 2$, which is possible only for $n=4$ with $n_1=n_2=n_3=n_4=2$ (i.e. for a rectangle) and for $n=3$:

n_1	n_2	n_3	P
1	∞	∞	Strip
2	2		Semi-strip
2	3	6	Right triangle
2	4	4	Isosceles right triangle
3	3	3	Equilateral triangle

$$\frac{\pi(w-a)}{h}$$

1225. 1) $w = \frac{h}{\pi} \ln z + a$, $z = e^{\frac{\pi(w-a)}{h}}$ (a a real parameter), $z(w)$ is a periodic function with period $\omega = 2hi$; the group G is generated by the transformation $T(w) = w + \omega$: its fundamental domain B consists of a doubled strip and one

of its boundary sides; 2) $w = \frac{h}{\pi} \ln \frac{1+z}{1-z} + a$, $z = \tanh \frac{\pi(w-a)}{2h}$ (a a real parameter); $z(w)$ is a periodic function with period $\omega = 2hi$; the group G and its fundamental domain B are the same as in item (1). 1226. $w = \arcsin z$, $z = \sin w$; $z(w)$ is a periodic function with period $\omega = 2\pi$; the group G is generated by the transformations $T(w) = w + \omega$, $S(w) = -w$; its fundamental domain B consists of a strip $0 < u < \pi$ and boundary half-lines $u=0$, $u=\pi$, $v \leq 0$.

$$1227^1) 1) w = C \int_0^z z^{-\frac{5}{6}} (1-z)^{-\frac{1}{2}} dz, \text{ where } C = \frac{\omega}{B\left(\frac{1}{6}, \frac{1}{2}\right)} \quad \left(B(p, q) = \int_0^1 x^{p-1} (1-x)^{q-1} dx \text{ is Euler's integral of the first kind} \right); z(w) \text{ is a doubly}$$

periodic function with periods 2ω and $2\omega e^{\frac{2\pi i}{6}}$; the group G is generated by

$$\text{the transformations } T(w) = w + 2\omega, S(w) = we^{\frac{2\pi i}{6}}; \text{ its fundamental domain } B \text{ consists of a doubled triangle and two dissimilar boundary sides; 2) } w = C \int_0^z z^{-\frac{3}{4}} (1-z)^{-\frac{1}{2}} dz, \text{ where } C = \frac{\omega}{B\left(\frac{1}{4}, \frac{1}{2}\right)}; z(w) \text{ is a doubly periodic}$$

function with periods 2ω and $2\omega i$; the group G is generated by the transformations $T(w) = w + 2\omega$, $S(w) = iw$; its fundamental domain B consists of a square with side ω and two boundary sides of one of the triangles making up

$$\text{the square; 3) } w = C \int_0^z z^{-\frac{2}{3}} (1-z)^{-\frac{2}{3}} dz, \text{ where } C = \frac{\omega}{B\left(\frac{1}{3}, \frac{1}{3}\right)}; z(w) \text{ is a}$$

doubly periodic function with periods $2hi$ and $2he^{\frac{\pi i}{6}}$, where $h = \frac{\omega\sqrt{3}}{2}$; the

group G is generated by the transformations $T(w) = w + 2hi$, $S(w) = we^{\frac{\pi i}{3}}$; its fundamental domain B consists of a doubled triangle and two dissimilar boundary sides. 1228. 1) A "triangle" with two vertices at points $w=0$, $w=d = B(\alpha, \beta)$ and angles $\pi\alpha$, $\pi\beta$ at these vertices. If $\alpha + \beta < 1$, then the third vertex is finite; if $\alpha + \beta \geq 1$, then the third vertex lies at infinity, if $\alpha + \beta = 1$,

then $d = \frac{\pi}{\sin \pi\alpha}$, and the "triangle" has the shape of an oblique semi-strip,

provided $\alpha \neq \beta$; in the case $\alpha + \beta = 2$ the sides of the "triangle" emanating from the vertices at the base are parallel and oppositely directed, and

$d = \frac{\pi(\alpha-1)}{\sin \pi(\alpha-1)}$; if $\alpha = \beta = \frac{3}{2}$, then the "triangle" represents the exterior of

a right semi-strip (Fig. 93). 2) A "triangle" with one finite vertex at the point $w=0$, with an angle $\pi\alpha$ and two vertices at infinity. Two sides of the "triangle" represent rays emanating from the origin, the third being a straight line at a

distance $h = \frac{\sin \pi\beta}{\beta} \frac{\Gamma(\alpha)\Gamma(\beta+1)}{\Gamma(\alpha+\beta)}$ from the origin. If $\alpha=1$ we have a strip of

width π ; in the case $\alpha + \beta = 1$ two sides are parallel and $h = \pi$; in the case $\alpha = 2$ we have a half-plane with a cut along the real positive axis and

¹⁾ Diagrams of fundamental domains are given in Fig. 62 (page 235)

$h = \frac{\sin \pi \beta}{\beta(\beta+1)}$, in particular, $h=4$ if $\beta = -\frac{1}{2}$, and $h=\pi$ if $\beta = -1$ (see Fig. 93). 1229. 1) See Fig. 94, 1; $w_1 = w(1) = -i\pi \left(\frac{1}{2} - \lambda \right)$; $w_\lambda = w(\lambda)$. 2) See Fig. 94, 2. $w_\lambda = w(\lambda)$. 1230. 1) $w = \frac{2}{\pi} [\arcsin \sqrt{z} - (1-2z) \sqrt{z-z^2}]$;

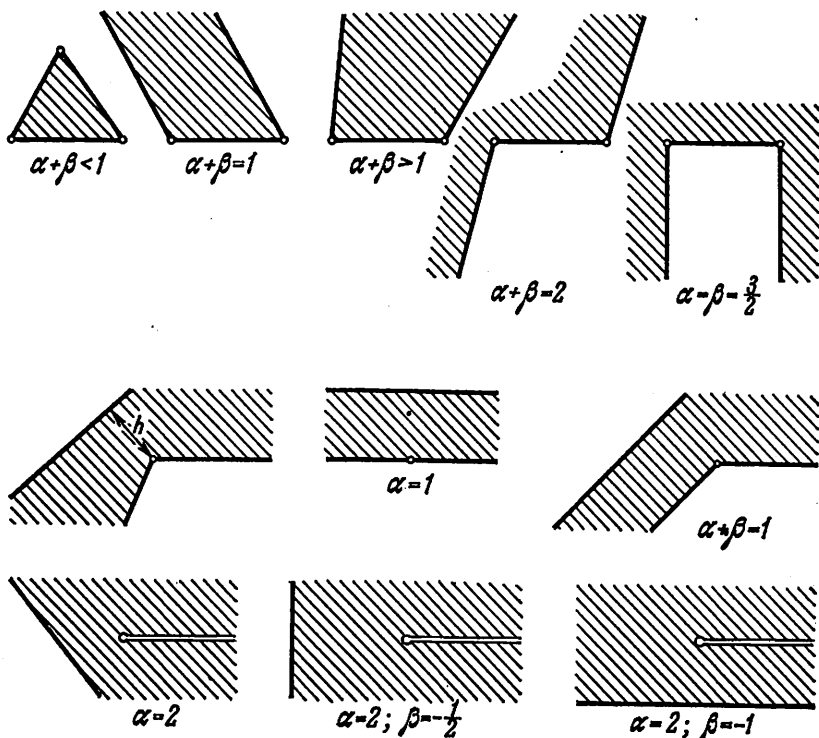


FIG. 93

$$\begin{aligned}
 2) \quad w &= \frac{2}{\pi} [\arcsin \sqrt{z} - \sqrt{z-z^2}]; & 3) \quad w &= \frac{h}{\pi} \left(\ln \frac{1+\sqrt{z}}{1-\sqrt{z}} - 2\sqrt{z} \right) = \\
 &= \frac{2h}{\pi} (\operatorname{artanh} \sqrt{z} - \sqrt{z}); & 4) \quad w &= \frac{2h}{\pi} (\arcsin \sqrt[4]{z} + \operatorname{artanh} \sqrt[4]{z} - 2\sqrt[4]{z}); \\
 5) \quad w &= ia \left(-\sqrt{z} \frac{z-3}{2} - 1 \right) & 1231. \quad 1) \quad w &= -\frac{a}{\pi} \int_1^z \frac{dz}{z^{1-\beta} (z-1)^\beta}. \quad \text{If } \theta = \frac{p}{q}, \\
 \text{then } w &= \frac{a}{\pi} \sum_{v=0}^{q-1} \frac{1}{t_v^p} \ln \left(1 - \frac{t}{t_v} \right), \quad \text{where } t = \left(\frac{z-1}{z} \right)^{\frac{1}{q}} \text{ and } t_v = e^{\frac{2v\pi i}{q}} \quad (v=0,
 \end{aligned}$$

1, ..., $q-1$). 2) $w = \frac{a}{\pi\theta} \int_1^z \left(\frac{z-1}{z}\right)^\theta dz$. If $\theta = \frac{p}{q}$, then $w = \frac{a}{\pi} \left[-\frac{t^p}{\theta(t^q-1)} + \sum_{v=0}^{q-1} t_v^p \ln \left(1 - \frac{t}{t_v}\right) \right]$, where t and t_v have the same values as in item (1).

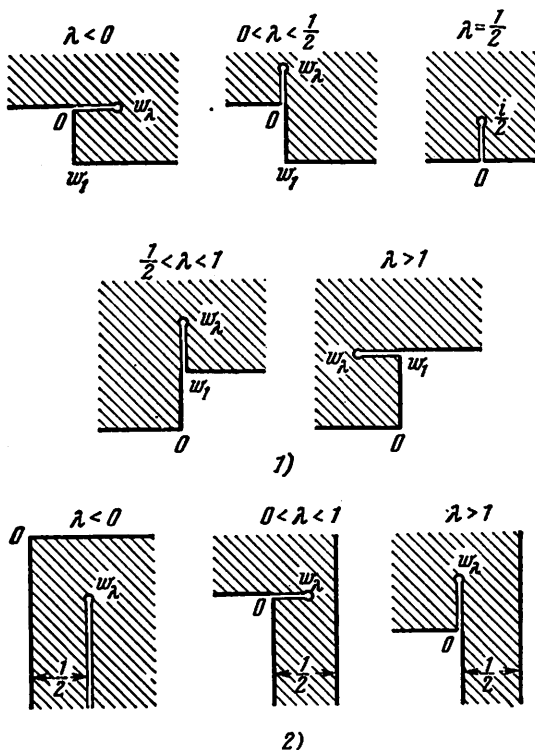


FIG. 94

1232. $w = \frac{a}{\xi}$, where $\xi = \frac{1}{\pi} \left(\frac{2t}{1-t^2} + \ln \frac{1-t}{1+t} \right)$, $t = \sqrt{\frac{z-1}{z}}$.

1233. $w = \frac{2h}{\pi} \left(\sqrt{z^2-1} + \arcsin \frac{1}{z} \right)$. 1234. $w = \frac{2d}{B \left(\frac{\alpha}{2}, 1-\alpha \right)} \int_0^z t^{\alpha-1} (1-t^2)^{-\alpha} dt$.

1235. 1) $w = \frac{2h}{\pi} \left(H \arctan \sqrt{\frac{z}{a-z}} + h \operatorname{arctanh} \frac{h}{H} \sqrt{\frac{z}{a-z}} \right)$, where $a = 1 + \frac{h^2}{H^2}$; 2) $w = \frac{2}{\pi} \left(H \arctan \frac{z}{\sqrt{a^2-z^2}} + h \operatorname{arctanh} \frac{h}{H} \frac{z}{\sqrt{a^2-z^2}} \right)$, where

$a^2 = 1 + \frac{h^2}{H^2}$; 3) $w = C \int_0^z \frac{\sqrt{t} dt}{(t-1)(t-a)(t+b)}$, where C, a, b are determined

from the equations $\frac{C\pi}{(a-1)(a+b)} = h_1$, $\frac{C\pi\sqrt{a}}{(a-1)(a+b)} = h_2$, $\frac{C\pi\sqrt{b}}{(b+1)(a+b)} = h_3$.

$$1236. \quad 1) \quad w = \frac{n}{B\left(\frac{1}{n}, 1 - \frac{2}{n}\right)} \int_0^z (1-t^n)^{-\frac{2}{n}} dt; \quad 2) \quad w = C \int_z^1 \frac{(1-t^n)^{\frac{2}{n}}}{t^2} dt + 1,$$

$$c_{-1} = C = \frac{2^{-\frac{2}{n}} n \sin \frac{\pi}{n}}{B\left(\frac{1}{n} + \frac{1}{2}, \frac{1}{2}\right)}. \quad 1237. \quad 1) \quad w = \frac{5 \times 2^{\frac{2}{5}}}{B\left(\frac{1}{10}, \frac{1}{5}\right)} \int_0^z (1-t^5)^{-\frac{4}{5}} (1+t^5)^{\frac{2}{5}} dt;$$

$$2) \quad w = C \int_z^1 \frac{(1-t^5)^{\frac{4}{5}} (1+t^5)^{-\frac{2}{5}}}{t^2} dt + 1, \quad c_{-1} = C = \frac{2^{-\frac{2}{5}} \Gamma\left(\frac{7}{10}\right)}{\Gamma\left(\frac{9}{10}\right) \Gamma\left(\frac{4}{5}\right)}.$$

1238. Onto a many-pointed star with angles $\pi - \frac{2\pi}{n} - \lambda\pi$ and $\pi + \lambda\pi$ alternately, with centre at the origin and with one of the vertices of the angles of the first kind at the point

$$w(1) = \frac{2^{-\frac{2}{n}} \Gamma\left(\frac{1-\lambda}{2} - \frac{1}{n}\right) \Gamma\left(\frac{1+\lambda}{2}\right) \sin \frac{\pi(1+\lambda)}{2}}{n \sin \frac{\pi}{n}}.$$

$$1239. \quad w = CC_0 \int_i^z \left\{ z \prod_{k=1}^{m-1} \left(z^2 - \tan^2 \frac{\pi k}{n} \right) \right\}^{-\frac{2}{n}} dz \quad \text{if } n = 2m, \quad \text{and}$$

$$w = CC_1 \int_i^z \left\{ z \prod_{k=1}^m \left(z^2 - \tan^2 \frac{\pi k}{n} \right) \right\}^{-\frac{2}{n}} dz \quad \text{if } n = 2m+1, \quad \text{where}$$

$$C = \frac{n}{B\left(\frac{1}{n}, 1 - \frac{2}{n}\right)}, \quad C_0 = -2i(2n)^{-\frac{2}{n}} e^{-i\frac{\pi}{n}} \quad \text{and} \quad C_1 = -2i \times 2^{-\frac{2}{n}}.$$

1240. The parameters are determined with the aid of equation (3) for b_k from the equalities $|f(b_k)| = l_k$ ($k=1, 2, \dots, n$), and from the direction of one of the sides of the star. One value of a_k is chosen arbitrarily.

$$1241. \quad w = \frac{C(z-1)^{2\alpha}(z+1)^{2\alpha-2}}{z}, \quad C = \frac{h}{4} \alpha^{-2\alpha} (1-\alpha)^{2\alpha-2}. \quad 1243. \quad \text{The parameters are determined from the values of } |f(b_i)|, |f(d_s)| \text{ taken from the stipulated value of } P \text{ and from the direction of one of the sides of } P. \text{ Three of the parameters } (a_k, b_i, c_j, d_s) \text{ are chosen arbitrarily. If one of the parameters } a_k \text{ or } c_j \text{ is equal to } \infty, \text{ then (4) and (5) hold, provided we discard the factor and summand with this parameter. If one of the parameters } b_i \text{ or } d_s \text{ is equal to } \infty, \text{ then (4) and (5) hold true without change.}$$

$$1245. \quad w = C(z+1)^{\alpha_1} (z-1)^{\alpha_2}, \quad C = h(b+1)^{-\alpha_1} (1-b)^{-\alpha_2}, \quad b = \frac{\alpha_1 - \alpha_2}{\alpha_1 + \alpha_2}. \quad 1246. \quad w = C \sqrt{\frac{z^2 - a^2}{1 - a^2 z^2}}, \quad C = \sqrt{hH},$$

$$a = \sqrt{\frac{h}{H}}. \quad 1247. \quad w = Cz^{1-2\alpha} (z^2 - 1)^{\alpha}, \quad C = hb^{2\alpha-1} (1-b^2)^{-\alpha}, \quad b = \sqrt{1-2\alpha}.$$

$$1248. \quad w = \sqrt{hH} \left(\frac{z+a}{1+az} \right)^{\alpha_1} \left(\frac{z-b}{1-bz} \right)^{\alpha_2}, \quad \text{where } a \text{ and } b \text{ are determined from the}$$

system of equations $a^{\alpha_1} b^{\alpha_2} = \sqrt{\frac{h}{H}}$, $\alpha_1 \left(\frac{1}{a} - a \right) = \alpha_2 \left(\frac{1}{b} - b \right)$.

1249. 1) $w = [T_n(z)]^{\frac{1}{n}}$, where $T_n(z) = \frac{1}{2} [(z + \sqrt{z^2 - 1})^n + (z - \sqrt{z^2 - 1})^n]$ are

Chebyshev's polynomials; 2) $w = \left[\frac{T'_n(z) \sqrt{z^2 - 1}}{n} \right]^{\frac{1}{n}}$. 1250. The parameters

are determined from the values of $\operatorname{Re} f(b_i)$ and $\operatorname{Re} f(d_s)$ taken from the stipulated value of P and from the position of one of the sides of P . Three of the parameters (a_k, b_i, c_j, d_s) are chosen arbitrarily. If one of the parameters a_k or c_j is equal to ∞ , then (6) and (7) hold, provided the appropriate summands are discarded. If one of the parameters b_i or d_s is equal to ∞ , then (6) and (7) hold true without change (see the answer to Problem 1243). 1251. The parameters are determined from the values of $\operatorname{Re} f(b_i)$, $\operatorname{Re} f(d_s)$. Three of the parameters a_k, b_i, c_j, d_s are chosen arbitrarily. In formula (10) two of the parameters a_k, b_i, c_j, d_s are chosen arbitrarily. 1252. The parameters are determined from the values of $\operatorname{Re} f(b_i)$ and from the position of one of the sides of P . Two parameters are chosen arbitrarily. 1253. The parameters are determined from the values of $\operatorname{Re} f(b_i)$ and $\operatorname{Re} f(d_s)$ and from the position of one of the sides of P . Three parameters are chosen arbitrarily. In formula (11) two parameters corresponding to the vertices are chosen arbitrarily. 1254. 1) $w = \frac{h_1}{\pi} \ln(z+1) +$

$$+ \frac{h_2}{\pi} \ln(z-1) + C, \quad C = -\frac{h_1}{\pi} \ln(b+1) - \frac{h_2}{\pi} \ln(1-b), \quad b = \frac{h_1 - h_2}{h_1 + h_2};$$

$$2) \quad w = \frac{h_1}{\pi} \ln(z^2 - 1) + \frac{h_2}{\pi} \ln z + C, \quad C = -\frac{h_1}{\pi} \ln(1 - b^2) - \frac{h_2}{\pi} \ln b, \quad b =$$

$$= \sqrt{\frac{h_2}{2h_1 + h_2}}; \quad 3) \quad w = \frac{h_1}{\pi} \ln \frac{z + a_1}{1 + a_1 z} + \frac{h_2}{\pi} \ln \frac{z - a_2}{1 - a_2 z}. \quad \text{The parameters } a_1, a_2$$

are determined from the system of equations $a_1 h_1 a_2 h_2 = e^{-\pi}$, $h_1 \left(\frac{1}{a_1} - a_1 \right) =$

$$= h_2 \left(\frac{1}{a_2} - a_2 \right); \quad 4) \quad w = f \left[\frac{1}{2} \left(z + \frac{1}{z} \right) \right], \quad \text{where } f(z) \text{ is the mapping of}$$

item 3; 5) $w = \frac{h}{\pi} \ln T_n(z)$, where $T_n(z)$ are Chebyshev's polynomials (see the

answer to Problem 1249); 6) $w = \frac{h}{\pi} [\ln(z-1) + z]$; 7) $w = \ln \frac{z+1}{1-z} + Az$,

$A = \frac{2}{b_2 - 1}$, where b is determined from the equation $\ln \frac{b+1}{b-1} + \frac{2b}{b^2 - 1} = d$;

8) $w = \ln(z+1) - Az^2 - z + \text{const}$, where $A = \frac{1}{2a}$ and a is determined from the

equation $\ln a + \frac{1}{2} \left(a - \frac{1}{a} \right) + d = 0$. In particular, if $d = 0$, then $a = 1$;

9) $w = \frac{h}{\pi} \ln(z-a) + \frac{C}{z-a} + Az + \text{const}$, where $A = \frac{h}{2\pi a}$, $C = \frac{h}{2\pi a} (1 - a^2)$ and

a is determined from the equation $\ln \frac{1-a}{1+a} + \frac{2}{a} = \frac{\pi d}{h}$. In particular, if $h = 0$,

then $a = 0$, $A = C = \frac{d}{4}$.

1255. The correspondence between the planes u, z and φ is shown in Fig. 95.

The affixes of the points C and C' in the φ -plane are equal to $\pm \frac{\pi}{2} + i \ln \frac{1+k'}{k}$.

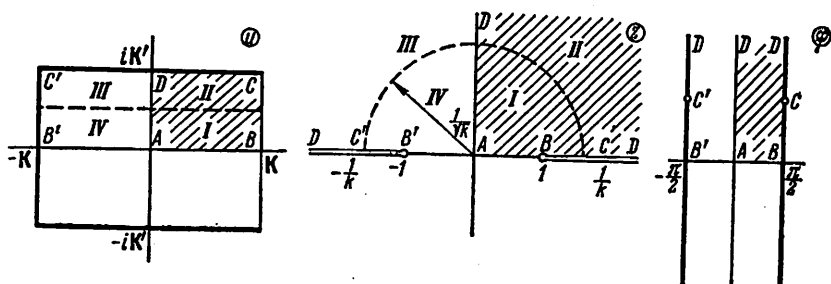


FIG. 95

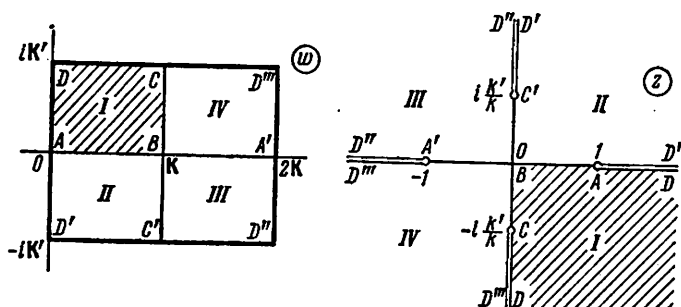


FIG. 96

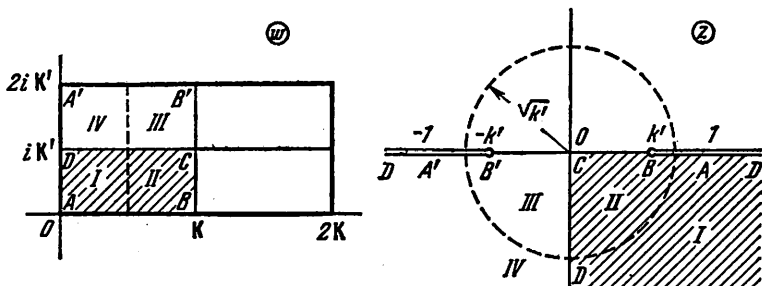


FIG. 97

The dashed line in the u -plane corresponds to the dashed semicircle of radius $1/\sqrt{k}$ in the z -plane.

$$1256. \quad \omega = \lambda \int_0^z \frac{dz}{V(1-z^2)(1-k^2z^2)}, \quad z = \operatorname{sn} \left(\frac{\omega}{\lambda}, k \right), \quad \text{where } \lambda \text{ and } k \text{ are}$$

determined from the relations $\frac{K'}{K} = \frac{b}{a}$ and $a = \lambda K$. 1257. See Fig. 96. 1258. See Fig. 97. 1259. The mapping of the z -plane onto the u -plane is shown in Fig. 98 (the v -plane in the same figure refers to the answer to Problem 1260).

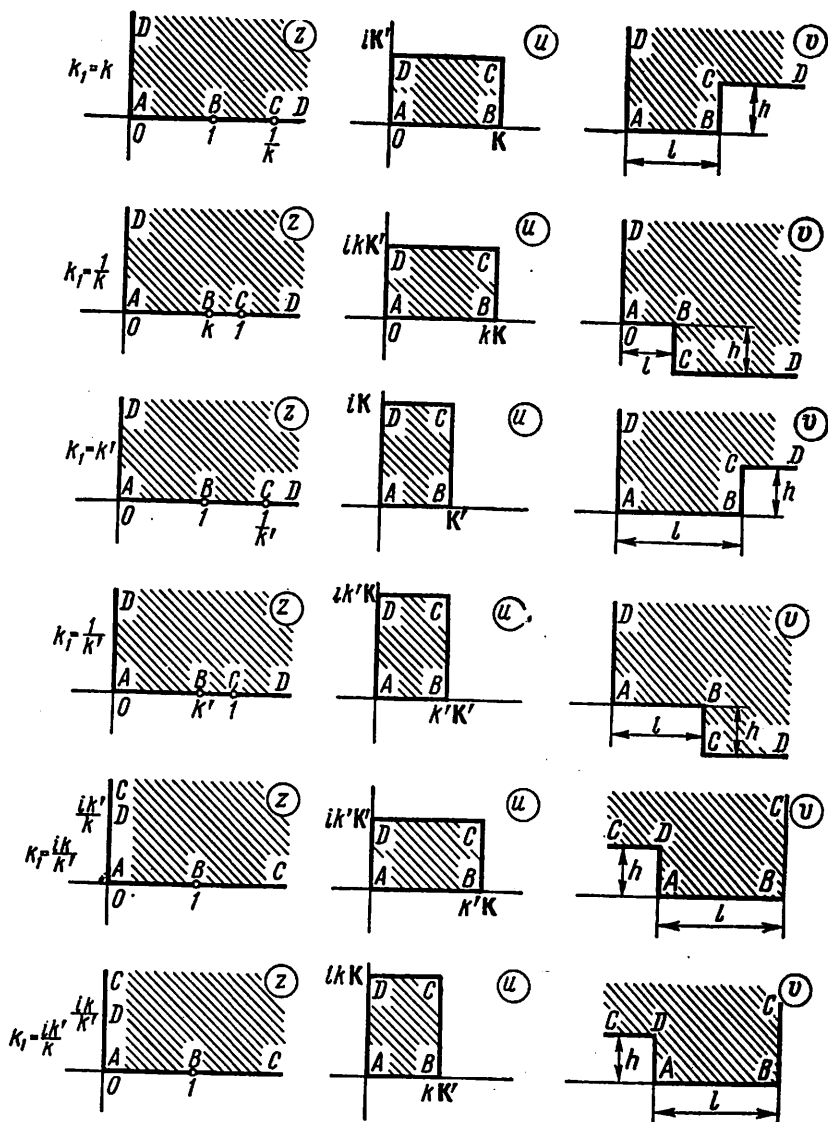


FIG. 98

1260. The correspondence between the z - and v -planes is shown in Fig. 98. The expressions for l and h are given in Table I (page 298). 1261. *Solution.* We have three equations for determining the parameters C_1 , k and b :

$$1) \omega(1) = a \text{ or } C_1 [(k^2 b^2 - 1)K + E] = a,$$

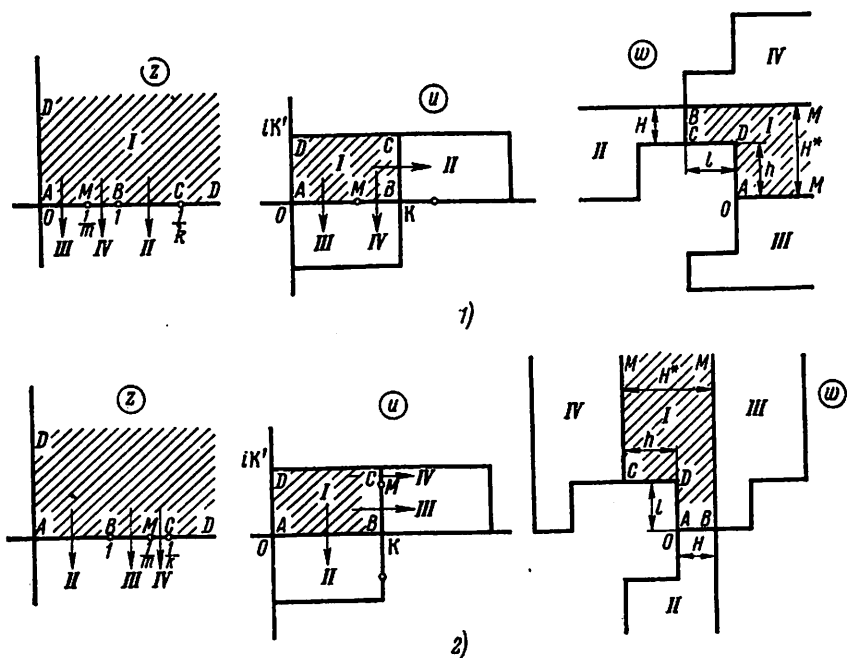


FIG. 99

2) $w(1) = w\left(\frac{1}{k}\right)$ or $(k^2b^2 - 1)K + E = (k^2b^2 - 1)(K + iK') + E + i(K' - E')$
(see the hint to Problem 1260).

3) $w(b) - w(1) = ih$.

From equation 2) we have $b = \frac{1}{k} \sqrt{\frac{E'}{K'}}$. Substituting in equation 1), we obtain $C_1 \left[\frac{E'K - KK' + EK'}{K'} \right] = a$, i.e. $C_1 = \frac{2aK'}{\pi}$ (see formula (10), page 180).

TABLE 1

k_1	l	h
k	E	$K' - E'$
$\frac{1}{k}$	$\frac{1}{k} (E - k'^2 K)$	$\frac{1}{k} (E' - k^2 K')$
k'	E'	$K - E$
$\frac{1}{k'}$	$\frac{1}{k'} (E' - k^2 K')$	$\frac{1}{k'} (E - k'^2 K)$
$\frac{ik}{k'}$	$\frac{1}{k'} E$	$\frac{1}{k'} (K' - E')$
$\frac{ik'}{k}$	$\frac{1}{k} E'$	$\frac{1}{k} (K - E)$

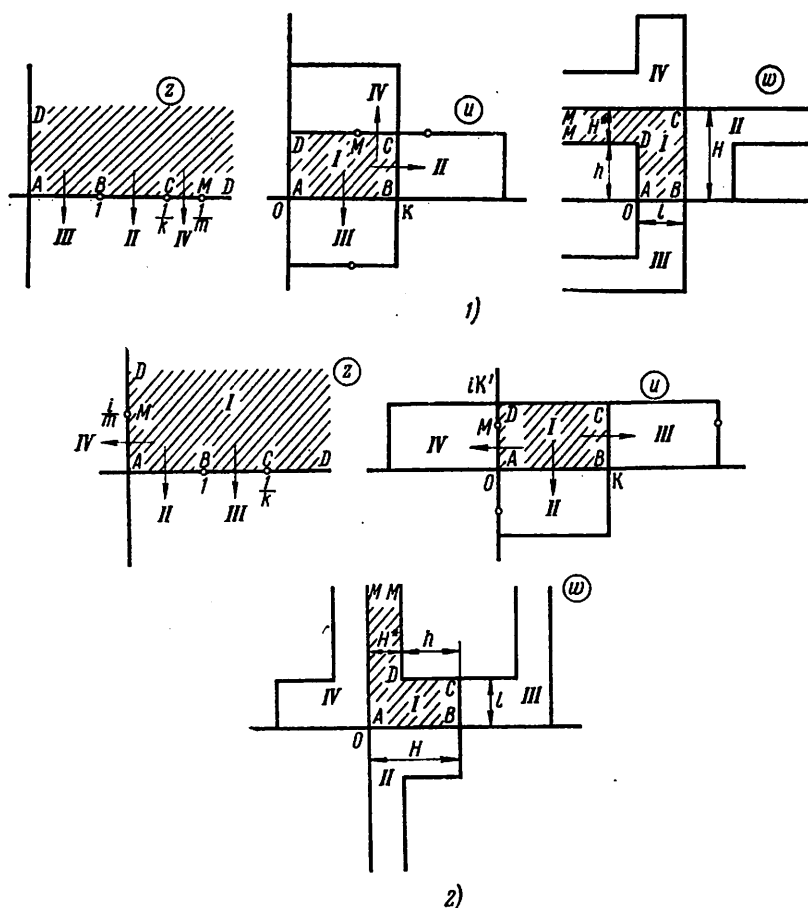


FIG. 100

Then for determining k we get from 3) the transcendental equation

$$(E' - K')u \left(\frac{1}{k} \sqrt{\frac{E'}{K'}}, k \right) + K'v \left(\frac{1}{k} \sqrt{\frac{E'}{K'}}, k \right) = \frac{\pi}{2} (a + ih). \quad \text{For more detail see A. Betz, Konforme Abbildung, Berlin, 1948.}$$

1263. The cases 1) and 2), 3) and 4) are illustrated in Figs. 99 and 100, respectively. For comparison, in all cases we give the mapping onto the u -plane with the aid of the normal elliptic integral of the first kind. Continuation of the mapping of the first quadrant (I) of the z -plane according to the symmetry principle leads in the w -plane to a strip with a rectangular recess (see domains I + II in Fig. 99, I, 2), to a strip with a rectangular projection (see domains I + II in Fig. 100, I, 2), and also to other domains (some of them are shown in the accompanying figures). The main dimensions involved (H , L , h) are given in Table 2. Note that the second case reduces to the first, and the fourth to the third by substituting t for z : $k^2 z^2 + k'^2 t^2 = 1$. In the cases 2) and 4), in

place of v , k there appear the values $v' = -\frac{v}{k^2+v} k'^2$, k' , and the corresponding w -figures are obtained from the w -figures for the cases 1) and 3) through entire linear transformations with the stretching factors $\frac{(\text{length})_2}{(\text{length})_1} = -\frac{k^2+v}{k^2}$, $\frac{(\text{length})_4}{(\text{length})_3} = \frac{k^2+v}{k^2}$ (the subscript indicates the case). For the cases $v = -1$ and $v = -k^2$ Fig. 101 gives the relations between the u - and w -planes. Using Table 2, we obtain for $v = -1$:

$$w = \frac{1}{k'^2} \left[k'^2 u - E(u) + \frac{\text{sn } u \text{ dn } u}{\text{cn } u} \right] = -\frac{i}{k'} E \left(ik' u, \frac{1}{k'} \right),$$

$$l = \frac{1}{k'^2} (E - k'^2 K), \quad h = \frac{1}{k'^2} (E' - k'^2 K')$$

and for $v = -k^2$:

$$w = \frac{1}{k'^2} \left[E(u) - k^2 \frac{\text{sn } u \text{ cn } u}{\text{dn } u} \right], \quad l = \frac{1}{k'^2} (K' - E'), \quad H = \frac{1}{k'^2} E.$$

1264. *Solution.* From the condition $\Delta > 0$ it follows that e_1, e_2, e_3 are real and distinct, and $g_2 > 0$. Assume $e_1 > e_2 > e_3$. The upper half-plane $\text{Im } z > 0$ is mapped onto a rectangle with vertices $0, \omega, \omega - \omega', -\omega'$ (it is assumed that

TABLE 2

v	H^*	H	l	h
$v < -1$	$\frac{\pi}{2} \sqrt{\frac{ v }{ v+1 v+k^2 }}$	$-\frac{k^2}{k^2+v} \times \prod_1(v', k')$	$-\prod_1(v, k)$	$H^* - H$
$-1 < v < -k^2$	$\frac{\pi}{2} \sqrt{\frac{ v }{ v+1 v+k^2 }}$	$\prod_1(v, k)$	$\frac{k^2}{k^2+v} \times \prod_1(v', k')$	$H^* - H$
$-k^2 < v < 0$	$\frac{\pi}{2} \sqrt{\frac{ v }{ v+1 v+k^2 }}$	$\frac{k^2}{k^2+v} \times \prod_1(v', k')$	$\prod_1(v, k)$	$H - H^*$
$v > 0$	$\frac{\pi}{2} \sqrt{\frac{ v }{ v+1 v+k^2 }}$	$\prod_1(v, k)$	$\frac{k^2}{k^2+v} \times \prod_1(v', k')$	$H - H^*$
$v = -1$	—	—	$\frac{1}{k'^2} (E - k'^2 K)$	$\frac{1}{k'^2} (E' - k'^2 K')$
$v = -k^2$	—	$\frac{1}{k'^2} E$	$\frac{1}{k'^2} (K' - E')$	—

Note. In column " l " the integrals are taken in the sense of the principal value. The quantity $v' = -\frac{v}{k^2+v} k'^2$.

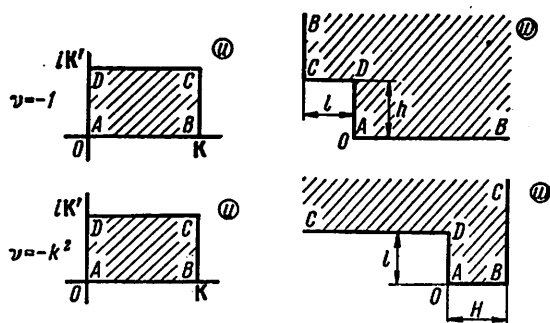


FIG. 101

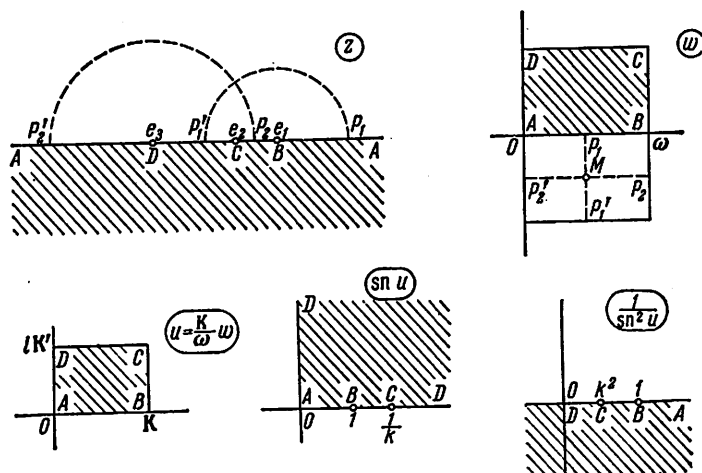


FIG. 102

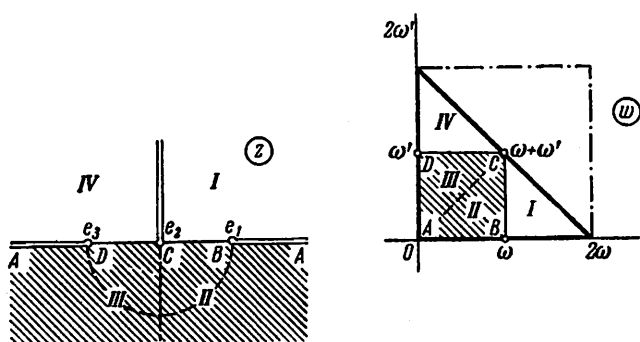


FIG. 103

$\operatorname{Im} \frac{\omega'}{\omega} > 0$) corresponding to the points ∞, e_1, e_2, e_3 . To the median lines of the rectangle there correspond two semicircles (Fig. 102): the first one with centre at the point e_3 , with respect to which e_2 and e_1 are symmetrical (hence, its radius is $\sqrt{(e_1 - e_3)(e_2 - e_3)}$), and the second one with centre at the point e_1 , with respect to which e_2 and e_3 are symmetrical (its radius is $\sqrt{(e_1 - e_2)(e_1 - e_3)}$). Continuing the mapping of $p(w)$ in accordance with the symmetry principle, we find the half-periods ω, ω' of this function:

$$\omega = \int_{e_1}^{\infty} \frac{dx}{\sqrt{(x - e_1)(x - e_2)(x - e_3)}},$$

$$\omega' = i \int_{-\infty}^{e_3} \frac{dx}{\sqrt{(e_1 - x)(e_2 - x)(e_3 - x)}}.$$

From the consideration of Fig. 102 we find

$$p(w) = e_3 + \frac{e_1 - e_3}{\operatorname{sn}^2\left(\frac{K}{\omega} w, k\right)}, \quad k^2 = \frac{e_2 - e_3}{e_1 - e_3}.$$

If $g_3 > 0$, then $e_3 < e_2 < 0 < e_1$ and $\omega < |\omega'|$ (since $k < k'$ and hence $K < K'$) if $g_3 < 0$, then $e_3 < 0 < e_2 < e_1$ and hence $\omega > |\omega'|$. If $g_3 = 0$, then $e_2 = 0$

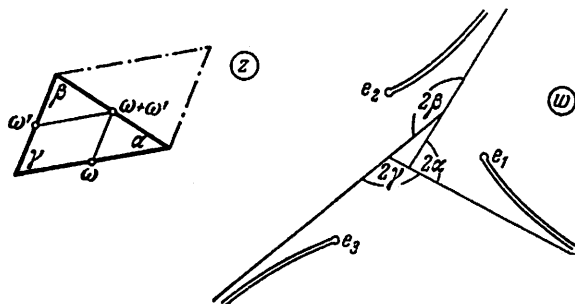


FIG. 104

$e_3 = -e_1$ and $\omega = |\omega'|$. In this case the mapping is symmetrical with respect to the vertical axis as well. To the entire z -plane with cuts $(-\infty, e_3], [e_1, \infty), [0, i\infty)$ there corresponds a triangle $(0, 2\omega, 2\omega')$ constituting half of the parallelogram of periods (this time it is a square) (Fig. 103). Also note that in case of arbitrary e_1, e_2, e_3 ($e_1 + e_2 + e_3 = 0$) to half of the parallelogram of periods there corresponds the z -plane with cuts (curvilinear in the general case) emanating from e_1, e_2, e_3 and going to ∞ (see the sketch in Fig. 104). 1265. *Solution.* The main mapping is shown in Fig. 105. It is obtained from the mapping of the semicircle II with the aid of the symmetry principle. Noting that

$$dw = - \frac{dz}{\sqrt{(z - e_1)(z - e_2)(z - e_3)}} \text{ and putting } \operatorname{Arg}(-1) = \pm \pi, \text{ we get } \operatorname{Arg} dw =$$

$$= \pm \pi + \operatorname{Arg} dz - \frac{1}{2} \sum_{k=1}^3 \operatorname{Arg}(z - e_k), \text{ whence it follows that } \operatorname{Arg} dw \text{ on the sides}$$



FIG. 105

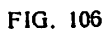


FIG. 106

of the "quadrilateral" $PBMC$ has values $-\frac{\pi}{2}$, 0 , $\frac{\pi}{2}$, $-\pi$, respectively, which leads to the mapping shown in Fig. 105. Thus, for instance, on the arc $\widetilde{PB}$ we have (Fig. 106) $\arg dw = -\pi + \left(\alpha_2 + \frac{\pi}{2}\right) - \frac{1}{2} \left(-\alpha_1 + \alpha_2 + \alpha_3\right) = -\frac{\pi}{2} + \frac{\alpha_1 + \alpha_2 - \alpha_3}{2} = -\frac{\pi}{2}$ and so on. For determining the complex conjugates of the half-periods ω and ω' of the function p we have

$$\omega + \omega' = \int_{e_1}^{\infty} \frac{dx}{\sqrt{(x-e_1)(x-e_2)(x-e_3)}},$$

$$\omega - \omega' = \frac{i}{2} \int_{-\infty}^{e_2} \frac{dx}{\sqrt{(e_1-x)(e_2-x)(e_3-x)}}.$$

If $g_2=0$, then e_1, e_2, e_3 are the vertices of an equilateral triangle, and the parallelogram of periods has the shape of a rhombus with an angle of 60° at zero if $g_3 < 0$ (then $e_2 < 0$), and with an angle of 120° at zero if $g_3 > 0$ (then $e_2 > 0$). In both cases half the parallelogram of periods corresponds to the z -plane with symmetrical radial cuts emanating from the points e_1, e_2, e_3 (see Figs.

107, 108). 1266. 1) $w = -\left[\frac{p(z)}{e_1}\right]^3$, half-periods $\omega, i\omega$; 2) $w = \frac{p'(z)}{2|e_2|^2}$, half-

periods $\omega = he^{-i\frac{\pi}{6}}$, $\omega' = he^{i\frac{\pi}{6}}$ where $h = \frac{a\sqrt{3}}{2}$; 3) $w = \frac{p'^2(z)}{4e_2^3}$, half-periods

are the same as in item (2). 1267. 1) $z = \operatorname{sn}(u, k)$, $w = e^{\frac{\pi u}{K'}}$, $\ln \mu = 2\pi \frac{K}{K'}$; $|\operatorname{Re} u| < K$, $|\operatorname{Im} u| < K'$; 2) the mapping function is the same as in item (1), but $0 < \operatorname{Re} u < K$, $|\operatorname{Im} u| < K'$, $\ln \mu = \pi \frac{K}{K'}$; 3) is reduced to the

case (1) with the aid of linear transformation; here $k = \frac{\sqrt{\lambda} - \sqrt{\lambda-1}}{\sqrt{\lambda} + \sqrt{\lambda-1}}$, where $\lambda = (a, b, c, d)$ is an anharmonic ratio of the indicated points. 4) is reduced to item (3) with the aid of the mapping $t = \sqrt{z^2 + h^2}$; 5) $z = \operatorname{sn}(u, k)$,

$w = ie^{\frac{i\pi u}{2K}}$, $\ln \mu = \frac{\pi}{2} \frac{K'}{K}$; $-3K < \operatorname{Re} u < K$, $0 < \operatorname{Im} u < K'$; 6) is reduced to

item (5) with the aid of the mapping $t = \sqrt{1+z^2}$; here $k = \cos \alpha$; 7) the mapping function is the same as in item (5), but one time $0 < \operatorname{Im} u < \frac{1}{2} K'$, the

other time $\frac{1}{2} K' < \operatorname{Im} u < K'$; $\ln \mu = \frac{\pi}{4} \frac{K'}{K}$; 8) $z = k \operatorname{sn}^2(u, k)$, $w = e^{-i\frac{\pi u}{K}}$,

$\ln \mu = \pi \frac{K'}{K}$; $|\operatorname{Re} u| < K$, $0 < \operatorname{Im} u < K'$; 9) the mapping function is the same

as in (8), but one time $0 < \operatorname{Im} u < \frac{1}{2} K'$, the other time $\frac{1}{2} K' < \operatorname{Im} u < K'$;

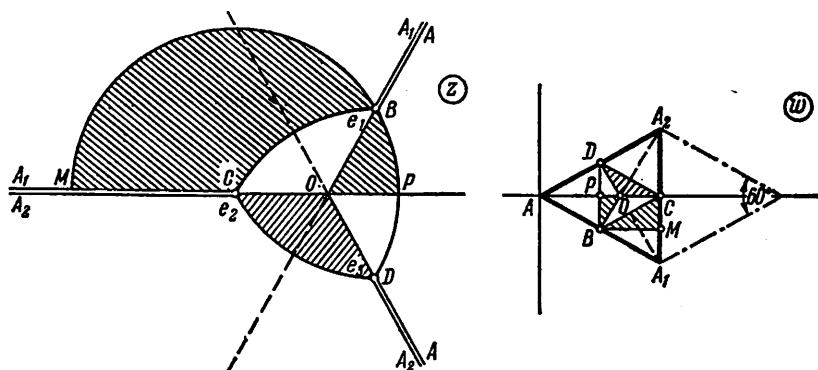


FIG. 107

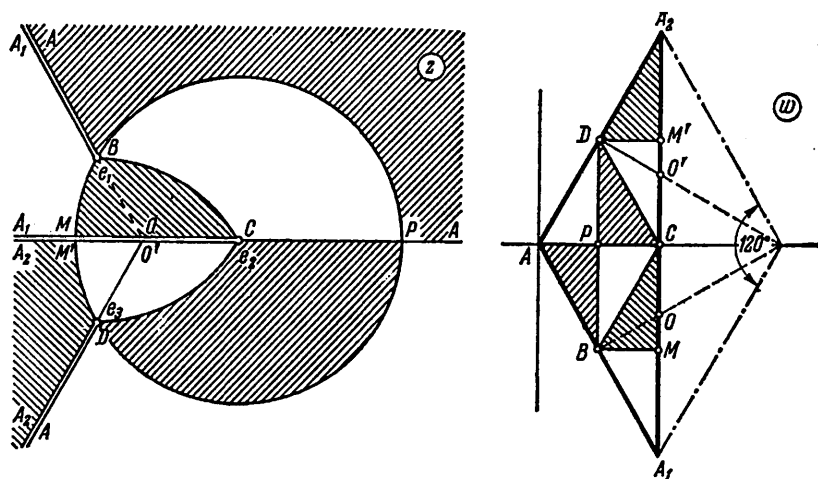


FIG. 108

$\ln \mu = \frac{\pi}{2} \frac{K'}{K}$; 10) is reduced to item (8) with the aid of the mapping $t = \frac{z+1}{\sqrt{1+\rho}}$; here $k = \frac{1}{\sqrt{1+\rho}}$; 11) is reduced to item (7) with the aid of the mapping $t = \frac{1-z}{1+z}$; here $k = \sin \frac{\alpha}{2}$; 12) is reduced to (2) with the aid of the mapping $t = -i \sin z$; here $k = \frac{\sinh H_1}{\sinh H_2}$; 13) is reduced to (7) with the aid of the mapping $t = \sin z$; here $k = \cos \tanh H$; 14) is reduced to (5) with the aid of the mapping $t = \sin z$ followed by a linear transformation; here $k = \frac{\sqrt{\lambda} - \sqrt{\lambda-1}}{\sqrt{\lambda} + \sqrt{\lambda-1}}$, where $\lambda = \frac{(1+\sin \beta)(1-\sin \alpha)}{2(\sin \beta - \sin \alpha)}$; 15) $z = C \left[Z(u) + \frac{dn u \operatorname{cn} u}{\operatorname{sn} u} \right]$,

$w = e^{-i\frac{u\pi}{K}}$, $\ln \mu = 2\pi \frac{K'}{K}$, where k is determined from the equation $Z(\beta) \tan \alpha = \frac{\pi}{2K}$, $\operatorname{dn}^2 \beta = \frac{E}{K}$. 1268. $t = \frac{1 - \operatorname{cn} u}{\operatorname{sn} u}$. Vertices $K \pm iK'$, $-K \pm iK'$ go into the points $e^{\pm i\alpha}$, $-e^{\pm i\alpha}$, where $\cos \alpha = k$. 1269. Is reduced to Problem 1268 with the aid of the mapping $u = \frac{2K}{\pi} \arcsin z$; here k is determined from the equation $q = e^{-\pi \frac{K'}{K}} = \frac{1}{(a+b)^2}$, where $b = \sqrt{a^2 - 1}$. 1270. $t = \frac{1 - \operatorname{dn} u}{k \operatorname{sn} u}$, where $u = \frac{2K}{\pi} \arcsin z$, $|\operatorname{Re} u| < K$, $|\operatorname{Im} u| < K'$. The parameter k has the same value as in the answer to Problem 1269. The foci $z = \pm 1$ go into the points $\pm \frac{k}{1+k'}$. 1271. 1) $z = C \left[Z(u) + \frac{\operatorname{dn} u \operatorname{cn} u}{\operatorname{sn} u} \right]$, $t = \frac{1 + \operatorname{cn} u}{\operatorname{sn} u}$; 2) $z = C \left[Z(u) + \frac{\pi u}{2KK'} \right]$, $t = \frac{1 + \operatorname{cn} u}{\operatorname{sn} u}$; 3) $z = \frac{a}{hk} \left[E(u) - i(E' - k^2 K') - k'^2 u + \frac{\operatorname{cn} u \operatorname{dn} u}{\operatorname{sn} u} \right]$. For determination of the constants see A. Betz, *Konforme Abbildung*, Berlin, 1948.

Chapter X

1272. Translational motion with a velocity $V = \alpha - i\beta$. At ∞ a doublet with moment $p = 2\pi c$. Streamlines $\beta x + \alpha y = C$; equipotential lines $\alpha x - \beta y = C$. 1273. At the point $z=0$ a critical point (branch-point); at ∞ a multiplet of order $2n$ (also a branch-point); $r^n \cos n\varphi = C$ are equipotential lines, $r^n \sin n\varphi = C$ are streamlines ($z = re^{i\varphi}$); $V = n\bar{z}^{n-1}$. 1274. At the points $z=0$ and $z=\infty$, vortex sources: $(0; Q, \Gamma)$, $(\infty; -Q, -\Gamma)$; $\ln r = -\frac{\Gamma}{Q} \varphi + C$ are equipotential lines; $\ln r = \frac{Q}{\Gamma} \varphi + C$ are streamlines. Both families of lines are logarithmic spirals; in the cases $\Gamma=0$ or $Q=0$ one of these families represents circles $r=C$, the other, rays $\varphi=C$. Velocity $V = \frac{|\Gamma + iQ|}{2\pi r} e^{i\left(\varphi - \arctan \frac{Q}{\Gamma} + \frac{\pi}{2}\right)}$ ($z = re^{i\varphi}$). 1275. At the points a, b , vortex sources: $(a; Q, \Gamma)$; $(b; -Q, \Gamma)$; the field lines are logarithmic spirals around the points a and b (Fig. 109); $\ln \rho = -\frac{\Gamma}{Q} \theta + C$ are equipotential lines; $\ln \rho = \frac{Q}{\Gamma} \theta + C$ are streamlines ($\frac{z-a}{z-b} = \rho e^{i\theta}$). Velocity $V = \frac{\Gamma - iQ}{-2\pi i} \frac{\bar{a} - \bar{b}}{(z-a)(\bar{z}-\bar{b})}$. 1276. At the point $z=0$, a doublet with moment $p = 2\pi$; $r = C \cos \varphi$ are equipotential lines; $r = C \sin \varphi$ are streamlines. $V = -\frac{e^{2i\varphi}}{r^2}$, $V_\infty = 0$. At the points $2 \pm i$ the velocity equals $-\frac{3 \pm 4i}{25}$. 1277. 1) and 2) At the points 0 and ∞ , doublets with moments $\pm 2\pi R^2$ and 2π (the upper sign refers to case 1), the lower to case 2). The field lines are curves of the 3rd order: $x \pm \frac{R^2 x}{x^2 + y^2} = C$ are equipotential lines, $y \mp \frac{Ry}{x^2 + y^2} = C$ are streamlines.

$V = 1 \mp \frac{R^2}{r^2} e^{2i\varphi}$, $V_\infty = 1$. The points $z = \pm R$ for the case 1) and $z = \pm Ri$ for the case 2) are critical points (see Figs. 110 and 111). 1278. At the point $z = 0$, a quadruplet, $r^2 = C \cos 2\varphi$ are equipotential lines; $r^2 = C \sin 2\varphi$ are streamlines. $V = -\frac{2}{r^3} e^{3i\varphi}$, $V_\infty = 0$ (Fig. 112¹⁾).

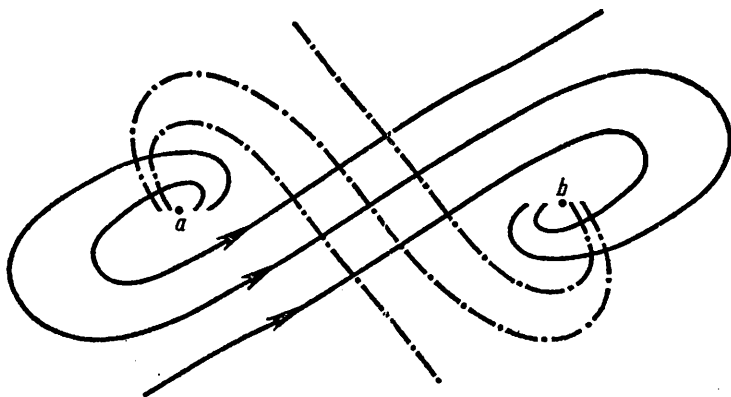


FIG. 109

1279. At the points $\pm a$, sources of strength 2π ; at ∞ , a source of strength -4π . The equipotential lines are lemniscates $|z^2 - a^2| = C$, the streamlines are hyperbolas $x^2 - \frac{2}{C}xy - y^2 = a^2$ centred at the origin and passing through the points $\pm a$.

$V = \frac{2\bar{z}}{z^2 - a^2}$, $V_{\pm ia} = \pm \frac{i}{a}$; the origin is a critical point (Fig. 113). 1280. At the points $\pm a$, $\pm ai$, sources: $(\pm a; 2\pi)$, $(\pm ai; -2\pi)$; $r^4 + Ca^2r^2 \cos 2\varphi + a^4 = 0$ ($|C| > 2$) are equipotential lines (including straight lines $y = \pm x$); $r^4 + Ca^2r^2 \sin 2\varphi - a^4 = 0$ are streamlines (including the axes of coordinates).

$V = \frac{4az^2}{z^4 - a^4}$, $V_\infty = 0$, $z = 0$ is a critical point (Fig. 114). 1281. At the points

$\pm 1, 0$ and at ∞ , sources: $(\pm 1; Q)$ $(0; -Q)$, $(\infty; -Q)$; $r^2 + \frac{1}{r^2} = C + 2 \cos 2\varphi$ ($C > 0$) are equipotential lines (for large values of C the lines are close to circles $r = \sqrt{C}$ and $r = \frac{1}{\sqrt{C}}$); $r = \sqrt{\frac{C + \tan \varphi}{C - \tan \varphi}}$ are streamlines (including

the axes of coordinates and the circle $r = 1$). $V = \frac{Q}{2\pi} \left[\frac{z^2 + 1}{z(z^2 - 1)} \right]$, $V_\infty = 0$. The points $\pm i$ are critical points (Fig. 115). 1282. At the points $\pm i, 0$, sources: $(\pm i; 2\pi)$, $(0; -4\pi)$; $Cr^4 - 2r^2 \cos 2\varphi - 1 = 0$ ($C > -1$) are equipotential lines (for $C = 0$, a hyperbola $y^2 - x^2 = \frac{1}{2}$); $r = \sqrt{C \sin 2\varphi - \cos 2\varphi}$ are streamlines

¹⁾ Here and elsewhere, where the flow is symmetrical with respect to the coordinate axes, it is shown only in the first quadrant.

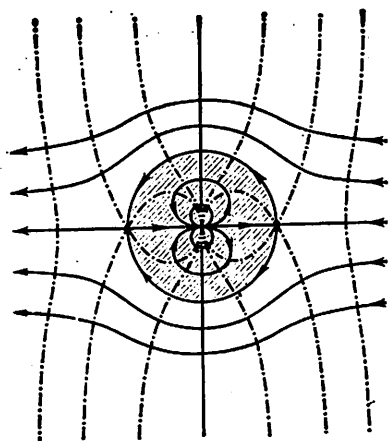


FIG. 110

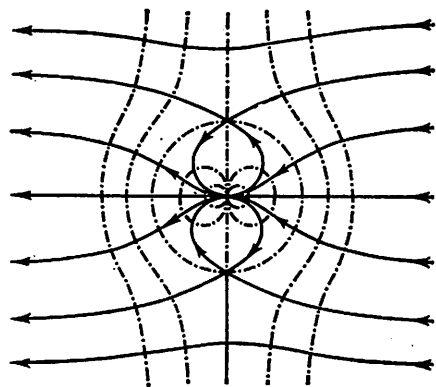


FIG. 111



FIG. 112

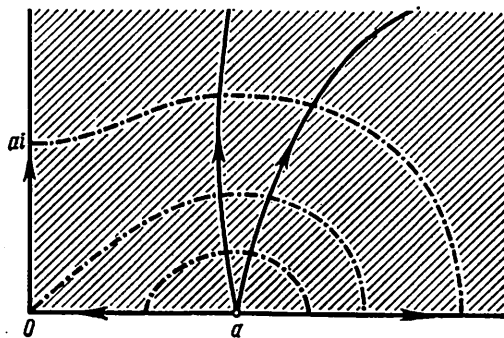


FIG. 113

(for $C=0$, Bernoulli's lemniscate), including the axes of coordinates. $V = -\frac{2}{z(z^2+1)}$; $V_\infty = 0$ (Fig. 116). 1283. At the points $\pm \frac{1 \pm i}{\sqrt{2}}$, sources of strength 2π , at the point $z=0$, a source of strength -4π ; at ∞ , a source of strength -4π ; $r^4 + \frac{1}{r^4} = C - 2 \cos 4\varphi$ ($C > 0$) are equipotential lines (for $C < 4$ the lines decompose

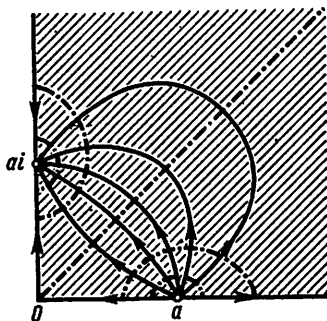


FIG. 114



FIG. 115

into four components, for $C > 4$, into two; for large values of C they are almost circles $r = \sqrt[4]{C}$ and $r = \frac{1}{\sqrt[4]{C}}$; $\tan 2\varphi = C \frac{r^4-1}{r^4+1}$ are streamlines (including the

coordinate axes, bisectors of coordinate angles and the circle $r=1$). $V = 2 \left[\frac{z^4-1}{z(z^4+1)} \right]$, $V_\infty = 0$. The points ± 1 , $\pm i$ are critical points (Fig. 117). 1284. At the point $z=0$, a source (0 ; Q), at the point $z=\infty$, a doublet and source (∞ ; $-Q$); $y^2 = e^{\frac{4\pi}{Q}(C-ax)} - x^2$ (in polar coordinates: $ar \cos \varphi + \frac{Q}{2\pi} \ln r = C$) are equi-

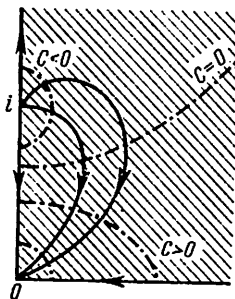


FIG. 116

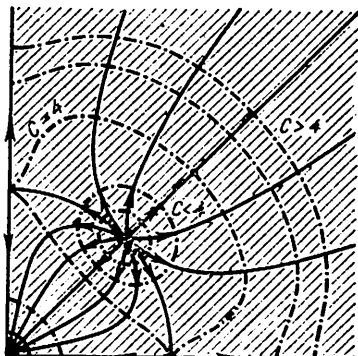


FIG. 117

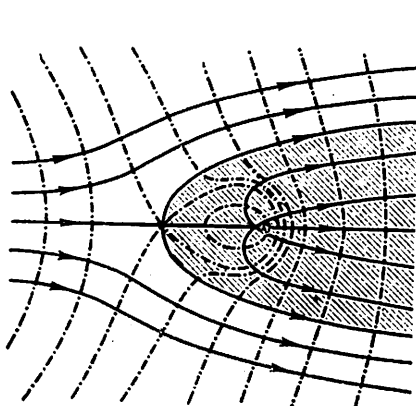


FIG. 118

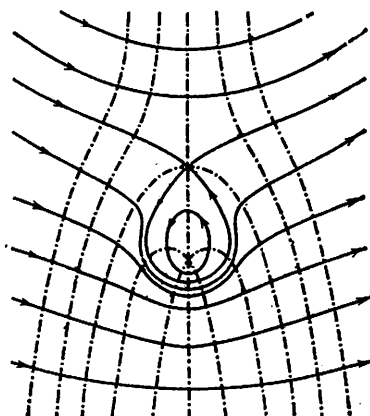


FIG. 119

potential lines; $r = \frac{C - \frac{Q}{2\pi} \varphi}{a \sin \varphi}$ are streamlines; the streamlines have horizontal

asymptotes: $y_{x \rightarrow +\infty} \rightarrow \frac{C}{a}$, $y_{x \rightarrow -\infty} \rightarrow \frac{C - \frac{Q}{2}}{a}$, $V = a + \frac{Qe^{i\varphi}}{2\pi r}$; $V_{\infty} = a$; the critical point $z = -\frac{Q}{2\pi a}$ (Fig. 118). 1285. At the point $z=0$, a vortex $(0; \Gamma)$;

at ∞ a doublet and a vortex $(\infty; -\Gamma)$; $r = \frac{C - \frac{\Gamma\varphi}{2\pi}}{a \cos \varphi}$ are equipotential lines;

$x^2 + y^2 = e^{\frac{4\pi}{\Gamma}(ay - C)} - y^2 \left(ar \sin \varphi - \frac{\Gamma \ln r}{2\pi} = C \right)$ are streamlines; $V = a + \frac{\Gamma}{2\pi r} e^{i\left(\varphi + \frac{\pi}{2}\right)}$.

$V_x = a$; the critical point $z = \frac{\Gamma i}{2\pi a}$ (Fig. 119). 1286. The fluid streamlines the circle of radius R ; $V_\infty = a$, circulation Γ ; the critical points are determined by the equality $z_{cr} = \frac{1}{4\pi a} (\Gamma i \pm \sqrt{16\pi^2 a^2 R^2 - \Gamma^2})$. If $\Gamma < 4\pi a R$, then $|z_{cr}| = R$, i.e. both critical points lie on the circle $|z| = R$; if $\Gamma = 4\pi a R$, then the critical points coincide; if $\Gamma > 4\pi a R$, then $|z_{cr}| > R$ (in this case the second critical point lies inside the circle $|z| = R$). 1287. $w(z) = V e^{-i\alpha} z +$

$+\sum_{k=1}^n \frac{\Gamma_k + iQ_k}{2\pi i} \ln(z - a_k)$. At ∞ , a doublet with moment $2\pi V e^{-i\alpha}$ and a vortex

source of strength $Q_\infty = \sum_{k=1}^n Q_k$ and intensity $\Gamma_\infty = -\sum_{k=1}^n \Gamma_k$. 1288. 1) No;

2) yes; 3) yes (for example, the flow $w = \frac{1}{z} + \frac{\Gamma}{2\pi i} \ln z$ has streamlines emanating

from the origin). 1289. Under a univalent conformal mapping a vortex source goes into a vortex source of the same strength and intensity. A multiplet is transformed into a set of multiplets up to the same order inclusive. A doublet goes into a doublet with the following law of variation of moment: 1) $(a; p) \rightarrow$

$\rightarrow (\alpha; p c_1)$, 2) $(\infty; p) \rightarrow (\alpha; p c_{-1})$, 3) $(a; p) \rightarrow \left(\infty; \frac{p}{c_{-1}}\right)$, 4) $(\infty; p) \rightarrow$

$\rightarrow \left(\infty; \frac{p}{c_1}\right)$. 1290. Under an n -valent conformal mapping a vortex source is

transformed into a vortex source with the strength and intensity reduced n times. 1291. The law of variation of the vortex source (a^* is a point symmetrical to the point a) is: $(a; Q, \Gamma) \rightarrow (a^*; Q, -\Gamma)$ in the case of a streamline, $(a; Q, \Gamma) \rightarrow (a^*; -Q, \Gamma)$ in the case of an equipotential line. The law of variation of a doublet is more complicated. In the case of a rectilinear streamline it is: $(a; p) \rightarrow (a^*; p')$, where the vectors p, p' , drawn through a and a^* , respectively, are symmetrical with respect to the streamline. In the case of a circular

streamline $|z| = R$: $(a; p) \rightarrow \left(a^*; -\frac{R^2}{a^2} \bar{p}\right)$ if $a \neq 0$, and $(a; p) \rightarrow \left(\infty; \frac{\bar{p}}{R^2}\right)$ if

$a = 0$. In the case of a rectilinear and a circular equipotential line, using the same notation; we have $(a; p) \rightarrow (a^*; -p')$, $(a; p) \rightarrow \left(a^*; \frac{R^2}{a^2} \bar{p}\right)$, $(0; p) \rightarrow$

$\rightarrow \left(\infty, -\frac{\bar{p}}{R^2}\right)$, respectively. 1292. 1) In all cases flow singularities must be

located symmetrically with respect to the circle $|z| = R$ (see Problem 1291). In particular, the axes of the doublets must be situated on this circle, they must be tangential to it. The sum of strengths must equal zero, for which

$\sum Q_k + \frac{1}{2} \sum Q'_i = 0$, where Q_k are the strengths of the sources inside $|z| = R$

and Q'_i , the strengths of the sources on $|z| = R$. There must be no vortices on $|z| = R$; 2) Flow singularities must be situated symmetrically with respect to the circle $|z| = R$. In particular, the axes of doublets situated on the circle $|z| = R$ must be orthogonal to it. The sum of intensities must equal zero, for

which $\sum \Gamma_k + \frac{1}{2} \sum \Gamma'_i = 0$, where Γ_k are intensities of the vortices inside

$|z| = R$ and Γ'_i , intensities of the vortices on $|z| = R$; there must be no sources on $|z| = R$. 1293. 1) $w = Vz + c$ ($c = \text{const}$); 2) $w = \frac{\Gamma}{2\pi i} \ln \frac{z-a}{z-\bar{a}} + c$; 3) $w = \frac{Q}{2\pi} \times$

- $\times \ln [(z-a)(z-\bar{a})] + c$ (at ∞ the source of strength $-2Q$); 4) $w = \frac{p}{2\pi} \frac{1}{z-a} + \frac{\bar{p}}{2\pi} \frac{1}{z-\bar{a}} + c$; 5) $w = \sum_{k=1}^n \left[\frac{\Gamma_k + iQ_k}{2\pi i} \ln(z-a_k) + \frac{-\Gamma_k + iQ_k}{2\pi i} \ln(z-\bar{a}_k) \right] + \frac{p}{2\pi} \frac{1}{z-a} + \frac{\bar{p}}{2\pi} \frac{1}{z-\bar{a}} + Vz + c$ (at ∞ the source of strength $-\sum_{k=1}^n Q_k$); 6) Flow is possible only when $\Gamma=0$, $\text{Im } p=0$; then $w = \frac{p}{2\pi z} + \frac{Q}{2\pi} \ln z + c$.
1294. 1) $w = \frac{\Gamma}{2\pi i} \ln \frac{z-a}{R^2-\bar{a}z} + c$; 2) $w = \frac{p}{2\pi} \frac{1}{z-a} + \frac{p^*}{2\pi} \frac{1}{z-a^*} + c$, if $a \neq 0$ ($a^* = \frac{R^2}{\bar{a}}$, $p^* = -\frac{R^2}{\bar{a}^2} \bar{p}$), and $w = \frac{p}{2\pi z} + \frac{p^*}{2\pi} z + c$, if $a=0$ ($p^* = \frac{\bar{p}}{R^2}$).
1295. 1) $w = \sum_{k=1}^n \frac{Q_k}{2\pi} \ln [(z-a_k)(R^2-\bar{a}_k z)] + c$, if $\sum_{k=1}^n Q_k = 0$; 2) $w = \sum_{k=1}^n \frac{Q_k}{2\pi} \times \ln [(z-a_k)(R^2-\bar{a}_k z)] + \sum_{i=1}^m \frac{Q'_i}{2\pi} \ln(z-a'_i) + c$ if $\sum_{k=1}^n Q_k + \frac{1}{2} \sum_{i=1}^m Q'_i = 0$.
1296. 1) $w = \frac{\Gamma}{2\pi i} \ln \frac{z-a}{R^2-\bar{a}z} + c$; 2) $w = \frac{p}{2\pi} \frac{1}{z-a} + \frac{p^*}{2\pi} \frac{1}{z-a^*} + c$ ($a^* = \frac{R^2}{\bar{a}}$, $p^* = -\frac{R^2}{\bar{a}^2} \bar{p}$); 3) $w = Ve^{-iz} + \frac{R^2 V e^{iz}}{z} + c$; 4) $w = Ve^{-iz} + \frac{R^2 V e^{iz}}{z} + \frac{\Gamma}{2\pi i} \ln z + c$.
1297. $w = \frac{\Gamma}{2\pi i} \ln \frac{(z-ia)(az+i)}{(z+ia)(az-i)} + c$. 1298. $w = \frac{Q}{2\pi} \ln(z^6 + a^6) + c$. 1299. $w = \frac{Q}{2\pi} \ln(z^2 - 1) + c$. 1300. 1) $w = \frac{Q}{2\pi} \ln \frac{z^2-1}{z^2+1} + c$; 2) $w = \frac{Q}{2\pi} \ln \left(1 + \frac{4}{z^4}\right) + c$.
1301. $w = \sum_{k=1}^n \left[\frac{\Gamma_k + iQ_k}{2\pi i} \ln(z-a_k) + \frac{\Gamma_k - iQ_k}{2\pi i} \ln(z-\bar{a}_k) \right] + \frac{p}{2\pi} \frac{1}{z-a} - \frac{\bar{p}}{2\pi} \times \frac{1}{z-\bar{a}} + Ve^{-iz} + c$. Flow is possible if $\alpha = \pm \frac{\pi}{2}$ (if $\sum_{k=1}^n \Gamma_k \neq 0$, then at ∞ the vortex has an intensity of $-2 \sum_{k=1}^n \Gamma_k$). 1302. $w = \sum_{k=1}^n \left[\frac{\Gamma_k + iQ_k}{2\pi i} \ln(z-a_k) + \frac{\Gamma_k - iQ_k}{2\pi i} \ln(R^2 - \bar{a}_k z) \right] + Ve^{-iz} - \frac{R^2 V e^{iz}}{z} + c$. Flow is possible if $\sum_{k=1}^n \Gamma_k = 0$, $a=0$, and $p = -2\pi R^2 V e^{iz}$.
1303. Let $t = f(z)$ map D conformally onto the unit circle $|t| < 1$. Then $w = \Phi[f(z)]$, where
- $$\Phi(t) = \sum_{k=1}^n \left[\frac{\Gamma_k + iQ_k}{2\pi i} \ln(t-t_k) + \frac{-\Gamma_k + iQ_k}{2\pi i} \ln(1-\bar{t}_k t) \right] + c, \quad t_k = f(a_k) \quad \text{with}$$
- the indispensable condition that $\sum_{k=1}^n Q_k = 0$. 1304. Let $t = f(z)$ map D conformally onto a domain $|t| > 1$ with normalization $f(\infty) = \infty$, $f'(\infty) > 0$. Then,

provided $\sum_{k=1}^n Q_k = 0$, we have $w = \Phi[f(z)]$, where

$$\Phi(t) = \frac{Ve^{-i\alpha}}{f'(\infty)} t + \frac{Ve^{i\alpha}}{f'(\infty) \times t} + \sum_{k=1}^n \left[\frac{\Gamma_k + iQ_k}{2\pi i} \ln(t - t_k) + \frac{-\Gamma_k + iQ_k}{2\pi i} \times \right. \\ \left. \times \ln(1 - \bar{t}_k t) \right] + c, \quad t_k = f(a_k).$$

1305. Using the notation of Problem 1304, $w = \Phi[f(z)]$, where

$$\Phi(t) = \frac{Ve^{-i\alpha}}{f'(\infty)} t + \frac{Ve^{i\alpha}}{f'(\infty) \times t} + \frac{\Gamma}{2\pi i} \ln t + c.$$

For $\Gamma = 0$, $w(z)$ maps the exterior of C onto the exterior of the segment $\left[-\frac{2V}{f'(\infty)}, \frac{2V}{f'(\infty)} \right]$ of the real axis of the w -plane with normalization $w(\infty) = \infty$,

$$w'(\infty) = Ve^{-i\alpha}. \quad 1306. w(z) = \frac{V}{a-b} [(az-b\sqrt{z^2-c^2}) \cos \alpha + i(bz-a\sqrt{z^2-c^2}) \times$$

$$\times \sin \alpha] + \text{const} \quad (V_\infty = Ve^{i\alpha}, c = \sqrt{a^2-b^2}); \quad 2) w(z) = \frac{V}{a-b} [(az-b\sqrt{z^2-c^2}) \times$$

$$\times \cos \alpha + i(bz-a\sqrt{z^2-c^2}) \sin \alpha] + \frac{\Gamma}{2\pi i} \ln(z + \sqrt{z^2-c^2}) + \text{const}. \quad 1307. 1) w(z) =$$

$$= V(z \cos \alpha - i\sqrt{z^2-c^2} \sin \alpha) + \text{const} \quad (V_\infty = Ve^{i\alpha}); \quad 2) w(z) = V(z \cos \alpha -$$

$$-i\sqrt{z^2-c^2} \sin \alpha) + \frac{\Gamma}{2\pi i} \ln(z + \sqrt{z^2-c^2}) + \text{const}, \text{ where } \Gamma = -2\pi c V \sin \alpha \quad (c$$

is the point of departure). 1308. Let Zhukovsky's airfoil be obtained under the mapping $z = \frac{1}{2} \left(\zeta + \frac{1}{\zeta} \right)$ of the circle $|\zeta - \zeta_0| = |1 - \zeta_0| = R > 1$, $\zeta_0 = 1 -$

$-Re^{-i\beta}$ ($0 \leq \beta < \frac{\pi}{2}$). Then with circulation Γ and $V_\infty = Ve^{i\alpha}$:

$$w(z) = \frac{VR}{2} \left(\frac{z - \zeta_0 + \sqrt{z^2-1}}{R} e^{-i\alpha} + \frac{Re^{i\alpha}}{z - \zeta_0 + \sqrt{z^2-1}} \right) + \frac{\Gamma}{2\pi i} \ln(z - \zeta_0 + \\ + \sqrt{z^2-1}) + c;$$

here $\Gamma = -2\pi RV \sin(\alpha + \beta)$ (Γ is determined from the condition $w'(1) = 0$ in accordance with the Zhukovsky-Chaplygin postulate). 1309. $w(z) = \sqrt{z - \frac{p}{2}} + c$

is streamlining of the parabola from the outside; $w(z) = i \cosh \frac{\sqrt{2z-p}}{2\sqrt{p}}$ is stream-

lining of the parabola from the inside. 1310. $w(z) = \frac{1}{\sqrt{2}} \left[(z + \sqrt{z^2-c^2})^{\frac{\pi}{2\beta}} \times \right.$

$\times e^{-\frac{i\pi\alpha}{2\beta}} - (z - \sqrt{z^2-c^2})^{\frac{\pi}{2\beta}} e^{\frac{i\pi\alpha}{2\beta}} \Big] + \text{const}$ is streamlining of the right-hand branch of the hyperbola from the outside ($\tan \alpha = \frac{b}{a}$, $\beta = \pi - \alpha$, $c = \sqrt{a^2-b^2}$)

and $w(z) = \frac{i}{\sqrt{2}} \left[(z + \sqrt{z^2-c^2})^{\frac{\pi}{2\alpha}} + (z - \sqrt{z^2-c^2})^{\frac{\pi}{2\alpha}} \right] + \text{const}$ is streamlining

of the right-hand branch of the hyperbola from the inside. 1311. $w(z)$ is determined

from the equation $z = e^{\frac{\pi w}{v}} + \frac{\pi w}{\eta}$ (the values of the stream function on the

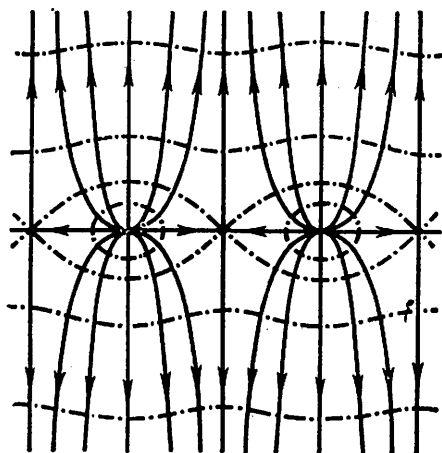


FIG. 120

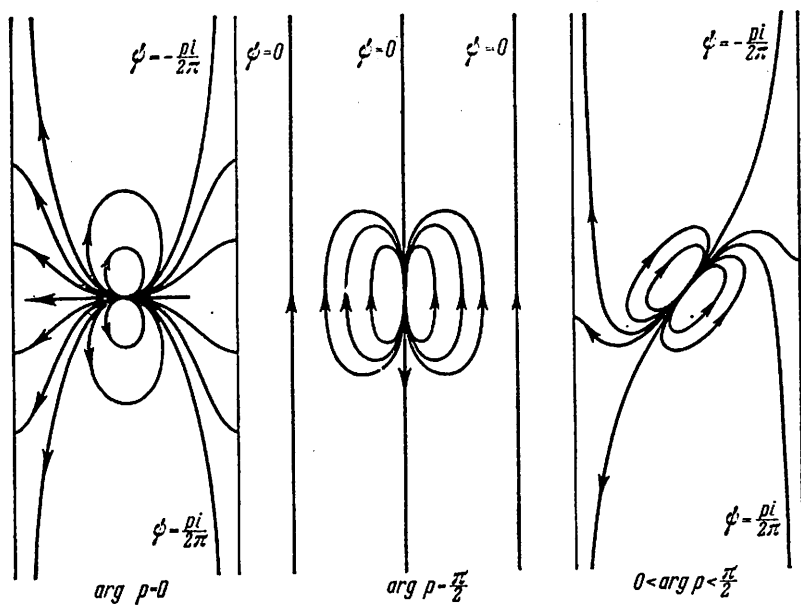


FIG. 121

streamlined half-lines are taken to be equal to $\pm v$). 1312. $\omega = \text{Arcosh } z = \text{Ln } (z + \sqrt{z^2 - 1})$ (the values of the stream function on the streamlined half-lines are taken to be equal to 0 and π). 1313. 1) A flow with a period π ; at points $k\pi$ (k an integer), sources of strength Q ; the points $\frac{\pi}{2} + k\pi$ are critical points. Ve-

locity at ∞ in the strip of periods $V_\infty = V(x \pm i\infty) = \pm \frac{Q}{2\pi} i$. The streamlines and equipotential lines are shown in Fig. 120, 2) The same, but in place of sources at points $k\pi$ there are vortices of intensity Γ and $V(x \pm i\infty) = \mp \frac{\Gamma}{2\pi}$.

To construct the field, the streamlines and the equipotential lines shown in Fig. 120 should be interchanged. 1314. A flow with period π ; at points $k\pi$, doublets with moment p ; velocity $V(x \pm i\infty) = 0$. The streamlines are shown in Fig. 121. 1315. A solution is possible for $V_1 = V - \frac{Q}{\omega}$;

$$\omega = \frac{\Gamma + iQ}{2\pi i} \ln \sin \frac{\pi(z-a)}{2\omega} + \frac{-\Gamma + iQ}{2\pi i} \ln \sin \frac{\pi(z+\bar{a})}{2\omega} + i \left(V - \frac{Q}{2\omega} \right) z + c.$$

1316. $\omega = \frac{p}{2\pi} \cot \frac{\pi(z-a)}{2\omega} - \frac{\bar{p}}{2\pi} \cot \frac{\pi(z+\bar{a})}{2\omega} + iVz + c$. 1317. Let $t = f(z)$ map S conformally onto a rectilinear strip S_t so that Ω_1 and Ω_2 go into points S_t at infinity. If there exist nonzero derivatives $f'(\Omega_1)$, $f'(\Omega_2)$, the velocities V_1 , V_2 are tangent to the boundary of S at infinity, and one of them is specified arbitrarily, then for S_t the problem reduces to Problems 1315—1316; there is a solution, which is unique. 1318. It is necessary and sufficient that M and C be real numbers. In this case, too, the lines $\text{Re } u = \pm \omega$ are equipotential lines. Fig. 122 shows the mapping of $t = f(u) = \zeta(u) + Cu$ for distinct real C . Fig. 122 corresponds to the case $\omega > |\omega'|$. According to the solution of Problem 1264, we have

$$e_3 < 0 < e_2 < e_1 \text{ and } k^2 > \frac{1}{2}. \quad 2) \quad f(u) = \frac{M}{4\pi\omega} \frac{\vartheta_1' \left(\frac{u-\alpha}{2\omega} \right)}{\vartheta_1 \left(\frac{u-\alpha}{2\omega} \right)} + c.$$

For $\alpha=0$ and $M=2\pi$ the mapping $t = f(u) = \zeta(u) - \frac{\eta}{\omega} u = \frac{1}{2\omega} \frac{\vartheta_1' \left(\frac{u}{2\omega} \right)}{\vartheta_1 \left(\frac{u}{2\omega} \right)}$

is shown in Fig. 123 (note that $e_3 < -\frac{\eta}{\omega} < e_2$). 1322. $f(u) = \frac{M}{2\pi} \zeta(u-\alpha) + \frac{N}{2\pi} \zeta(u-\beta) + Cu + c$. For the function $f(u)$ to be elliptical, it must be of the

form $f(u) = \frac{M}{2\pi} [\zeta(u-\alpha) - \zeta(u-\beta)] + c$. If $\text{Im } u = \pm \text{Im } \omega'$ are streamlines, then M must be real, and if they are equipotential lines, then it must be purely imaginary. For α and β , only the values 0, ω_k ($k=1, 2, 3$) are possible. For $\alpha=0$, $\beta=\omega_k$, $M=2\pi$

$$f(u) = \zeta(u) - \zeta(u-\omega_k) = \eta_k - \frac{1}{2} \frac{\wp'(u)}{\wp(u) - e_k} = \eta_k + \frac{\sigma_j(u) \sigma_j(u)}{\sigma(u) \sigma_k(u)}$$

(i, j, k permutations from 1, 2, 3). The points $u \equiv \frac{\omega_k}{2} \pmod{\omega, \omega'}$ are critical,

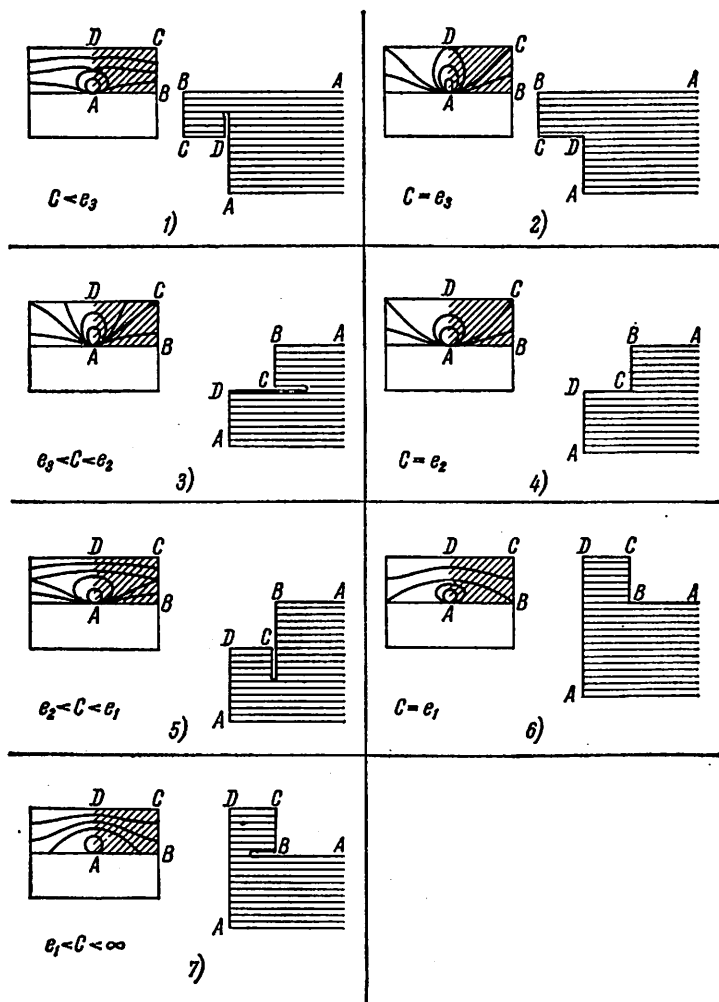


FIG. 122

i.e. at these points $f'(u)=0$. The main mappings are shown in Fig. 124. The rectangles depicted there are mapped onto a half-plane bounded by a horizontal straight line ($k=1$), a half-plane bounded by a vertical line ($k=2$), and onto a two-sheeted quadrant joined along a horizontal half-line corresponding to the dashed line on the rectangle ($k=3$). These mappings are continued according to the symmetry principle. 1323. The flow periods are $4K$ and $2iK'$, doublets $2mK + (2n+1)iK'$ with moments $2\pi \left(-\frac{1}{k}\right)^m$, the critical points are $(2m+1)K + niK'$ (m, n integers). The mapping is shown in Fig. 125. 1324. The flow periods are $4K$ and $2K + 2iK'$, the doublets are the same as in Problem 1323,

with moments $2\pi \frac{(-1)^{n+m}}{ik}$, the critical points are $2mK + 2niK'$ and $(2m+1)K + (2n+1)iK'$ (Fig. 126). 1325. The flow periods are $2K$ and $4iK'$, the doublets are the same as in Problem 1323, with moments $2\pi (-1)^{n+m}i$, the critical points are $mK + 2niK'$. The main mappings are shown in Fig. 127. 1326. $f(u) = \frac{\Gamma+iQ}{2\pi i} \ln \frac{\sigma(u-\alpha)}{\sigma(u-\beta)} + Cu + c$. In particular, for $\alpha=0$, ω , $\omega+\omega'$, and $\beta=\omega'$, omitting the additive constant and the factor $\frac{\Gamma+iQ}{2\pi i}$, changing C and transforming the σ -functions, we obtain, respectively, $\ln \frac{\sigma(u)}{\sigma_3(u)} + Cu$, $\ln \frac{\sigma_1(u)}{\sigma_3(u)} + Cu$,

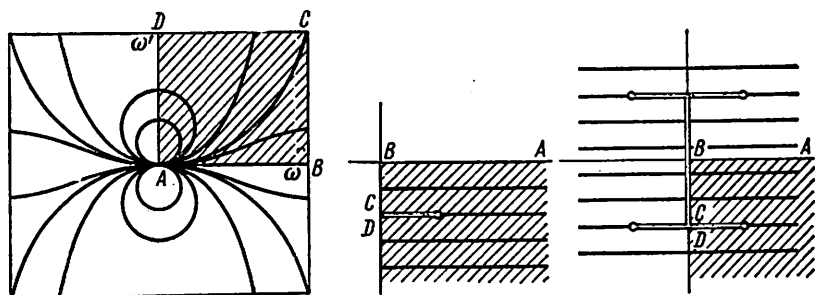


FIG. 123

$$\ln \frac{\sigma_2(u)}{\sigma(u)} + Cu. \quad \text{If } f(u+2\omega) = f(u), \quad \text{then} \quad f(u) = \frac{\Gamma+iQ}{2\pi i} \ln \frac{\vartheta_1\left(\frac{u-\alpha}{2\omega}\right)}{\vartheta_1\left(\frac{u-\beta}{2\omega}\right)} + c.$$

1327. 1), 2), 3) Doubly periodic flows with sources of strength 2π and -2π at zeros and poles of the functions $\operatorname{sn} u$, $\operatorname{cn} u$, $\operatorname{dn} u$ (Fig. 128). 1328. A doubly periodic flow with quadruplets at zeros of $p(u)$ (Fig. 129). 1329. Periodic flows with period 2ω (period of velocity!), with sources of strength 2π at zeros of $\vartheta_k(u)$ (Fig. 130, 1 for ϑ_4 and for ϑ_3 with displacement to the right by $1/2$; Fig. 130, 2 for ϑ_1 and for ϑ_2 with displacement to the right by $1/2$). 1330. 1) $f(z) = \frac{\Gamma}{2\pi i} \ln z + c$;

2) $f(z) = \frac{\Gamma}{2\pi i} \ln t(z) + c$, where $t(z)$ maps the domain D onto an annulus so that the directions of traversal of the boundary contours are preserved; 3) $f(z) = \frac{\Gamma}{2\pi i} \ln \frac{z-z_1}{z-z_2} + c$, where z_1, z_2 are mutually symmetrical with respect to each of the circles (i.e. are points of intersection of the circle orthogonal to the two given circles and the straight line joining their centres), the point z_1 lying inside the circle with circulation Γ ; 4) $f(z) = \frac{\Gamma}{2\pi i} \ln t(z) + c$, where the function $t(z)$ maps the domain D onto an annulus, preserving the direction of traversal of the contour with circulation Γ . 1331. $f(z) = \Phi\left(\frac{\omega}{\pi i} \ln z\right)$, where

$$\Phi(u) = -\frac{1}{4\pi^2 a} \left[\rho \frac{\vartheta_1'\left(\frac{u-\alpha}{2\omega}\right)}{\vartheta_1\left(\frac{u-\alpha}{2\omega}\right)} - \bar{\rho} \frac{\vartheta_1'\left(\frac{u+\alpha}{2\omega}\right)}{\vartheta_1\left(\frac{u+\alpha}{2\omega}\right)} \right] + c, \quad \alpha = \frac{\omega}{\pi i} \ln a.$$

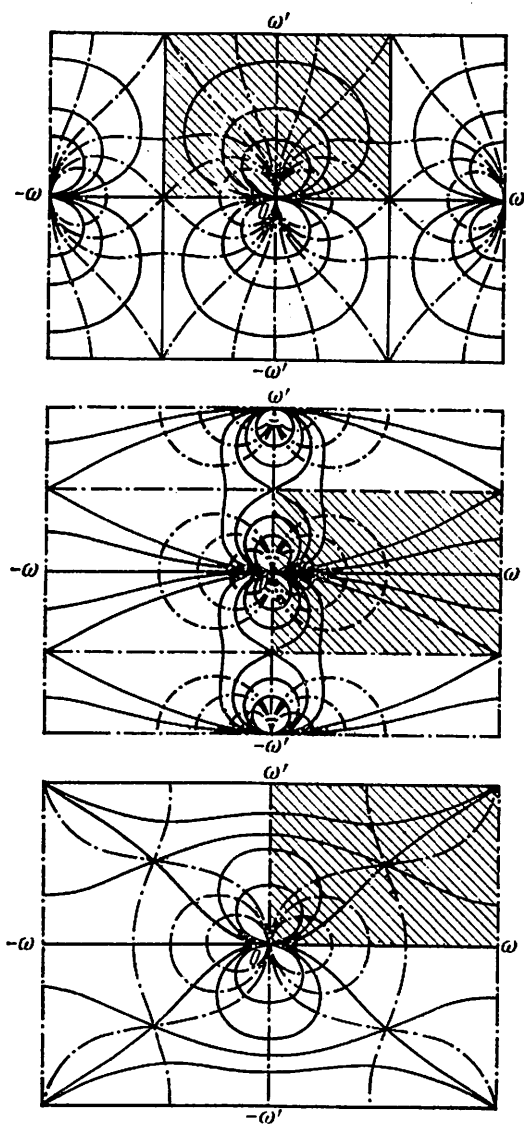


FIG. 124

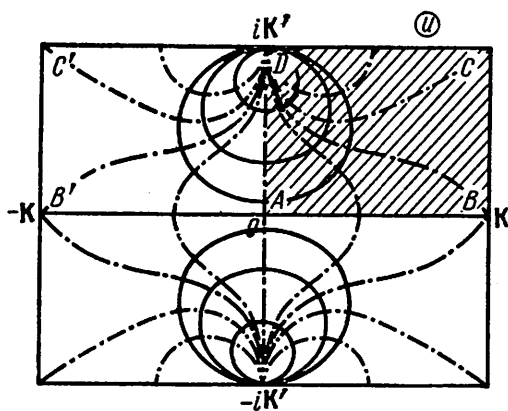


FIG. 125

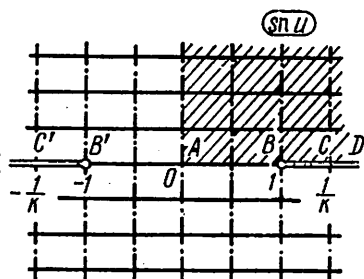


FIG. 126

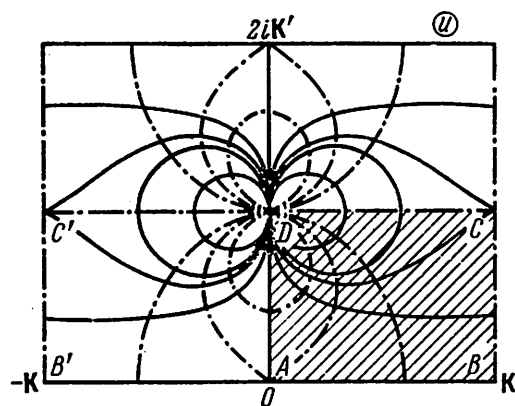
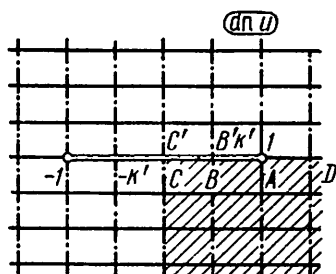
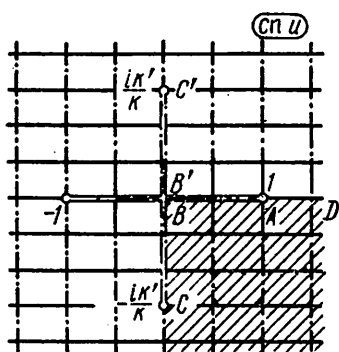


FIG. 127



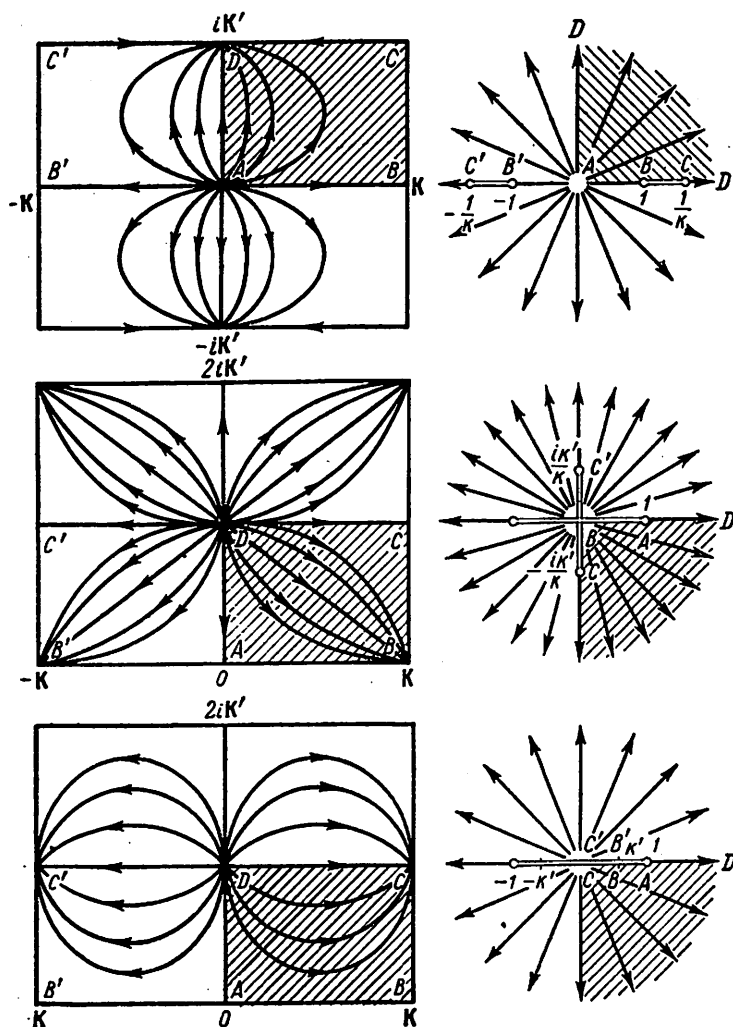


FIG. 128

The function $f(z)$ maps R onto the exterior of two parallel segments spaced at a distance of $\frac{|\operatorname{Re} ip|}{2\pi a}$ (Fig. 131). The end-points of the segments are determined from the condition $\Phi'(u)=0$. 1332. $f(z)=\Phi[u(z)]$, where

$$\Phi(u) = \frac{V}{2\omega} \left[ke^{i\alpha} \frac{\vartheta_1' \left(\frac{u-a}{2\omega} \right)}{\vartheta_1 \left(\frac{u-a}{2\omega} \right)} + \bar{k} e^{-i\alpha} \frac{\vartheta_1' \left(\frac{u-a}{2\omega} \right)}{\vartheta_1 \left(\frac{u-a}{2\omega} \right)} \right] + c,$$

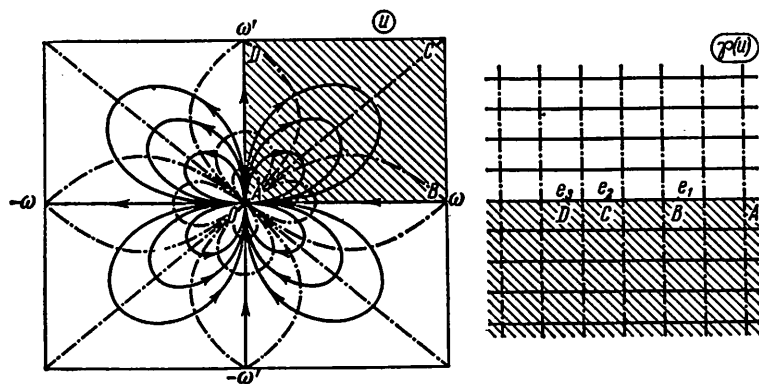


FIG. 129

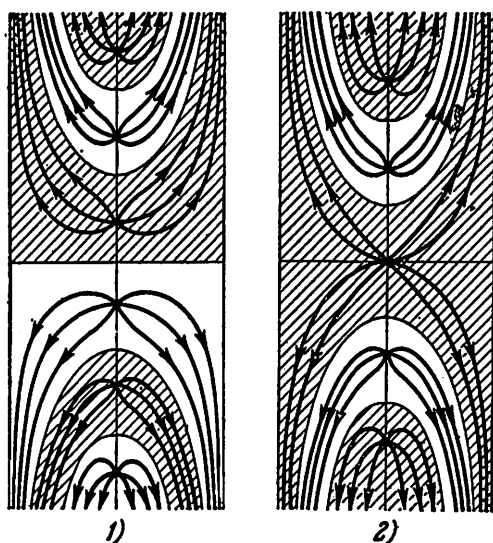


FIG. 130

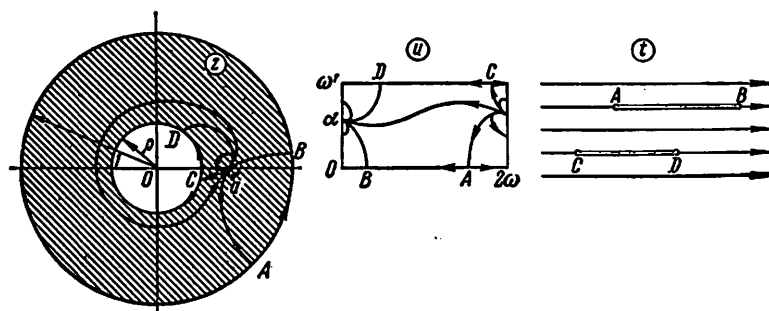


FIG. 131

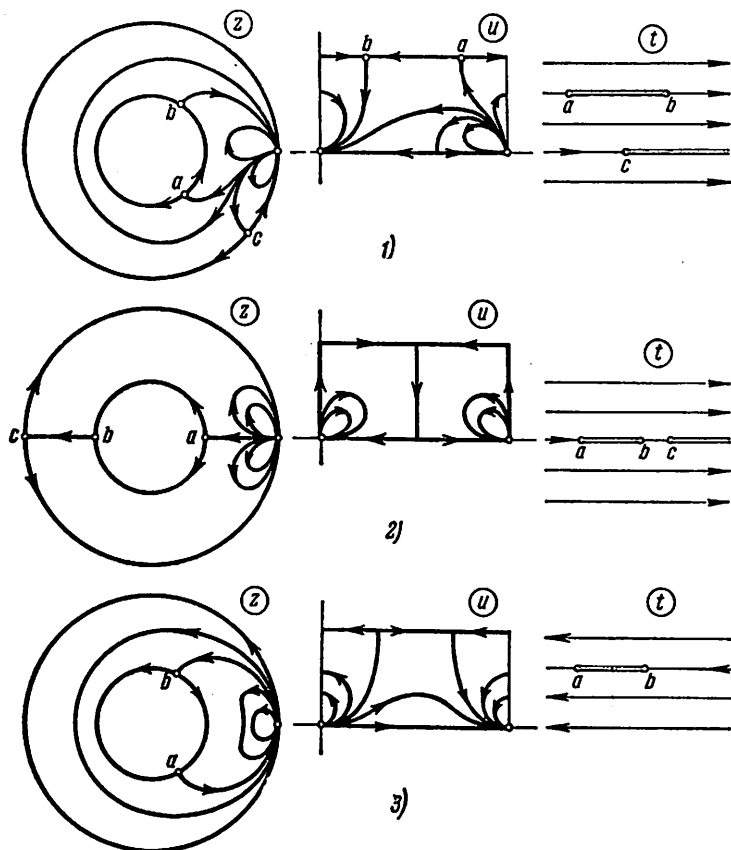


FIG. 132

and $u(z) = a + \frac{k}{z} + \dots$ is a function which maps the domain D onto a rectangle.

1333. $f(z) = \Phi\left(\frac{\omega}{\pi i} \ln z\right)$, where

$$\Phi(u) = A\wp(u) + \frac{B}{2\omega} \frac{\wp'_1\left(\frac{u}{2\omega}\right)}{\wp_1\left(\frac{u}{2\omega}\right)} + c,$$

$$A = -\frac{c_{-2}\omega^2}{\pi^2}, \quad B = \frac{i\omega}{\pi}(c_{-2} - c_{-1}).$$

The problem is possible if c_{-2} is a real number and the difference $c_{-2} - c_{-1}$ is purely imaginary. If $A \neq 0$, then the function $f(z)$ maps R onto the exterior of a horizontal ray and of a line-segment parallel to it and spaced at a distance of $\frac{|c_{-2} - c_{-1}|}{2}$ from it. The end-points of the line-segment and the starting point of the ray are determined from the condition $\Phi'(u) = 0$ (Fig. 132, 1) (Fig. 132, 2

illustrates the case $B=0$). But if $A=0$, i.e. there is only a doublet, then R is mapped onto a half-plane bounded by a horizontal straight line and with a cut along a horizontal line-segment passing at a distance of $\frac{|c-1|}{2}$ from the straight line (Fig. 132, 3). 1334. 1) A solution is possible if $\Gamma_1 - \Gamma_2 = \Gamma$; under this condition $f(z) = \Phi\left(\frac{\omega}{\pi i} \ln z\right)$, where

$$\Phi(u) = \frac{\Gamma}{2\pi i} \ln \frac{\vartheta_1\left(\frac{u-\alpha}{2\omega}\right)}{\vartheta_1\left(\frac{u+\alpha}{2\omega}\right)} + \frac{\Gamma_2}{2\omega} u + c, \quad \alpha = \frac{\omega}{\pi i} \ln a$$

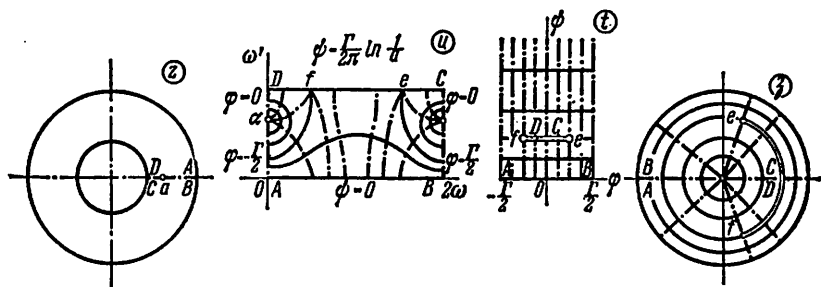


FIG. 133

(It should be kept in mind that with u varying from 0 to 2ω the increment of $\ln \frac{\vartheta_1\left(\frac{u-\alpha}{2\omega}\right)}{\vartheta_1\left(\frac{u+\alpha}{2\omega}\right)}$ is equal to $2\pi i$, and with u varying from $2\omega + i\omega'$ to $i\omega'$ it equals zero). Critical points of the flow are determined from the equation

$$\wp(u) = \wp(\alpha) + \frac{1}{2} \frac{\wp'(\alpha)}{\left[\zeta(\alpha) - \frac{\eta}{\omega} \alpha\right] + \lambda \left[\zeta(\omega') - \frac{\eta}{\omega} \omega'\right]},$$

where $\lambda = \frac{\Gamma_2}{\Gamma}$, and are situated on the sides of a rectangle with vertices $(0, \omega, \omega + \omega', \omega')$ and of rectangles symmetrical to it. In the case $\Gamma_2 = 0$ the function $\wp(z)$ maps R onto a circle with a cut along a circular arc (Fig. 133; $\Gamma > 0$).

In the case $\Gamma_2 = -\Gamma_1 = -\frac{\Gamma}{2}$ the function $\wp(z)$ maps R onto a two-sheeted domain formed by joining the exteriors of the circles $|\wp| > 1$ and $|\wp| > \frac{\rho}{a^2}$ along

cuts from $-\infty$ to $\wp_0 = -e^{\frac{4\pi\psi_0}{\Gamma}}$, where ψ_0 is the value of ψ at the critical point. The function $s(\wp)$ maps this two-sheeted domain onto the exterior of two lemniscates, and $f(z) = -\frac{\Gamma}{4\pi i} \ln [(s(z) - s_0)(s(z) + s_0)]$ (Fig. 134, $\Gamma > 0$, $a^2 < \rho$; the semi-strips in the t -plane should be joined along the common cut). In the general case $\psi = 0$ on the lower base of the rectangle in the u -plane, and φ

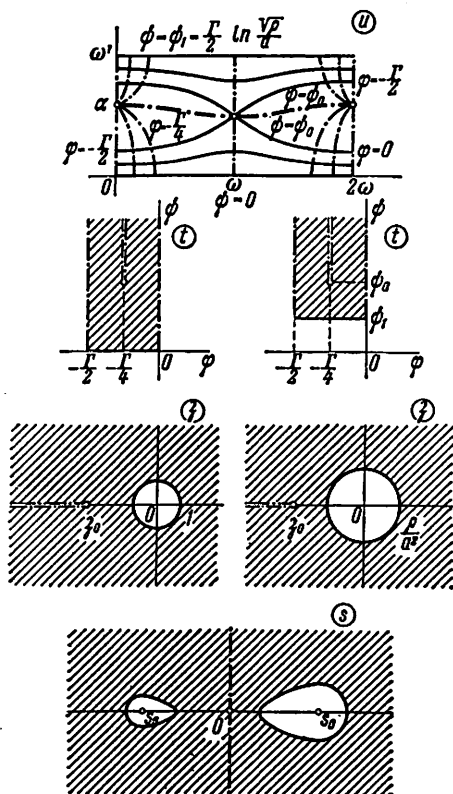


FIG. 134

varies from $-\frac{\Gamma}{2}$ to $\frac{\Gamma}{2} + \Gamma_2$, and on the upper base $\psi = \frac{\Gamma}{2\pi} \ln \frac{1}{a} + \frac{\Gamma_2}{2\pi} \ln \frac{1}{\rho}$ and φ varies from 0 to Γ_2 ($\Gamma > 0$). 2) $f(z) = \Phi[u(z)]$, where

$$\Phi(u) = \frac{V}{2\omega} \left[ke^{i\alpha} \frac{\vartheta_1' \left(\frac{u-a}{2\omega} \right)}{\vartheta_1 \left(\frac{u-a}{2\omega} \right)} + \bar{k} e^{-i\alpha} \frac{\vartheta_1' \left(\frac{u-\bar{a}}{2\omega} \right)}{\vartheta_1 \left(\frac{u-\bar{a}}{2\omega} \right)} \right] +$$

$$+ \frac{\Gamma_1 - \Gamma_2}{2\pi i} \ln \frac{\vartheta_1 \left(\frac{u-a}{2\omega} \right)}{\vartheta_1 \left(\frac{u-\bar{a}}{2\omega} \right)} + \frac{\Gamma_2}{2\omega} u + C,$$

and $u(z) = a + \frac{k}{z} + \dots$ maps the domain D onto an annulus. 1335. $u = \alpha x - \beta y$, $v = \beta x + \alpha y$, $E = -i\bar{c}$, a dipole $(\infty; -i\bar{c})$. 1336. $u = 2q\varphi$, $v = 2q \ln \frac{1}{r}$;

$E = \frac{2q}{r} e^{i\varphi}$; point charges $(a; 2q)$ and $(\infty; -2q)$. 1337. $u = 2q \arg \frac{z-b}{z-a}$,
 $v = 2q \ln \left| \frac{z-a}{z-b} \right|$; $E = \frac{2q(\bar{b}-\bar{a})}{(\bar{z}-\bar{a})(\bar{z}-\bar{b})}$; point charges $(b; 2q)$ and $(a; -2q)$.
 1338. $u = -2q \arg(z^2 - a^2)$, $v = 2q \ln |z^2 - a^2|$; $E = -\frac{4q\bar{z}}{z^2 - a^2}$; point charges
 $(a, -2q)$, $(-a; -2q)$ and $(\infty; 4q)$ (see Fig. 113). 1339. $u = \frac{|p|}{r} \sin(\varphi - \alpha)$,

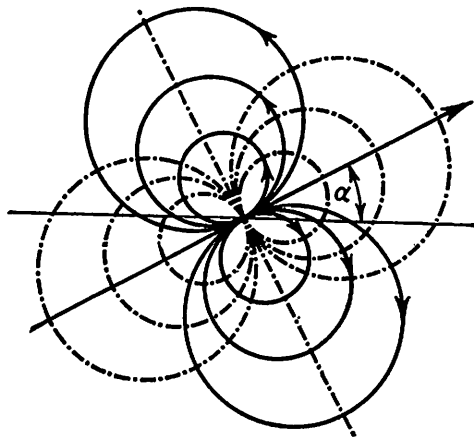


FIG. 135

$v = \frac{|p|}{r} \cos(\varphi - \alpha)$; $E = \frac{|p|}{r^2} e^{i(2\varphi - \alpha)}$; a dipole $(0; p)$ (Fig. 135). 1340. $u =$
 $= \left(r \pm \frac{R^2}{r}\right) \cos \varphi$, $v = \left(r \pm \frac{R^2}{r}\right) \sin \varphi$; $E = -i \left(1 \mp \frac{R^2}{r^2} e^{2i\varphi}\right)$; dipoles
 $(0; \mp iR^2)$ and $(\infty; -i)$ (see Figs. 110, 111). 1341. $u = -py + 2q\varphi$, $v = px +$
 $+ 2q \ln \frac{1}{r}$; $E = -p + \frac{2q}{r} e^{i\varphi}$; point charges $(0; 2q)$ and $(\infty; -2q)$; dipole (∞, p)

(cf. Fig. 118). 1342. $u = -py + \sum_{k=1}^n 2q_k \varphi_k$, $v = px + \sum_{k=1}^n 2q_k \ln \frac{1}{r_k}$, $E = -p +$
 $+ \sum_{k=1}^n \frac{2q_k}{r_k} e^{i\varphi_k}$, where $z - a_k = r_k e^{i\varphi_k}$; point sources $(a_k, 2q_k)$; a dipole $(\infty; p)$

(Fig. 136). 1343. 1) The magnitude of the point charge remains unchanged; the law of variation of the dipole moment is the same as in Problem 1289; 2) the sign of the charge is reversed; the variation law of the dipole moment is the same as in Problem 1291 when continued across the streamline. 1344. $v = 2qg(z, a)$.

1345. $w = 2qi \ln \frac{z - \bar{z}_0}{z - z_0} + c$. 1346. 1) and 2) $w = 2qi \ln \frac{R^2 - z_0 \bar{z}}{R(z - z_0)} + c$

1347. $w = 2qi \ln \frac{1}{f(z)} + \text{const}$, where $f(z) = \frac{z - \sqrt{z^2 - c^2}}{a - b}$, $c^2 = a^2 - b^2$.

1348. $w = 2qi \ln \frac{1}{f(z)} + c$, where $f(z) = \frac{z - \sqrt{z^2 - R^2}}{R}$. 1349. $w = 2qi \ln \frac{1}{f(z)}$,

where $t = f(z)$ is determined from the equation $z = \frac{2d}{B \left(\frac{1}{2}, \frac{3}{4} \right)} \int_1^t \frac{\sqrt{1-t^4}}{t^3} dt + \frac{d}{2}$

(see Problem 1236 for $n=4$ and 1271). 1350. $w = 2qi \ln \frac{1}{f(z)} + c$, where

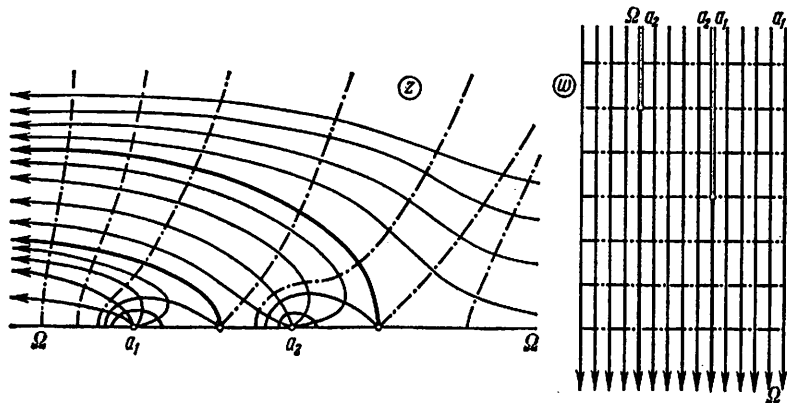


FIG. 136

$f(z) = \frac{1 - \operatorname{cn} \left(\frac{K}{a} z, k \right)}{\operatorname{sn} \left(\frac{K}{a} z, k \right)}$ and k is determined from the equation $\frac{K'}{K} = \frac{b}{a}$ (see

Problem 1268). 1351. $w = 2qi \ln \frac{1}{f(z)} + c$, where $f(z) = \frac{\vartheta_1 \left(\frac{z-z_0}{4a} \right) \vartheta_1 \left(\frac{z-z_2}{4a} \right)}{\vartheta_1 \left(\frac{z-z_1}{4a} \right) \vartheta_1 \left(\frac{z-z_3}{4a} \right)}$

and $\omega = 2a$, $\omega' = 2ib$, $z_1 = (4a - x_0) + iy_0$, $z_3 = (4a - x_0) + i(4b - y_0)$, $z_2 = x_0 + i(4b - y_0)$. 1352. $w = \frac{pi}{z-a} + \frac{p^*i}{z-a^*} + c$ ($a \neq 0$, $a^* = \frac{R^2}{a}$, $p^* = \frac{R^2 \bar{p}}{a^2}$),

$w = \frac{pi}{z} - \frac{\bar{p}i}{R^2} z + c$ ($a=0$), c is a real number. Compare with Problem 1294(2).

1353. $w = \frac{pi}{z-a} + \frac{p^*i}{z-a^*} + c$ ($a \neq \infty$, $a^* = \frac{R^2}{a}$, $p^* = \frac{R^2 \bar{p}}{a^2}$), $w = \frac{pi}{z} - \frac{\bar{p}i}{R^2} z + c$

($a = \infty$). See Problem 1296(2). 1354. $w = \rho (z \cos \alpha + i \sin \alpha \sqrt{z^2 - R^2}) + \text{const.}$

1355. $w = \frac{\rho}{a-b} [(az - b \sqrt{z^2 - c^2}) \cos \alpha - i(bz - a \sqrt{z^2 - c^2}) \sin \alpha] + \text{const.}$

where $c^2 = a^2 - b^2$. 1356. $w = \frac{2K\rho}{a \operatorname{sn} u} (\cos \alpha + i \sin \alpha \operatorname{cn} u)$, where $u = \frac{K}{a} z$ and

$\frac{K'}{K} = \frac{b}{a}$ (see Problem 1350.) 1357. 1) If $pi = \rho e^{i\alpha}$, then

$$f(z) = \rho t'(a) \left\{ \left[\frac{1}{t(z)} + t(z) \right] \cos \alpha + i \left[\frac{1}{t(z)} - t(z) \right] \sin \alpha \right\} + c;$$

$$2) f(z) = \rho i t(z) - \frac{R^2 \bar{p} i}{t(z)} + c = \rho \left\{ \left[t(z) + \frac{R^2}{t(z)} \right] \cos \alpha + i \left[t(z) - \frac{R^2}{t(z)} \right] \sin \alpha \right\} + c,$$

where $pi = \rho e^{i\alpha}$. The functions in square brackets effect normalized conformal mappings of D onto the exterior of a horizontal (or vertical) line segment.

$$1358. w = \sum_{k=1}^n 2q_k i \ln \frac{1}{f(z, a_k)} + i f'(a, a) \left[\frac{p}{f(z, a)} - \bar{p} f(z, a) \right] + c, \quad \text{where}$$

$f(z, a_k)$ and $\bar{f}(z, a)$ conformally map D onto the unit circle with normalization $f(a_k, a_k) = \bar{f}(a, a) = 0$, $f'(a, a) > 0$ and c is a real number.

$$1360. v(z, a) = \begin{cases} -\ln \frac{1}{|z - \bar{a}|} & \text{if } \operatorname{Im} z \geq 0, \\ -\ln \frac{1}{|z - a|} & \text{if } \operatorname{Im} z \leq 0; \end{cases}$$

$$\rho(x, a) = -\frac{1}{\pi} \frac{\beta}{(x - \alpha)^2 + \beta^2} \quad (a = \alpha + i\beta).$$

1361. 1) Inside the circle

$$v(z, a) = \ln \frac{|R^2 - \bar{a}z|}{R} = \begin{cases} -\left[\ln \sqrt{\frac{1}{|z - \frac{R^2}{\bar{a}}|}} + \ln \frac{|a|}{R} \right] & \text{if } a \neq 0, \\ \ln R, & \text{if } a = 0. \end{cases}$$

Outside the circle $v(z, a) = -\ln \frac{1}{|z - a|}$. Density

$$\rho(Re^{i\theta}, a) = -\frac{1}{2\pi R} \frac{R^2 - |a|^2}{R^2 - 2R|a|\cos(\theta - \alpha) + |a|^2} \quad (a = |a|e^{i\alpha}).$$

In particular, for $a=0$ it has a constant value $-\frac{1}{2\pi R}$ and induces a layer potential of constant value $\ln R$ inside the circle and $\ln |z|$ outside the circle.

$$2) v(z, a) = \begin{cases} \ln \frac{|R^2 - \bar{a}z|}{R}, & a \neq \infty, |z| \geq R, \\ -\ln \frac{1}{|z - a|}, & a \neq \infty, |z| \leq R; \end{cases}$$

$$v(z, \infty) = \begin{cases} \ln |z| & \text{if } |z| \geq R, \\ \ln R & \text{if } |z| \leq R; \end{cases}$$

$$\rho(Re^{i\theta}, a) = -\frac{1}{2\pi R} \frac{|a|^2 - R^2}{R^2 - 2R|a|\cos(\theta - \alpha) + |a|^2} \quad (a = |a|e^{i\alpha} \text{ and } a \neq \infty).$$

If $a = \infty$, the same potential is induced as in the preceding case for $a=0$.

$$1362. v(z, \infty) = \ln \frac{|z + \sqrt{z^2 - R^2}|}{2}, \quad \rho(x, \infty) = -\frac{1}{2\pi \sqrt{R^2 - x^2}} (|x| < R).$$

$$1363. v(z, \infty) = \ln \frac{|z + \sqrt{z^2 - c^2}|}{2} \quad \text{outside the ellipse, } v(z, \infty) = -\ln 2(\alpha - \beta)$$

inside the ellipse. Density $\rho(z, \infty) = -\frac{1}{2\pi \sqrt{|\zeta^2 - c^2|}}$ (ζ on the ellipse, $c^2 = \alpha^2 - \beta^2$). 1364. $\rho(\zeta) = \frac{1}{2\pi} \frac{\partial g(\zeta, \infty)}{\partial n}$. 1365. $\rho(\zeta) = \frac{1}{2\pi R}$. 1366. $\rho(x) = \frac{1}{2\pi \sqrt{R^2 - x^2}}$ ($|x| < R$). 1367. $\rho(\zeta) = \frac{1}{2\pi \sqrt{|\zeta^2 - c^2|}}$ ($c^2 = a^2 - b^2$).

1368. R . 1369. $\frac{1}{2R}$. 1370. $\frac{1}{2(a-b)}$. 1371. a . 1374. If $\omega(z; \Delta)$, the harmonic measure of the interval Δ of the real axis at the point z with respect to the upper half-plane: $\omega(z; \Delta) = \frac{1}{\pi} \int_{\Delta} \frac{\partial}{\partial n} \ln \frac{1}{|t-z|} dt$ (see Problems 1065 and

1373), then (omitting real additive constant): 1) $w = \frac{\varphi}{\pi} \ln(z-a)$, $v = \varphi \omega(z; \Delta)$, $\Delta = (-\infty, a)$; 2) $w = \frac{\varphi}{\pi} \ln \frac{z-b}{z-a}$, $v = \varphi \omega(z; \Delta)$, $\Delta = (a, b)$; 3) $w = \frac{1}{\pi} \left[\varphi_1 \ln(z-a_1) + \varphi_2 \ln \frac{z-a_2}{z-a_1} + \dots + \varphi_n \ln \frac{z-a_n}{z-a_{n-1}} \right]$,

$v = \sum_{k=1}^n \varphi_k \omega(z; \Delta_k)$, $\Delta_k = (a_{k-1}, a_k)$, $a_{-1} = -\infty$; 4) w and v are obtained from the expressions given in the answer to case 3) by substituting $\varphi_k - \varphi_0$ for φ_k , adding $i\varphi_0$ to w and φ_0 to v . The potential can also be represented in

the form $\sum_{k=0}^n \varphi_k \omega(z; \Delta_k)$, $\Delta_0 = (a_n, \infty)$. 1375. 1) $w(z) = \frac{id}{\ln \mu} \ln z + \text{const}$,

$v(z) = \frac{d}{\ln \mu} \ln r + \text{const}$, where $\mu = \frac{r_2}{r_1}$; 2) $w(z) = \frac{id}{\ln \mu} \ln t(z) + \text{const}$,

$v(z) = \frac{d}{\ln \mu} \ln |t(z)| + \text{const}$, where μ is the modulus of domain D (see page 38) and $t(z)$ maps D conformally onto an annulus.

1377. $w(z)$ is defined by the formula given in Problem 1376, where:

$$1) t(z) = \sqrt{\mu} \frac{z - \sqrt{a^2 - R^2}}{z + \sqrt{a^2 - R^2}}, \quad \mu = \frac{a}{R} + \sqrt{\left(\frac{a}{R}\right)^2 - 1};$$

$$2) \text{ and } 3) t(z) = \sqrt{\mu} \frac{z - x_1}{z - x_2}, \quad \mu = \left(\frac{R - x_1}{R - x_2}\right)^2, \quad x_1, x_2 \text{ are determined from}$$

the equation $x^2 + \frac{r^2 - R^2 - a^2}{a} x + R^2 = 0$ ($x_1 < x_2$);

$$4) t(z) = \frac{1}{c} (z + \sqrt{z^2 - c^2}), \quad \mu = \frac{a+b}{c}, \quad c = \sqrt{a^2 - b^2};$$

$$5) t = e^{\frac{\pi(K+u)}{K'}}, \quad z = \text{sn}(u, k), \quad \mu = e^{\frac{2\pi K}{K'}} (|\text{Re } u| < K, |\text{Im } u| \leq K' \text{ (see Problem 1267(1))});$$

$$6) t = e^{\frac{\pi(K+u)}{K'}}, \quad z = \frac{2Kh}{\pi} \left[Z(u) + \frac{\text{dn } u \text{ cn } u}{\text{sn } u} \right], \quad \mu = e^{\frac{2\pi K}{K'}}, \text{ where } k \text{ is determi-}$$

ned from the equations $KZ(\mu) = \frac{\pi h}{2a}$, $\operatorname{dn}^2 \beta = \frac{E}{K}$ [see Problem 1267(15)].

$$1379. 1) \omega_1(z) = 1 - \frac{\ln|z|}{\ln \mu}, \quad \omega_2(z) = \frac{\ln|z|}{\ln \mu};$$

$$\rho_1(\zeta) = -\rho_2(\zeta) = \begin{cases} \frac{1}{2\pi\mu \ln \mu} & \text{onto } |z|=1, \\ -\frac{1}{2\pi\mu \ln \mu} & \text{onto } |z|=\mu; \end{cases} \quad \rho_{11} = \rho_{22} = -\rho_{12} = -\rho_{21} = \frac{1}{\ln \mu};$$

$$2) \omega_1(z) = 1 - \omega_2(z), \quad \omega_2(z) = \frac{\ln|t(z)|}{\ln \mu};$$

$$\rho_1(\zeta) = -\rho_2(\zeta) = \begin{cases} \left| \frac{t'(\zeta)}{t(\zeta)} \right| \frac{1}{\ln \mu} & \text{onto } \Gamma_1, \\ -\left| \frac{t'(\zeta)}{t(\zeta)} \right| \frac{1}{\ln \mu} & \text{onto } \Gamma_2; \end{cases} \quad \rho_{11} = \rho_{22} = -\rho_{12} = -\rho_{21} = \frac{1}{\ln \mu},$$

where $t(z)$ maps D conformally onto the annulus $1 < |t| < \mu$; here the contour Γ_1 goes into the circle $|t|=1$. 1381. $v(z) = \operatorname{Im} f(z) - \sum_{k=1}^n \alpha_k \omega_k(z)$, where $f(z)$ maps D conformally onto a plane with horizontal cuts, and

$$f(z) = \begin{cases} \frac{pi}{z-a} + \dots & \text{if } a \neq \infty, \\ piz + \dots & \text{if } a = \infty. \end{cases}$$

$$1382. \quad v(z) = 2qg(z, a) + \sum_{k=1}^n \alpha_k \omega_k(z), \quad \omega_k(z) = -\frac{1}{2\pi} \int_{\Gamma_k} \frac{\partial g(\zeta, z)}{\partial n} ds. \quad 1383. \quad \text{If}$$

$f(a) = \infty$, then the field is generated by a dipole $(a; p)$, where p is determined from the expansion of $f(z)$ in the neighbourhood of the point a :

$$f(z) = \begin{cases} \frac{pi}{z-a} + \dots & \text{if } a \neq \infty, \\ piz + \dots & \text{if } a = \infty. \end{cases}$$

1384. 1) If $f(a)=0$, $f(b)=\infty$, then the field is generated by point charges $(a; 2q)$, $(b; -2q)$, the flux of the intensity vector across each boundary contour being equal to zero; 2) if $f(a)=0$, then the field is generated by a point charge $(a; 2q)$; here the flux of the intensity vector across the boundary contour corresponding to a circle, in the direction of a normal external to the domain D , is equal to $4\pi q$, and across every other contour, to zero; 3) the field is regular everywhere. The flux of the intensity vector across boundary contours turning into circles, in the direction of normal external to the domain D , equals $\pm 4\pi q$ (+ for contour going into external circle), and across every other contour, zero.

1385. See Problems 1332 and 1334. 1386. 1) $v(z) = \sum_{k=1}^{n-1} \alpha_k \omega_k(z) + c$, where α_k are determined uniquely from system

$$\sum_{k=1}^{n-1} p_{ik} \alpha_k = 2q_i \quad (i=1, 2, \dots, n-1)$$

(see Problem 1077) and c is an arbitrary real number. The problem is equivalent to constructing a flow in D streamlining the boundary contours Γ_k with circulations $4\pi q_k$ ($k=1, 2, \dots, n$) if $\infty \in D$, and circulations $4\pi q_k$ ($k=1, 2, \dots, n-1$), $-4\pi q_n$ if $\infty \notin D$ (Γ_n is the external contour); 2) $v(z) = v_0(z) -$

$-2qg(z, a)$, where $v_0(z)$ is defined as in case 1) from the plate charges of the layer $2q_k + 2q'_k$, where $q'_k = -\frac{q}{2\pi} \int_{\Gamma_k} \frac{\partial g(\zeta, a)}{\partial n} ds$. 1387. $v(z) =$

$$= 2q_1 \ln \frac{r_2}{|z|} + c \text{ if } q=0, \text{ and}$$

$$v(z) = 2(q_1 - \lambda q) \ln \frac{r_2}{|z|} - 2qg(z, a) + c, \text{ where}$$

$$\lambda = \frac{\ln a}{\ln \rho}, \quad g(z, a) = |z|^\lambda \ln \left| \frac{\vartheta_1 \left(\frac{\ln z + \ln a}{2\pi i} \right)}{\vartheta_1 \left(\frac{\ln z - \ln a}{2\pi i} \right)} \right|$$

for $\tau = \frac{\ln \rho}{\pi i}$, where $\rho < a < 1$ if $q \neq 0$ (in terms of Problem 1334 Green's function $g(z, a) = \text{Im} \Phi \left(\frac{\omega}{\pi i} \ln z \right)$ for $\Gamma = 2\pi$ and $\Gamma_2 = -2\pi$; the latter from the condition $\psi = 0$ on the boundary of the annulus). 1388. The source ($a; q$) passes into the source ($a^*; -q$), where a^* is a point symmetrical to a . The function

$$u = \frac{q}{2\pi} \ln \frac{1}{|f(z, a)|} + c,$$

where $f(z, a)$ maps the domain D conformally onto the unit circle (here and elsewhere the coefficient of heat conductivity k is taken as unity).

$$1389. \quad u = \frac{q}{2\pi} \ln \left| \frac{z - \bar{a}}{z - a} \right| + c.$$

$$1390. \quad u = \frac{q}{2\pi} \ln \left| \frac{R^2 - \bar{a}z}{R(z - a)} \right| + c.$$

$$1391. \quad u = \frac{q}{2\pi} \ln \left| \frac{\sin \frac{\pi z}{2a} + i \sinh \frac{\pi h}{2a}}{\sin \frac{\pi z}{2a} - i \sinh \frac{\pi h}{2a}} \right| + c.$$

$$1392. \quad u = \frac{q}{2k} \ln \left| \frac{t + \frac{i}{\sqrt{k}}}{t - \frac{i}{\sqrt{k}}} \right| + c,$$

$$t = \text{sn} \left[\frac{K}{a} (z + ib), u \right], \text{ where } k \text{ is determined from the ratio } \frac{K'}{K} = \frac{2b}{a}.$$

1393. 1) Green's function $g(z, a)$ of the domain D may be considered as the temperature generated in D by a heat source ($a; 2\pi$) when the temperature on

the boundary is equal to zero; 2) $u(z) = \frac{q}{2\pi} g(z, a) + \sum_{k=1}^n u_k \omega_k(z)$, where $\omega_k(z)$

is a harmonic measure of Γ_k . 1394. $u = \frac{q}{2\pi} g(z, a) + \frac{u_2 - u_1}{\ln \frac{r_2}{r_1}} \ln \frac{|z|}{r_1} + u_1$, where

$g(z, a)$ is Green's function (see the answer to Problem 1387).

Chapter XI

$$1397. 1) \text{ and } 2) A = \frac{1}{2} [(a_1 + b_2) + i(a_2 - b_1)],$$

$$B = \frac{1}{2} [(a_1 - b_2) + i(a_2 + b_1)],$$

$$C = c_1 + ic_2, \quad A_1 = \frac{1}{2\Delta} [(a_1 + b_2) + i(b_1 - a_2)],$$

$$B_1 = \frac{1}{2\Delta} [(b_2 - a_1) - i(a_2 + b_1)],$$

$$C_1 = \frac{1}{\Delta} [(c_2 b_1 - c_1 b_2) + i(c_1 a_2 - c_2 a_1)].$$

4) Mapping (2) is reduced to the projection of z -points onto a straight line $w = \mu e^{i\frac{\beta-\alpha}{2}}$ ($-\infty < \mu < \infty$), rotation through an angle β about the origin and a similarity transformation with coefficient $|A|$. Here the entire z -plane is transformed into a straight line $w = \lambda e^{i\frac{\alpha+\beta}{2}}$ ($-\infty < \lambda < \infty$). 1407. It is possible to

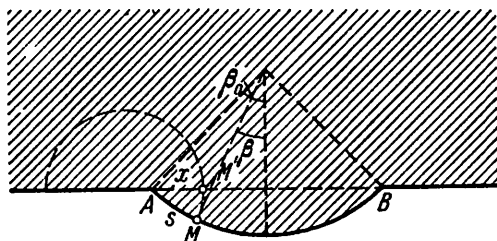


FIG. 137

construct the desired quasi-conformal transformation with characteristic $p \leq \leq \frac{R+|a|}{R-|a|}$. 1408. $u = \frac{x}{\cos \alpha}$, $v = y - x \tan \alpha$, $p = \frac{1 + \sin \alpha}{\cos \alpha}$. 1409. Solution.

With the aid of the function $\zeta = \ln \frac{z-a}{z-b}$ (the values $z=a$ and $z=b$ correspond to the points A and B) map the given lune conformally onto a strip of width $\pi + \beta_0$. Contract this strip to obtain a strip of width π (a quasi-conformal mapping with characteristic $p = 1 + \frac{\beta_0}{\pi}$), and, using the inverse function $z(\zeta)$, map the latter strip onto the strip of width π . The arc AM (Fig. 137) of length s will then occupy the position of line segment AM' of length x ; the point M moves along a circle of Apollonius with respect to the points A and B (shown dashed), and

$$\frac{dx}{ds} = \frac{\cos^2 \frac{\beta_0}{2}}{\cos^2 \frac{\beta}{2}} \geq \cos^2 \frac{\beta_0}{2} \quad (\text{prove this!}).$$

In the half-plane obtained expand the vertical semi-strip resting upon the seg-

ment AB up to the length of the arc $\widetilde{AB}$ so as to preserve the lengths throughout the arc $\widetilde{AB}$. The resulting quasi-conformal mapping has the characteristic

$$\rho \leq \left(1 + \frac{\beta_0}{\pi}\right) \sec^2 \frac{\beta_0}{2}.$$

$$1411. \quad A = \frac{1}{4} [(a-d) + i(c+b)], \quad B = \frac{1}{4} [(a+d) + i(c-b)], \quad F = \frac{1}{2} (f + ig).$$

$$1412. \quad b = q_2(z) \bar{a} \text{ or } b = \frac{1}{q_2(z)} \bar{a}. \text{ In particular, we may take } w = \omega + q_2(z) \bar{\omega}.$$

Transformation $w = a\omega + b\bar{\omega}$ is nondegenerate if $|q_2(z)| \neq 1$. 1413. $b = \lambda \bar{a}$, where λ is determined from the equation $\bar{q}_2 \lambda^2 - \lambda(1 + |q_2|^2 - |q_1|^2) + q_2 = 0$ or $(\lambda - q_2) \left(\frac{1}{\lambda} - \bar{q}_2\right) = |q_1|^2$. The transformation is nondegenerate if $|q_1| + |q_2| < 1$ or $\|q_1\| - \|q_2\| > 1$. 1414. The higher-degree terms of the equation of the problem define a quasi-conformal mapping with two pairs of characteristics $(\rho, \theta), (\rho_1, \theta_1)$ depending on $q_1(z)$ and $q_2(z)$. The transformation $\zeta(z)$ is a quasi-conformal mapping with one pair of characteristics ρ, θ ; $q_1^* = -\frac{\rho-1}{\rho+1} e^{2i\varphi}$, $q_2^* = \frac{\rho_1-1}{\rho_1+1} e^{2i\varphi_1}$ (see Problems 1401 and 1403). 1422. $F(z) = -2z \ln \ln \frac{1}{r} + 2z \ln \ln \frac{1}{R}$;

$$\frac{\partial F}{\partial x} = -2 \ln \ln \frac{1}{r} - 2 \frac{e^{i\varphi} \cos \varphi}{\ln r} + 2 \ln \ln \frac{1}{R};$$

$$\frac{\partial F}{\partial y} = -2i \ln \ln \frac{1}{r} - 2 \frac{e^{i\varphi} \sin \varphi}{\ln r} + 2i \ln \ln \frac{1}{R},$$

$$\frac{\partial F}{\partial z} = -2 \ln \ln \frac{1}{r} - \ln \frac{1}{r} + 2 \ln \ln \frac{1}{R},$$

$$\frac{\partial F}{\partial \bar{z}} = \varphi(z) = \frac{e^{2i\varphi}}{\ln \frac{1}{r}}.$$

TO THE READER

Mir Publishers would be grateful for your comments on the content, translation and design of this book. We would also be pleased to receive any other suggestions you may wish to make.

Our address is:
USSR, 129820, Moscow, I-110, GSP
Pervy Rizhsky Pereulok, 2
MIR PUBLISHERS

Printed in the Union of Soviet Socialist Republics